FlightSafety international

KING AIR C90A/B/GT PILOT TRAINING MANUAL

VOLUME 2
AIRCRAFT SYSTEMS

REVISION 1.3

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CHAPTER 1
AIRCRAFT GENERAL

INTRODUCTION

A good basic understanding of the airplane will help in studying the individual systems and their operation. This chapter provides basic and background information needed to learn the details of airplane operation and performance to be studied in other chapters.

GENERAL

This chapter of the training manual presents an overall view of the airplane. This includes external familiarization, cabin arrangements, and cockpit layout.

In this chapter of the training manual you will find diagrams and data describing the airplane in general and its systems that are not included in the *Pilot's Operating Handbook (POH)*.

Reference material in this training manual is organized into 15 chapters (with two unused tabs) covering all airplane systems. Each chapter is complete and independent and can be referred to in any sequence.

Following are brief descriptions of the subject matter in each chapter. All material is discrete to the Beechcraft King Air C90A and C90B.

AIRPLANE SYSTEMS

GENERAL

The "Systems Description" section of the *POH* gives a brief description of all the systems incorporated in the King Air C90A and C90B. Additional description and details of these systems are included in separate chapters of this training manual. The *POH* information is updated as required and always supersedes any information in this training manual.

CHAPTERS

Aircraft General

Chapter 1—"Aircraft General" presents an overall view of the airplane. This includes external familiarization, cabin arrangement, and cockpit layout. In this chapter you will find diagrams and data describing the airplane in general that are not included in the *Pilot's Operating Handbook*.

Electrical Power Systems

Chapter 2—"Electrical Power Systems" describes the airplane electrical system and its components. The electrical system is discussed to the extent necessary for pilot management of all normal and emergency operations. The location and purpose of switches, indicators, lights, and circuit breakers are noted. DC and AC generation and distribution are described. This chapter also includes electrical system limitations and a discussion of potential electrical system faults.

Lighting

Chapter 3—"Lighting" discusses cockpit lighting, cabin lighting, and exterior lighting. All lights are identified and located. The location and use of controls for the lighting system are also included.

Master Warning System

Chapter 4—"Master Warning System" presents a description and discussion of the warning, caution, and advisory annunciator panels. Each annunciator is described in detail, including its purpose and associated cause for illumination. Emphasis is on corrective action required by the pilot if an annunciator is illuminated.

Fuel System

Chapter 5—"Fuel System" presents a description and discussion of the fuel system. The physical layout of fuel cells are described. Correct use of the boost pumps, transfer pumps, crossfeed, and firewall shutoff valves are discussed. Locations and types of fuel drains and correct procedures for taking and inspecting fuel samples are detailed. This chapter includes a list of approved fuels and procedures for fuel servicing.

Powerplant

Chapter 7—"Powerplant" presents a discussion of the Pratt and Whitney PT6A turboprop engines. Engine theory and operating limitations are described, and normal pilot procedures are detailed. Crewmembers must have sufficient knowledge of the PT6A series engines to understand all normal and emergency procedures.

This chapter also describes the propeller system. Location and use of propeller controls, principle of operation, reversing, and feathering are discussed.

Fire Protection

Chapter 8—"Fire Protection" describes the fire warning and protection systems. Operation and testing information for the fire detection and fire-extinguishing systems is included.

Pneumatics

Chapter 9—"Pneumatics" presents a discussion of pneumatic and vacuum systems. Sources and operation of pneumatic and vacuum air are described. Acceptable gage readings and normal and abnormal system indications are outlined.

Ice and Rain Protection

Chapter 10—"Ice and Rain Protection" presents a description and discussion of the anti-ice and deice systems. All of the anti-ice, deice, and rain protection systems in this airplane are described, showing location, controls, and how they are used. The purpose of this chapter is to acquaint the pilot with all the systems available for flight in icing or heavy rain conditions and their controls. Procedures in case of malfunction in any system are included. This also includes information concerning preflight deicing and defrosting.

Air Conditioning

Chapter 11—"Air Conditioning" presents a description of the air-conditioning, heating, and fresh air systems. Each subsystem discussion includes general description, principle of operation, controls, and emergency procedures.

Pressurization System

Chapter 12—"Pressurization" presents a description of the pressurization system. The function of various major components, their physical location, and operation of the pressurization system controls are discussed. Where necessary, references are made to the environmental system as it affects pressurization.

Landing Gear and Brakes

Chapter 14—"Landing Gear and Brakes" presents a description and discussion of the landing gear system, landing gear controls, and operating limitations. The indicator system and emergency landing gear extension are also described.

This chapter also discusses the wheel brake system. Correct use of the brakes and parking brakes, along with brake system description, and what to look for when inspecting brakes are detailed.

Flight Controls

Chapter 15—"Flight Controls" describes the four-segment Fowler-type flap system. System controls and limitations are considered, with reference to operation as outlined in the *Pilot's Operating Handbook*.

This chapter also describes the rudder boost system. This system is designed to reduce pilot effort if single-engine flight is encountered.

Avionics

Chapter 16—"Avionics" describes the standard avionics installation for the King Air C90A and C90B. The avionics controls, along with the weather radar, are mounted on an isolation panel in the center of the instrument panel so that it is easily available to the pilot or copilot. Individual audio switches, across the top of the panel, control audio to the speakers or headphones for the pilot and copilot. There are separate sets of controls for pilot and copilot so that each can select audio from any nav or comm receiver.

A glossary of avionics terminology is included in an Appendix at the back of this training manual.

This chapter also presents a discussion of the dual pitot-static system, which is vital to airspeed indications in the airplane. The principle of operation, sources of static and pitot pressure, instruments that depend on the system, and the pilot's alternate static air source are covered.

Miscellaneous Systems

Chapter 17—"Miscellaneous Systems" presents a summary of the oxygen system and its components. General description, principle of operation, system controls, and emergency procedures are included. Use of the oxygen duration chart involves working simulated problems under various flight conditions. FAR requirements for crew and passenger oxygen needs are part of the discussion, as well as the types and availability of oxygen masks. Local servicing procedures referenced in the *Pilot's Operating Handbook* are also included.

CHANGES DISTINGUISHING MODEL C90B FROM MODEL C90A

The following are significant changes that differentiate the Model C90B from the Model C90A. The C90B serial numbers are LJ-1288, LJ-1295, LJ-1302, LJ-1303, LJ-1305 through LJ-1308, LJ1311, LJ-1312, LJ-1314 through LJ-1316, LJ1318, LJ-1320 and subsequent.

- Significantly reduced cabin sound and vibration levels.
- Four-blade dynamically balanced 90inch-diameter McCauley or Hartzell propellers. Includes:
  - Improved low-friction hub to improve propeller synchrophasing.
  - New streamlined, more aerodynamic propeller spinners.
  - A gated ground fine power lever position which provides improved ground handling, as well as reduced accelerate-stop and landing distances. The ground fine position allows a flatter propeller blade angle to be used for taxi and for deceleration during accelerate-stop and landing.
  - Dynamic propeller balancing to reduce propeller vibration and associated airframe vibration to improve comfort and reduce fatigue.
- Hardwire installation on airplane to simplify in-field dynamic propeller balancing. Canon plug for Chadwick-Helmuth dynamic propeller balancing analyzer is located just aft of copilot's seat on cockpit sidewall.
- Super King Air 350 follow-up type flap selector switch. Flaps follow position of flap selector with three positions: up, approach, and down.
- Super King Air 300/350 type approach chart holder on pilot's and copilot's control wheels (optional).
- Avionics compartment moisture barrier to prevent infiltration of water into the avionics bay through the avionics bay access door. Improves avionics reliability and life. This same barrier is used on the B200 and 350.
- Digital outside air temperature system similar to Super King Air 300/350.
- Cockpit flashlight and flashlight holder mounted on control pedestal similar to Super King Air Model 350.
- Changed chip detect warning annunciation to caution annunciation-also changed chip detect procedures in *Pilot's Operating Handbook* and checklist. New annunciator and checklist eliminate need for emergency shutdown of affected engine. Procedure now simply requires the pilot to monitor engine performance.
- Completely Redsigned Interior
  - Incorporates 26 electronically tuned dynamic vibration absorbers mounted in strategic locations on specific fuselage frames. Provides 12 to 15 dB reduction in specific frequency ranges to significantly reduce propeller induced sound and vibration.
  - New lightweight composite sound absorbing headliner, sidewall panels, and floorboards.
  - New thermal and acoustic insulating materials in cockpit and cabin sidewalls (bagged insulation and Scotch damp).
  - Window reveals covered with stretched wool headliner material.
  - Indirect fluorescent lighting behind window reveals now has bright and dim lighting intensities.
  - Sidewall panels incorporate integral sidewall armrest and Super King Air 350 style tables. Integral sidewall armrests allow for greater comfort by offering greater seat width between armrests when seat is laterally tracked. New tables offer greater strength and improved operation. Adjustable set

screws are provided to allow easy adjustment to ensure the table is level in the extended position.

- Redesigned side facing seat with partition. Redesign of the side facing seat allowed the vertical partition to be moved three inches further aft to provide an additional three inches of legroom for right side of the club.
- Redesigned and restyled cabin chairs. New chairs feature Super King Air 350 styling. The cushion on the seat backs were retailored and reshaped. The new shape allows the occupant to sit 1 1/2 inches further back in the seat. This offers a total of three inches of additional legroom between the seats in the club for greater passenger comfort.
- Combined seat redesign and relocation of aft partition increases legroom in club by a total of six inches. Standard C90B seat pitch is 59 inches. B200 seat pitch in club is 57 and C90A was 53 inches.
- New carpet installation covers all but seven inches of the seat tracks to allow for seat travel. Covered seat tracks provide greater comfort through reduced noise and cold that radiates from the exposed track. Cabin image is also improved.
- Seat belt chime and no smoking sign relocated and updated to Super King Air 350 type. Now located on forward and aft partitions, for better visibility.
- Standard no smoking configuration removes cigarette lighters and ashtrays. No smoking sign will remain illuminated during all operations. Ashtrays and cigarette lighters are packaged in a smoking option.
- Forward right side cabinet now includes, in addition to the hot beverage dispenser, a cold beverage dispenser. This unit has a serving light similar to the Super King Air refreshment centers.
- A new cabinet (optional) has been designed for the aft right side of the cabin. This cabinet features the four liquor decanters and an insulated ice drawer.
- Forward sliding door partition is now standard.
- All cabinet drawer slides feature roller-bearing-type guides to improve operation of drawers. C90A slides were friction-type guides.

- Optional quick-disconnect second door cable allows for greater ease in loading baggage/cargo but retains second cable for strength.
- New stylized C90B logo under cockpit D-windows provides common image and styling with Super King Air 350.
- New updated and distinctive paint scheme to establish C90B as new and different from all previous King Airs.
- Significant performance improvements substantially improve safety.
  - Reduced V MCA now 80 KIAS was 90 KIAS. V MCA was reduced because the new four-blade propellers with reduced diameter produce lower thrust at high power settings and low speeds.
  - Accelerate-stop distance reduced 9.7%. Now 3,650 feet, was 4,042 feet. (Sea level, standard day, no wind at maximum weight.)
  - Accelerate-go distance reduced 18.8%. Now 3,650 feet over 35-foot obstacle, was 4,500 feet over 50-foot obstacle (sea level, standard day, no wind at maximum weight).
  - Landing distance reduced 6.3%. Now 2,290 feet, was 2,443 feet (sea level, standard day, no wind at maximum weight).
- New *POH* features Abnormal and Emergency Sections with bold-face action items. The *POH* has been revised to reflect all the changes to the airplane, as well as to make the *POH* more consistent with the other King Air models.

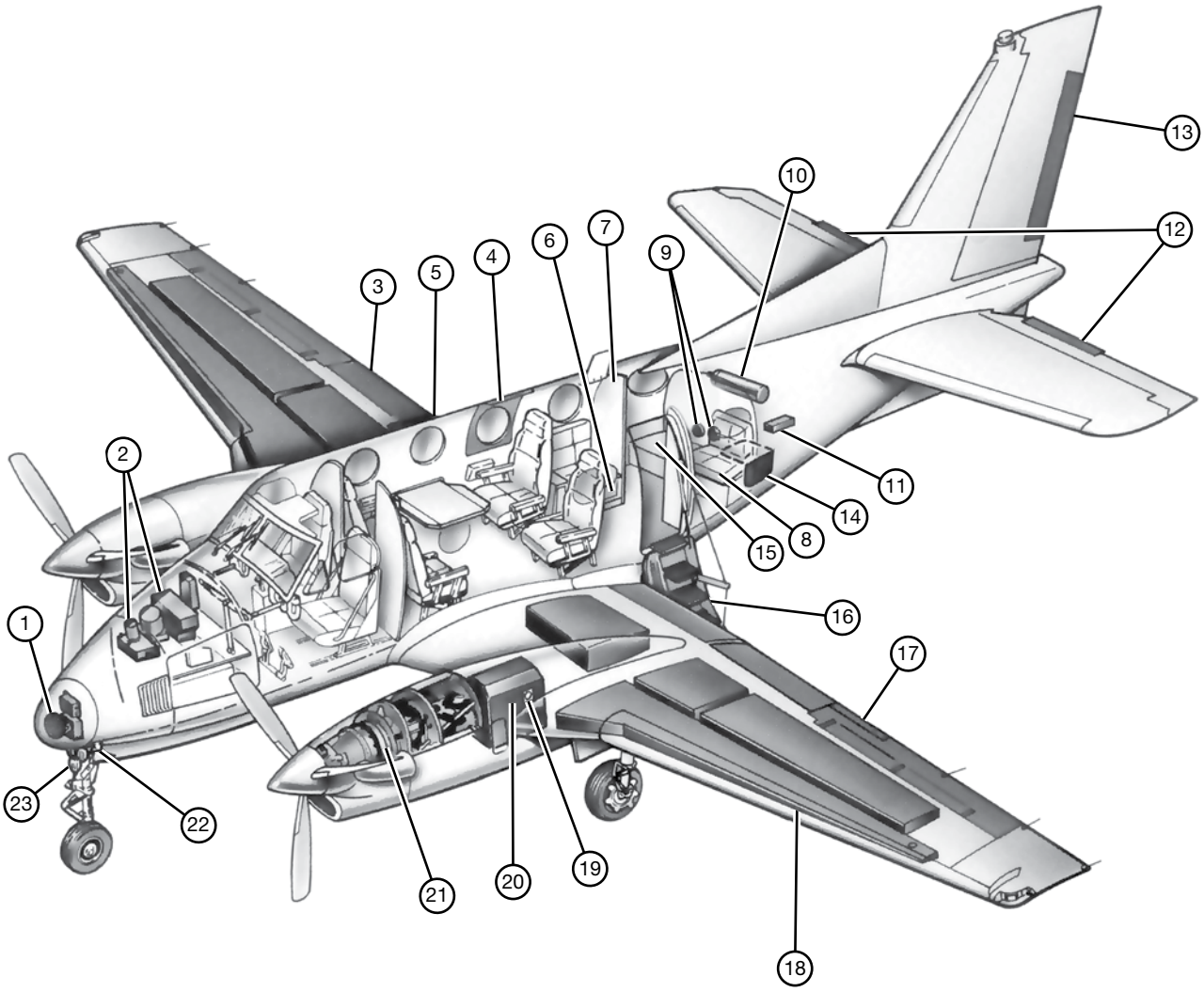
BEECHCRAFT KING AIR C90A AND C90B DESCRIPTION

The Beechcraft King Air C90A and C90B are high-performance, conventional tail, pressurized, twin-engine turboprop airplanes (Figure 1-1 through Figure 1-5). They are designed and equipped for flight in IFR conditions, day or night, into high-density air traffic zones, and into known or forecast icing conditions. They are also capable of operating in and out of small unimproved airports within the *POH* operating limits.

The King Air design is a blend of a highly efficient airframe with proven current technology components, providing a reliable, economical, versatile, and cost-productive airplane.

The structure is all-metal, low-wing monoplane. It has fully cantilevered wings and a conventional-tail empennage. The wings are an efficient, high-aspect ratio design. The airfoil section provides an excellent combination of low drag for cruise conditions, and easy handling for the low-speed terminal conditions or small airport operations.

Figure 1-1. Beechcraft King Air C90A



1. WEATHER RADAR ANTENNA
2. COMMUNICATIONS, NAVIGATION AND RADAR EQUIPMENT
3. OUTBOARD FLAP SECTION
4. GROUND ESCAPE HATCH
5. INBOARD FLAP SECTION
6. LIQUID STORAGE CABINET
7. LAVATORY PRIVACY CURTAIN
8. BELTED LAVATORY
9. PRESSURIZATION SAFETY AND OUTFLOW VALVES
10. OXYGEN BOTTLE
11. EMERGENCY LOCATOR TRANSMITTER
12. ELEVATOR TRIM TABS
13. RUDDER TRIM TAB
14. REAR FUSELAGE ACCESS DOOR
15. BAGGAGE AREA
16. AIRSTAIR DOOR
17. AILERON TRIM TAB
18. LEADING EDGE FUEL TANKS
19. WING ICE CHECK LIGHT
20. NACELLE FUEL TANK
21. PT6A-21 TURBOPROP ENGINE
22. HEATED PITOT MAST
23. LANDING AND TAXI LIGHTS

Figure 1-2. General Arrangement—C90A

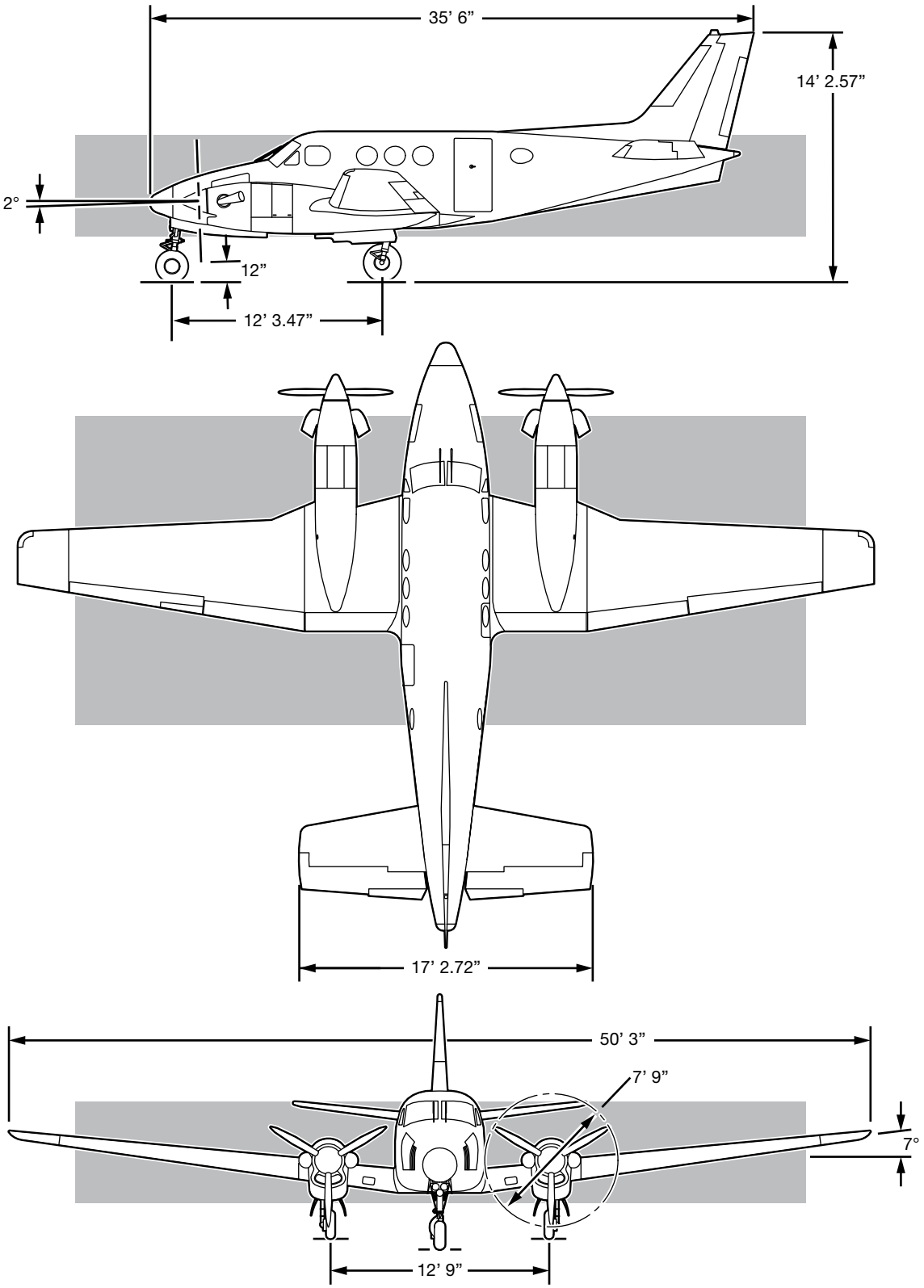
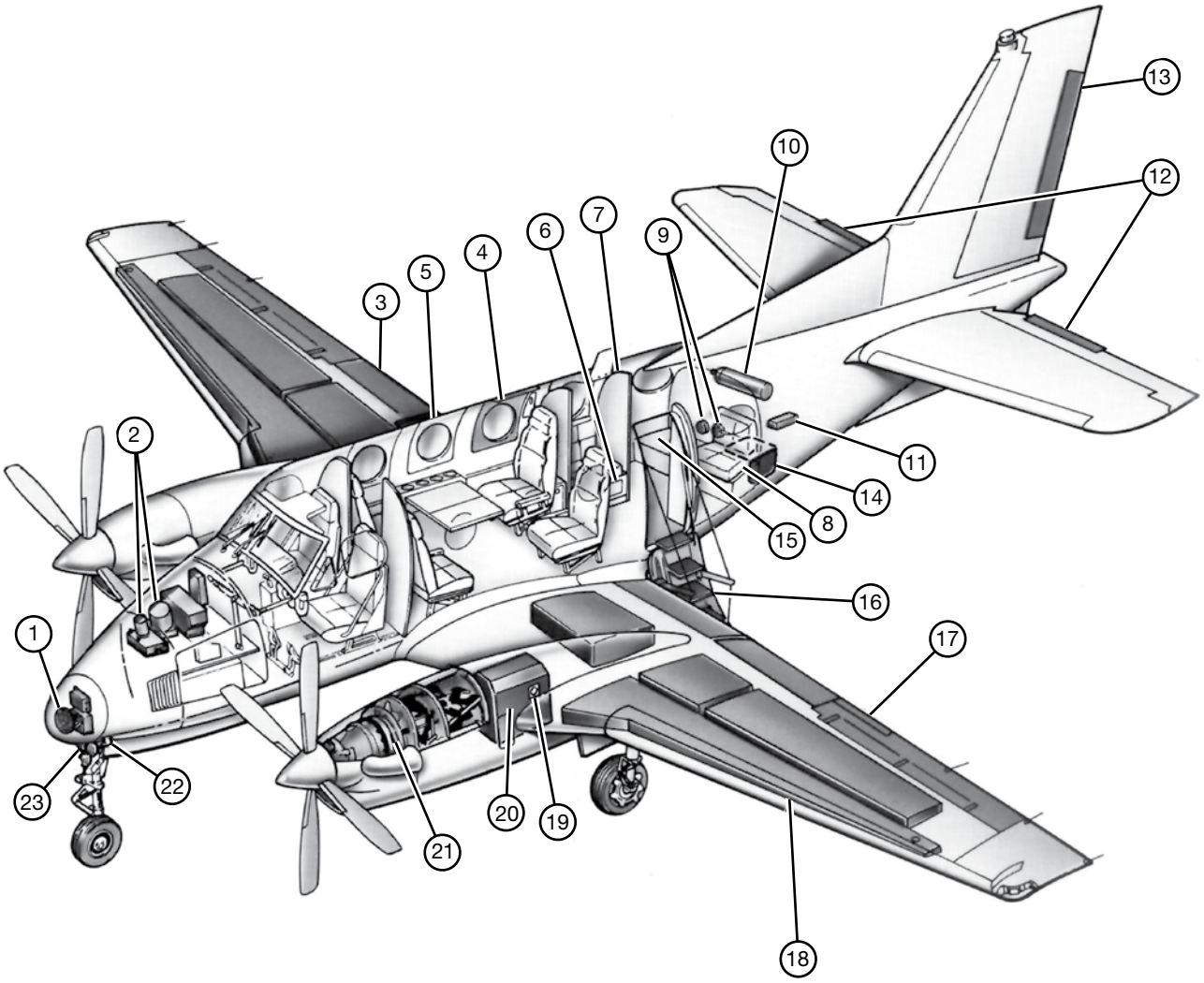


Figure 1-3. Three-View Diagram—C90A



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10. OXYGEN BOTTLE
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12. ELEVATOR TRIM TABS
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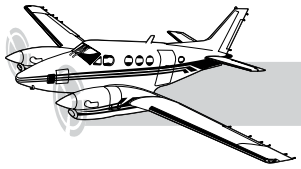
16. AIRSTAIR DOOR
17. AILERON TRIM TAB
18. LEADING EDGE FUEL TANKS
19. WING ICE CHECK LIGHT
20. NACELLE FUEL TANK
21. PT6A-21 TURBOPROP ENGINE
22. HEATED PITOT MAST
23. LANDING AND TAXI LIGHTS

Figure 1-4. General Arrangement—C90B



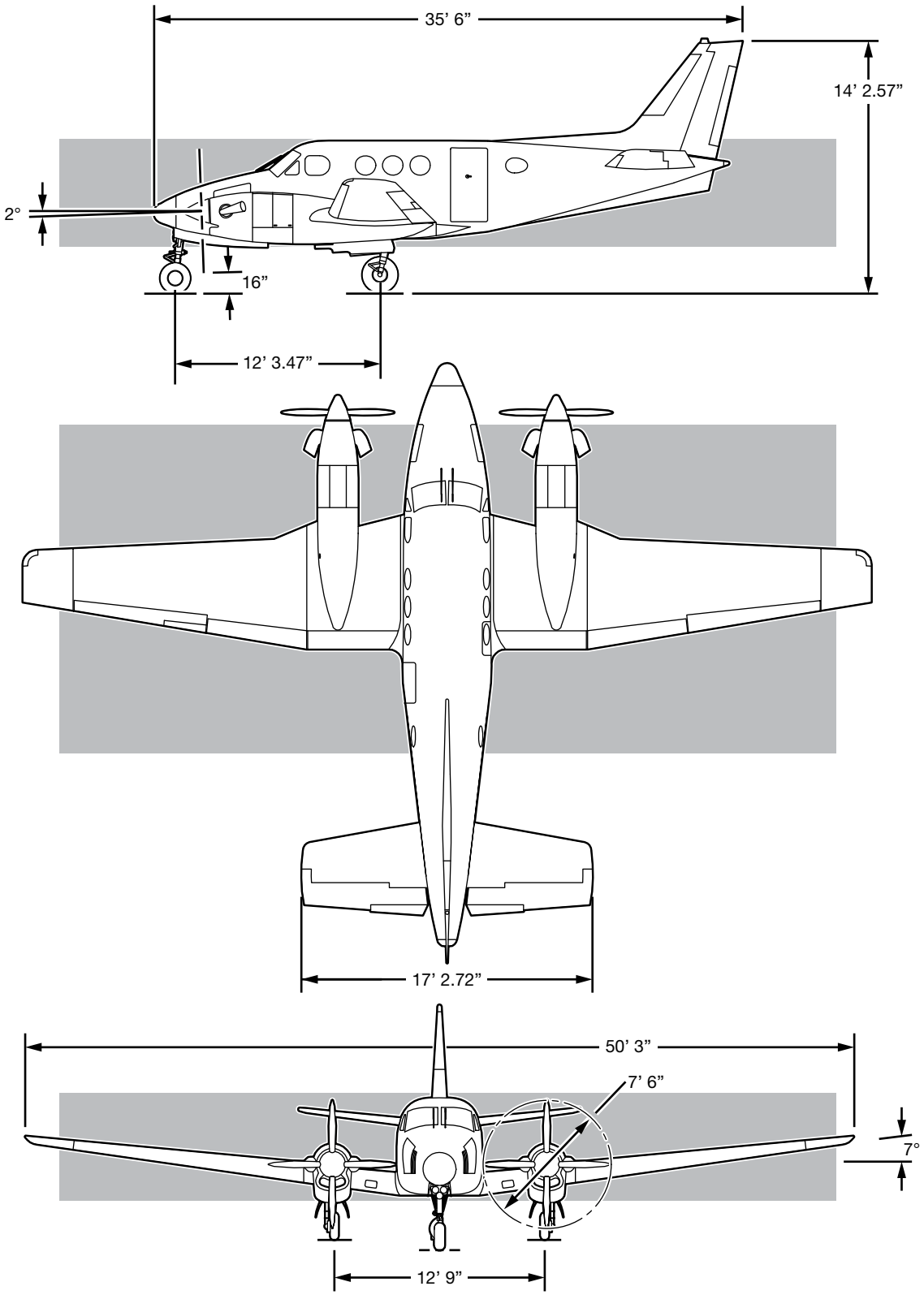


Figure 1-5. Three-View Diagram—C90B

A faired, oval, minimum frontal area nacelle is installed on each side of the wing center section to house both the engine and landing gear. The "pitot" type intakes (Figure 1-6) boost performance by reducing drag, and the exhaust stacks are shaped for smaller frontal area to reduce drag. The nacelles are designed and located to maximize propeller/ground clearance, minimize chain noise, and provide a low-drag installation of the powerplants on the wing.

Figure 1-6. Engine Air Inlet

The fuselage is conventional monocoque structure using high-strength aluminum alloys. The basic cross-sectional shape of the cabin is a favorable compromise between passenger comfort and efficient cruise performance. The cabin profile is squared-oval, not round. Passengers can sit comfortably without leaning their heads to accommodate sloping walls. The floors are flat from side to side for passenger ease in entering and leaving the cabin. The Beechcraft King Air C90A and C90B are certificated for up to 10 people (Figure 1-7). The most popular configuration provides comfortable seating for six passengers and a crew of two. Almost any arrangement is possible.

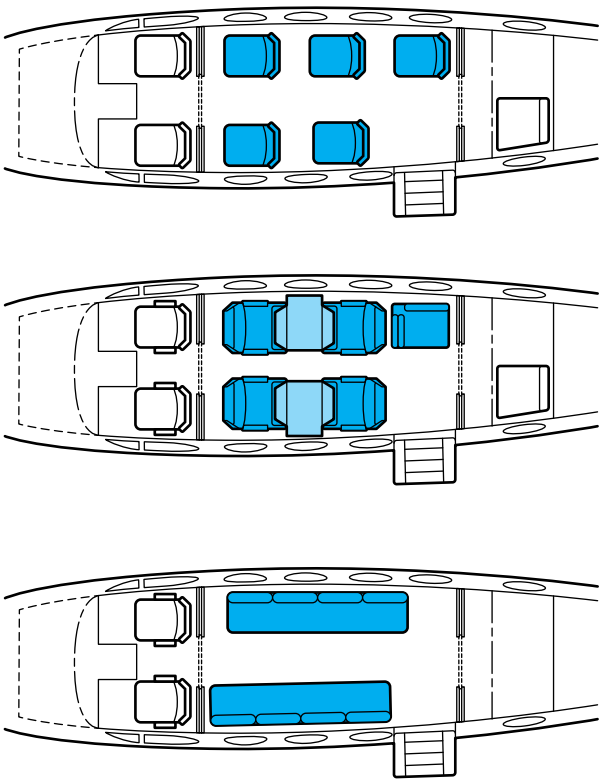


Figure 1-7. Optional Cabin Seating Arrangement

KING AIR C90A AND C90B CONFIGURATION

The King Air C90A and C90B are powered by Pratt & Whitney, 550 shp (flat-rated) PT6A-21 turboprop engines. In addition to the standard airplane configurations, Beechcraft offers many optional items which are available at additional cost and weight. The basic configurations, dimensions, weights, and specifications are summarized in Tables 1-1 and 1-2. Refer to the respective airplane *POH* for detailed, up-to-date information.

Table 1-1. SPECIFICATIONS—C90A

MODEL DESIGNATION - PASSENGER..... C90A
MINIMUM CREW 1
OCCUPANTS - MAX. FAA CERT. (INCL. CREW)..... 13
PASSENGERS - NORMAL CORP. CONFIG. 6
ENGINES - P&W TURBOPROP 2 PT6A-21
PROPELLERS - 3-BLADE, CONSTANT-SPEED,
FULL-FEATHERING, COUNTER-WEIGHTED,
HYDRAULICALLY-ACTUATED..... 2 MCCAULEY
LANDING GEAR - RETRACTABLE, TRICYCLE..... HYDRAULIC
WING AREA ..... 293.94 SQ. FT.

MAXIMUM CERTIFICATED WEIGHTS

CERTIFICATED WEIGHTS	LJ1138 AND LJ1063-1137 AND 1146	SUBSEQUENT EXCEPT 1146
MAXIMUM RAMP WEIGHT	9,710 POUNDS	10,160 POUNDS
MAXIMUM TAKE-OFF WEIGHT	9,650 POUNDS	10,100 POUNDS
MAXIMUM LANDING WEIGHT	9,168 POUNDS	9,600 POUNDS
MAXIMUM ZERO FUEL WEIGHT	NO STRUCTURAL LIMITATION	

MAXIMUM WEIGHT IN BAGGAGE COMPARTMENT:
REAR BAGGAGE COMPARTMENT .....350 POUNDS
NOSE BAGGAGE COMPARTMENT
(BAGGAGE AND AVIONICS)..... 350 POUNDS

CABIN AND ENTRY DIMENSIONS

CABIN WIDTH (MAXIMUM)..... 54 INCHES
CABIN LENGTH (PARTITION TO PARTITION)..... 155 INCHES
CABIN LENGTH (MAXIMUM BETWEEN PRESSURE BULKHEADS).....214 INCHES
CABIN HEIGHT (MAXIMUM) .....57 INCHES
AIRSTAIR ENTRANCE DOOR WIDTH (MINIMUM) ..... 27 INCHES
AIRSTAIR ENTRANCE DOOR HEIGHT (MINIMUM)..... 51.6 INCHES
SILL HEIGHT (MAXIMUM) 48 INCHES
PRESSURIZED COMPARTMENT VOLUME.....313.6 CUBIC FEET

REAR BAGGAGE COMPARTMENT VOLUME.....53.5 CUBIC FEET

NOSE AVIONICS/BAGGAGE COMPARTMENT VOLUME..... 16 CUBIC FEET

SPECIFIC LOADINGS

	LJ1063-1137 AND 1146	LJ1138 AND SUBSEQUENT EXCEPT 1146
WING LOADING	32.8 POUNDS/FT^2	34.4 POUNDS/FT^2
POWER LOADING	8.8 POUNDS/H.P	9.2 POUNDS/H.P.

Table 1-2. SPECIFICATIONS—C90B

MODEL DESIGNATION - PASSENGER C90B

CREW - FAA CERTIFIED 1

OCCUPANTS - MAX. FAA CERT. (INCL. CREW) 13

PASSENGERS - NORMAL CORP. CONFIG. 6

ENGINES - P&WTURBOPROP 2 PT6A-21

PROPELLERS - 4-BLADE, CONSTANT-SPEED, FULL-FEATHERING, COUNTER-WEIGHTED, HYDRAULICALLY-ACTUATED ..... 2 MCCAULEY (FULL REVERSING) HARTZELL AFTER LJ 1542

LANDING GEAR - RETRACTABLE, TRICYCLE ..... HYDRAULIC

WING AREA ..... 293.94 SQ. FT.

MAXIMUM CERTIFICATED WEIGHTS

MAXIMUM RAMP WEIGHT ..... 10,160 POUNDS

MAXIMUM TAKE-OFF WEIGHT ..... 10,100 POUNDS

MAXIMUM LANDING WEIGHT ..... 9,600 POUNDS

MAXIMUM ZERO FUEL WEIGHT ..... NO STRUCTURAL LIMITATION

MAXIMUM WEIGHT IN BAGGAGE COMPARTMENT:

REAR BAGGAGE COMPARTMENT ..... 350 POUNDS

NOSE BAGGAGE COMPARTMENT (BAGGAGE AND AVIONICS ..... 350 POUNDS

CABIN AND ENTRY DIMENSIONS

CABIN WIDTH (MAXIMUM) ..... 54 INCHES

CABIN LENGTH (PARTITION TO PARTITION) ..... 155 INCHES

CABIN LENGTH (MAXIMUM BETWEEN PRESSURE BULKHEADS) ..... 214 INCHES

CABIN HEIGHT (MAXIMUM) ..... 57 INCHES

AIRSTAIR ENTRANCE DOOR WIDTH (MINIMUM) ..... 27 INCHES

AIRSTAIR ENTRANCE DOOR HEIGHT (MINIMUM) ..... 51.6 INCHES

SILL HEIGHT (MAXIMUM) ..... 48 INCHES

PRESSURIZED COMPARTMENT VOLUME ..... 313.6 CUBIC FEET

REAR BAGGAGE COMPARTMENT VOLUME ..... 53.5 CUBIC FEET

NOSE AVIONICS/BAGGAGE COMPARTMENT VOLUME ..... 16 CUBIC FEET

SPECIFIC LOADINGS

WING LOADING ..... 32.8 POUNDS PER SQUARE FOOT

POWER LOADING ..... 8.8 POUNDS PER SHAFT HORSEPOWER

C90A OPERATING SPEEDS

The Beechcraft King Air C90A (Figure 1-8) qualifies as one of the most maneuverable corporate airplanes in the world. Insistence on handling ease in all flight regimes and tough construction techniques contribute to the following KIAS data (calculated at maximum takeoff weight of 9,650 pounds):

Maximum operating speed (V_{MO}) ..... 226 knots

Maneuvering speed (V_A) ..... 153/169 knots

Maximum landing gear operating speed (V_{LO}):

Extensions/extended ..... 182 knots

Retraction..... 163 knots

Maximum flap extension/extended (V_{FE}):

Approach..... 184 knots

Down..... 148 knots

Stall (100% flaps, power off)..... 76 knots

Air minimum control (V_{MCA})..... 90 knots

Figure 1-8. King Air C90A Front Three-Quarter View (Engines Primary)

C90B OPERATING SPEEDS

The Beechcraft King Air C90B (Figure 1-9) qualifies as one of the most maneuverable corporate airplanes in the world. Insistence on handling ease in all flight regimes and tough construction techniques contribute to the following KIAS data (calculated at maximum takeoff weight of 10,100 pounds):

Maximum operating

speed (V_{MO})..... 226 KIAS

Maneuvering speed (V_A) ..... 169 KIAS

Maximum landing gear operating speed (V_{LO}):

Extension/extended..... 182 KIAS

Retraction..... 163 KIAS

Maximum flap extension/extended (V_{FE}):

Approach..... 184 KIAS

Down..... 148 KIAS

Stall (100% flaps, power off)..... 78 KIAS
(with four-blade propeller installed)

Air minimum control (V_{MCA})..... 80 KIAS
(with four-blade propeller installed)

Figure 1-9. King Air C90B in Flight

CABIN ENTRY AND EXITS

The cabin entry airstair door is on the left side of the fuselage, just aft of the wing (Figure 1-10). A swing-down door, hinged at the bottom, provides a convenient stairway for entry and exit.

Two of the four steps are movable and automatically fold flat against the door in the closed position. A self-storing platform automatically folds down over the door sill when the door opens to provide a stepping platform for door seal protection.

A plastic-encased cable provides support for the door in the open position, a handhold for passengers, and a means of closing the door from inside the airplane. A hydraulic dampener permits the door to lower gradually during opening. It is important that not more than one person be on the airstair door at a time as excessive weights could cause structural damage to the door.

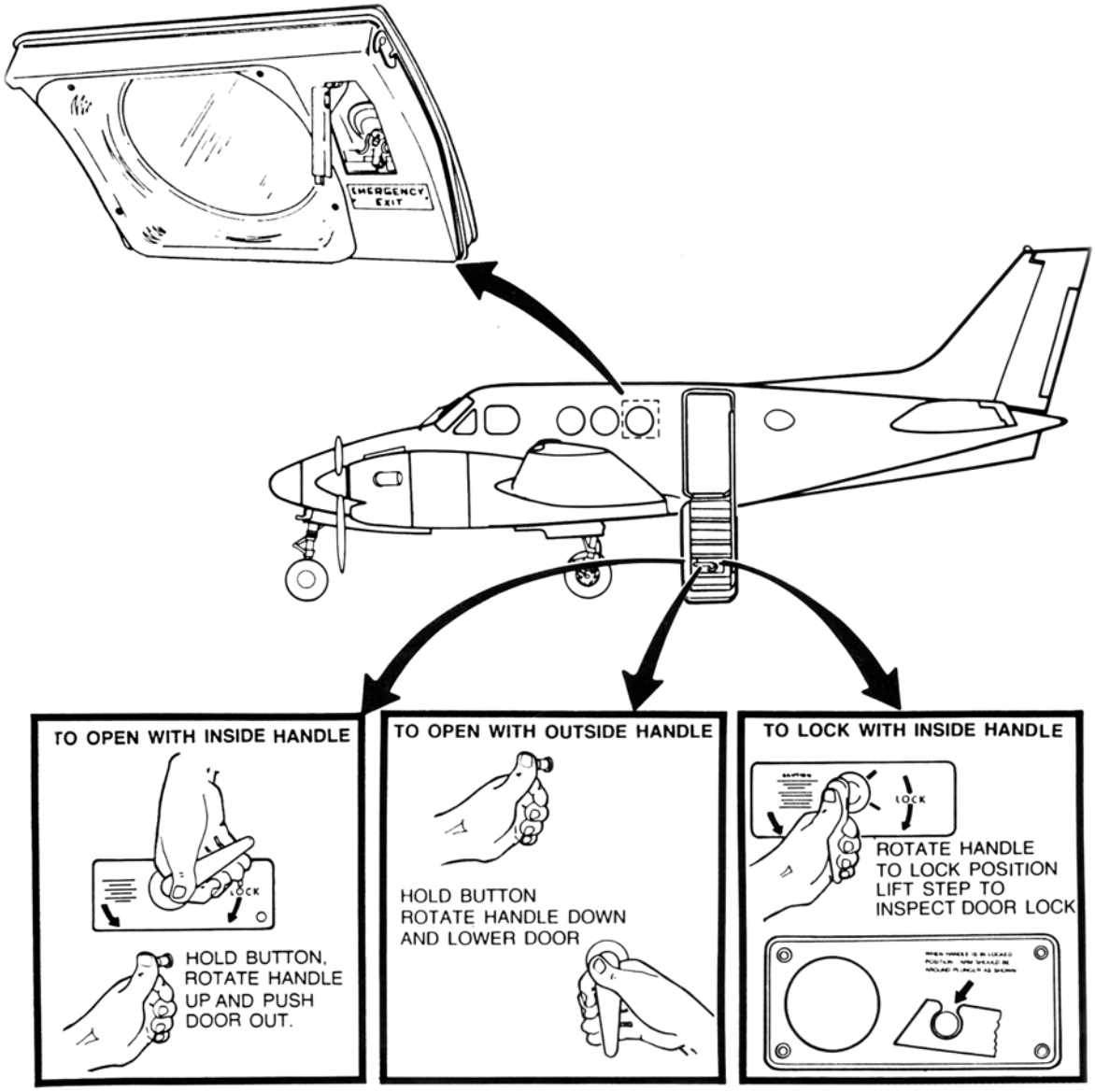


Figure 1-10. Entrance and Exit Provisions

Dual Door Cables with One Detachable (Optional)

Dual stair assist cables are available as an option on the C90B (Figure 1-11). Door assist cables provide passengers a way to stabilize themselves when going up or down the stairs. The forward assist cable is easily detachable to provide more room for loading large baggage or cargo into the airplane.

Figure 1-11. Dual Door Cables

Airstair Locking Mechanism

The door locking mechanism is operated by either of the two vertically staggered handles, one inside and the other outside the door. The inside and outside handles are mechanically interconnected.

When either handle is rotated per placard instructions, two latch bolts at each side of the door, and two latch hooks at the top of the door, lock into the doorframe to secure the airstair door. A button adjacent to the door handle must be depressed before the handle can be rotated to open the door. For security of the airplane on the ground, the door can be locked with a key.

To secure the airstair door inside, rotate the handle clockwise as far as it will go. The release button should pop out, and the handle should be pointing down. Check the security of the airstair door by attempting to rotate the handle counterclockwise without depressing the release button; the handle should not move.

Next lift the folded stairstep that is just below the door handle. Ensure the safety lock is in position around the diaphragm shaft when the handle is in the locked position.

To observe this area, depress a red switch near the window that illuminates a lamp inside the door. If the arm is properly positioned around the shaft, proceed to check the indication in each of the visual inspection ports located near each corner of the door (see Figure 1-10). Ensure the green stripe on the latch bolt is aligned with the black pointer in the visual inspection port.

WARNING

Never attempt to unlock or check the security of the door in flight. If the CABIN DOOR annunciator illuminates in flight, or if the pilot has any reason to suspect that the door may not be securely locked, the cabin pressure should be reduced to zero differential, and all occupants instructed to remain seated with their seat belts fastened. After the airplane has made a full-stop landing, only a crewmember should check the security of the airstair door.

EMERGENCY EXIT

The emergency exit door is located at the third cabin window on the right side of the fuselage (see Figure 1-10). A placard at the window gives instructions for access to the release mechanism.

The door is released from the inside with two hooks, a trigger button, and a latch-release pull-up handle. A placard on the emergency exit hatch release cover lists proper opening procedures.

A pressure lock prevents the door from being opened when the cabin is pressurized. If pressurized, pulling the hooks overrides the pressure lock and allows the trigger button to be depressed. This releases the latch-release handle. When the handle is pulled up and the securing latches are released, a hinge at the bottom allows the hatch to swing outward and downward for emergency exit.

NOSE BAGGAGE DOOR (OPTIONAL)

Prior to LJ-1531 the King Air C90A and C90B have an optional 16 cubic-foot nose baggage compartment which is accessible through a door located on the left side of the nose (Figure 1-12). This compartment is limited to 350 pounds, which includes the weight of the avionics equipment within the compartment.

The baggage door is hinged at the top to allow the door to swing upward. A flush-mounted door handle with a push-to-release button activates three bayonet-type latching bolts that, when engaged, will hold the door securely closed.

When not engaged, a switch at the forward latching bolt will close, and the BAG DOOR OPEN annunciator will illuminate. In addition, the door is equipped with a secondary safety latch to hold the door in a partially closed position in the event the primary latching bolt is not engaged.

The push-to-release button, adjacent to the door handle, will prevent the door inadvertently opening. For security of the unattended airplane, the nose baggage compartment door features a key-lock latch.

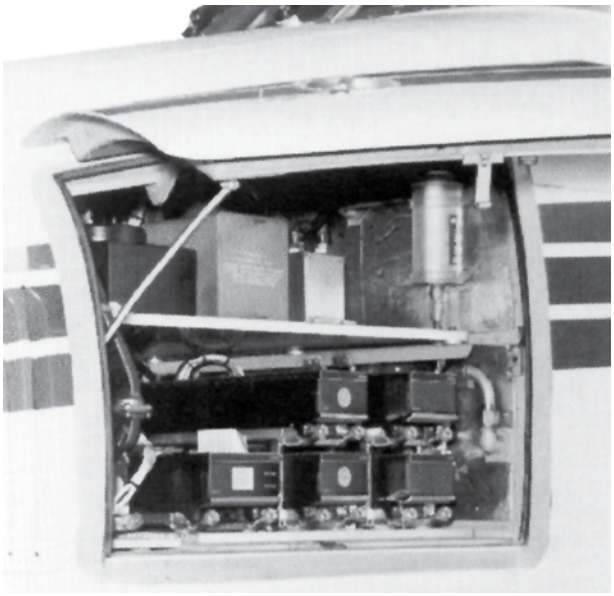


Figure 1-12. Nose Compartment Door (Baggage Compartment Not Shown)

CABIN COMPARTMENTS

The pressurized cabin interior consists of the flight deck, passenger seating area, and an aft baggage area (Figure 1-13). The flight deck provides side-by-side seating for the pilot and copilot.

Typically for corporate use, the cabin is arranged in a five-passenger club seating and aisle-facing cabinet seat layout (Figure 1-14).

A lavatory area is located in the aft compartment, with a padded seat which can be used as the sixth passenger seat.

Aft of the cabin area is the baggage area. This pressurized area is capable of holding 53.5 cubic feet of luggage, cargo, or clothing (all accessible in flight). The location of the baggage area next to the airstair door makes loading and unloading easy.

If an operation requires, some or all of the seats, wall partitions, and lavatory can be quickly removed to configure the airplane for cargo transport.

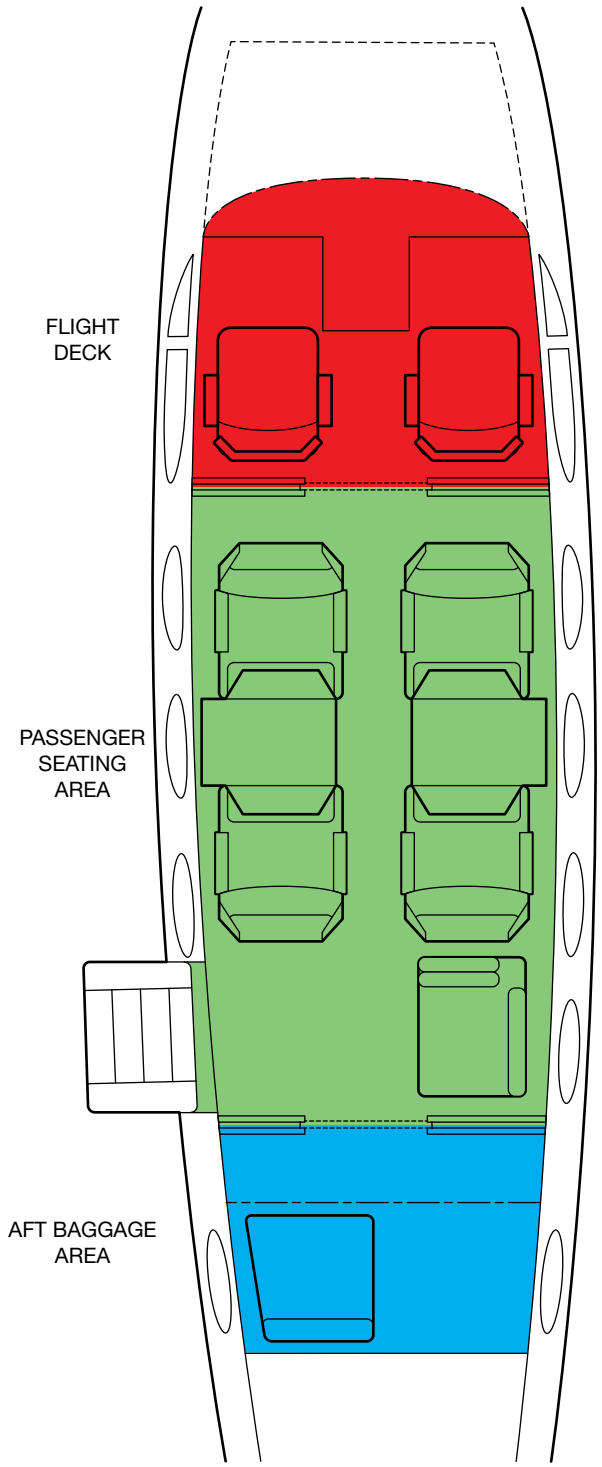


Figure 1-13. Cabin Areas

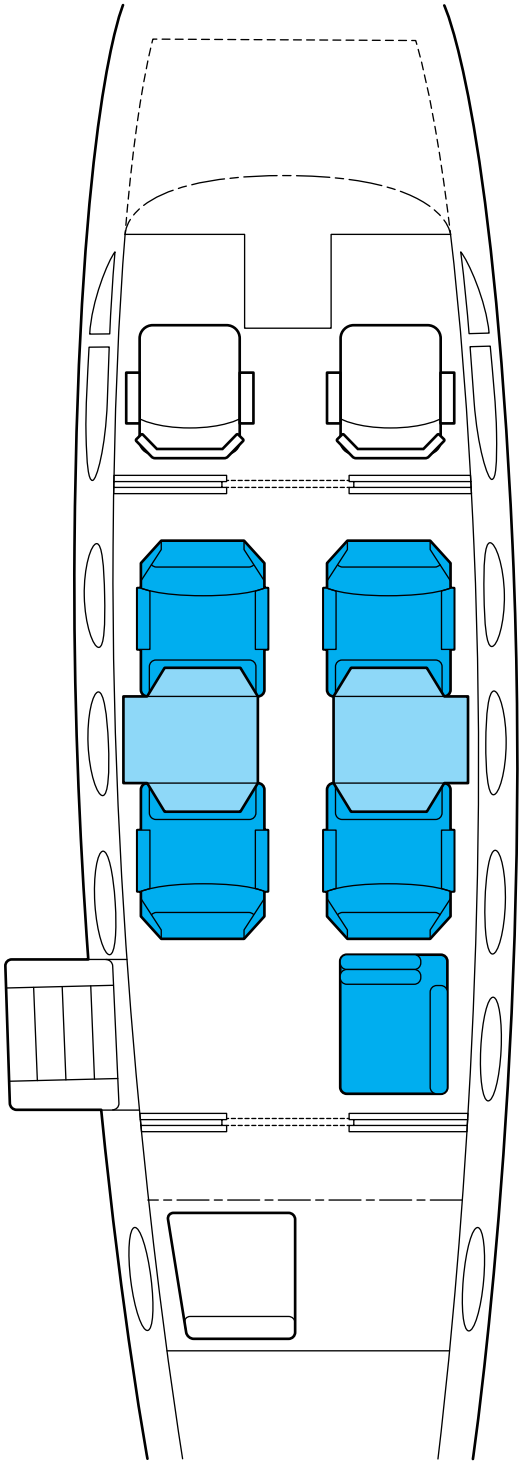


Figure 1-14. Cabin Seating Layout (Typical)

FLIGHT DECK

The flight deck layout is a time-proven design that has optimized crew efficiency and comfort (Figure 1-15). The pilot and copilot sit side-by-side in individual chairs, separated by the control pedestal. The seats are adjustable fore and aft as well as vertically. Seat belts and inertia-type shoulder harnesses are provided for each seat.

The general layout of the flight deck shows the location of the instruments and controls. Conventional dual controls are installed so that the airplane can be flown by either pilot (Figure 1-16). The controls and instruments are arranged for convenient single-pilot operation or for a pilot and copilot crew.

The fuel control panel (Figure 1-16) is located on the left sidewall, next to the pilot. Fuel quantity gages and switches, firewall valve switches, and circuit breakers are located on this panel.

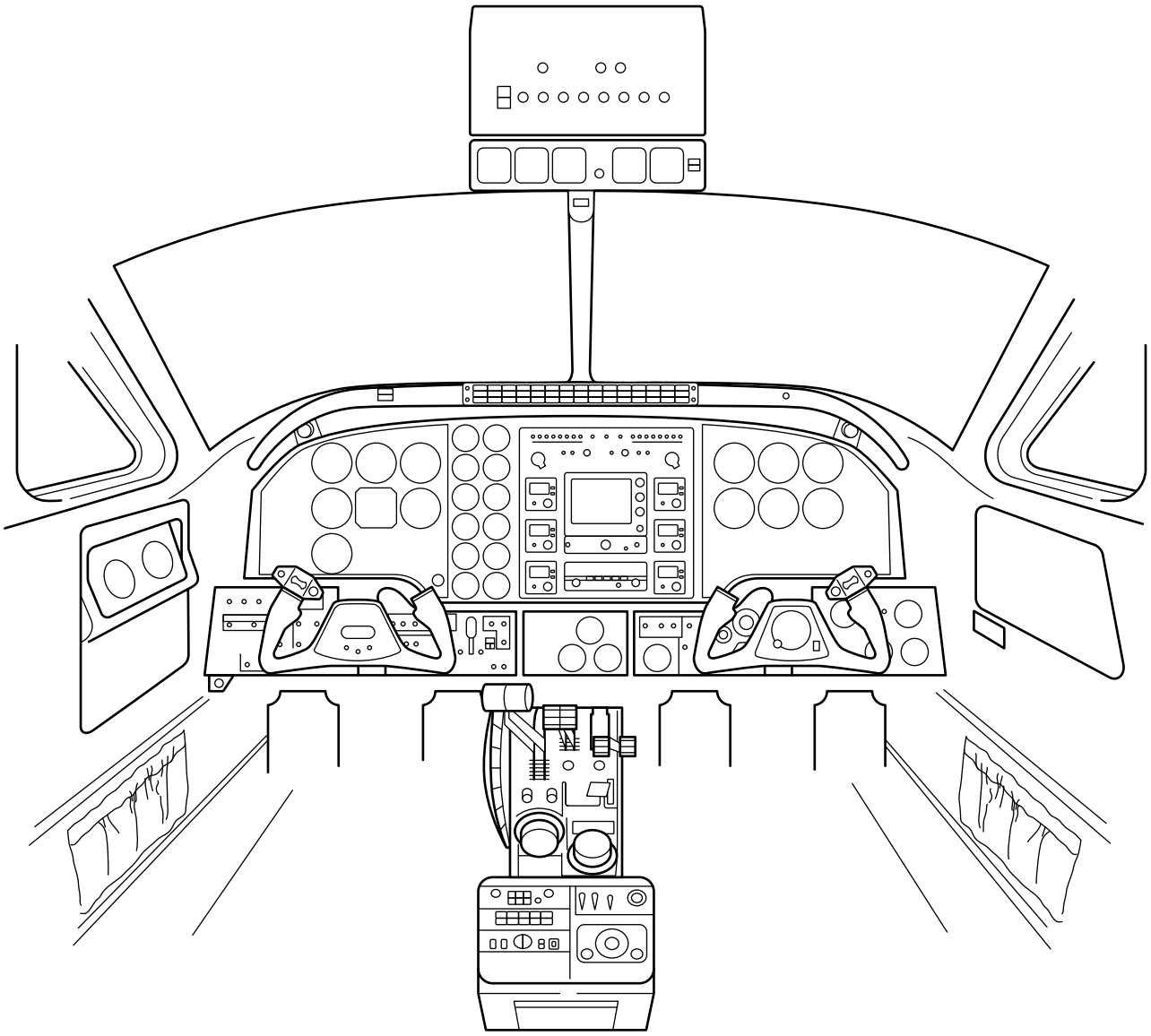


Figure 1-15. Flight Deck Layout

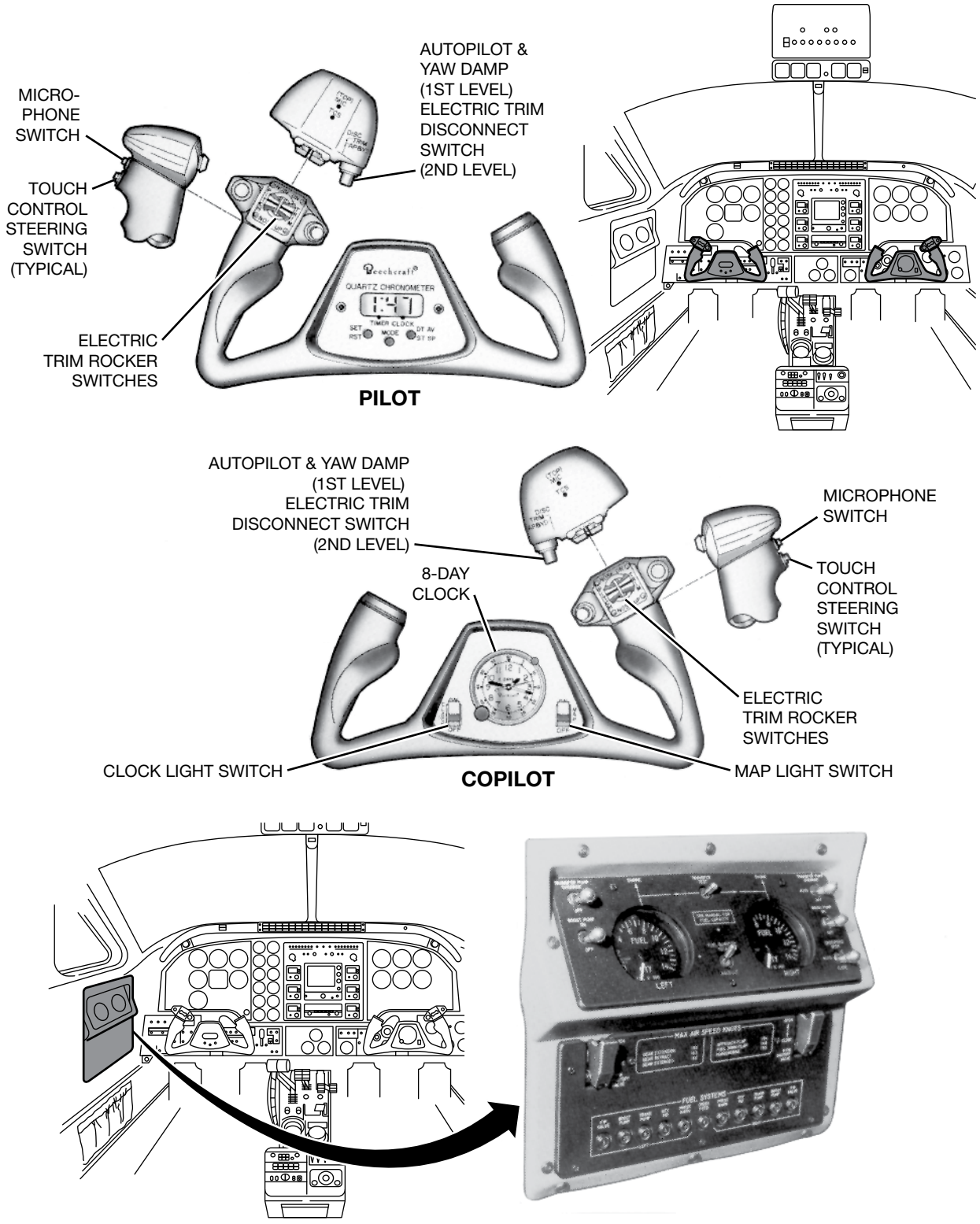


Figure 1-16. Control Wheels and Fuel Control Panel

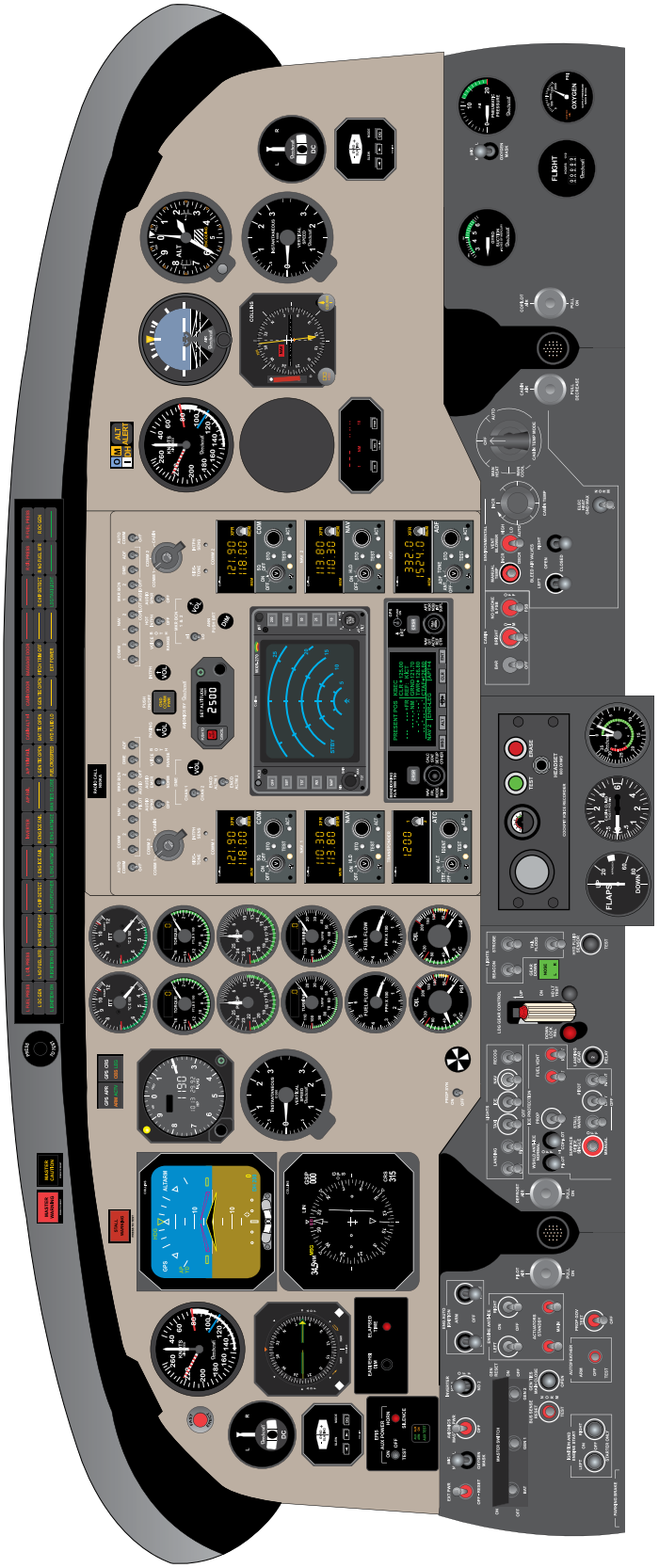
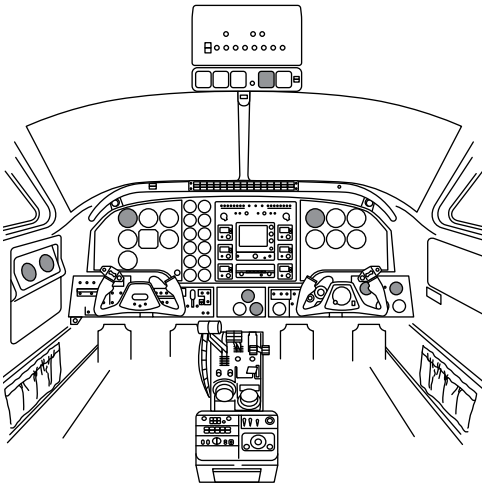


Figure 1-17. Instrument Panels

The instrument panels (Figure 1-17) contain the flight instruments, engine instruments, and avionics panel. The airspeed indicator and other miscellaneous system gages which provide limitations markings are shown in Figure 1-18.



C90A AIRSPEED INDICATOR MARKINGS

- 226 KIAS MAXIMUM (V_{MO}) OR .46 MACH, WHICHEVER IS LOWER

NOTE

MAXIMUM ALLOWABLE AIRSPEED (RED STRIPED) POINTER IS SELF ADJUSTING WITH ALTITUDE

- 90 KIAS MINIMUM SINGLE-ENGINE CONTROL SPEED (V_{MCA})
- 107 KIAS SINGLE-ENGINE CONTROL SPEED (V_{MCA})
- 76-89 KIAS FULL FLAP OPERATING RANGE
- 89-148 KIAS (NARROW) FLAPS UP STALL SPEED (V_S) TO MASIMUM LIMIT FLAPS MORE THAN 30%
- 76-89 KIAS MAXIMUM FLAPS TO/AT APPROACH SPEED

C90B AIRSPEED INDICATOR MARKINGS

- 226 KIAS MAXIMUM (V_{MO}) OR .46 MACH, WHICHEVER IS LOWER

NOTE

MAXIMUM ALLOWABLE AIRSPEED (RED STRIPED) POINTER IS SELF ADJUSTING WITH ALTITUDE

- 80 KIAS MINIMUM SINGLE-ENGINE CONTROL SPEED (V_{MCA})
- 108 KIAS SINGLE-ENGINE CONTROL SPEED (V_{MCA})
- 78-88 KIAS FULL FLAP OPERATING RANGE
- 88-148 KIAS (NARROW) FLAPS UP STALL SPEED (V_S) TO MASIMUM LIMIT FLAPS MORE THAN 30%
- 184 KIAS MAXIMUM FLAPS TO/AT APPROACH SPEED

PNEUMATIC PRESSURE

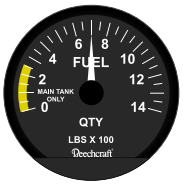
- 12-20 PSI NORMAL OPERATING RANGE
- 20 PSI MAXIMUM

GYRO SUCTION

- HG 2.8-4.3 NORMAL OPERATING RANGE 30,000-15,000 FT.
- HG 4.3-5.9 NORMAL OPERATING RANGE 15,000 - SEA LEVEL

PROPELLER DEICER AMMETER

- 14-18 AMPERES NORMAL OPERATION (C90A)
- 18-24 AMPERES NORMAL OPERATION (C90B)

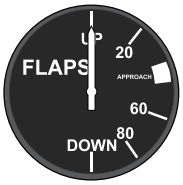


FUEL QUANTITY

- 0-265 LBS NO TAKEOFF RANGE

CABIN ALTIMETER AND PRESSURE DIFFERENTIAL

- 0-5 PSI (APPROVED OPERATING RANGE)
- 5.1 PSI TO END OF SCALE (UNAPPROVED OPERATING RANGE)



FLAP POSITION INDICATOR

- 32.5% APPROACH

Figure 1-18. Instrumentation

The engine instruments (Figure 1-19 and Figure 1-20) are mounted in a vertical double row next to the avionics panel. The avionics panel in the center contains the nav/comm controls and weather radar unit.

INTERSTAGE TURBINE TEMPERATURE GAGE

- 400-695°C NORMAL OPERATING RANGE
- 695°C MAXIMUM CONTINUOUS LIMIT
- 1090°C MAXIMUM STARTING ONLY LIMIT

TORQUE METER

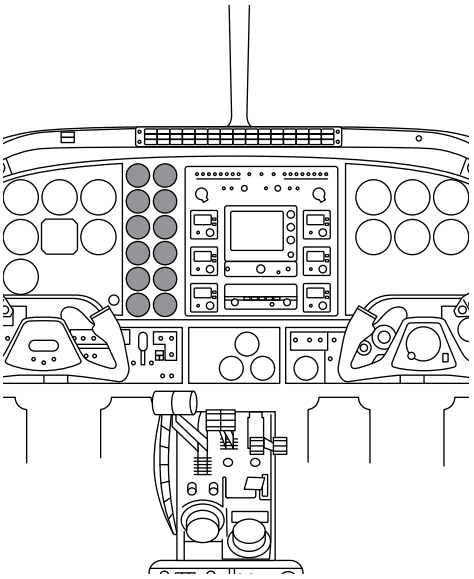
- 400-1315 FT-LBS NORMAL OPERATING RANGE
- 1315 FT-LBS MAXIMUM

PROPELLER TACHOMETER (N_2 SPEED)

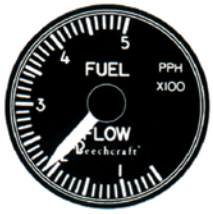
- 1800-2200 RPM NORMAL OPERATING RANGE
- 2200 RPM MAXIMUM LIMIT

GAS GENERATOR TACHOMETER (N_1 SPEED)

- 101.5% MAXIMUM LIMIT



NOTE
ALL ILLUSTRATION NEEDLES MAY NOT REFLECT NORMAL INDICATIONS



NO LIMITATION MARKINGS

OIL TEMPERATURE SCALE

- 10-99°C NORMAL OPERATING RANGE
- 99°C MAXIMUM OIL TEMPERATURE LIMIT

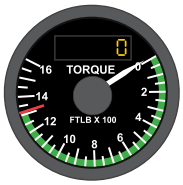
OIL PRESSURE SCALE

- 40 PSI MINIMUM LIMIT
- 80-100 PSI NORMAL OPERATING RANGE
- 100 PSI MAXIMUM LIMIT

Figure 1-19. Engine Instruments—Prior to LJ-1361

INTERSTAGE TURBINE TEMPERATURE GAGE

- 400-695°C NORMAL OPERATING RANGE
- 695°C MAXIMUM CONTINUOUS LIMIT
- 1090°C MAXIMUM STARTING ONLY LIMIT

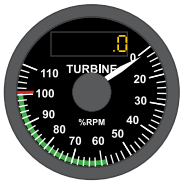


TORQUE METER

- 0-1315 FT-LBS NORMAL OPERATING RANGE
- 1315 FT-LBS MAXIMUM

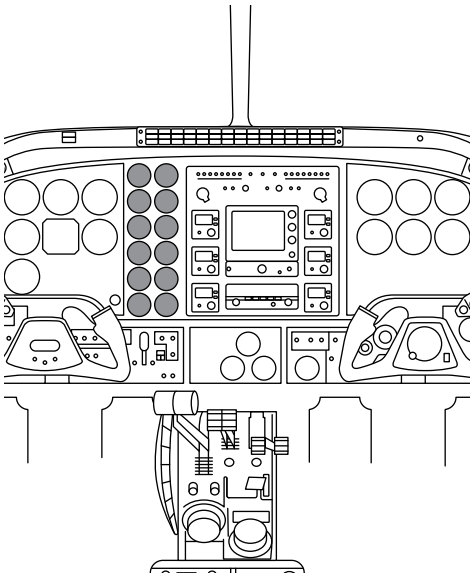
PROPELLER TACHOMETER (N_2 SPEED)

- 1100-2200 RPM NORMAL OPERATING RANGE
- 2200 RPM MAXIMUM LIMIT

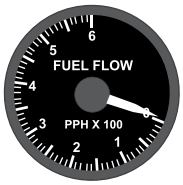


GAS GENERATOR TACHOMETER (N_1 SPEED)

- 58-101.5% NORMAL OPERATING RANGE
- 101.5% MAXIMUM LIMIT



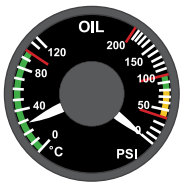
NOTE
ALL ILLUSTRATION NEEDLES MAY NOT REFLECT NORMAL INDICATIONS



NO LIMITATION MARKINGS

OIL TEMPERATURE SCALE

- 0-99°C NORMAL OPERATING RANGE
- 99°C MAXIMUM OIL TEMPERATURE LIMIT



OIL PRESSURE SCALE

- 40 PSI MINIMUM LIMIT
- 40-80 PSI CAUTION RANGE
- 80-100 PSI NORMAL OPERATING RANGE
- 100 PSI MAXIMUM LIMIT

Figure 1-20. Engine Instruments—LJ-1361, LJ-1363, and After

Extending aft from the center subpanel is the engine control quadrant and pedestal (Figure 1-21). Engine controls, flap control handle, rudder and aileron trim knobs, and pressurization controls are mounted on this pedestal. The flight director and autopilot systems are also installed here.

On the right side panel next to the copilot is the main circuit-breaker panel (Figure 1-21), where the majority of the system circuit breakers are located. The static air selector handle is mounted just below the circuit-breaker panel.

Just below the instrument panel are the pilot's (left) and copilot's (right) subpanels (Figure 1-22). Aircraft system controls, engine switches, master switches, and landing gear controls are located on these subpanels.

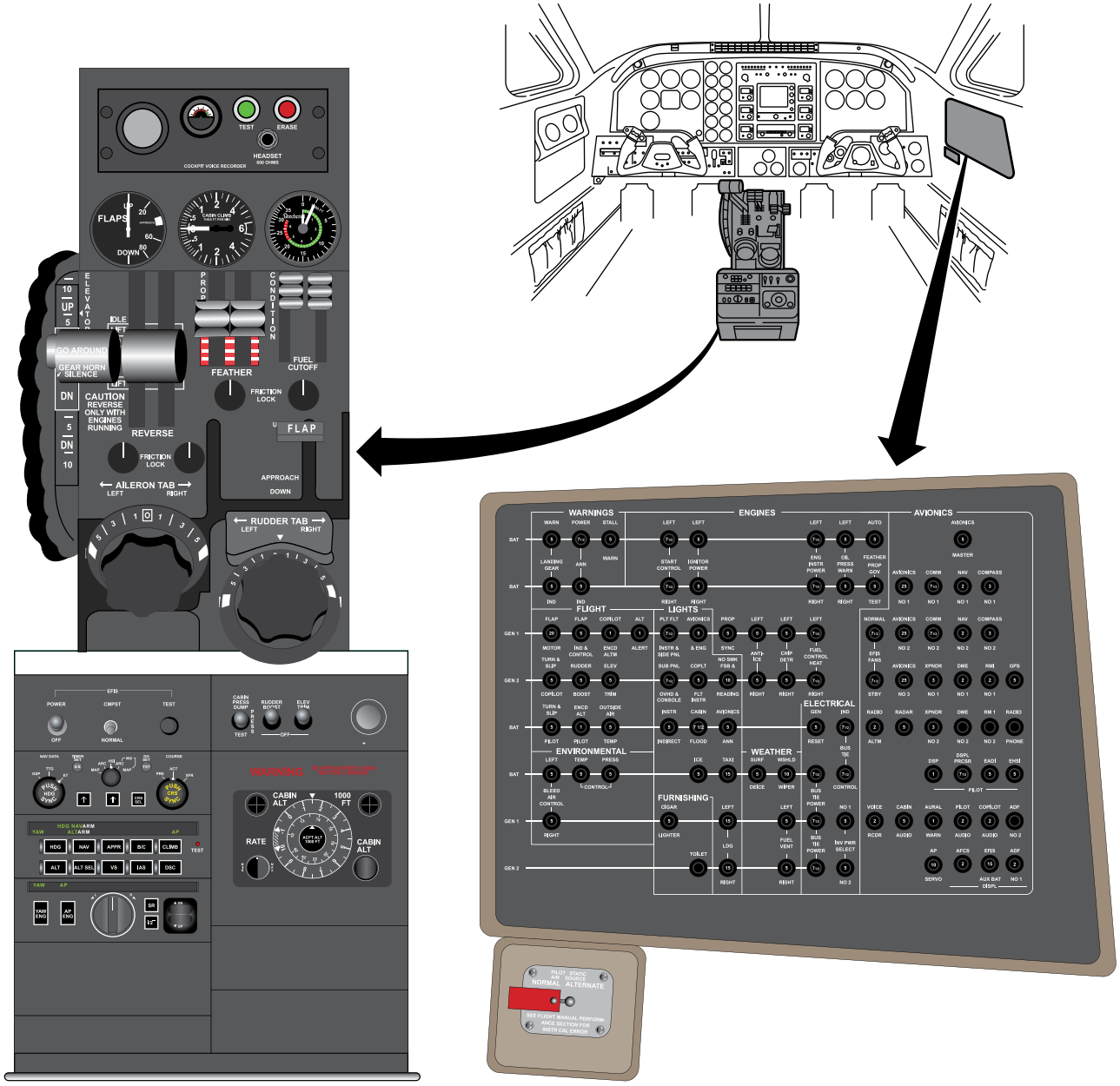


Figure 1-21. Pedestal and Right Side Panel

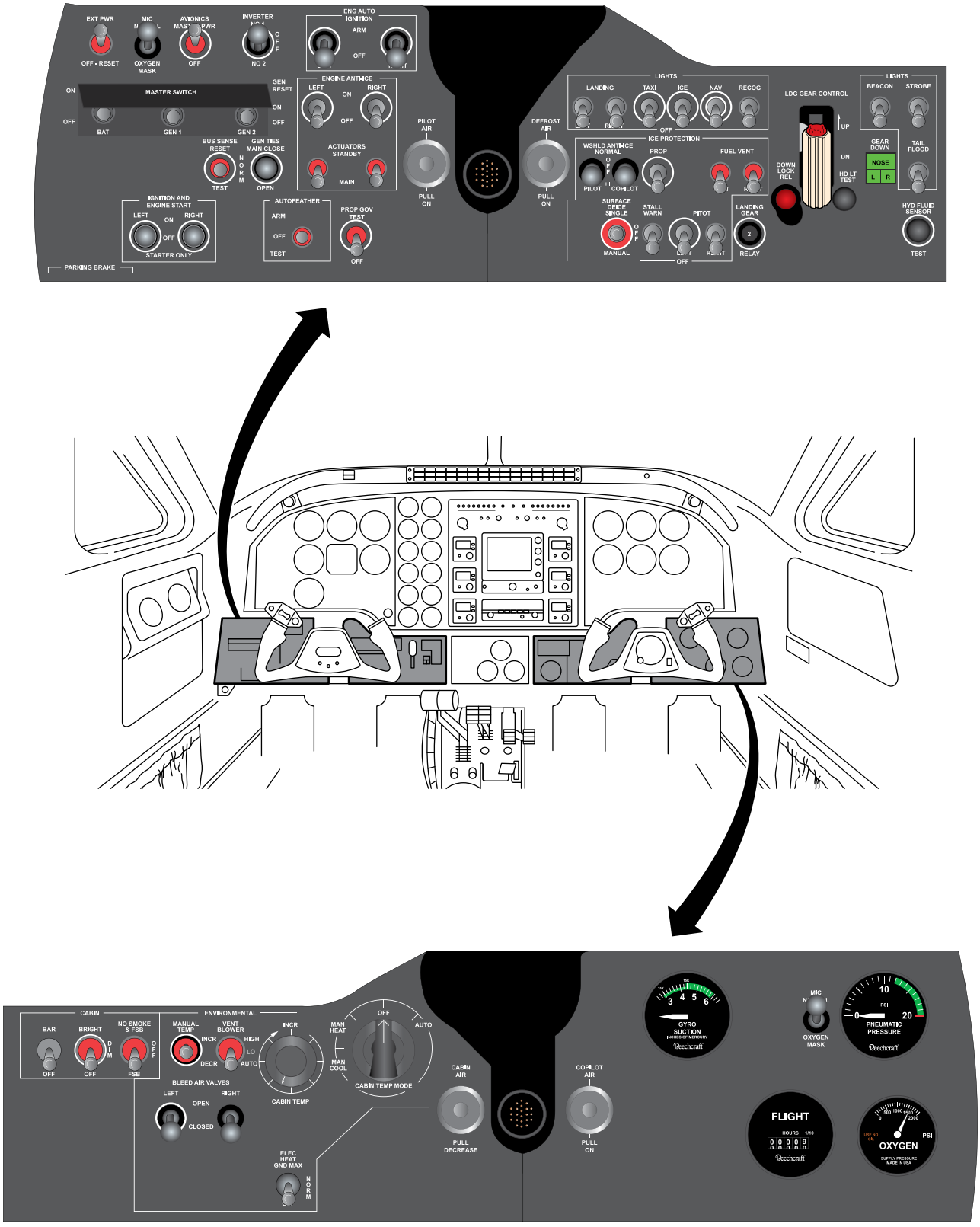


Figure 1-22. Pilot's and Copilot's Subpanels

The annunciator system (Figure 1-23) consists of an annunciator panel centrally located in the glareshield, an annunciator panel dimming control, a press-to-test switch, and a fault warning light. The annunciators are word-readout type.

Whenever a condition covered by the annunciator system occurs, a signal is generated, and the appropriate annunciator is illuminated.

The illumination of a green or yellow annunciator light will not trigger the fault warning system, but a red annunciator will actuate the fault warning flasher. After LJ-1353, a yellow light will trigger a MASTER CAUTION flasher.

In the overhead area, between the pilot and copilot, is the lighting control panel (Figure 1-24). The various rheostat controls for the flight deck and instrument lighting are mounted on this panel, convenient to both pilot and copilot.

Also mounted on this panel are the windshield wiper control, the generator load and voltage gages, the deice amps gage, and the inverter monitoring gage. Certain operation limitations are also placarded on this panel.

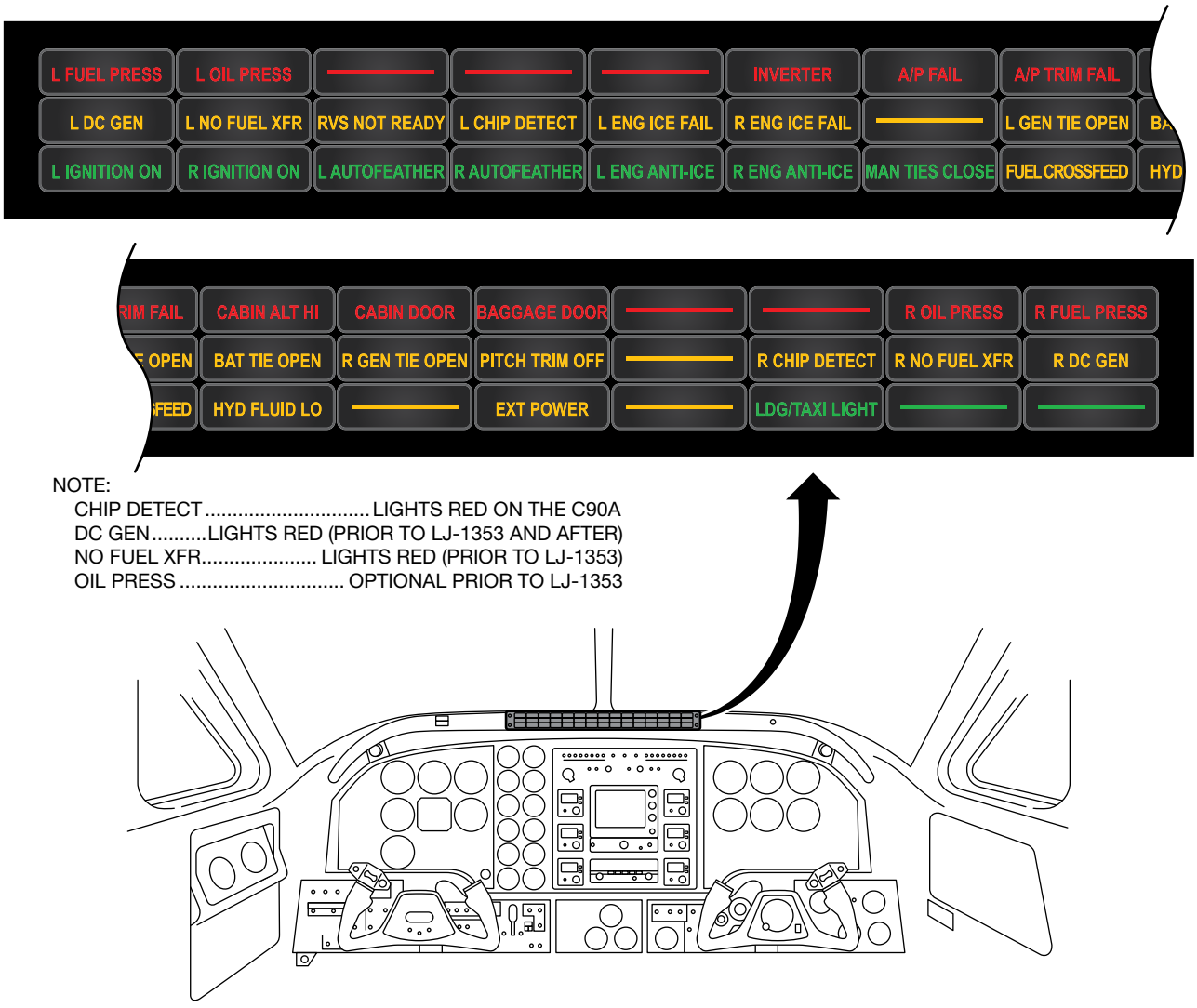


Figure 1-23. Annunciators

C90A INSTRUMENT PANEL/AVIONICS

The C90A panel features as standard equipment a Collins Pro Line II avionics package, including an EFIS HSI (EHSI-74).

The airplane is approved for single-pilot operation; however, a full set of copilot's flight instruments are installed as standard equipment.

Optionally available are Bendix/King Gold Crown and Silver Crown avionics packages. Two RNAV/LORAN systems are available as options: the Bendix/King KLN 88 (Figure 1-25) and the Foster LNS616B.

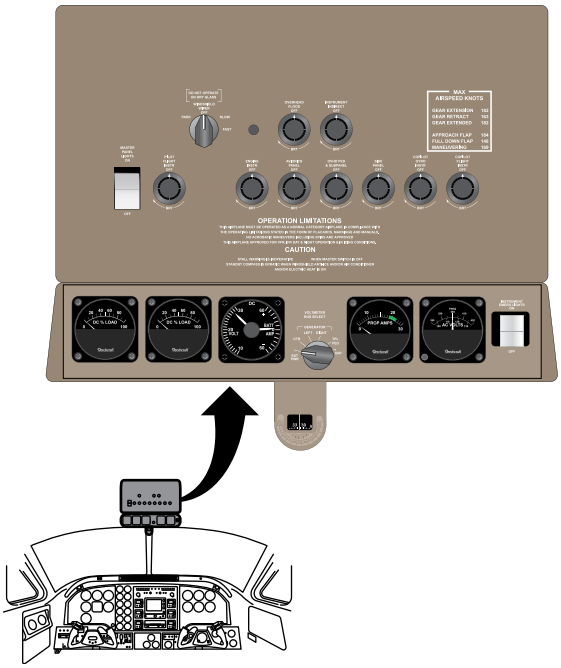


Figure 1-24. Overhead Light Control Panel

Figure 1-25. Avionics Installation (C90A)

CONTROL SURFACES

The King Air C90A and C90B are equipped with conventional ailerons, elevators, and rudder (Figure 1-26). The control surfaces are pushrod- and cable-operated by conventional dual controls in the flight deck.

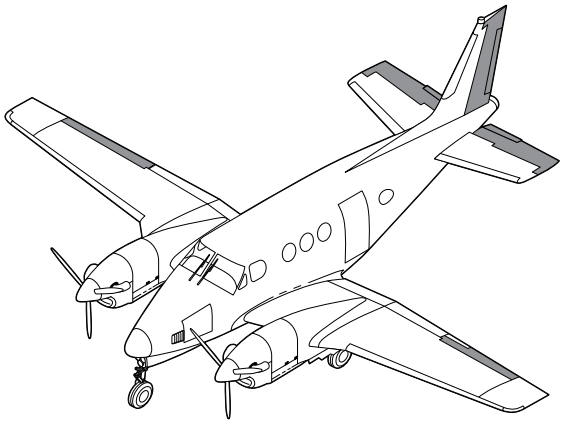


Figure 1-26. Flight Control Surfaces

Any time the airplane is parked overnight or in windy conditions, the rudder gust pin and control locks should be installed to prevent damage to the control surfaces and hinges or to the controls (Figure 1-27). Two items require particular attention: the parking brake handle mounted just under the left corner of the subpanel, and the rudder gust lock bar mounted between the pilot's rudder pedals.

Before towing the airplane, the parking brake must be released (brake handle pushed in), and the rudder gust lock bar must be removed from between the rudder pedals. Serious damage to the tires, brakes, and steering linkage can result if these items are not released.

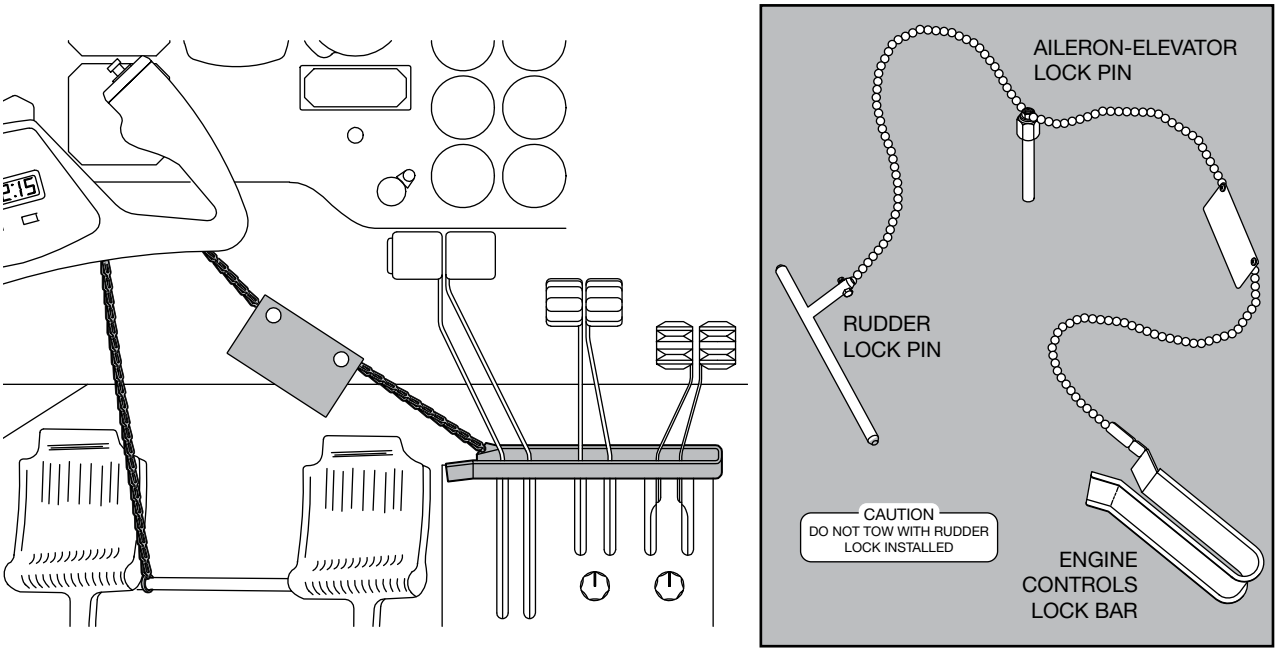


Figure 1-27. Flight Control Locks

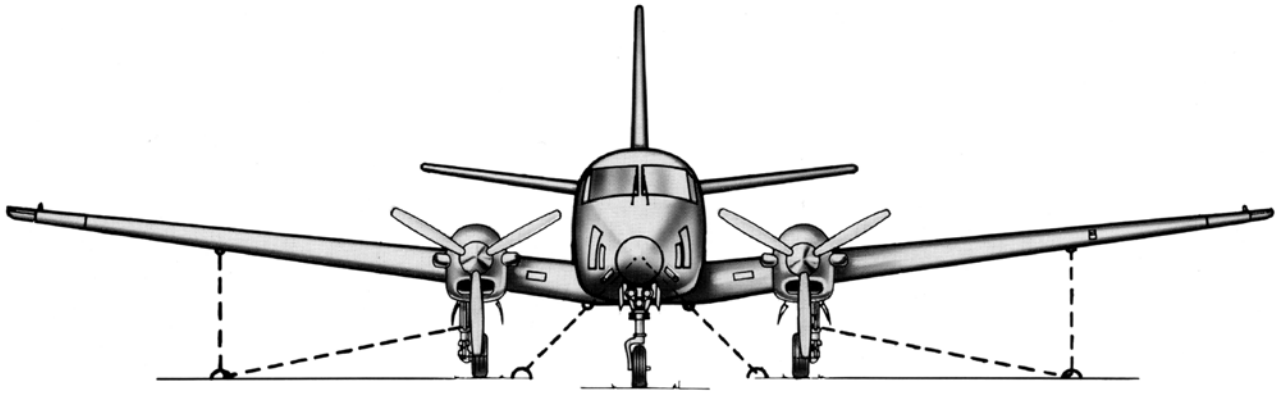


Figure 1-28. Tiedowns

TIEDOWN AND SECURING

When the airplane is parked overnight or during high winds, it should be securely moored with protective covers in place (Figure 1-28). Place wheel chocks fore and aft of the main gear wheels and nosewheel. In severe conditions the parking brake should be set.

Using the airplane mooring points, tie the airplane down with suitable chain or rope. Install the control surface lock, and be sure the flaps are up. Secure the propellers with appropriate tiedown boots (one blade down) to prevent wind-milling (Figure 1-29).

This airplane has free spinning propellers that could be hazardous if not restrained. Windmilling gears and bearings without lubrication is not good practice. When there is blowing dust or rain, install the pitot mast cover, as well as the engine inlet and exhaust covers.

Two items require particular attention: the parking brake handle mounted just under the left corner of the pilot's subpanel and the rudder pedal gust lock. Before towing the airplane, the parking brake must be released (brake knob pushed in) and the rudder gust lock removed. Serious damage to tires, brakes, and steering linkage can result if these items are not released.

Figure 1-29. Propeller Boots

TAXIING

The ground turning radii are predicated on the use of partial braking action, differential power, and the nosewheel fully castored in the direction of the turn (Figure 1-30). Locking the inside brake can cause tire or strut damage. When turning the airplane, if the wingtip clears obstacles the tail will also. The turning radius for the wingtip is 35 feet 6 inches. While turning, the pilot should be aware of vertical stabilizer clearance, which is 14 feet 3 inches.

When taxiing, turning, and starting the engines, there is an area directly to the rear of the engines where the propeller windstream can be hazardous to persons or parked airplanes (Figure 1-31). While the velocities and temperatures cannot be accurately measured, reasonable care should be taken to prevent incidents within these danger areas.

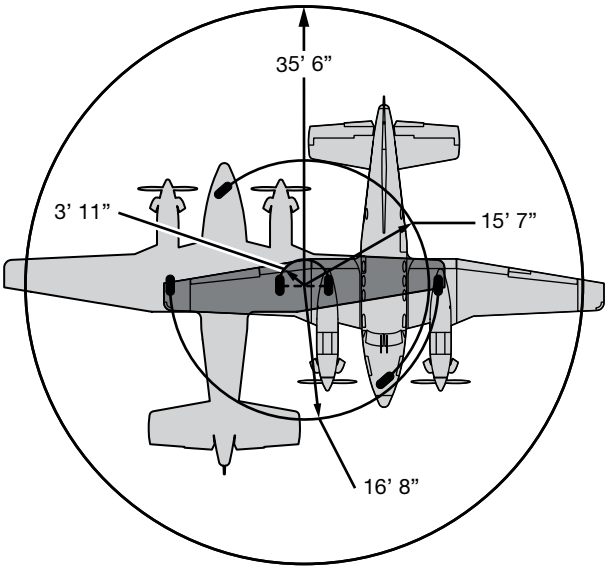


Figure 1-30. Turning Radius

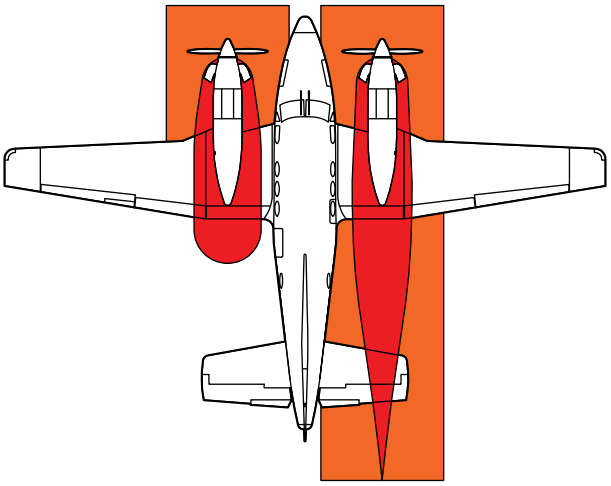
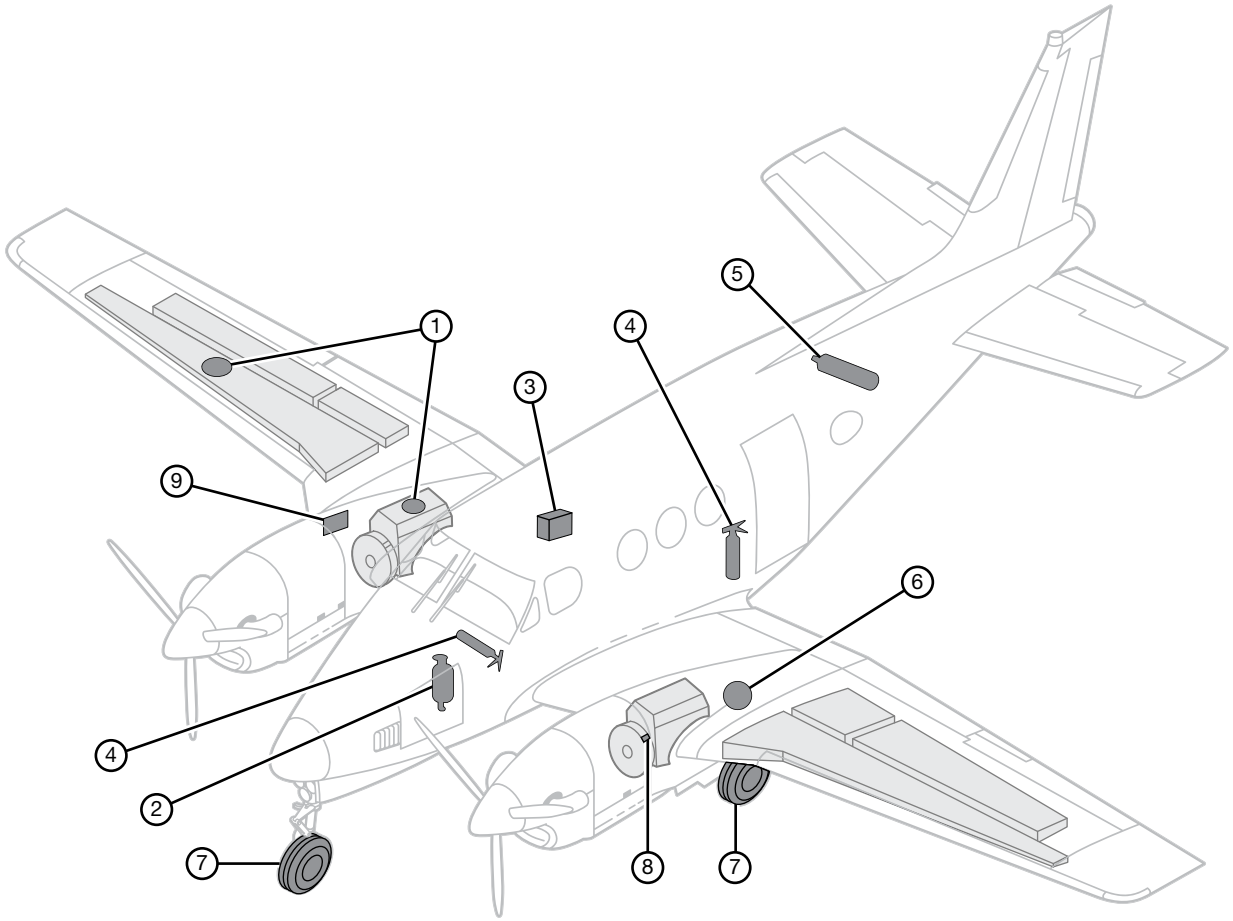


Figure 1-31. Danger Areas

SERVICING DATA

The “Handling, Servicing, and Maintenance” section of the *POH* outlines to the Owner and Operator the requirements for maintaining the King Air C90A and C90B in a condition equal to that of its original manufacture. This information sets time intervals at which the airplane should be taken to a Beechcraft Aviation Center for periodic servicing or preventive maintenance. All limits, procedures, safety practices, time limits, servicing and maintenance requirements contained in the *POH* are mandatory. This section of the *POH* includes a Consumable Materials chart which lists approved and recommended materials for servicing the airplane (Figure 1-32). The “Servicing Schedule and Lubrication Schedule” lists and illustrates servicing points and materials required.



(1) FUEL TANK FILLER CAPS (TYPICAL LEFT & RIGHT)
APPROVED FUEL GRADES AND ADDITIVES
RECOMMENDED ENGINE FUELS
COMMERCIAL GRADES:
JET A
JETA-1
JET B
MILITARY GRADES:
JP-4
JP-5
JP-8
EMERGENCY ENGINE FUELS
AVIATION GASOLINE GRADES:
80 (RED) (FORMERLY 80/87)
100LL (BLUE)*
100 (GREEN) (FORMERLY 100/130)
115/145 (PURPLE)

(2) HYDRAULIC FLUID RESERVOIR (BRAKE)
SPECIFICATION MIL-H-5606,
(REF. MAINTENANCE MANUAL)

(3) BATTERY (LEAD ACID LJ1534 & AFTER)
24 VOLT, 34 AMPERE-HOUR, NICKEL-CADMIUM

(4) FIRE EXTINGUISHERS (HAND TYPE) HALON 1301

(5) OXYGEN SUPPLY CYLINDER
OXYGEN SPECIFICATION: MIL-0-27210
AVIATORS BREATHING OXYGEN 22, 49, OR
66 CU. FT.

(6) ENGINE FIRE EXTINGUISHER
(TYPICAL LEFT & RIGHT)
EXTINGUISHING AGENT: MIL-E-52031
2.5 LBS. CF3BR, 450 PSI (DRY NITROGEN)

(7) TIRE SIZE:
MAIN WHEELS - 8.50 X 10 (TUBELESS,
8-OR 10-PLY)
NOSE WHEEL - 6.50 X 10 (TUBELESS, 6-PLY)
TIRE PRESSURE:
MAIN WHEELS - 52-58 PSI
NOSE WHEEL - 50.55 PSI

(8) ENGINE OIL DIPSTICK (TYPICAL LEFT & RIGHT)
OIL SPECIFICATION: P & W SERVICE BULLETIN
NO. 1001, 14 US QUARTS

(9) DC EXTERNAL POWER RECEPTACLE

*IN SOME COUNTRIES, THIS FUEL IS
COLORED GREEN AND DESIGNATED "100L."

Figure 1-32. Servicing Data

PRODUCT SUPPORT

Beech Aircraft has established service facilities throughout the world, which are fully equipped and professionally staffed to provide total support for the Super King Airs.

These facilities are listed in the Beechcraft Quality Service Center Directory (USA) and the International Service Facility Directory, copies of which are provided to each new Beechcraft owner. To support this worldwide service organization, Beech Aircraft, through its Parts and Equipment Marketing Wholesalers and International Distributors, provides a computer-controlled parts service that assures rapid shipment of equipment on a 24-hour basis.

PREFLIGHT INSPECTION

The preflight inspection procedure in the *POH* has been divided into five areas, as shown in Figure 1-33. The inspection begins in the flight compartment, proceeds aft, then moves clockwise around the aircraft, discussing the left wing, landing gear, left engine and propeller, nose section, etc.

Exterior Inspection

1. Cockpit check
2. Left wing, landing gear, engine, nacelle and propeller
3. Nose section
4. Right wing, landing gear, engine, nacelle and propeller
5. Empennage and tail

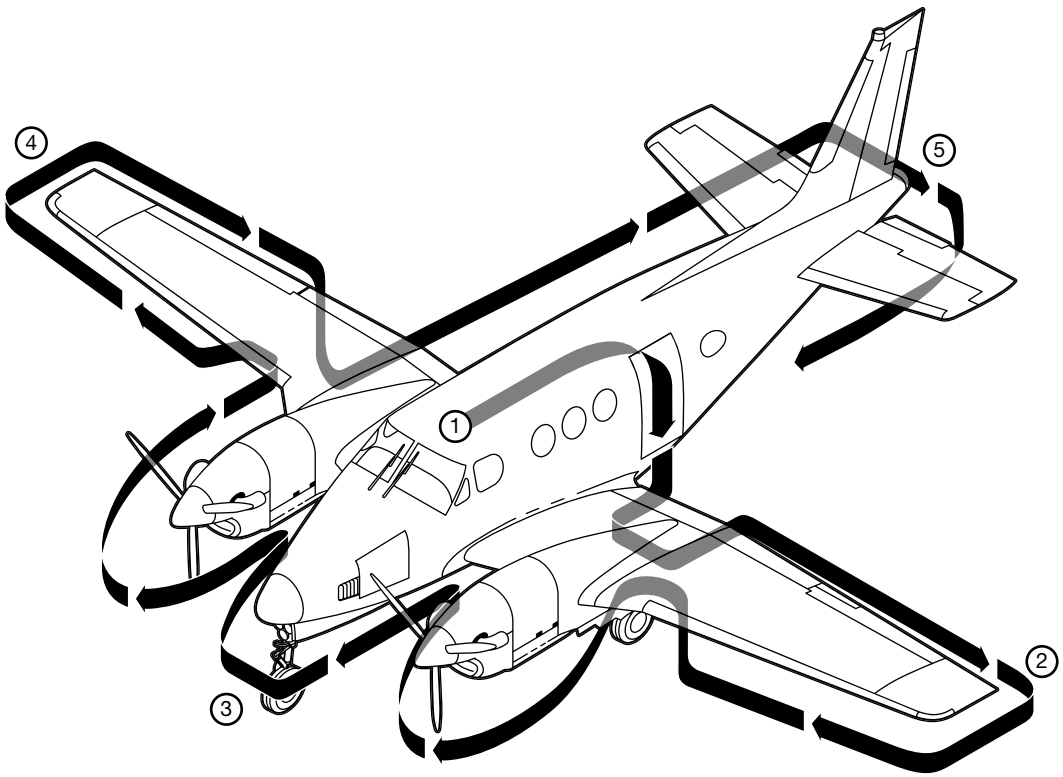


Figure 1-33. Exterior Inspection

CHAPTER 2
ELECTRICAL POWER SYSTEMS

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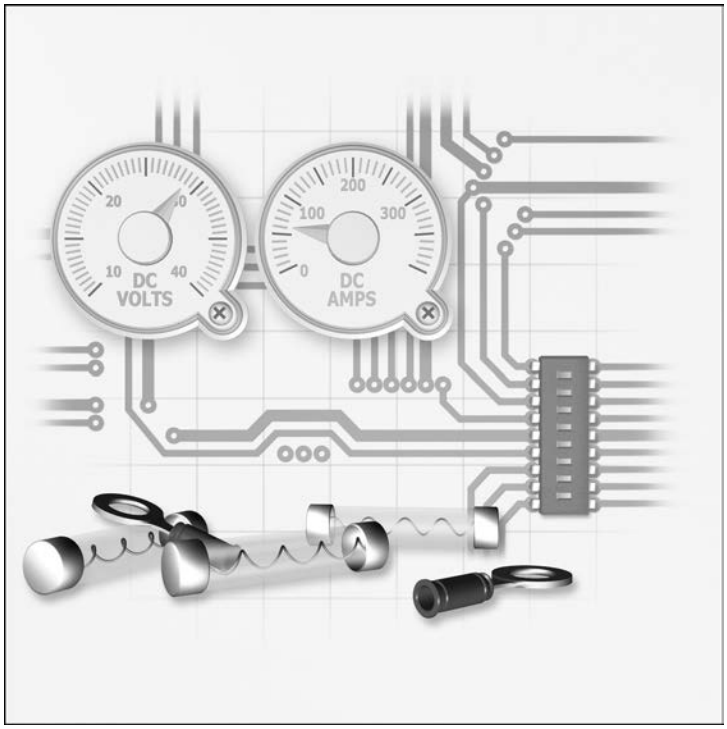
CHAPTER 2
ELECTRICAL POWER SYSTEMS



INTRODUCTION

Familiarity with, and an understanding of, the airplane electrical system will ease pilot workload in normal operations in case of an electrical system or component failure. The pilot should be able to locate and identify switches and circuit breakers quickly, and should also be familiar with appropriate corrective actions in emergency situations.

GENERAL

The Electrical System section of the training manual presents a description and discussion of the airplane electrical system and components. The electrical system is discussed to the extent necessary for the pilot to cope with normal and emergency operations. The location and purpose of switches, indicators, and circuit breakers, along with DC and AC generation and distribution are described. This section also includes some of the limits of, and possible faults with, systems or components.

L	=	LEFT
R	=	RIGHT
BT	=	BATTERY
LC	=	LINE CONTACTOR
SB	=	SUB BUS
SR	=	STARTER RELAY
EPR	=	EXTERNAL POWER RELAY
STR/GEN	=	STARTER GENERATOR
GEN CONT	=	GENERATOR CONTROL
EXT PWR	=	EXTERNAL POWER
CRT BUS	=	CENTER BUS
RG	=	RIGHT GENERATOR
LG	=	LEFT GENERATOR
RCSR	=	RIGHT CROSS START RELAY
LCSR	=	LEFT CROSS START RELAY

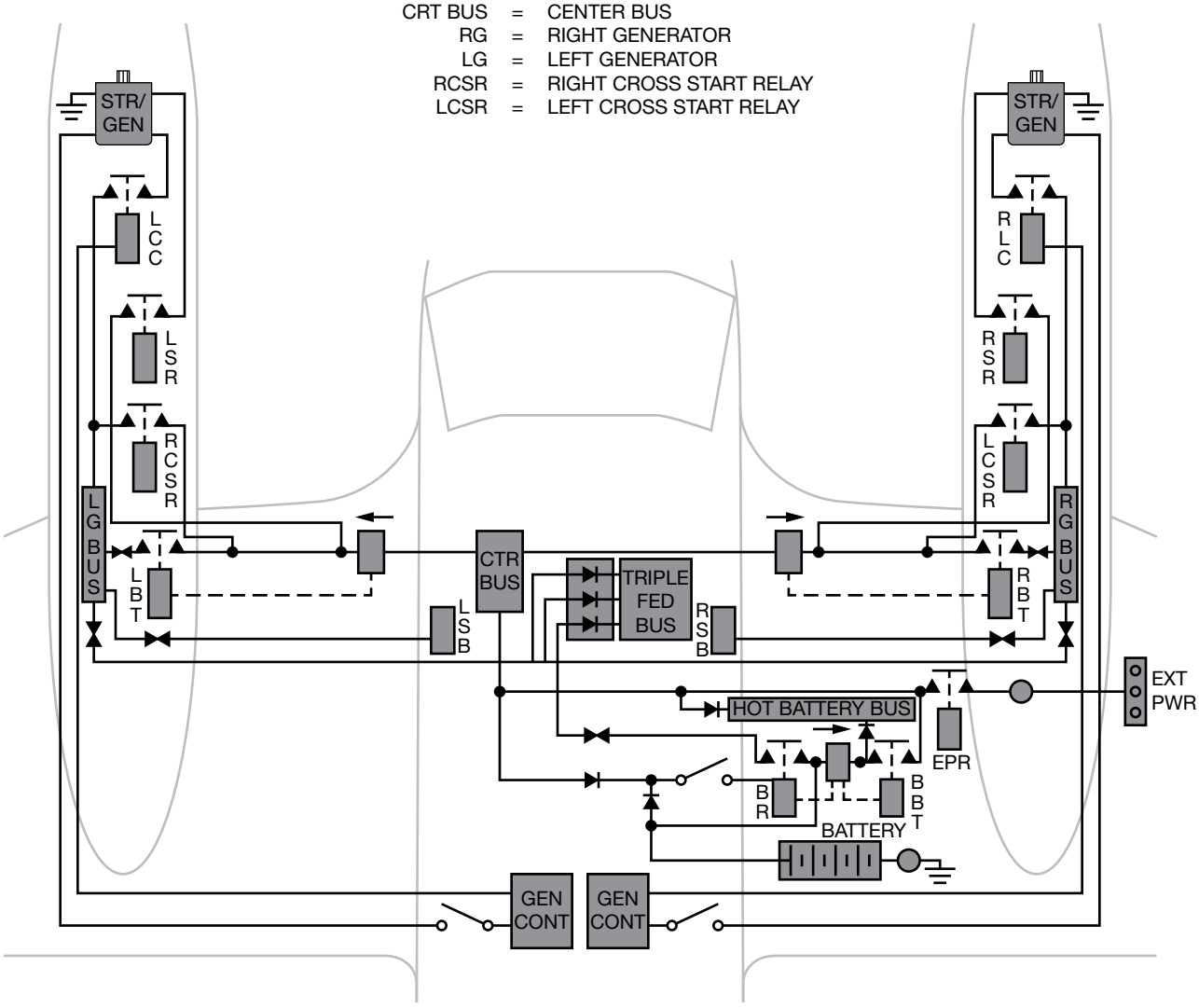


Figure 2-1. Electrical System Component Locations

BATTERY AND GENERATOR

The airplane electrical system is a 28-VDC (nominal) system with the negative lead of each power source grounded to the main airplane structure. DC electrical power is provided by one 34-ampere-hour, air-cooled, 20-cell, nickel-cadmium battery (airplanes prior to LJ-1534) or one 42-ampere-hour, sealed, lead-acid battery (airplanes LJ-1534 and after), and two 250-ampere starter/generators connected in parallel. Basic electrical symbols are shown in Figure 2-2.

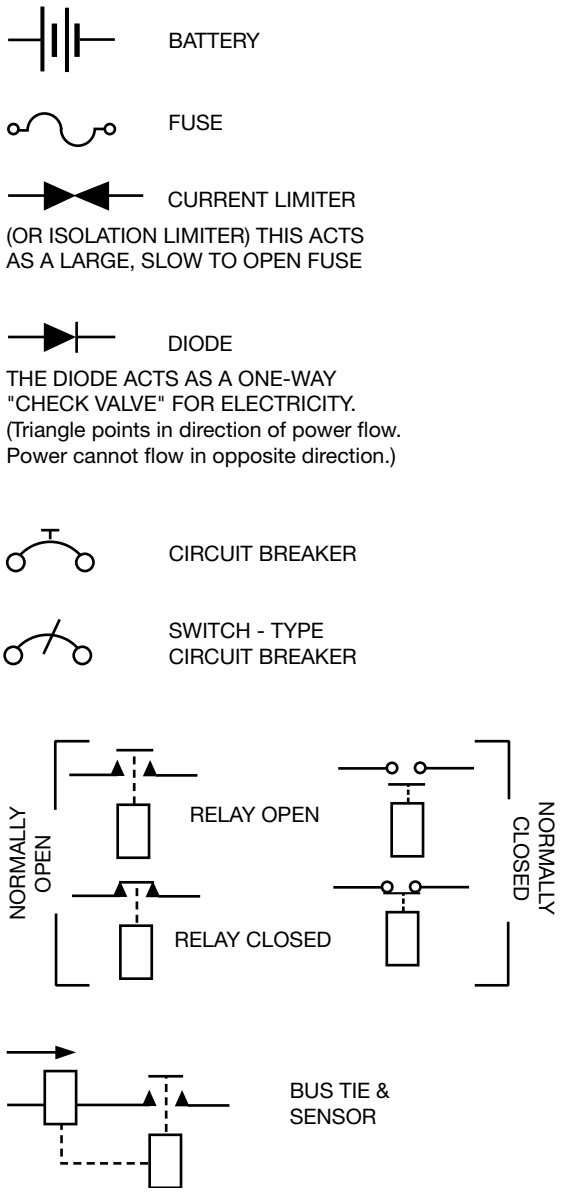


Figure 2-2. Basic Electrical Symbols

This system is capable of supplying power to all subsystems necessary for normal operation of the airplane. The battery and generator switches on the pilot's left subpanel are used to control power from the battery and generators into the airplane electrical system.

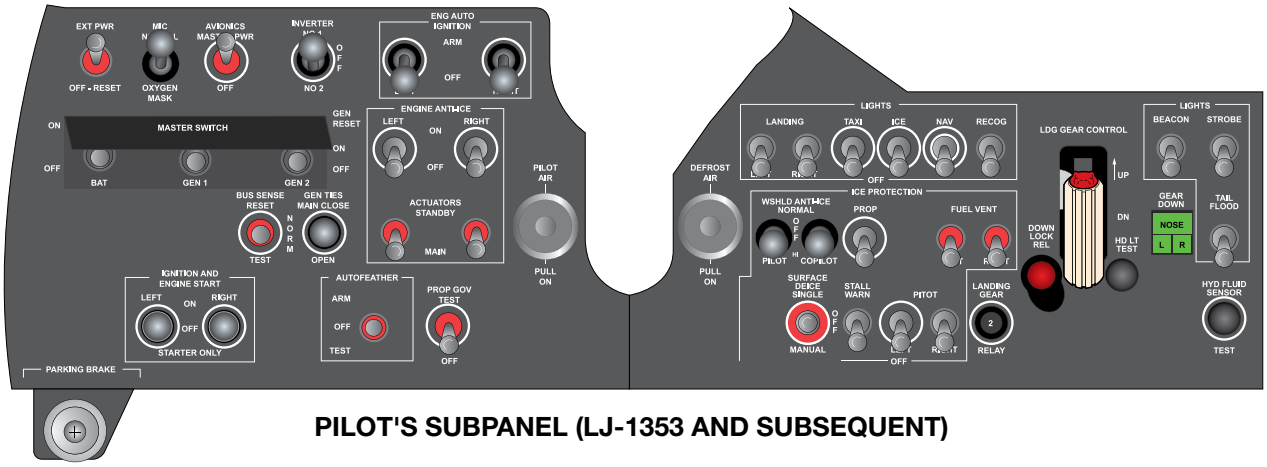
The battery is always connected to the hot battery bus (Figure 2-21). Both are located in the right wing center section. Operation of equipment on the hot battery bus does not depend on the position of the battery switch. The battery switch, on the pilot's left subpanel, closes a battery bus tie and a battery relay which connect the battery to the rest of the electrical system.

The generators are controlled by individual generator control panels which allow constant voltage to be presented to the buses during variations in engine speed and electrical load requirements. The load on each generator is indicated by left and right loadmeters located on the overhead meter panel (Figure 2-4). A normal system potential of 28.25 ±0.25 volts maintains the battery at full charge.

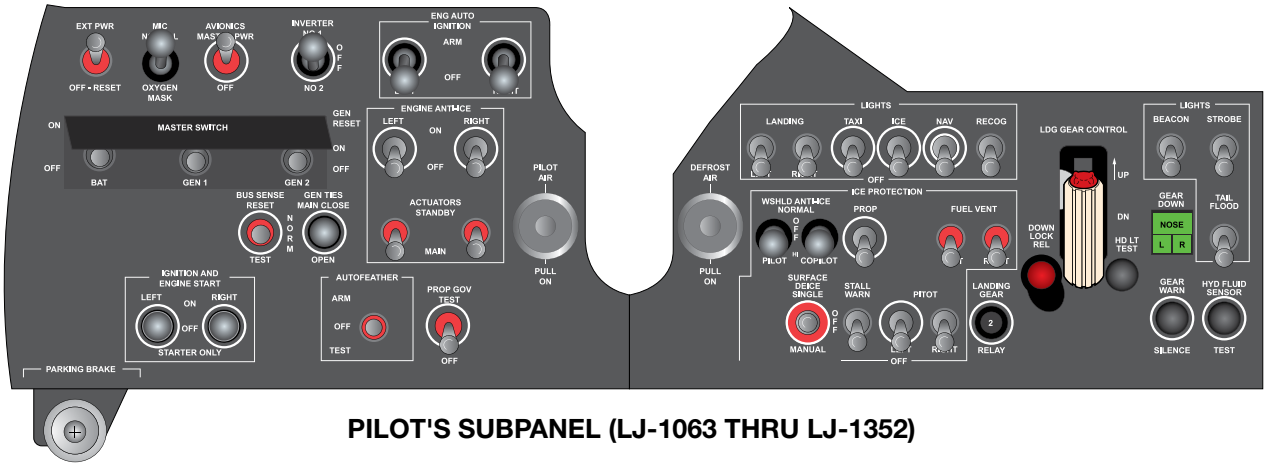
This airplane utilizes a multi-bus system. The main buses are the left and right generator buses, center bus, triple-fed bus, and battery emergency bus. Switches in the cockpit which receive power from the center and triple-fed buses are identified by a white ring on the panel around the switch.

Electrical loads are divided among the buses as noted on the Electrical System Buses and Feeders chart (Figure 2-5). Equipment on the buses is arranged so that all items with duplicate functions (such as right and left landing lights) are connected to different buses. Circuit breakers on the same feeder or sub-bus are connected by white lines on the right circuit breaker panel face board (Figure 2-6).

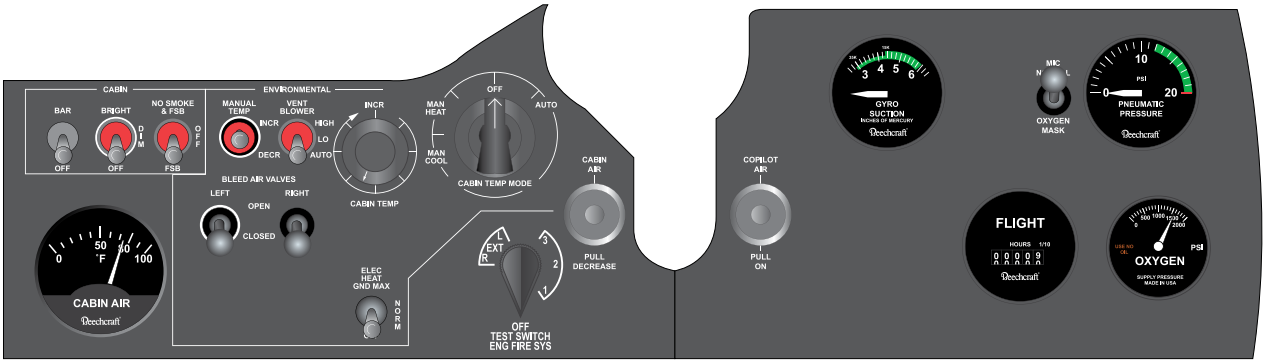
In normal operation, all buses are automatically tied into a single-loop system where all sources supply power through individual protective devices. The triple-fed bus is powered from the battery and both generator buses. The left and right generators supply power to their respective left and right generator buses.



PILOT'S SUBPANEL (LJ-1353 AND SUBSEQUENT)

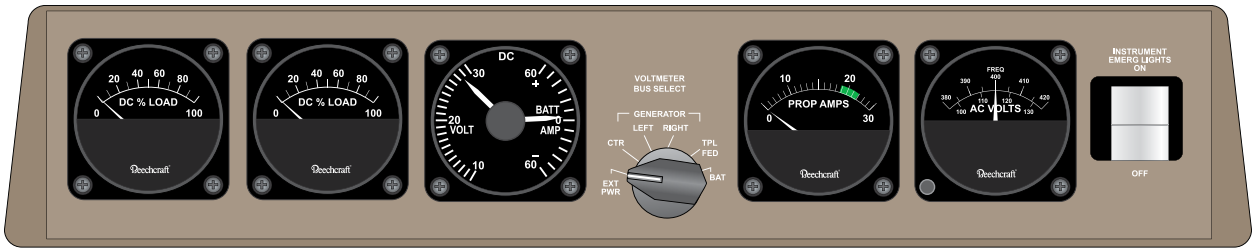


PILOT'S SUBPANEL (LJ-1063 THRU LJ-1352)

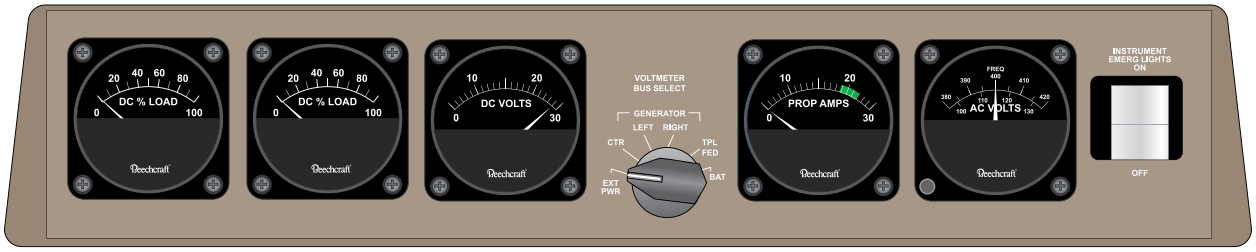


COPILOT'S SUBPANEL (C90A AND B)

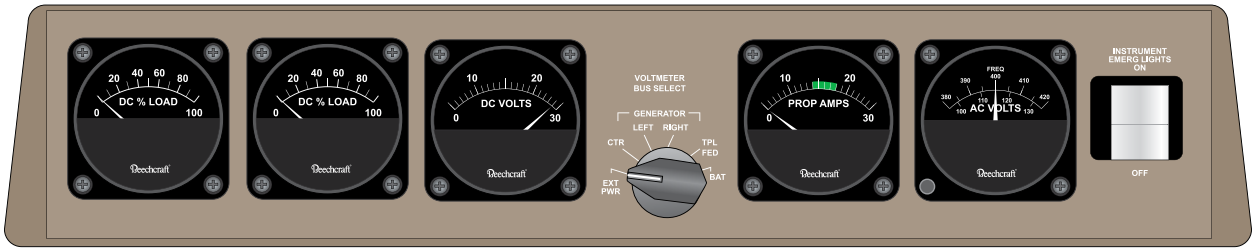
Figure 2-3. Pilot and Copilot Subpanels



OVERHEAD METER PANEL (LJ-1534 AND AFTER)



OVERHEAD METER PANEL (LJ-1353 TO LJ-1533)



OVERHEAD METER PANEL (PRIOR TO LJ-1353)

Figure 2-4. Overhead Meter Panel

Table 2-1. ELECTRICAL SYSTEM BUSES AND FEEDERS (1 of 2)

LEFT GENERATOR BUS	CENTER BUS	RIGHT GENERATOR BUS
(Gen No. 1) Flap Motor Flap Control & Indicator L Generator Bus (Bus Tie & Meter Indication) Flight Instrument Lights Avionics & Engine Instrument Lights Prop Synchrophaser (Opt) L Main Anti-ice (Ice Vane) L Chip Detector L Engine Fuel Control Heater R Bleed Air Control Cigarette Lighter L Landing Light L Fuel Vent Heat No. 2 Avionics Bus L Generator Control Panel (1) L Generator Field & Sense (1) L Generator Loadmeter Pilot Windshield Anti-ice (1) R Standby Anti-ice (Ice Vane) Vent Blower Rotating Beacon Lights Tail Flood Lights (Opt) R Firewall Valve R Fuel Boost Pump Crossfeed No. 1 Inverter Power Control (1) No. 1 Inverter Power Select (1) (Avionics)* Comm 2 ADF 1 Copilot Audio Autopilot Transponder 2 DME 2 VLF/OMEGA Weather Radar Paging Amplifier Check List (Radar) Data Nav VNAV EFIS	AVIONICS ANNUNCIATOR GENERATOR RESET PNEUMATIC SURFACE DEICE WINDSHIELD WIPER TAXI LIGHT ICING LIGHT NO. 1 INVERTER POWER CONTROL NO. 2 INVERTER POWER CONTROL PROP DEICE POWER PROP DEICE CONTROL AIR CONDITIONER NORMAL HEAT (ELECTRIC) MAX HEAT (ELECTRIC) LANDING GEAR MOTOR (1) NO. 1 & NO. 2 INVERTER POWER	(Gen No. 2) Trim Tab (Opt) R Generator Bus (Bus Tie & Meter Indication) Overhead, Subpanel & Pedestal Lights Side Panel Lights Cabin Reading Lights & Sign Chime R Main Anti-ice (Ice Vane) R Chip Detector R Engine Fuel Control Heater Rudder Boost Electric Toilet (Opt) R Landing Light R Fuel Vent Heat No. 3 Avionics Bus R Generator Control Panel (1) R Generator Field & Sense (1) R Generator Loadmeter (1) Copilot Windshield Anti-ice (1) L Standby Anti-ice (Ice Vane) Furnishing - (Refreshment Bar) R Landing Lights Strobe Lights (Opt) R Pitot Heat Stall Warning Heat L Firewall Valve L Fuel Boost Pump Crossfeed No. 2 Inverter Power Control (1) No. 2 Inverter Power Select (1) (Avionics)* Nav 2 Glideslope 2 Transponder 1 DME 1 Compass 2 Flight Director 2 ADF 2 Marker Beacon 2 HF Radar Altimeter Radio Telephone RMI 1 Stereo Interphone Flight Path Advisory Air Data Encoder GPS Integrated Hazard Avoidance System Multifunction Display

Table 2-1. ELECTRICAL SYSTEM BUSES AND FEEDERS (2 of 2)

TRIPLE-FED OR BATTERY BUS		HOT BATTERY BUS
Landing Gear Warning Horn	R Generator Overheat (CAA)	L Engine Fire Extinguisher (1)
Annunciator Power	R Starter Control	R Engine Fire Extinguisher (1)
Stall Warning System	R Ignitor Power	RNAV Memory (1)
L Generator Overheat (CAA)	R Fire Detection (CAA)	(Ground Comm Power)
L Starter Control	R Oil Temp & Oil Press	Entrance & Aft Dome Lights (1)
L Ignitor Power	R Oil Pressure Warning (CAA)	Stereo (1)
Fire Detection	Prop Governor Test	Battery Relay Control (1)
L Fire Detection (CAA)	R Fuel Flow Indicator	L Fuel Boost Pump (1)
L Oil Temperature & Oil Pressure	L Bleed Air Control	R Fuel Boost Pump (1)
L Oil Pressure Warning (CAA)	Cabin Pressure Loss (CAA)	Fuel Crossfeed (1)
Autofeather System (Opt)	Bus Tie Control	
L Fuel Flow Indicator	No. 1 Avionics Bus	(1) The circuit breaker in this circuit is not accessible to the pilot in flight.
L Pitot Heat	R Firewall Fuel Valve	
Landing Gear Control	R Boost Pump	
Pilot Turn & Slip	R Fuel Quantity Indicator	
Navigation Lights	R Fuel Pressure Warning	
Pilot Encoder & Altimeter		
Avionics Master Control	(Avionics)*	
Instrument Indirect Lights	Comm 1	
Cabin Fluorescent Lights	Nav 1	
Triple-Fed Bus	Glideslope 1	
(Bus Tie & Meter Indication)	Radio Relays	
L Firewall Fuel Valve	Compass 1	
L Boost Pump	Flight Director 1	
L Transfer Pump	Pilot Audio	
L Fuel Quantity Indicator	Marker Beacon 1	
L Fuel Pressure Warning	Servo Altimeter	
Crossfeed Fuel Valve	RNAV	
Landing Gear	RMI 2	
Position Indicator		
Annunciator Indicator		

*Optional avionics busing. Check avionics circuit breaker panel or wiring diagram for specific busing configuration.

The center bus is fed by two generator buses and the battery, which automatically connects those components whenever the bus ties are closed. The power distribution schematics (Figures 2-14 through 2-28) show how buses are interconnected.

Voltage on each bus may be monitored on the voltmeter (located in the overhead panel) by selecting the desired bus using the VOLTMETER BUS SELECT switch, adjacent to the voltmeter. The electrical system provides maximum protection against loss of electrical power should a ground fault occur. High current (Hall effect) sensors, bus tie relays and current limiters are provided to isolate a fault from its power source. The electrical system bus arrangement is designed to provide multiple power sources for all circuits

DC POWER DISTRIBUTION

The DC power distribution system is commonly called a “triple-fed” system. In normal operation, all buses are automatically tied into a single loop system in which all sources collectively supply power through individual protective devices.

Three in-flight DC power sources are available:

- One 24-volt, 34-ampere-hour nickel-cadmium battery (or one 24-volt, 42-ampere hour, lead acid battery for LJ-1534 and later)
- Two 28-volt, 250-ampere starter/generators

When the battery switch is turned ON, the battery relay and the battery bus tie relays close (Figure

2-16). Battery power is routed through the battery relay to the triple-fed bus, and through the battery bus tie relay to the center bus and to both starter relays. Neither generator bus is powered since the generator bus ties are normally open, however, battery power is available to permit starting either engine.

After either engine has been started and the generator switch has been moved to RESET, the generator control unit (GCU) will bring the generator up to voltage. Releasing the spring-loaded switch to the center ON position closes the generator line contactor, thereby powering the generator bus, and closing both generator ties automatically. This action distributes power through the 250-amp current limiters and the generator bus tie relays. Generator output will then be routed through the center bus to permit battery charging. In addition, the opposite generator bus and triple-fed bus will be powered by the generator, supplying 28-VDC power to the five primary airplane buses (Figure 2-19) When both generators are operating, each generator directly feeds its respective generator bus.

The generator buses, hot battery bus, and battery are tied together by the center bus. The triple-fed bus is powered by the battery and each generator bus through 60-amp limiters and through diodes providing fault isolation protection between the power sources.

BUS TIE SYSTEM

The electrical system is protected from excessively high current flow by the bus tie system. Three current sensors, consisting of Hall effect devices and solid-state circuitry, are used to sense current flow through the portion of the circuit being monitored. Two bus tie sensors and their relays are located between the generator buses and the center bus, and a third is between the battery and the center bus.

With no power applied to the aircraft electrical system, all three bus tie relays are open. When the BAT switch is turned ON, hot battery bus voltage energizes the coil circuit of the battery bus tie relay, thereby closing it. This action has no effect on the generator bus ties.

A similar action occurs when a generator or external power is brought on-line. When either generator is brought on-line, voltage from the generator control panel energizes the coil circuit of both generator bus tie relays. This switches voltage from the L and R GEN TIE OPEN annunciators to the relays, causing the annunciators to extinguish and the bus tie relays to close. When external power is brought on-line, the only difference is the source of generator bus tie coil voltage, which is the small pin of the external power receptacle. Neither generator or external power affect the battery bus tie circuitry unless the battery switch is also turned ON.

Activation of an internal, solid-state switch within the sensor by a current of at least 275 ±5 amperes will open the coil circuit of the relay, causing it to deenergize and open the associated bus tie relay. The coil circuit of the bus tie relay is latched open to prevent the bus tie relay from closing. Deenergizing the bus tie relay will illuminate the appropriate BUS TIE OPEN annunciator. When the bus tie relay has been opened by excessively high current flow through the Hall effect sensor (i.e. a bus fault), it can only be reset by momentarily activating the BUS SENSE switch on the pilot's left subpanel to RESET. The Hall effect sensors are unidirectional. They only sense overcurrent in the direction of the arrow on the symbol.

Two switches located on the pilot's left subpanel control the bus tie system. One switch, placarded BUS SENSE–TEST–RESET, is spring loaded to the center (NORM) position. Momentarily activating it to TEST connects bus voltage to all three current sensor test circuits (Figure 2-23). This voltage simulates the condition resulting from a high current through each bus tie relay. The solid state switches of each sensor are thus activated to de-energize (open) their respective relays, thereby opening the bus tie relays and activating the annunciator readouts. Once activated, the test circuitry latches the bus ties open, preventing their automatic closing.

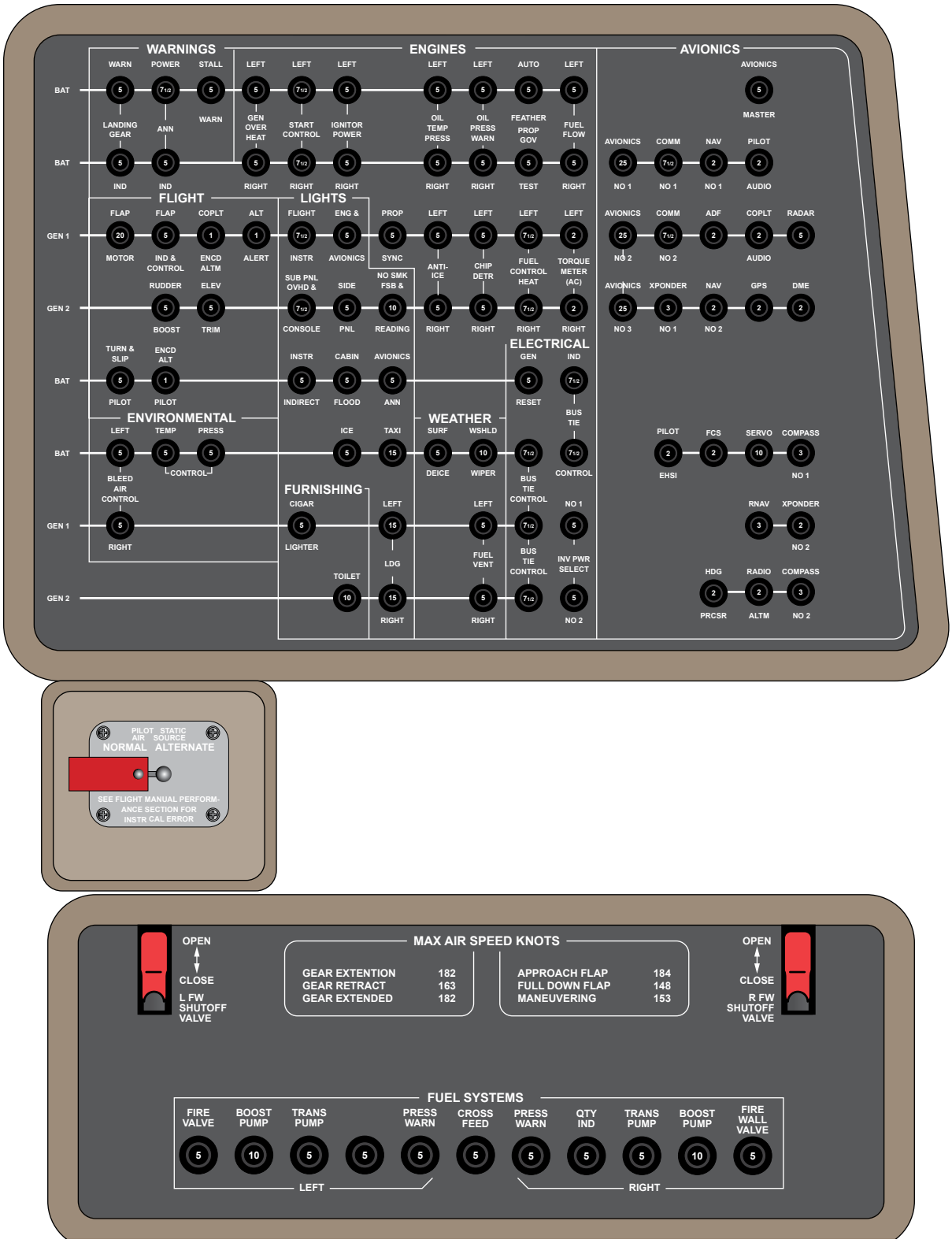


Figure 2-5. Right Side and Fuel Management Circuit Breaker Panels (LJ-1063 Thru LJ-1360, LJ-1362

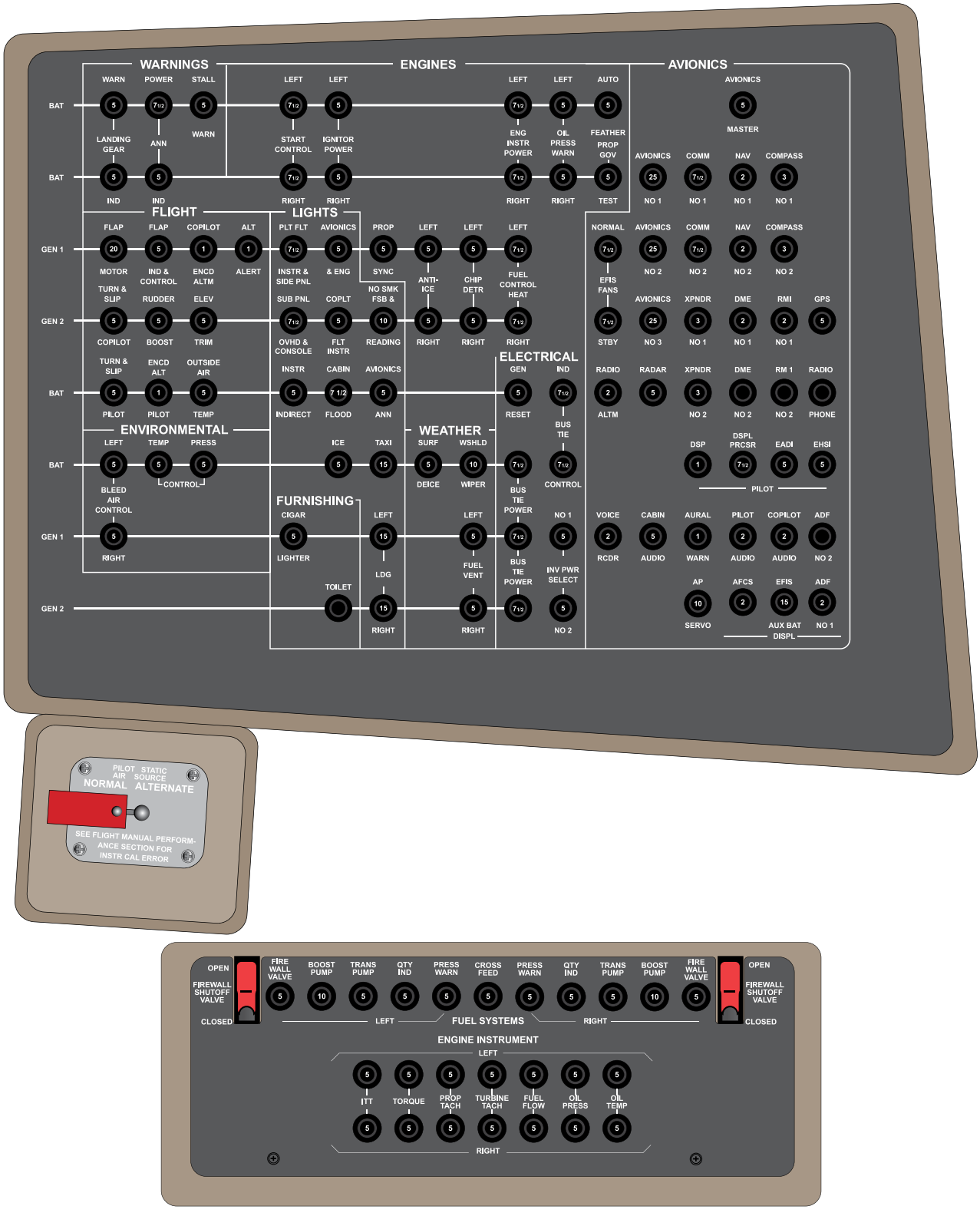


Figure 2-6. Right Side and Fuel Management Circuit Breaker Panels (LJ-1361, LJ-1363 and After)

Current sensor reaction time is approximately 0.010 seconds for the generator current sensors and 0.012 seconds for the battery current sensor. Once activated, the relays latch open, and reaction time for the system is limited to reaction time for the relays. Therefore, only momentary activation of the TEST switch is required. Prolonged activation of this switch will damage or destroy the sensor modules and should be avoided.

Momentary activation of the switch to RESET powers the coil of the bus tie relays, unlatching the test circuits and, permitting the bus ties to energize (close). Voltage is transferred from the annunciator readouts to the coils, closing the bus tie relays. Since high-current sensing is latched out when the switch is in RESET, only momentary activation is desirable. This prevents accidental welding of the bus tie relay contacts and/or opening a 250-amp current limiter by a bus ground fault.

The second switch on the pilot's left subpanel controls the bus tie system and is placarded GEN TIES–MAN CLOSE–NORM–OPEN. This switch must be lifted (lever-lock) to move it from center to OPEN. This switch is spring loaded to MAN CLOSE.

Only the generator bus tie relays may be manually opened or closed with this switch. Manually closing the generator bus tie relays will connect the generator buses to the center bus and power to the entire system (Figure 2-17). Momentarily placing the switch in CLOSE applies bus voltage to the coil of the generator bus tie relays, completes a latching circuit, activates the MAN TIES CLOSE annunciator and closes the bus tie relays. The latching circuit is completed through the normally closed contacts of the control relay for the generator line contactors. A generator bus tie relay cannot be manually closed if a fault opened the tie; the BUS SENSE switch must be momentarily activated to RESET, which resets the tie.

When the generator ties are closed, the GEN TIES switch can open the generator bus ties as certain normal/abnormal procedures may dictate. When the GEN TIES switch is positioned to OPEN, the ground is removed from the relay circuit which allows the relay to spring open.

BUS ISOLATION

Bus isolation is one of the features of the multi-bus electrical system. The two generator buses and the center bus are protected by high-current sensing (Hall effect) devices. In case of excessive current draw on one bus, the sensors will isolate the affected bus by opening its bus tie, allowing the other buses to continue operating as a system. During cross-generator engine starts, the high current sensors and current limiters are bypassed by cross-start relays to allow the required high current flow to pass from the power sources to the starter generator without causing the bus ties to open. Battery starts are routed through the battery bus tie, which is desensitized for starting.

A 250-amp current limiter (slow to open fuse) is also located in the circuitry between the center bus and each of the generator buses. Since the Hall effect devices sense high current in only one direction, the current limiters provide protection in the opposite direction. If an overcurrent situation causes a current limiter to open, it also will cause bus isolation.

The current protection for the triple-fed bus is provided exclusively by 60-amp current limiters. Triple-fed bus isolation will occur only if all three of these limiters open.

For typical examples of bus isolation, refer to Figure 2-25 (generator bus), Figure 2-26 (center bus), and Figure 2-27 (triple-fed bus).

LOAD SHEDDING

Load shedding is another highly beneficial feature of the multi-bus electrical system. The electrical system will automatically remove excess loads (generator buses), when the power source is reduced to battery only. When both generators are off line, the generator bus ties open and the generator bus loads are "shed" (Figure 2-24). The battery will continue to power the center, triple-fed, and hot-battery buses. If necessary, power to the generator buses can be restored by closing the generator ties manually with the GEN TIES switch (Figure 2-17). When load shedding occurs in flight, land as soon as practical, unless the situation can be remedied and at least one generator brought back on-line.

WARNING

Closing the generator bus ties manually in flight with a loss of both generators will cause the battery to discharge at a faster rate. If it becomes necessary to close the generator ties in this situation, they should be opened as soon as possible since battery power should be conserved. Without an operable generator, the battery cannot be recharged in flight. Land as soon as practical.

BATTERY

The nickel-cadmium (Ni-Cad) battery is located in the right wing center section in an air-cooled box (Figure 2-7). The battery relay, charge monitor shunt, and air-cooling thermostat are mounted in the battery compartment immediately forward of the battery. Power to the main electrical buses is routed from the battery via the battery relay and battery bus tie, which are controlled by the BAT-ON-OFF switch on the pilot's left subpanel.

The hot battery bus provides power directly to a few aircraft systems (Figure 2-15). These systems may be operated without turning the battery switch ON. Care should be taken, however, to insure that utilization of these systems is minimal when the generators are inoperative and/or the aircraft is secured to prevent excessive discharge of the battery.

The lead acid battery box is not air cooled and does not include the charge monitor shunt, or the air-cooling thermostat. A battery charge monitor system (airplanes prior to LJ-1534) advises the pilot of battery charge rate. The system is disabled on aircraft that have been converted to a lead-acid battery. A charge of more than 7 amperes, for six or more seconds, will trigger the yellow BATTERY CHARGE annunciator, indicating excessive charge rate.

The BATTERY CHARGE annunciator may occasionally illuminate for short intervals when heavy electrical draw items are cycled. For example, following a battery-powered engine start, the battery recharge current is very high and causes illumination of the BATTERY CHARGE annunciator, thus providing an automatic self-test of the battery monitor system.

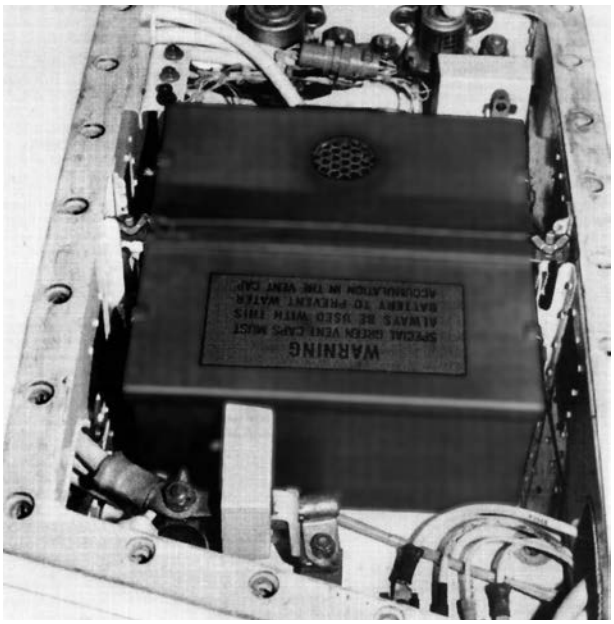


Figure 2-7. Battery Installation

As the battery approaches a full charge and the charge current decreases to a satisfactory level. the annunciator will extinguish. This will normally occur within a few minutes after an engine start, but may require a longer time if the battery has a low state of charge, low charge voltage per cell (20-cell battery), or low battery temperature.

STARTER/GENERATORS

The starter/generators are dual-purpose, engine-driven units (Figure 2-8). The same unit is used as a starter to drive the engine during engine start and as a generator to provide electrical power when driven by the engine. A series starter winding is used during starter operation and a shunt field winding is used during generator operation. The generator shunt field winding is disabled when the series starter winding is activated by the start switch. The regulated output of the generator is 28.25 ±0.25 volts with a maximum continuous load of 250 amperes.

In addition to the starter/generators, the generator system consists of control switches, generator control units (GCU), line contactors and loadmeters.

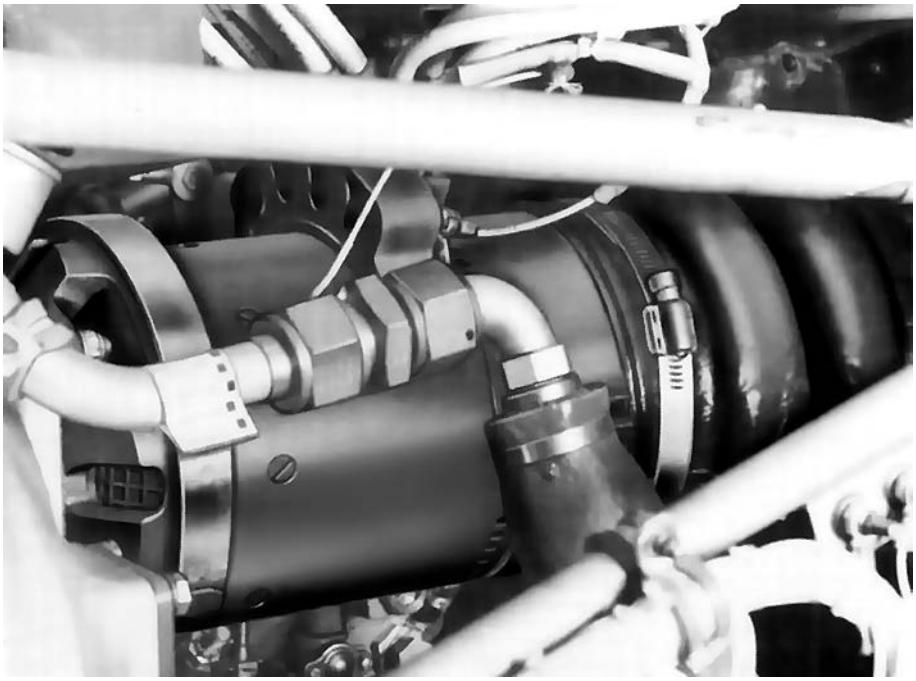


Figure 2-8. Starter/Generator Installation

Starter power to each individual starter/generator is provided by the battery, or by the operating generator for cross-starts. The start cycle is controlled by a three-position switch, one for each engine, placarded: IGNITION AND ENGINE START–LEFT–RIGHT–ON–OFF STARTER ONLY, located on the pilot's left sub-panel (Figure 2-3).

Selecting a start switch to either the STARTER only position or ON activates the starter and disables the respective generator. The starter drives the compressor section of the engine through the accessory gearbox.

During engine starts, the battery is connected to the starter/generator by the starter relay. With one engine running and its generator on the line, the opposite engine can by started with power from the battery and operating generator through the starter relay and the cross-start relay. This is called a cross-start. Normally one engine is started on battery power alone, and the second engine is cross-started.

During a cross generator start, (Figure 2-20) the operating generator control panel closes the cross-start relay, bypassing the generator bus, current limiter and bus tie relay. This assures the 250-amp current limiter will not open due to transient surges, since the generator would normally provide the current required for the start. In addition, while a starter is selected the bus tie sensors are disabled to prevent them from opening their respective bus tie relays.

CAUTION

Do not exceed the starter motor operating time limits of 40 seconds ON, 1 minute off, 40 seconds ON, 1 minute off, 40 seconds ON, then 30 minutes off.

DC Generation

The generator phase of operation is controlled by the generator switches, located in the pilot's left subpanel, next to the BAT switch under the MASTER SWITCH gang bar (Figure 2-3). The switches provide OFF, ON, and RESET capabilities. The generating system is self-exciting and does not require electrical power from the aircraft electrical system for operation.

Generator operation is controlled through two generator control units (GCU) mounted below the center aisle floor, that make constant voltage available to the buses during variations in engine speed and electrical load requirements. The generators are manually connected to the GCUs by GEN 1 and GEN 2 control switches located on the pilot's left subpanel. The load on each generator is indicated by the respective left and right loadmeters located on the overhead panel (Figure 2-3).

The generator control units are designed to control the generators and the load shared within 2.5 percent.

The generator control units (GCU) provide the following functions:

- Voltage regulation and line contactor control
- Overvoltage and overexcitation protection
- Paralleling/load sharing
- Reverse-current protection
- Cross-start relay activation

Voltage Regulation and Line Contactor Control

The generators are normally regulated to 28.25 ±.25 VDC. When the generator control switch is held to RESET, generator residual voltage is applied through the GCU to the generator shunt field causing the generator output voltage to rise. When the switch is released to ON, the 28-volt regulator circuit takes over and begins controlling the generator shunt field in order to maintain a constant output voltage. The voltage regulator circuit varies shunt field excitation as required to maintain a constant 28-volt output from the generator for all rated conditions of generator speed, load, and temperature.

When the generator switch is released to ON generator voltage is applied to the GCU to enable the line contactor control circuit. The GCU compares the generator output voltage with aircraft bus voltage. If the generator output voltage is within 0.5 volts of the aircraft bus voltage, the GCU sends a signal to the line contactor which closes and connects the generator to the aircraft bus (Figure 2-21) and closes both generator ties to connect the center bus and the generator buses. This allows the generator to recharge the aircraft battery and power all aircraft electrical loads.

During single-generator operation, the GCU opens the line contactor and isolates the inoperative generator from its bus.

Overvoltage and Overexcitation Protection

The GCU provides overvoltage protection to prevent excessive generator voltage from being applied to the aircraft equipment. If a generator output exceeds the maximum allowable 32volts, the overexcitation circuits of the GCU will detect which generator is producing excessive voltage output and attempting to absorb all the aircraft electrical loads. The GCU overexcitation circuit will then disconnect the generator from the electrical system.

Paralleling/Load Sharing

The paralleling circuit averages the output of both generators to equalize load levels. The paralleling circuits of both GCUs become operative when both generators are on the line. The paralleling circuits sense the interpole winding voltages of both generators to provide an indication of the load on each generator.

The voltage regulator circuits are then biased up or down as required to increase or decrease generator loads until both generators share the load equally. The GCUs are designed to balance loads to within 2.5 percent.

Reverse-Current Protection

Reverse-current protection is provided by the GCU. When a generator becomes underexcited or cannot maintain bus voltage, i.e., low generator speed during engine shutdown, it will begin to draw current (reverse current) from the aircraft electrical system. The GCU senses the reverse current by monitoring the generator interpole voltage and opens the line contactor to protect the generator.

Cross-Start Relay Activation

During cross-start, the operating generator helps to start the second engine. The cross-start relay on the operating generator circuit closes to allow starting current to bypass the generator bus, current limiter, and bus tie relay. The current flows through the center bus, to the Hall effect sensor on the opposite generator bus.

During start, the Hall effect sensors are disabled, so no bus isolation takes place. The current is routed to the starter physically between the Hall effect sensor and the bus tie relay, so if the bus tie opened, it wouldn't effect engine start. The current is then made available to the start relay for engine start.

AC POWER DISTRIBUTION

AC power for the avionics equipment and the AC-powered engine instruments is supplied by two inverters (Figure 2-9). Either one may be used at the pilot's discretion through the inverter selector switch. Each inverter provides two levels of power: 115 volts, 400 Hz for the avionics equipment and 26 volts, 400 Hz for the applicable engine instruments, and some avionics. Output of the standard inverter is rated at 250 volt-amps. An optional inverter is rated at 300 volt-amps.

The inverters are installed in the wing, immediately outboard of each nacelle. Inverter operation is controlled by the INVERTER NO. 1–OFF NO. 2 select switch on the pilot's left subpanel. Selection of either inverter actuates the inverter power relay installed nearby to supply it with DC power. An inverter select relay provides the necessary switching to permit the operating inverter to supply 26 VAC avionics and instrument power, and 115 VAC avionics and test jack power. The inverter select relay is energized when the number one inverter is selected. It is de-energized when the inverter switch selects either OFF or NO. 2.

Dual sources of DC input power are provided for each inverter. The power select relay for each inverter is automatically selected to provide inverter power from the adjacent generator bus, or from the center bus if the generator bus is de-energized (Figure 2-10, Figure 2-11, and Figure 2-12). When battery power is applied to the center bus prior to engine start (Figure 2-16), inverter power is routed through a circuit breaker and the normally-closed contacts of an inverter power-select relay to the power relay of each inverter. When the generator buses are powered (Figure 2-21), voltage is also routed through a circuit breaker on the copilot's circuit breaker panel to the coil of each inverter power-select relay, causing the inverter to be powered by its generator bus.

During normal operation, an inverter bus select relay is energized and power is supplied from the generator bus. Should a fault occur that would interrupt power to that bus, the bus select relay would de-energize and inverter input power would be taken from the center bus of the airplane, precluding the possibility of loss of an inverter due to failure of the generator bus.

Inadequate inverter output power is indicated by the illumination of the INVERTER annunciator. This could happen due to loss of input power, or an inverter failure. Other indications of inverter loss would be erratic behavior, or loss of AC powered instruments (torque gage), or AC avionics.

During inverter power up (after start and taxi) both inverters should be checked for the following:

- Using the AC volt/frequency meter
  - 115 VAC
  - 400 Hz
  - Inverter annunciator light out
- When cycling inverters, check the AC volt/frequency meter drops to zero and the annunciator light comes on when the switch is in the center or OFF position

EXTERNAL POWER

The external power receptacle, under the right wing outboard of the nacelle, connects an external power unit to the electrical system when the airplane is parked. The power receptacle is designed for a standard three prong AN plug.

When external power is connected, a relay in the external power sensor will close only if the polarity of the voltage being supplied to the external power receptacle is correct (Figure 2-28).

Whenever an external power plug is connected to the receptacle and the BAT switch is ON, the yellow EXT PWR annunciator will illuminate, whether or not the external power unit is ON. If the EXT PWR annunciator is flashing–and the external power unit is connected–then one of three conditions exists: EXT PWR Switch is OFF, EXT PWR voltage is low, or EXT PWR voltage is too high.

External power voltage can be monitored any time, even before the EXT PWR switch on the pilot's left subpanel is switched ON, by turning the VOLTMETER BUS SELECT switch in the overhead panel (Figure 2-3) to the EXT PWR position and reading the voltage on the voltmeter.

A high-voltage sensor will lock out the external power relay if external power is above 31 ±0.5 volts DC.

When the EXT PWR–ON–OFF–RESET switch is switched ON, the external power relay closes. As external power enters the aircraft. the left and right generator bus tie relays close, permitting power to reach all buses. Consequently, the entire electrical system can be operated.

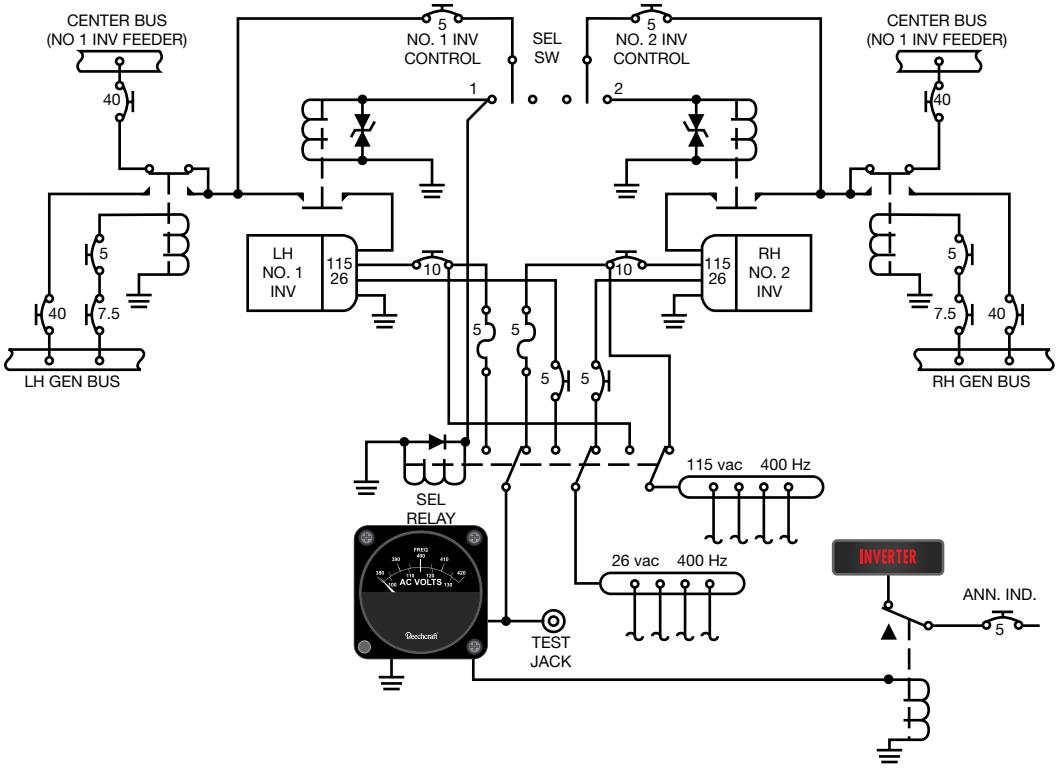


Figure 2-9. Simplified Inverter Schematic

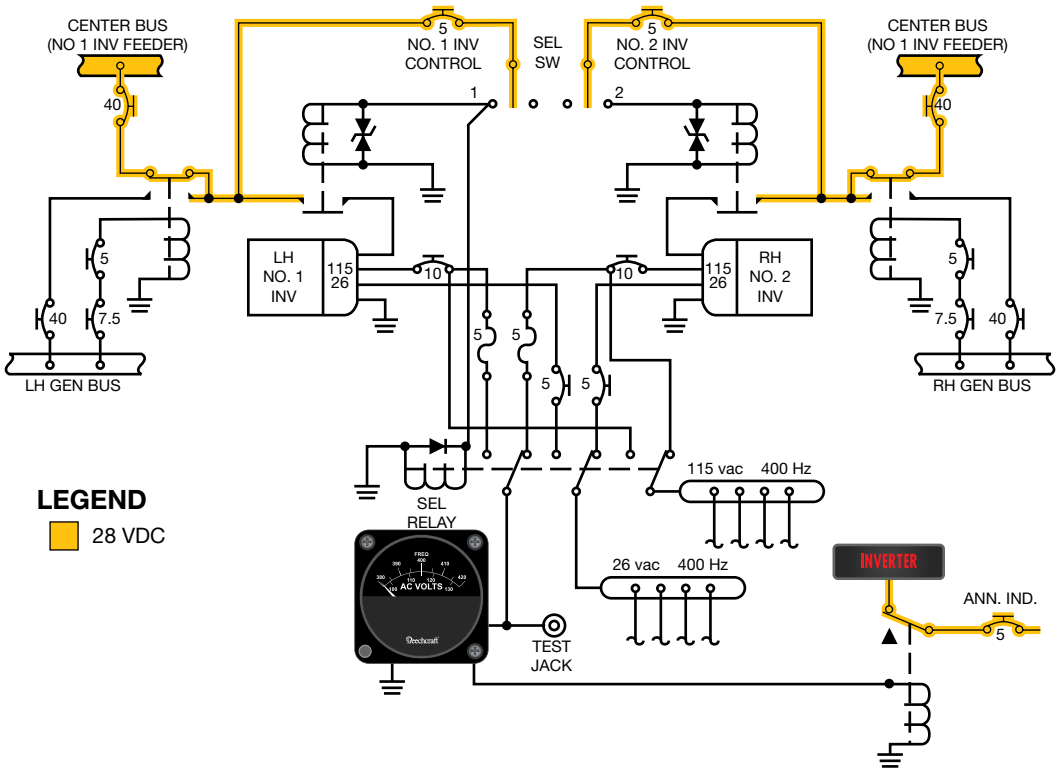


Figure 2-10. Inverter Schematic—Condition 1

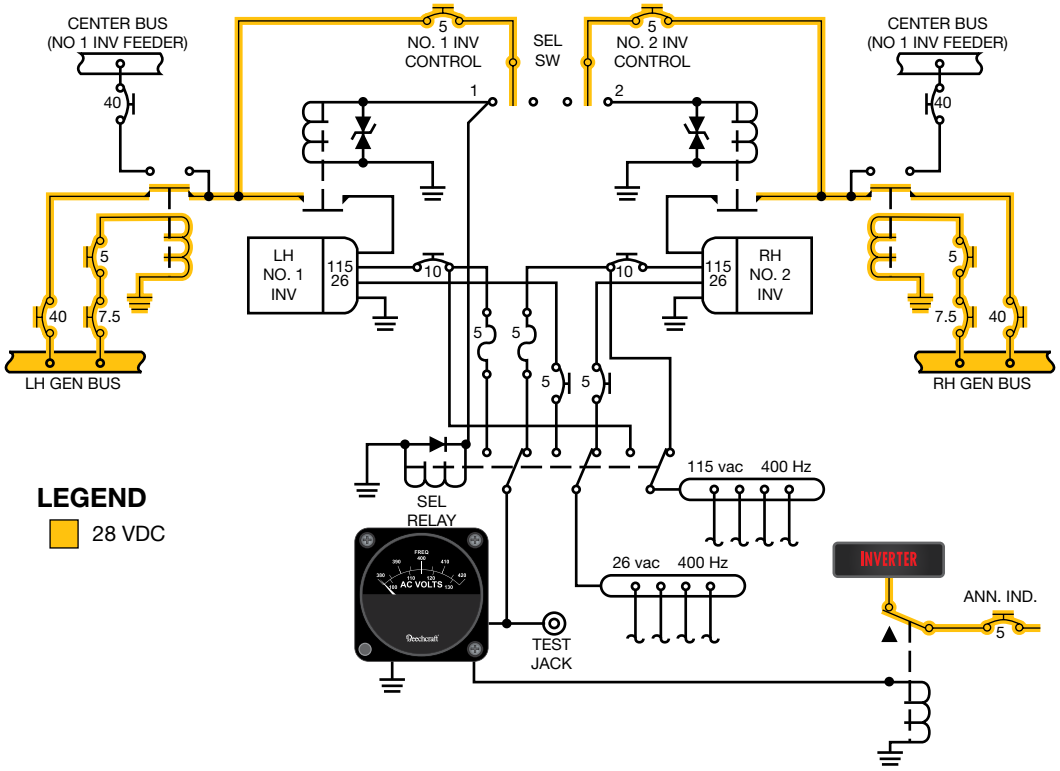


Figure 2-11. Inverter Schematic—Condition 2

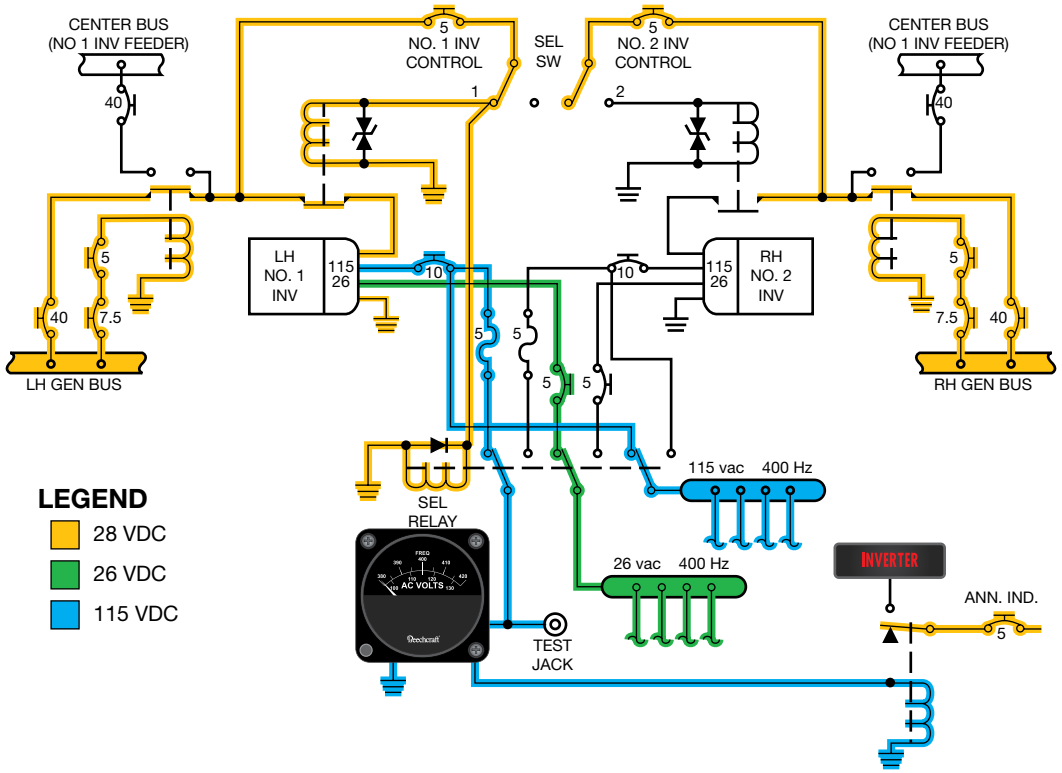


Figure 2-12. Inverter Schematic—Condition 3

Observe the following precautions when using an external power source:

CAUTION

THE RECOMMENDED MINIMUM INDICATED BATTERY VOLTAGE PRIOR TO CONNECTING EXTERNAL POWER IS 23 VOLTS. HOWEVER, NEVER CONNECT AN EXTERNAL POWER SOURCE TO THE AIRPLANE UNLESS A BATTERY INDICATING A CHARGE OF AT LEAST 20 VOLTS IS IN THE AIRCRAFT. If the battery voltage is less than 20 volts, the battery must be recharged, or replaced with a battery indicating at least 20 volts, before connecting external power.

Only use an external power source fitted with an AN-type plug. The auxiliary power unit must be regulated at 28.25 volts DC and be capable of supplying at least 1000 amperes for at least 1 second (300 amperes maximum continuous) at a minimum of 16 volts DC during the start cycle.

Voltage is required to energize the avionics master power relays to remove the power from the avionics equipment. Therefore, never apply external power to the airplane without first applying battery voltage.

The battery may be damaged if exposed to voltages higher than 30 volts for extended periods of time.

To preclude damage to the external power unit, disconnect external power from the airplane before applying generator power to the electrical buses.

Refer to the "Normal Procedures" section of the *POH* for procedural details of using external power.

AVIONICS MASTER POWER

The avionics systems installed on each airplane usually consist of individual nav/com units, each having its own ON–OFF switch. Avionics packages will vary on different airplane installations. Due to the large number of individual receivers and transmitters, a Beech avionics master switch placarded AVIONICS MASTER POWER is installed on the pilot's left subpanel. An Avionics Master Power Schematic diagram is shown in Figure 2-13. Refer to the Avionics chapter of this training manual for details of the avionics system.

CIRCUIT BREAKERS

Both AC and DC power are distributed to the various aircraft systems via two separate circuit breaker panels which protect most of the components in the airplane. The smaller one is located below the fuel management panel, to the left of the pilot (Figure 2-6). The large panel is located to the right of the copilot's position. Each of the circuit breakers has its amperage rating printed on it.

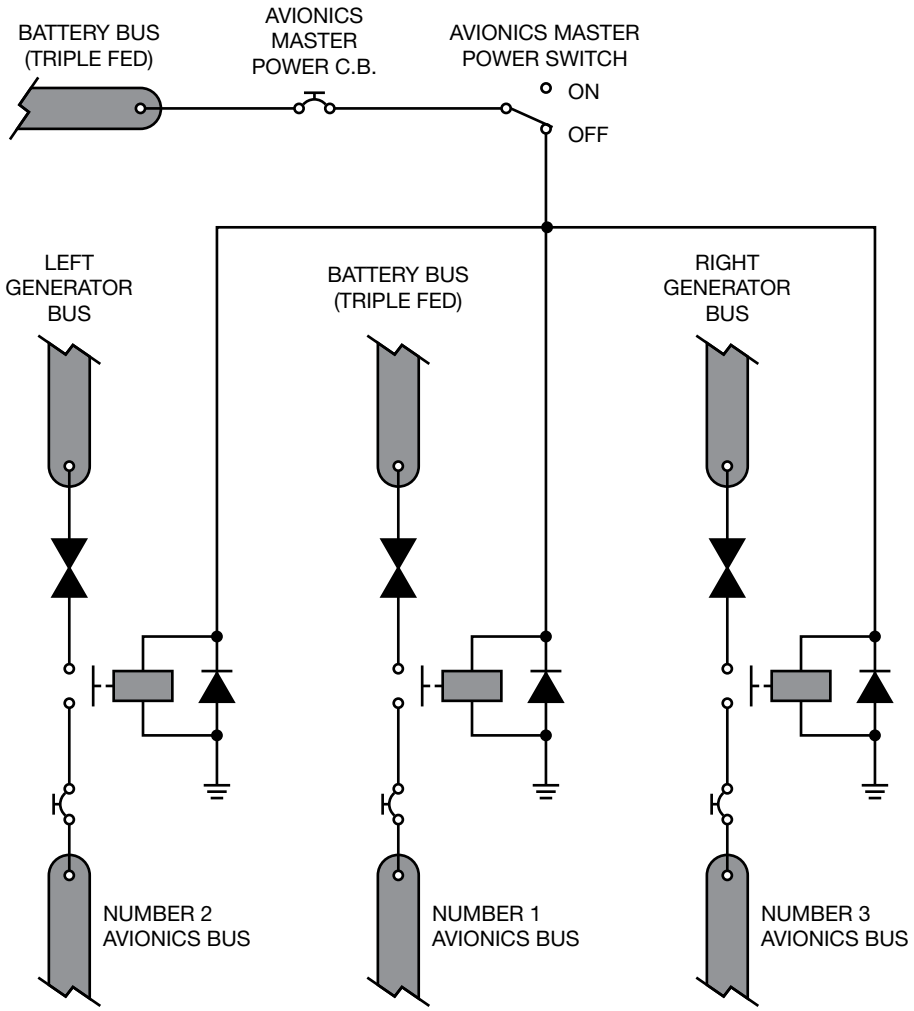


Figure 2-13. Avionics Master Power Schematic

The small circuit breaker panel, on the lower portion of the fuel panel, contains the circuit breakers for the fuel system. (On LJ-1361, LJ1363 and after, engine instrument circuit breakers are also included here.) (See Figure 2-6)

The large circuit breaker panel is located on the copilot's side of the cockpit. This panel contains the breakers for the remaining electrical systems, which include engine-related systems, all avionics components, the environmental system, lights, annunciator warning systems, and other systems. The circuit breakers for the electrical distribution system are also located on this panel.

Procedures for tripped circuit breakers, and other related electrical system warnings, can be found in the "Emergency" section of the Pilot's Operating Handbook. If a non-essential circuit breaker on either of the two circuit breaker panels trips while in flight, do not reset it. Resetting a tripped breaker can cause further damage to the component or system.

If an essential system circuit breaker trips, however, wait 30 seconds and then reset it. If it fails to reset, DO NOT attempt to reset it again. Take corrective action according to the procedures in the "Emergency" section of your *POH*.

If all the avionics equipment drops off-line but does not trip the circuit breaker, the trouble may be in the AVIONICS MASTER switch. The switch can be bypassed, and your radios returned to service, by pulling the AVIONICS MASTER circuit breaker on the copilot's circuit breaker panel.

The various power distribution configurations for the electrical system are as follow:

- Power Distribution-Battery OFF (Figure 2-15)
- Power Distribution-Battery ON (Figure 2-16)
- Power Distribution-Battery ON (Generator Ties Manually Closed) (Figure 2-17)
- Power Distribution-Right Engine Start (Generator Ties Normal) (Figure 2-18)
- Power Distribution-Right Generator ON (Figure 2-19)
- Power Distribution-Left Engine Cross-start (Right Engine Running) (Figure 2-20)
- Power Distribution-Both Generators ON (Figure 2-21)
- Power Distribution-Both Generators ON (Generator Ties Open) (Figure 2-22)
- Bus Sense Test-Both Generators ON (Figure 2-23)
- Both Generators Failed-Load Shedding (Figure 2-24)
- Right Generator Bus Short-Bus Isolation (Figure 2-25)
- Center Bus Short-Bus Isolation (Figure 2-26)
- Triple-Fed Bus Short-Bus Isolation (Figure 2-27)
- Power Distribution-External Power (External Power and Battery Switches ON) (Figure 2-28)

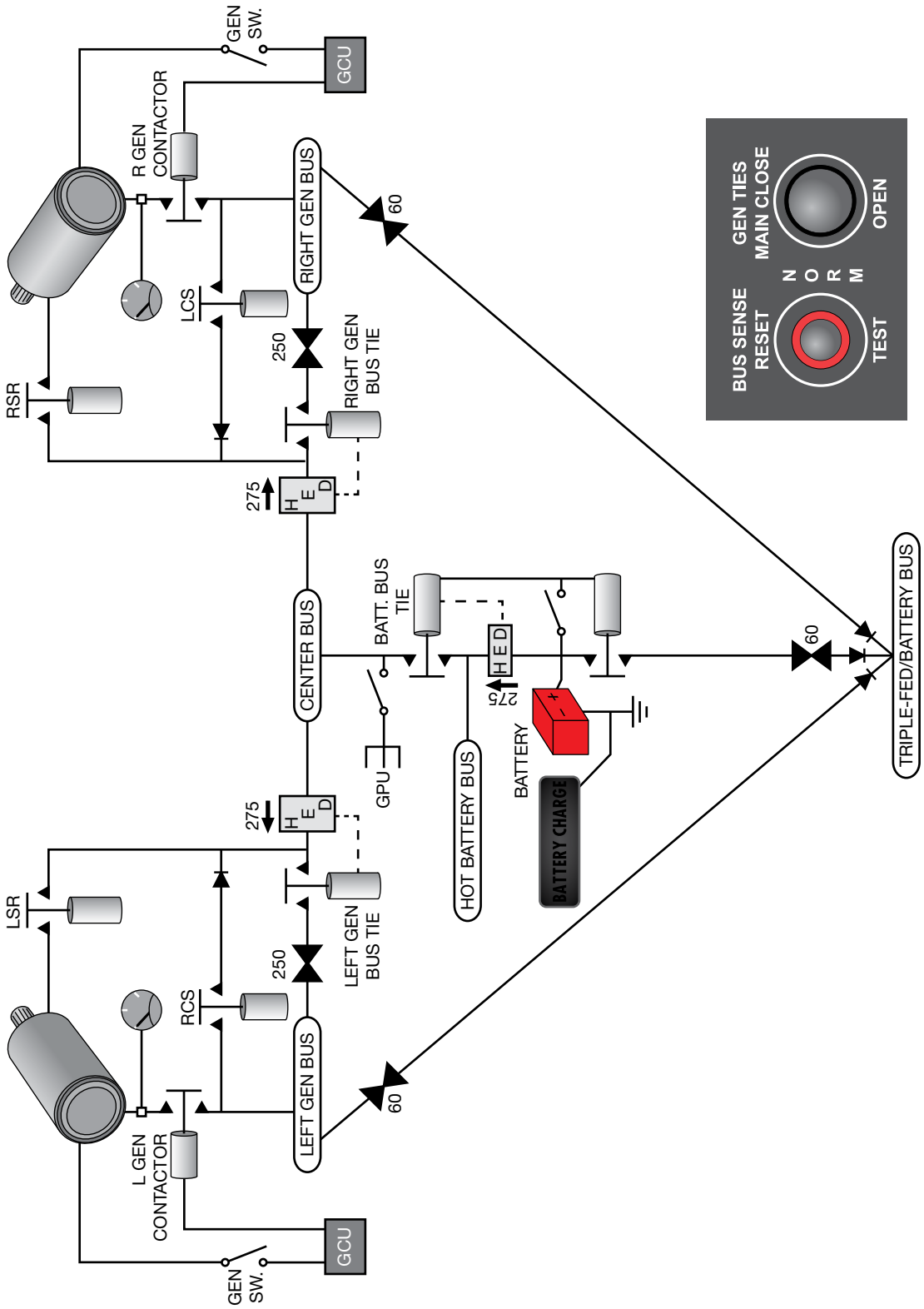


Figure 2-14. Power Distribution Schematic

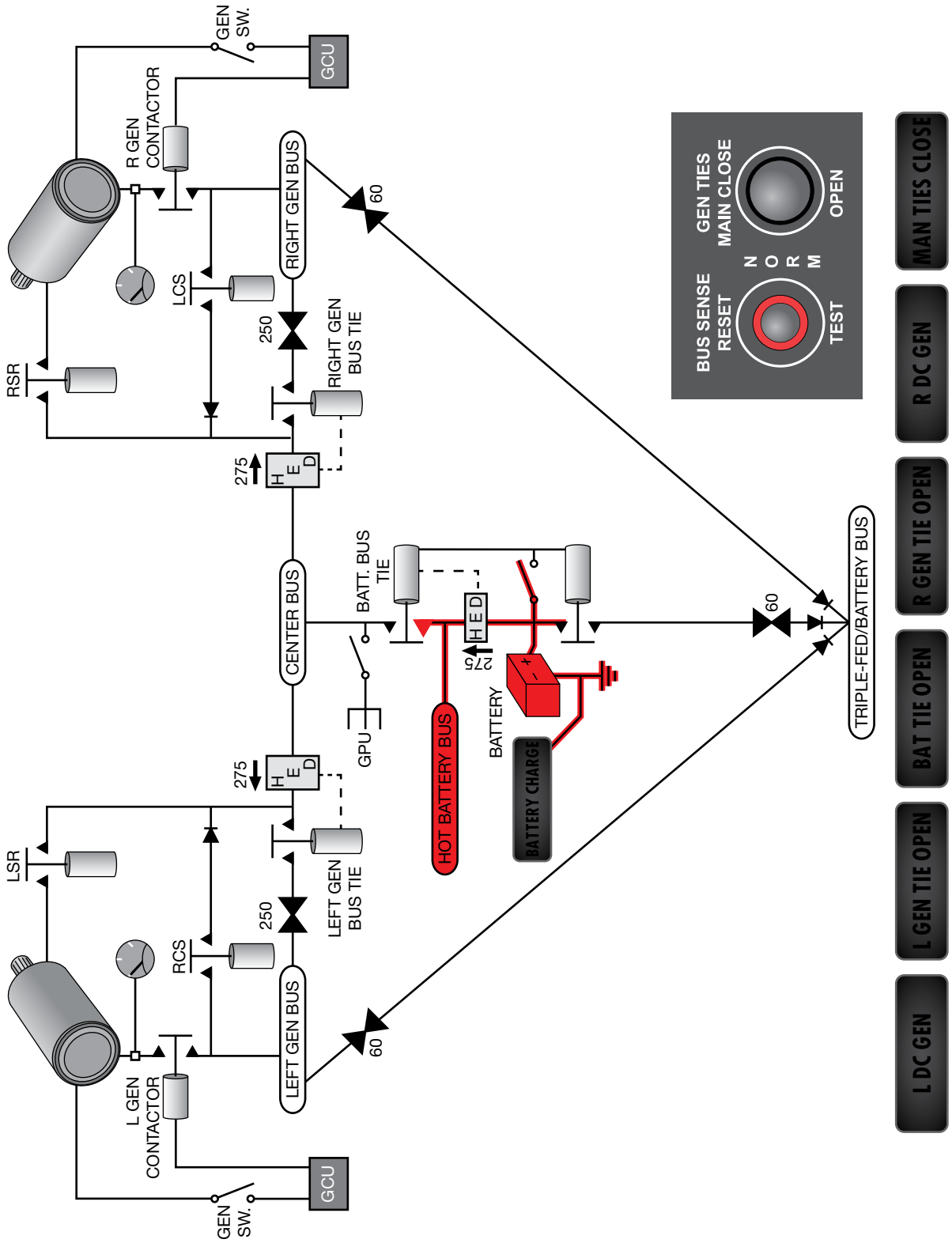


Figure 2-15. Power Distribution—Battery OFF

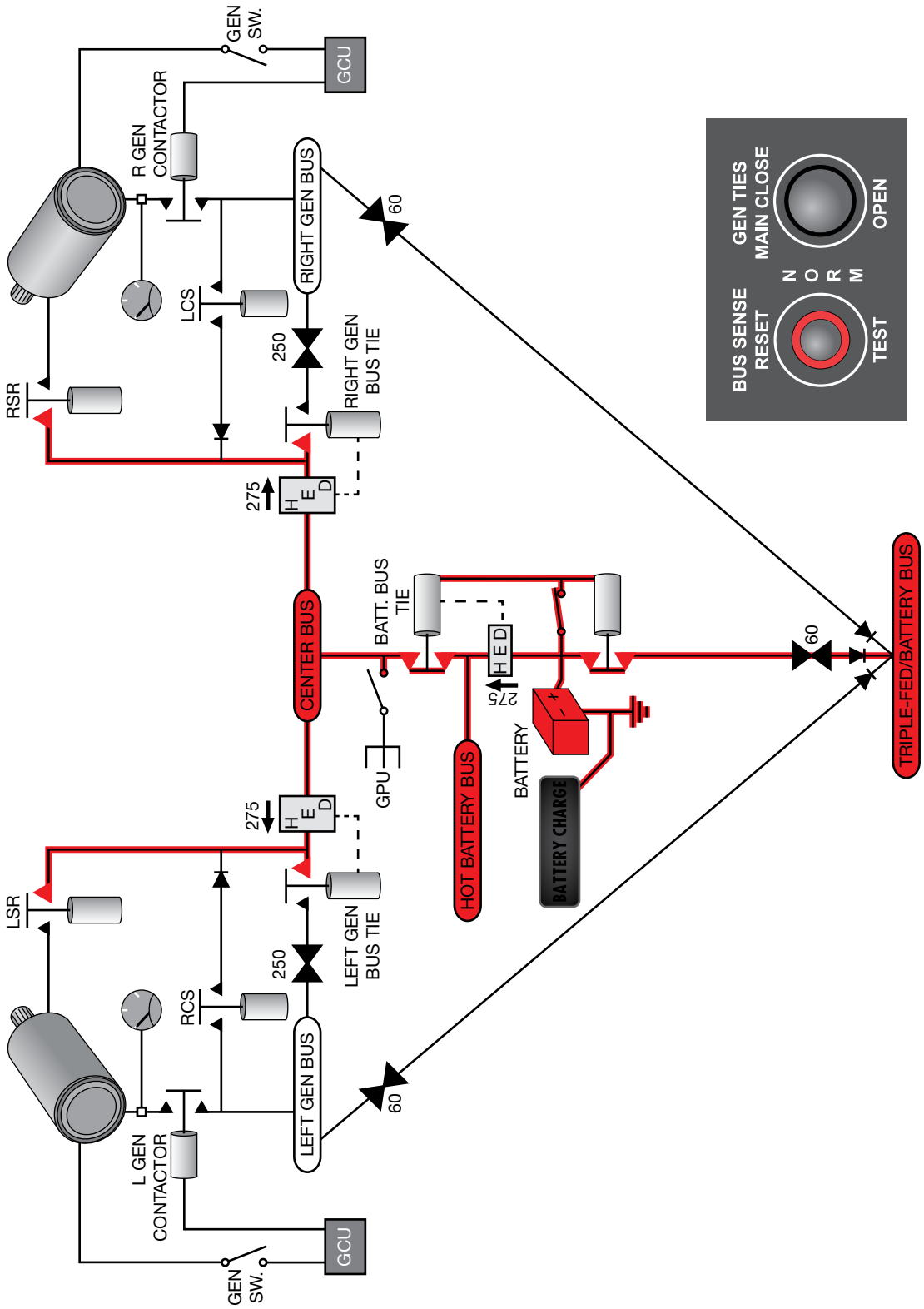


Figure 2-16. Power Distribution—Battery ON

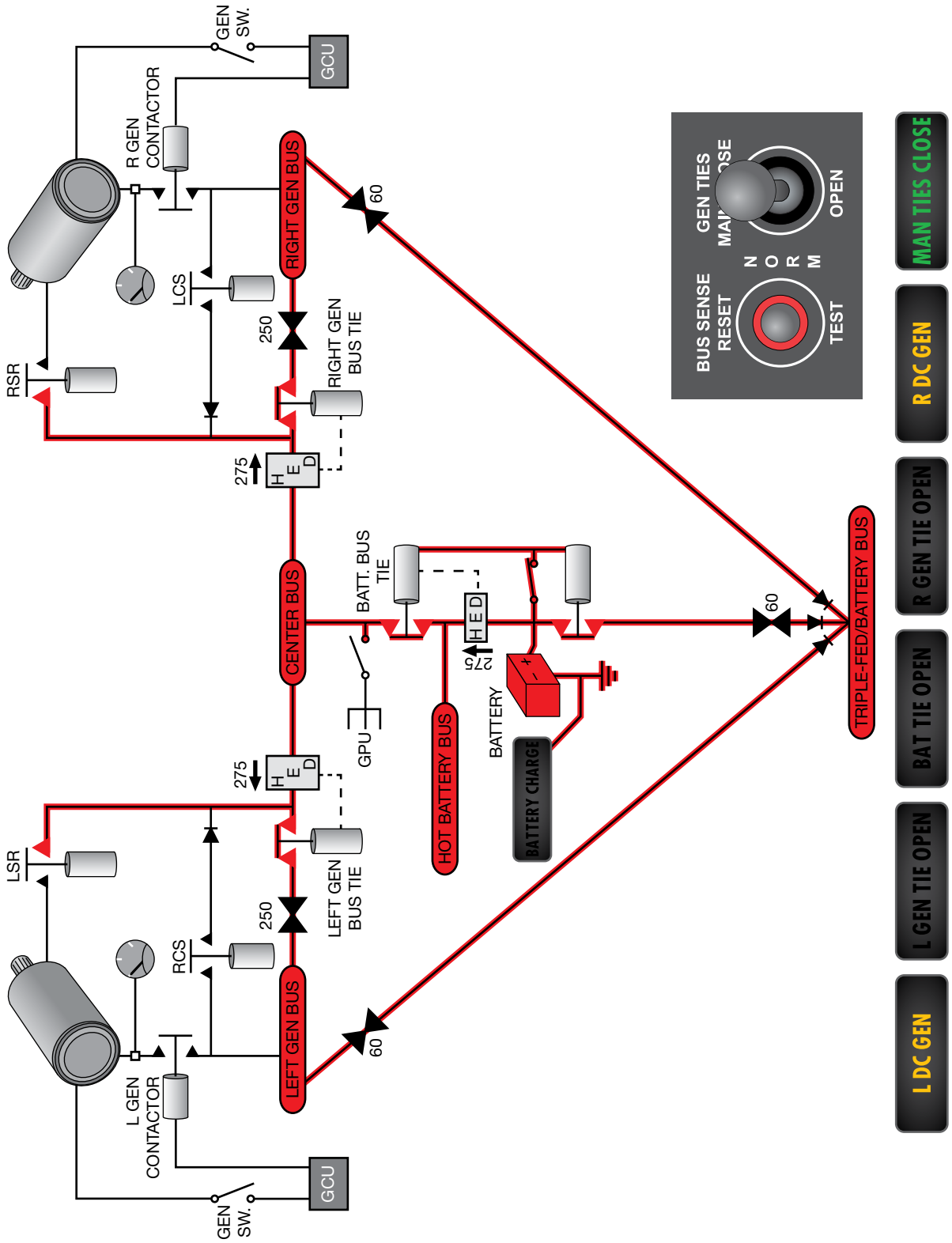


Figure 2-17. Power Distribution—Battery ON (Generator Ties Manually Closed)

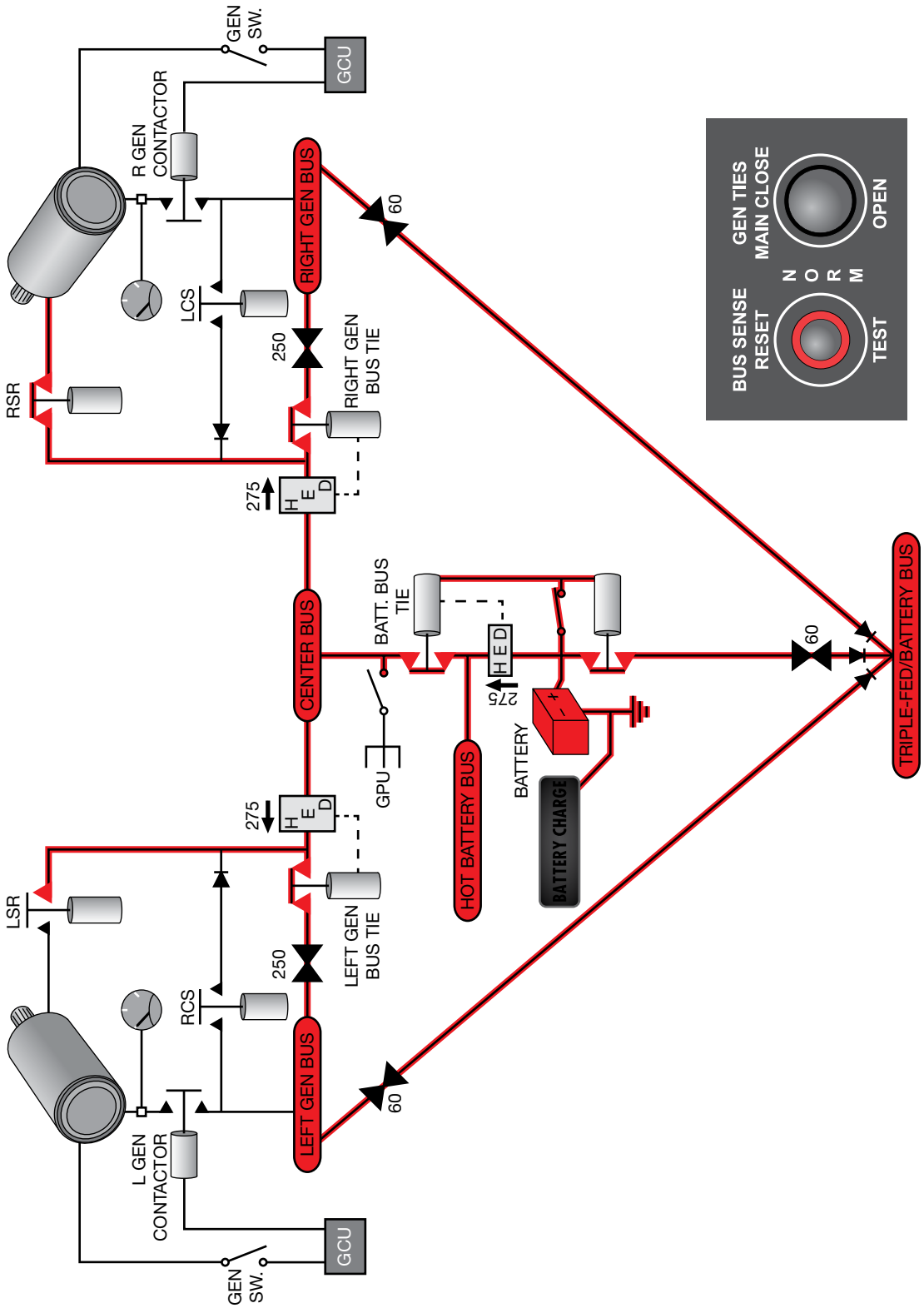


Figure 2-18. Power Distribution—Right Engine Start (Generator Ties Normal)

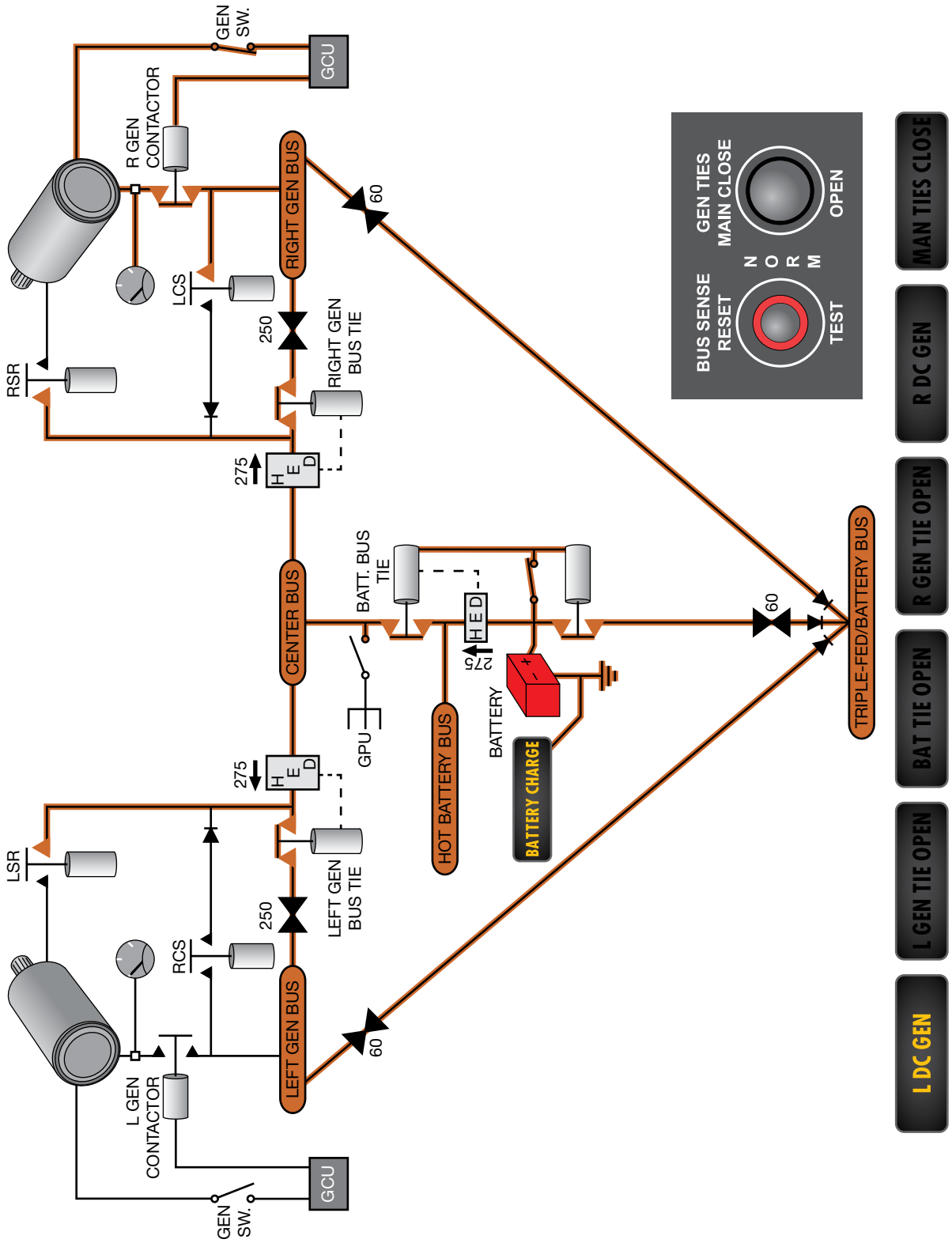


Figure 2-19. Power Distribution—Right Generator ON

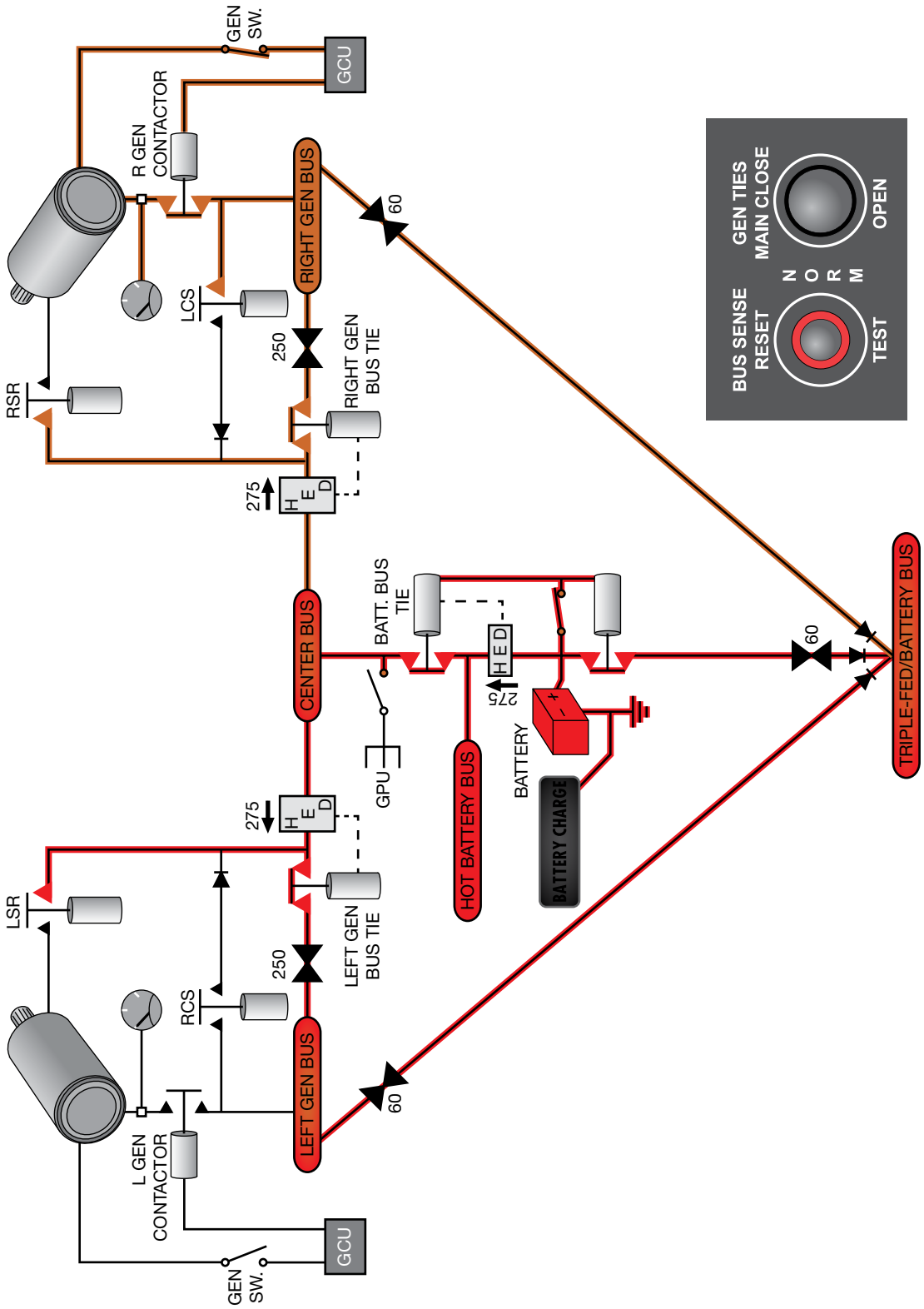


Figure 2-20. Power Distribution—Left Engine Cross-Start (Right Engine Running)

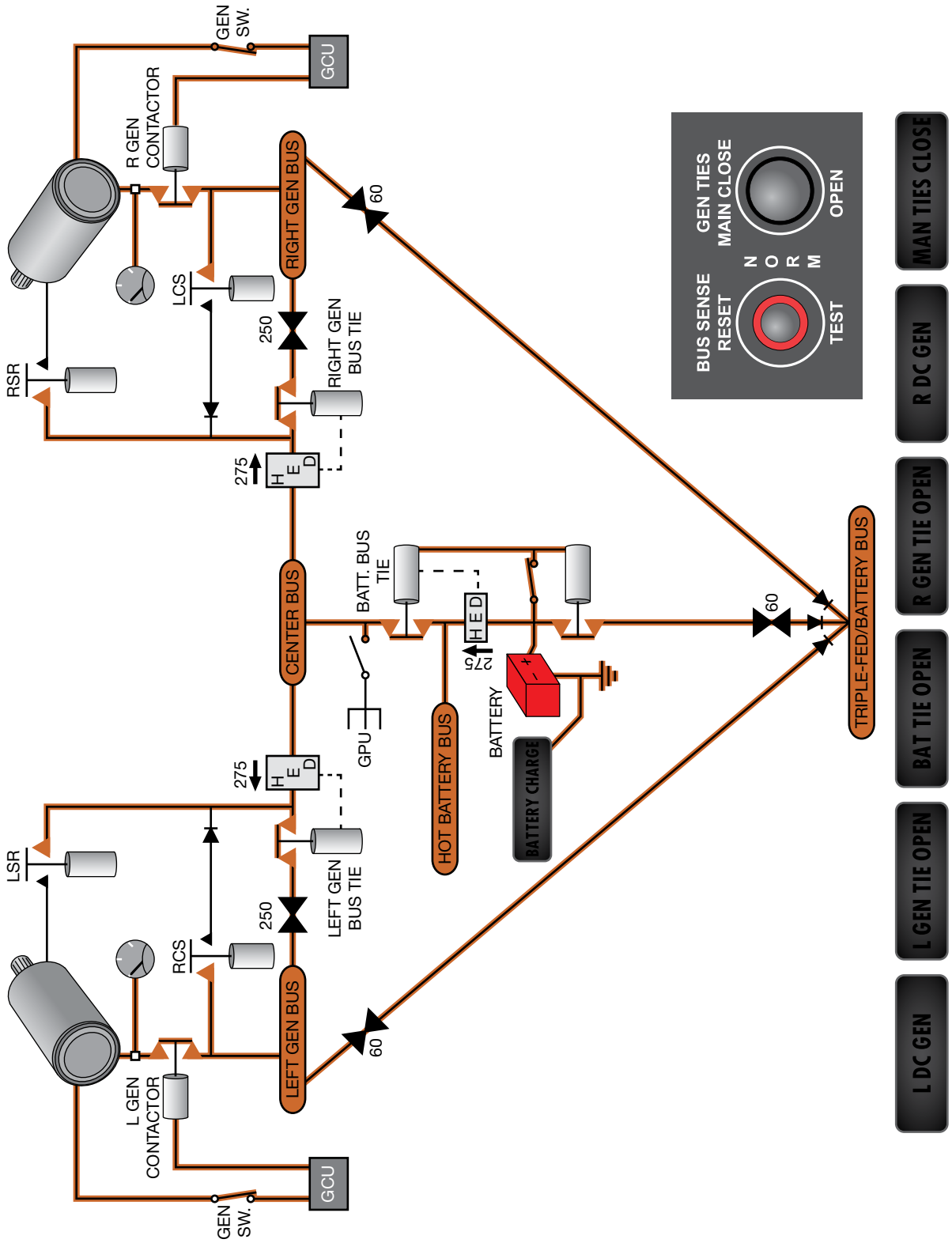


Figure 2-21. Power Distribution—Both Generators ON

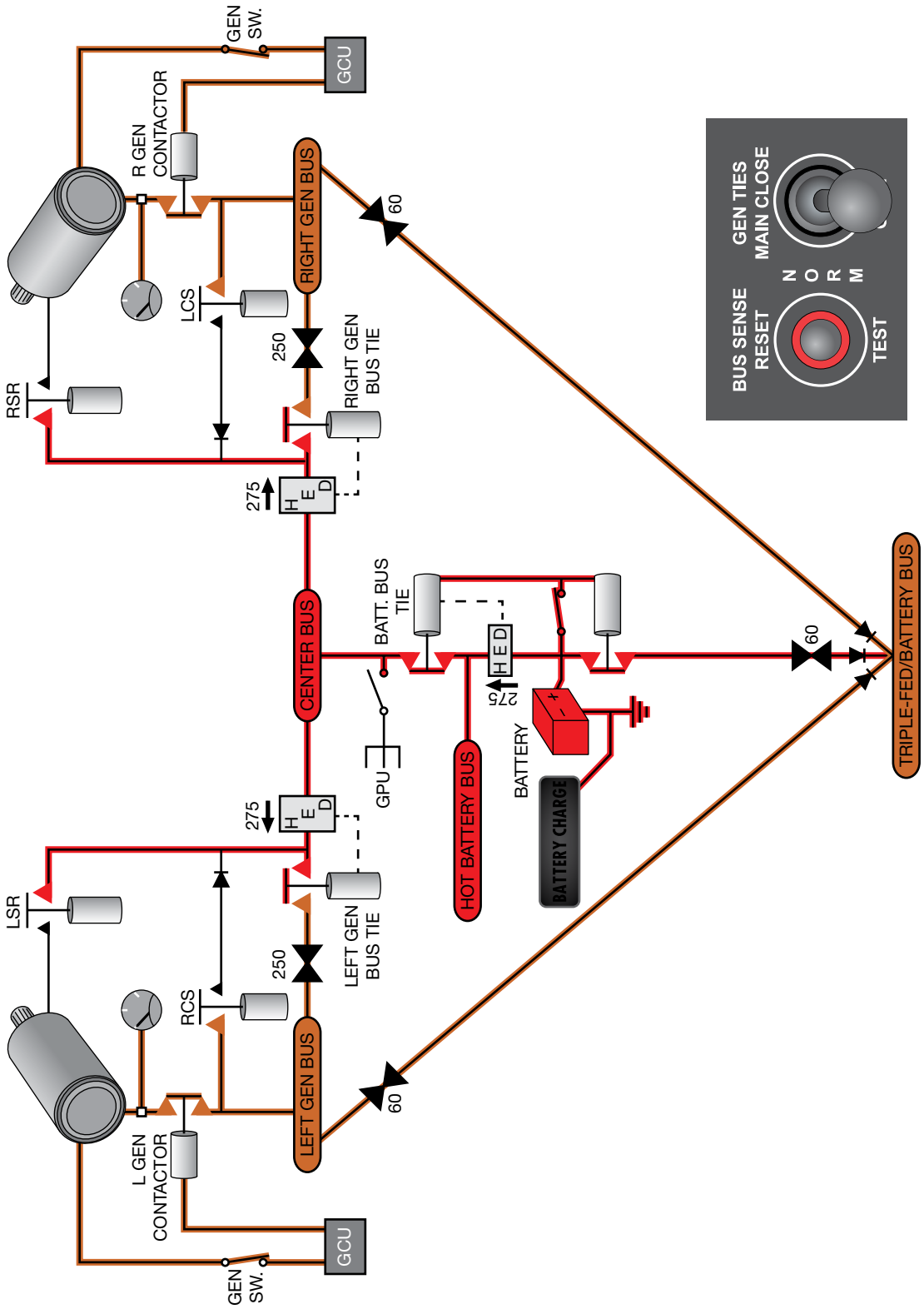


Figure 2-22. Power Distribution—Both Generators ON (Generator Ties Open)

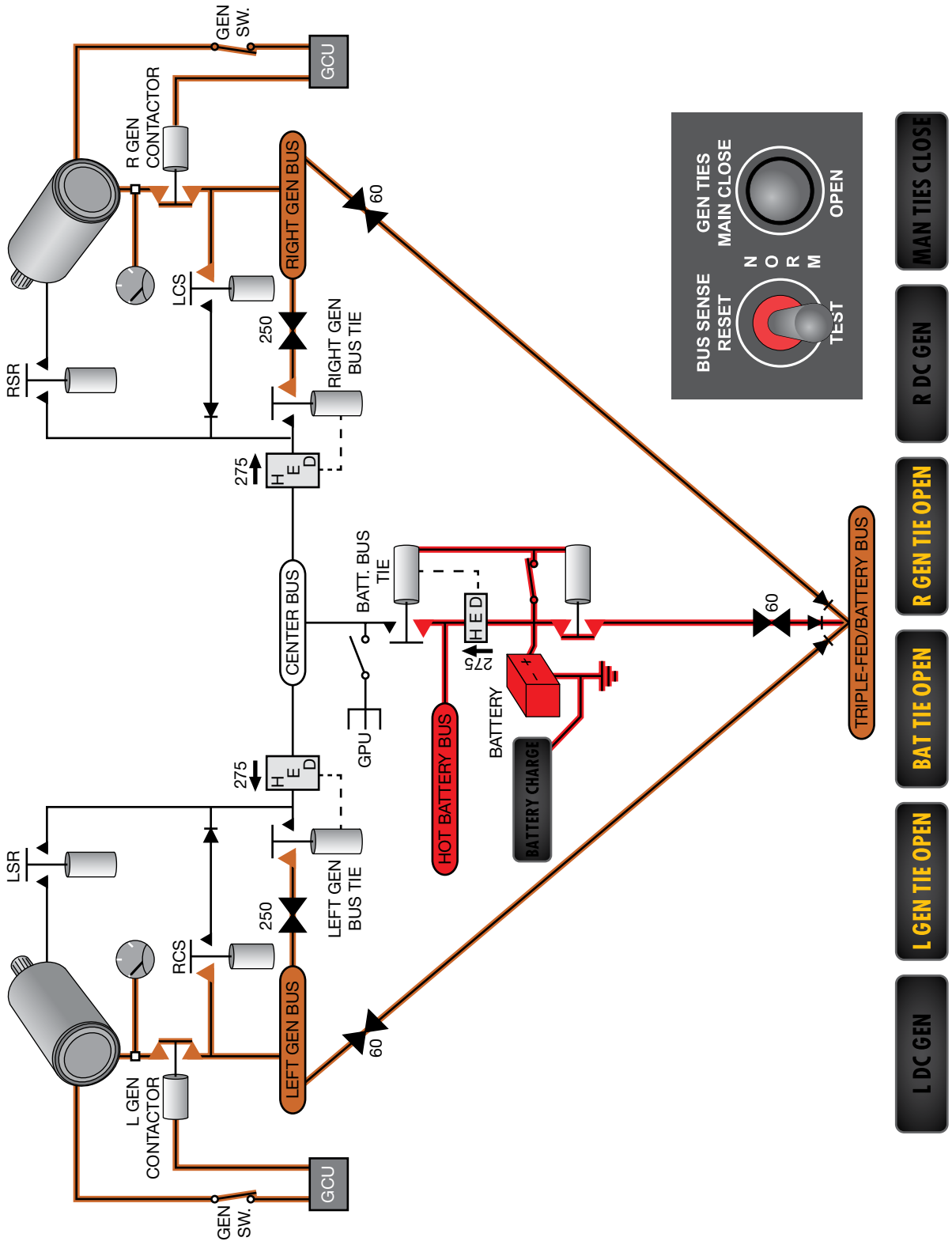


Figure 2-23. Bus Sense Test—Both Generators ON

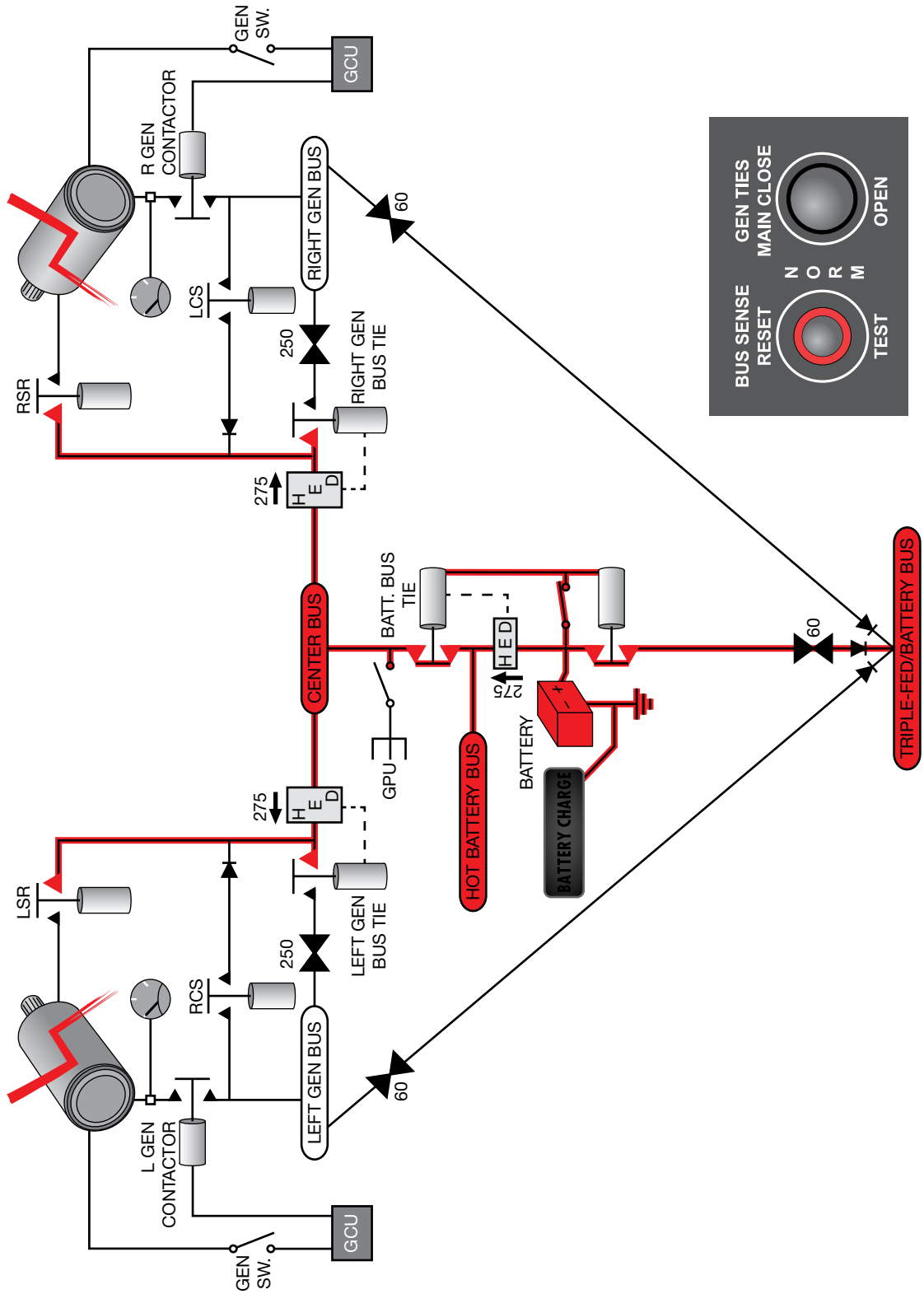


Figure 2-24. Both Generators Failed—Load Shedding

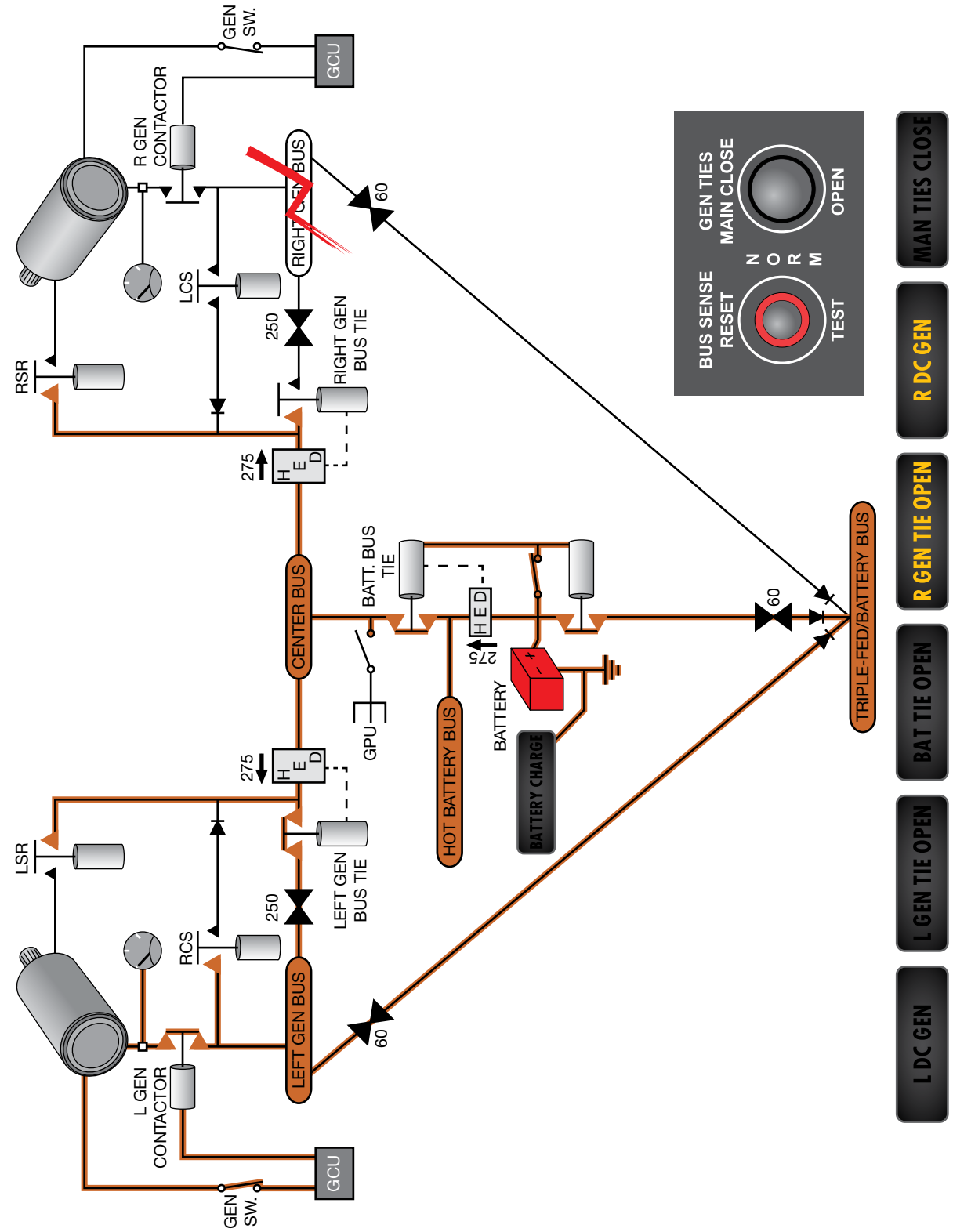


Figure 2-25. Right Generator Bus Short—Bus Isolation

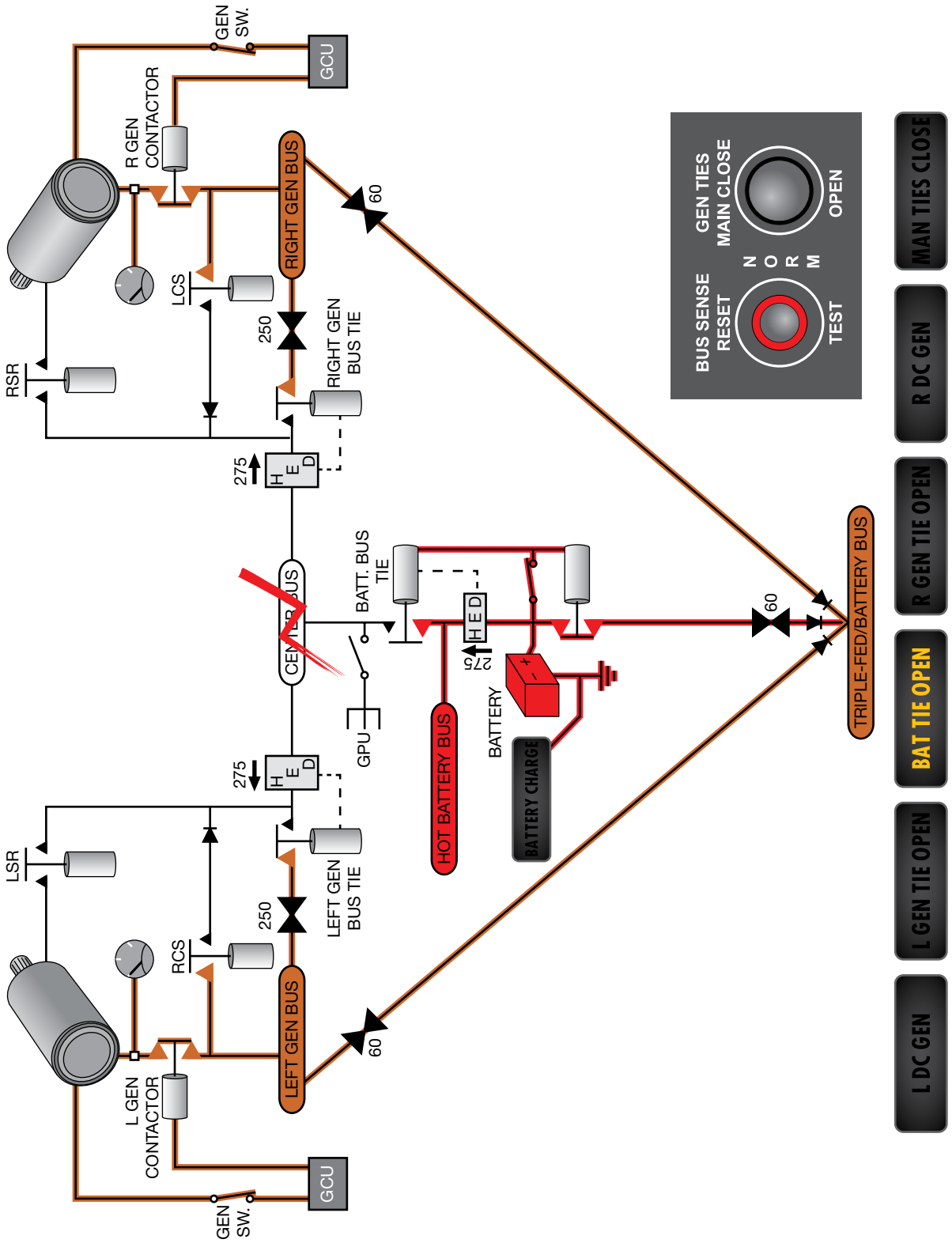


Figure 2-26. Center Bus Short—Bus Isolation

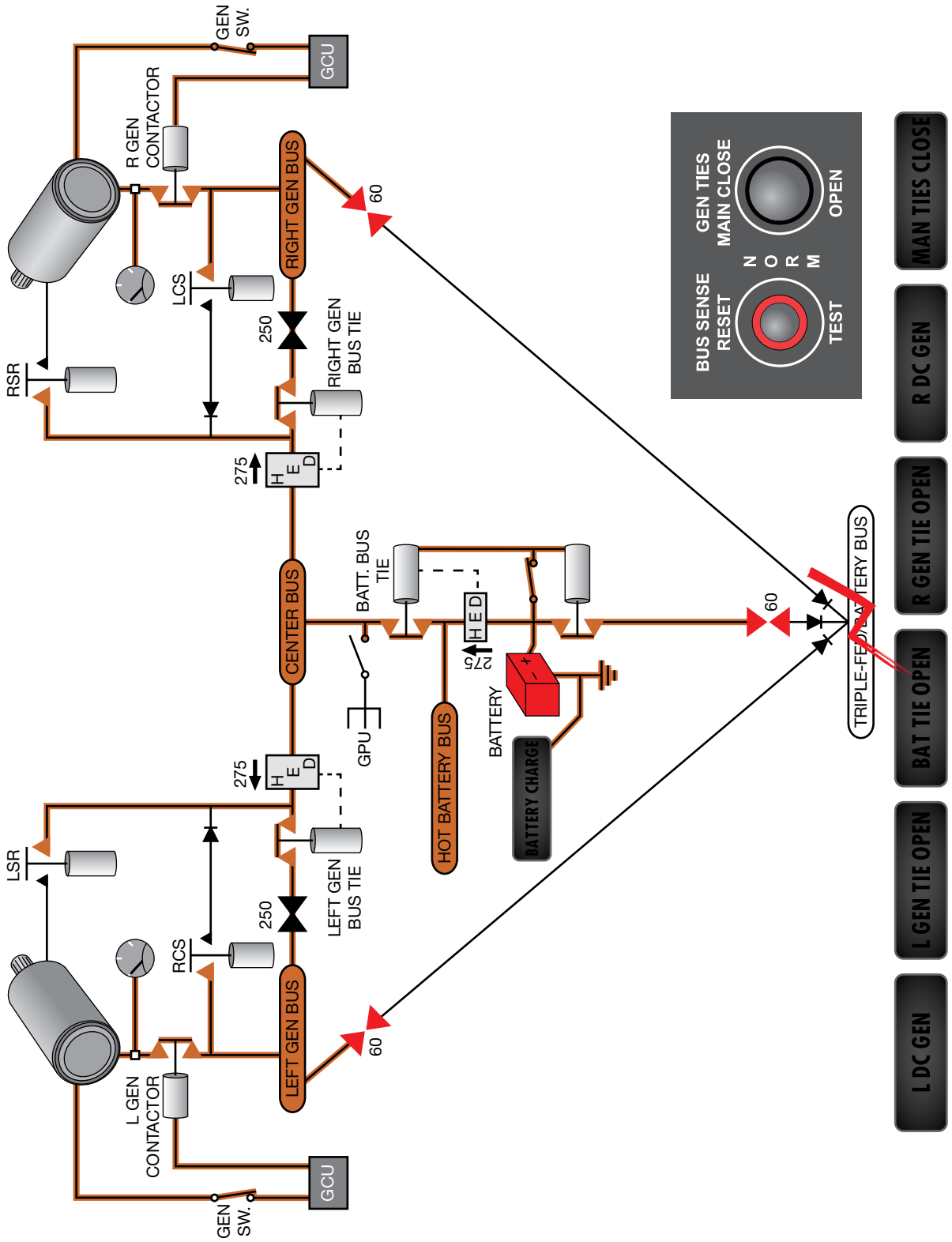


Figure 2-27. Triple-Fed Bus Short—Bus Isolation

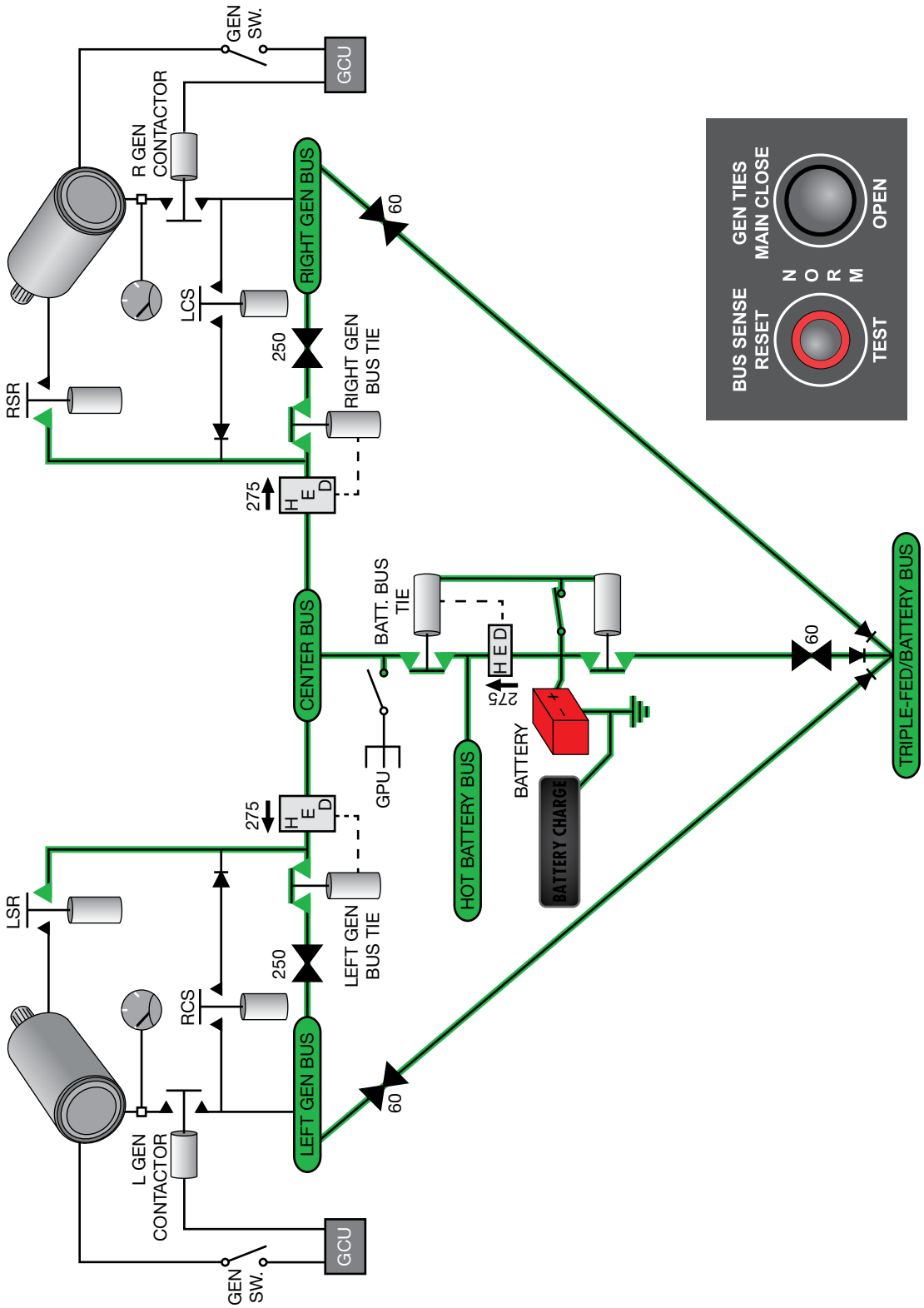


Figure 2-28. Power Distribution—External Power (External Power and Battery Switches ON)

QUESTIONS

1. Where is the battery located?

A. In the left wing center section
B. In the aft compartment
C. In the right wing center section
D. In the nose compartment

2. What is the individual generator rating?

A. 30 volt, 200 ampere
B. 24 volt, 300 ampere
C. 28 volt, 250 ampere
D. 24 volt, 250 ampere

3. Where are the generator switches located?

A. Under a gang bar on the overhead
B. On the center instrument panel
C. On the co-pilot's subpanel
D. Under a gang bar on the pilot's left subpanel

4. How is a generator turned on?

A. Move the generator switch to OFF, then to ON
B. Hold the generator switch to RESET for one second and release to ON
C. Move the generator switch to the ON position
D. Hold the generator switch to ON for one second

5. How much continuous current should the external power unit be capable of supplying?

A. 100 amperes
B. 250 amperes
C. 300 amperes
D. 1,000 amperes

6. What indication is provided to alert the operator that an external power plug is connected to the airplane?

A. An audible tone
B. A flashing EXT PWR annunciator
C. A master warning flashing annunciator
D. Fluctuating generator meters

7. What is the minimum required battery voltage before using an external power unit?

A. 20 volts
B. 23 volts
C. 24 volts
D. 31.5 volts

8. After starting the right engine and turning the right generator on, what should the load meter reading decrease to before starting the left engine?

A. 25%
B. 50%
C. 75%
D. 100%

9. Where is the INVERTER switch located?

A. On the copilot's left subpanel
B. On the overhead panel
C. On the pilot's right subpanel
D. On the pilot's left subpanel

10. When a generator is offline, what indication is present?

A. The lights go dim is the only way to tell
B. An amber DC GEN annunciator is on with the load meter indicating no output
C. A red DC GEN annunciator is on with the load meter indicating no output
D. No indication is indicated unless both generators go off line

11. Where is the external power receptical located?

A. Under the left wing, outboard of the engine nacelle
B. On the left aft portion of the fuselage
C. Under the right wing, outboard of the engine nacelle
D. On the right forward portion of the fuselage

12. What is one reason to leave the battery on during an external power start?

A. To charge the battery during the start.
B. To power the load meters
C. To protect against voltage spikes
D. To power the engine instruments during engine start

13. What is the rating of each inverter?

A. 24 volts and 28 volts, 120 Hz
B. 120 volts and 28 volts, 400 Hz
C. 115 volts and 26 volts, 400 Hz
D. 120 volts and 26 volts, 300 Hz

14. What are the starter limits?

A. 30 sec ON, 60 sec OFF, 30 sec ON, 60 sec OFF, 30 sec ON, 30 min OFF
B. 40 sec ON, 60 sec OFF, 40 sec ON, 60 sec OFF, 40 sec ON, 30 min OFF
C. 60 sec ON, 60 sec OFF, 60 sec ON, 60 sec OFF, 60 sec ON, 30 min OFF
D. 40 sec ON, 40 sec OFF, 40 sec ON, 40 sec OFF, 40 sec ON, 30 min OFF

15. If the BAT TIE annunciator illuminates, and the center bus voltage reads 28 volts, the center bus:

A. Has a short and the BAT TIE should not be reset.
B. Is isolated and all items on the bus are not available.
C. Is okay and the BAT TIE can be reset by the bus sense switch.
D. Is shorted out and the landing gear will have to be manually extended.

CHAPTER 3
LIGHTING

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CHAPTER 3
LIGHTING

INTRODUCTION

The aircraft lighting system consists of cockpit-controlled interior and exterior lights. Interior lights are in the cockpit and passenger cabin and consists of navigation lights, entry and exit threshold lights, and baggage area lights. Exterior lighting consists of navigation lights, rotating beacons, strobe lights, landing and taxi lights, ice lights, and recognition lights.

DESCRIPTION

The Lighting chapter of the training manual presents a description and discussion of the airplane lighting system and components. The location and purpose of switches, indicators, lights, and circuit breakers are described.

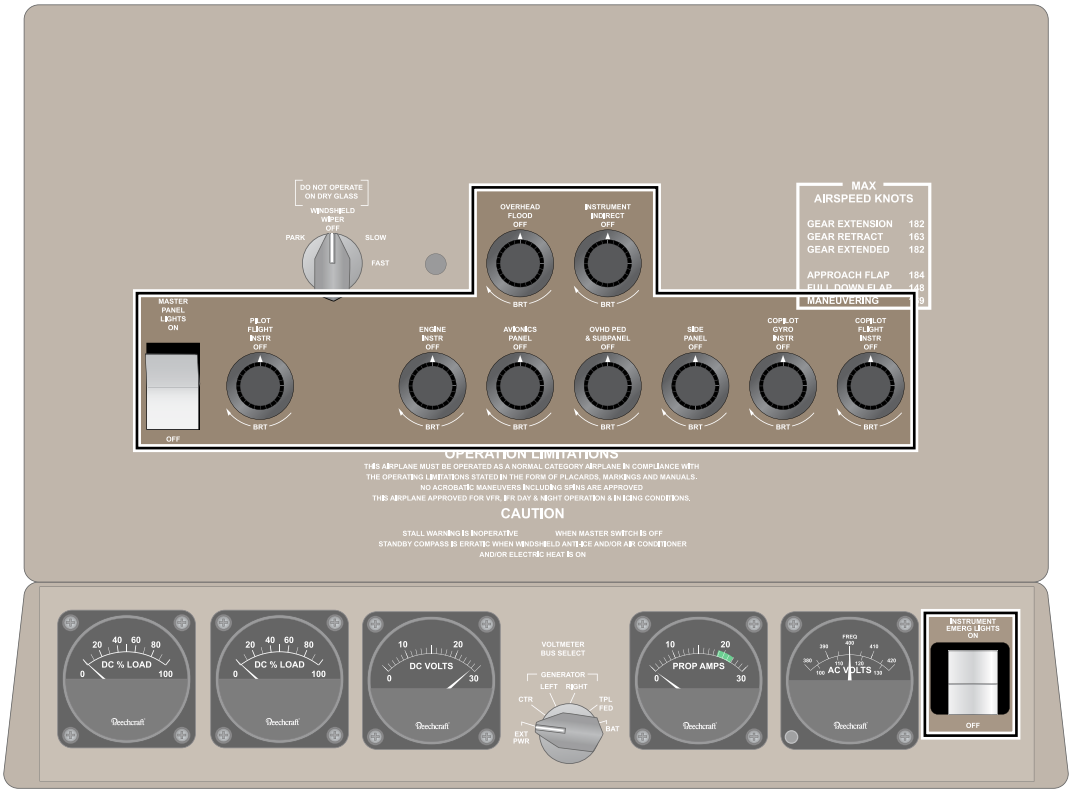

Figure 3-1. Overhead Lighting Control Panel

COCKPIT LIGHTING

An overhead light control panel, easily accessible to both pilot and copilot, incorporates a functional arrangement of all lighting systems in the cockpit (Figure 3-1). Each light group has its own rheostat switch placarded BRT–OFF.

The MASTER PANEL LIGHTS–ON/OFF switch is the master switch for: PILOT & COPILOT FLIGHT INSTR, PILOT & COPILOT GYRO INSTR, ENGINE INSTR, AVIONICS PANEL, OVHD, PED & SUBPANEL, and SIDE PANEL. The indirect instrument lighting and map (overhead) lights are controlled by rheostat switches mounted on the overhead panel.

CABIN LIGHTING

A three-position switch on the copilot's left sub-panel light control panel, placarded CABIN START/BRIGHT–DIM–OFF on the C90A and CABIN–BRIGHT–DIM–OFF on the C90B, controls the indirect fluorescent cabin lights (Figure 3-2). A switch to the right of the interior light switch activates the cabin NO SMOKING/ FASTEN SEAT BELT signs and accompanying chimes. This three-position switch is placarded NO SMK & FSB–OFF–FSB.

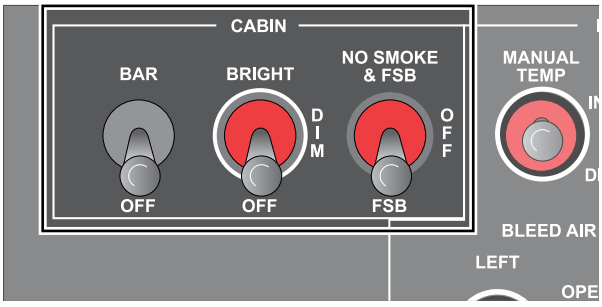

Figure 3-2. Cabin Lighting Controls

A hot-wired threshold light is mounted on the left side of the entryway at floor level. Optional airstair door lights mounted under each step may be installed. These lights share the same controls;

a slide type switch (Figure 3-3) mounted adjacent to the threshold light, and a microswitch mounted in the door lock. Whenever the slide switch is in the ON position and the door is open, the lights will come on.

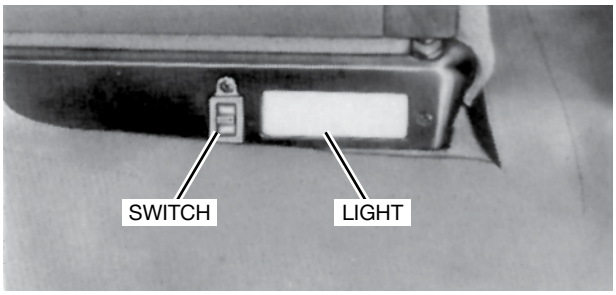

Figure 3-3. Threshold Light Switch

To turn the lights OFF, either use the threshold light switch, or fully close and lock the cabin door. The microswitch in the door lock will turn off the lights when the threshold switch is left on.

The lights will not go out if the door is simply latched, the door handle must be in the fully locked position.

When the battery master switch is on, the individual reading lights along the top of the cabin may be turned on or off by the passengers with the pushbutton switch adjacent to each light.

The light in the baggage compartment may be turned on or off by the adjacent push-button switch regardless of the position of the battery master switch. This baggage compartment light is connected to the hot battery bus.

EXTERIOR LIGHTING

Switches for the landing lights, taxi lights, wing ice lights, navigation lights, recognition lights, rotating beacons, and wingtip and tail flood lights are located on the pilot's subpanel Figure 3-4. They are appropriately placarded as to their function.

Tail floodlights, if installed, are incorporated into the horizontal stabilizers and are designed to illuminate both sides of the vertical stabilizer. A switch for these lights, placarded LIGHTS TAIL FLOOD–OFF, is located on the pilot's subpanel (Figure 3-2).

CIRCUIT BREAKERS

Lighting system circuit breakers are shown in Figure 3-5

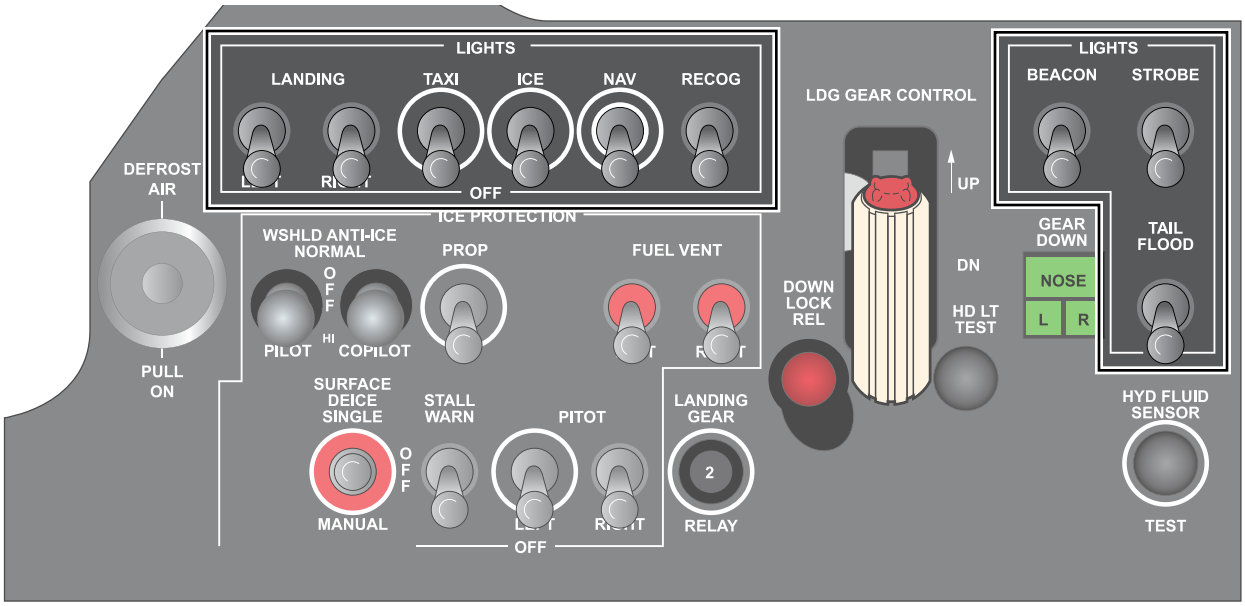

Figure 3-4. Exterior Light Controls

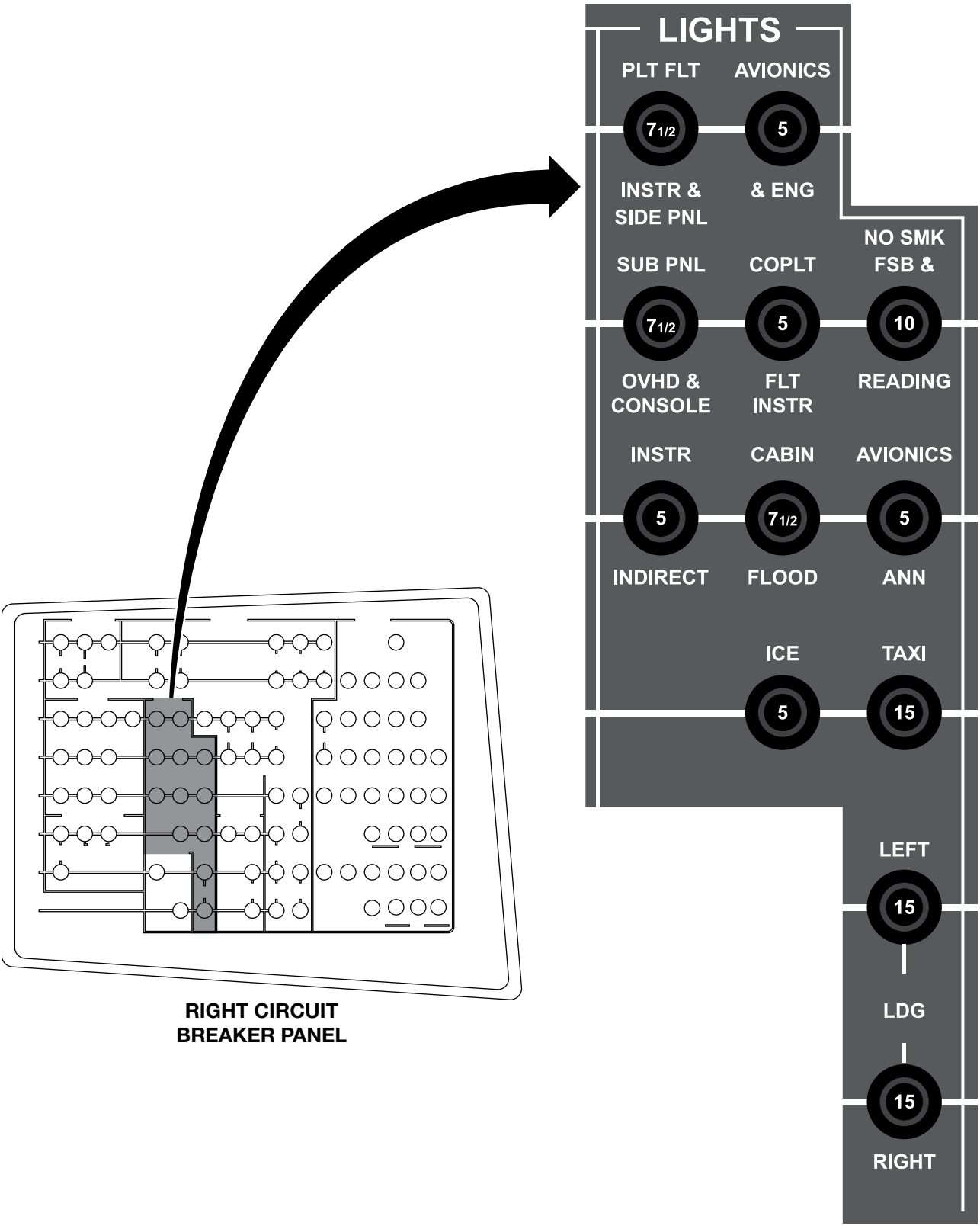

Figure 3-5. Light System Circuit Breakers

QUESTIONS

1. What bus does the baggage light receive power from?

A. The triple fed bus
B. The center bus
C. The left generator bus
D. The hot battery bus

2. Where is the switch for the strobe lights located?

A. On the overhead panel
B. On the copilot's side panel
C. On the pilot's right subpanel
D. On the pilot's left subpanel

3. Where are the recognition lights mounted?

A. In each wingtip
B. In the nose of the fuselage
C. On the landing gear under the landing lights
D. In the engine nacelles

4. Where are the ice lights mounted?

A. On the nose
B. On the wingtip
C. On the outside of the engine nacelles
D. On the wing root

5. What bus powers the INSTRUMENT EMERG LIGHTS?

A. The center bus
B. The triple fed bus
C. The hot battery bus
D. The battery bus

6. After takeoff, the landing lights are extinguished:

A. Automatically as the gear doors close
B. Automatically as the airplane lifts off
C. By turning off the LANDING light switches
D. By turning off the TAXI light switch

CHAPTER 4
MASTER WARNING SYSTEM

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CHAPTER 4
MASTER WARNING SYSTEM

INTRODUCTION

Warning and caution indicators can be the first indication of trouble or malfunction in some system or component of the airplane. Crewmembers should have complete familiarity with these indicators and the related action necessary to correct the problem or cope with the situation until a safe landing can be made. In the case of an on-ground indication, the problem should be corrected before flight.

GENERAL

This chapter presents a description and discussion of the warning, caution, and advisory annunciator panel.

The annunciator panel is described in detail, including each annunciator, its purpose, and the associated cause for illumination.

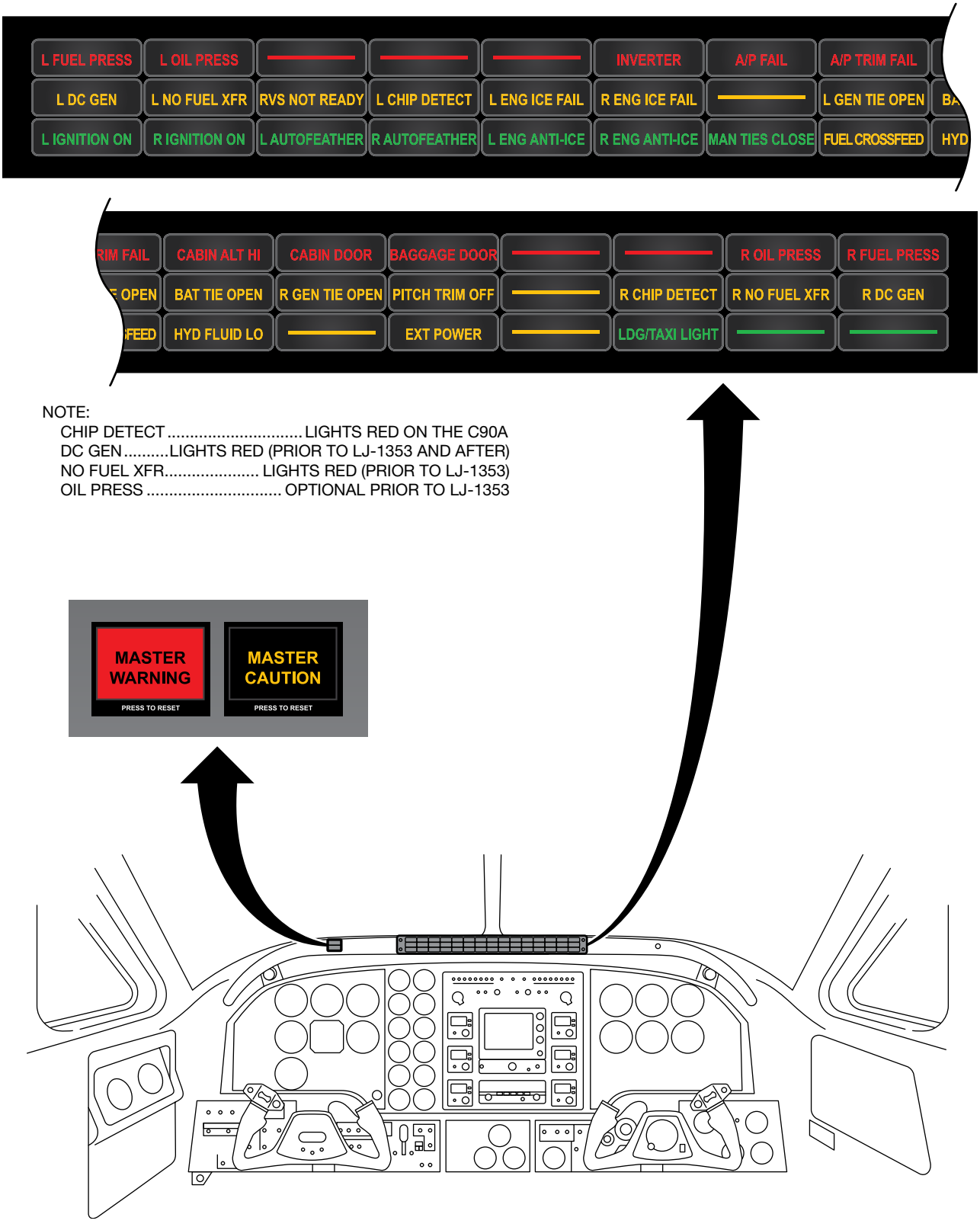

Figure 4-1. Annunciator System

ANNUNCIATOR SYSTEM

The annunciator system (Figure 4-1) consists of an annunciator panel centrally located in the glareshield, a PRESS-TO-TEST switch, and a FAULT WARNING flasher (Figure 4-2). The red FAULT WARNING flasher (and yellow MASTER CAUTION flasher [LJ-1353 and after]) is located in the glareshield in front of the pilot, and the PRESS-TO-TEST switch is located immediately to the right of the annunciator panel. The annunciators are of the word-readout type. Whenever a fault condition covered by the annunciator system occurs, a signal is generated, and the appropriate annunciator is illuminated.

Whenever an annunciator-covered condition occurs that requires the pilot's attention but not his immediate reaction, the appropriate yellow caution annunciator (Figure 4-1) in the annunciator panel illuminates (as well as the MASTER CAUTION flasher on LJ-1353 and after).

The annunciator panel also contains green advisory annunciators. There are no fault warning flashers associated with advisory annunciators.

An illuminated caution annunciator on the annunciator panel will remain on until the fault condition is corrected, at which time it will extinguish. An annunciator can be extinguished only by correcting the condition indicated on the illuminated lens.

The illumination of a green annunciator light will not trigger the fault warning system, but a red annunciator will actuate the FAULT WARNING flasher. Yellow annunciators will actuate the yellow MASTER CAUTION flasher.

FAULT WARNING FLASHER

If the fault requires the immediate attention and reaction of the pilot, the appropriate red warning annunciator (Figure 4-1 in the annunciator panel illuminates, and the FAULT WARNING flasher begins flashing.

Any illuminated red lens in the annunciator panel will remain on until the fault is corrected. The FAULT WARNING flasher can be extinguished by depressing the face of the FAULT WARNING flasher, even if the fault is not corrected. In such a case, the FAULT WARNING flasher will again be activated if an additional warning annunciator illuminates. When a warning fault is corrected, the affected warning annunciator will extinguish, but the FAULT WARNING flasher will continue flashing until it is depressed.

DIMMING

The warning, caution, and advisory annunciators feature both a bright and a dim mode of illumination intensity. The dim mode will be selected automatically whenever all of the following conditions are met:

- A generator is on line.
- The OVERHEAD FLOODLIGHT is OFF.
- The MASTER PANEL LIGHTS switch is ON.
- The PILOT FLIGHT LIGHTS are ON.
- The ambient light level in the cockpit (as sensed by a photoelectric cell located in the overhead light control panel) is below a preset value.

MASTER WARNING AND MASTER CAUTION FLASHERS (LJ-1353 AND AFTER)

FAULT WARNING FLASHER (AIRPLANES PRIOR TO LJ-1353)

Figure 4-2. Master Caution and Fault Warning Flashers

Unless all these conditions are met, the bright mode will be selected automatically. The FAULT WARNING flasher does not have a dim mode. A warning or caution annunciator will cause the dim mode to be bright.

TESTING AND LAMP REPLACEMENT

The lamps in the annunciator system should be tested before every flight and any time the integrity of a lamp is in question. Depressing the PRESS-TO-TEST button, located to the right of the annunciator panel in the glareshield, illuminates all the annunciator lights and the FAULT WARNING flasher. Any lamp that fails to illuminate when tested should be replaced.

The annunciator panel style allows each annunciator to be removed from the panel (Figure 4-3). Each readout annunciator contains two lamps. To replace any annunciator lamp, first depress the center of the annunciator with your finger. Release your finger, and the annunciator will pop out slightly. Pull the annunciator from the panel, and remove the lamp from the rear of the annunciator. Replace the failed lamp with a spare lamp contained in an unused annunciator. Depress the annunciator until it locks in place.

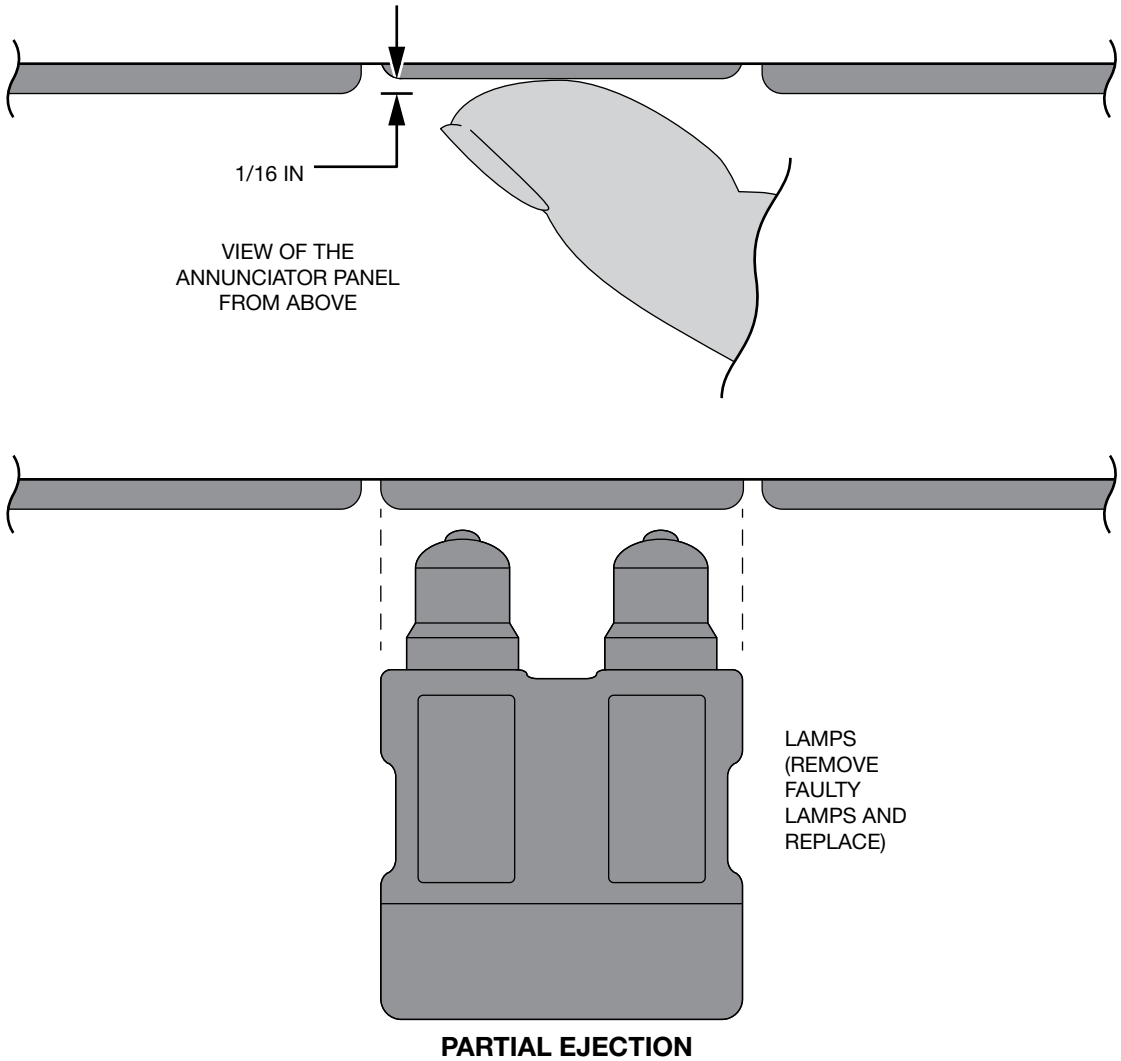

Figure 4-3. Lamp Replace

ANNUNCIATOR PANEL DESCRIPTION

Table 4-1, Table 4-2 and Table 4-3 list all the warning, caution, and advisory annunciators on the King Air C90A/B. The cause for illumination is included beside each annunciator.

Table 4-1. WARNING ANNUNCIATORS

NOMENCLATURE	CAUSE FOR ILLUMINATION
L FUEL PRESS	Low fuel pressure on left side; check boost pump, crossfeed.
L OIL PRESS	(LJ-1353 and after)
INVERTER	The inverter selected is inoperative, or both inverters are off.
A/P FAIL	Autopilot is disconnected by switching other than pilot's disconnect button
A/P TRIM FAIL	Improper trim or no trim command from autopilot.
CABIN ALT HI	Cabin altitude exceeds10,000 feet (12,500 feet on LJ-1353 and later) pressure altitude.
CABIN DOOR	Cabin door is open or not secure.
BAGGAGE DOOR	Nose baggage door is not secure (Prior to LJ-1531).
R OIL PRESS	(LJ-1353 and after)
R FUEL PRESS	Low fuel pressure on right side; check boost pump, crossfeed.
L ENG FIRE	Fire in left engine compartment.
R ENG FIRE	Fire in right engine compartment.
* Optional equipment	

Table 4-2. CAUTION ANNUNCIATORS

NOMENCLATURE	CAUSE FOR ILLUMINATION
L DC GEN	Left generator is off line (Red prior to LJ-1353 and after).
L NO FUEL XFR	Left wing tank is empty or transfer pump failed.
RVS NOT READY	Propeller levers are not in the high rpm position with the landing gear extended.
L CHIP DETECT	Metal contamination is detected in left engine oil, probable engine shutdown (red for C90A; yellow for C90B).
L ENG ICE FAIL	Left engine anti-ice vanes in transit or inoperative.
R ENG ICE FAIL	Right engine anti-ice vanes in transit or inoperative.
L GEN TIE OPEN	Left generator bus is isolated from the center bus.
BAT TIE OPEN	Battery is isolated from the generator buses and center bus.
R GEN TIE OPEN	Right generator bus is isolated from the center bus.
FUEL CROSSFEED	Crossfeed valve is receiving power.
HYD FLUID LO	Hydraulic fluid in the hydraulic fluid reservoir is low.
BATTERY CHARGE	Charge rate on the battery exceeds 7 amps for 6 seconds (Airplanes prior to LJ-1534).
PITCH TRIM OFF	Pitch trim deenergized by a trim disconnect switch on the control wheel with the system power switch on the pedestal turned on.
R CHIP DETECT	Metal contamination is detected in right engine oil, probable engine shutdown (red for C90A; yellow for C90B).
R NO FUEL XFR	Right wing tank is empty or transfer pump failed.
R DC GEN	Right generator is off line (Yellow for LJ-1353 and after).
EXT PWR	External power connector is plugged in.

Table 4-3. ADVISORY ANNUNCIATORS

NOMENCLATURE	CAUSE FOR ILLUMINATION
L IGNITION ON	System is armed and left engine torque is below 400 ft-lb, or the left ignition and engine start switch is ON.
R IGNITION ON	System is armed and right engine torque is below 400 ft-lb, or the right ignition and engine start switch is ON..
L AUTOFEATHER*	Left autofeather is armed with power levers advanced above 90% N1 position, or autofeather test switch is in test.
R AUTOFEATHER*	Right autofeather is armed with power levers advanced above 90% N1 position, or autofeather test switch is in test.
L ENG ANTI-ICE	Left engine anti-ice vanes are in position for icing conditions.
R ENG ANTI-ICE	Right engine anti-ice vanes are in position for icing conditions.
MAN TIES CLOSE	Manually closed generator bus ties.
LDG/TAXI LIGHT	Landing lights or taxi light is on with landing gear UP.

QUESTIONS

1. How are the annunciators dimmed?

A. By using the BRT DIM switch
B. With the overhead control rheostats
C. Automatically relative to cockpit light intensity
D. With the CAUTION switch on the copilot's subpanel

2. How can the annunciator lights be tested?

A. By depressing each light individually
B. By moving the CAUTION switch to ON
C. With the APPROACH PLATE rheostat
D. With the PRESS TO TEST switch

3. To extinguish a WARNING flasher, what action must be taken?

A. Move the CAUTION switch to OFF
B. Depress the WARNING flasher
C. Depress the PRESS TO TEST button
D. Clear the illuminating fault

4. When will a red annunciator light extinguish?

A. When the indicated fault is cleared
B. When the WARNING flasher is pressed
C. When the RESET button is depressed
D. When the TEST button is depressed

CHAPTER 5
FUEL SYSTEM

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CHAPTER 5
FUEL SYSTEM

INTRODUCTION

A complete understanding of the fuel system is essential to competent and confident operation of the aircraft. Management of fuel and fuel system components is a major everyday concern of the pilot. This section gives the pilot the information he needs for safe, efficient fuel management.

DESCRIPTION

The Fuel System section of the training manual presents a description and discussion of the fuel system. The physical layout of the fuel cells and fuel system are described in this section. Correct use of the boost pumps, transfer pumps, crossfeed, and firewall shutoff valves are discussed. Fuel drains, their location, and type are described with correct procedure for taking and inspecting samples of fuel. Approved fuels and tank filling sequence are included.

FUEL SYSTEM

The Beechcraft King Air fuel system is designed to simplify flight procedures in the cockpit, and provide easy access on the ground (Figure 5-11). There are two separate wing fuel systems, one for each engine, connected by a valve-controlled crossfeed system. Each fuel system consists of a nacelle tank and four interconnected wing tanks, electrical boost and transfer pumps and an electrically operated crossfeed valve. Total usable fuel capacity is 384 gallons.

Three modes of operation are available, each of which is described briefly.

1. Normal operation-Each engine receives fuel from its corresponding fuel cells and boost pump. The boost pump is required to provide fuel under pressure to the engine driven high pressure pump.
2. Automatic crossfeed operation-In the event of a boost pump failure, boost pressure is obtained by supplying fuel to both engines, through the crossfeed valve, from one boost pump. A drop in output pressure from the failed pump is sensed by a pressure switch, which automatically opens the crossfeed valve when the pressure drops below about 10 psi, and illuminates the low fuel pressure annunciator. The fuel pressure annunciator will then extinguish as pressure is restored by the boost pump on the opposite engine.
3. Suction feed-This mode of operation may be employed after a boost pump has failed, and allows the use of fuel from tanks on the side with the failed pump. Suction feed operation is obtained by moving the crossfeed valve control switch from the AUTO position to the CLOSED position. Vacuum created by the engine-driven fuel pump draws fuel from the nacelle fuel tank. Suction feed is limited to ten hours cumulative between engine-driven fuel pump overhauls.

FUEL TANK SYSTEM

The fuel system (Figure 5-2) in each wing consists of one wing leading-edge bladder-type tank (40 gallons), two outboard-wing panel bladder-type tanks (23 gallons and 25 gallons), one center section bladder-type tank (44 gallons), and the nacelle tank (61 gallons). The total usable fuel capacity of each wing fuel system is 192 gallons. The outboard wing tanks supply the center section and nacelle tanks by gravity flow. Since the center section tank is lower than the other wing tanks and the nacelle tank, the fuel is transferred to the nacelle tank by the fuel transfer pump in the low point of the center section tank. Fuel for each engine is pumped directly from its nacelle fuel tank by an electric boost pump. Each system has two filler cap openings; one in the top of the nacelle tank and one mid-wing in the leading edge tank. An anti-siphon valve is installed at each filler port to prevent the loss of fuel or collapse of fuel-tank bladder in the event the filler cap is improperly secured.

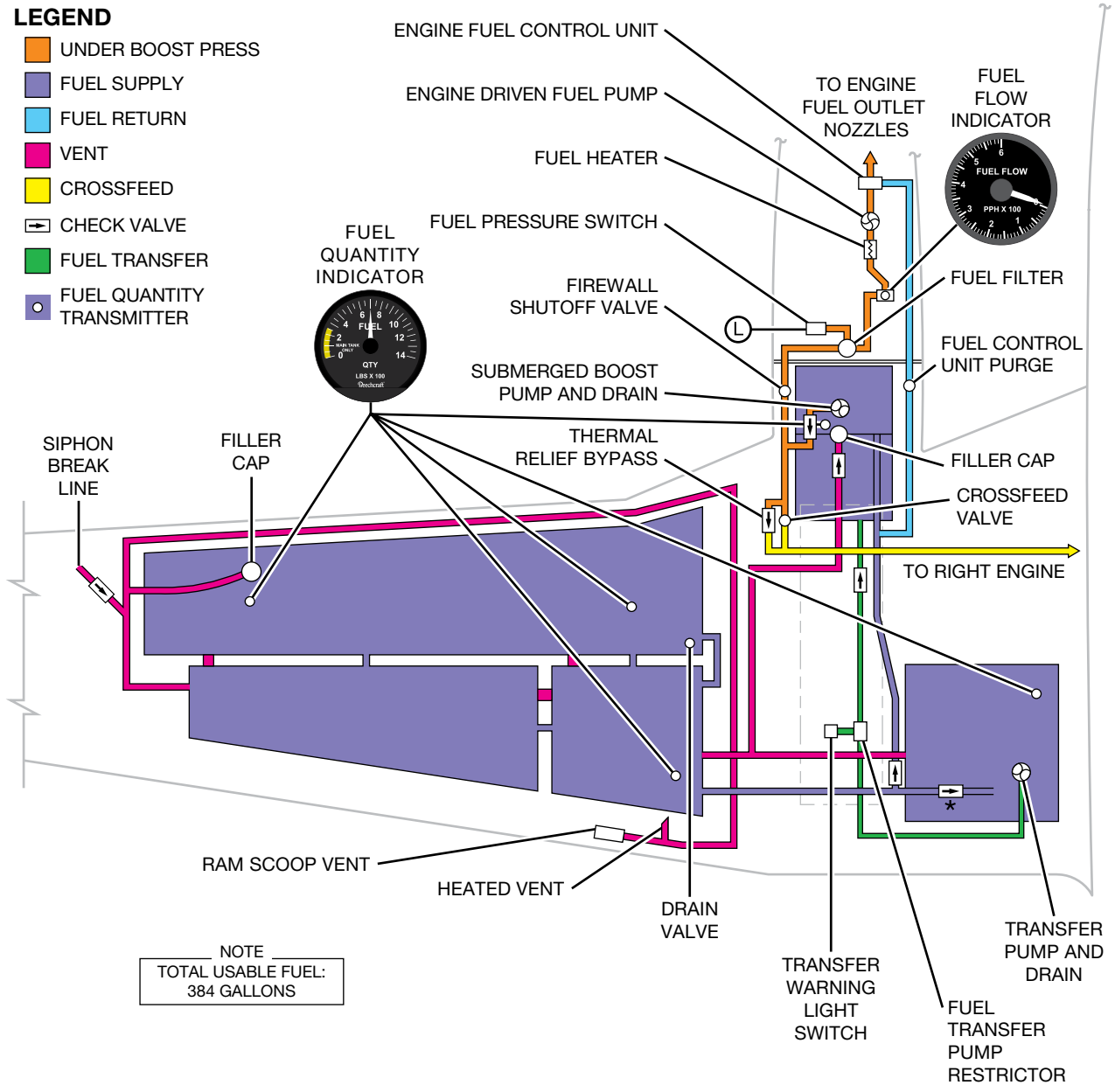
There is a check valve between the nacelle tank and the wing tank. Fuel can flow only into the nacelle tank, not back into the wing tank. If a full fuel load is needed, fill the nacelle tank first, then fill the wing tank.

The heated fuel vent and the NASA integral ram scoop vent work together to prevent the bladders from collapsing as fuel is drawn out of them.

Each nacelle tank is connected to the engine on the opposite side by a crossfeed line for single-engine or failed boost pump operation. Crossfeed operation is automatic depending on the boost pump selected in the feeding nacelle tank. This system makes it possible for fuel in either wing system to be available to either engine, or both engines simultaneously.

BOOST PUMPS

Each system has a submerged boost pump in the nacelle tank. This pump supplies a pressure of about 30 psi to the engine-driven fuel pump. The boost pumps are submerged, rotary, vane-type impeller pumps, and are electrically-driven. A 10-amp circuit breaker for each boost pump is located on the fuel panel. Two red FUEL PRESS annunciators are associated with the boost pumps. When illuminated, there is low fuel pressure on the side indicated. Check the boost pumps prior to flight.



NOTE

RIGHT SYSTEM IS IDENTICAL TO LEFT SYSTEM EXCEPT THAT THE LATTER CONTAINS THE CROSS-FEED VALVE. IT SHOULD ALSO BE NOTED THAT THE PURGE VALVE AND FUEL LINE ARE LOCATED ON THE INBOARD SIDE OF THE NACELLE AND THAT THERE IS A THERMAL RELIEF VALVE AND LINE FROM THE CROSSFEED LINE IN THE RIGHT FUEL SYSTEM.

* VALVE HAS HOLES FOR FLOW OUT AT REDUCED RATE. 28 GALLON WILL NOT GRAVITY FEED TO NACELLE.

NOTE

A FUEL CAPACITANCE GAGING SYSTEM UTILIZES A SINGLE FUEL QUANTITY GAGE FOR EACH WING FUEL SYSTEM. THIS GAGE CAN BE SWITCHED TO DESIGNATE THE AMOUNT OF FUEL IN THE NACELLE TANK OR THE TOTAL FUEL IN THE SYSTEM.

Figure 5-1. Fuel System Schematic Diagram

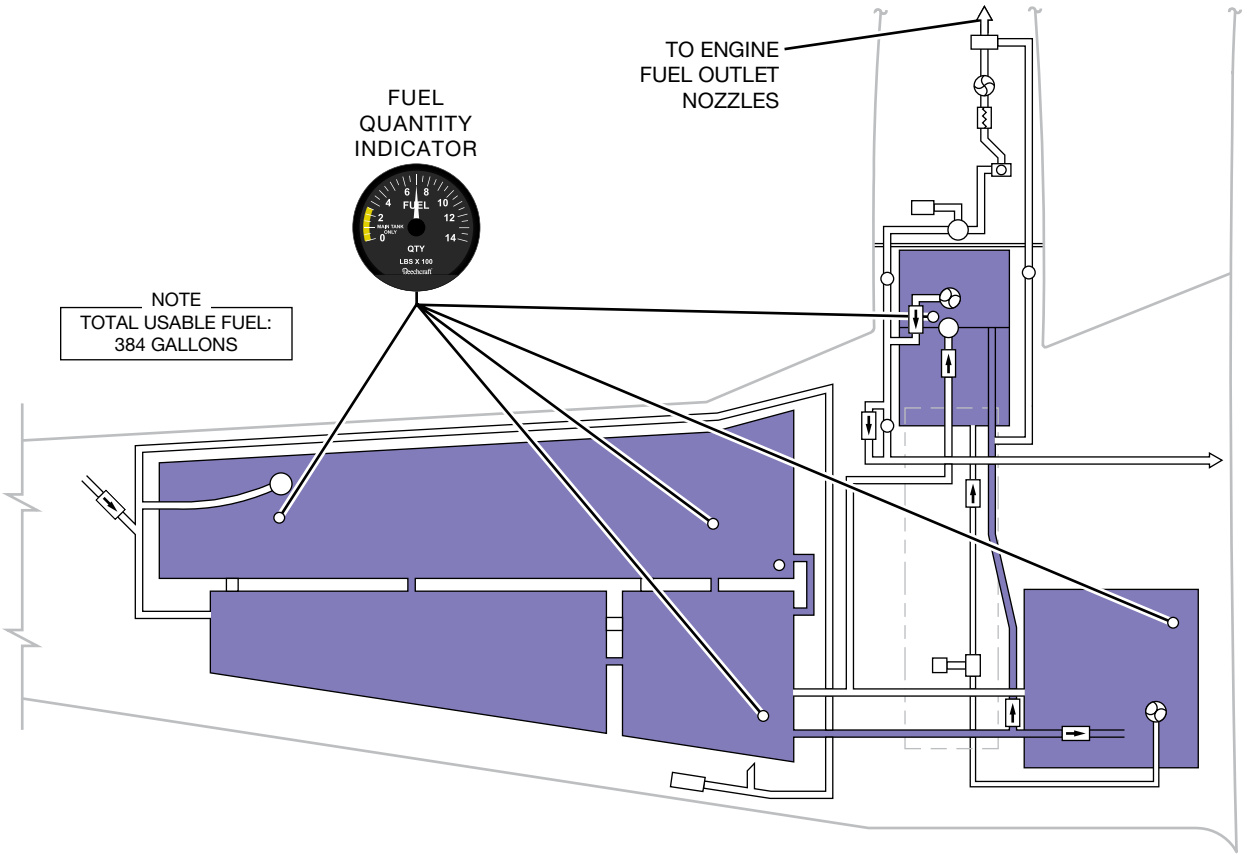


Figure 5-2. Fuel Tank System

With crossfeed in AUTO, a boost pump failure will be denoted by the momentary illumination of the FUEL PRESS annunciator and fault warning flasher, then the illumination of the FUEL CROSSFEED annunciator. To identify the failed boost pump, momentarily place the crossfeed in the CLOSED position. The FUEL PRESS annunciator on the side of the failed boost pump will illuminate. Place the crossfeed switch in the OPEN position. The FUEL PRESS annunciator will then extinguish.

In the event of a boost pump failure during any phase of flight, the system will begin to crossfeed automatically. If the boost pump fails , the cross-feed switch may be closed and the flight continued, relying on the engine-driven high pressure pump. In some instances the pilot may elect to continue the flight with the remaining pump and the crossfeed system in operation.

CAUTION

Operation with the FUEL PRESS annunciator on is limited to 10 hours, after which the engine-driven high pressure pump must be overhauled or replaced. When operating with Aviation Gasoline base fuels, operation on the engine-driven high pressure pump alone is permitted up to 8,000 feet for a period not to exceed 10 hours. Operation above 8,000 feet requires boost or operation of crossfeed.

The following Fuel Management Limitations, listed in the Limitations section of the C90A and C90B *POH*, pertain to fuel system boost pumps.

Both boost pumps must be operable prior to takeoff.

Operation is limited to 8000 feet when operating on aviation gasoline with boost pumps inoperative.

Operation with the FUEL PRESS annunciator on is limited to 10 hours between main engine-driven fuel pump overhaul or replacement.

FUEL TRANSFER PUMPS

Fuel level in the nacelle tank is automatically maintained at near full capacity during normal operation by a fuel transfer system, whenever the fuel level in the nacelle tank drops by approximately 10 gallons. Submerged, electrically-driven, impeller pumps located in the wing center section tanks provide the motive force for fuel transfer from wing tanks to nacelle tanks. The transfer pumps are controlled by float-operated switches on the nacelle tank fuel quantity transmitters.

Fuel is transferred automatically when the TRANSFER PUMP switches are placed in AUTO, unless the nacelle tanks are full. As the engines burn fuel from the nacelle tanks (61 gallon capacity each tank), fuel from the wing tanks is transferred into the nacelle tanks each time the nacelle tank levels drop approximately 10 gallons. The nacelle tanks will fill until the fuel reaches the upper transfer limit and a float switch turns the TRANSFER PUMP off.

A pressure switch, located in the fuel transfer line, will automatically turn off the transfer pump if a preset pressure is not obtained within approximately 30 seconds after the pump is turned on, or if the transfer pump pressure drops below a preset pressure due to empty wing tanks or pump failure. For example, when 132 gallons of fuel (each side) are used from the wing tanks (132 gallons usable each side), the pressure sensing switch reacts to a pressure drop in the fuel transfer line as the wing tanks are exhausted of fuel. After-30 seconds, the transfer pump shuts off and the respective yellow (red on prior to LJ1353) NO FUEL XFR annunciator on the annunciator panel illuminates.

The NO FUEL XFR annunciators will illuminate for the reasons mentioned: no pressure after 30 second time delay due to empty wing tanks or transfer pump failure. The NO FUEL XFR annunciator also functions as an operation indicator for the transfer pump during preflight. A TRANSFER TEST switch (placarded ENGINE L and ENGINE R) is provided to verify the operation of each pump when its nacelle tank is full. Holding the Transfer Test switch in the test position (either L or R) will activate the transfer pump and pressure sensor. In the test mode, the 30-second delay is by-passed, resulting in immediate indications. The NO FUEL XFR annunciator will momentarily illuminate and the Fault Warning Flasher will also begin flashing. The NO FUEL XFR annunciator will extinguish when fuel pressure to the sensor reaches a minimum pressure of 2.5 psi. If the transfer pump is operating, use of the transfer test will not be possible.

The fuel transfer system may be monitored by periodically checking the nacelle tank quantity against the total tank quantity.

If the NO FUEL XFR does not illuminate and the transfer test indicates a working pump, the flow switches may be suspect. Using the transfer test will begin the fill-up cycle, however, fuel quantity in the nacelle will drop below the lower level without activating the transfer pump. Proceed by moving the transfer pump switch (Figure 5-3) to the OVERRIDE position. In this mode, the transfer pump will run continuously until the transfer pump switch is returned to the OFF position. When the nacelle tank becomes full, excess fuel will be returned to the center section wing tank through the vent line.

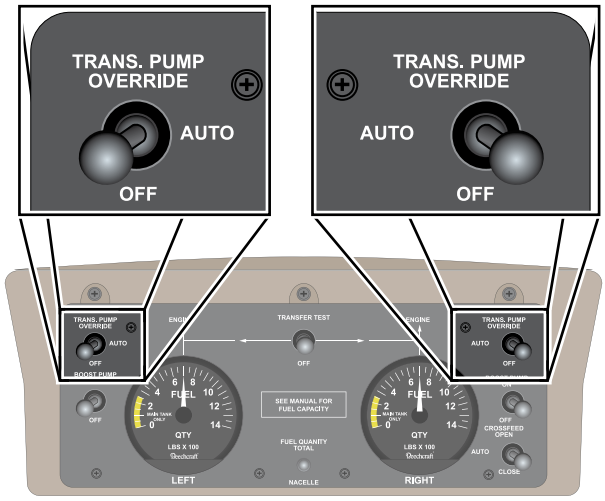


Figure 5-3. Fuel Transfer Pump Switch

Illumination of the NO FUEL XFR annunciator may indicate a normal or abnormal situation. During normal operation, when the fuel in the wing tanks is exhausted, the NO FUEL XFR annunciator indicates that the wing tanks are empty and the fuel transfer switch should be turned off.

If the transfer pump fails to operate during flight, gravity feed will perform the transfer. When the nacelle tank level drops to approximately 150 pounds, or approximately 22 gallons, the gravity port in the nacelle tank opens and gravity flow from the wing tank starts. All wing fuel, except 28 gallons from the center section tank, will transfer during gravity feed.

FUEL CAPACITY

The fuel quantity system is a capacitance gaging system with one quantity indicator per wing (Figure 5-4). A toggle switch selector allows the pilot to check total system or just the nacelle tank quantity. The system has a total capacity of 387 gallons, and a maximum usable fuel quantity of 384 gallons. The fuel quantity gages and the engine fuel flow indicators read in pounds times 100. At 6.7 pounds per gallon, 2572.8 pounds of usable fuel are available in the system, 1286.4 pounds per side.

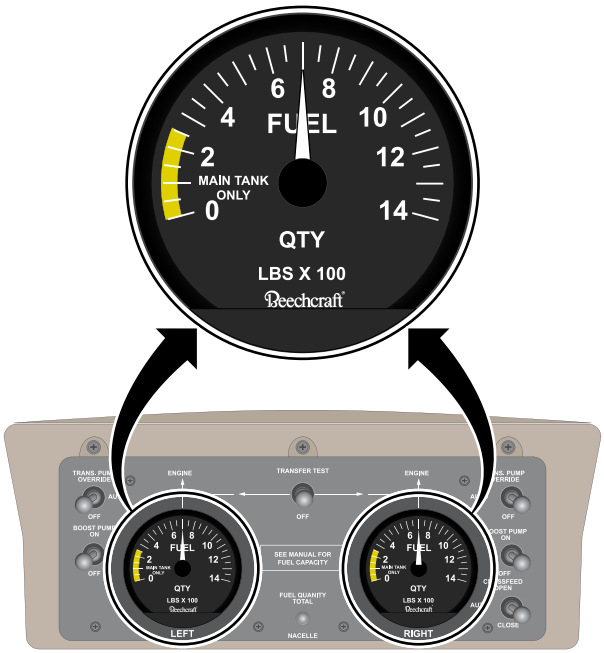


Figure 5-4. Fuel Control Panel

There is no structural limitation for which a Maximum Zero Fuel Weight must be set.

FUEL TANK VENTS

The fuel system is vented through a recessed ram scoop vent, coupled to a heated external vent, located on the underside of the wing, adjacent to the nacelle (Figure 5-5). One vent is recessed to prevent icing. The external vent is heated to prevent icing. Each vent serves as a backup for the other should one or the other become plugged.

In each wing fuel system, the wing panel tanks, the leading edge tank, the center section tank, and the nacelle tank are all crossvented with one another.

The line from the vent valve in the outboard wing panel fuel tank is routed forward along the leading edge of the wing, inboard to the nacelle, and aft through a check valve to the heated ram vent. Another line tees off from the heated vent line and extends to a recessed or ram scoop vent. The heated vent is described in the Anti-Ice Section of this manual. A suction relief valve is installed in the line from the float-operated vent valve to the siphon break line.

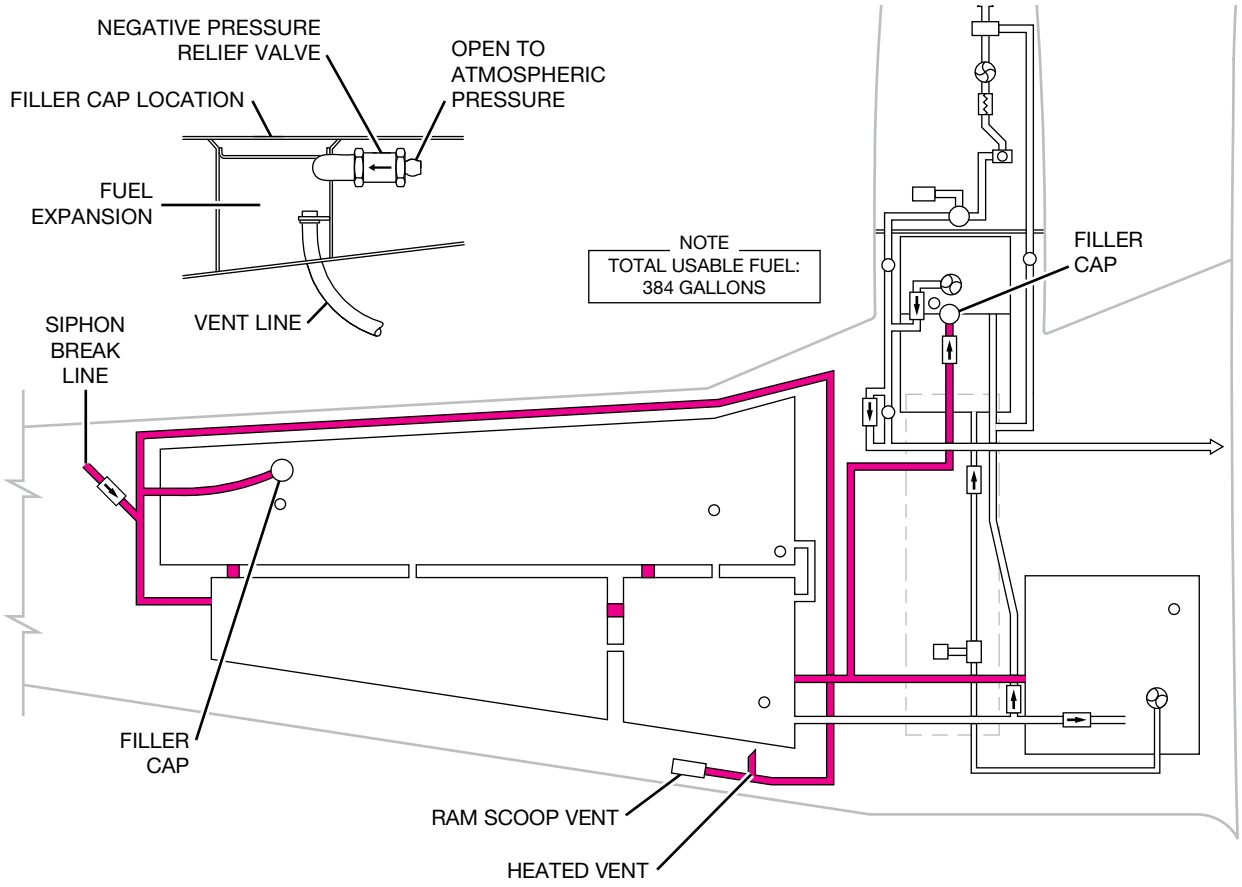


Figure 5-5. Fuel Vent System

FUEL SYSTEM OPERATION

Fuel flow from each wing tank system and nacelle tank is automatic without pilot action (Figure 5-6). The wing tanks gravity feed into the center section tank through a line extending from the aft inboard wing tank to be outboard side of the center section tank. A flapper-type check valve in the end of the gravity feed line prevents any backflow of fuel into the wing tanks.

The fuel pressure required to operate the engine is provided by an engine-driven fuel pump mounted in conjunction with the fuel control unit on the accessory case. Fuel is pumped to the high pressure fuel pump by an electrically-driven boost pump submerged in the nacelle tank.

The supply line from the nacelle tank is routed from the outboard side of the nacelle tank, forward to the engine-driven fuel pump through a motored firewall shutoff valve installed in the fuel line immediately behind the engine firewall.

The firewall shutoff valve for each engine fuel system is actuated by its respective FW SHUTOFF VALVE switch on the pilot's fuel control panel. When the FW SHUTOFF VALVEswitch is closed, its respective firewall shutoff valve closes to shut off the flow of fuel to the engine. From the firewall shutoff valve, fuel is routed to the fuel strainer filter and drain on the lower center of the engine firewall, the fuel pressure switch, the fuel flow indicator transmitter, the fuel heater, and then to the engine-driven fuel pump and engine fuel control unit. The 20 micron filter incorporates a bypass valve to permit fuel flow in case of plugging

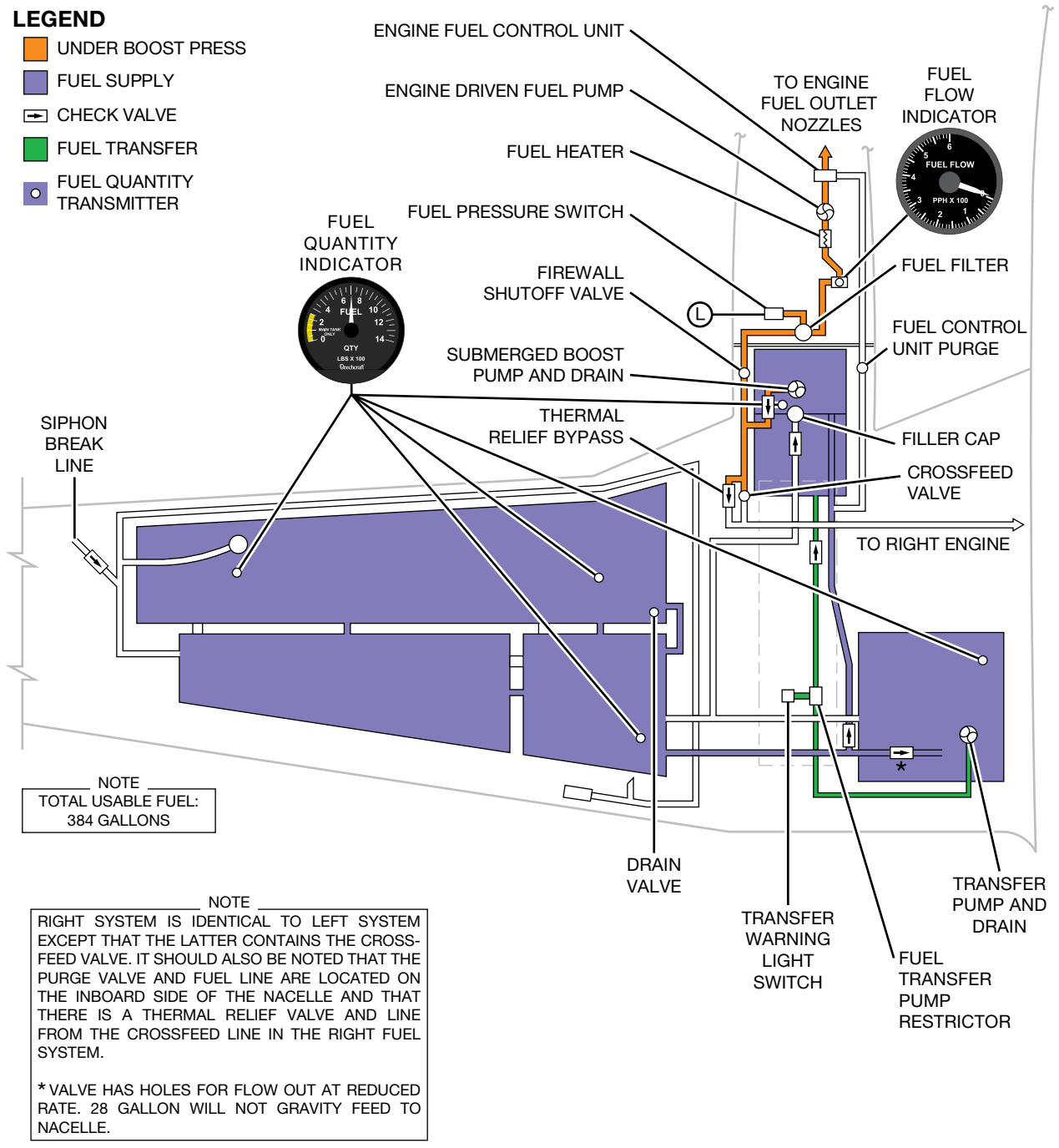


Figure 5-6. Fuel Flow Diagram

and a drain valve used to drain the filter prior to each flight. A pressure switch mounted directly above the filter senses boost pump fuel pressure at the filter. At a pressure, about 10 psi, the switch closes and actuates the red FUEL PRESS light in the annunciator panel.

CAUTION

Operation with the FUEL PRESS light ON is limited to 10 hours between overhaul or replacement of the engine-driven fuel pump. Such operation is restricted to 10 hours at altitudes not to exceed 8000 feet when aviation gasoline is being used. Windmilling time is not equivalent to operation of the engine at high power with respect to the effects of cavitation on fuel pump components; consequently, windmilling time is not to be included in the 10-hour limit on engine operation without a boost pump.

The red FUEL PRESS light will go out at about 10 psi of increasing fuel pressure. From the fuel strainer and filter, fuel is routed through the fuel flow transmitter mounted on the firewall, inboard of the pressure switch. Fuel from the transmitter is routed through the fuel heater, which utilizes heat from the engine oil to warm the fuel. The fuel is then routed to the fuel control unit that monitors the flow of fuel to the engine fuel nozzles. A heater boot is also installed on the governor control line of each engine. Each air line heater is protected by a 7.5 ampere, push-pull circuit breaker mounted in the circuit breaker panel beside the copilot. The heaters are controlled by switches installed on the pedestal and activated by the condition levers.

The engine-driven fuel pump is mounted on the accessory case of the engine in conjunction with the fuel control unit. This pump is protected against fuel contamination by an internal, 200mesh strainer. The primary fuel boost pump is an electrically-driven pump located in the bottom of each nacelle tank. The electrically-driven boost pump is capable of supplying fuel to the engine-driven fuel pump at the minimum pressure requirements of the engine manufacturer.

CAUTION

Should the boost pumps fail, suction feed operation may be employed; however, suction feed operation is restricted to 10 hours total time between fuel pump overhaul periods. If the engine-driven pump is operated on suction feed beyond the 10-hour limit, overhaul or replacement of the pump is necessary.

The electrically-driven boost pump also provides the pressure required for the crossfeed of fuel from one side of the aircraft to the other.

The electrical power with which the boost pumps are operated is controlled by lever-lock toggle switches on the fuel control panel. One source of power to the boost pumps is supplied from the triple-fed bus that supplies the circuit breakers. This circuit is protected by two 10-ampere circuit breakers located on the fuel panel. Power from this circuit is available only when the master switch is on.

The other source of power to the boost pumps is directly from the battery through the battery emergency bus. During shutdown, both boost pump switches and crossfeed must be turned off to prevent discharge of the battery.

FIREWALL SHUTOFF VALVES

The firewall shutoff valves (Figure 5-7), located between the engine-driven fuel pump and the nacelle tank, are controlled by guarded switches in the cockpit (Figure 5-8). There is one switch on each side of the fuel system circuit breaker panel on the fuel panel. These switches have two positions. The OPEN position allows uninterrupted fuel flow to the engine. The CLOSE position cuts off all fuel to the engine. When the red guard closes, it forces the switch into the open position and protects it in the open position.

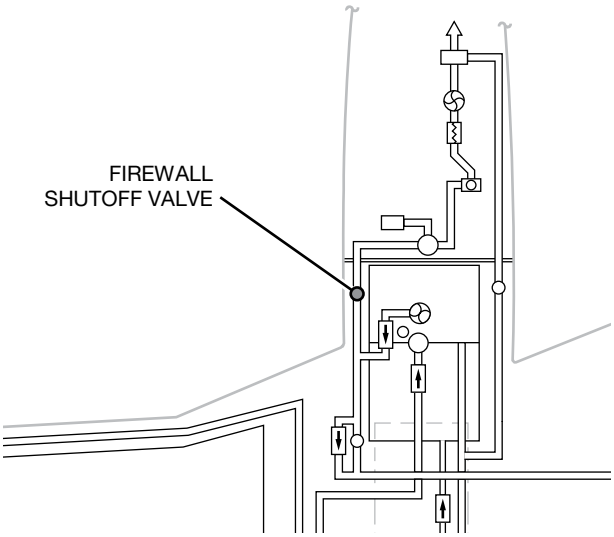


Figure 5-7. Firewall Shutoff Valve

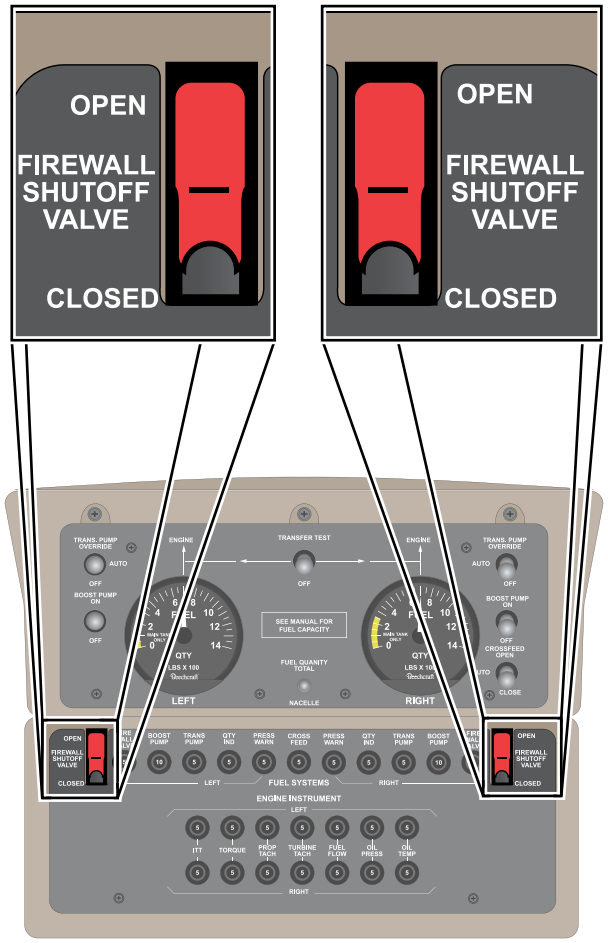


Figure 5-8. Firewall Shutoff Valve Switches

Each firewall shutoff valve receives electric power through its own 5-amp breaker on the fuel panel which brings electric power from the triple-fed bus as well as the generator bus. This source of power is available only when the battery and/or generator switches are on. The only pilot action necessary to ensure main fuel system operation is to have the firewall shutoff valves in the OPEN position.

CROSSFEED OPERATION

Crossfeeding fuel is authorized only in the event of engine failure or electric boost pump failure.

Each nacelle tank is connected to the engine in the opposite wing by a crossfeed line routed from the side of the nacelle, aft to the center section, and across to the side of the opposite nacelle. The crossfeed line is controlled by a valve (Figure 5-9). With the crossfeed valve OPEN, one system can supply fuel to the other engine. The system uses the electric boost pump in the nacelle tank. This pump supplies the pressure to transfer fuel as well as fuel boost to one or both engines. With one engine inoperative, the crossfeed system allows fuel from the inoperative side to be supplied to the operating engine.

The crossfeed system is controlled by a three-position switch placarded: CROSSFEED OPEN, AUTO, and CLOSED. The valve can be manually opened or closed, but under normal flight conditions it is left in the AUTO position. In the AUTO position, the fuel pressure switches are connected into the crossfeed control circuit.

In the event of a boost pump failure, causing a drop in fuel pressure, these switches open the crossfeed valve allowing the remaining boost pump to supply fuel to both engines.

In the event of a boost pump failure during takeoff, the system will begin to crossfeed automatically allowing the pilot to complete the takeoff without an increase in workload at a crucial time. After the takeoff is completed, or if the boost pump fails after takeoff, the crossfeed switch may be closed and the flight continued relying on the engine-driven high pressure pump without boosted pressure. In some instances, the pilot may elect to continue the flight with the remaining boost pump and the crossfeed system in operation.

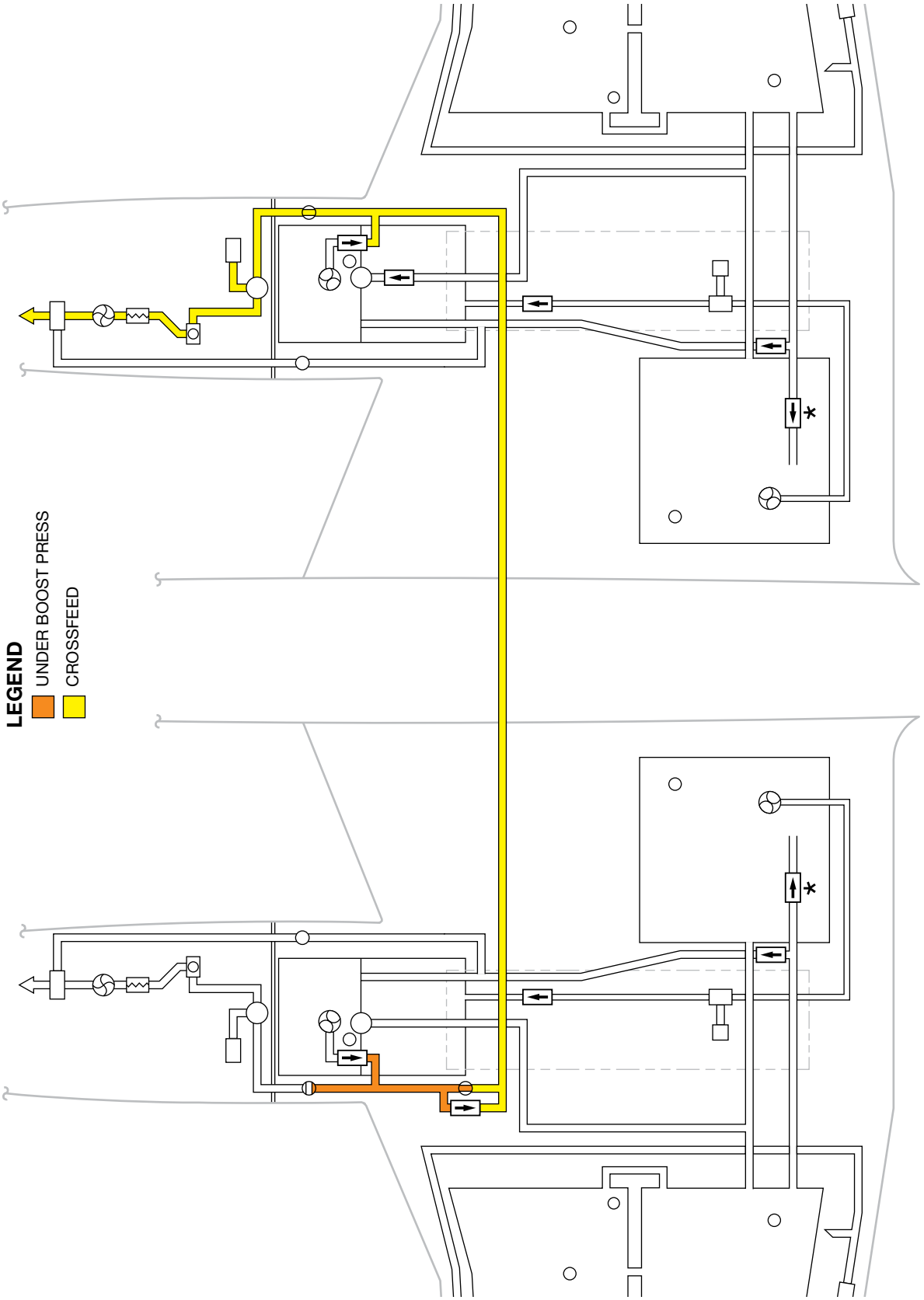


Figure 5-9. Crossfeed Schematic

When the crossfeed switch on the fuel control panel is actuated, power is drawn from a 5-ampere circuit breaker on the fuel control panel to the solenoid that opens the crossfeed valve. The crossfeed is also powered through the hot battery bus through a 5-amp fuse.

When the crossfeed valve is receiving power, the yellow FUEL CROSSFEED light on the annunciator panel will illuminate. The crossfeed will not transfer fuel from one wing to another; its function is to supply fuel from one side to the opposite engine during a boost pump failure or an engine-out condition. If the boost pumps on both sides are operating and the crossfeed valve is open, fuel will be supplied to the engines in the normal manner because the pressure on each side of the crossfeed valve should be equal.

FUEL DRAIN PURGE SYSTEM

The fuel purge system (Figure 5-10) is designed to assure that any residual fuel in the fuel manifolds is consumed during engine shutdown. During engine starting, fuel manifold pressure closes the fuel manifold poppet valve, allowing P3 air to pressurize the purge tank. During engine operation, engine compressor air (P3 air) is routed through a filter and check valve and maintains pressurization of the small purge tank. Upon engine shutdown, fuel manifold pressure subsides, thus allowing the engine fuel manifold poppet valve to open. The pressure differential between the purge tank and fuel manifold causes air to be discharged from the purge tank, forcing residual fuel out of the engine fuel manifold lines, through the nozzles, and into the combustion chamber. As the fuel is burned, a momentary surge in (Nl) gas generator rpm should be observed. The entire operation is automatic and requires no input from the crew.

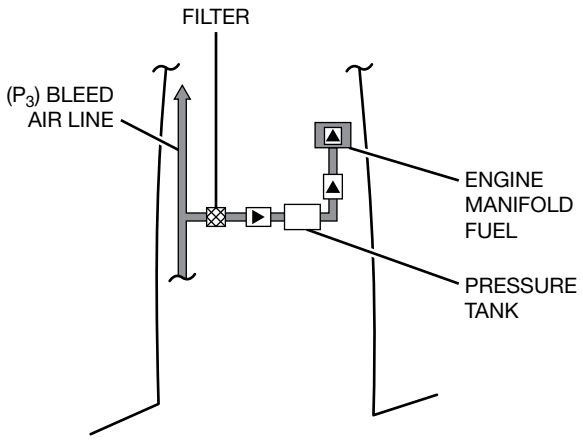


Figure 5-10. Fuel Drain Purge System Schematic

FUEL GAGING SYSTEM

The airplane is equipped with a capacitance-type fuel quantity indication system (Figure 5-11). It automatically compensates for fuel temperature density variations. The left fuel quantity indicator, on the fuel control panel, indicates the amount of fuel remaining in the left-side fuel system tanks when the FUEL QUANTITY select switch is in the TOTAL (upper) position, and the amount of fuel remaining in the left-side nacelle fuel tank when the FUEL QUANTITY select switch is in the NACELLE (lower) position. The right fuel quantity indicator indicates the same information for the right-side fuel systems, depending upon the position of the FUEL QUANTITY switch. The gages are marked in pounds.

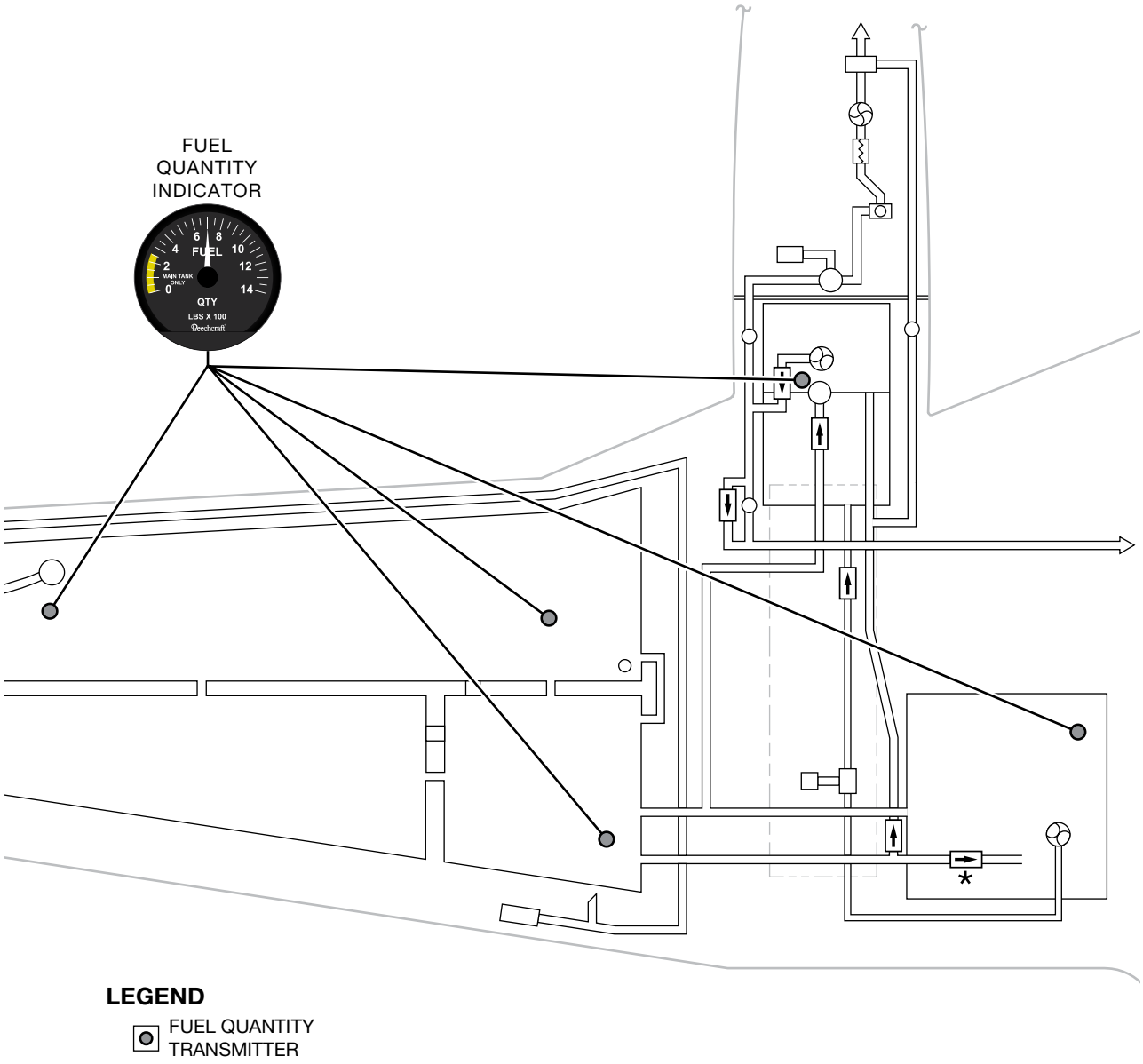


Figure 5-11. Fuel Quantity Indication System

The fuel quantity indicating system is a capacitance type that is compensated for specific gravity and reads in pounds on a linear scale. An electronic circuit in the system processes the signals from the fuel quantity (capacitance) probes (Figure 5-12) in the various fuel cells for an accurate readout by the fuel quantity indicators. A selector switch, located between the fuel quantity indicators in the fuel panel beside the pilot, may be set in either the TOTAL or NACELLE positions to determine whether the gages indicate the pounds of fuel in the nacelle and wing fuel cells of the fuel system, or the pounds of fuel in only the nacelle fuel cell.

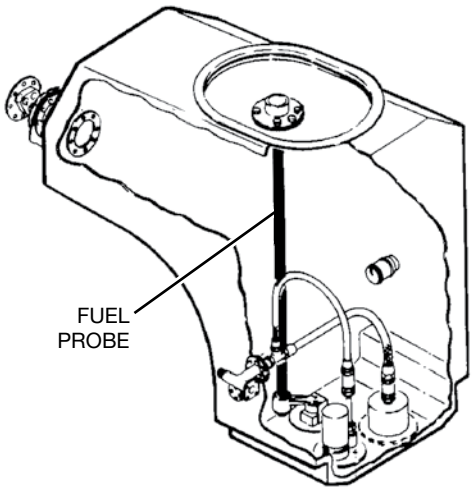


Figure 5-12. Fuel Probe

COMPONENTS AND OPERATION

Each side of the airplane has an independent gaging system consisting of a fuel quantity (capacitance) probe in the nacelle fuel cell, one in the aft-inboard fuel cell, two in the leading-edge fuel cell, and one in the center-section fuel cell.

When the fuel selector switch is left in its TOTAL position, power is supplied from a 5-ampere circuit breaker (on the fuel panel) through the fuel quantity indicator to all of the capacitance probes in the fuel system. When the fuel selector switch is placed in the NACELLE position, power is then supplied through the fuel quantity indicator to the capacitance probe in the nacelle fuel cell only.

Fuel density and electrical dielectric constantly vary with respect to temperature, fuel type, and fuel batch. The capacitance gaging system is designed to sense and compensate for these variables. The fuel quantity probe is simply a variable capacitor comprised of two concentric tubes. The inner tube is profiled by changing the diameter as a function of height so that the capacitance between the inner and outer tube is proportional to the tank volume. The tubes serve as fixed electrodes and the fuel of the tank in the space between the tubes acts as the dielectric of the fuel quantity probe.

The capacitance of the fuel quantity probe varies with respect to the change in the dielectric that results from the ratio of fuel-to-air in the fuel cell. As the fuel level between the inner and outer tubes rises, air with a dielectric constant of one is replaced by fuel with a dielectric constant of approximately two, thus increasing the capacitance of the fuel quantity probe. This variation in the volume of fuel contained in the fuel cell produces a capacitance variation that actuates the fuel quantity indicator.

FUEL DRAINS

During each preflight, the fuel sumps on the tanks, pumps and filters or strainers should be drained to check for fuel contamination. There are four sump drains and one filter drain or strainer drain in each wing (Figure 5-13 and Table 5-1).

The leading edge tank sump has a drain on the underside of the outboard wing just forward of the main spar. The flush drain valve for the firewall fuel strainer drain is accessible on the underside of the engine cowling. The boost pump sump drain is at the bottom center of the nacelle, just forward of the wheel well. The wheel well sump drain is inside the wheel well on the gravity feed line. The drain for the transfer pump sump is just outboard of the wing root, forward of the flap.

When draining the flush-mounted drains, do not turn the draining tool. Turning or twisting of the draining tool will unseat the O-ring seal and cause a leak.

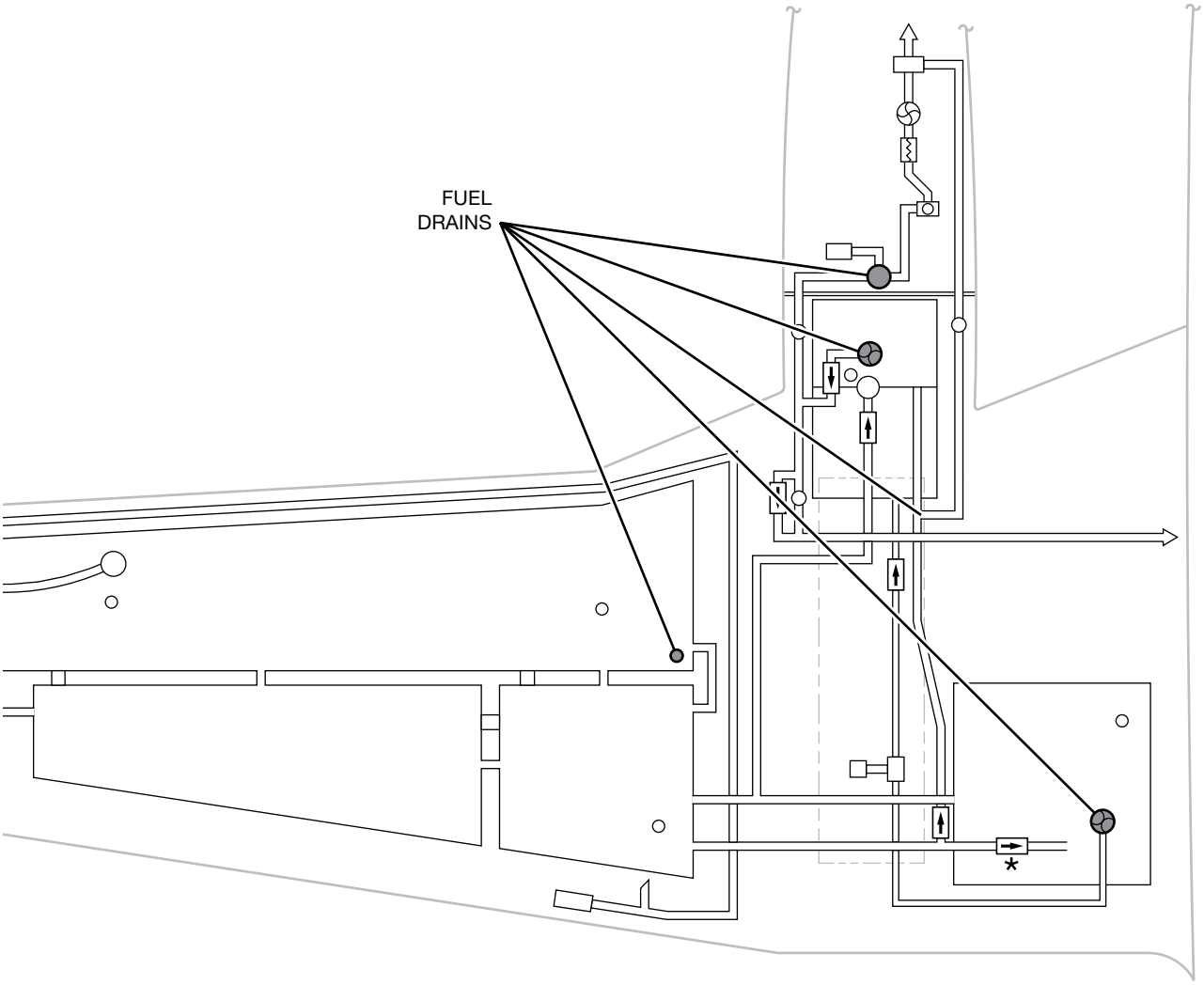


Figure 5-13. Fuel Drains

The flush valve attached to the base of the fuel strainer can be opened or closed with a coin, a screw driver, or a fuel drain tool making it possible to drain fuel from the fuel strainer for preflight check.

Since jet fuel and water are of similar densities, water does not settle out of jet fuel as easily as from aviation gasoline. For this reason, the airplane must sit perfectly still, with no fuel being added, for approximately three hours prior to draining the sumps if water is to be removed. Although turbine engines are not so critical as reciprocating engines regarding water ingestion, water should still be removed periodically to prevent formations of fungus and contamination induced inaccuracies in the fuel gaging system.

Table 5-1. FUEL DRAIN LOACATIONS

NUMBER	DRAINS	LOCATION
1	Leading Edge Tank Sump	On underside of outboard wing just forward of main spar
1	Firewall Fuel Filter (Strainer) Drian	Flush drain valve is accessible on underside of engine cowling
1	Boost Pump Sump	Bottom center of nacelle forward of wheel well
1	Transfer Pump Sump Drain	Just outbard of wing root, forward of flap
1	Gravity Feed Line	Inside wheel well

FUEL HANDLING PRACTICES

Takeoff is prohibited when the fuel-quantity indicator needles are in the yellow arc, with the selector in the total position, or when there is less than 265 pounds of fuel in each wing system.

The King Air C90A and C90B require that both boost pumps must be operable prior to takeoff.

All hydrocarbon fuels contain some dissolved and some suspended water. The quantity of water contained in the fuel depends on temperature and the type of fuel. Kerosene, with its higher aromatic content, tends to absorb and suspend more water than aviation gasoline. In addition to water, it will suspend rust, lint and other foreign materials longer. Given sufficient time, these suspended contaminants will settle to the bottom of the tank.

The settling time for kerosene is five times that of aviation gasoline; therefore, jet fuels require good fuel-handling practices to assure that the airplane is serviced with clean fuel. If recommended ground procedures are carefully followed, solid contaminants will settle and free water can be reduced to 30 parts per million (ppm), a value that is currently accepted by the major airlines.

Since most suspended matter can be removed from the fuel by sufficient settling time and proper filtration, it is not a major problem. Dissolved water has been found to be the major fuel contamination problem. Its effects are multiplied in aircraft operating primarily in humid regions and warm climates.

Dissolved water cannot be filtered from the fuel by micronic-type filters, but can be released by lowering the fuel temperature, which will occur in flight. For example, a kerosene fuel may contain 65 ppm (8 fluid ounces per 1000 gallons) of dissolved water at 80°F. When the fuel temperature is lowered to 15°F, only about 25 ppm will remain in solution. The difference of 40 ppm will have been released as supercooled water droplets which need only a piece of solid contaminant or an impact shock to convert them to ice crystals.

Tests indicate that these water droplets will not settle during flight and are pumped freely through the system. If they become ice crystals in the tank, they will not settle since the specific gravity of ice is approximately equal to that of kerosene. The 40 ppm of suspended water seems like a very small quantity, but when added to suspended water in the fuel at the time of delivery, it is sufficient to ice a filter. While the critical fuel temperature range is from 0 to -20°F, which produces severe system icing, water droplets can freeze at any temperature below 32°F.

Even if the fuel does not contain water or you have drained the water out, there is still the possibility of fuel icing at very low temperatures. The oil-to-fuel heat exchanger is used to heat the fuel prior to entering the fuel control unit. Since no temperature measurement is available for fuel prior to the heat exchanger, the temperature must be assumed to be the same as the outside air temperature.

The graph in the Limitations section of the *Pilot's Operating Handbook* is used as a guide in preflight planning, based on known or forecast conditions, to determine operating temperatures where icing at the fuel control unit could occur. Enter the graph with the known or forecast Outside Air Temperature and plot vertically to the given pressure altitude. In this example (Figure 5-14), Outside Air Temperature equals minus thirty degrees Celsius and pressure altitude equals 5000 feet. Next, plot horizontally to determine the minimum oil temperature required to prevent icing. In this example, the minimum oil temperature required is 38 degrees Celsius. If the plot should indicate that oil temperature versus Outside Air Temperature is such that ice formation could occur during takeoff or in flight, anti-icing additive must be mixed with the fuel.

The King Air maintains a constant oil temperature, however, this temperature varies from one aircraft to another. For most aircraft the oil temperature will be between 50 and 60 degrees Celsius. Compare the minimum oil temperature obtained from this graph with the oil temperature achieved by each particular airplane involved. When required, only anti-icing additive conforming to Specification MIL-I-27686 is approved. The anti-icing additive should be added during fueling.

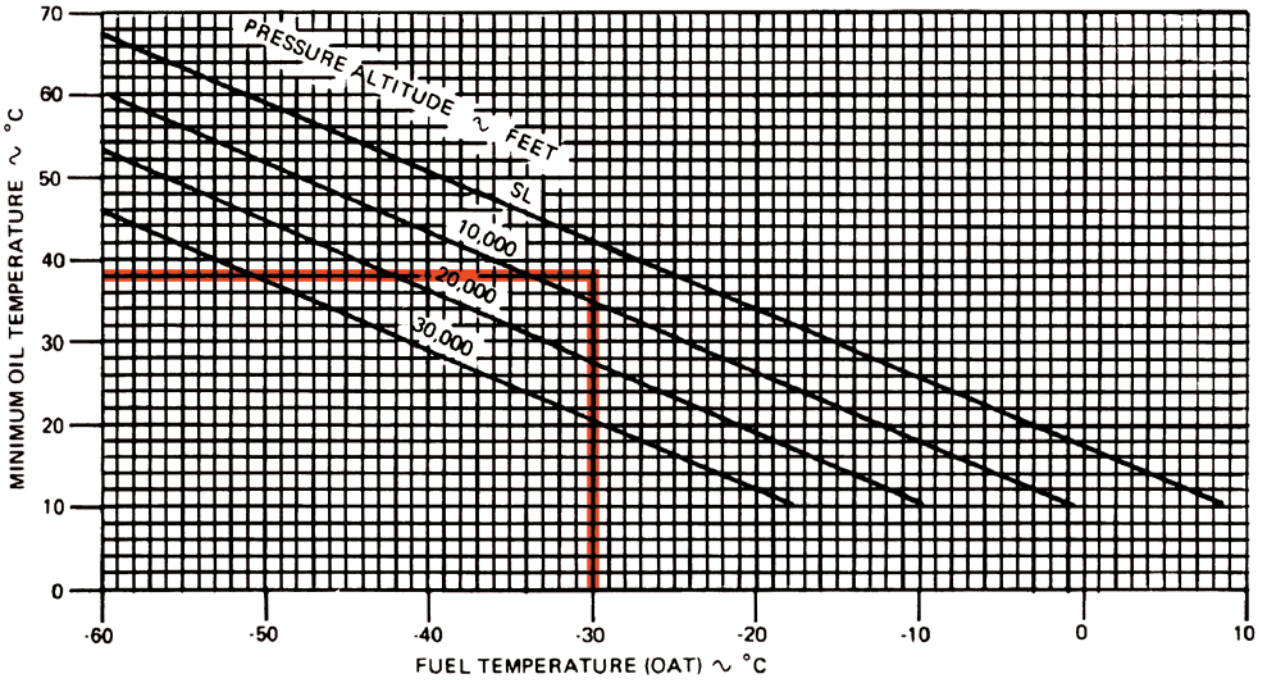


Figure 5-14. Fuel Temperature Graph

Water in jet fuel also creates an environment favorable to the growth of a microbiological "sludge" in the settlement areas of the fuel cells. This sludge, plus other contaminants in the fuel, can cause corrosion of metal parts in the fuel system as well as clogging of the fuel filters. Although this airplane uses bladder-type fuel cells, and all metal parts (except the boost pumps and transfer pumps) are mounted above the settlement areas, the possibility of filter clogging and corrosive attacks on fuel pumps exists if contaminated fuels are consistently used.

Fuel biocide-fungicide "BIOBORJF" in concentrations noted in the *POH* may he used in the fuel. BIOBORJF may be used as the only fuel additive or it may be used with the anti-icing additive conforming to MIL-I-27686 specification. Used together, the additives have no detrimental effect on the fuel system components.

The primary means of fuel contamination control by the owner/ operator is "good housekeeping." This applies not only to fuel supply, but to keeping the aircraft system clean. The following is a list of steps that may be taken to recognize and prevent contamination problems:

1. Know your supplier. It is impractical to assume that fuel free from contaminants will always be available, but it is feasible to exercise caution and be watchful for signs of fuel contamination.
2. Assure, as much as possible, that the fuel obtained has been properly stored, that it is filtered as it is pumped to the truck, and again as it is pumped from the truck to the aircraft.
3. Perform filter inspections to determine if sludge is present.
4. Maintain good housekeeping by periodically flushing the fuel tanks and systems. The frequency of flushing will be determined by the climate and the presence of sludge.
5. Aviation gas is an emergency fuel. The 150 hours maximum operation on aviation gasoline per a "Time Between Overhaul" should be observed.
6. Use only clean fuel-servicing equipment.
7. After refueling, allow a settling period of at least four hours whenever possible, then drain a small amount of fuel from each drain.

CAUTION

Remove spilled fuel from the ramp area immediately to prevent the contaminated surface from causing tire damage.

When fueling the King Air C90A or C90B, the nacelle fuel tanks should be filled first before any fuel is put in the wing tank system to insure that the wing tanks are completely full.

FUEL GRADES AND ADDITIVES

Aviation Kerosene Grades Jet A, Jet A-1, Jet B, JP-4, JP-5, and JP-8 may be mixed in any ratio. Aviation Gasoline Grades 80 (80/87), 100LL, 100 (100/130), and 115/145 are emergency fuels and may be mixed with the recommended fuels in any ratio; however, use of the lowest octane rating available is suggested. Operation on aviation gasoline shall be limited to 150 hours per engine during each Time Between Overhaul (TBO) period.

If the King Air C90A or C90B is fueled with aviation gasoline, some operational limitations, which are listed in the *POH*, must be observed. Maximum operation with aviation gasoline is limited to 150 hours between engine overhauls.

Use of aviation gas is limited to 150 hours due to lead deposits which form on the turbine wheels during aviation gas consumption, and which cause power degradation. Since the aviation gas will probably be mixed with jet fuel already in the tanks, it is important to record the number of gallons of aviation gas taken aboard for each engine. Determine the average fuel consumption for each hour of operation. If, for example, an engine has an average fuel consumption of 40 gallons per hour, each time 40 gallons of aviation gasoline are added, one hour of the 150 hour limitation is being used. In other words, using the 40 gph consumption rate as an example, the engine is allowed 6000 gallons of aviation gasoline between overhauls.

If the tanks have been serviced with aviation gas, flights are limited to 8,000 feet pressure altitude or below if either boost pump is inoperative. Because it is less dense, aviation gas delivery is much more critical than jet fuel delivery. Aviation gas feeds well under pressure feed but does not feed well on suction feed -particularly at high altitudes. For this reason, an alternate means of pressure feed must be available for aviation gas at high altitude. This alternate means is crossfeed from the opposite side. Thus, a crossfeed capability is required for climbs above 8,000 feet pressure altitude. These limitations are found in the Limitations section of your *Pilot's Operating Handbook.*

The POH lists two approved fuel additives for the King Air C90A and C90B. Any anti-icing additive conforming to Specification MIL-L-127686 is approved as is the fuel biocide-fungicide BIOBORJF. Each additive may be used as the only fuel additive or they may be used together. It has been determined that, used together, the additives have no detrimental effect on the fuel system components.

Additive concentrations and blending procedures are found in the *King Air 90 Maintenance Manual*.

The FUEL BRANDS AND TYPE DESIGNATIONS chart in the Handling, Service & Maintenance section of the *POH* gives the fuel refiner's brand names, along with the corresponding designations established by the American Petroleum Institute (APT) and the American Society of Testing Material (ASTM). The brand names are listed for ready reference and are not specifically recommended by Beech Aircraft Corporation. Any product conforming to the recommended specification may be used.

FILLING THE TANKS

When filling the aircraft fuel tanks, always observe the following:

1. Make sure the aircraft is statically grounded to the servicing unit and to the ramp.
2. Service the nacelle tank on each side first. The nacelle tank filler caps are located at the top of each nacelle. The wing tank filler caps are located in the top of the wing, outboard of the nacelles.

NOTE

Servicing the nacelle tanks first prevents fuel transfer through the gravity feed interconnect lines from the wing tanks into the nacelle tanks during fueling. If wing tanks are filled first, fuel will transfer from them into the nacelle tank leaving the wing tanks only partially filled. Be sure the nacelle tanks are completely full after servicing the fuel system to assure proper automatic fuel transfer during flight operation.

3. Allow a four-hour settling period whenever possible, then drain a small amount of fuel from each drain point. Check fuel at each drain point for contamination.

DRAINING THE FUEL SYSTEM

Open each fuel drain daily to drain off any water or other contamination collected in the low places. Along with the drain on the firewall mounted fuel filter, there are four other drains: the nacelle tank fuel-pump drain, center-section tank transfer-pump drain, wheelwell drain, and the inboard end of the outboard-wing tank drain.

The fuel pump and tank drains are accessible from the underside of the airplane.

NOTE

The firewall shutoff valve has to be electrically opened to drain large quantities of fuel from the firewall fuel-filter drain.

Fuel may be drained from the tanks by gravity flow through the center-section transfer-pump drains into suitable containers. Fuel may also by pumped out of the tanks utilizing an external pump and suction hoses inserted into the filler openings. For the fastest means of draining the system see the procedures in the *Beechcraft King Air 90 Series Maintenance Manual*.

QUESTIONS

1. During normal operation the engine receives low-pressure fuel from:

A. Gravity feed.
B. The transfer pump.
C. The electric boost pump.
D. The engine-driven boost pump.

2. During normal operation the fuel CROSSFEED switch is in the:

A. CLOSED position.
B. OPEN position.
C. AUTO position.
D. MAN position.

3. Crossfeed operation is authorized:

A. For single-engine operation, or when a boost pump is inoperative.
B. For single-engine operation.
C. To maintain fuel balance.
D. Only after attaining cruising altitude.

4. Operation of the engine with a FUEL PRESS annunciator illuminated is limited to?

A. Ten hours of operation above 20,000 feet.
B. Unlimited operation below 20,000 feet.
C. Ten hours of operation before the engine-driven fuel pump must be overhauled or replaced.
D. The respective engine must be shut down with the FUEL PRESS light illuminated.

5. Fuel is heated prior to entering the fuel control unit by:

A. Bleed air from the engine's compressor.
B. Compressing the fuel with the electric boost pump.
C. An air-to-fuel heat exchanger prior to the fuel control unit.
D. Engine oil, through an oil-to-fuel heat exchanger.

6. A NO FUEL XFR annunciator will illuminate:

A. If the low level nacelle tank switch fails, or if the transfer pump fails.
B. If the transfer pump fails, or if the center wing tank is out of fuel.
C. Only if the transfer pump fails.
D. Only if the low level nacelle tank switch fails.

7. Which of the following is a limitation reguarding the use of aviation gasoline?

A. Operation is limited to 8,000 feet if a boost pump is inoperative
B. Operation is limited to 150 hours between engine overhauls
C. There are no limits on the use of aviation gasoline
D. Operation is limited to 10,000 feet if crossfeed is inoperative

8. Suction feed may be used (within limitations) to supply fuel to an engine:

A. When the engine-driven fuel pump is inoperative.
B. For takeoff and climb out.
C. When the boost pump is inoperative.
D. When the transfer pump is inoperative.

9. The transfer pump is used:

A. As a backup, in the event of boost pump failure.
B. To transfer fuel from the center section tank to the nacelle tank.
C. In crossfeed operations.
D. All the above.

CHAPTER 6
AUXILIARY POWER UNIT

The information normally contained in this chapter is not applicable to this particular airplane.

CHAPTER 7
POWERPLANT

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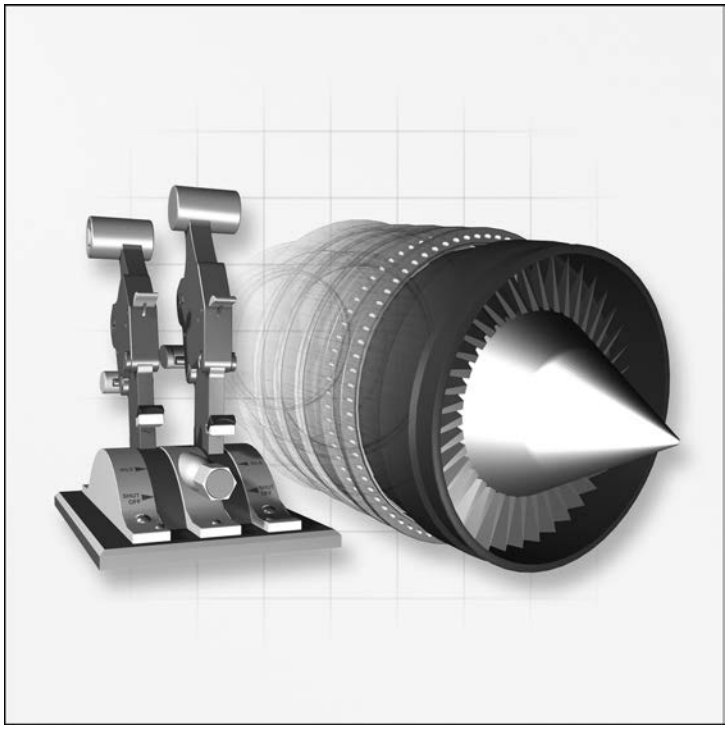
CHAPTER 7
POWERPLANT



INTRODUCTION

In-depth knowledge of the powerplants is essential to good power management by the pilot. Knowing and operating within safe parameters of the powerplant and propeller system extends engine life and ensures safety. This chapter describes the basic sections of the engine and its operational limits and preflight checks.

In-depth knowledge of the propeller system is also essential to proper operation of the engine power system. Operating within safe parameters of the powerplant and propeller systems extends engine life and ensures safety. This chapter also describes the propeller system and its operational limits and preflight checks.

GENERAL

The Engines section of this chapter presents a description and discussion of the Pratt and Whitney PT6A turboprop engines. The engines used on these airplanes will be described in sufficient detail for flight crewmembers to understand normal operational practices and limitations. The purpose of this section is to give the participants a sufficient understanding of the engine so that they will be familiar with normal and emergency procedures.

The Propellers section of this chapter presents a description and discussion of the propeller system. Location and use of propeller controls, principle of operation, reversing, and feathering are included.

ENGINES

GENERAL

The powerplants chosen by Beech designers for the King Airs are Pratt and Whitney Series PT6A free-turbine turboprop engines (Figure 7-1 and Figure 7-2). The King Air C90A and C90B use PT6A-21 engines. The PT6A-21 engine is flat-rated to 550 shaft horsepower.

The engines are equipped with conventional three-blade (C90A) or four-blade (C90B), full-feathering, reversing, variable-pitch propellers mounted on the output shaft of the engine reduction gearbox. The propeller pitch and speed are controlled by engine oil pressure through single-action, engine-driven propeller governors. The propellers will feather automatically when the engines are shut down on the ground, and will unfeather when the engines are started.

When reference is made to the right or left side of the airplane or engine, it is always looking from the rear to the front.

TURBOPROP ENGINE RATINGS

In turboprop engines, power is measured in Equivalent Shaft Horse Power (ESHP) and Shaft Horse Power (SHP). SHP is determined by propeller rpm and torque applied to turn the propeller shaft. The hot exhaust gases also develop some kinetic energy as they leave the engine, similar to a turbojet engine. This jet thrust amounts to about 10% of the total engine power. ESHP is the term applied to total power delivered, including the jet thrust. Turboprop engine specifications usually show both ESHP and SHP, along with limiting ambient temperatures. The engine specifications in Table 7-1 Primary Governor Diagram show the engine ratings and temperatures.

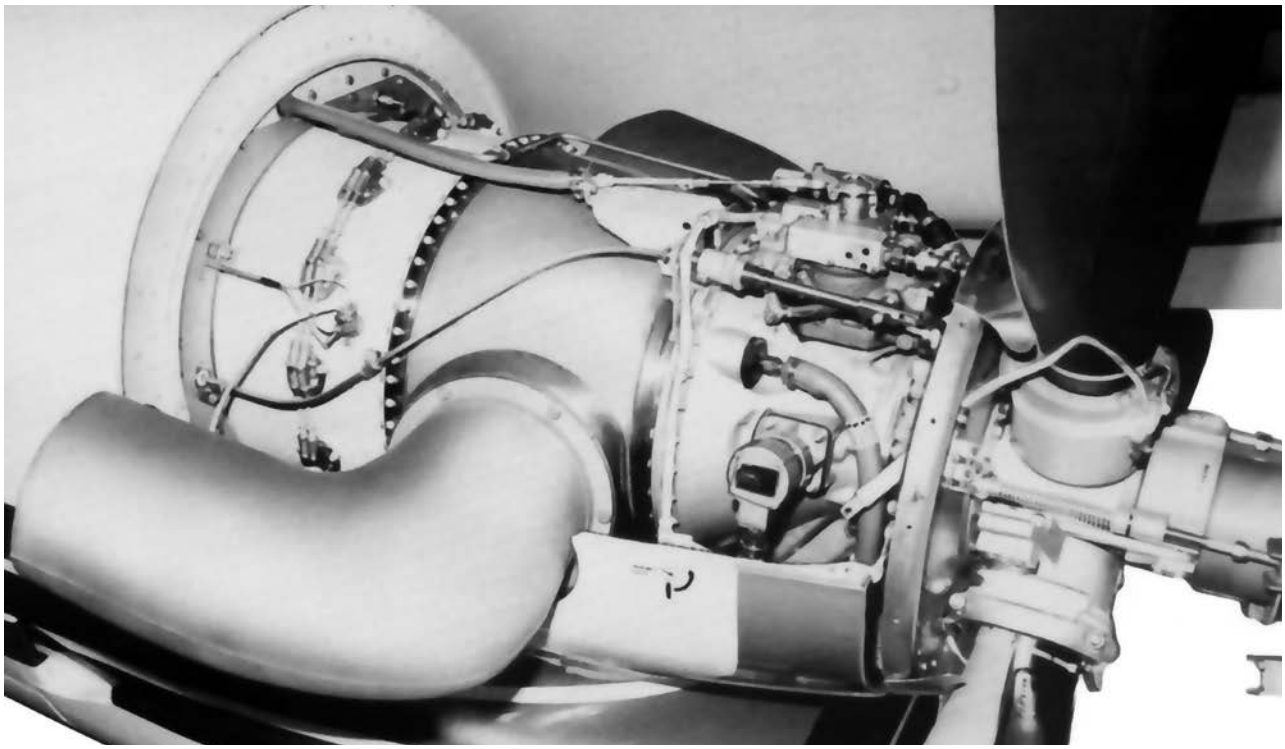


Figure 7-1. Powerplant Installation

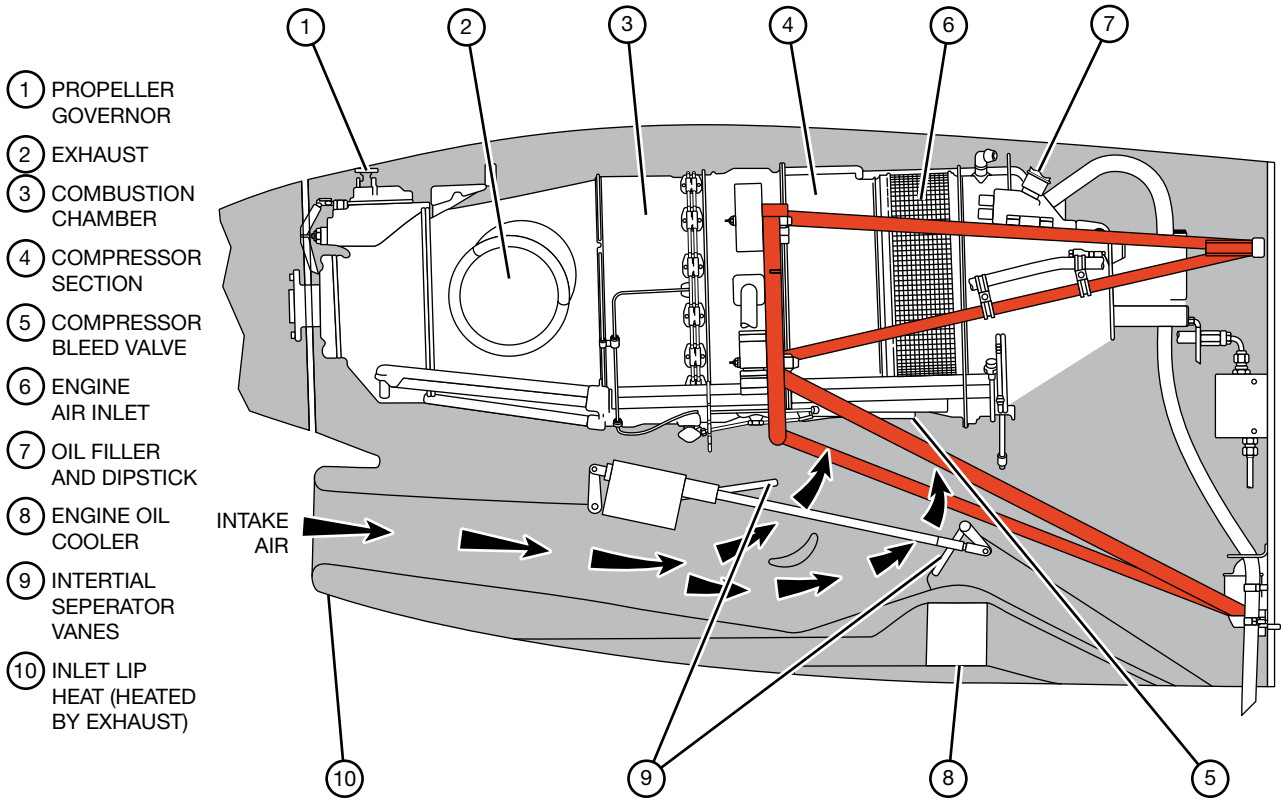


Figure 7-2. Engine Installation

Table 7-1. PT6A-21 SPECIFICATIONS

(Sea Level Static Output - Standard Day)

OPERATING CONDITION	ESHP	SHP	PROPELLER RPM	JET THRUST (lbs.)	JP-4 FUEL CONSUMPTION (lb./eshp/hr.) at 15° C (59°F)
Takeoff & Max. Continuous/ Enroute Emergency	580	550*	2200***	75	0.630
Max. Cruise +	580	550**	2200***	75	0.630
Max. Climb +	568	538	2200***	72	0.633
Normal Cruise +	523	495	2200***	70	0.649
Max. Performance Cruise +	505	475	1900***	70	.0677

* Avilable to +32.8°C (+91°F) Ambient
** Available to +27.8°C (+82°F) Ambient
*** Beechcraft Performace Charts are based on 1900 rpm
+ Not Cerification Ratings

(a) Corresponding Rotor Speed: Power Turbine - 33,000 rpm

Engine Type ..... Free Turbine
Type of Combustion Chamber ..... Annular
Compression Ratio 7.0:1
Compressor Shaft Rotation (looking forward) ..... Counterclockwise
Propeller Shaft Rotation (looking forward) ..... Clockwise
Propeller Shaft Gear Ratio 15:1
Oil Consumption, Maximum Average ..... 0.2 lb./hr (0.091 kg./hr.)
Gas Generator Speed 100% N_1 ..... 37,500 rpm
Max. Continuous Gas Generator Speed 101.5% N_1 ..... 38,100 rpm

ENGINE TERMS

To properly understand the operation of the PT6A series engines, there are several basic terms you should know: N_1 or N_G-Gas generator rpm is percent of turbine speed

- N_2 or N_P-Propeller rpm
- NF-Power turbine rpm (not indicated on engine instruments)
- P_3-Air pressure at station three (the source of bleed air)
- ITT or T_5-Interstage Turbine Temperature in degrees of temperature at station 5

Review and remember these terms. They will be used often to describe PT6A engines.

FREE-TURBINE REVERSE-FLOW PRINCIPLE

The Pratt and Whitney PT6 family of engines consists basically of free-turbine, reverse-flow engines driving a propeller through planetary gearing (Figure 7-3, Figure 7-4, Figure 7-5, and Figure 7-6). The term "free-turbine" refers to the design of the turbine sections of the engine. There are two turbine sections: one, called the compressor turbine, which drives the engine compressor and accessories; and the other, consisting of a single power turbine, which drives the power section and propeller. The power turbine section has no physical connection to the compressor turbine at all. These turbines are mounted on separate shafts and are driven in opposite directions by the gas flow across them. The term "reverse flow" refers to airflow through the engine. Inlet air enters the compressor at the aft end of the engine, moves forward through the combustion section and the turbines, and is exhausted at the front of the engine.

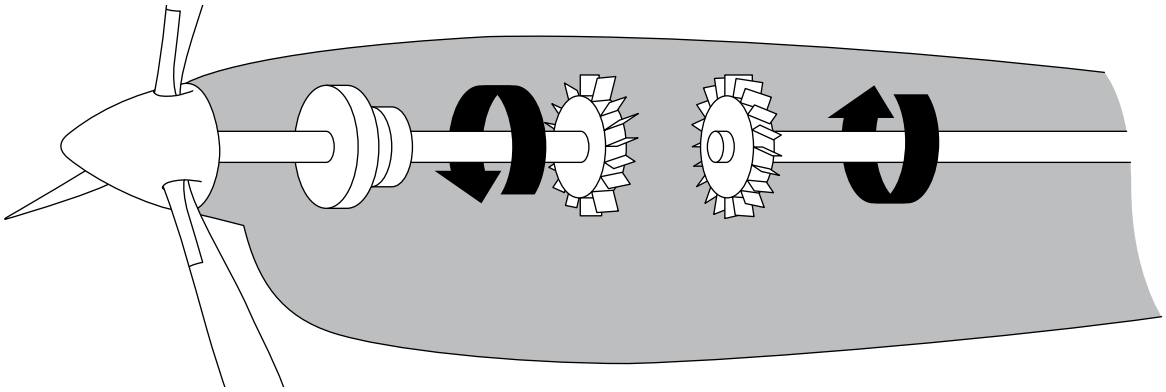


Figure 7-3. Free Turbine

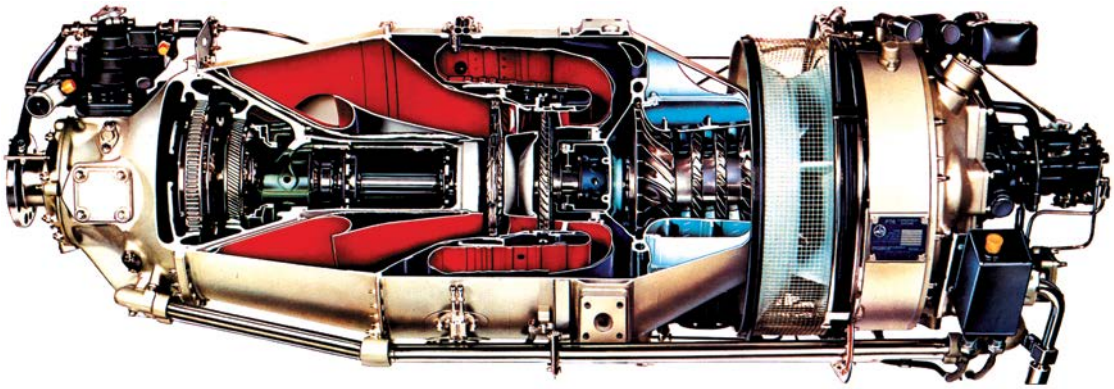


Figure 7-4. Engine Cutaway

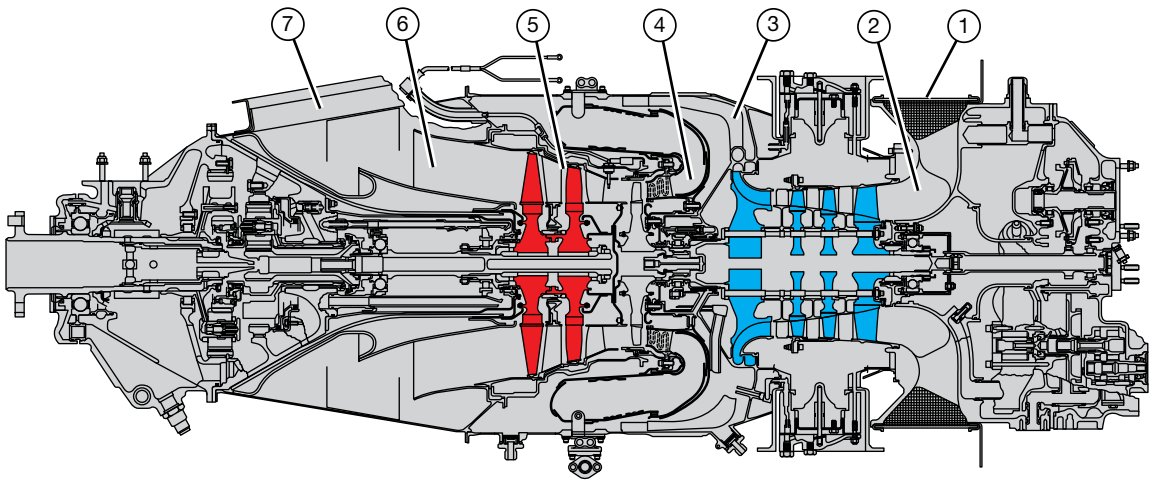


Figure 7-5. Engine Stations

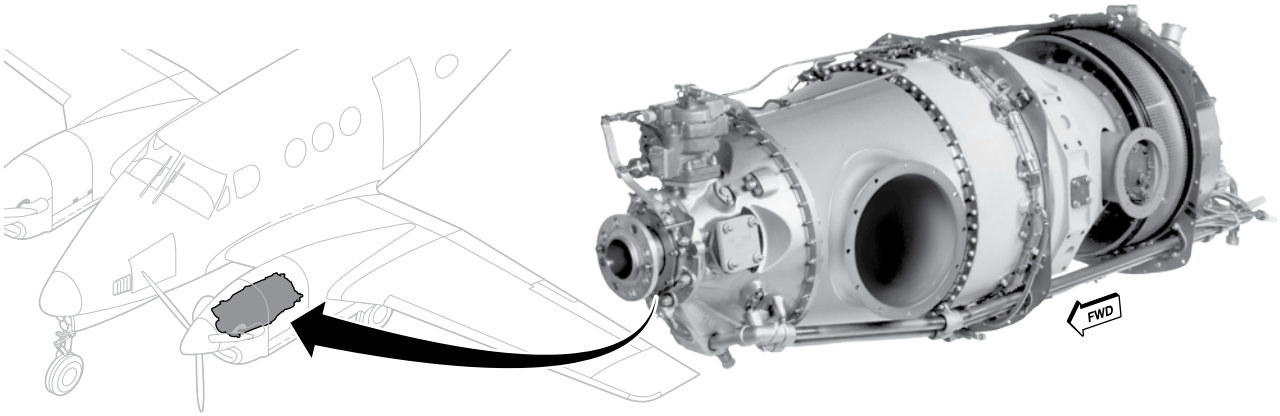


Figure 7-6. Engine Orientation

ENGINE AIRFLOW

Inlet air enters the engine through an annular plenum chamber, formed by the compressor inlet case, where it is directed forward to the compressor (Figure 7-7, Figure 7-8, and Figure 7-9). The compressor consists of three axial stages combined with a single centrifugal stage, assembled as an integral unit.

A row of stator vanes, located between each stage of compression, diffuses the air, raises its static pressure, and directs it to the next stage of compression. The compressed air passes through diffuser tubes, which turn the air through 90° in direction and convert velocity to static pressure. The diffused air then passes through straightening vanes to the annulus surrounding the combustion chamber liner.

The combustion chamber liner has varying size perforations which allow entry of compressor delivery air. Approximately 25% of the air mixes with fuel to support combustion. The remaining 75% centers the flame in the combustion chamber and provides internal cooling for the engine. As it enters the combustion area and mixes with fuel, the flow of air changes direction 180°. The fuel/air mixture is ignited, and the resultant expanding gases are directed to the turbines. The location of the liner eliminates the need for a long shaft between the compressor and the compressor turbine, thus reducing the overall length and weight of the engine.

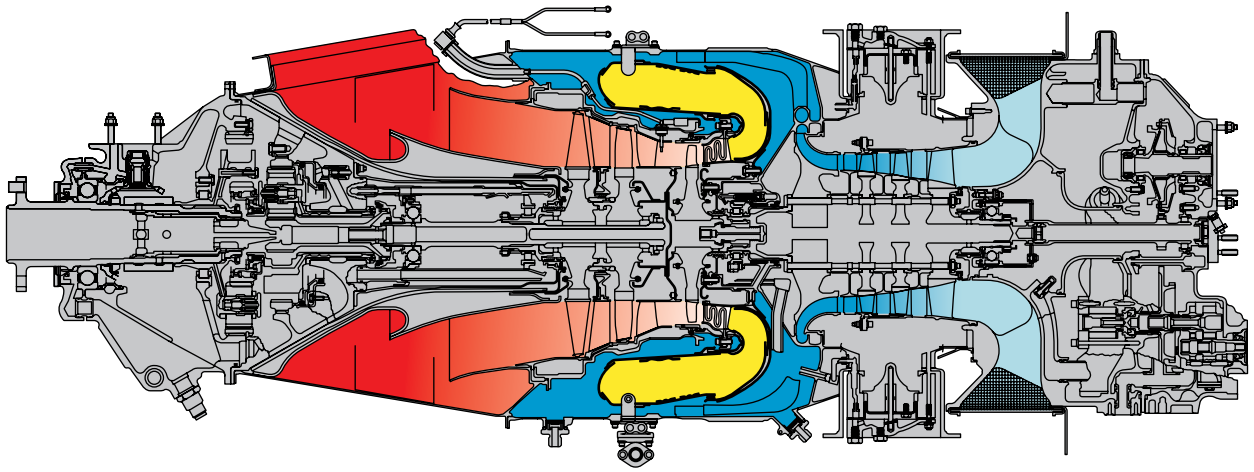


Figure 7-7. Engine Gas Flow

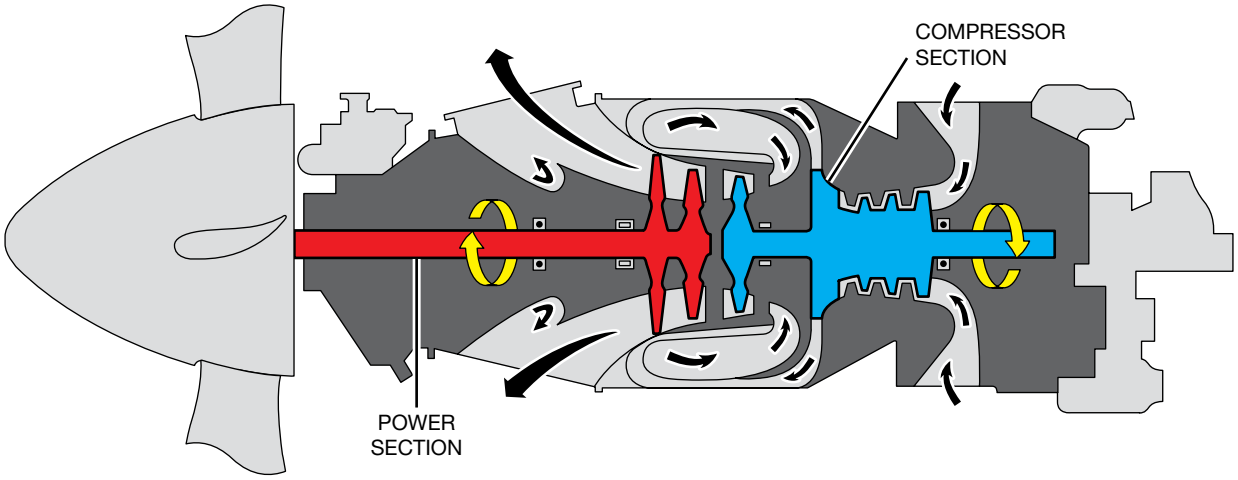


Figure 7-8. Power and Compressor Sections

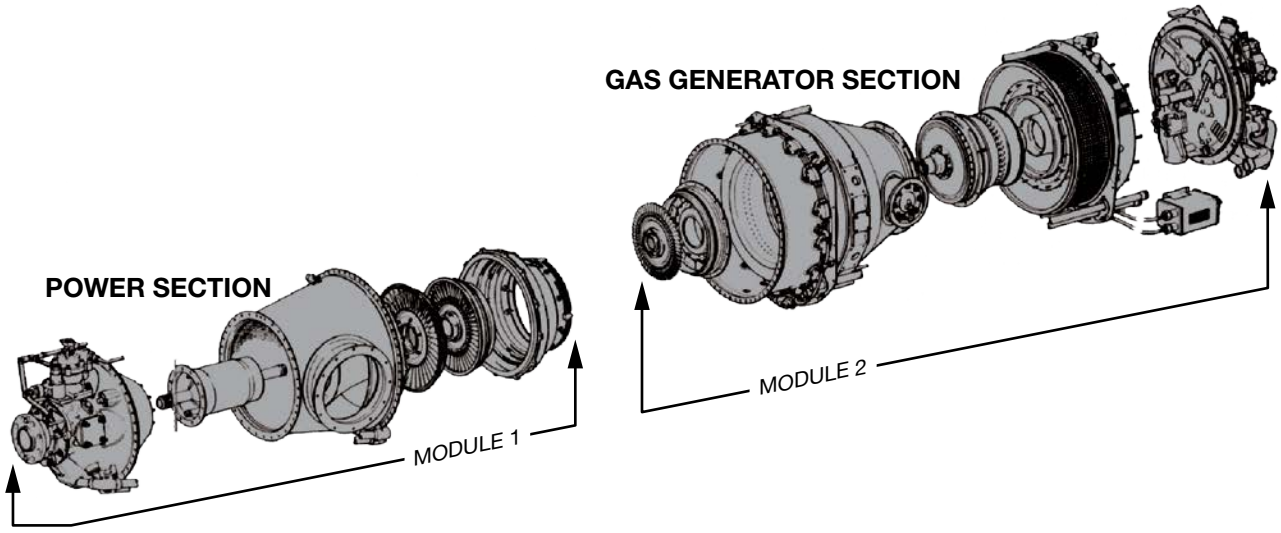


Figure 7-9. Engine Construction

During normal operation, fuel is injected into the combustion chamber liner through 14 simplex nozzles, which are supplied by a dual manifold consisting of primary and secondary transfer tubes and adapters. During starting, the fuel/air mixture is ignited by two spark igniters which protrude into the liner. After starting, the igniters are turned off, since combustion is self-sustaining. The resultant gases expand from the liner, reverse direction in the exit duct zone, and pass through the compressor turbine inlet guide vanes to the single-stage compressor turbine. The guide vanes ensure that the expanding gases impinge on the turbine blades at the correct angle, with minimum loss of energy. The expanding gases are then directed forward to drive the power turbine section.

The single-stage power turbine, consisting of an inlet guide vane and turbine, drives the propeller shaft through a reduction gearbox.

The compressor and power turbines are located in the approximate center of the engine, with their respective shafts extending in opposite directions. This feature simplifies the installation and inspection procedures. The exhaust gas from the power turbine is directed through an annular exhaust plenum to atmosphere through twin opposed exhaust ports provided in the exhaust duct.

ENGINE STATIONS

To identify points in the engine, it is common practice to establish engine station numbers at various points (Figure 7-5). To refer to pressure or temperature at a specific point in the engine airflow path, the appropriate station number is used, such as P_3 for the Station 3 pressure or T_5 for the gas temperature at Station 5. For instance, temperature of the airflow is measured between the compressor turbine and the power turbine at Engine Station Number 5. This is called Inter-stage Turbine Temperature (ITT) or T_5. Bleed air is taken off the engine after the centrifugal compressor stage and prior to entering the combustion chamber. This air, commonly referred to as P_3 air, is used for cabin heat, pressurization, and the pneumatic system.

ENGINE MODULAR CONCEPT

With the modular free-turbine design, the engine is basically divided into two modules: a gas generator section and a power section (Figure 7-10). The gas generator section includes the compressor and the combustion section. Its job is to draw air into the engine, add energy to it in the form of burning fuel, and produce the gases necessary to drive the compressor and power turbines.

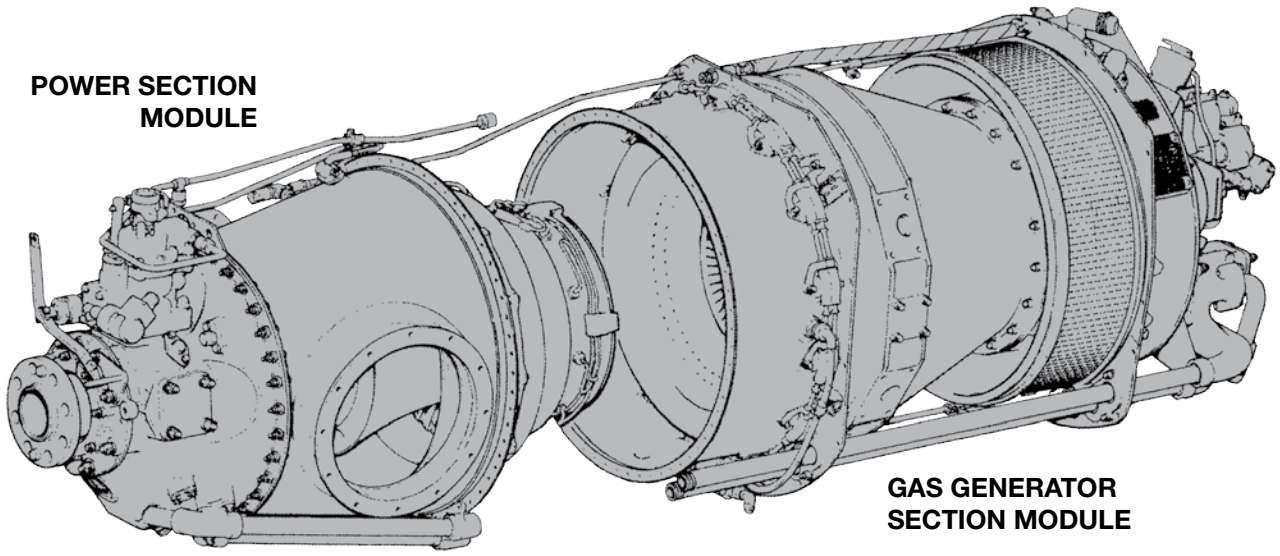


Figure 7-10. Typical Engine Modular Construction

The power section's job is to convert the gas flow from the gas generator section into mechanical action to drive the propeller. This is done through an integral planetary gearbox, which converts the high speed and low torque of the power turbine to the low speed and high torque required at the propeller. The reduction ratio from power turbine shaft rpm to propeller rpm is approximately 15:1.

COMPRESSOR BLEED VALVE

At low N_1 rpm, the axial compressors produce more compressed air than the centrifugal compressor can effectively handle (accept). A compressor bleed valve compensates for this excess airflow at low rpm by opening, to relieve this pressure. As compressor speed increases, the valve closes proportionally until, at 80% N_1, the valve is fully closed (Figure 7-11). This pressure relief helps prevent compressor stall of the centrifugal stage.

The compressor bleed valve is a pneumatic piston which references the pressure differential between the axial and centrifugal stages. Looking forward, the valve is located at the 6 o'clock position. The function of this valve is to prevent compressor stalls and surges in the low N_1 rpm range (75 to 80% N_1).

At low N_1 rpm, the valve is in the open position. At takeoff and cruise N_1 rpm, above approximately 80%, the bleed valve will be closed. If the compressor bleed valve sticks closed, a compressor stall will result. If the valve sticks open, the ITT would be noticably higher as the power lever is advanced above 80% N_1.

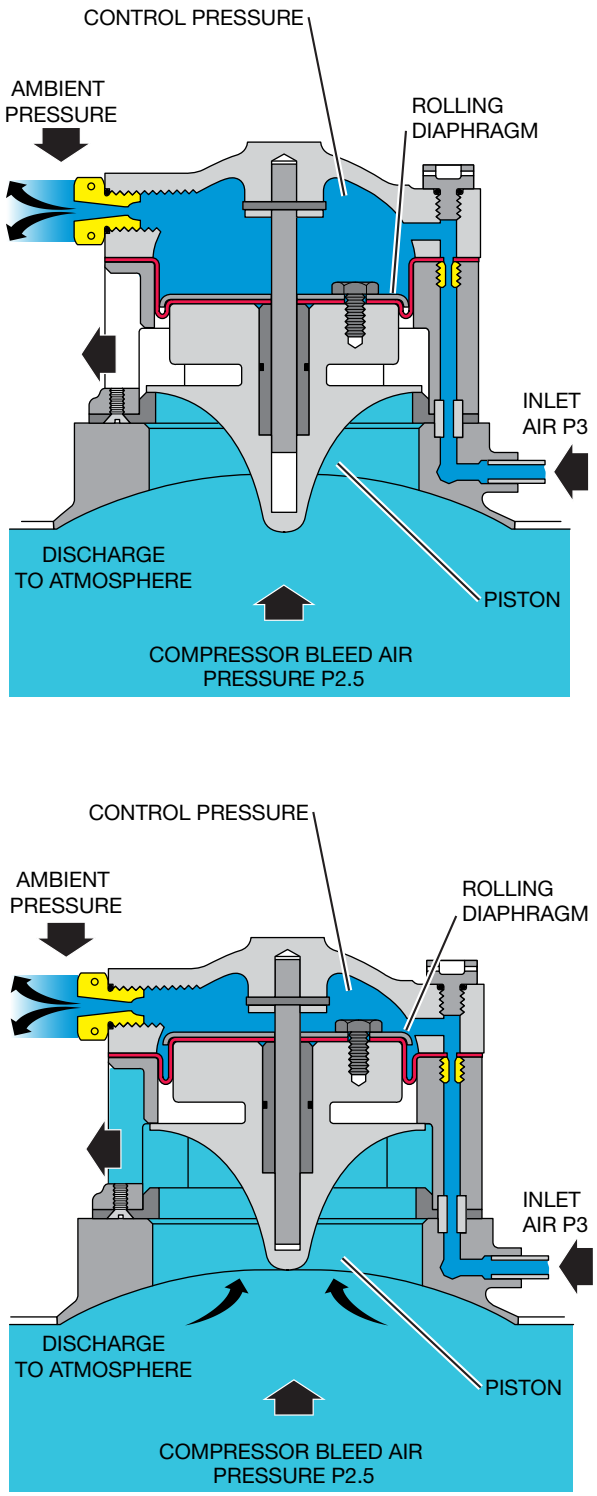


Figure 7-11. Compressor Bleed Valve

IGNITERS

The engine start switches are located on the pilot's left subpanel (Figure 7-12). This subpanel contains the IGNITION AND ENGINE START switches and ENG AUTO IGNITION switches.

The IGNITION AND ENGINE START switches have three positions: ON, OFF, and STARTER ONLY. The ON position is lever-locked and activates both the starter and igniters. The STARTER ONLY position is a momentary hold-down position of the spring-loaded-to-center OFF position. It provides for motoring only to clear the engine of unburned fuel. With the switch in this position, there is no ignition.

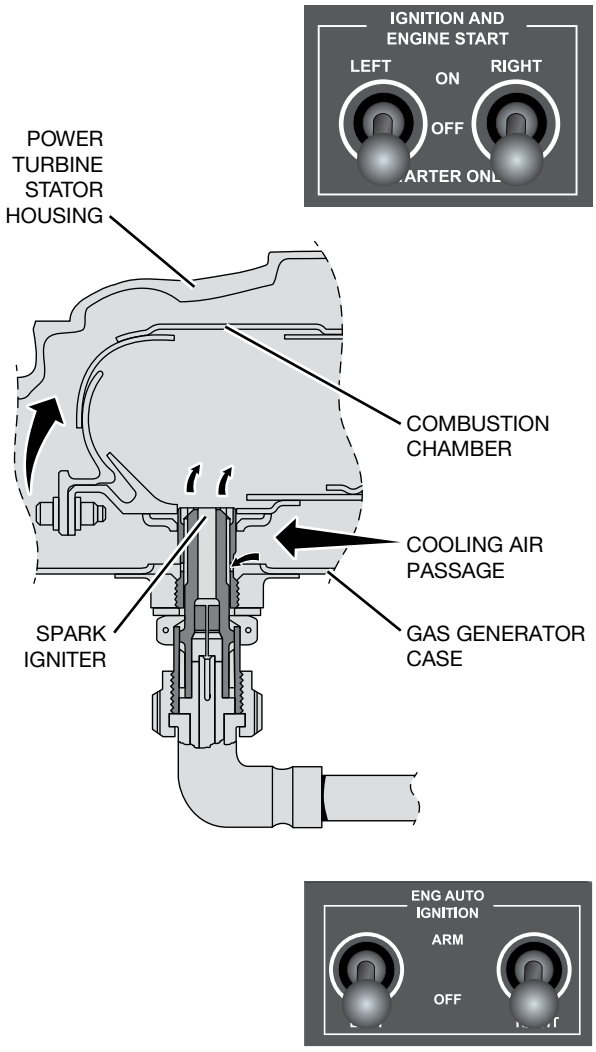


Figure 7-12. Engine Start and Ignition Switches

The combustion chamber has two spark-type igniters to provide positive ignition during engine start. While the engine is equipped with two igniters, it will start with only one. The system is designed so that if one igniter is open or shorted, the remaining igniter will continue to function. Once the engine is started, the igniters are de-energized, since the combustion is self-sustaining.

The ignition system features an automatic backup function for emergencies. This backup system is called "autoignition." The ENG AUTO IGNITION switches should be moved to the ARM position just prior to takeoff. If engine torque falls below approximately 400 ft-lb, the igniter will automatically energize, attempting to restart the engine. The IGNITION ON annunciator will be illuminated.

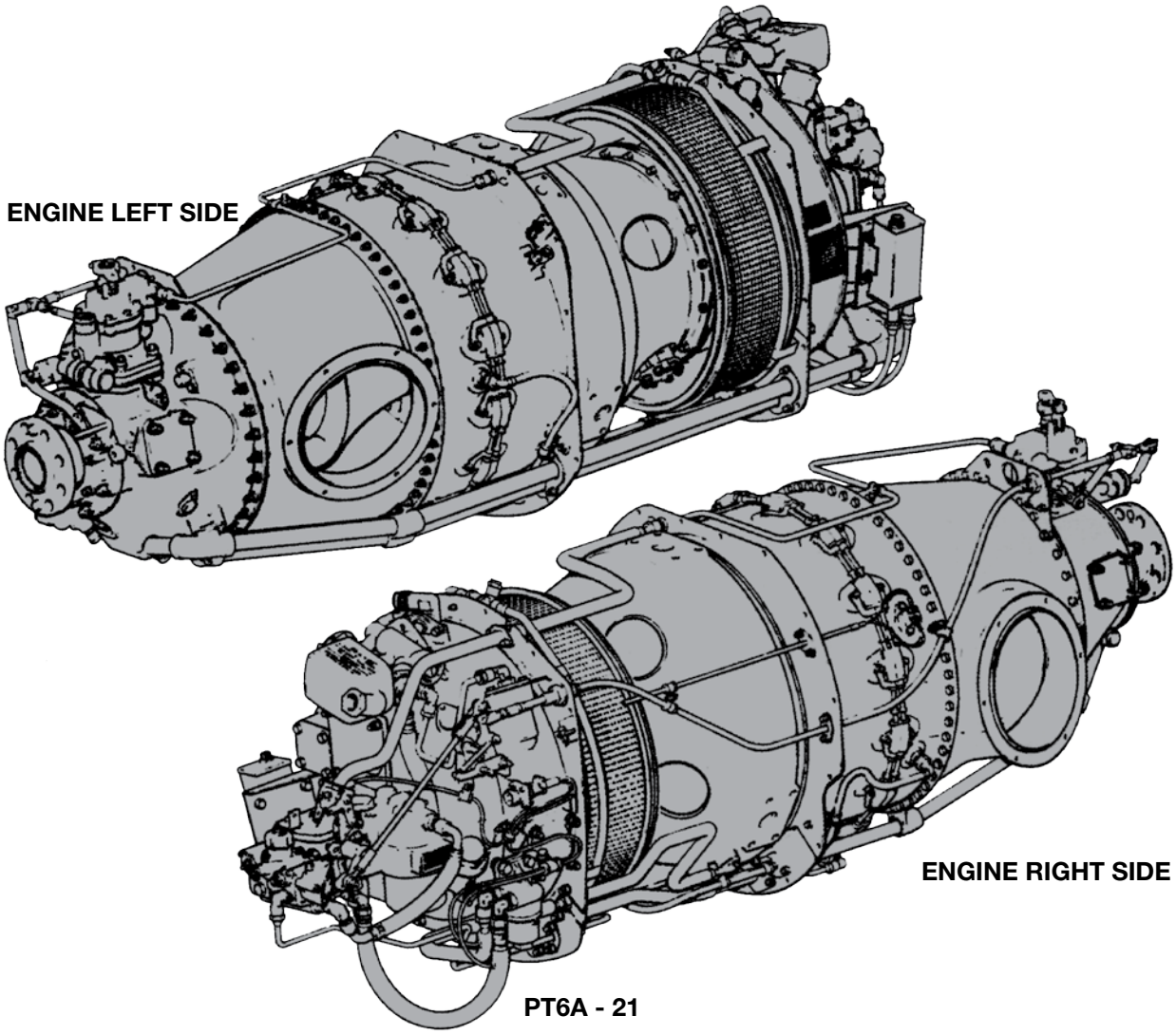
The spark ignition provides the engine with an ignition system capable of quick light-ups over a wide temperature range. The system consists of an airframe-mounted ignition exciter, two individual high-tension cable assemblies, and two spark igniters. It is energized from the aircraft nominal 28-VDC supply and will operate in the 9- to 30-volt range. The igniter control box produces up to 3,500 volts. The ignition exciter is energized only during the engine starting sequence and emergencies to initiate combustion in the combustion chamber.

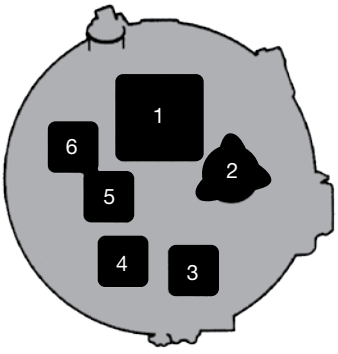
ACCESSORY SECTION

Most of the engine-driven accessories, except the propeller governors and propeller tach generator, are mounted on the accessory gearbox located at the rear of the engine (Figure 7-13). The accessories are driven from the compressor shaft through a coupling shaft.

The lubricating and scavenge oil pumps are mounted inside the accessory gearbox, with the exception of the two scavenge pumps which are externally mounted.

The starter-generator, high-pressure fuel pump, N_1 tachometer generator, and other optional accessories are mounted on pads on the rear of the accessory drive case. There are seven such mounting pads, each with its own different gear ratio.





AS VIEWED FROM REAR

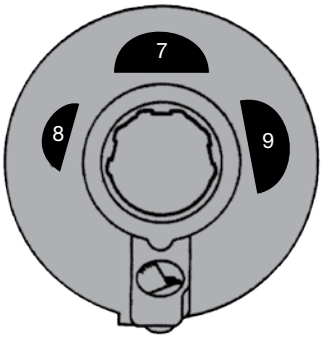
ACCESSORY SECTIONS

REAR ACCESSORY DRIVES

1. STARTER-GENERATOR
2. FUEL PUMP/FCU
3. TACHOMETER-GENERATOR (N_G)
4. VACUUM AIR PUMP (OPTIONAL)
5. OPTIONAL ACCESSORY DRIVE
6. OPTIONAL ACCESSORY DRIVE

FRONT ACCESSORY DRIVES

7. PROPELLER GOVERNOR
8. TACHOMETER-GENERATOR (N_F)
9. PROPELLER OVERSPEED GOVERNOR



AS VIEWED FROM FRONT

Figure 7-13. Typical PT6A Engine

LUBRICATION SYSTEM

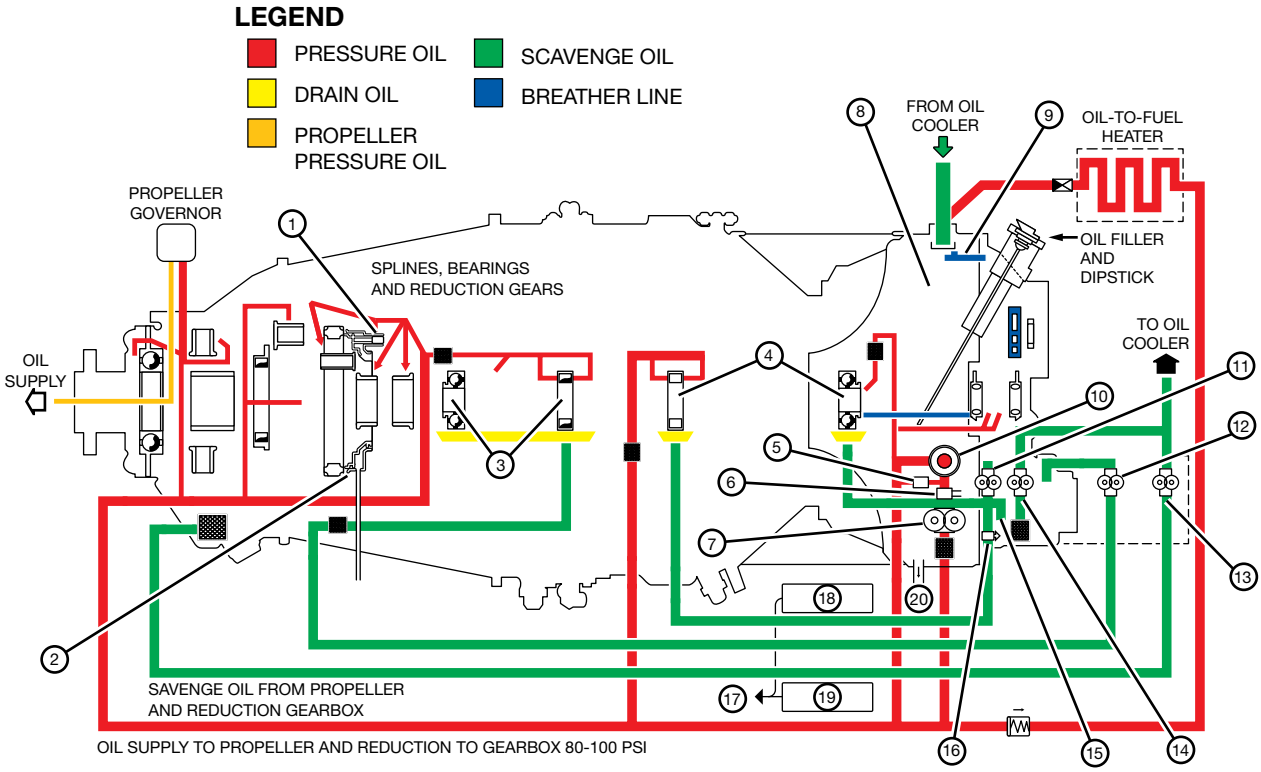
The PT6A engine lubrication system has a dual function (Figure 7-14). Its primary function is to cool and lubricate the engine bearings and bushings. Its second function is to provide oil to the propeller governor and propeller reversing control system.

The main oil tank houses a gear-type engine-driven pressure pump, oil pressure regulator, and oil filter. The engine oil tank is an integral part of the compressor inlet case and is located in front of the accessory gearbox.

The oil tank is provided with a filler neck and integral quantity dipstick housing. The cap and dipstick are secured to the filler neck, which passes through the gearbox housing and accessory diaphragm and into the tank. The markings on the dipstick indicate the number of U.S. quarts of oil less than full (Figure 7-15).

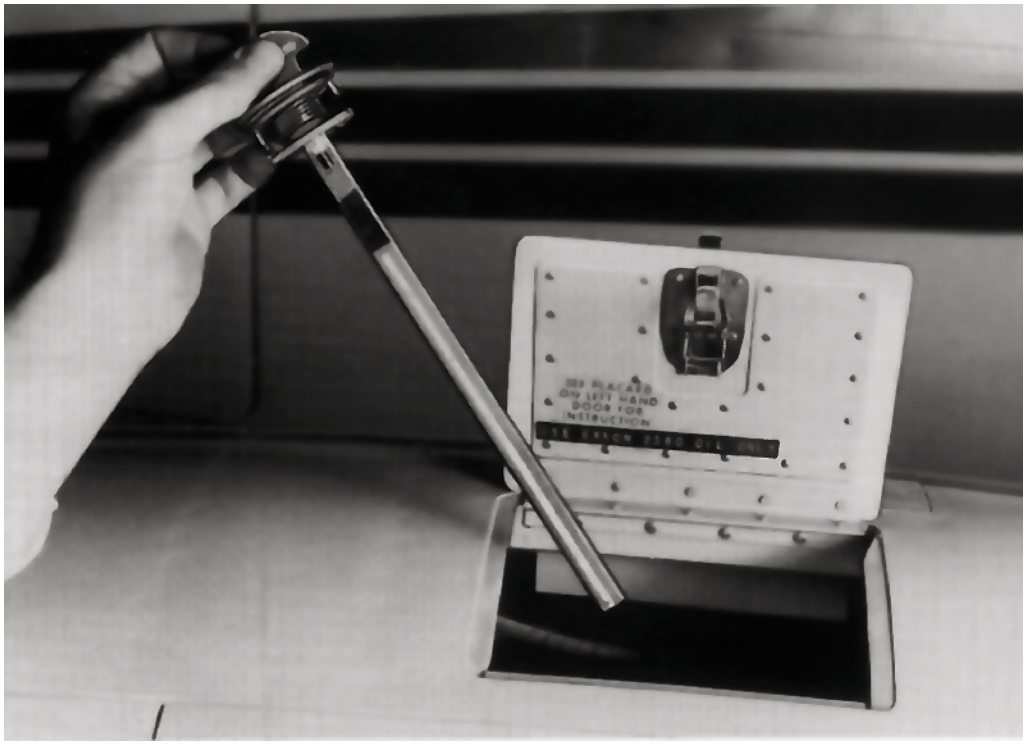
The engine oil system has a total capacity of 3.5 U.S. gallons, including the 2.3-gallon oil tank. Maximum oil consumption is one quart every 10 hours of operation. Normal oil consumption may be as little as 1 quart per 50 hours of operation.

The dipstick will indicate 1 to 2 1/2 quarts below full when the oil level is normal. Do not overfill.



1. TORQUEMETER OIL CONTROL VALVE
2. TORQUEMETER
3. POWER TURBINE BEARINGS
4. COMPRESSOR BEARINGS
5. FILTER BYPASS VALVE
6. RELEIF VALVE RETURN TO OIL TANK
7. OIL PRESSURE PUMP
8. INTERNAL OIL TANK CAPACITY 2.3 GALLONS
9. OIL TANK BREATHER
10. OIL FILTER AND CHECK VALVE ASSEBMBLY
11. NO. 2 BEARING SCAVENGE PUMP
12. POWER TURBINE BEARINGS SCAVENGE PUMP
13. REDUCTION CASE SCAVENGE PUMP
14. ACCESSORY CASE SCAVEGE PUMP
15. GRAVITY DRAIN TO ACCESSORY GEARBOX
16. BYBASS VALVE
17. TO COCKPIT INSTRUMENTS
18. PRESSURE TRANSMITTER
19. TEMPERATURE BULB
20. OIL TANK DRAIN

Figure 7-14. Engine Lubrications Diagram



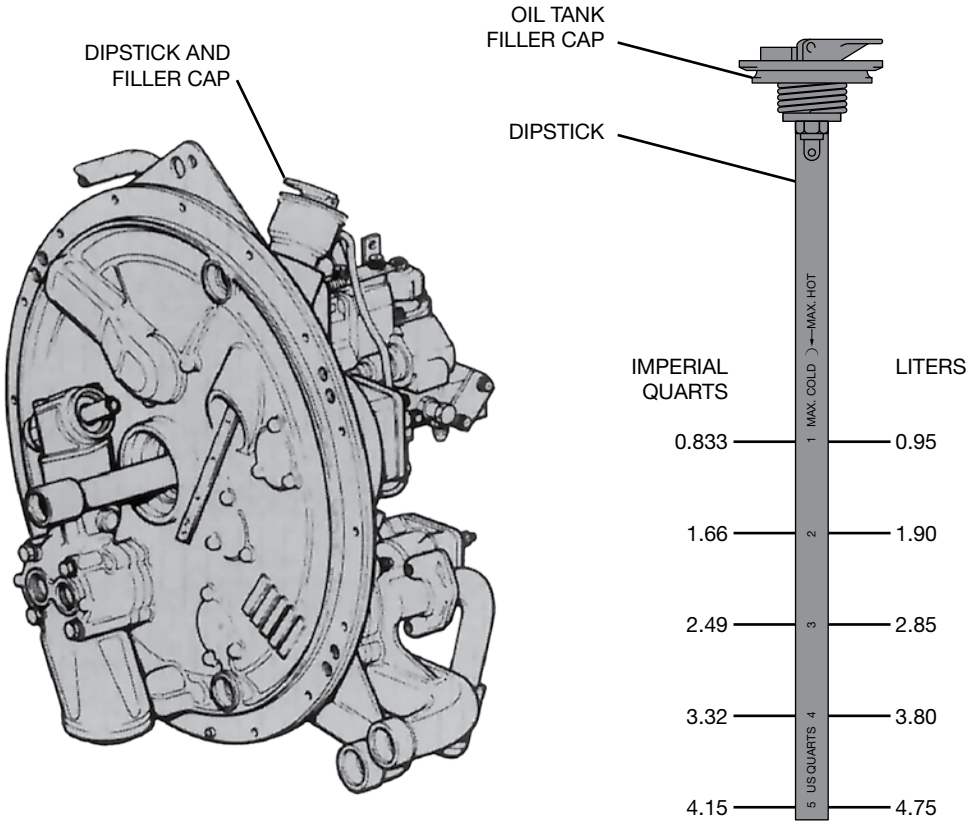


Figure 7-15. Engine Oil Dipstick

When adding oil between oil changes, do not mix types or brands of oil due to the possibility of chemical incompatibility and loss of lubricating qualities.

A placard inside the engine cover shows the brand and type of oil used in that particular engine. Although the preflight checklist calls for checking the oil level, which is required, the best time to check oil quantity is shortly after shutdown, since oil levels are most accurately indicated at that time.

Oil level checks during preflight may require motoring the engine for a brief time for an accurate level reading. Each engine tends to seek its own oil level. The pilot should monitor the oil level to ensure proper operation.

As pressure oil leaves the tank, it passes through the pressure and temperature-sensing bulbs mounted on or near the rear accessory case. The oil then proceeds to the various bearing compartments and nose case through an external oil transfer line below the engine. Scavenge oil returns from the nose case and the bearing compartments to the gear-type oil scavenge pumps in the accessory case through external oil transfer lines, and through the external oil cooler below the engine.

The oil cooler is thermostatically controlled to maintain the desired oil temperature. Another externally mounted unit, the oil-fuel heat exchanger, uses hot engine oil to heat fuel before it enters the engine fuel system. When gas generator speeds are above 72% N_1, and oil temperatures are between 60 and 70°C, normal oil pressure is between 80 and 100 psi.

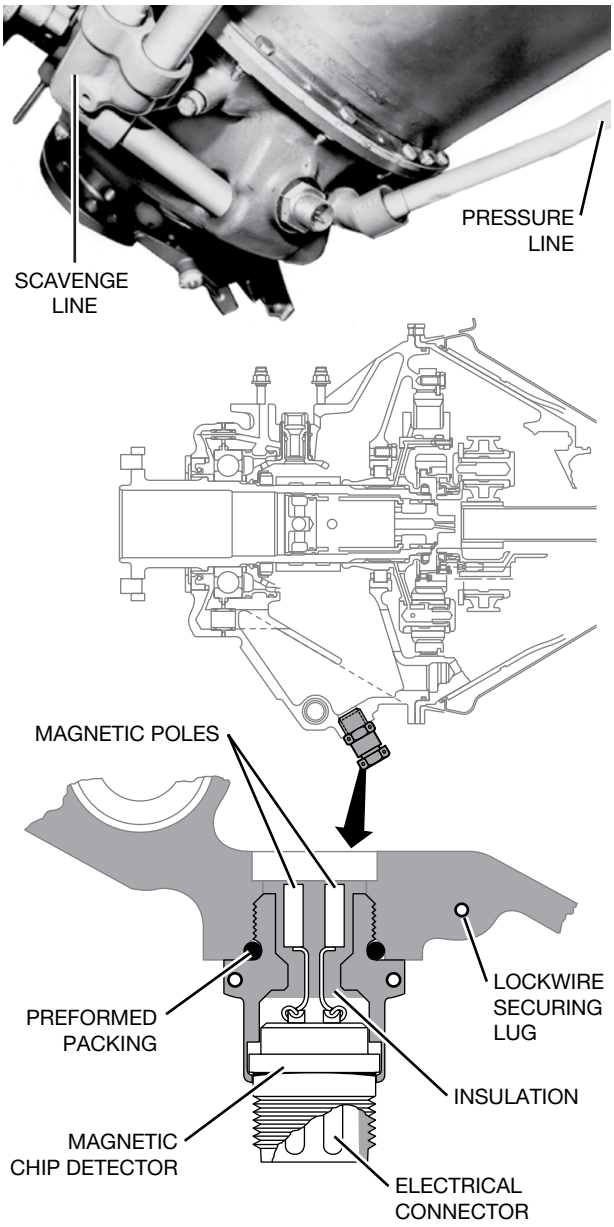


Figure 7-16. Magnetic Chip Detector

Magnetic Chip Detector

A magnetic chip detector is installed in the bottom of each engine nose gearbox (Figure 7-16). This detector will activate a yellow light on the annunciator panel, L CHIP DETECT or R CHIP DETECT, to alert the pilot of oil contamination.

C90B aircraft, engine parameters should be monitored for abnormal indications. If such indications are observed, appropriate check list action should be taken.

C90A aircraft are equipped with red "CHIP DETECT" annunciator panel lights. A steady "CHIP DETECT" light requires the engine be shut down to prevent serious internal damage.

When a CHIP DETECT annunciator light comes on and stays on, timely action is required to prevent serious damage to the internal engine components. The chip detector indicates the presence of ferrous particles in the propeller gearbox.

ENGINE FUEL SYSTEM

The fuel control system for PT6A engines is essentially a fuel governor that increases or decreases fuel flow to the engine to maintain selected engine operating speeds. At first glance, the system may appear quite complicated. The engine fuel control system consists of the main components shown in the block diagram (Figure 7-17). They are the electric low-pressure boost pump, oil-to-fuel heat exchanger, high-pressure fuel pump, fuel control unit, fuel cutoff valve, flow divider, and dual fuel manifold with 14 simplex nozzles.

The PT6A-21 engine uses an electric low-pressure boost pump to supply a 30-psi head pressure to the high-pressure engine-driven fuel pump. This head pressure prevents fuel cavitation at the high-pressure pump. The fuel is also used for cooling and lubricating the pump. The oil-to-fuel heat exchanger uses warm engine oil to maintain a desired fuel temperature at the fuel pump inlet to prevent icing at the pump filter. This is done with automatic temperature sensors and requires no action by the pilot.

Fuel enters the engine fuel system through the oil-to-fuel heat exchanger, and then flows into the high-pressure engine-driven fuel pump and on into the fuel control unit (FCU).

The high-pressure fuel pump is an engine-driven gear-type pump with an inlet and outlet filter. Flow rates and pressures will vary with gas generator (N_1) rpm. Its primary purpose is to provide sufficient pressure at the fuel nozzles for a proper spray pattern during all modes of engine operation. The high-pressure pump supplies fuel at approximately 800 psi to the fuel side of the FCU.

Two valves included in the FCU ensure consistent and cool engine starts. When the ignition or start system is energized, the purge valve is electrically opened to clear the FCU of vapors and bubbles.

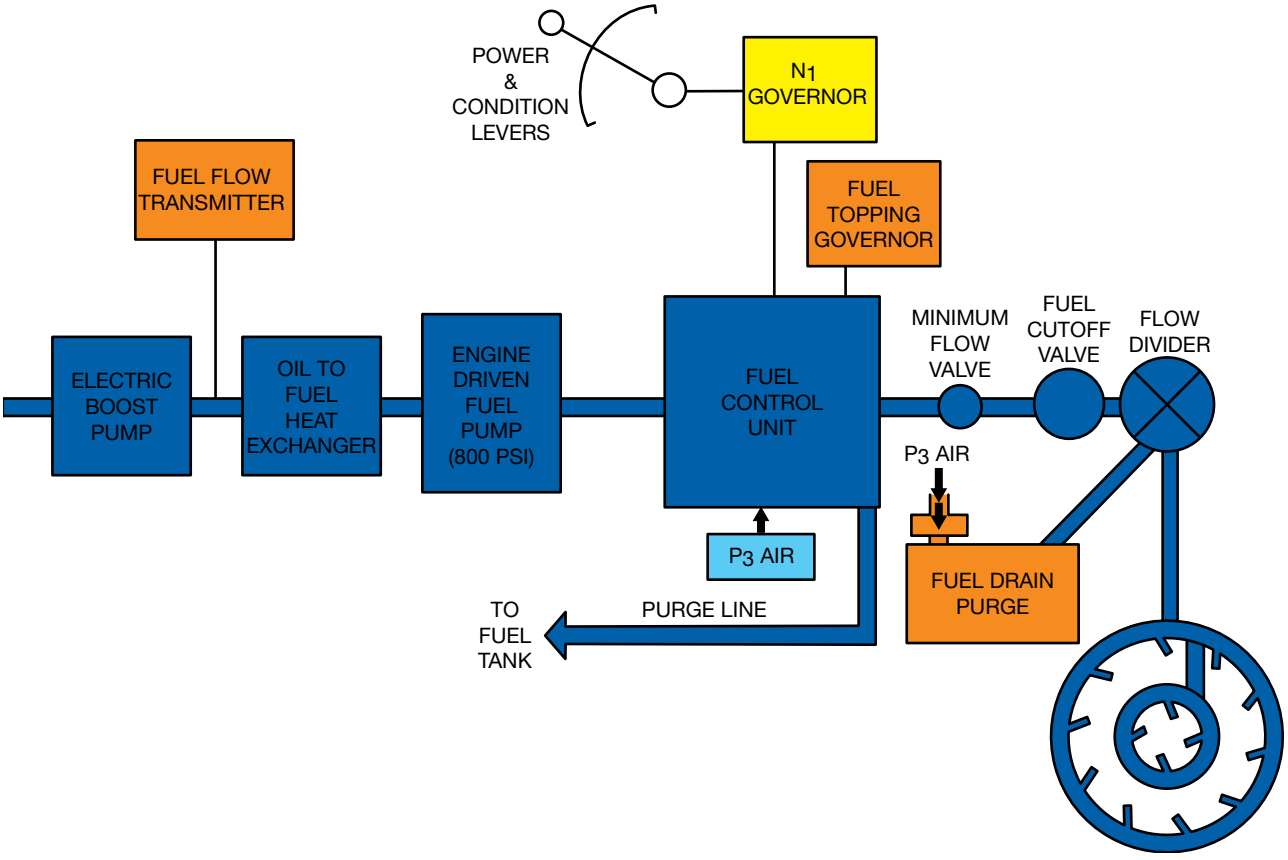


Figure 7-17. Simplified Fuel System Diagram

The excess fuel flows back to the nacelle fuel tanks. The spill valve, referenced to atmospheric pressure, adjusts the fuel flow for cooler high-altitude starts.

Between the FCU fuel valve and the engine combustion chamber, the minimum pressurizing valve in the FCU remains closed during starting until fuel pressure builds sufficiently to maintain a proper spray pattern in the combustion chamber. About 80 psi is required to open the minimum pressurizing valve. If the high pressure fuel pump should fail, the valve would close, and the engine would flame out.

The fuel cutoff valve is located downstream from the minimum pressurizing valve in the FCU. This valve is controlled by the condition lever, either open or closed. There is no intermediate position of this valve. For starting, fuel flows initially through the flow divider to the 10 primary fuel nozzles in the combustion chamber. As the engine accelerates through approximately 40% N_1, fuel pressure is sufficient to open the flow divider to the 4 secondary fuel nozzles. At this time all 14 nozzles are delivering atomized fuel to the combustion chamber. This progressive sequence of primary and secondary fuel nozzle operation provides cooler starts. During engine starting, there is a noticable increase in ITT when the secondary fuel nozzles are activated.

During engine shutdown, any fuel left in the manifold is forced out through the fuel nozzles and into the combustion chamber by purge tank pressure. As the fuel is burned, a momentary increase in N_1 rpm may be observed. The entire operation is automatic and requires no input from the crew.

Fuel Control Unit

The fuel control unit (Figure 7-18), which is referred to as the FCU, has multiple functions, but its primary purpose is to meter proper fuel amounts to the fuel nozzles in all modes of engine operation.

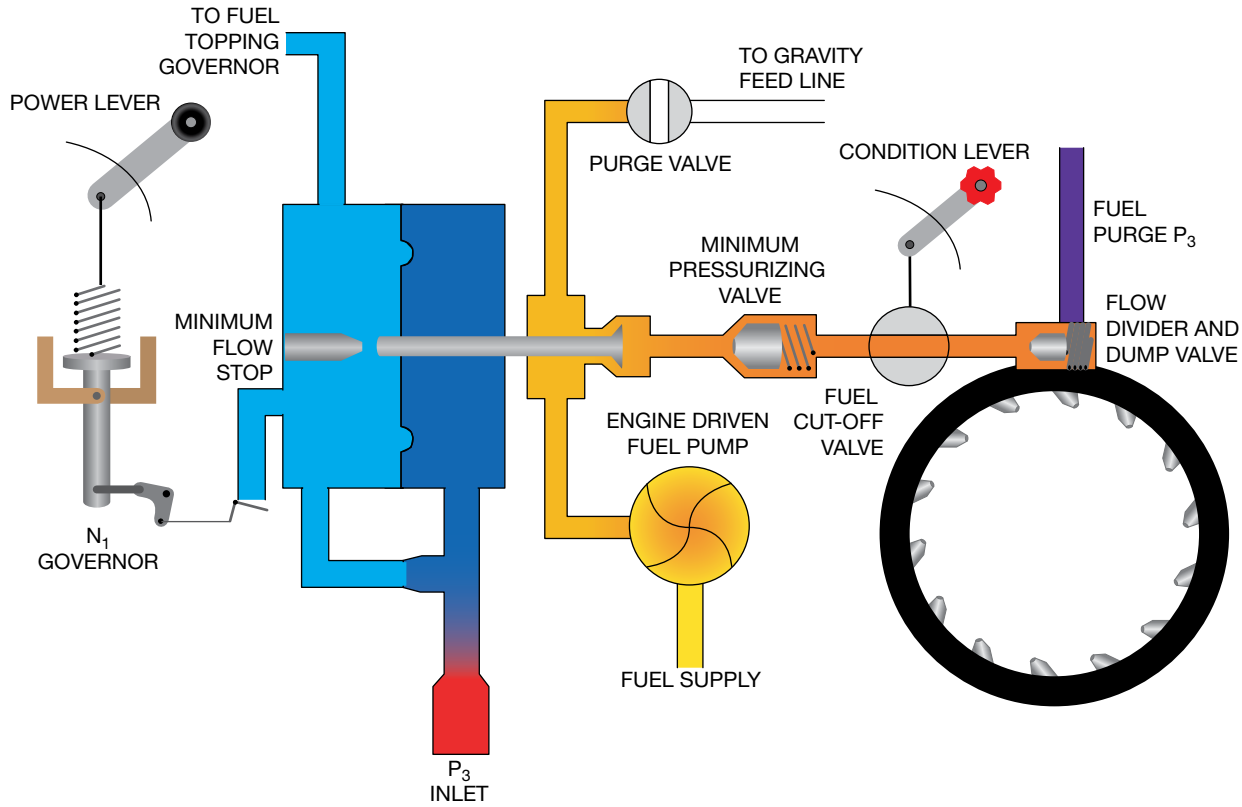


Figure 7-18. Simplified Fuel Control System

FCU operation will be simplified and described briefly here. For detailed description and operation, refer to the *Pratt & Whitney Maintenance Manual* which applies to this engine.

The condition lever selects idle speeds between LOW IDLE (51% to 58% N_1) to HIGH IDLE (70% N_1), while the power lever selects speeds between idle and maximum, 101.5% N_1. These control levers influence the N_1 governor and control N_1 speed. The governor uses pneumatic air (P_3) pressure to control engine speed. The governor controls the air pressure in the fuel control unit by varying the P_3 leak rate.

The P_3 air chamber and fuel chamber are separated by a diaphragm, which has a needle valve mounted on it which is called the metering valve. As the diaphragm is influenced by varying air/fuel pressures, the metering valve is repositioned to achieve the desired fuel flow. The N_1 governor controls fuel flow by allowing some P_3 pressure to be leaked off at varying rates, depending on the desired fuel flow.

In an underspeed condition, the N_1 governor acts to increase P_3 air pressure. This repositions the metering valve, allowing more fuel to enter the combustion chamber, increasing N_1.

In an overspeed condition, the N_1 governor allows the P_3 pressure to be reduced in the FCU, which repositions the metering valve reducing the fuel flow into the combustion chamber, decreasing N_1.

Should the P_3 air pressure be lost, due to a malfunction, the metering valve will be positioned to the minimum flow stop. Minimum flow power would be approximately 48% N_1. The power lever and condition lever would then have no effect on engine speed.

Fuel Pressure Indicators

In the event of an electric boost pump failure, the respective FUEL PRESS annunciator (Figure 7-19) will illuminate and the master warning light will flash. The FUEL PRESS light illuminates when outlet pressure at the boost pump decreases below about 10 psi. If the crossfeed switch is in the AUTO position, the automatic crossfeed feature will open the valve extinguishing the annunciator.

Figure 7-19. Fuel Pressure Annunciators

In the event of an engine-driven fuel pump (highpressure) failure, the engine will flame out.

CAUTION

Engine operation with the FUEL PRESS light on is limited to ten hours between overhaul or replacement of the engine-driven high-pressure fuel pump.

Fuel Flow Indicators

Fuel flow information is sensed by a transmitter in the engine fuel supply line, between the boost pump and the engine-driven high-pressure pump, and indicated on the fuel flow gage on the instrument panel (Figure 7-20). The gage indicates fuel flow in pounds-per-hour units times 100. Therefore when the needle indicates 2 on the dial, fuel flow is 200 pounds per hour. The fuel flow gages are DC-powered.

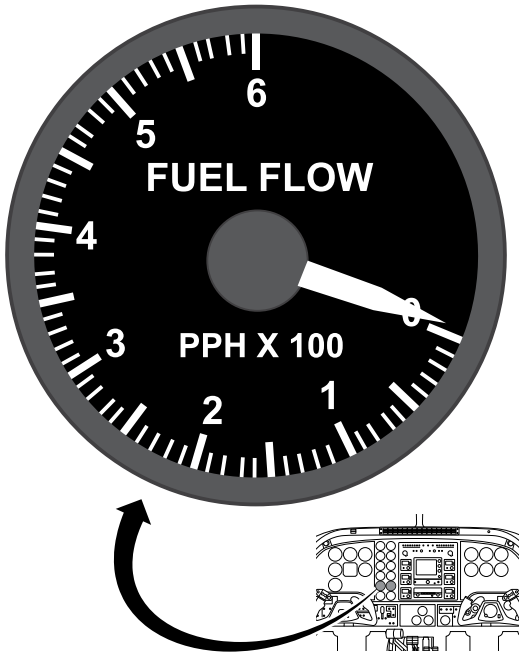


Figure 7-20. Fuel Flow Indicator

Anti-icing Fuel Additive

Engine oil is used to heat the fuel prior to entering the FCU. Since no temperature measurement is available for the fuel at this point, it must be assumed to be the same as the Outside Air Temperature. The Minimum Oil Temperature chart is supplied for use as a guide in preflight planning, based on known or forecast operating conditions, to indicate operating temperatures where icing at the FCU could occur. If the plot should indicate that oil temperature versus OAT is such that ice formation could occur during takeoff or in flight, anti-icing additive per MIL-1-27686 should be mixed with the fuel at refueling to ensure safe operation. Refer to the *King Air Maintenance Manual* for procedures to follow when blending anti-icing additive with the airplane fuel.

Anti-icing additive conforming to Specification MIL-1-27686 is the only approved fuel additive.

ENGINE POWER CONTROL

The propeller lever adjusts the propeller governor to the desired propeller speed (Figure 7-21). The propeller will maintain the set speed by varying the blade angle. Torque is controlled by the power lever acting on the N_1 governor. When the power lever is advanced, the N_1 governor causes the FCU to increase fuel flow, resulting in an increase in engine speed.

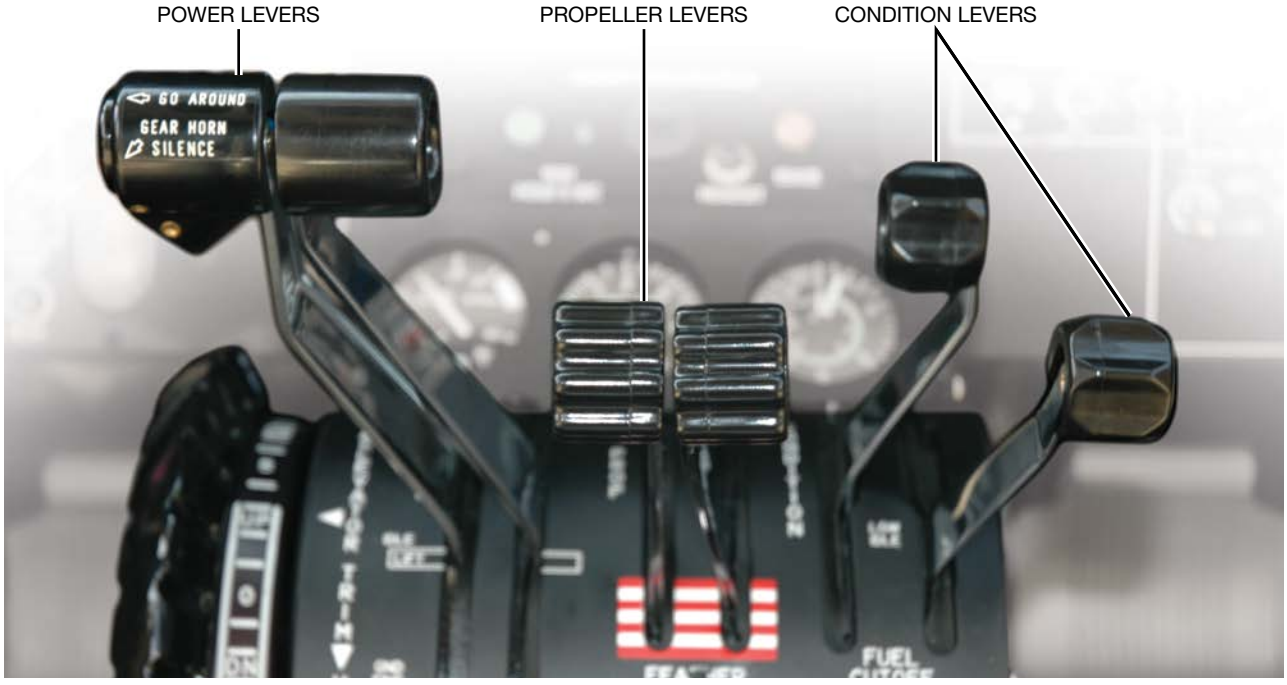


Figure 7-21. Control Levers

ITT AND TORQUEMETERS

Power management is relatively simple, with two primary operating limitations. The engines are temperature and torque limited. During operation requiring maximum engine performance, engine torque and ITT operating parameters are affected by ambient temperature and altitude: at cold temperature or low altitude, torque limits power; at hot temperature or high altitude, ITT limits power. Whichever limit is reached first, determines the power available.

ITT GAGE

The ITT gage monitors the interstage turbine temperature at station 5 (Figure 7-22). ITT is a prime limiting indicator of the amount of power available from the engine under varying ambient temperature and altitude conditions. The normal operating range, indicated by the green arc on the gage, is 400 to 695°C. These limits also apply to maximum continuous power. The maximum starting temperature of 1,090°C is indicated by the dashed red line on the instrument, or a red diamond on LJ-1361, 1363 and after. This starting limit of 1,090°C is limited to two seconds. The ITT gages are self-energizing and do not require electrical power (LJ-1361, LJ-1363, and after are DC-powered). The engines will be damaged if limiting temperatures indicated on the ITT gage are exceeded.

TORQUEMETER

The torquemeter, which is marked in ft-lb, constantly measures rotational force applied to the propeller shaft (Figure 7-22). The maximum permissible sustained torque is 1,315 ft-lb, the red radial at the top of the green arc on the instrument. A transient torque limit of 1,500 ft-lb is time-limited to two seconds. Cruise torques vary with altitude and temperature.

Torque is measured by a hydromechanical torquemeter in the first stage of the reduction gearcase. Rotational force on the first-stage ring gear allows oil pressure to change in the torquemeter chamber. The difference between the torquemeter chamber pressure and reduction gear internal pressure accurately indicates the torque being produced at the propeller shaft. The torque transmitter measures this torque and sends an AC signal to the instrument on the instrument panel (DC signal on LJ-1361, LJ1363, and after).

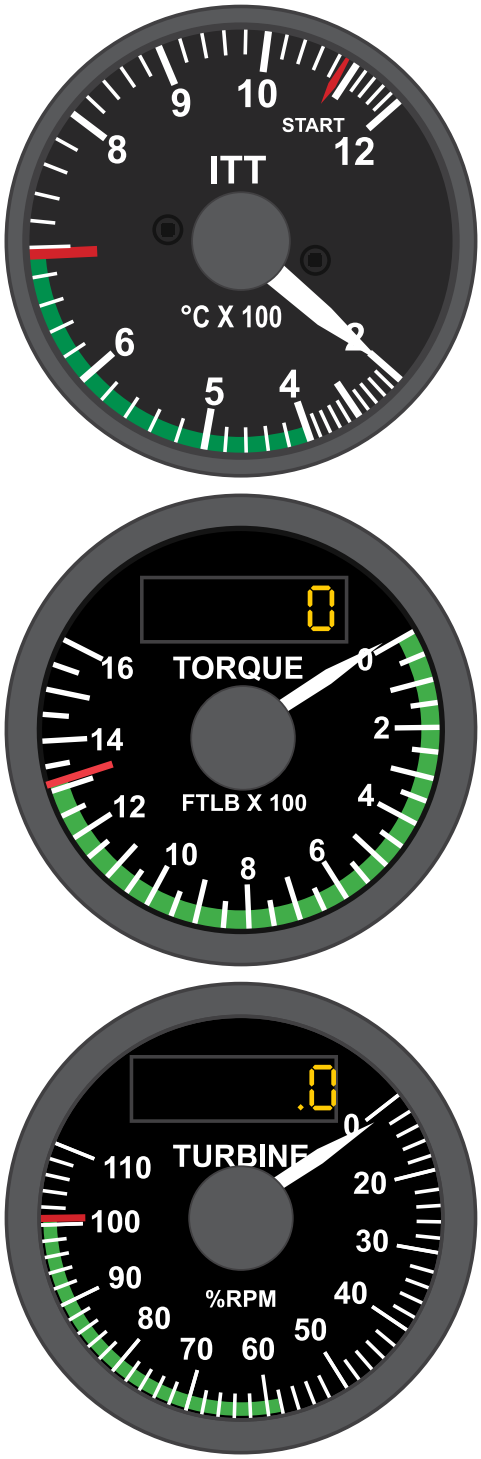


Figure 7-22. Engine Instrument Markings

GAS GENERATOR TACHOMETER (N_1)

The N_1 gas generator tachometer measures the rotational speed of the compressor shaft, in percent of rpm, based on 37,500 rpm at 100% (Figure 7-22). The face of this instrument consists of two dials: a smaller dial labeled from 0 to 9, and a larger dial labeled from 0 to 100. The smaller dial is calibrated in 1% increments, and the larger dial in 10% increments. Between 30 and 100% on the larger dial, the increments are in gradations of 2%.

The N_1 indicator is self-generating (LJ-1361, LJ1363, and after are DC-powered). The tachometer generator sensing unit, located in the engine accessory section, is geared down to supply N_1 speed information to the instrument panel to indicate the percent of N_1 revolutions.

Maximum continuous gas generator speed is limited to 38,100 rpm, which is 101.5% on the N_1 indicator. A transient speed up to 102.6%, 38,500 rpm, is time-limited to 2 seconds, to provide a buffer for surges during engine acceleration.

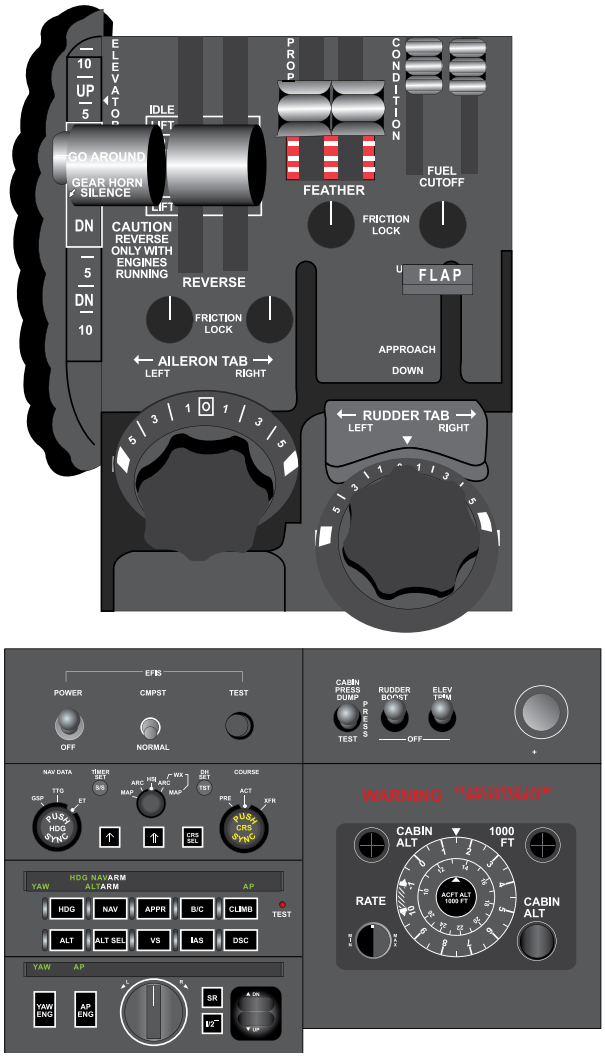


Figure 7-23. Control Pedestal

CONTROL PEDESTAL

The control pedestal extends between pilot and copilot (Figure 7-23). The three sets of control levers are left to right: the power levers, propeller levers, and the condition levers.

Power Levers

The power levers (Figure 7-21) control engine power, from idle to maximum power, by operation of the N_1 governor in the fuel control unit. Increasing N_1 rpm results in increased engine power. The power levers have three control ranges: flight, Beta, and reverse. The bottom of the flight range is at IDLE. When the levers are lifted over the IDLE detent and pulled back, they control engine power through the Beta and reverse ranges. A selectable ground fine (or zero thrust) power lever gate position is provided on the C90B.

Condition Levers

The condition levers have multiple positions: FUEL CUTOFF and LO IDLE through HI IDLE (Figure 7-21). At the FUEL CUTOFF position, fuel flow to its respective engine is cut off.

At LO IDLE, engine gas generator speed (N_1) is a minimum of 51% on the C90A or 58% on the C90B; at HI IDLE it is 70%. The levers can be set anywhere between LOW IDLE and HIGH IDLE

Propeller Levers

The propeller levers are conventional in setting the propeller rpm for takeoff, climb and cruise (Figure 7-21). The normal governing range is 1,800 to 2,200 rpm. This airplane is equipped

with both manual and automatic propeller feathering systems. To feather a propeller manually, pull the propeller lever back past the friction detent into the red and white striped section of the quadrant. To unfeather, push the lever forward of the detent into the governing range. The propellers go to feathered position when the engines shut down because of the loss of oil pressure in the propeller dome.

Control Lever Operation

The engines are controlled from the cockpit by using the propeller, power, and condition levers. Both the power and condition levers are connected to the N_1 governing section of the FCU. Either lever will reset the FCU to maintain a new N_1 rpm. For starting, the power levers are at the IDLE position, and the condition levers are moved to the LO IDLE position to open the fuel cutoff valves and set the governor at LO IDLE. The condition levers are continuously variable from LO IDLE to HI IDLE. This variable operating speed with power levers at IDLE enhances engine cooling by maintaining a steady airflow through the engines. With the condition levers at LO IDLE, the power levers will select N_1 rpm from LOW IDLE to 101.5%, the maximum for takeoff. However, if the condition levers are at HI IDLE, the power levers can select N_1 rpm only from 70 to 101.5%.

Moving the power or condition levers most directly affects N_1 rpm. As the power or condition levers are advanced, ITT, torque, and fuel flow increases. These indicators are by-products of the N_1 speed maintained by the FCU. With the power levers in a fixed position, N_1 remains constant even in a climb or descent. However, ITT, torque, and fuel flow will vary with altitude, ambient air temperature, and propeller setting.

ENGINE LIMITATIONS

Airplane and engine limits are described in the "Limitations" section of the *POH* (Table 7-2). These limitations have been approved by the Federal Aviation Administration, and must be observed in the operation of the Beechcraft King Air C90A and C90B. The Engine Operating Limits chart gives the major operating limits. The Power Plant Instrument Markings chart lists the minimum, normal, and maximum limits.

During engine start, temperature is the most critical limit. The ITT starting limit of 1,090°C, represented on the ITT gage by a dashed red line, is limited to two seconds. During any start, if the indicator needle approaches the limit, the start should be aborted before the needle passes the dashed red line. For this reason, it is helpful during starts to keep the condition lever out of the LO IDLE detent so that the lever can be quickly pulled back to FUEL CUTOFF.

Monitor oil pressure and oil temperature. During the start, oil pressure should come up to the minimum red line at 40 psi quickly, but should not exceed the maximum at 100 psi. During normal operation the oil temperature and pressure gages should be in the green arc normal operating range. The green arc extends from 80 to 100 psi.

Oil pressure between 40 and 80 psi is undesirable; it should be tolerated only for completion of the flight, and then only at a reduced power setting.

Oil pressure below 40 psi is unsafe; it requires that either the engine be shut down or that a landing be made as soon as possible, using minimum power required to sustain flight.

For increased service life of engine oil, an oil temperature between 74 and 80°C is recommended. A minimum oil temperature of 55°C is recommended for oil-to-fuel heater operation at takeoff power. Oil temperature limits are -40 and +99°C. During extremely cold starts, oil pressure may reach 200 psi. Refer to the Engine Limits chart in the *POH* for minimum oil temperature operation limitations.

During ground operations, ITT temperatures are critical. With the condition levers at LO IDLE, high ITT can be corrected by reducing the DC generator and other N_1 loads, then increasing the N_1 rpm by advancing the condition levers to HI IDLE. The air conditioner, for example, draws a heavy load on both engines, and may have to be temporarily turned off. At approximately 70% N_1 rpm, the HI IDLE condition lever position will normally reduce the ITT. At any N_1 below 70%, there is an idle ITT restriction of 660°C maximum. If an ITT above 660°C is observed when running N_1 below 70%, the generator load

Table 7-2. ENGINE LIMITS CHART

BEECHCRAFT **Section II**

King Air C90A **Limitations**

POWER PLANT LIMITATIONS

NUMBER OF ENGINES 2

ENGINE MANUFACTURER ..... Pratt & Whitney Aircraft of Canada Ltd.

ENGINE MODEL PT6A-21

C90 ENGINE OPERATING LIMITS

The following limitations shall be observed. Each column presents limitations. The limits presented do not necessarily occur simultaneously. Refer to Pratt & Whitney Engine Maintenance Manual for specific actions required if limits are exceeded.

OPERATING CONDITION	SHP	TORQUE FT-LBS (1)	MAXIMUM OBSERVED ITT°C	GAS GENERATOR RPM N_1		PROP RPM N_2	OIL PRESS. PSI (2)	OIL TEMP °C (3)
				RPM	%			
STARTING	---	---	1090 (4)	---	---	---	---	-40 (min)
LOW IDLE	---	---	660 (5)	---	51 (min)	---	40 (min)	-40 to 99
HIGH IDLE	---	---	---	---	70 (approx)	---	---	0 to 99
TAKEOFF AND MAX CONT	550	1315	695	38,100	101.5	2200	80 to 100	10 to 99
CRUISE CLIMB AND MAX CRUISE	538	1315 (6)	695	38,100	101.5	2200	80 to 100	0 to 99
MAX REVERSE (7)	---	---	695	---	88	2100	80 to 100	0 to 99
TRANSIENT	---	1500 (4)	825 (4) (8)	38,500	102.6	2420	---	0 to 99

C90B ENGINE OPERATING LIMITS

The following limitations shall be observed. Each column presents limitations. The limits presented do not necessarily occur simultaneously. Refer to Pratt & Whitney Engine Maintenance Manual for specific actions required if limits are exceeded.

OPERATING CONDITION	SHP	TORQUE FT-LBS (1)	MAXIMUM OBSERVED ITT°C	GAS GENERATOR RPM N_1		PROP RPM N_2	OIL PRESS. PSI (2)	OIL TEMP °C (3)
				RPM	%			
STARTING	---	---	1090 (4)	---	---	---	---	-40 (min)
LOW IDLE	---	---	660 (5)	---	51 (min)	1100 (min)	40 (min)	-40 to 99
HIGH IDLE	---	---	---	---	70 (approx)	---	---	0 to 99
TAKEOFF AND MAX CONT	550	1315	695	38,100	101.5	2200	80 to 100	10 to 99
CRUISE CLIMB AND MAX CRUISE	538	1315 (6)	695	38,100	101.5	2200	80 to 100	0 to 99
MAX REVERSE (7)	---	---	695	---	88	2100	80 to 100	0 to 99
TRANSIENT	---	1500 (4)	825 (4) (8)	38,500	102.6	2420	---	0 to 99

FOOTNOTES:

(1) Maximum permissible sustained torque is 1315 ft-lbs. Propeller speeds (N_2) must be set so as not to exceed power limitation.

(2) When gas generator speeds are above 72% N_1 and oil temperatures are between 60°C and 70°C, normal oil pressure is between 80 and 100 psi. Oil pressure between 40 and 80 psi is undesirable; it should be tolerated only for the completion of the flight, and then only at a reduced power setting. Oil pressure below 40 psi is unsafe; it requires that either the engine be shut down, or that a landing be made as soon as possible, using the minimum power required to sustain flight.

(3) For increased service life of engine oil, an oil temperature of between 74° to 80°C is recommended.

(4) These values are time-limited to two seconds.

(5) High ITT at ground idle may be corrected by reducing accessory load and or increasing N_1 rpm.

(6) Cruise torque values vary with altitude and temperature.

(7) Reverse power operation is limited to one minute.

(8) High generator loads at low N_1 speeds may cause the ITT transient temperature limit to be exceeded. Observe generator load limits.

should be reduced and the N_1 speed increased before re-introducing a load on the engines.

At N_1 speeds of 70% or more, the 660°C restriction is removed, as airflow through the engine is sufficient.

In the climb, torque will decrease and ITT may increase slightly. The cruise climb and recommended normal cruise ITT limit is not placarded on the indicator. At altitude, the Performance Chart numbers may not be attainable due to altitude and temperature variations.

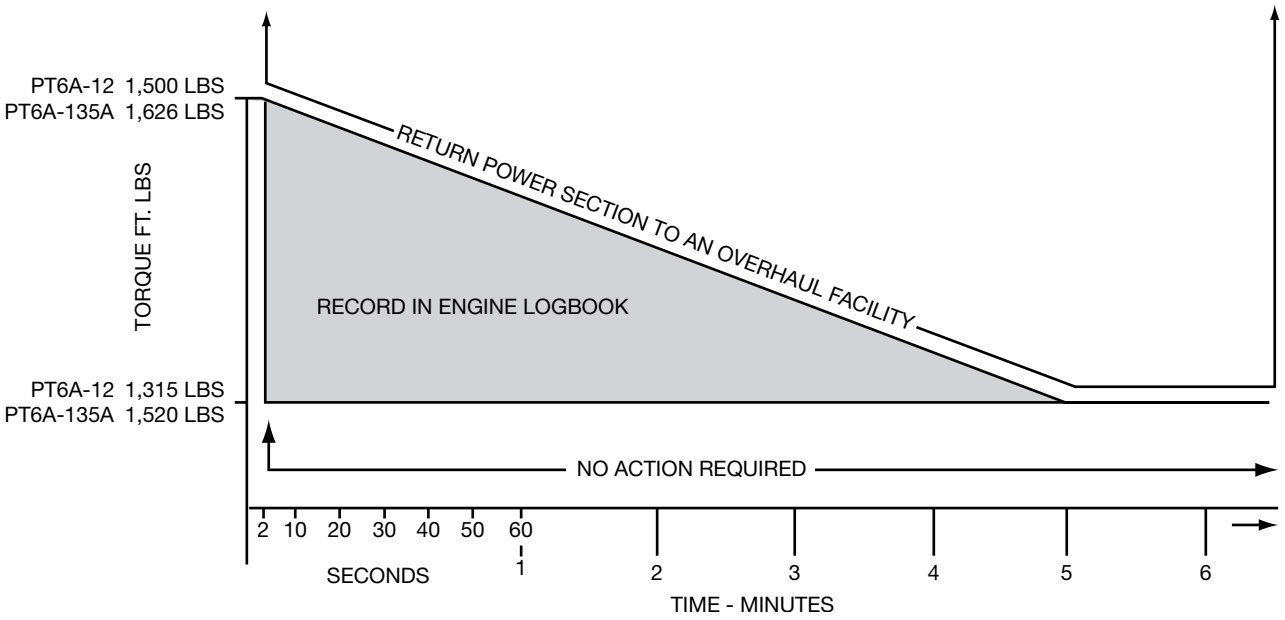
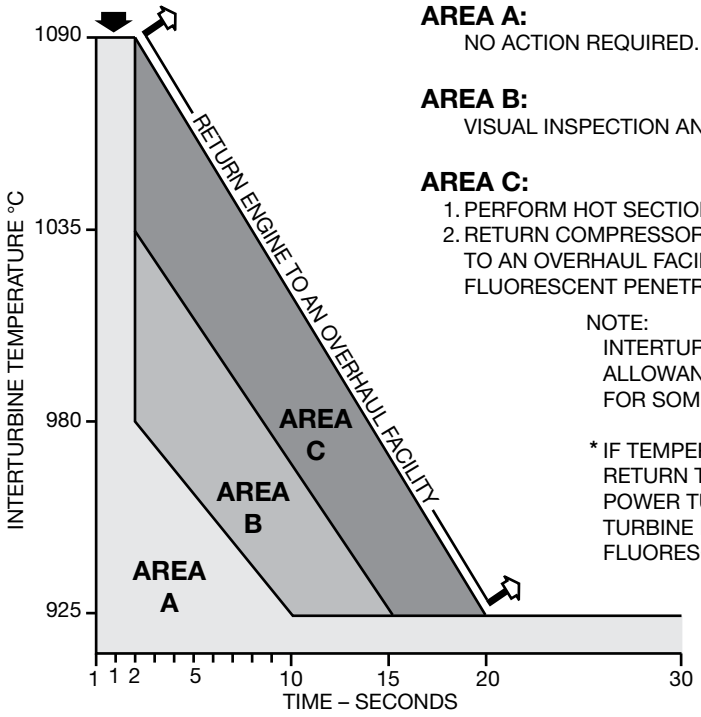


Figure 7-24. Overtorque Limits Chart



AREA A:
NO ACTION REQUIRED.

AREA B:
VISUAL INSPECTION AND RECORD IN ENGINE LOGBOOK.

AREA C:
1. PERFORM HOT SECTION INSPECTION.
2. RETURN COMPRESSOR TURBINE BLADE AND DISK ASSEMBLY TO AN OVERHAUL FACILITY FOR STRETCH CHECK AND FLUORESCENT PENETRANT INSPECTION.

NOTE:
INTERTURBINE TEMPERATURES SHOWN MAKE NO ALLOWANCE FOR INSTRUMENT ERRORS, BUT DO ALLOW FOR SOME TYPICAL INSTRUMENT LAG.

* IF TEMPERATURE EXCEEDS THIS VALUE FOR 2 SECONDS RETURN TO AN OVERHAUL FACILITY. COMPRESSOR AND POWER TURBINE BLADES MUST BE DISCARDED AND BOTH TURBINE DISKS SUBJECTED TO STRETCH CHECK AND FLUORESCENT PENETRANT INSPECTION.

Figure 7-25. Overtemperature Limits (Starting)

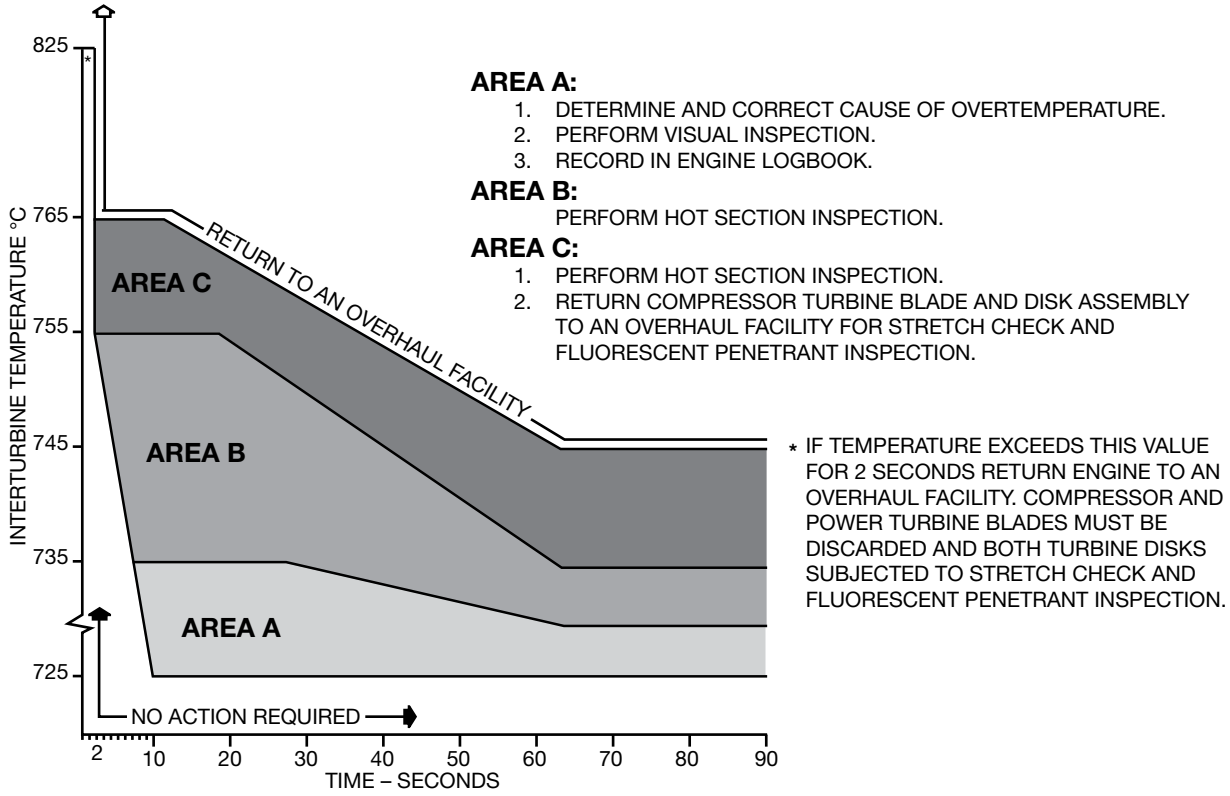


Figure 7-26. Overtemperature Limits (Except Starting)

Transient limits provide buffers for surges during engine acceleration. Torque and ITT have an allowable excursion duration of two seconds. A momentary peak of 1,500 ft-lb and 825°C is allowed for torque and ITT respectively during acceleration.

The Overtorque Limits Chart (Figure 7-24) shows actions required if torque limits are exceeded under all conditions. If the torque limits are exceeded for more than a few minutes, the gearbox can be damaged. The chart shows the specific limits and action required if they are exceeded.

The Overtemperature Limits charts (Figure 7-25 and Figure 7-26) show the specific actions required if ITT limits are exceeded during Starting Conditions and All Conditions Except Starting. For area A (Figure 7-26. Overtemperature Limits (Except Starting)), determine and correct the cause of overtemperature. If it was during a start, have the engine visually inspected through the exhaust duct (Figure 7-27), then record the action in the engine logbook.

Overtemperature in area B will require that a hot section inspection be performed. During a hot section inspection, the components forward of the combustion chamber are examined and replaced. Parts may be repaired or replaced as necessary. In area C overtemperatures may require that the engine be returned for overhaul. Exceeding ITT limits in this area for more than a few seconds may cause extensive engine damage.

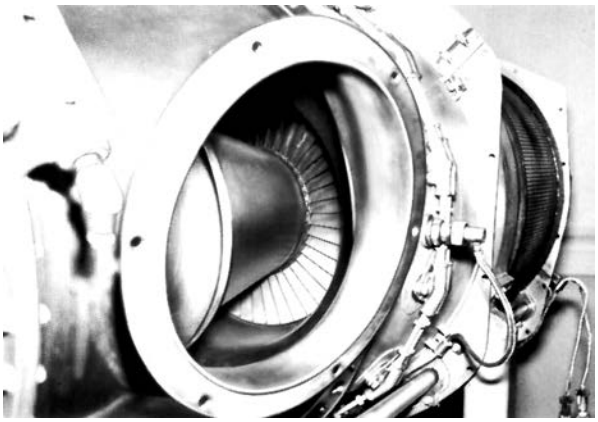


Figure 7-27. View through Exhaust Duct

STARTER OPERATING TIME LIMITS

The engine starters are time-limited during the starting cycle if for any reason multiple starts are required in quick sequence. The starter is limited to 40 seconds ON then 60 seconds OFF for cooling before the next sequence of 40 seconds ON, 60 seconds OFF. After the third cycle of 40 seconds ON, the starter must stay OFF for 30 minutes. If these limits are not observed, overheating may damage the starter.

Trend Monitoring

During normal operations, gas turbine engines are capable of producing rated power for extended periods of time. Engine operating parameters, such as output torque, interstage turbine temperature, compressor speed, and fuel flow for individual engines are predictable under specific ambient conditions. On PT6A engines, these predictable characteristics may be taken advantage of by establishing and recording individual engine performance parameters. These parameters can then be compared periodically to predicted values to provide day-to-day visual confirmation of engine efficiency.

The Engine Condition Trend Monitoring System, recommended by Pratt and Whitney, is a process of periodically recording engine instrument readings such as torque, interstage turbine temperature, compressor speed, and fuel flow, correcting the readings for altitude, outside air temperature, and airspeed, if applicable, and then comparing them to a set of typical engine characteristics. Such comparisons produce a set of deviations in interstage turbine temperature, compressor speed, and fuel flow.

DATA COLLECTION FORM

The trend monitoring procedure used specifies that flight data be recorded on each flight day, every five flight hours, or other flight period. Select a flight with long established cruise, preferably at a representative altitude and airspeed. With engine power established and stabilized for a minimum of five minutes, record the following data on a form similar to the in-flight engine data log shown in (Figure 7-28):

Indicated airspeed (IAS) ..... In knots

Outside air temperature (OAT).....In °C

Pressure altitude (ALT).....In feet

Propeller speed (NP)..... In rpm

Torque (TQ) ..... In foot-pounds

Gas generator speed
(N_G or N_1).....In %NG or N_1

Interturbine temperature (ITT).....In °C

Fuel Flow (FF) .....In pph

DATE	OAT (°)	PRESS ALT	IAS (KTS)	PROP SPEED		TORQUE (FT/LBS)	N_1 (%)	ITT	FUEL FLOW	DELTA* NG	DELTA* ITT	DELTA* FF	OIL TEMP	OIL PRESS	ELECT LOAD
					LEFT										
					RIGHT										
					LEFT										
					RIGHT										
					LEFT										
					RIGHT										
					LEFT										
					RIGHT										
					LEFT										
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					RIGHT										
					LEFT										
					RIGHT										

Figure 7-28. In-Flight Engine Data Log

PROPELLERS

GENERAL

This section describes the propellers and the associated system. Location and use of propeller controls, principle of operation, reversing, and feathering are included in this discussion.

PROPELLER SYSTEM

This section on the operation and testing of the propeller system on the Beechcraft King Air C90A and C90B is directed at increasing the pilot's understanding of the theory of operation of a constant-speed, full-feathering, reversing propeller, and helping him better understand the propeller system checks conducted as outlined in the Before Takeoff (Runup) checklist in the *Pilot's Operating Handbook.*

Each engine is equipped with a conventional three-blade (C90A) or four-blade (C90B), full-feathering, constant-speed, counterweighted, reversing, variable-pitch propeller mounted on the output shaft of the reduction gearbox (Figure 7-29).

The propeller pitch is controlled by engine oil pressure boosted through a governor pump integral within the propeller governor. Centrifugal counterweights and feathering springs move the propeller blades toward high pitch and into the feathered position. Without oil pressure to counteract the counterweights and feathering springs, the propeller blades would move into feather. An oil pump, which is part of the propeller governor, boosts engine oil pressure to move the propeller to low pitch and reverse. The propeller feathers after engine shutdown.

Propeller tiedown boots (Figure 7-30) are provided to prevent windmilling at zero oil pressure when the airplane is parked.

Low pitch propeller position is determined by the primary low pitch stop, which is a mechanically actuated hydraulic stop. Beta and reverse blade angles are controlled by the power levers in the Beta and reverse range.

HARTZELL 3-BLADE PROPELLER—C90A

McCAULEY 4-BLADE PROPELLER

Figure 7-29. Propellers

Figure 7-30. Propeller Tiedown Boot Installed

Two governors, a primary governor and a backup overspeed governor, control the propeller rpm. The propeller control lever adjusts the governor's setting (1,800 to 2,200 rpm). The overspeed governor will limit the propeller to 2,288 rpm should the primary governor malfunction. However, if the propeller exceeds 6% above the selected rpm of the primary governor, usually the fuel topping governor will limit the rpm by reducing engine power. In the reverse range, the fuel topping governor is reset to limit the propeller rpm to 95% of selected rpm.

MCCAULEY AND HARTZELL FOUR-BLADE PROPELLERS

The C90B is equipped with Hartzell on LJ-1542 and after (McCauley on C90B prior to LJ-1542), 90-inch- diameter, four-blade, full-reversing, dynamically balanced propellers. The main advantages of the four-blade propellers are that they have lower tip speeds (and thus generate less noise), create less airframe vibration, and provide generous propeller tip-to-ground clearance. Dynamic vibration absorbers mounted inside the cockpit and cabin (a total of 26 absorbers) are used in conjunction with the four-blade propellers to reduce noise and vibration even more.

BLADE ANGLE

Blade angle is the angle between the chord of the propeller and the propeller's plane of rotation. Blade angle is different near the hub than it is near the tip, due to the normal twist which is incorporated in a blade to increase its efficiency. In the propellers used on the C90A and C90B King Air, the blade angle is measured at the chord 30 inches out from the propeller's center. This position is referred to as the "30-inch station." All blade angles given in this section are approximate (Figure 7-31).

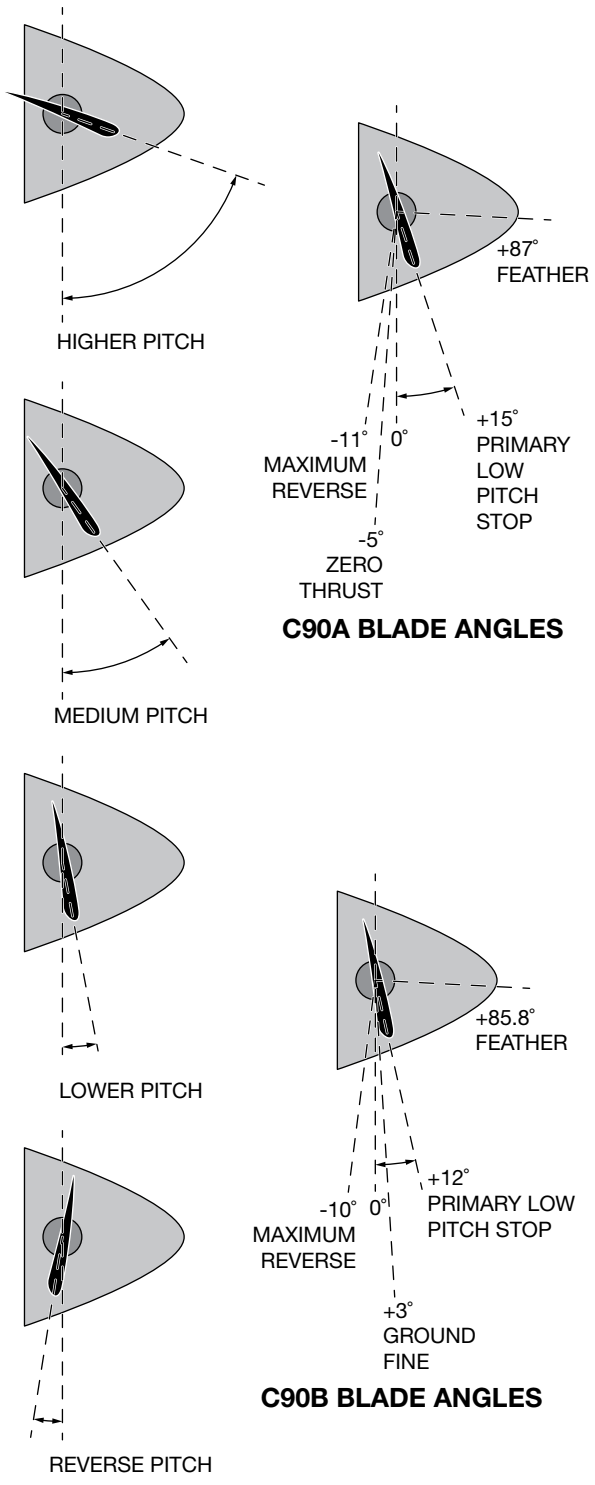

Figure 7-31. Blade Angle Diagram

PRIMARY GOVERNOR

The primary governor (Figure 7-32) is needed to convert a variable-pitch propeller into a constant-speed propeller. It does this by changing blade angle to maintain the propeller speed the operator has selected. The primary governor can maintain any selected propeller speed from approximately 1,800 rpm to 2,200 rpm.

Suppose an airplane is in normal cruising flight with the propeller turning 1,900 rpm. If the pilot trims the airplane down into a descent without changing power, the airspeed will increase. This decreases the angle of attack of the propeller blades, causing less drag on the propeller, thus beginning to increase its rpm. Since this propeller has a variable-pitch capabilities and is equipped with a governor set at 1,900 rpm, the governor will sense this "overspeed" condition and increases blade angle to a higher pitch. The higher pitch increases the blade's angle of attack, slowing it back to 1,900 rpm, or "onspeed."

Likewise, if the airplane moves from cruise to climb airspeeds without a power change, the propeller rpm tends to decrease, but the governor responds to this "underspeed" condition by decreasing blade angle to a lower pitch, and the rpm returns to its original value. Thus the governor gives "constant-speed" characteristics to the variable-pitch propeller.

Power changes, as well as airspeed changes, cause the propeller to momentarily experience overspeed or underspeed conditions, but again the governor reacts to maintain the onspeed condition.

There are times, however, when the primary governor is incapable of maintaining selected rpm. For example, imagine an airplane approaching to land with its governor set at 1,900 rpm. As power and airspeed are both reduced, underspeed conditions exist which cause the governor to decrease blade angle to restore the onspeed condition. If blade angle could decrease all the way, to 0°or reverse, the propeller would create so much drag on the airplane that the aircraft control would be dramatically reduced. The propeller, acting as a large disc, would blank the airflow

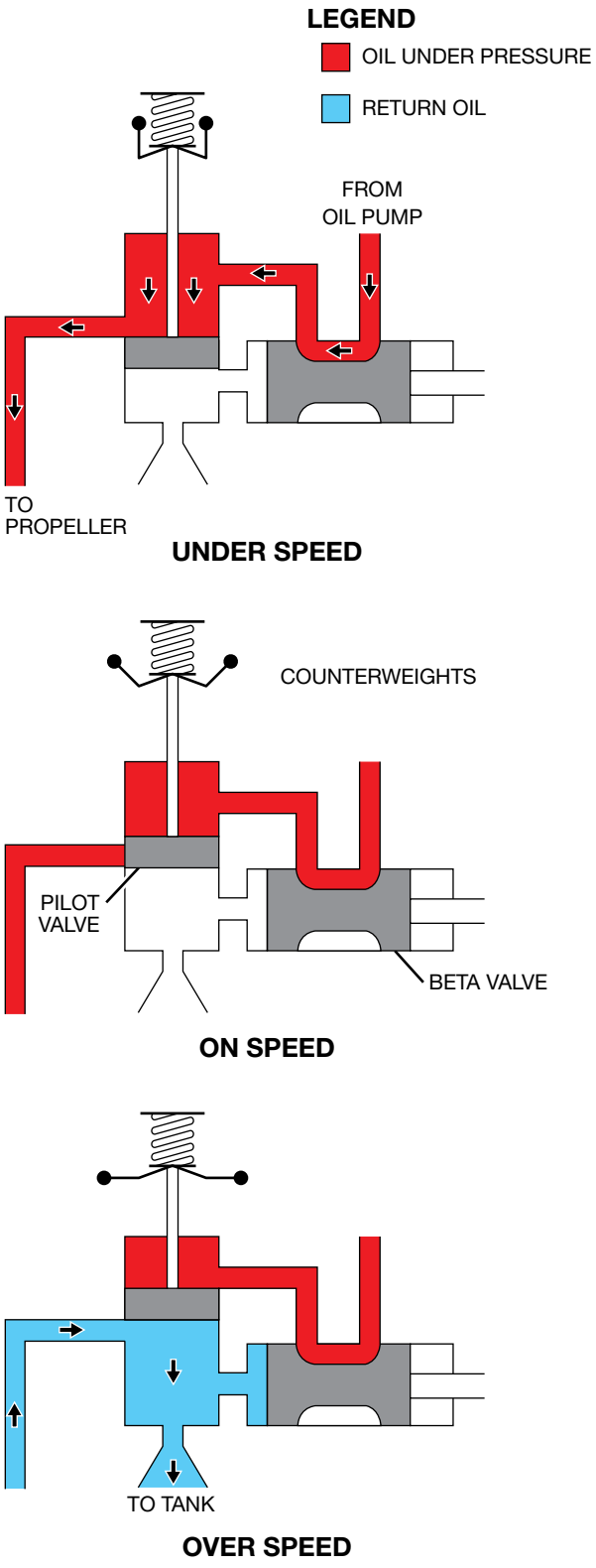

Figure 7-32. Primary Governor Diagram

around the tail surfaces, and a rapid nosedown pitch change would result.

To prevent these unwanted aerobatics, some device must be provided to stop the governor from selecting blade angles that are too low for safety. As the blade angle is decreased by the governor, eventually the low pitch stop is reached, and now the blade angle becomes fixed and cannot continue to a lower pitch. The governor is therefore incapable of restoring the onspeed condition, and propeller rpm falls below the selected governor rpm setting.

PRIMARY GOVERNOR OPERATION

The propeller levers adjust the primary propeller governor between 1,800 rpm and 2, 200 rpm. The primary propeller governor, mounted at the top of the engine reduction gearbox, has two functions: it can select any constant propeller rpm within the range of 1,800 to 2,200, and it can also feather the propeller. The primary propeller governor adjusts propeller rpm by controlling the oil supply to the propeller dome.

An integral part of the primary propeller governor is the governor pump. This pump is driven by the N2 shaft and raises the engine oil pressure from normal to approximately 375 psi. The greater the oil pressure sent to the propeller dome, the lower the propeller pitch. The oil pressure is always trying to maintain a low pitch; however, the feathering springs and centrifugal counterweights are trying to send the propeller into the feathered position. Propeller control is a balancing act of opposing forces. A transfer gland is located on the propeller shaft. This transfer gland allows the oil to enter and exit the propeller dome area. Thus, the transfer gland is always replenishing the oils supply to the propeller pitch mechanism with fresh warm oil.

The primary propeller governor uses a set of rotating flyweights that are geared to the propeller shaft. The flyweights act as a comparison to a desired reference speed of how fast the propeller is turning. These flyweights are connected to a free-floating pilot valve. The slower the flyweights are turning in relation to the desired reference speed, the lower the position of the pilot valve. If the propeller and the flyweights turn faster, the additional centrifugal force makes the pilot valve rise inside the governor. The pilot valve position determines how much oil pressure is being sent to the propeller pitch mechanism. Here are a few examples.

If a propeller rpm of 1,900 is selected and the propeller is actually turning at 1,900, the flyweights are in their center or “onspeed” condition (Figure 7-33). The pilot valve is in the middle position. This maintains a constant oil pressure to the propeller pitch mechanism, which creates a constant pitch and a constant rpm.

If the airplane enters a descent, without any change to the cockpit controls, there will be a tendency for the airspeed to increase and the propeller to turn faster (Figure 7-34). The flyweights will, in turn, rotate faster. The additional centrifugal force will make the pilot valve rise. Notice that oil can now escape via the pilot valve. Lower oil pressure will result in a higher pitch and a reduction of propeller rpm. The propeller will then return to its original rpm setting. The flyweights will then slow down, and the pilot valve will return to the equilibrium position to maintain the selected propeller rpm.

If the airplane enters a climb without any change in the cockpit controls, the airspeed will decrease and the propeller will tend to slow (Figure 7-35). The flyweights in the propeller governor will slow down, because of a loss in centrifugal force, and the pilot valve will lower. This will allow more oil pressure to the propeller pitch mechanism. High oil pressure will result in a lower pitch. This in turn will cause an increase in propeller rpm. The propeller will increase to its original rpm setting, the flyweights will then speed up, and the pilot valve will return to its equilibrium or “onspeed” position, such as torque, interstage turbine temperature, compressor speed, and fuel flow, correcting the held constant by changing the propeller blade angles. The cockpit propeller lever adjusts where the equilibrium or “onspeed” condition will occur. The pilot can select any constant propeller rpm from 1,800 to 2,200 rpm. Normally 2,200 is used for takeoff and 2,000 rpm for climb. Cruise rpm is 1,900 rpm.

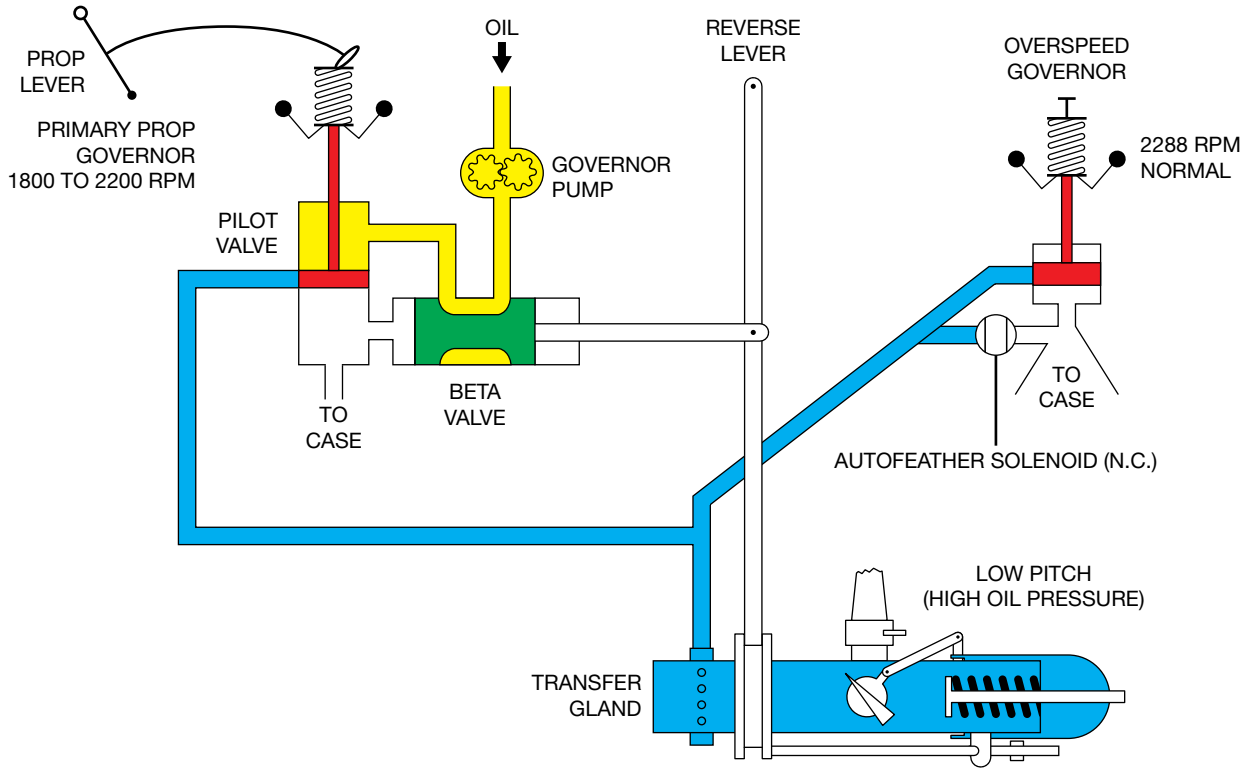

Figure 7-33. Propeller Onspeed Diagram

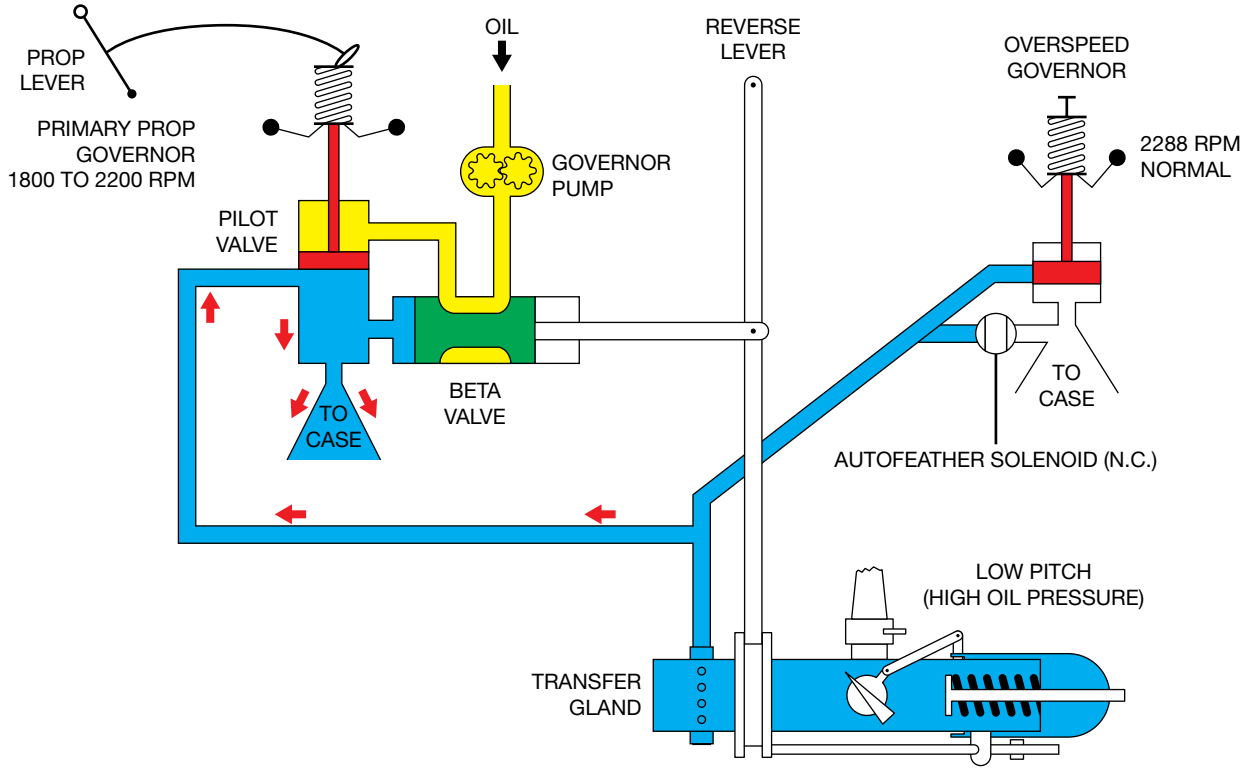

Figure 7-34. Propeller Overspeed Diagram

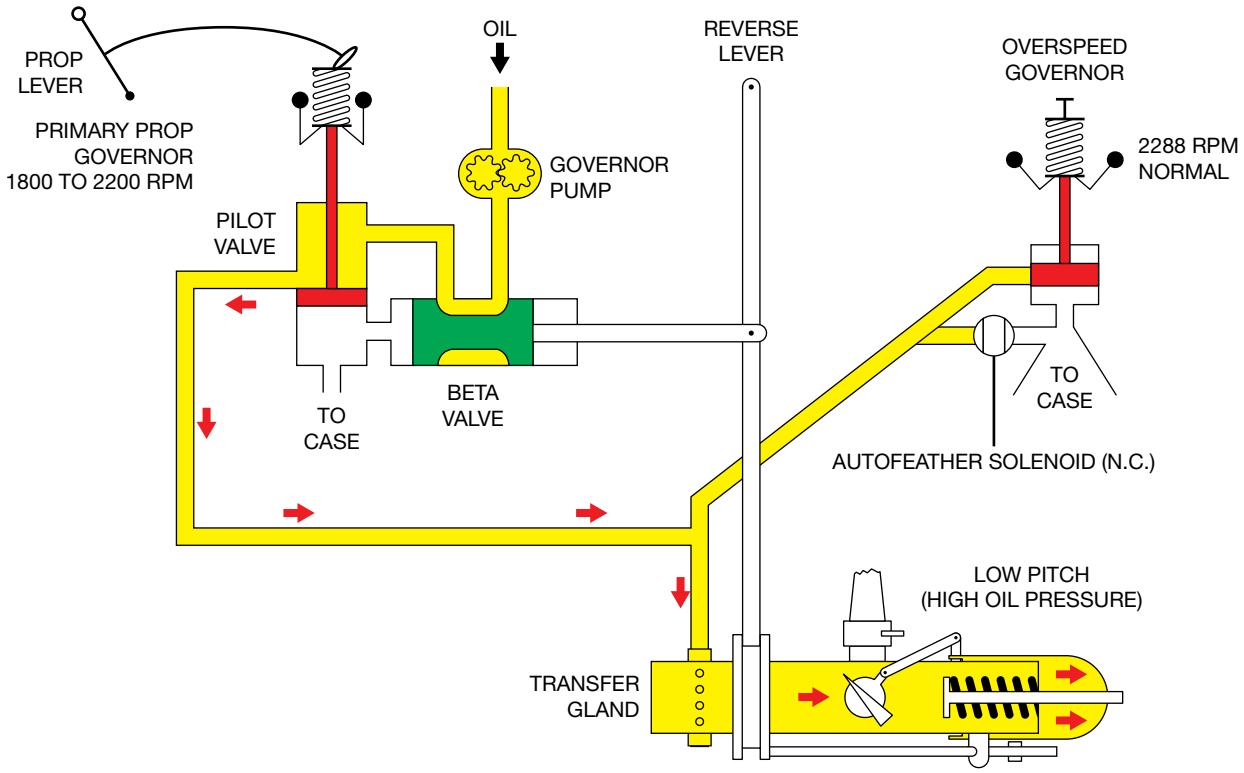

Figure 7-35. Propeller Underspeed Diagram

LOW PITCH STOP

It is easy for the pilot to determine when the propeller blade angle is at the low pitch stop. Assuming the propeller is not feathered or in the process of being feathered, ***whenever the propeller rpm is below the selected governor rpm, the propeller blade angle is at the low pitch stop.***

This assumes that momentary periods of underspeed are not being considered. Rather, the propeller rpm is below and staying below the selected governor rpm.

For example, if the propeller control is set at 1,900 rpm but the propeller is turning at less than 1,900 rpm, the blade angle is at the low pitch stop.

On many types of airplanes, the low pitch stop is simply at the low pitch limit of travel, determined by the propeller's construction. But with a reversing propeller, the extreme travel in the low pitch direction is past 0°, into reverse or negative blade angles (Figure 7-36). Consequently, the low pitch stop on this propeller must be designed in such a way that it can be repositioned when reversing is desired.

The low pitch stop is created by mechanical linkage sensing the blade angle. The linkage causes a valve to close, which stops the flow of oil pressure coming into the propeller dome. Since this pressure causes low pitch and reversing, once it is blocked, a low pitch stop has been created. The low pitch stop is commonly referred to as the "Beta" valve. Furthermore, the valve is spring-loaded to cause the propeller to feather in the event of mechanical loss of Beta valve control.

The position of the low pitch stop is controlled from the cockpit by the power lever. Whenever the power lever is at IDLE or above, this stop is set at approximately 15°for the C90A or approximately 12°for the C90B. But bringing the power lever aft of IDLE progressively repositions the stop to lesser blade angles.

Before reversing can take place, the propeller must be on the low pitch stop. As the propellers reach

approximately 15°for the C90A or approximately 12°for the C90B, the Beta valve is repositioned, creating the low pitch stop. The primary governor is sensing an underspeed and is directing oil pressure into the propeller dome. The Beta valve is controlling oil flow into the primary governor, and is defining the low pitch stop through oil pressure.

When blade angles less than approximately 15°for the C90A or approximately 12°for the C90B are requested, the linkage pulls the Beta valve actuator, readjusting the propeller blade angle as the Beta valve allows more oil into the propeller dome. The slip ring moves with the prop dome and will define the low pitch stop at a lower, or negative, blade angle. If blade angles less than approximately 15°for the C90A or approximately 12°for the C90B are requested before the propeller blades are on the low pitch stop, the slip ring will not move, and the reversing cable and linkage may be damaged.

The region from 15° to –11°(C90A) or 12° to –10° (C90B) blade angle is referred to as the Beta range. On the C90A, the range from 15° to –5°, the engine's compressor speed (N_1) remains at the value it had when the power lever was at IDLE (low idle to high idle) based on condition lever position. From –5° to –11° blade angle, the N_1 speed progressively increases to a maximum value at –11° blade angle of approximately 85% ±3%. This region, designated by red and white stripes on the power lever gate, is referred to as the "Beta Plus Power" range or Reverse, and ends at maximum reverse.

On the C90B, the Ground Fine range extends from +12° to +3°, and the engine's compressor speed (N_1) remains at the value it had when the power lever was at IDLE (low idle to high idle) based on condition lever position. From +3° to –10° blade angle, the N_1 speed progressively increases to a maximum value at –10° blade angle of approximately +85% ±3%.

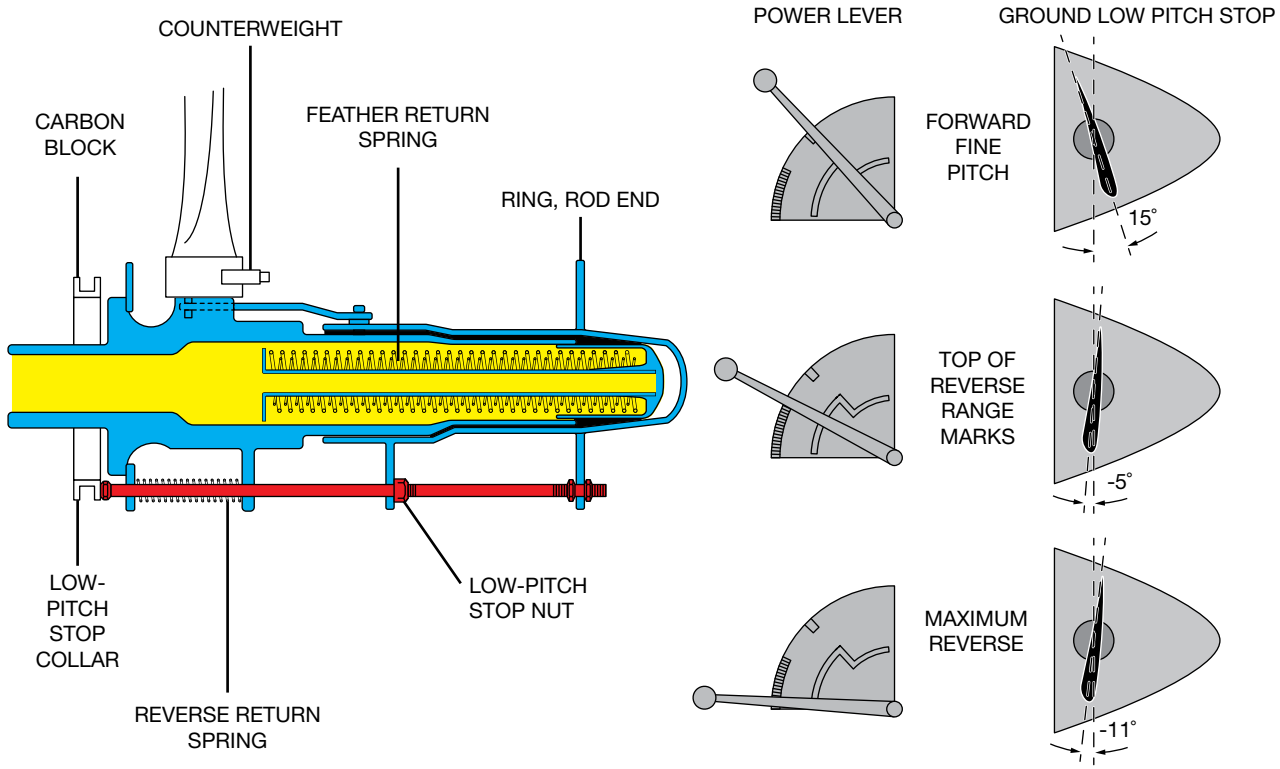

Figure 7-36. Low Pitch Stop Diagram

Low Pitch Stop Operation

During non-reversing operations, the low pitch stop prevents the propeller blades from reducing the airflow over the empennage of the aircraft.

The low pitch stop uses a mechanical linkage to hydraulically control propeller blade angle. As the propeller blades reduce angle through approximately 20° of pitch, the flange mounted on the propeller dome contacts the nuts located on the rods mounted on the slip ring. The propeller dome moves the slip ring forward, which in turn activates the Beta valve, which controls oil pressure into the propeller dome.

Riding in the slip ring is linkage which connects the Beta valve with the slip ring, and the power levers via a cable. As the slip ring moves, the linkage pivots about the end with the cable attached to it, with the Beta valve in the middle. For reversing, the pilot repositions the linkage with the power levers, which resets the low pitch stop.

When the Beta valve is controlling blade angle, oil pressure supplied from the governor oil pump is supplying pressure through the Beta valve to the propeller dome. The Beta valve modulates the amount of pressure entering the propeller dome, controlling the blade angle. The primary governor must be in the underspeed condition, allowing all of the pressure flowing from the Beta valve into the propeller dome. If the underspeed condition did not exist when lower blade angles are requested, the Beta valve could not fully control the propeller blade angle, and the slip ring would not move without help from the propeller blades. Since the propeller blades only contact the slip ring when the blades are at the low pitch stop, the request for lower blade angles when the propellers are not on the low pitch stop will result in damage to the control cable, as it cannot effect these changes alone.

BETA AND REVERSE CONTROL

The geometry of the power lever linkage through the cam box is such that power lever increments from idle to full forward thrust have no effect on the position of the Beta valve. When the power lever is moved from idle into the reverse range, it positions the Beta valve to direct governor oil pressure to the propeller piston, decreasing blade angle through zero into a negative range. The travel of the propeller servo piston is fed back to the Beta valve to null its position and, in effect, provide infinite negative blade angles all the way to maximum reverse. The opposite will occur when the power lever is moved from full reverse to any forward position up to idle, therefore providing the pilot with manual blade angle control for ground handling.

Beta and Reverse Control Operation

When the blade angle reaches approximately 20°, the flange extending from the dome makes contact with the Beta nuts (Figure 7-37). As the propeller pitch angle continues to decrease, each flange on the propeller dome pushes the nut and the attached Beta rod forward. As the rod moves forward, it pulls the slip ring forward. In turn, a Beta valve inside the governor is pulled into the oil pressure cutoff position. The linkage is set to control the oil pressure supply to the dome when the blade angle reaches low pitch stop.

If this system were fixed at the low pitch stop, the propeller could not be reset throughout the Beta range. However, the low pitch stop can be adjusted to allow access to the Beta and reverse range on the ground. The hydraulic low pitch stop can be reset to allow the propeller to operate in the Beta and reverse ranges while the aircraft is on the ground and the engines are operating.

When the power levers are lifted up and over the idle detent into the Beta range, the Beta valve is repositioned. As the Beta arm moves back, the Beta valve is opened, re-establishing oil flow to the propeller dome. This allows the propeller blade to move to a flatter pitch. As the propeller blades move to a flatter pitch, the propeller dome and slip ring continue forward, eventually moving the Beta valve back into position to stop propeller blades. In summary, the position of the low pitch stop is controlled by the power levers. When the power levers are set at idle or above, the stop is set at approximately 15° on the C90A or approximately 12° on the C90B. When the power levers are moved aft of idle, however, the low pitch stop is repositioned to lesser blade angles.

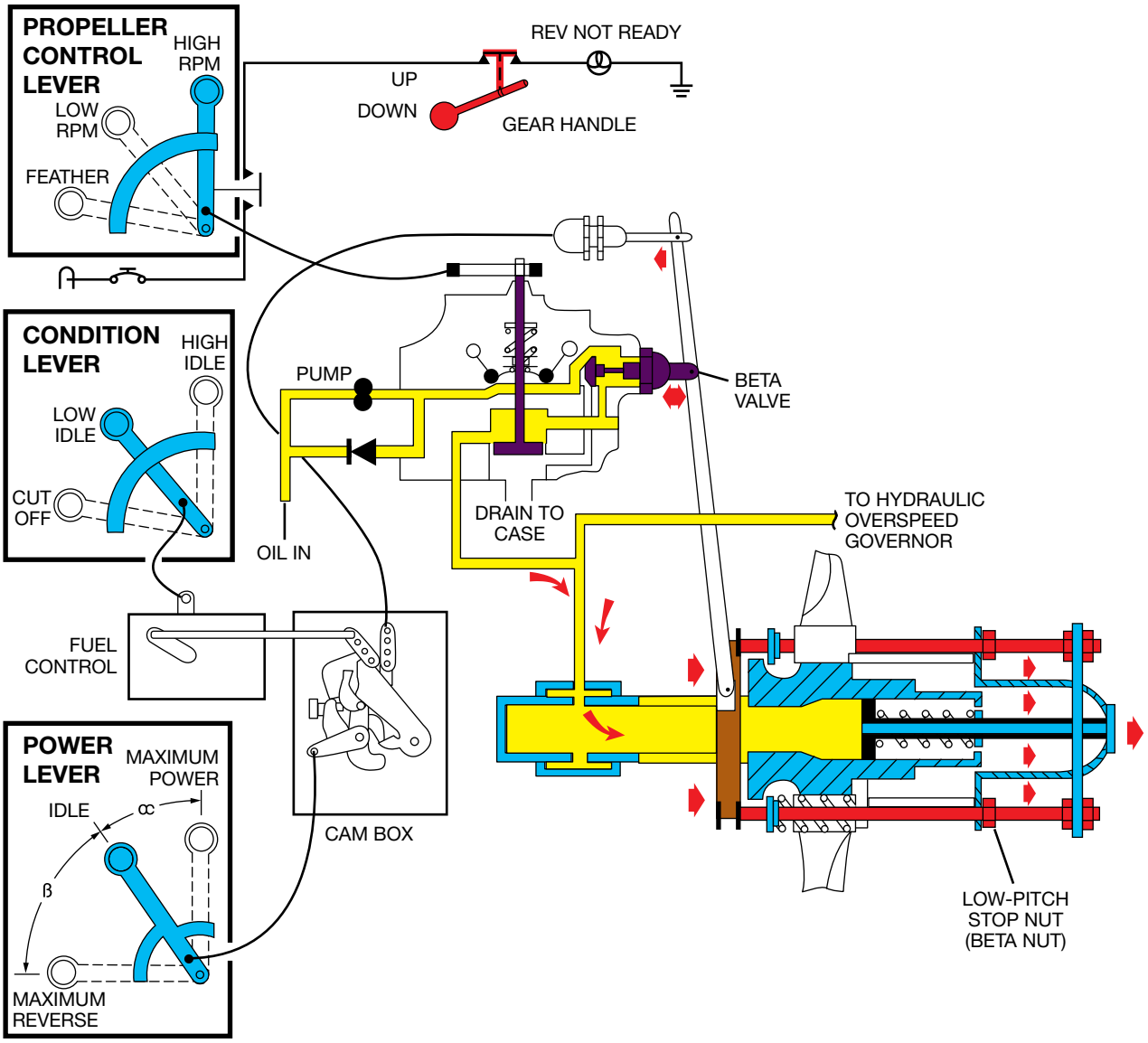

Figure 7-37. Beta Range and Reverse Diagram

The propeller can be feathered by moving the propeller lever full aft past the detent into the feather range. The feathering action raises the pilot valve to the full up position. The oil pressure is released from the propeller pitch mechanism and the propeller feathers. In this type of turbine engine, the propeller shaft and N_1 shaft are not connected. Thus, the propeller can be feathered with the engine running at idle power. Without an autofeather system, in flight, the propeller will maintain rpm unless it is manually feathered when the engine is shut down.

There are situations where the propeller primary governor cannot maintain the selected propeller rpm, such as final approach where power and airspeed are being reduced. With the progressive reduction of power and airspeed on final, the propeller and rotating counterweights will tend to go to the underspeed condition. In the underspeed condition the pilot valve will open, increasing oil pressure to the dome, and the propeller pitch will decrease as power and airspeed are reduced. Since the reversible propeller is capable of decreasing past 0° into negative or reverse blade angles, the low pitch stop prevents the blade angle from

decreasing beyond a predetermined value. When the propeller governor becomes incapable of maintaining the onspeed condition, the propeller rpm will fall below the selected governor rpm setting.

Assuming the propeller is not feathered, whenever the propeller rpm is below the selected governor setting, the propeller blade angle is at the low pitch stop. The low pitch stop mechanism is created by linkage that references the actual blade angle.

Moving the power lever within the Beta range on the C90A or the ground fire range on the C90B adjusts propeller pitch. Moving the power levers within the reverse range adjusts propeller pitch and N_1, up to the maximum N_1 in reverse of 88%. Attempting to pull the power levers in reverse with the propellers in feather will cause damage to the reversing linkage of the power lever. Also, pulling the power levers into the reverse position on the ground with the engines shut down will damage the reversing system.

OVERSPEED GOVERNOR

The overspeed governor provides protection against excessive propeller speed in the event of primary governor malfunction. Since the PT6's propeller is driven by a free turbine (independent of the engine's), overspeed could occur if the primary governor were to fail.

The operating point of the overspeed governor is set at 2,288 rpm. If an overspeeding propeller's speed reached 2,288 rpm, the overspeed governor would control the oil pressure and pitch to prevent the rpm from continuing its rise. From a pilot's point of view, a propeller tachometer stabilized at approximately 2,288 would indicate failure of the primary governor and proper operation of the overspeed governor. The overspeed governor can be reset to approximately 2,000 rpm for test purposes.

OVERSPEED GOVERNOR OPERATION

If the primary propeller governor failed, an overspeed condition could occur. However, several safety devices in the systems come into play in the event of a primary governor failure. A hydraulic overspeed governor (Figure 7-38) is located on the left side of the propeller reduction gearbox. It has a set of flyweights and a pilot valve similar to those of the primary governor. If a runaway propeller's speed were to reach 2,288 rpm, the overspeed governor flyweights would make its pilot valve rise. This would decrease the oil pressure at the propeller dome. The blade angle would increase as necessary to prevent the rpm from continuing its rise. Testing of the overspeed governor at approximately 2,000 rpm is accomplished during runup by using the propeller governor test switch on the pilot's left subpanel.

FUEL TOPPING GOVERNOR

The fuel topping governor can also control an overspeed condition and is set at 6% above the primary governor's selected speed. In an overspeed condition, the fuel topping governor will limit propeller rpm by decreasing pneumatic pressure to the fuel control unit, reducing fuel flow and engine speed as means of controlling propeller rpm. In reverse, the fuel topping governor is reset to 95% of selected rpm to insure that the propeller will not reach the selected rpm. The fuel topping governor will only prevent an over-speed if the primary governor's flyweight's are still operational.

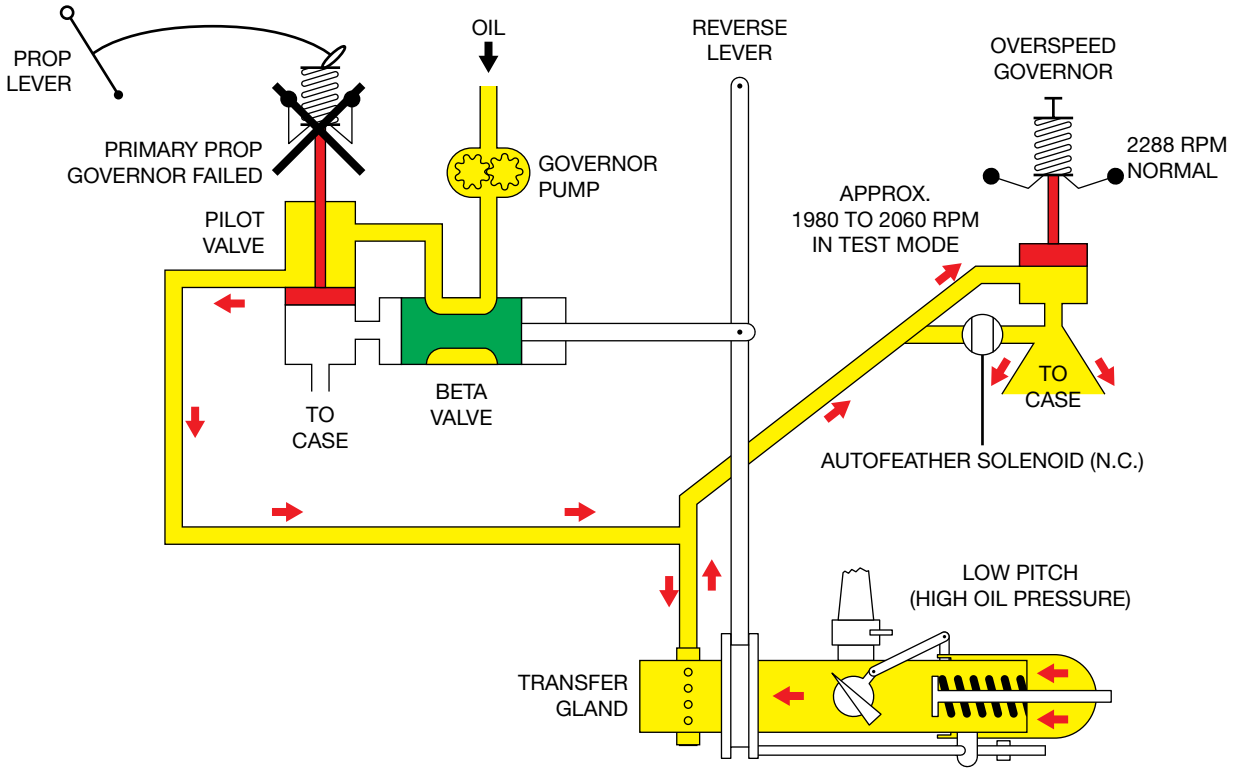

Figure 7-38. Overspeed Governor Diagram

POWER LEVERS

The power levers (Figure 7-39) are located on the power lever quadrant (first two levers on the left side) on the center pedestal. They are mechanically interconnected through a cam box to the fuel control unit, the Beta valve and follow-up mechanism, and the fuel topping (NP) governor. The power lever quadrant permits movement of the power lever from idle to maximum thrust and in the Beta/reverse range from idle to maximum reverse. A gate in the power lever quadrant at the IDLE position prevents inadvertent movement of the lever into the Beta/reverse range. The pilot must lift the power levers up and over this gate to select Beta or reverse. On the C90B, there is a second gate labeled "GROUND FINE," to distinguish between GROUND FINE and REVERSE.

The function of the power levers is to establish a gas generator rpm through the gas generator governor (N_G) and a fuel flow that will produce and maintain the selected N_1 rpm. In the Beta or GROUND FINE range, the power levers are used to change the propeller blade angle, thus changing propeller thrust.

In the REVERSE range, the power lever:

- Selects a blade angle proportionate to the aft travel of the lever
- Selects an N_1 that will sustain the selected reverse power
- Resets the fuel topping governor from its normal setting of 106% to approximately 95% of the primary governor setting

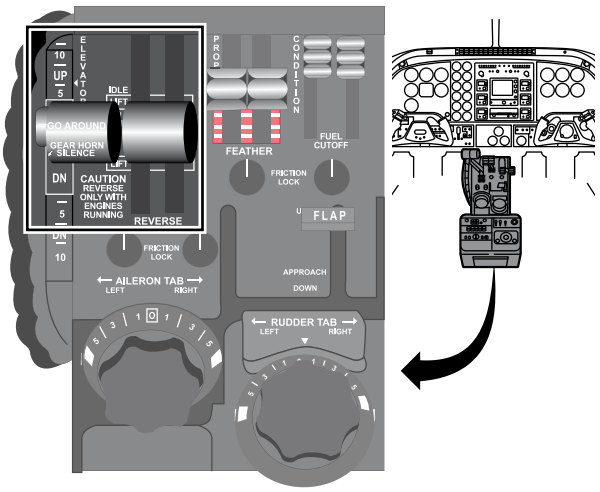

Figure 7-39. Power Levers

Propeller Control Levers

Propeller rpm, within the primary governor range of 1,800 to 2,200 rpm, is set by the position of the propeller control levers (Figure 7-40). These levers, one for each propeller, are located between the power levers and the condition levers on the center pedestal quadrant. The full forward position sets the primary governor at 2,200 rpm.

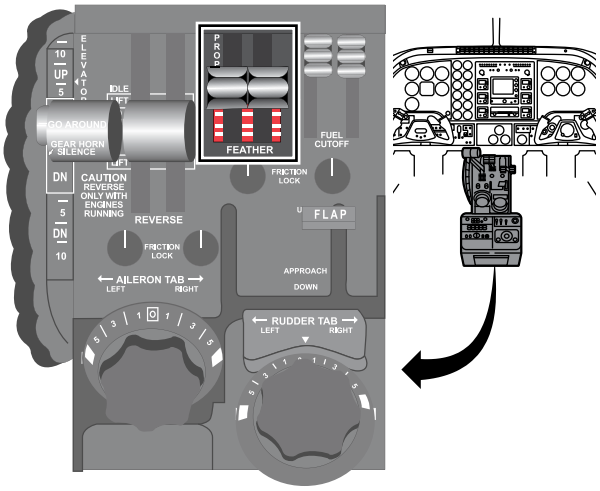

Figure 7-40. Propeller Control Levers

In the full aft position at the feathering detent, the primary governor is set at 1,750 rpm. Intermediate propeller rpm positions can be selected by moving the propeller levers to the corresponding position, to select the desired rpm as indicated on the propeller tachometer. These tachometers read directly in revolutions per minute.

A detent at the low rpm position prevents inadvertent movement of the propeller lever into the feather position, indicated by the red and white stripes across the lever slots in the quadrant. At the full feather position, the levers position the governor pilot valve to dump oil pressure from the propeller hub, and allow the counterweights and springs to position the propeller blades to the feather position.

AUTOFEATHER SYSTEM

The automatic feathering system provides a means of immediately dumping oil pressure from the propeller hub, thus enabling the feathering spring and counterweights to start the feathering action of the blades in the event of an engine failure (Figure 7-41). Although the system is armed by a switch on the subpanel, placarded "AUTOFEATHER" and "ARM-OFF-TEST," the completion of the arming phase occurs when both power levers are advanced above 90% N_1, at which time both the right and left indicator lights on the annunciator panel indicate a fully armed system (Figure 7-42). The annunciator panel

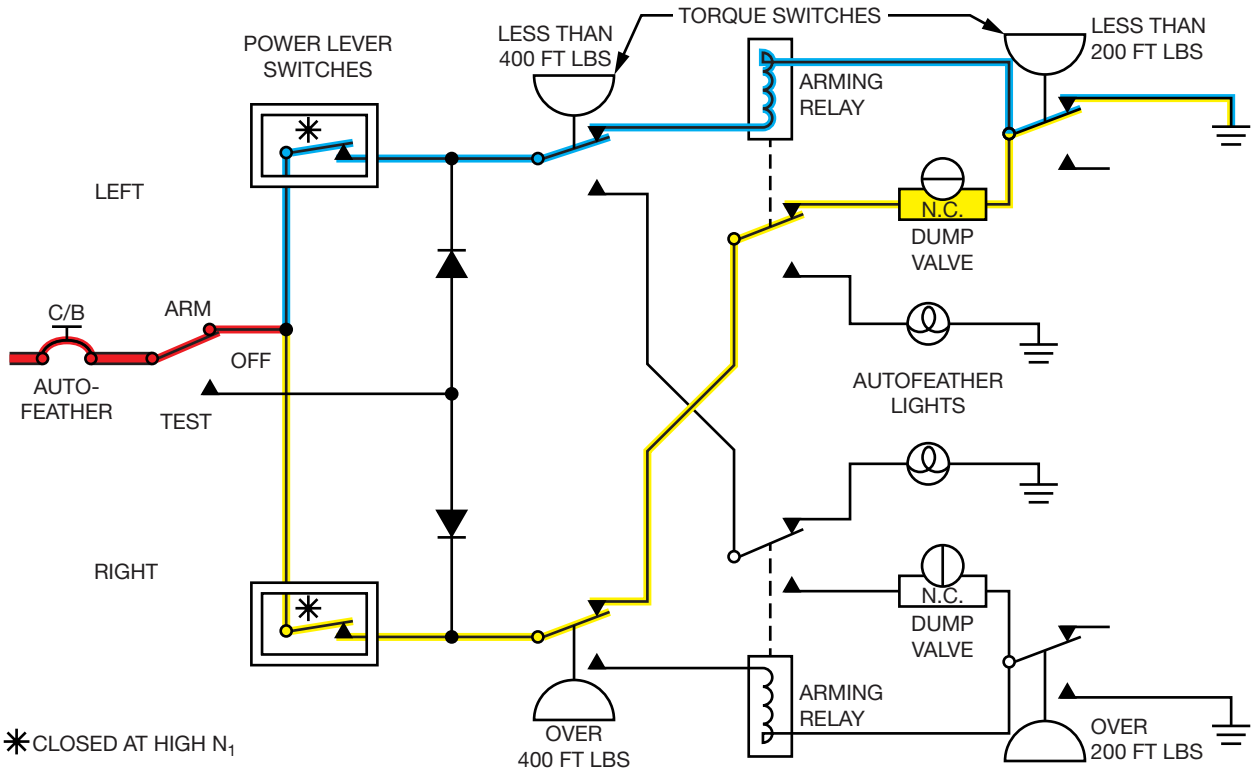

Figure 7-41. Autofeather System Diagram—Left Engine Failed and Feathering

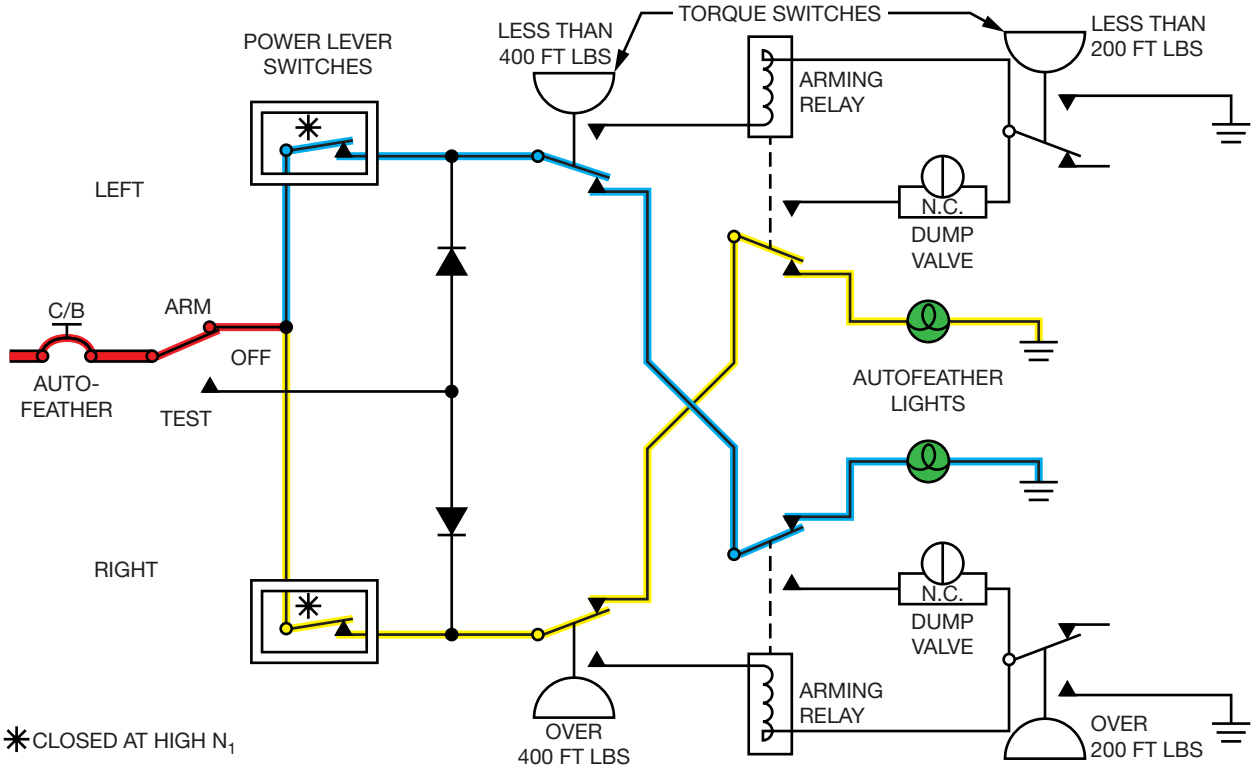

Figure 7-42. Autofeather System Diagram—Armed

lights are green, placarded "L AUTOFEATHER" and "R AUTOFEATHER." The system will remain inoperative as long as either power lever is retarded below 90% N_1 position. The system is designed for use only during takeoff, climb, and missed approach and should be turned off when establishing cruise. With the system armed, if torquemeter oil pressure on either engine drops below a prescribed setting, the oil is dumped from the servo, the feathering spring starts the blades toward feather, and the autofeather system of the other engine is disarmed. Disarming of the autofeather portion of the operative engine is further indicated when the annunciator indicator light for that engine extinguishes. Autofeather System Test

The autofeather test is accomplished with the power below 90% N_1. Therefore, the autofeather switch must be held to TEST so that the power lever switches are bypassed to complete the autofeather circuit (Figure 7-43).

PROPELLER SYNCHROPHASER SYSTEM

A Type II synchrophaser system is installed in the King Air C90A and C90B. The propeller synchrophaser automatically matches the rpm of the two propellers and maintains the blades of one propeller at a predetermined relative position with the blades of the other propeller. The purpose of the system is to reduce propeller beat and cabin noise from unsynchronized propellers.

Synchrophaser Operation

The Type II synchrophaser system (Figure 7-44) is an electronic system, certificated for takeoff and landing. It is not a master-slave system, and it functions to match the rpm of both propellers and establish a blade phase relationship between the left and right propellers to reduce cabin noise to a minimum.

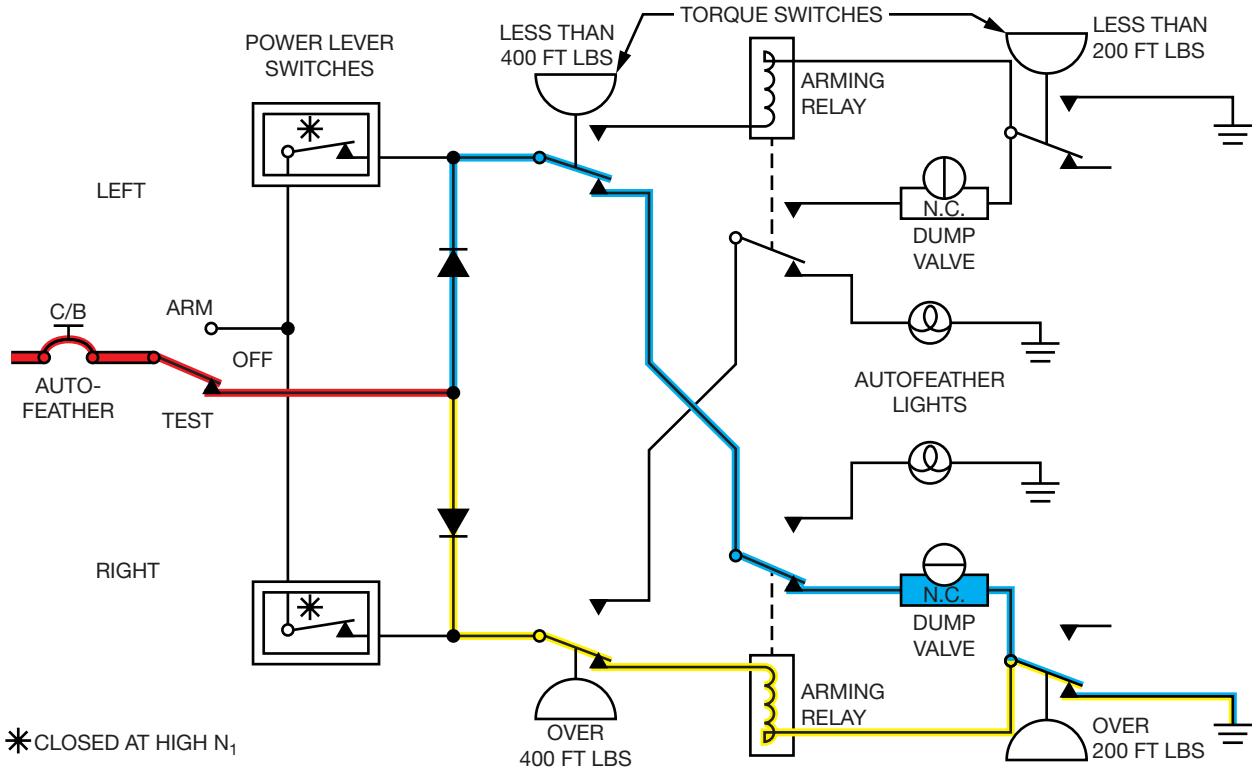

Figure 7-43. Autofeather Test Diagram

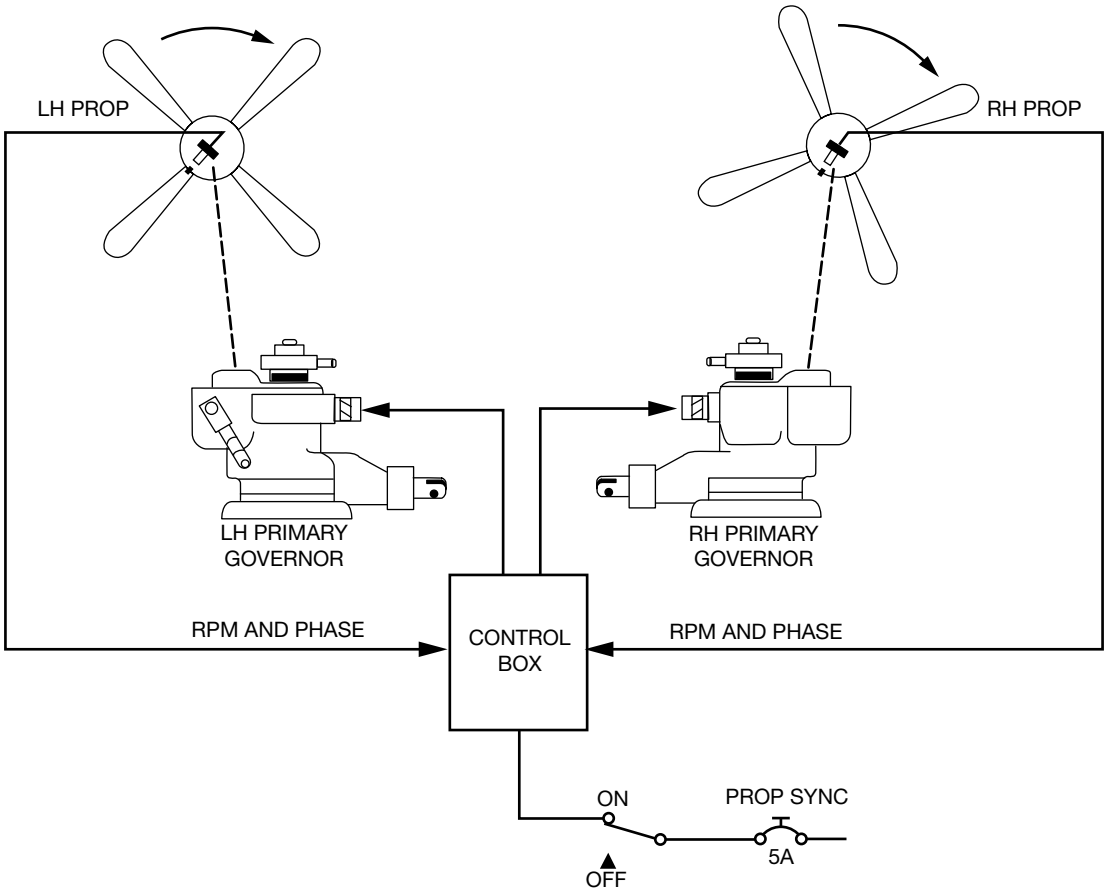

Figure 7-44. Propeller Synchrophaser

The system cannot reduce rpm of either propeller below the datum selected by the propeller control lever. Therefore, there is no indicating annunciator light associated with the Type II system.

To prevent either propeller from losing excessive rpm if the other propeller is feathered while the synchrophaser is on, the synchrophaser has a limited range of authority from the manual governor setting. In no case will the rpm fall below that selected by the propeller control lever. Normal governor operation is unchanged, but the synchrophaser will continuously monitor propeller rpm and reset either governor as required. Propeller rpm and position is sensed by a magnetic pickup mounted adjacent to each propeller spinner bulkhead. This magnetic pick-up will transmit electrical pulses once per revolution to a control box installed forward of the pedestal.

The control box converts any pulse rate differences into correction commands, which are transmitted to coils mounted close to the flyweights of each primary governor. By varying the coil voltage, the governor speed settings are biased until the prop rpm's exactly match. A toggle switch installed adjacent to the synchroscope turns the system on. In the synchrophaser OFF position, the governors operate at the manual speed settings selected by the pilot. To operate the synchrophaser system, synchronize the propellers manually or establish a maximum of 20 rpm difference between the engines, then turn the synchrophaser on. The system may be on for takeoff and landing.

To change rpm with the system on, adjust both propeller controls at the same time. If the synchrophaser is on but does not adjust the prop rpm to match, the system has reached the end of its range. Increasing the setting of the slow prop, or reducing the setting of the fast prop, will bring

the speeds within the limited synchrophaser range. If preferred, turn the synchrophaser switch off, resynchronize manually, and turn the synchrophaser on.

Propeller Synchroscope

A propeller synchroscope is located to the left of the oil pressure/temperature indicators and gives the status of propeller synchronization. The face of the synchroscope has a black and white cross pattern which can spin either left or right. If the right propeller rpm is greater than the left, the face turns clockwise or right. With the left propeller rpm greater than the right, the face turns counterclockwise or left. No rotation of the face indicates that both propellers are synchronized.

QUESTIONS

1. The PT6A engine used on the King Air C90A/B/GT is defined as a:

A. Twin-spool straight-flow gas generator.
B. Reverse-flow, dual shaft, free-turbine turboprop engine.
C. Reverse-flow, single-shaft turboprop engine.
D. Single-spool, axial-type gas generator.

2. The PT6A engine power section consists of:

A. One compression stage and four turbine stages.
B. A two-stage turbine and a centrifugal compressor.
C. A single-stage reaction turbine.
D. One compressor and one power turbine.

3. The function of the reduction gear system is to provide gear reduction:

A. Between the compressor and the compressor turbine.
B. Between the compressor and the power turbine.
C. For the airplane's accessory drive section.
D. For the propeller.

4. If a CHIP DETECT annunciator illuminates, the pilot should:

A. Perform the EMERGENCY ENGINE SHUTDOWN checklist.
B. Reduce the power to 500 foot-pounds of torque for the remainder of the flight.
C. Check oil temperature and if normal, no action is required.
D. Monitor the engine instruments, and if normal, no action is required.

5. During ground operation at LO IDLE, you note that the ITT is exceeding 660°. Which of the following actions would you consider best to reduce ITT?

A. Move the propeller control lever to the low rpm position
B. Reduce accessory load or increase N_1 rpm
C. Move the power lever into the ground fine (Beta)/reverse range
D. Shut down and have the propeller LO IDLE stops checked

6. During a ground start of the right engine, the IGNITION ON annunciator should illuminate:

A. At 10% N_1 rpm.
B. When the condition lever is moved to LO IDLE.
C. When the N_1 is stabilized at 12% or more.
D. When the start switch is moved to the IGNITION and ENGINE START position.

7. When the AUTO-IGNITION switch is in the ARM position, ignition is:

A. Continuous.
B. Inactive but armed if torque is greater than 400 foot-pounds.
C. Inactive but armed if torque is less than 400 foot-pounds.
D. Continuous when torque is greater than 400 foot-pounds.

8. After lift-off with the AUTOFEATHER armed, if an autofeather is initiated, the immediate requirement is to:

A. Continue to fly the airplane and allow the propeller to feather.
B. Move the power lever to idle.
C. Move the condition lever to cutoff.
D. Feather the propeller manually.

9. Of the following, the most accurate definition of engine torque readout is:

A. Power developed by the gas generator.
B. Thrust supplied by the propeller.
C. Ratio of compressor inlet to exhaust outlet.
D. Power delivered to the propeller.

CHAPTER 7
POWERPLANT (C90GT)

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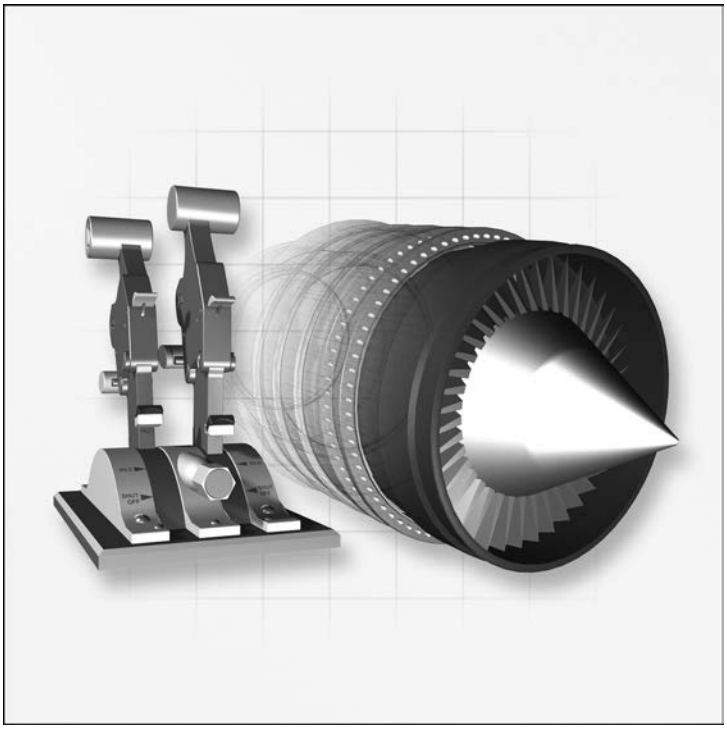
CHAPTER 7
POWERPLANT (C90GT)



INTRODUCTION

In-depth knowledge of the powerplants is essential to good power management by the pilot. Knowing and operating within safe parameters of the powerplant and propeller system extends engine life and ensures safety. This chapter describes the basic sections of the engine and its operational limits and preflight checks.

In-depth knowledge of the propeller system is also essential to proper operation of the engine power system. Operating within safe parameters of the powerplant and propeller systems extends engine life and ensures safety. This chapter also describes the propeller system and its operational limits and preflight checks.

GENERAL

The Engines section of this chapter presents a description and discussion of the Pratt and Whitney PT6A turboprop engines. The engines used on these airplanes will be described in sufficient detail for flight crewmembers to understand normal operational practices and limitations. The purpose of this section is to give the participants a sufficient understanding of the engine so that they will be familiar with normal and emergency procedures.

The Propellers section of this chapter presents a description and discussion of the propeller system. Location and use of propeller controls, principle of operation, reversing, and feathering are included.

ENGINES

General

The powerplants chosen by Beech designers for the King Airs are Pratt and Whitney Series PT6A free-turbine turboprop engines (Figure 7 GT-1 and Figure 7 GT-2). The King Air C90GT uses PT6A-135A engines. The PT6A-135A engine is Flat Rated to 550 shaft horsepower.

The engines are equipped with conventional four-blade, full-feathering, reversing, variable-pitch propellers mounted on the output shaft of the engine reduction gearbox. The propeller pitch and speed are controlled by engine oil pressure through single-action, engine-driven propeller governors. The propellers will feather automatically when the engines are shut down on the ground, and will unfeather when the engines are started.

When reference is made to the right or left side of the airplane or engine, it is always looking from the rear to the front.

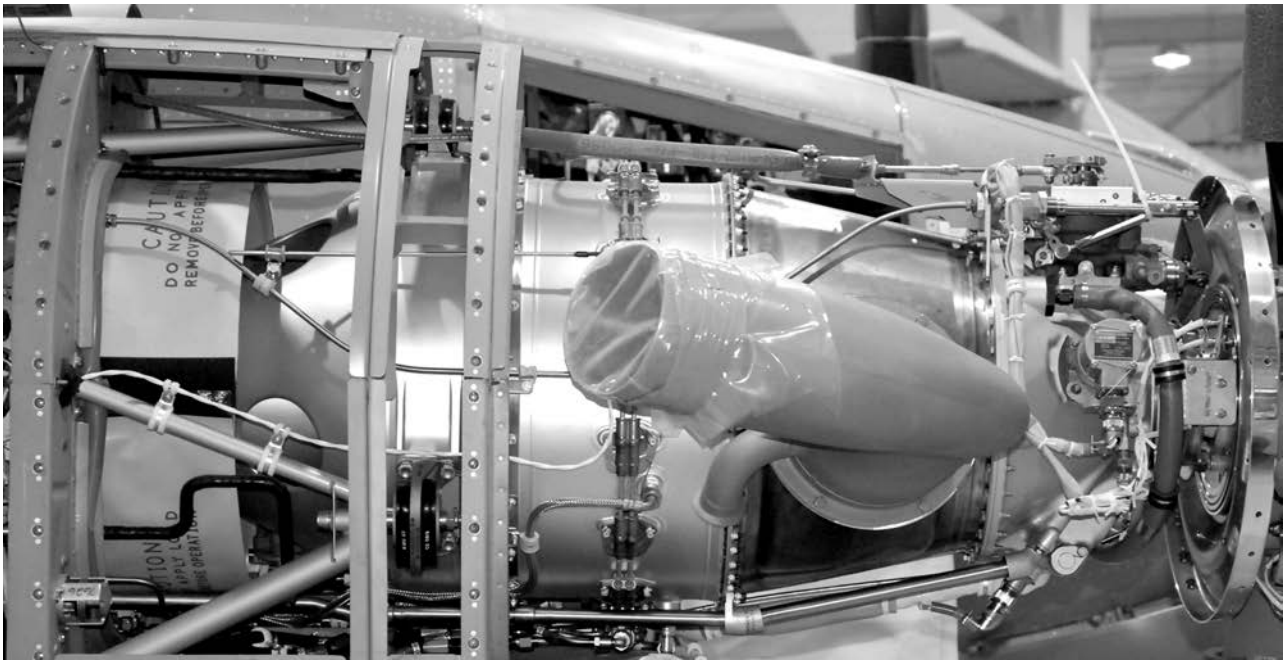


Figure 7 GT-1. Powerplant Installation

TURBOPROP ENGINE RATINGS

In turboprop engines, power is measured in Equivalent Shaft Horse Power (ESHP) and Shaft Horse Power (SHP). SHP is determined by propeller rpm and torque applied to turn the propeller shaft. The hot exhaust gases also develop some kinetic energy as they leave the engine, similar to a turbojet engine. This jet thrust amounts to about 10% of the total engine power. ESHP is the term applied to total power delivered, including the jet thrust. Turboprop engine specifications usually show both ESHP and SHP, along with limiting ambient temperatures.

ENGINE TERMS

To properly understand the operation of the PT6A series engines, there are several basic terms you should know:

N_1 or N_G-Gas generator rpm is percent of turbine speed

- N_2 or N_P-Propeller rpm
- N_F-Power turbine rpm (not indicated on engine instruments)
- P_3-Air pressure at station three (the source of bleed air)
- ITT or T_5-Interstage Turbine Temperature in degrees of temperature at station 5

Review and remember these terms. They will be used often to describe PT6A engines.

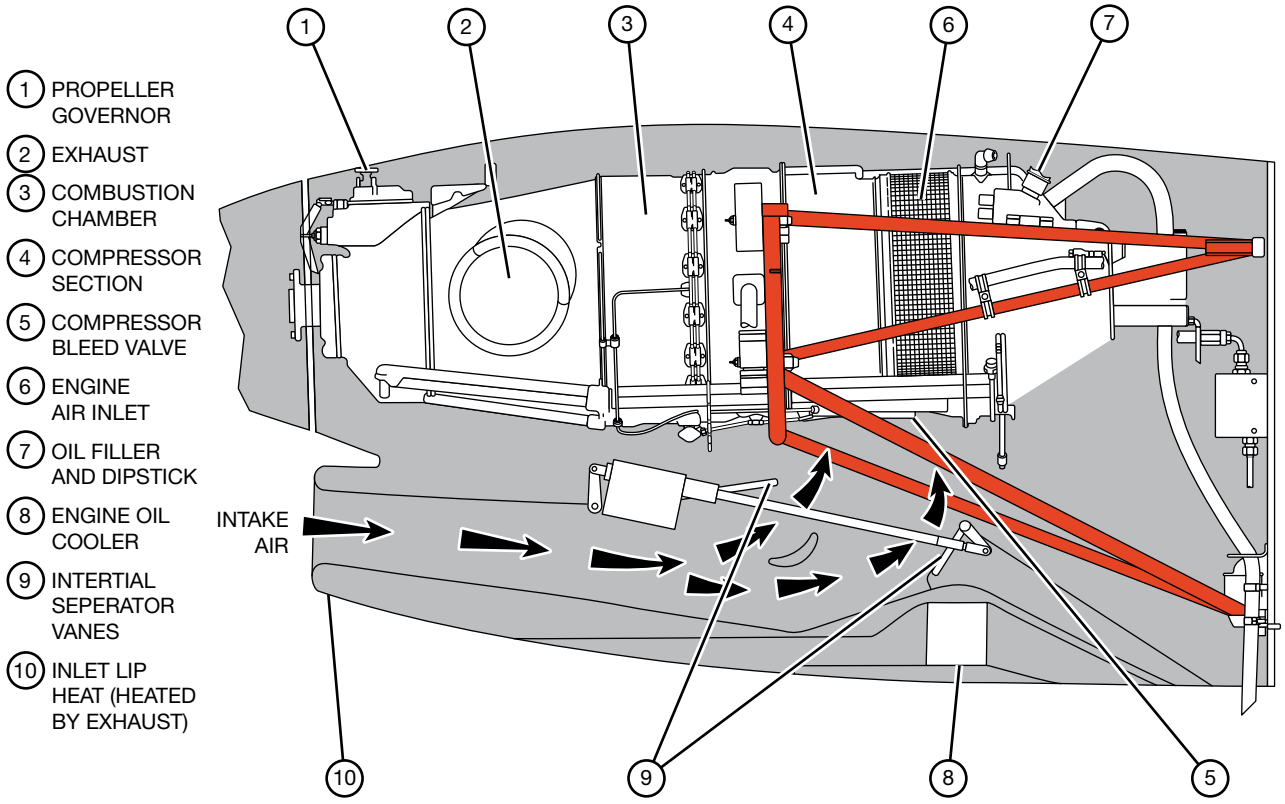


Figure 7 GT-2. Engine Installation

FREE-TURBINE REVERSE-FLOW PRINCIPLE

The Pratt and Whitney PT6 family of engines consists basically of free-turbine, reverse-flow engines driving a propeller through planetary gearing (Figure 7 GT-3, Figure 7 GT-4, Figure 7 GT-5, and Figure 7 GT-6). The term "free-turbine" refers to the design of the turbine sections of the engine. There are two turbine sections: one, called the compressor turbine, which drives the engine compressor and accessories; and the other, consisting of a single power turbine, which drives the power section and propeller. The power turbine section has no physical connection to the compressor turbine at all. These turbines are mounted on separate shafts and are driven in opposite directions by the gas flow across them. The term "reverse flow" refers to airflow through the engine. Inlet air enters the compressor at the aft end of the engine, moves forward through the combustion section and the turbines, and is exhausted at the front of the engine.

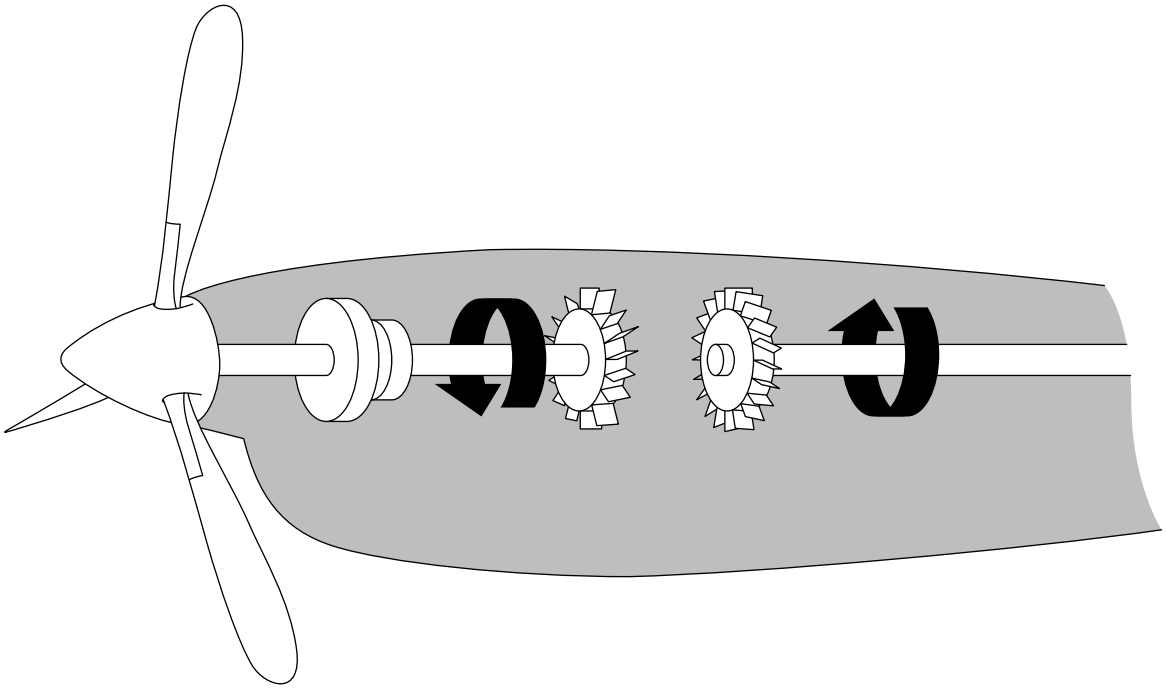


Figure 7 GT-3. Free Turbine

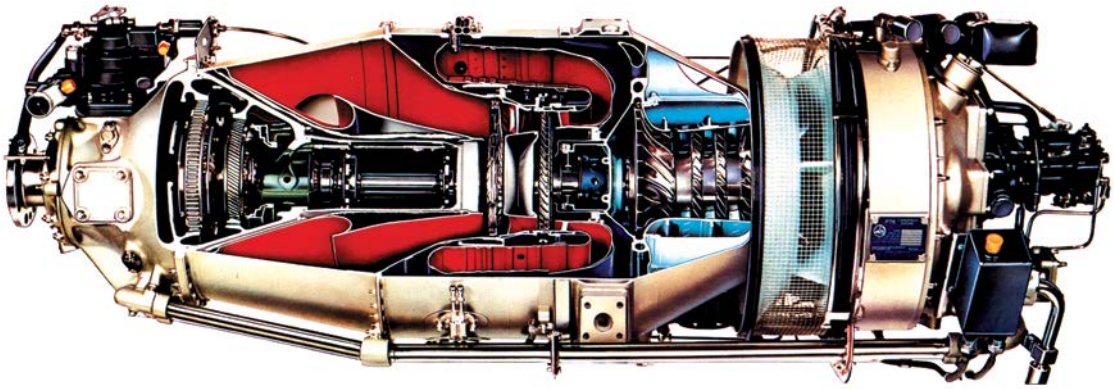


Figure 7 GT-4. Engine Cutaway

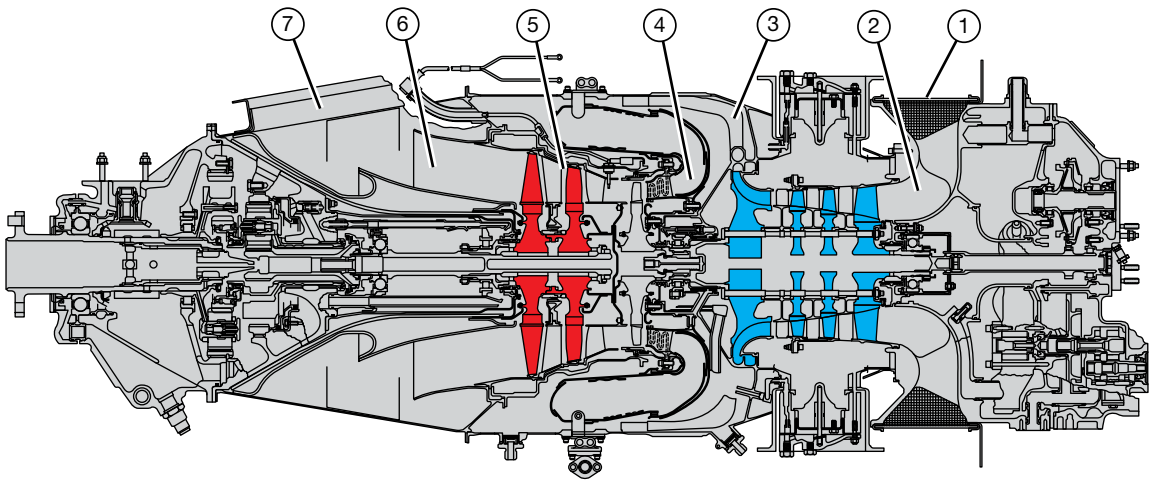


Figure 7 GT-6. Engine Stations

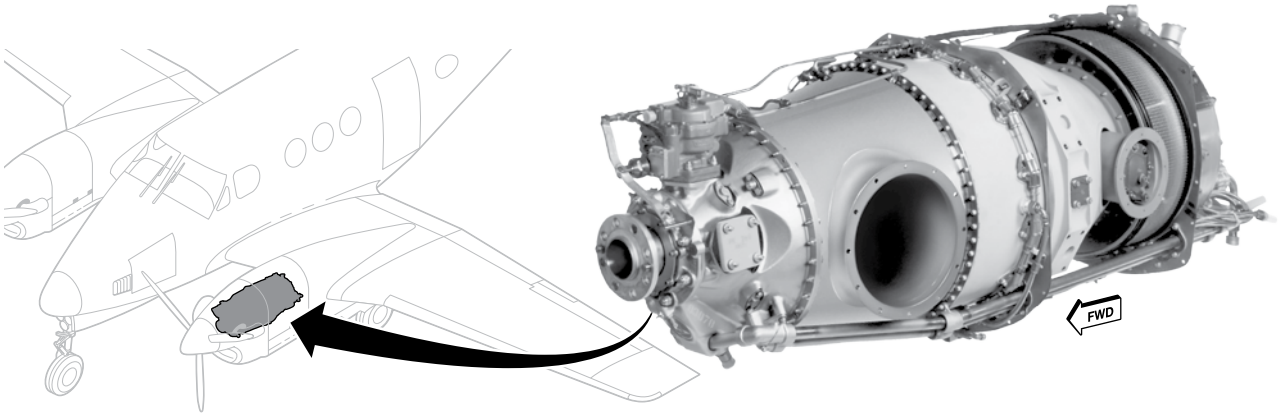


Figure 7 GT-5. Engine Orientation

ENGINE AIRFLOW

Inlet air enters the engine through an annular plenum chamber, formed by the compressor inlet case, where it is directed forward to the compressor (Figure 7 GT-7, and Figure 7 GT-8). The compressor consists of three axial stages combined with a single centrifugal stage, assembled as an integral unit.

A row of stator vanes, located between each stage of compression, diffuses the air, raises its static pressure, and directs it to the next stage of compression. The compressed air passes through diffuser tubes, which turn the air through 90° in direction and convert velocity to static pressure. The diffused air then passes through straightening vanes to the annulus surrounding the combustion chamber liner.

The combustion chamber liner has varying size perforations which allow entry of compressor delivery air. Approximately 25% of the air mixes with fuel to support combustion. The remaining 75% centers the flame in the combustion chamber and provides internal cooling for the engine. As it enters the combustion area and mixes with fuel, the flow of air changes direction 180°. The fuel/air mixture is ignited, and the resultant expanding gases are directed to the turbines. The location of the liner eliminates the need for a long shaft between the compressor and the compressor turbine, thus reducing the overall length and weight of the engine.

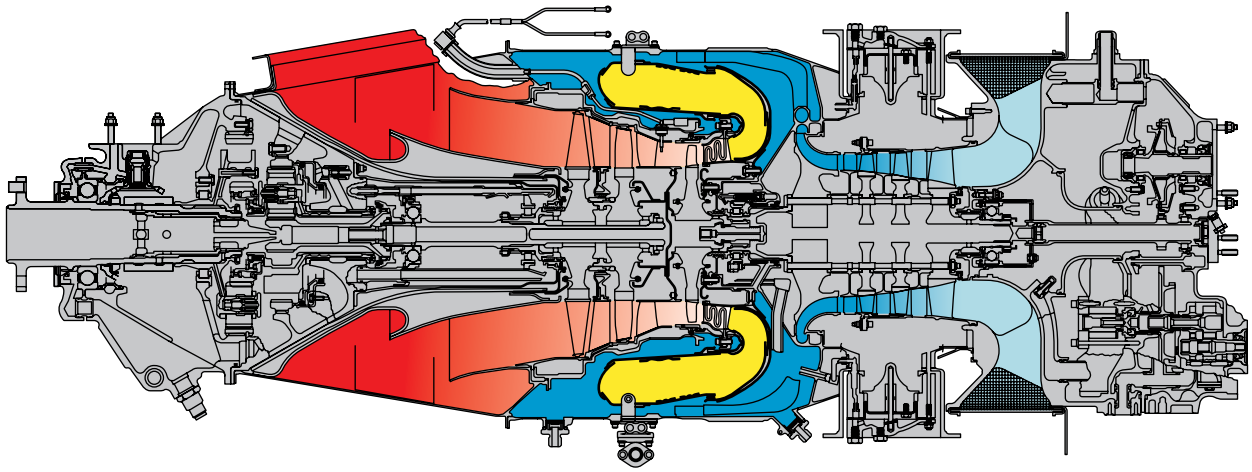


Figure 7 GT-7. Engine Gas Flow

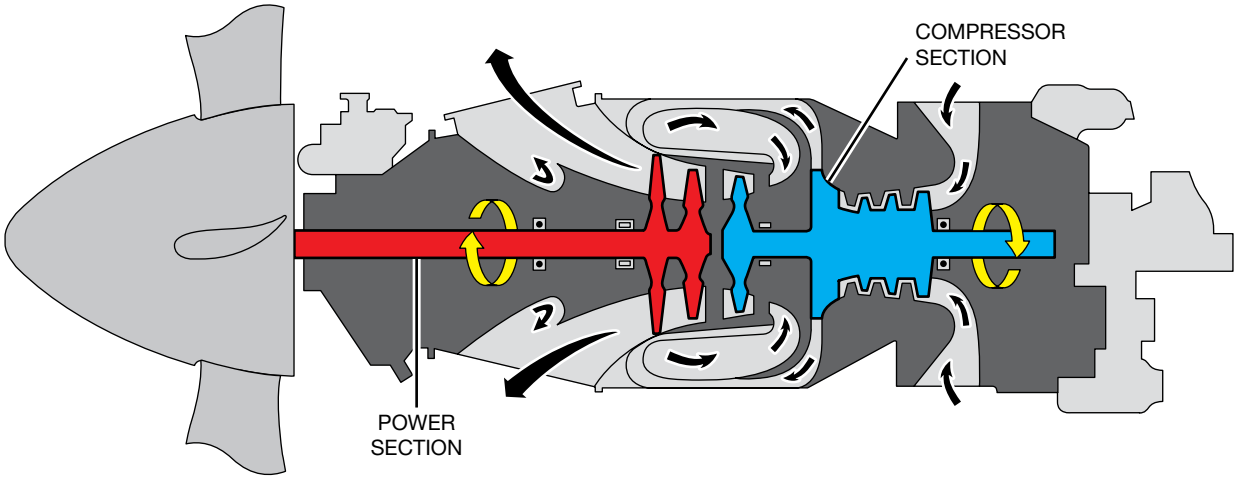


Figure 7 GT-8. Power and Compressor Sections

During normal operation, fuel is injected into the combustion chamber liner through 14 simplex nozzles, which are supplied by a dual manifold consisting of primary and secondary transfer tubes and adapters. During starting, the fuel/air mixture is ignited by two spark igniters which protrude into the liner. After starting, the igniters are turned off, since combustion is self-sustaining. The resultant gases expand from the liner, reverse direction in the exit duct zone, and pass through the compressor turbine inlet guide vanes to the single-stage compressor turbine. The guide vanes ensure that the expanding gases impinge on the turbine blades at the correct angle, with minimum loss of energy. The expanding gases are then directed forward to drive the power turbine section.

The single-stage power turbine, consisting of an inlet guide vane and turbine, drives the propeller shaft through a reduction gearbox.

The compressor and power turbines are located in the approximate center of the engine, with their respective shafts extending in opposite directions. This feature simplifies the installation and inspection procedures. The exhaust gas from the power turbine is directed through an annular exhaust plenum to atmosphere through twin opposed exhaust ports provided in the exhaust duct.

ENGINE STATIONS

To identify points in the engine, it is common practice to establish engine station numbers at various points (Figure 7 GT-5). To refer to pressure or temperature at a specific point in the engine airflow path, the appropriate station number is used, such as P_3 for the Station 3 pressure or T_5 for the gas temperature at Station 5. For instance, temperature of the airflow is measured between the compressor turbine and the power turbine at Engine Station Number 5. This is called Inter-stage Turbine Temperature (ITT) or T_5. Bleed air is taken off the engine after the centrifugal compressor stage and prior to entering the combustion chamber. This air, commonly referred to as P_3 air, is used for cabin heat, pressurization, and the pneumatic system.

ENGINE MODULAR CONCEPT

With the modular free-turbine design, the engine is basically divided into two modules: a gas generator section and a power section (Figure 7 GT-9). The gas generator section includes the compressor and the combustion section. Its job is to draw air into the engine, add energy to it in the form of burning fuel, and produce the gases necessary to drive the compressor and power turbines.

The power section's job is to convert the gas flow from the gas generator section into mechanical action to drive the propeller. This is done through an integral planetary gearbox, which converts the high speed and low torque of the power turbine to the low speed and high torque required at the propeller. The reduction ratio from power turbine shaft rpm to propeller rpm is approximately 15:1.

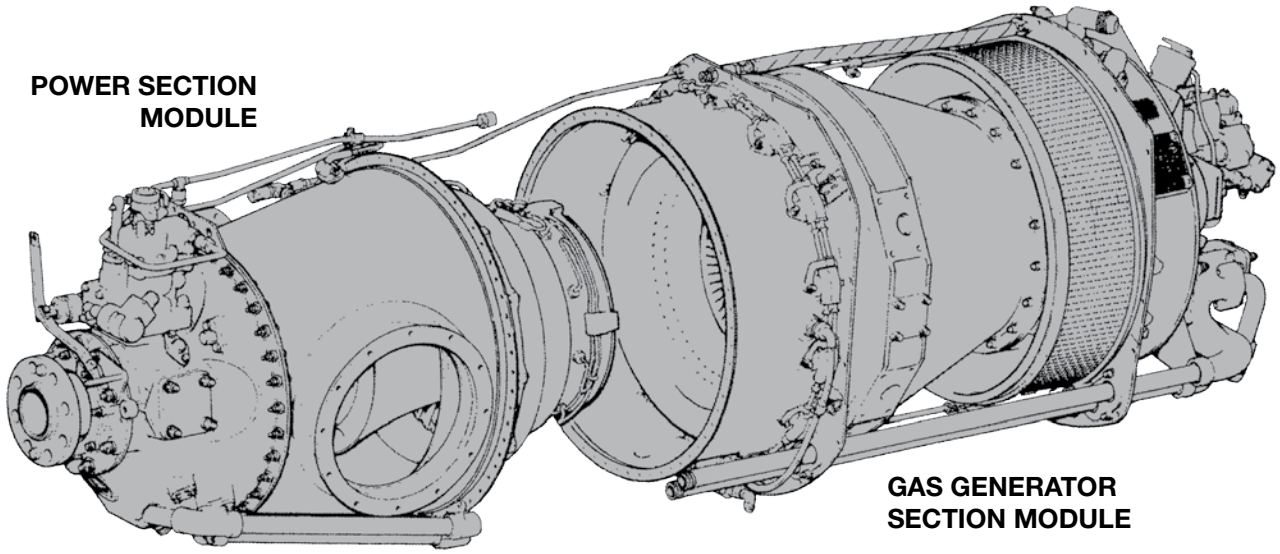


Figure 7 GT-9. Typical Engine Modular Construction

COMPRESSOR BLEED VALVE

At low N_1 rpm, the axial compressors produce more compressed air than the centrifugal compressor can effectively handle (accept). A compressor bleed valve compensates for this excess airflow at low rpm by opening, to relieve this pressure. As compressor speed increases, the valve closes proportionally until, at 80% N_1, the valve is fully closed (Figure 7 GT-10). This pressure relief helps prevent compressor stall of the centrifugal stage.

The compressor bleed valve is a pneumatic piston which references the pressure differential between the axial and centrifugal stages. Looking forward, the valve is located at the 6 o'clock position. The function of this valve is to prevent compressor stalls and surges in the low N_1 rpm range (75 to 80% N_1).

At low N_1 rpm, the valve is in the open position. At takeoff and cruise N_1 rpm, above approximately 80%, the bleed valve will be closed. If the compressor bleed valve sticks closed, a compressor stall will result. If the valve sticks open, the ITT would be noticably higher as the power lever is advanced above 80% N_1.

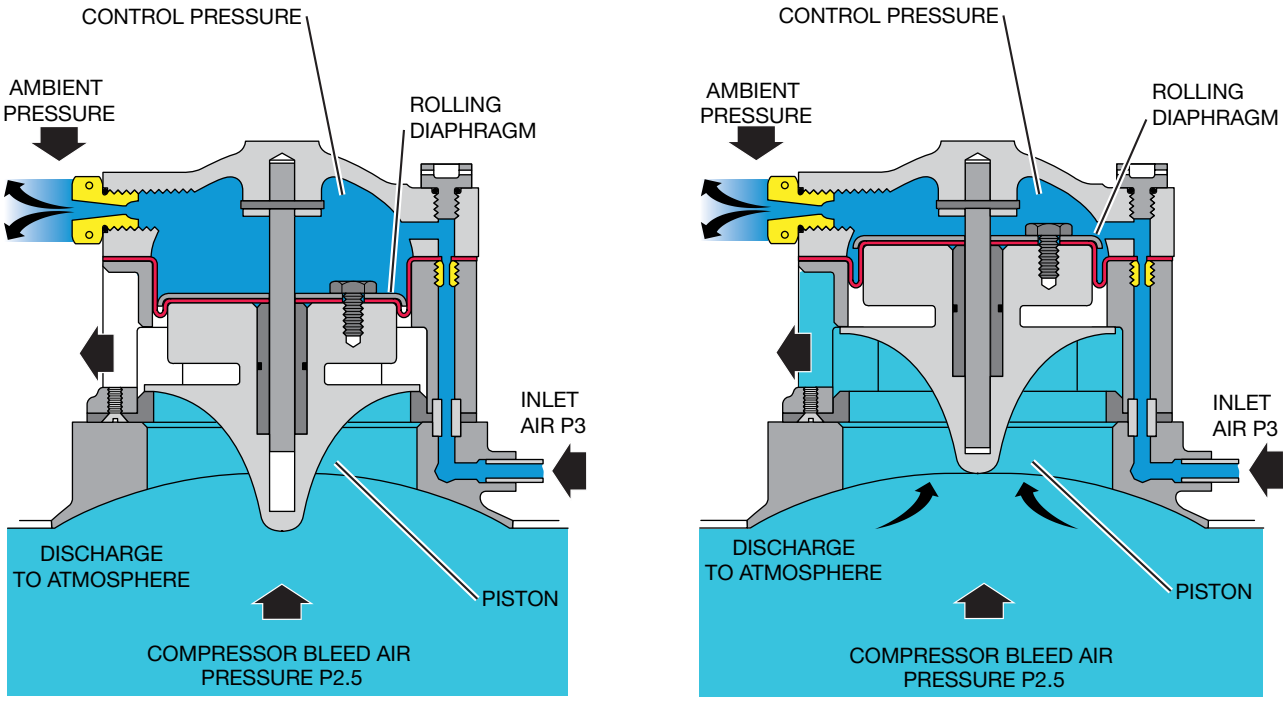


Figure 7 GT-10. Compressor Bleed Valve

IGNITERS

The engine start switches are located on the pilot's left subpanel (Figure 7 GT-11). This subpanel contains the IGNITION AND ENGINE START switches and ENG AUTO IGNITION switches.

The IGNITION AND ENGINE START switches have three positions: ON, OFF, and STARTER ONLY. The ON position is lever-locked and activates both the starter and igniters. The STARTER ONLY position is a momentary hold-down position of the spring-loaded-to-center OFF position. It provides for motoring only to clear the engine of unburned fuel. With the switch in this position, there is no ignition.

The combustion chamber has two spark-type igniters to provide positive ignition during engine start. While the engine is equipped with two igniters, it will start with only one. The system is designed so that if one igniter is open or shorted, the remaining igniter will continue to function. Once the engine is started, the igniters are de-energized, since the combustion is self-sustaining.

The ignition system features an automatic backup function for emergencies. This backup system is called "autoignition." The ENG AUTO IGNITION switches should be moved to the ARM position just prior to takeoff. If engine torque falls below approximately 400 ft-lb, the igniter will automatically energize, attempting to restart the engine. The IGNITION ON annunciator will be illuminated.

The spark ignition provides the engine with an ignition system capable of quick light-ups over a wide temperature range. The system consists of an airframe-mounted ignition exciter, two individual high-tension cable assemblies, and two spark igniters. It is energized from the aircraft nominal 28-VDC supply and will operate in the 9- to 30-volt range. The igniter control box produces up to 3,500 volts. The ignition exciter is energized only during the engine starting sequence and emergencies to initiate combustion in the combustion chamber.

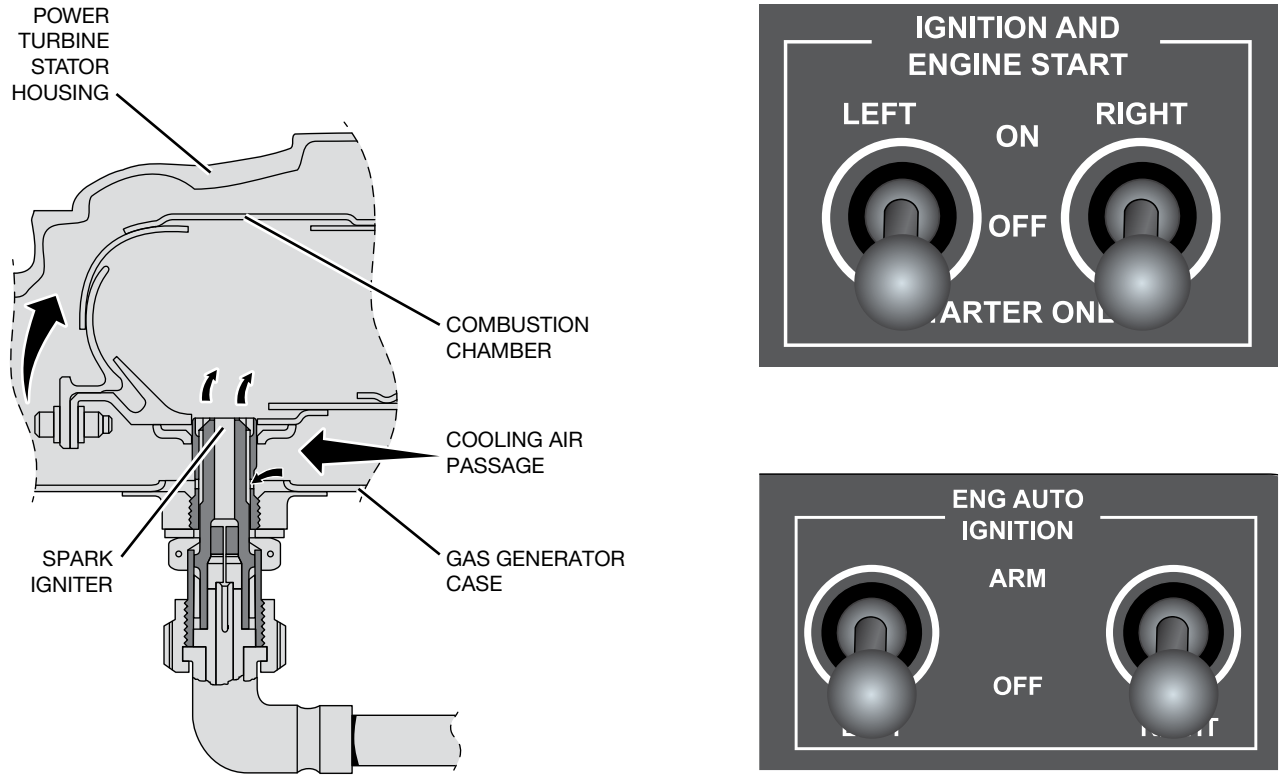
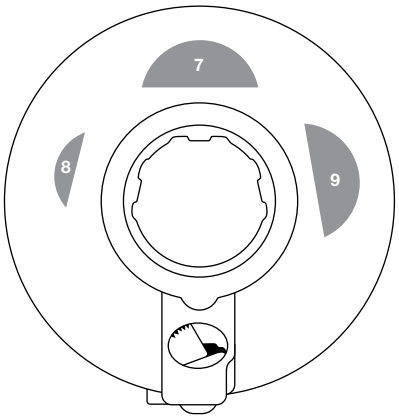


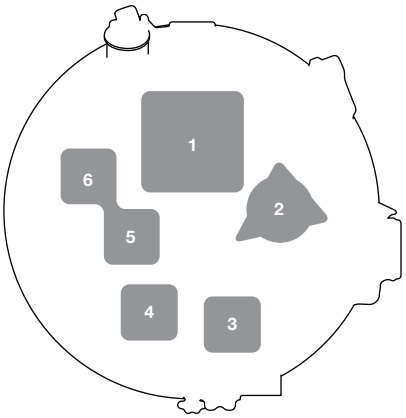
Figure 7 GT-11. Engine Start and Ignition Switches

ACCESSORY SECTION

Most of the engine-driven accessories, except the propeller governors and propeller tach generator, are mounted on the accessory gearbox located at the rear of the engine (Figure 7 GT-12). The accessories are driven from the compressor shaft through a coupling shaft.



AS VIEWED FROM FRONT



AS VIEWED FROM REAR

ACCESSORY SECTIONS

REAR ACCESSORY DRIVES

1. STARTER-GENERATOR
2. FUEL PUMP/FCU
3. TACHOMETER-GENERATOR (N_G)
4. VACUUM AIR PUMP (OPTIONAL)
5. OPTIONAL ACCESSORY DRIVE
6. OPTIONAL ACCESSORY DRIVE

FRONT ACCESSORY DRIVES

7. PROPELLER GOVERNOR
8. TACHOMETER-GENERATOR (N_F)
9. PROPELLER OVERSPEED GOVERNOR

Figure 7 GT-12. Typical PT6A Engine

The lubricating and scavenge oil pumps are mounted inside the accessory gearbox, with the exception of the two scavenge pumps which are externally mounted.

The starter-generator, high-pressure fuel pump, N_1 tachometer generator, and other optional accessories are mounted on pads on the rear of the accessory drive case. There are seven such mounting pads, each with its own different gear ratio.

LUBRICATION SYSTEM

The PT6A engine lubrication system hasa dual function (Figure 7 GT-13). Its primary function is to cool and lubricate the engine bearings and bushings. Its second function is to provide oil to the propeller governor and propeller reversing control system.

The main oil tank houses a gear-type engine-driven pressure pump, oil pressure regulator, and oil filter. The engine oil tank is an integral part of the compressor inlet case and is located in front of the accessory gearbox.

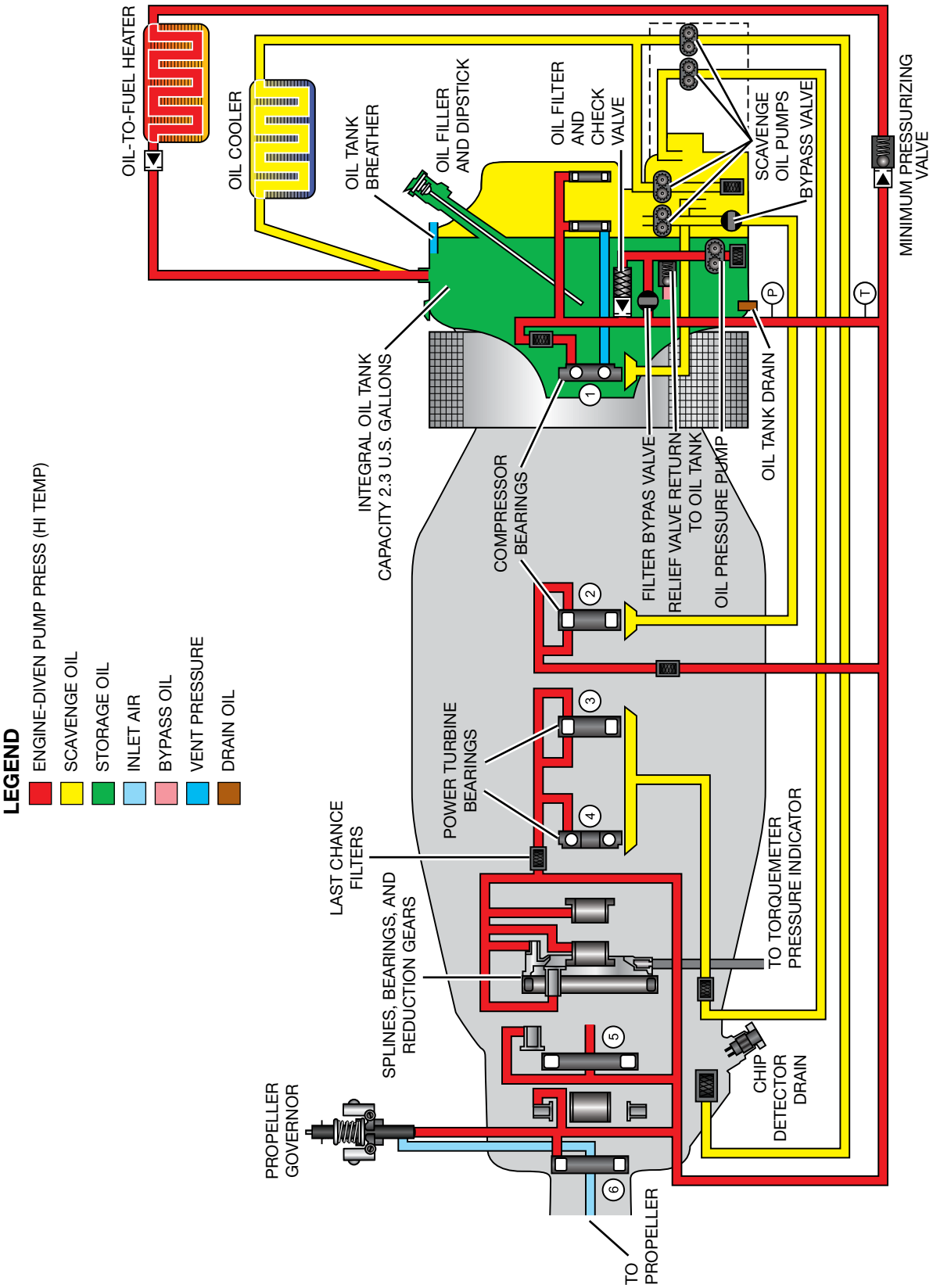


Figure 7 GT-13. Engine Lubrications Diagram

The oil tank is provided with a filler neck and integral quantity dipstick housing. The cap and dipstick are secured to the filler neck, which passes through the gearbox housing and accessory diaphragm and into the tank. The markings on the dipstick indicate the number of U.S. quarts of oil less than full (Figure 7 GT-14).

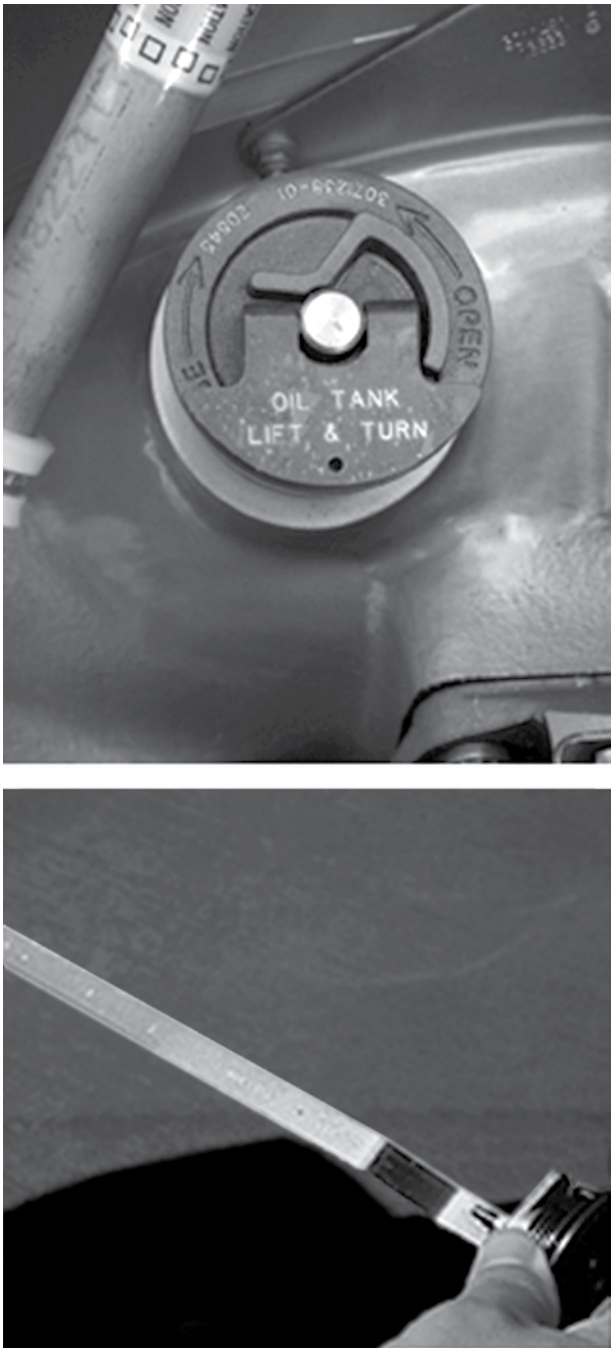


Figure 7 GT-14. Engine Oil Dipstick

The engine oil system has a total capacity of 3.5 U.S. gallons, including the 2.3-gallon oil tank. Maximum oil consumption is one quart every 10 hours of operation. Normal oil consumption may be as little as 1 quart per 50 hours of operation.

The dipstick will indicate 1 to 2 1/2 quarts below full when the oil level is normal. Do not overfill. When adding oil between oil changes, do not mix types or brands of oil due to the possibility of chemical incompatibility and loss of lubricating qualities.

A placard inside the engine cover shows the brand and type of oil used in that particular engine. Although the preflight checklist calls for checking the oil level, which is required, the best time to check oil quantity is shortly after shutdown, since oil levels are most accurately indicated at that time.

Oil level checks during preflight may require motoring the engine for a brief time for an accurate level reading. Each engine tends to seek its own oil level. The pilot should monitor the oil level to ensure proper operation.

As pressure oil leaves the tank, it passes through the pressure and temperature-sensing bulbs mounted on or near the rear accessory case. The oil then proceeds to the various bearing compartments and nose case through an external oil transfer line below the engine. Scavenge oil returns from the nose case and the bearing compartments to the gear-type oil scavenge pumps in the accessory case through external oil transfer lines, and through the external oil cooler below the engine.

The oil cooler is thermostatically controlled to maintain the desired oil temperature. Another externally mounted unit, the oil-fuel heat exchanger, uses hot engine oil to heat fuel before it enters the engine fuel system. When gas generator speeds are above 72% N_1, and oil temperatures are between 60 and 70°C, normal oil pressure is between 85 and 105 psi.

Magnetic Chip Detector

A magnetic chip detector is installed in the bottom of each engine nose gearbox (Figure 7 GT-15). This detector will activate a yellow light on the

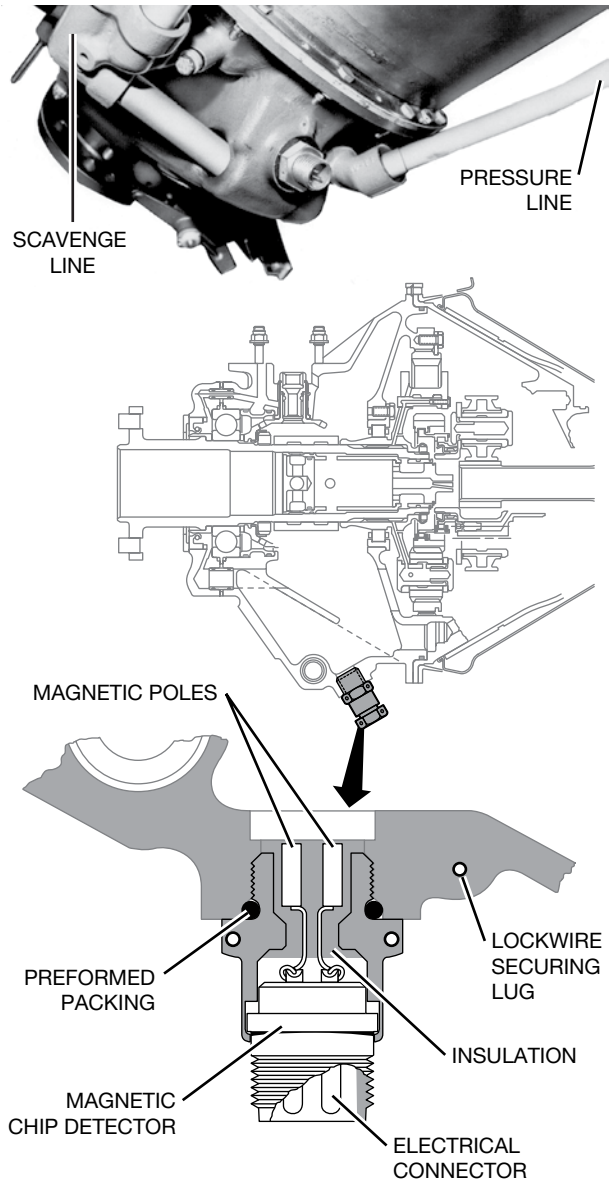


Figure 7 GT-15. Magnetic Chip Detector

annunciator panel, L CHIP DETECT or R CHIP DETECT, to alert the pilot of oil contamination.

C90GT aircraft, engine parameters should be monitored for abnormal indications. If such indications are observed, appropriate check list action should be taken.

A steady "CHIP DETECT" light requires the engine be shut down to prevent serious internal damage.

When a CHIP DETECT annunciator light comes on and stays on, timely action is required to prevent serious damage to the internal engine components. The chip detector indicates the presence of ferrous particles in the propeller gearbox.

ENGINE FUEL SYSTEM

The fuel control system for PT6A engines is essentially a fuel governor that increases or decreases fuel flow to the engine to maintain selected engine operating speeds. At first glance, the system may appear quite complicated. The engine fuel control system consists of the main components shown in the block diagram (Figure 7 GT-16). They are the electric low-pressure boost pump, oil-to-fuel heat exchanger, high-pressure fuel pump, fuel control unit, fuel cutoff valve, flow divider, and dual fuel manifold with 14 simplex nozzles.

The PT6A-135A engine uses an electric low-pressure boost pump to supply a 30-psi head pressure to the high-pressure engine-driven fuel pump. This head pressure prevents fuel cavitation at the high-pressure pump. The fuel is also used for cooling and lubricating the pump. The oil-to-fuel heat exchanger uses warm engine oil to maintain a desired fuel temperature at the fuel pump inlet to prevent icing at the pump filter. This is done with automatic temperature sensors and requires no action by the pilot.

Fuel enters the engine fuel system through the oil-to-fuel heat exchanger, and then flows into the high-pressure engine-driven fuel pump and on into the fuel control unit (FCU).

The high-pressure fuel pump is an engine-driven gear-type pump with an inlet and outlet filter. Flow rates and pressures will vary with gas generator (N_1) rpm. Its primary purpose is to provide sufficient pressure at the fuel nozzles for a proper spray pattern during all modes of engine operation. The high-pressure pump supplies fuel at approximately 800 psi to the fuel side of the FCU.

Two valves included in the FCU ensure consistent and cool engine starts. When the ignition or start system is energized, the purge valve is electrically opened to clear the FCU of vapors and bubbles.

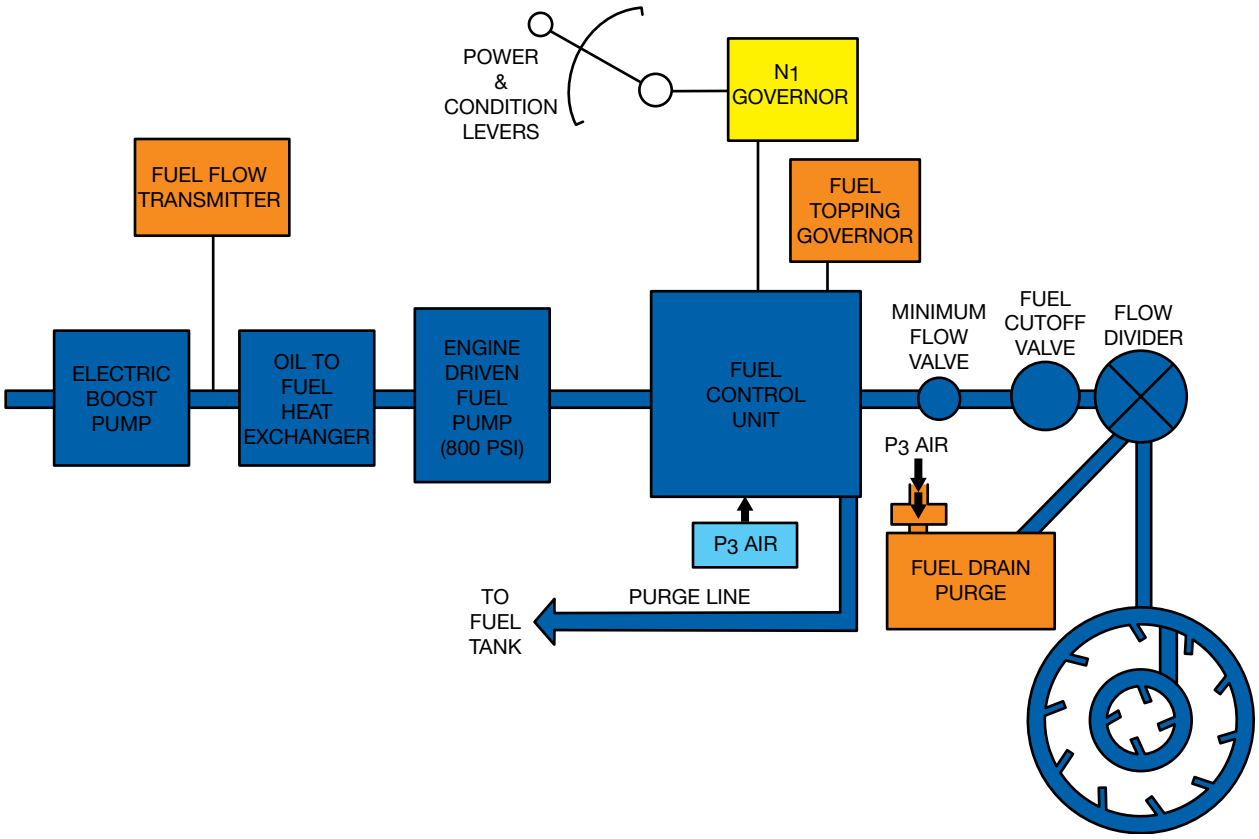


Figure 7 GT-16. Simplified Fuel System Diagram

The excess fuel flows back to the nacelle fuel tanks. The spill valve, referenced to atmospheric pressure, adjusts the fuel flow for cooler high-altitude starts.

Between the FCU fuel valve and the engine combustion chamber, the minimum pressurizing valve in the FCU remains closed during starting until fuel pressure builds sufficiently to maintain a proper spray pattern in the combustion chamber. About 80 psi is required to open the minimum pressurizing valve. If the high pressure fuel pump should fail, the valve would close, and the engine would flame out.

The fuel cutoff valve is located downstream from the minimum pressurizing valve in the FCU. This valve is controlled by the condition lever, either open or closed. There is no intermediate position of this valve. For starting, fuel flows initially through the flow divider to the 10 primary fuel nozzles in the combustion chamber. As the engine accelerates through approximately 40% N_1, fuel pressure is sufficient to open the flow divider to the 4 secondary fuel nozzles. At this time all 14 nozzles are delivering atomized fuel to the combustion chamber. This progressive sequence of primary and secondary fuel nozzle operation provides cooler starts. During engine starting, there is a noticable increase in ITT when the secondary fuel nozzles are activated.

During engine shutdown, any fuel left in the manifold is forced out through the fuel nozzles and into the combustion chamber by purge tank pressure. As the fuel is burned, a momentary increase in N_1 rpm may be observed. The entire operation is automatic and requires no input from the crew.

FUEL CONTROL UNIT

The fuel control unit (Figure 7 GT-17), which is referred to as the FCU, has multiple functions, but its primary purpose is to meter proper fuel amounts to the fuel nozzles in all modes of engine operation.

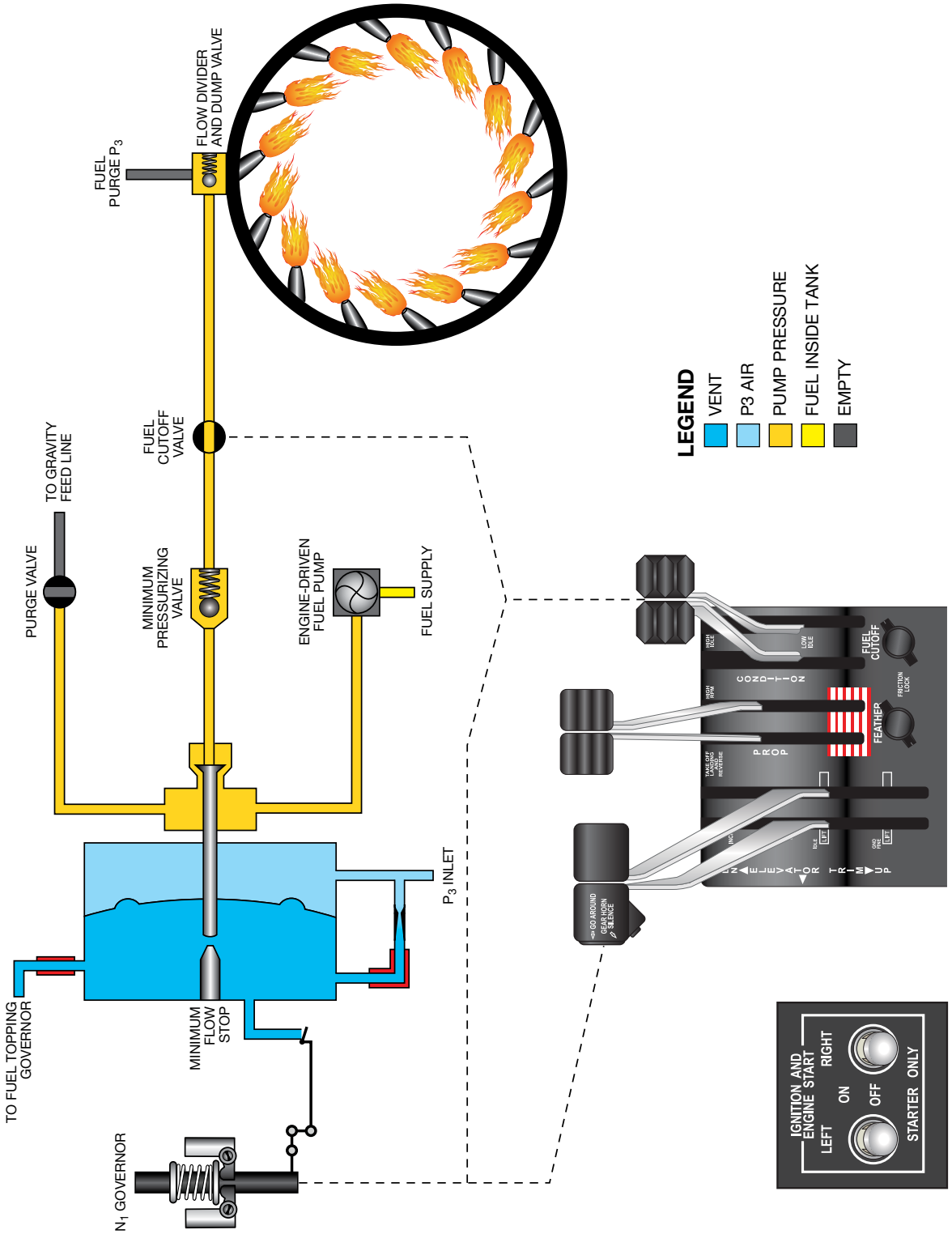


Figure 7 GT-17. Simplified Fuel Control System

FCU operation will be simplified and described briefly here. For detailed description and operation, refer to the *Pratt & Whitney Maintenance Manual* which applies to this engine.

The condition lever selects idle speeds between LOW IDLE (58% to 62% N_1) to HIGH IDLE (70% N_1), while the power lever selects speeds between idle and maximum, 101.5% N_1. These control levers influence the N_1 governor and control N_1 speed. The governor uses pneumatic air (P_3) pressure to control engine speed. The governor controls the air pressure in the fuel control unit by varying the P_3 leak rate.

The P_3 air chamber and fuel chamber are separated by a diaphragm, which has a needle valve mounted on it which is called the metering valve. As the diaphragm is influenced by varying air/fuel pressures, the metering valve is repositioned to achieve the desired fuel flow. The N_1 governor controls fuel flow by allowing some P_3 pressure to be leaked off at varying rates, depending on the desired fuel flow.

In an underspeed condition, the N_1 governor acts to increase P_3 air pressure. This repositions the metering valve, allowing more fuel to enter the combustion chamber, increasing N_1.

In an overspeed condition, the N_1 governor allows the P_3 pressure to be reduced in the FCU, which repositions the metering valve reducing the fuel flow into the combustion chamber, decreasing N_1.

Should the P_3 air pressure be lost, due to a malfunction, the metering valve will be positioned to the minimum flow stop. Minimum flow power would be approximately 48% N_1. The power lever and condition lever would then have no effect on engine speed.

FUEL PRESSURE INDICATORS

In the event of an electric boost pump failure, the respective FUEL PRESS annunciator (Figure 7 GT-18) will illuminate and the master warning light will flash. The FUEL PRESS light illuminates when outlet pressure at the boost pump decreases below about 10 psi. If the crossfeed switch is in the AUTO position, the automatic crossfeed feature will open the valve extinguishing the annunciator.

Figure 7 GT-18. Fuel Pressure Annunciators

In the event of an engine-driven fuel pump (highpressure) failure, the engine will flame out.

CAUTION

Engine operation with the FUEL PRESS light on is limited to ten hours between overhaul or replacement of the engine-driven high-pressure fuel pump.

FUEL FLOW INDICATORS

Fuel flow information is sensed by a transmitter in the engine fuel supply line, between the boost pump and the engine-driven high-pressure pump, and indicated on the fuel flow gauges in (Figure 7 GT-19). The indication of fuel flow is in pounds-per-hour.

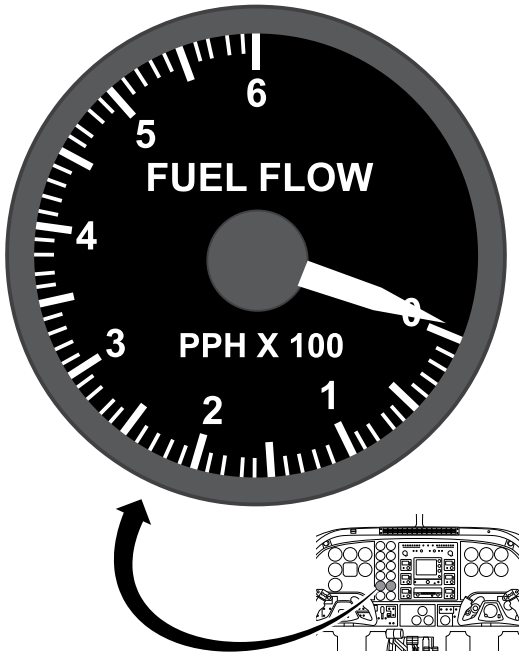


Figure 7 GT-19. Fuel Flow Indicator

ANTI-ICING FUEL ADDITIVE

Engine oil is used to heat the fuel prior to entering the FCU. Since no temperature measurement is available for the fuel at this point, it must be assumed to be the same as the Outside Air Temperature. The Minimum Oil Temperature chart is supplied for use as a guide in preflight planning, based on known or forecast operating conditions, to indicate operating temperatures where icing at the FCU could occur. If the plot should indicate that oil temperature versus OAT is such that ice formation could occur during takeoff or in flight, anti-icing additive per MIL-I-27686 or MIL-I-85470 should be mixed with the fuel at refueling to ensure safe operation. Refer to the *King Air Maintenance Manual* for procedures to follow when blending anti-icing additive with the airplane fuel.

Anti-icing additive conforming to Specification MIL-1-27686 is the only approved fuel additive.

ENGINE POWER CONTROL

The propeller lever adjusts the propeller governor to the desired propeller speed (Figure 7 GT-20). The propeller will maintain the set speed by varying the blade angle. Torque is controlled by the power lever acting on the N_1 governor. When the power lever is advanced, the N_1 governor causes the FCU to increase fuel flow, resulting in an increase in engine speed.

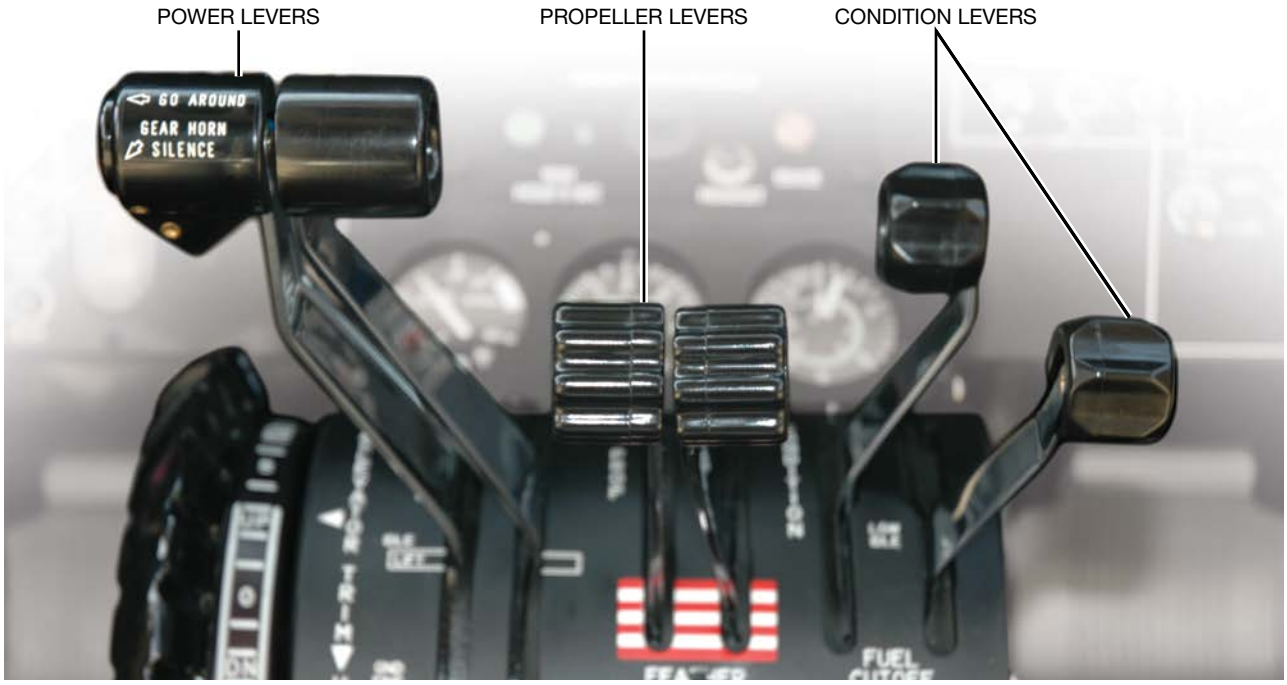


Figure 7 GT-20. Control Levers

ITT AND TORQUEMETERS

Power management is relatively simple, with two primary operating limitations. The engines are temperature and torque limited. During operation requiring maximum engine performance, engine torque and ITT operating parameters are affected by ambient temperature and altitude: at cold temperature or low altitude, torque limits power; at hot temperature or high altitude, ITT limits power. Whichever limit is reached first, determines the power available. These indications can be seen on the Engine Indicating System (EIS) (Figure 7 GT-21).

ITT GAUGE

The ITT gauge (Figure 7 GT-21), monitors the interstage turbine temperature at station 5. ITT is a prime limiting indicator of the amount of power available from the engine under varying ambient temperature and altitude conditions. The normal operating range, is 400 to 805°C. These limits also apply to maximum continuous power. The maximum starting temperature of 1,090°C is indicated by the secondary red line on the instrument. This starting limit of 1,090°C is limited to two seconds. The engines will be damaged if limiting temperatures indicated on the ITT gauge are exceeded.

TORQUEMETER

The torquemeter, (Figure 7 GT-21), which is indicated in ft-lb, constantly measures rotational force applied to the propeller shaft . The maximum permissible sustained torque is 1,520 ft-lb, the red radial on the instrument. A transient torque limit of 1,626 ft-lb is time-limited to twenty seconds. Cruise torques vary with altitude and temperature.

Torque is measured by a hydromechanical torquemeter in the first stage of the reduction gearcase. Rotational force on the first-stage ring gear allows oil pressure to change in the torquemeter chamber. The difference between the torquemeter chamber pressure and reduction gear internal pressure accurately indicates the torque being produced at the propeller shaft. The torque transmitter measures this torque and sends a signal to the instrument on the instrument panel.

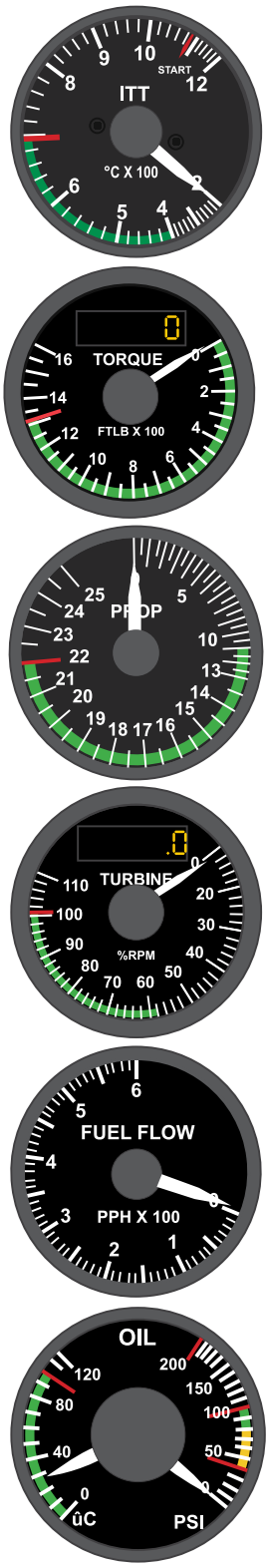


Figure 7 GT-21. Engine Indicating System

GAS GENERATOR TACHOMETER (N_1)

The N_1 gas generator tachometer (Figure 7 GT-21), measures the rotational speed of the compressor shaft, in percent of rpm, based on 37,500 rpm at 100%.

The N_1 indicator is self-generating. The tachometer generator sensing unit, located in the engine accessory section, is geared down to supply N_1 speed information to the instrument panel to indicate the percent of N_1 revolutions.

Maximum continuous gas generator speed is limited to 38,100 rpm, which is 101.5% on the N_1 indicator. A transient speed up to 102.6%, 38,500 rpm, is time-limited to 2 seconds, to provide a buffer for surges during engine acceleration.

CONTROL PEDESTAL

The control pedestal extends between pilot and copilot (Figure 7 GT-22). The three sets of control levers are left to right: the power levers, propeller levers, and the condition levers.

Power Levers

The power levers (Figure 7 GT-20) control engine power, from idle to maximum power, by operation of the N_1 governor in the fuel control unit. Increasing N_1 rpm results in increased engine power. The power levers have three control ranges: Flight, Ground Fine, and Reverse. The bottom of the flight range is at IDLE. When the levers are lifted over the IDLE detent and pulled back, they control engine power through the ground fine and REVERSE ranges.

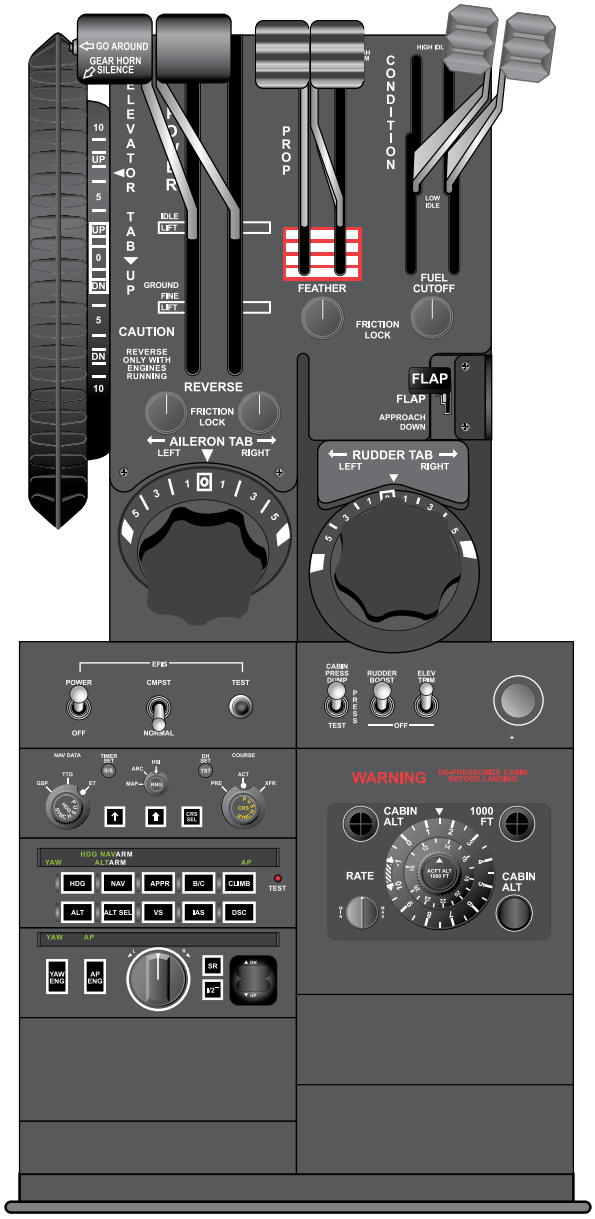


Figure 7 GT-22. Control Pedestal

Condition Levers

The condition levers have multiple positions: FUEL CUTOFF and LO IDLE through HI IDLE (Figure 7 GT-22). At the FUEL CUTOFF position, fuel flow to its respective engine is cut off.

At LO IDLE, engine gas generator speed (N_1) is a minimum of 58% on the C90GT; at HI IDLE it is 70%. The levers can be set anywhere between LOW IDLE and HIGH IDLE.

Propeller Levers

The propeller levers are conventional in setting the propeller rpm for takeoff, climb and cruise (Figure 7 GT-22). The normal governing range is 1,600 to 1,900 rpm. This airplane is equipped with both manual and automatic propeller feathering systems. To feather a propeller manually, pull the propeller lever back past the friction detent into the red and white striped section of the quadrant. To unfeather, push the lever forward

of the detent into the governing range. The propellers go to feathered position when the engines shut down because of the loss of oil pressure in the propeller dome.

Control Lever Operation

The engines are controlled from the cockpit by using the propeller, power, and condition levers. Both the power and condition levers are connected to the N_1 governing section of the FCU. Either lever will reset the FCU to maintain a new N_1 rpm. For starting, the power levers are at the IDLE position, and the condition levers are moved to the LO IDLE position to open the fuel cutoff valves and set the governor at LO IDLE. The condition levers are continuously variable from LO IDLE to HI IDLE. This variable operating speed with power levers at IDLE enhances engine cooling by maintaining a steady airflow through the engines. With the condition levers at LO IDLE, the power levers will select N_1 rpm from LOW IDLE to 101.5%, the maximum for takeoff. However, if the condition levers are at HI IDLE, the power levers can select N_1 rpm only from 70 to 101.5%.

Moving the power or condition levers most directly affects N_1 rpm. As the power or condition levers are advanced, ITT, torque, and fuel flow increases. These indicators are by-products of the N_1 speed maintained by the FCU. With the power levers in a fixed position, N_1 remains constant even in a climb or descent. However, ITT, torque, and fuel flow will vary with altitude, ambient air temperature, and propeller setting.

ENGINE LIMITATIONS

Airplane and engine limits are described in the "Limitations" section of the *POH* (Figure 7 GT-23). These limitations have been approved by the Federal Aviation Administration, and must be observed in the operation of the Beechcraft King Air C90GT. The Engine Operating Limits chart gives the major operating limits. The Power Plant Instrument Markings chart lists the minimum, normal, and maximum limits.

During engine start, temperature is the most critical limit. The ITT starting limit of 1,090°C, represented on the ITT gauge by a red line, is limited to two seconds. During any start, if the indicator needle approaches the limit, the start should be aborted before the needle passes the secondary red line. For this reason, it is helpful during starts to keep the condition lever out of the LO IDLE detent so that the lever can be quickly pulled back to FUEL CUTOFF.

Monitor oil pressure and oil temperature. During the start, oil pressure should come up to the minimum of 40 psi quickly, but should not exceed the maximum at 105 psi. During normal operation the oil temperature and pressure indications should be in the green normal operating range. The green range is from 85 to 105 psi.

Oil pressure between 40 and 85 psi is undesirable; it should be tolerated only for completion of the flight, and then only at a reduced power setting.

Oil pressure below 40 psi is unsafe; it requires that either the engine be shut down or that a landing be made as soon as possible, using minimum power required to sustain flight.

For increased service life of engine oil, an oil temperature between 74 and 80°C is recommended. A minimum oil temperature of 55°C is recommended for oil-to-fuel heater operation at takeoff power. Oil temperature limits are -40 and +99°C. During extremely cold starts, oil pressure may reach 200 psi. Refer to the Engine Limits chart in the *POH* for minimum oil temperature operation limitations.

During ground operations, ITT temperatures are critical. With the condition levers at LO IDLE, high ITT can be corrected by reducing the DC generator and other N_1 loads, then increasing the N_1 rpm by advancing the condition levers to HI IDLE. The air conditioner, for example, draws a heavy load on both engines, and may have to be temporarily turned off. At approximately 70% N_1 rpm, the HI IDLE condition lever position will normally reduce the ITT. At any N_1 below 70%, there is an idle ITT restriction of 685°C maximum. If an ITT above 685°C is observed when running N_1 below 70%, the generator load should be reduced and the N_1 speed increased before re-introducing a load on the engines.

At N_1 speeds of 70% or more, the 685°C restriction is removed, as airflow through the engine is sufficient.

Figure 7 GT-23. Engine Limits Chart

OPERATING CONDITION	SHP	TORQUE FT-LBS (1)	MAXIMUM OBSERVED ITT°C	GAS GENERATOR RPM N_1		PROP RPM N_2	OIL PRESS. PSI (2)	OIL TEMP °C (3)
				RPM	%			
STARTING	---	---	1090 (4)	---	---	---	---	-40 (min)
LOW IDLE	---	---	685 (5)	---	---	1,100 (min)	40 (min)	-40 to 99
HIGH IDLE	---	---	---	---	72	---	---	0 to 99
TAKEOFF AND MAX CONT	550	1,520 (13)	805	38,100	101.5	1,900 (12)	85 to 105	10 to 99
CRUISE CLIMB AND MAX CRUISE	538	1,520 (6) (13)	805	38,100	101.5	1,900 (12)	85 to 105	0 to 99
MAX REVERSE (7)	---	---	805	---	88	1,825	85 to 105	0 to 99
TRANSIENT	---	1,626 (10)	880 (4) (8)	38,500	102.6	2,090	---	104 (11)

NOTES:

1. MAXIMUM PERMISSIBLE SUSTAINED TORQUE IS 1520 FT-LBS. PROPELLER SPEEDS (N_2) MUST BE SET SO AS NOT TO EXCEED POWER LIMITATION.
2. WHEN GAS GENERATOR SPEEDS ARE ABOVE 72% N_1 AND OIL TEMPERATURES ARE BETWEEN 60°C AND 70°C, NORMAL OIL PRESSURE IS BETWEEN 85 AND 105 PSI. OIL PRESSURES BETWEEN 40 AND 85 PSI ARE UNDESIRABLE; THEY SHOULD BE TOLERATED ONLY FOR THE COMPLETION OF THE FLIGHT, AND THEN ONLY AT A REDUCED POWER SETTING. OIL PRESSURES BELOW 40 PSI ARE UNSAFE; THEY REQUIRE THAT EITHER THE ENGINE BE SHUT DOWN, OR THAT A LANDING BE MADE AT THE NEAREST SUITABLE AIRPORT, USING THE MINIMUM POWER REQURIED TO SUSTAIN FLIGHT.
3. FOR INCREASED SERVICE LIFE OF ENGINE OIL, AN OIL TEMPERATURE OF BETWEEN 74° TO 80°C IS RECOMMENDED. A MINIMUM OIL TEMPERATURE OF 55°C IS RECOMMENDED FOR FUEL HEATER OPERATION AT TAKE-OFF POWER.
4. THESE VALUES ARE TIME-LIMITED TO TWO SECONDS.
5. HIGH ITT AT GROUND IDLE MAY BE CORRECTED BY REDUCING ACCESSORY LOAD AND/OR INCREASING N_1 RPM.
6. CRUISE TORQUE VALUES VARY WITH ALTITUDE AND TEMPERATURE.
7. REVERSE POWER OPERATION IS LIMITED TO ONE MINUTE.
8. HIGH GENERATOR LOADS AT LOW N1 SPEEDS MAY CAUSE THE ITT TRANSIENT TEMPERATURE LIMIT TO BE EXCEEDED. OBSERVE GENERATOR LOAD LIMITS.
9. STABILIZED PROPELLER OPERATION ON THE GROUND BETWEEN 500 AND 1,100 RPM IS PROHIBITED. OPERATION IN THIS RANGE CAN GENERATE HIGH PROPELLER STRESSES, WHICH CAN CAUSE PROPELLER DAMAGE AND RESULT IN PROPELLER FAILURE AND LOSS OF CONTROL OF THE AIRCRAFT. THE PROPELLER MAY BE OPERATED WHEN FEATHERED AT OR BELOW 500 RPM.
10. THE VALUE IS TIME-LIMITED TO 20 SECONDS.
11. THIS VALUE IS TIMED-LIMITED TO 10 MINUTES.
12. TO ACCOUNT FOR POWER SETTING ACCURACY AND STEADY STATE FLUCTUATIONS, INADVERTENT PROPELLER EXCURSIONS UP TO 1,938 RPM ARE TIME-LIMITED TO 7 MINUTES.
13. TO ACCOUNT FOR POWER SETTING ACCURACY AND STEADY STATE FLUCTUATIONS, INADVERTENT TORQUE EXCURSIONS UP TO 1,550 FT-LIBS ARE TIME-LIMITED TO 7 MINUTES.

In the climb, torque will decrease and ITT may increase slightly. The cruise climb and recommended normal cruise ITT limit is not placarded on the indicator. At altitude, the Performance Chart numbers may not be attainable due to altitude and temperature variations.

Transient limits provide buffers for surges during engine acceleration. Torque has an allowable excursion duration of twenty seconds while the ITT has an allowable excursion duration of two seconds. A momentary peak of 1,626 ft-lb and 880°C is allowed for torque and ITT respectively during acceleration.

STARTER OPERATING TIME LIMITS

The engine starters are time-limited during the starting cycle if for any reason multiple starts are required in quick sequence. The starter is limited to 40 seconds ON then 60 seconds OFF for cooling before the next sequence of 40 seconds ON, 60 seconds OFF. After the third cycle of 40 seconds ON, the starter must stay OFF for 30 minutes. If these limits are not observed, overheating may damage the starter.

Trend Monitoring

During normal operations, gas turbine engines are capable of producing rated power for extended periods of time. Engine operating parameters, such as output torque, interstage turbine temperature, compressor speed, and fuel flow for individual engines are predictable under specific ambient conditions. On PT6A engines, these predictable characteristics may be taken advantage of by establishing and recording individual engine performance parameters. These parameters can then be compared periodically to predicted values to provide day-to-day visual confirmation of engine efficiency.

The Engine Condition Trend Monitoring System, recommended by Pratt and Whitney, is a process of periodically recording engine instrument readings such as torque, interstage turbine temperature, compressor speed, and fuel flow, correcting the readings for altitude, outside air temperature, and airspeed, if applicable, and then comparing them to a set of typical engine characteristics. Such comparisons produce a set of deviations in interstage turbine temperature, compressor speed, and fuel flow.

DATA COLLECTION FORM

The trend monitoring procedure used specifies that flight data be recorded on each flight day, every five flight hours, or other flight period. Select a flight with long established cruise, preferably at a representative altitude and airspeed. With engine power established and stabilized for a minimum of five minutes, record the following data on a form similar to the in-flight engine data log shown in (Figure 7 GT-24):

Indicated airspeed (IAS) ..... In knots

Outside air temperature (OAT)..... In °C

Pressure altitude (ALT)..... In feet

Propeller speed (NP)..... In rpm

Torque (TQ) ..... In foot pounds

Gas generator speed
(N_G or N_1)..... In %N_G or N_1

Interturbine temperature (ITT)..... In °C

Fuel Flow (FF) ..... In pph

DATE	OAT (°)	PRESS ALT	IAS (KTS)	PROP SPEED		TORQUE (FT/LBS)	N_1 (%)	ITT	FUEL FLOW	DELTA* NG	DELTA* ITT	DELTA* FF	OIL TEMP	OIL PRESS	ELECT LOAD
					LEFT										
					RIGHT										
					LEFT										
					RIGHT										
					LEFT										
					RIGHT										
					LEFT										
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					LEFT										
					RIGHT										

Figure 7 GT-24. In-Flight Engine Data Log

PROPELLERS

GENERAL

This section describes the propellers and the associated system. Location and use of propeller controls, principle of operation, reversing, and feathering are included in this discussion.

PROPELLER SYSTEM

This section on the operation and testing of the propeller system on the Beechcraft King Air C90GT is directed at increasing the pilot's understanding of the theory of operation of a constant-speed, full-feathering, reversing propeller, and helping him better understand the propeller system checks conducted as outlined in the Before Takeoff (Runup) checklist in the *Pilot's Operating Handbook.*

Each engine is equipped with a conventional four-blade, full-feathering, constant-speed, counterweighted, reversing, variable-pitch propeller mounted on the output shaft of the reduction gearbox (Figure 7 GT-25).

The propeller pitch is controlled by engine oil pressure boosted through a governor pump integral within the propeller governor. Centrifugal counterweights and feathering springs move the propeller blades toward high pitch and into the feathered position. Without oil pressure to counteract the counterweights and feathering springs, the propeller blades would move into feather. An oil pump, which is part of the propeller governor, boosts engine oil pressure to move the propeller to low pitch and reverse. The propeller feathers after engine shutdown.

Propeller tiedown boots (Figure 7 GT-26) are provided to prevent windmilling at zero oil pressure when the airplane is parked.

Low pitch propeller position is determined by the primary low pitch stop, which is a mechanically actuated hydraulic stop. Blade angles are controlled by the power levers in the Ground Fine and Reverse ranges.

Figure 7 GT-25. Propellers

Figure 7 GT-26. Propeller Tiedown Boot Installed

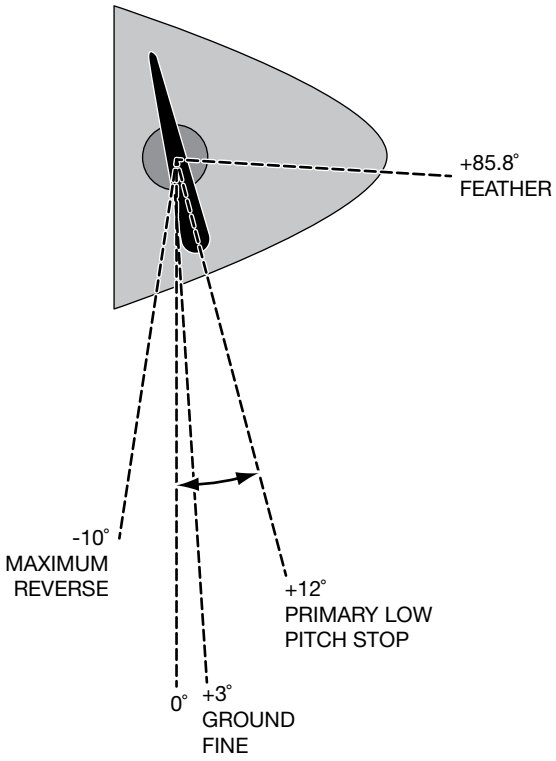
Two governors, a primary governor and a backup overspeed governor, control the propeller rpm. The propeller control lever adjusts the governor's setting (1,600 to 1,900 rpm). The overspeed governor will limit the propeller to 1,976 rpm should the primary governor malfunction. However, if the propeller exceeds 6% above the selected rpm of the primary governor, usually the fuel topping governor will limit the rpm by reducing engine power. In the Ground Fine and Reverse ranges, the fuel topping governor is reset to limit the propeller rpm to 95% of selected rpm.

HARTZELL FOUR-BLADE PROPELLERS

The C90GT is equipped with Hartzell 90-inch-diameter, four-blade, full-reversing, dynamically balanced propellers. The main advantages of the four-blade propellers are that they have lower tip speeds (and thus generate less noise), create less airframe vibration, and provide generous propeller tip-to-ground clearance. Dynamic vibration absorbers mounted inside the cockpit and cabin (a total of 26 absorbers) are used in conjunction with the four-blade propellers to reduce noise and vibration even more.

BLADE ANGLE

Blade angle is the angle between the chord of the propeller and the propeller's plane of rotation. Blade angle is different near the hub than it is near the tip, due to the normal twist which is incorporated in a blade to increase its efficiency. In the propellers used on the C90GT King Air, the blade angle is measured at the chord 30 inches out from the propeller's center. This position is referred to as the "30-inch station." All blade angles given in this section are approximate (Figure 7 GT-27).



C90GT PROPELLER BLADE ANGLES

Figure 7 GT-27. Blade Angle Diagram

PRIMARY GOVERNOR

The primary governor (Figure 7 GT-28) is needed to convert a variable-pitch propeller into a constant-speed propeller. It does this by changing blade angle to maintain the propeller speed the operator has selected. The primary governor can maintain any selected propeller speed from approximately 1,600 rpm to 1,900 rpm.

Suppose an airplane is in normal cruising flight with the propeller turning 1,900 rpm. If the pilot trims the airplane down into a descent without changing power, the airspeed will increase. This decreases the angle of attack of the propeller blades, causing less drag on the propeller, thus beginning to increase its rpm. Since this propeller has a variable-pitch capabilities and is equipped with a governor set at 1,900 rpm, the governor will sense this "overspeed" condition and increases blade angle to a higher pitch. The higher pitch increases the blade's angle of attack, slowing it back to 1,900 rpm, or "onspeed."

Likewise, if the airplane moves from cruise to climb airspeeds without a power change, the propeller rpm tends to decrease, but the governor responds to this "underspeed" condition by decreasing blade angle to a lower pitch, and the rpm returns to its original value. Thus the governor gives "constant-speed" characteristics to the variable-pitch propeller.

Power changes, as well as airspeed changes, cause the propeller to momentarily experience overspeed or underspeed conditions, but again the governor reacts to maintain the onspeed condition.

There are times, however, when the primary governor is incapable of maintaining selected rpm. For example, imagine an airplane approaching to land with its governor set at 1,900 rpm. As power and airspeed are both reduced, underspeed conditions exist which cause the governor to decrease blade angle to restore the onspeed condition. If blade angle could decrease all the way, to 0°or reverse, the propeller would create so much drag on the airplane that the aircraft control would be dramatically reduced. The propeller, acting as a large disc, would blank the airflow around the tail surfaces, and a rapid nosedown pitch change would result.

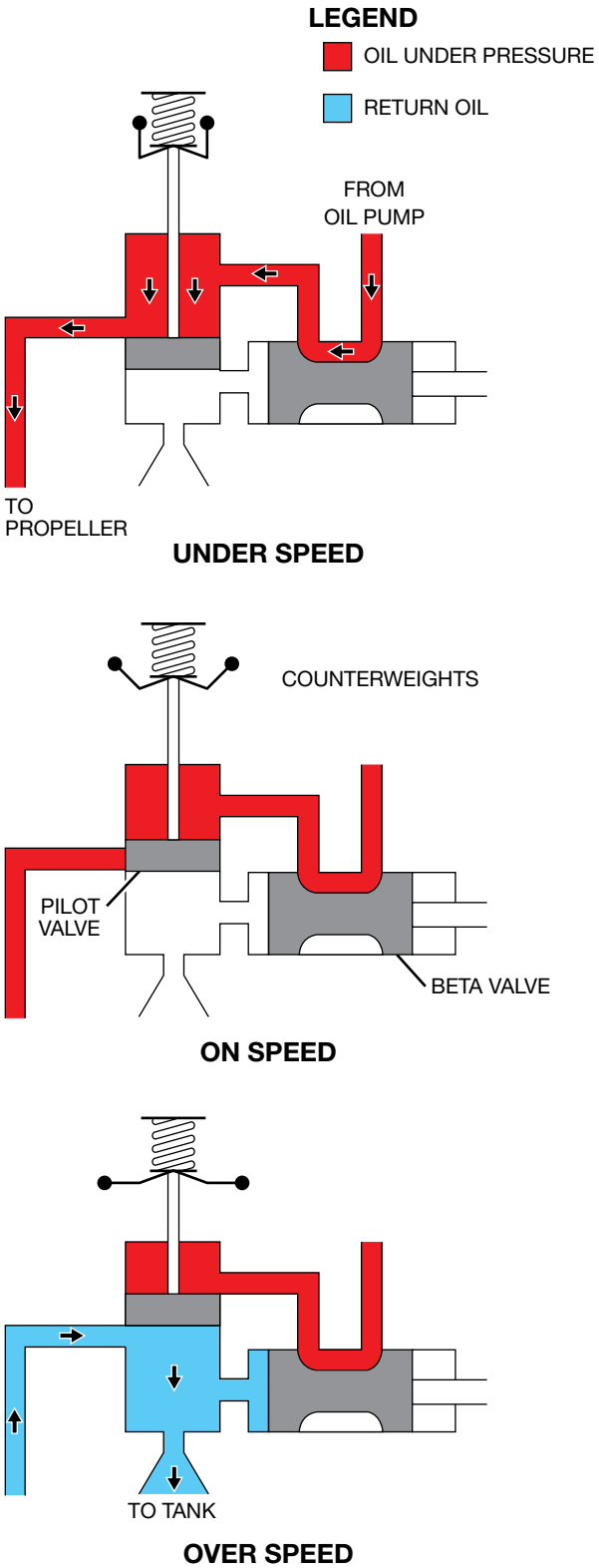


Figure 7 GT-28. Primary Governor Diagram

To prevent these unwanted aerobatics, some device must be provided to stop the governor from selecting blade angles that are too low for safety. As the blade angle is decreased by the governor, eventually the low pitch stop is reached, and now the blade angle becomes fixed and cannot continue to a lower pitch. The governor is therefore incapable of restoring the onspeed condition, and propeller rpm falls below the selected governor rpm setting.

Primary Governor Operation

The propeller levers adjust the primary propeller governor between 1,600 rpm and 1,900 rpm. The primary propeller governor, mounted at the top of the engine reduction gearbox, has two functions: it can select any constant propeller rpm within the range of 1,600 to 1,900, and it can also feather the propeller. The primary propeller governor adjusts propeller rpm by controlling the oil supply to the propeller dome.

An integral part of the primary propeller governor is the governor pump. This pump is driven by the N2 shaft and raises the engine oil pressure from normal to approximately 375 psi. The greater the oil pressure sent to the propeller dome, the lower the propeller pitch. The oil pressure is always trying to maintain a low pitch; however, the feathering springs and centrifugal counterweights are trying to send the propeller into the feathered position. Propeller control is a balancing act of opposing forces. A transfer gland is located on the propeller shaft. This transfer gland allows the oil to enter and exit the propeller dome area. Thus, the transfer gland is always replenishing the oils supply to the propeller pitch mechanism with fresh warm oil.

The primary propeller governor uses a set of rotating flyweights that are geared to the propeller shaft. The flyweights act as a comparison to a desired reference speed of how fast the propeller is turning. These flyweights are connected to a free-floating pilot valve. The slower the flyweights are turning in relation to the desired reference speed, the lower the position of the pilot valve. If the propeller and the flyweights turn faster, the additional centrifugal force makes the pilot valve rise inside the governor. The pilot valve position determines how much oil pressure is being sent to the propeller pitch mechanism. Here are a `few examples.

If a propeller rpm of 1,900 is selected and the propeller is actually turning at 1,900, the flyweights are in their center or “onspeed” condition (Figure 7 GT-29). The pilot valve is in the middle position. This maintains a constant oil pressure to the propeller pitch mechanism, which creates a constant pitch and a constant rpm.

If the airplane enters a descent, without any change to the cockpit controls, there will be a tendency for the airspeed to increase and the propeller to turn faster (Figure 7 GT-30). The flyweights will, in turn, rotate faster. The additional centrifugal force will make the pilot valve rise. Notice that oil can now escape via the pilot valve. Lower oil pressure will result in a higher pitch and a reduction of propeller rpm. The propeller will then return to its original rpm setting. The flyweights will then slow down, and the pilot valve will return to the equilibrium position to maintain the selected propeller rpm.

If the airplane enters a climb without any change in the cockpit controls, the airspeed will decrease and the propeller will tend to slow (Figure 7 GT-31). The flyweights in the propeller governor will slow down, because of a loss in centrifugal force, and the pilot valve will lower. This will allow more oil pressure to the propeller pitch mechanism. High oil pressure will result in a lower pitch. This in turn will cause an increase in propeller rpm. The propeller will increase to its original rpm setting, the flyweights will then speed up, and the pilot valve will return to its equilibrium or “onspeed” position, such as torque, interstage turbine temperature, compressor speed, and fuel flow, correcting the held constant by changing the propeller blade angles. The cockpit propeller lever adjusts where the equilibrium or “onspeed” condition will occur. The pilot can select any constant propeller rpm from 1,600 to 1,900 rpm, however, 1,900 rpm should be used for all operations.

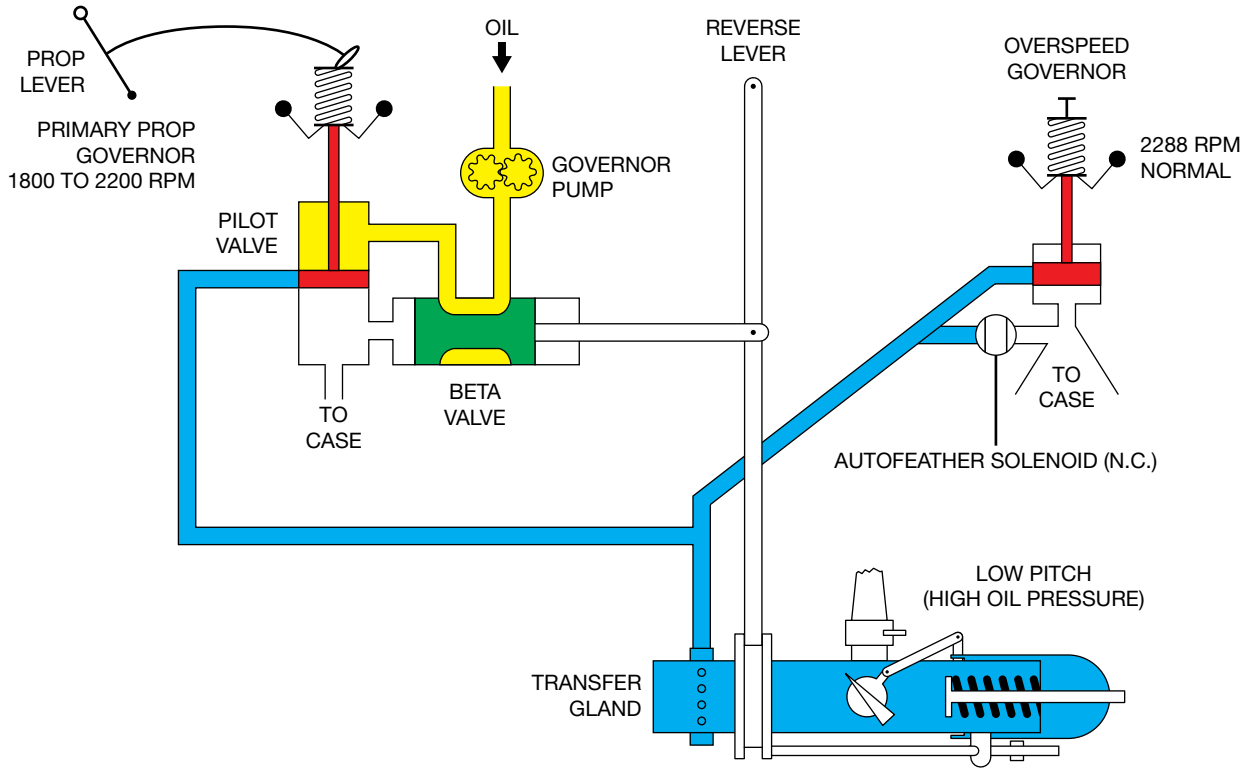


Figure 7 GT-29. Propeller Onspeed Diagram

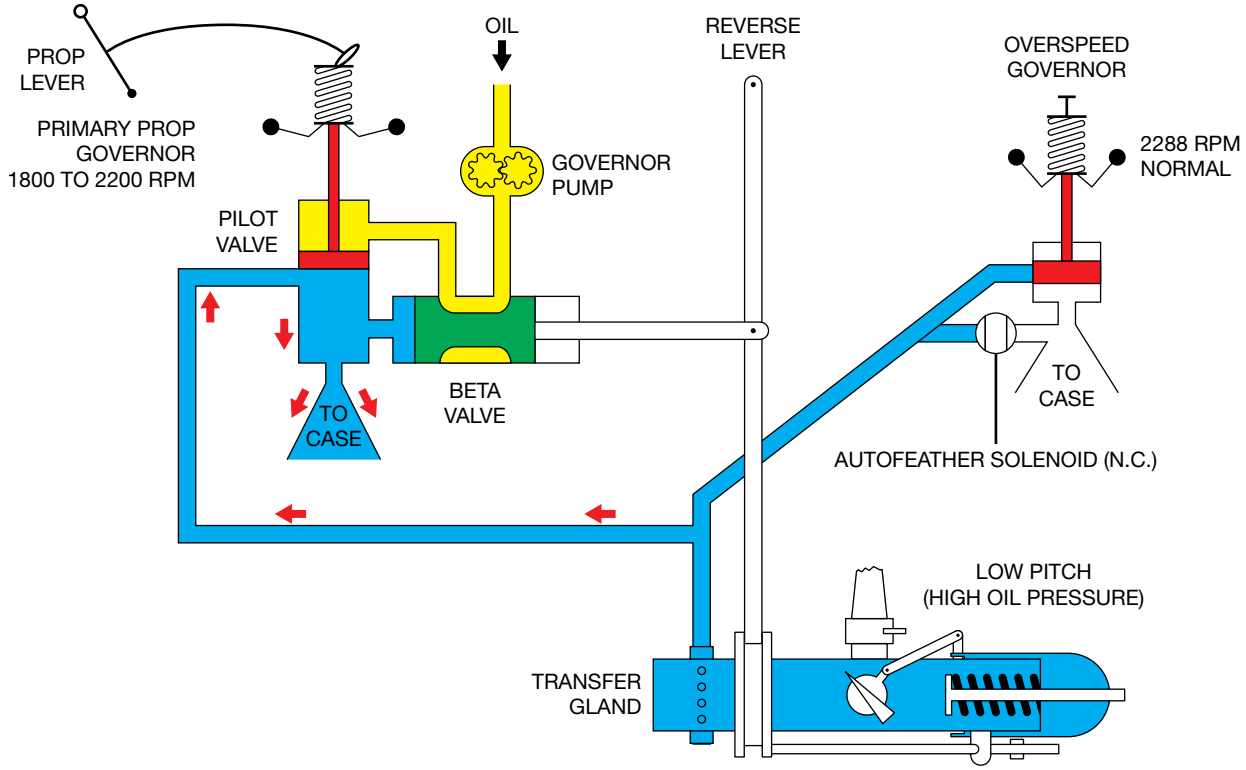


Figure 7 GT-30. Propeller Overspeed Diagram

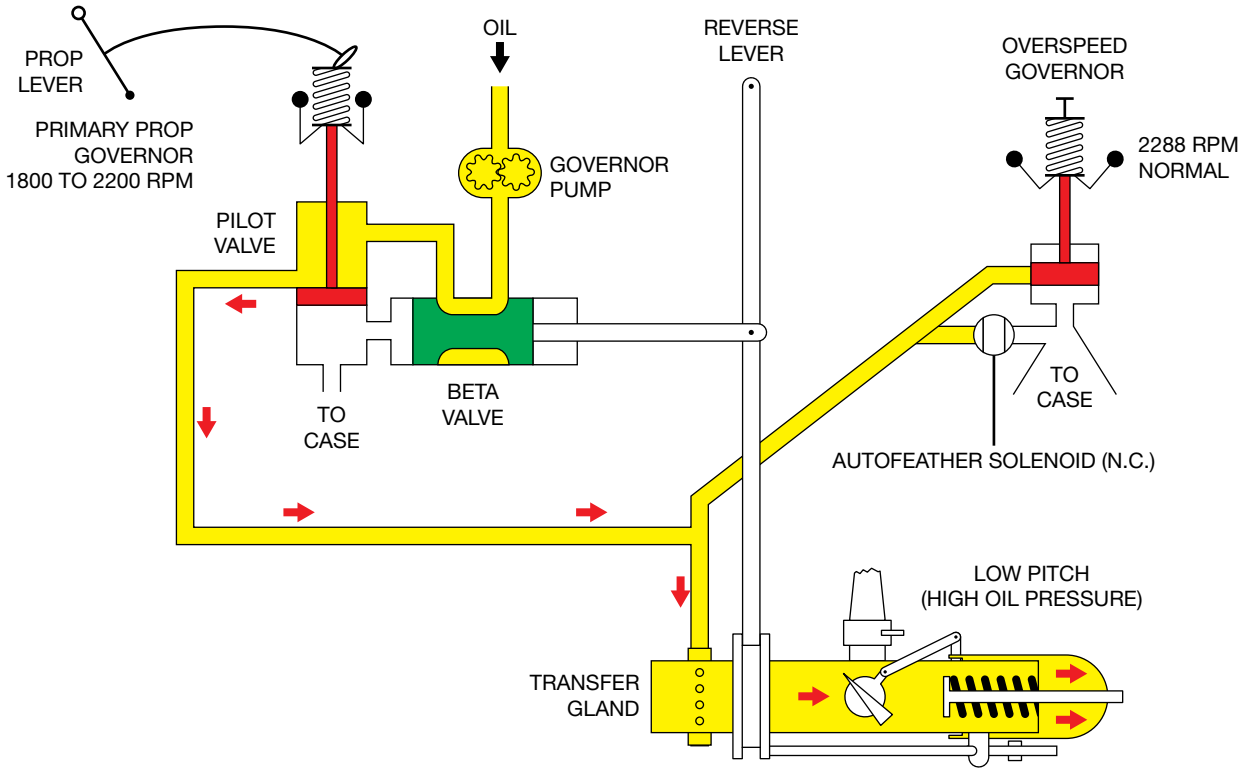


Figure 7 GT-31. Propeller Underspeed Diagram

LOW PITCH STOP

It is easy for the pilot to determine when the propeller blade angle is at the low pitch stop. Assuming the propeller is not feathered or in the process of being feathered, ***whenever the propeller rpm is below the selected governor rpm, the propeller blade angle is at the low pitch stop.***

This assumes that momentary periods of underspeed are not being considered. Rather, the propeller rpm is below and staying below the selected governor rpm.

For example, if the propeller control is set at 1,900 rpm but the propeller is turning at less than 1,900 rpm, the blade angle is at the low pitch stop.

On many types of airplanes, the low pitch stop is simply at the low pitch limit of travel, determined by the propeller's construction. But with a reversing propeller, the extreme travel in the low pitch direction is past 0°, into reverse or negative blade angles (Figure 7 GT-32). Consequently, the low pitch stop on this propeller must be designed in such a way that it can be repositioned when reversing is desired.

The low pitch stop is created by mechanical linkage sensing the blade angle. The linkage causes a valve to close, which stops the flow of oil pressure coming into the propeller dome. Since this pressure causes low pitch and reversing, once it is blocked, a low pitch stop has been created. The low pitch stop is commonly referred to as the "Beta" valve. Furthermore, the valve is spring-loaded to cause the propeller to feather in the event of mechanical loss of Beta valve control.

The position of the low pitch stop is controlled from the cockpit by the power lever. Whenever the power lever is at IDLE or above, this stop is set at approximately 12°. But bringing the power lever aft of IDLE progressively repositions the stop to lesser blade angles.

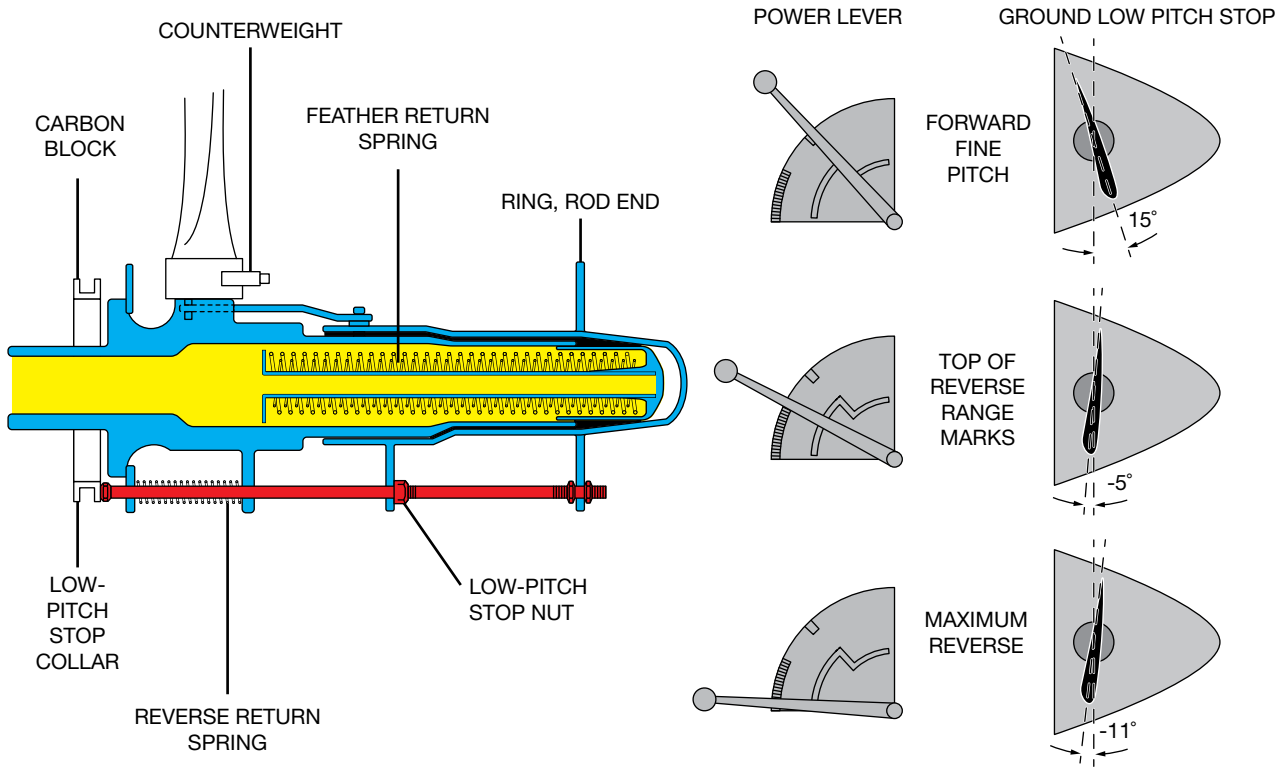


Figure 7 GT-32. Low Pitch Stop Diagram

Before reversing can take place, the propeller must be on the low pitch stop. As the propellers reach approximately 12°, the Beta valve is repositioned, creating the low pitch stop. The primary governor is sensing an underspeed and is directing oil pressure into the propeller dome. The Beta valve is controlling oil flow into the primary governor, and is defining the low pitch stop through oil pressure.

When blade angles less than approximately 12°, the linkage pulls the Beta valve actuator, readjusting the propeller blade angle as the Beta valve allows more oil into the propeller dome. The slip ring moves with the prop dome and will define the low pitch stop at a lower, or negative, blade angle. If blade angles less than approximately 12° are requested before the propeller blades are on the low pitch stop, the slip ring will not move, and the reversing cable and linkage may be damaged.

The region from 12° to –10° blade angle is referred to as the Beta range.

On the C90GT, the Ground Fine range extends from +12° to +3°, and the engine's compressor speed (N_1) remains at the value it had when the power lever was at IDLE (low idle to high idle) based on condition lever position. From +3° to –10° blade angle, the N_1 speed progressively increases to a maximum value at –10° blade angle of approximately +85% ±3%.

Low Pitch Stop Operation

During non-reversing operations, the low pitch stop prevents the propeller blades from reducing the airflow over the empennage of the aircraft.

The low pitch stop uses a mechanical linkage to hydraulically control propeller blade angle. As the propeller blades reduce angle through approximately 20° of pitch, the flange mounted on the propeller dome contacts the nuts located on the rods mounted on the slip ring. The propeller dome moves the slip ring forward, which in turn activates the Beta valve, which controls oil pressure into the propeller dome.

Riding in the slip ring is linkage which connects the Beta valve with the slip ring, and the power levers via a cable. As the slip ring moves, the linkage pivots about the end with the cable attached to it, with the Beta valve in the middle. For reversing, the pilot repositions the linkage with the power levers, which resets the low pitch stop.

When the Beta valve is controlling blade angle, oil pressure supplied from the governor oil pump is supplying pressure through the Beta valve to the propeller dome. The Beta valve modulates the amount of pressure entering the propeller dome, controlling the blade angle. The primary governor must be in the underspeed condition, allowing all of the pressure flowing from the Beta valve into the propeller dome. If the underspeed condition did not exist when lower blade angles are requested, the Beta valve could not fully control the propeller blade angle, and the slip ring would not move without help from the propeller blades. Since the propeller blades only contact the slip ring when the blades are at the low pitch stop, the request for lower blade angles when the propellers are not on the low pitch stop will result in damage to the control cable, as it cannot effect these changes alone.

GROUND FINE AND REVERSE CONTROL

The geometry of the power lever linkage through the cam box is such that power lever increments from idle to full forward thrust have no effect on the position of the Beta valve. When the power lever is moved from idle into the reverse range, it positions the Beta valve to direct governor oil pressure to the propeller piston, decreasing blade angle through zero into a negative range. The travel of the propeller servo piston is fed back to the Beta valve to null its position and, in effect, provide infinite negative blade angles all the way to maximum reverse. The opposite will occur when the power lever is moved from full reverse to any forward position up to idle, therefore providing the pilot with manual blade angle control for ground handling.

Ground Fine and Reverse Control Operation

When the blade angle reaches approximately 20°, the flange extending from the dome makes contact with the Beta nuts (Figure 7 GT-33). As the propeller pitch angle continues to decrease, each flange on the propeller dome pushes the nut and the attached Beta rod forward. As the rod moves forward, it pulls the slip ring forward. In turn, a Beta valve inside the governor is pulled into the oil pressure cutoff position. The linkage is set to control the oil pressure supply to the dome when the blade angle reaches low pitch stop.

If this system were fixed at the low pitch stop, the propeller could not be reset throughout the Beta range. However, the low pitch stop can be adjusted to allow access to the Ground Fine and Reverse ranges on the ground. The hydraulic low pitch stop can be reset to allow the propeller to operate in the Ground Fine and Reverse ranges while the aircraft is on the ground and the engines are operating.

When the power levers are lifted up and over the idle detent into the Ground Fine range, the Beta valve is repositioned. As the Beta arm moves back, the Beta valve is opened, re-establishing oil flow to the propeller dome. This allows the propeller blade to move to a flatter pitch. As the propeller blades move to a flatter pitch, the propeller dome and slip ring continue forward, eventually moving the Beta valve back into position to stop propeller blades. In summary, the position of the low pitch stop is controlled by the power levers. When the power levers are set at idle or above, the stop is set at approximately 12°. When the power levers are moved aft of idle, however, the low pitch stop is repositioned to lesser blade angles.

The propeller can be feathered by moving the propeller lever full aft past the detent into the feather range. The feathering action raises the pilot valve to the full up position. The oil pressure is released from the propeller pitch mechanism and the propeller feathers. In this type of turbine engine, the propeller shaft and N_1 shaft are not connected. Thus, the propeller can be feathered with the engine running at idle power. Without an autofeather system, in flight, the propeller will maintain rpm unless it is manually feathered when the engine is shut down.

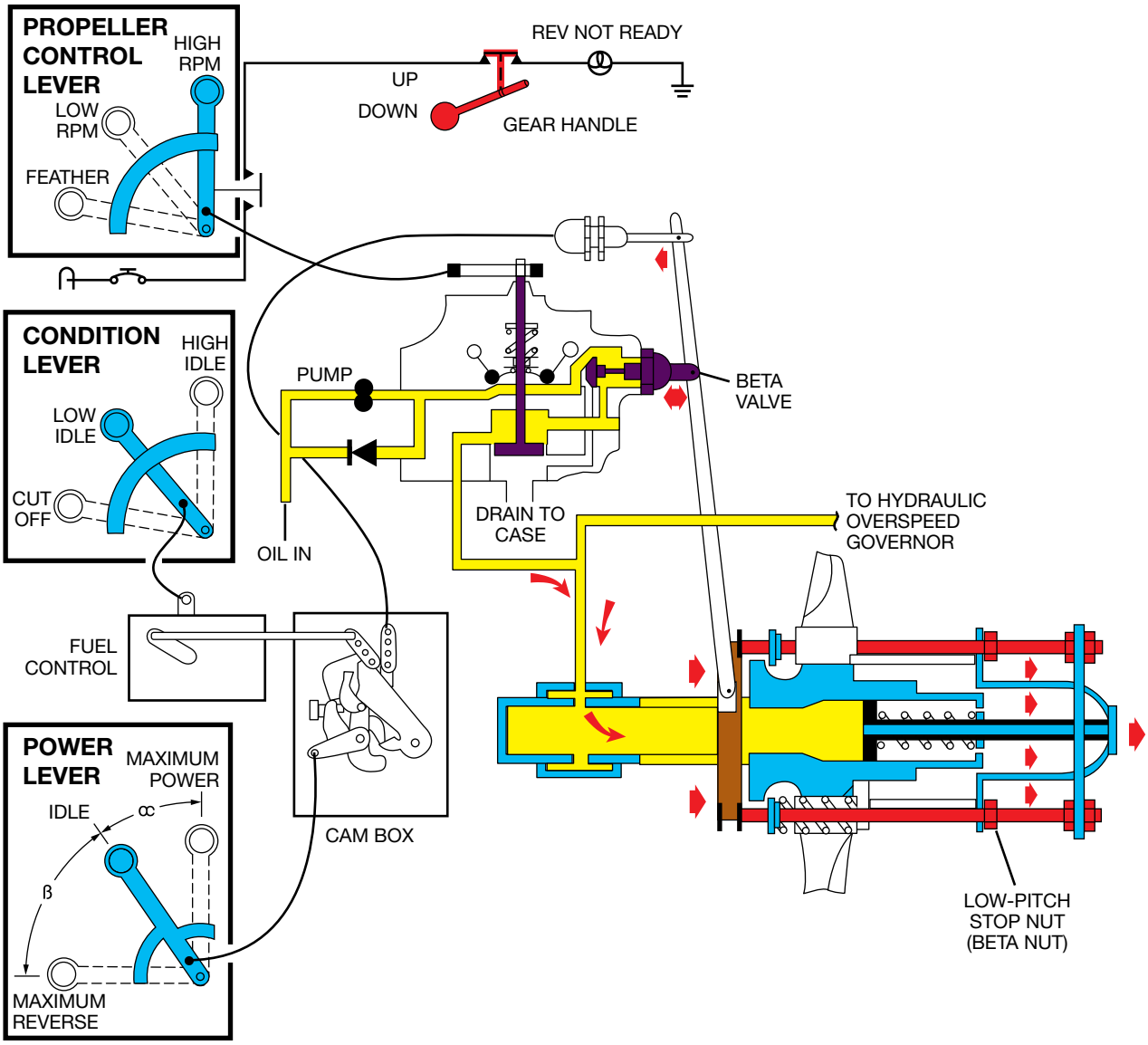


Figure 7 GT-33. Beta Range and Reverse Diagram

There are situations where the propeller primary governor cannot maintain the selected propeller rpm, such as final approach where power and airspeed are being reduced. With the progressive reduction of power and airspeed on final, the propeller and rotating counterweights will tend to go to the underspeed condition. In the underspeed condition the pilot valve will open, increasing oil pressure to the dome, and the propeller pitch will decrease as power and airspeed are reduced. Since the reversible propeller is capable of decreasing past 0° into negative or reverse blade angles, the low pitch stop prevents the blade angle from decreasing beyond a predetermined value. When the propeller governor becomes incapable of maintaining the onspeed condition, the propeller rpm will fall below the selected governor rpm setting.

Assuming the propeller is not feathered, whenever the propeller rpm is below the selected governor setting, the propeller blade angle is at the low pitch stop. The low pitch stop mechanism is created by linkage that references the actual blade angle.

Moving the power lever within the Ground Fine range on the C90GT adjusts propeller pitch. Moving the power levers within the reverse range adjusts propeller pitch and N_1, up to the maximum N_1 in reverse of 88%. Attempting to pull the power levers in reverse with the propellers in feather will cause damage to the reversing linkage of the power lever. Also, pulling the power levers into the reverse position on the ground with the engines shut down will damage the reversing system.

OVERSPEED GOVERNOR

The overspeed governor provides protection against excessive propeller speed in the event of primary governor malfunction. Since the PT6's propeller is driven by a free turbine (independent of the engine's), overspeed could occur if the primary governor were to fail.

The operating point of the overspeed governor is set at 1,976 rpm. If an overspeeding propeller's speed reached 1,976 rpm, the overspeed governor would control the oil pressure and pitch to prevent the rpm from continuing its rise. From a pilot's point of view, a propeller tachometer stabilized at approximately 1,976 would indicate failure of the primary governor and proper operation of the overspeed governor. The overspeed governor can be reset to approximately 1,750 rpm for test purposes.

Overspeed Governor Operation

If the primary propeller governor failed, an overspeed condition could occur. However, several safety devices in the systems come into play in the event of a primary governor failure. A hydraulic overspeed governor (Figure 7 GT-34) is located on the left side of the propeller reduction gearbox. It has a set of flyweights and a pilot valve similar to those of the primary governor. If a runaway propeller's speed were to reach 1,976 rpm, the overspeed governor flyweights would make its pilot valve rise. This would decrease the oil pressure at the propeller dome. The blade angle would increase as necessary to prevent the rpm from continuing its rise. Testing of the overspeed governor at approximately 1,750 rpm is accomplished during runup by using the propeller governor test switch on the pilot's left subpanel.

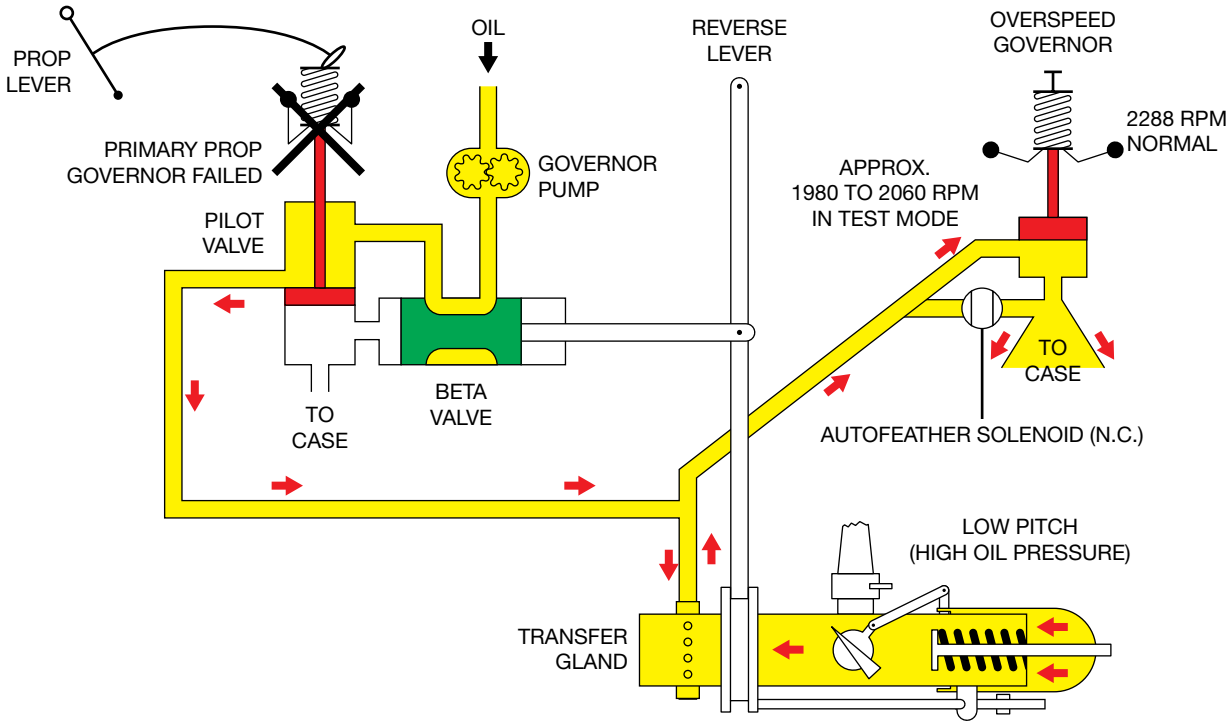


Figure 7 GT-34. Overspeed Governor Diagram

FUEL TOPPING GOVERNOR

The fuel topping governor can also control an overspeed condition and is set at 6% above the primary governor's selected speed. In an overspeed condition, the fuel topping governor will limit propeller rpm by decreasing pneumatic pressure to the fuel control unit, reducing fuel flow and engine speed as means of controlling propeller rpm. In reverse, the fuel topping governor is reset to 95% of selected rpm to insure that the propeller will not reach the selected rpm. The fuel topping governor will only prevent an over-speed if the primary governor's flyweight's are still operational.

POWER LEVERS

The power levers (Figure 7 GT-35) are located on the power lever quadrant (first two levers on the left side) on the center pedestal. They are mechanically interconnected through a cam box to the fuel control unit, the Beta valve and follow-up mechanism, and the fuel topping (NP) governor. The power lever quadrant permits movement of the power lever from idle to maximum thrust and in the Ground Fine and Reverse ranges from idle to maximum reverse. Two gates in the power lever quadrant aft of the IDLE position, prevent inadvertent movement of the power lever into the GROUND FINE or REVERSE ranges. The pilot must lift the power levers up and over the first gate to select GROUND FINE, and up and over the second gate to select REVERSE.

The function of the power levers is to establish a gas generator rpm through the gas generator governor (N_G) and a fuel flow that will produce and maintain the selected N_1 rpm. In the Beta or GROUND FINE range, the power levers are used to change the propeller blade angle, thus changing propeller thrust.

In the REVERSE range, the power lever:

- Selects a blade angle proportionate to the aft travel of the lever
- Selects an N_1 that will sustain the selected reverse power
- Resets the fuel topping governor from its normal setting of 106% to approximately 95% of the primary governor setting

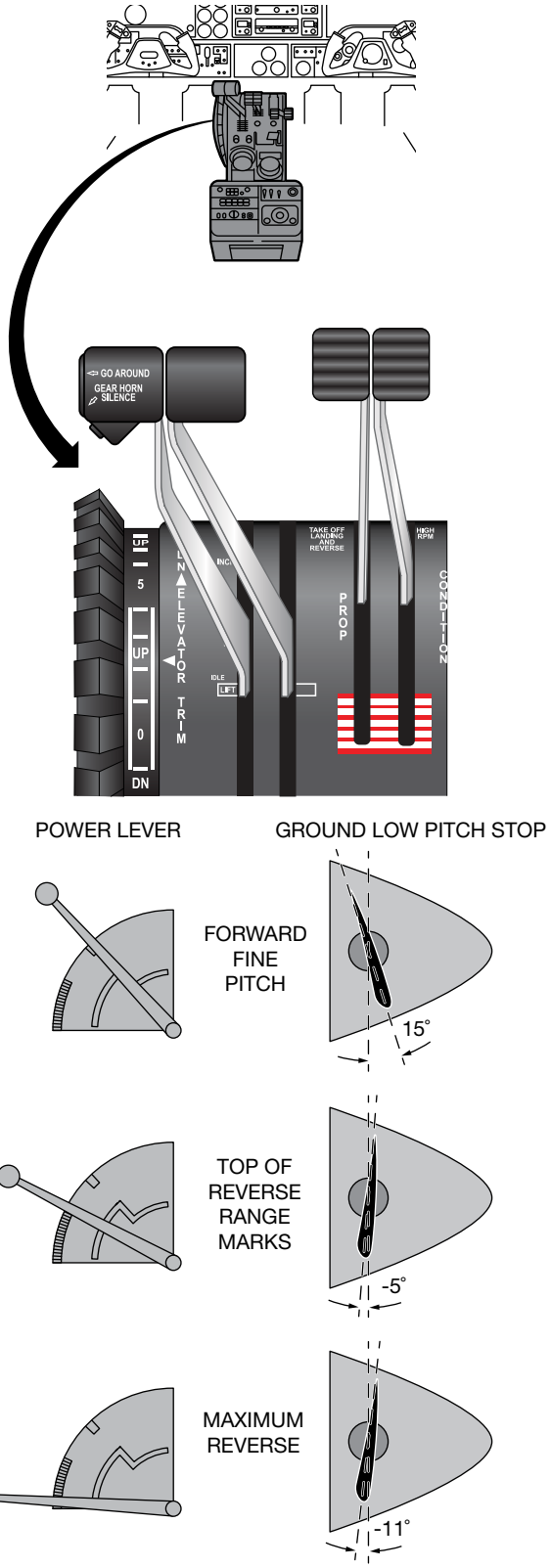


Figure 7 GT-35. Power Levers

Propeller Control Levers

Propeller rpm, within the primary governor range of 1,600 to 1,900 rpm, is set by the position of the propeller control levers (Figure 7 GT-36). These levers, one for each propeller, are located between the power levers and the condition levers on the center pedestal quadrant. The full forward position sets the primary governor at 1,900 rpm. In the full aft position at the feathering detent, the primary governor is set at 1,600 rpm. Intermediate propeller rpm positions can be selected by moving the propeller levers to the corresponding position, to select the desired rpm as indicated on the propeller tachometer. These tachometers read directly in revolutions per minute.

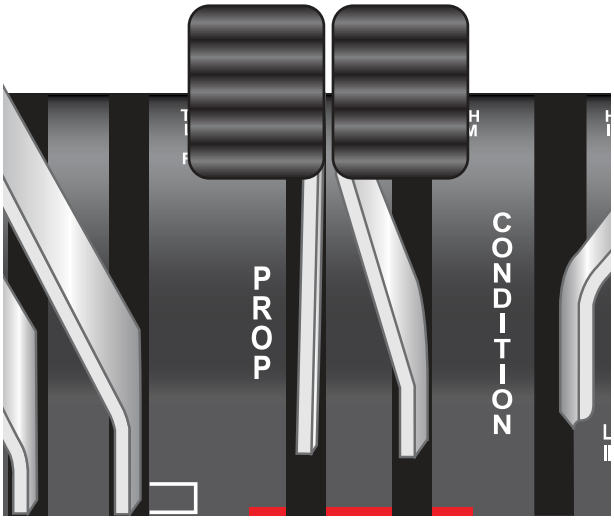


Figure 7 GT-36. Propeller Control Levers

A detent at the low rpm position prevents inadvertent movement of the propeller lever into the feather position, indicated by the red and white stripes across the lever slots in the quadrant. At the full feather position, the levers position the governor pilot valve to dump oil pressure from the propeller hub, and allow the counterweights and springs to position the propeller blades to the feather position.

AUTOFEATHER SYSTEM

The automatic feathering system provides a means of immediately dumping oil pressure from the propeller hub, thus enabling the feathering spring and counterweights to start the feathering action of the blades in the event of an engine failure (Figure 7 GT-37). Although the system is armed by a switch on the subpanel, placarded "AUTOFEATHER" and "ARM-OFF-TEST," the completion of the arming phase occurs when both power levers are advanced above 90% N_1, at which time green annunciators, placarded (L) and (R) AUTOFEATHER on the Caution/Advisory annunciator panel will illuminate, indicating the system is armed (Figure 7 GT-38). The system will remain inoperative as long as either power lever is retarded below 90% N_1 position. The system is designed for use only during takeoff, climb, and missed approach and should be turned off when establishing cruise. When the system is armed and the torque on a failing engine drops below approximately 400 ft-lbs, the autofeather system of the operative engine is disarmed causing its annunciators to extinguish. When the torque on the failing engine drops below approximately 260 ft-lbs, the oil is dumped from the servo, the feathering spring and counterweights feather the propeller, and the annunciators for the failed engine extinguish. The system may be tested on the ground using the spring-loaded TEST position of the switch. With the switch in the TEST position, the 90% N_1 switches are disabled and the system will arm with the power levers set at approximately 500 ft-lbs of torque. Retarding a single power lever will then simulate an engine failure and the resulting action of the autofeather system can be checked as described in Section 4, NORMAL PROCEDURES. Since an engine is not actually shut down during a test, the AUTOFEATHER annunciator for the engine being tested will cycle on and off as the torque oscillates above and below the 260 ft-lb setting. (Figure 7 GT-39).

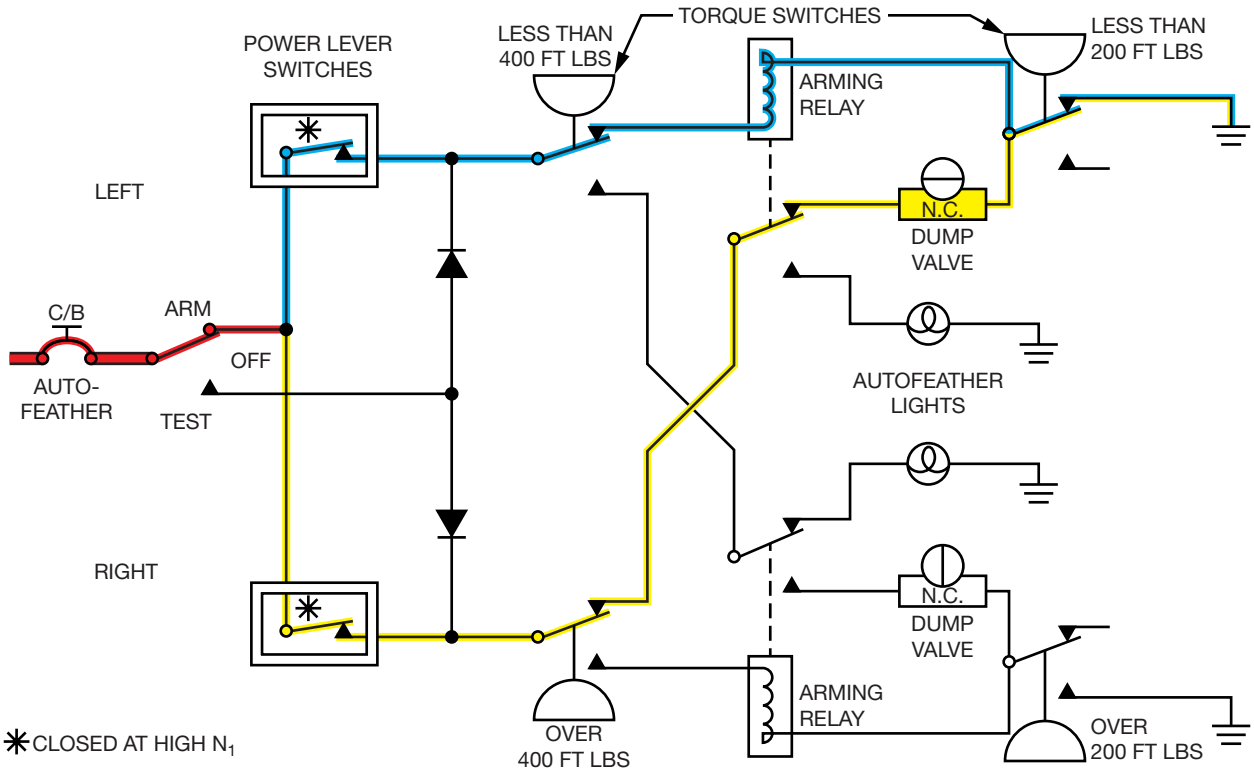


Figure 7 GT-37. Autofeather System Diagram—Left Engine Failed and Feathering

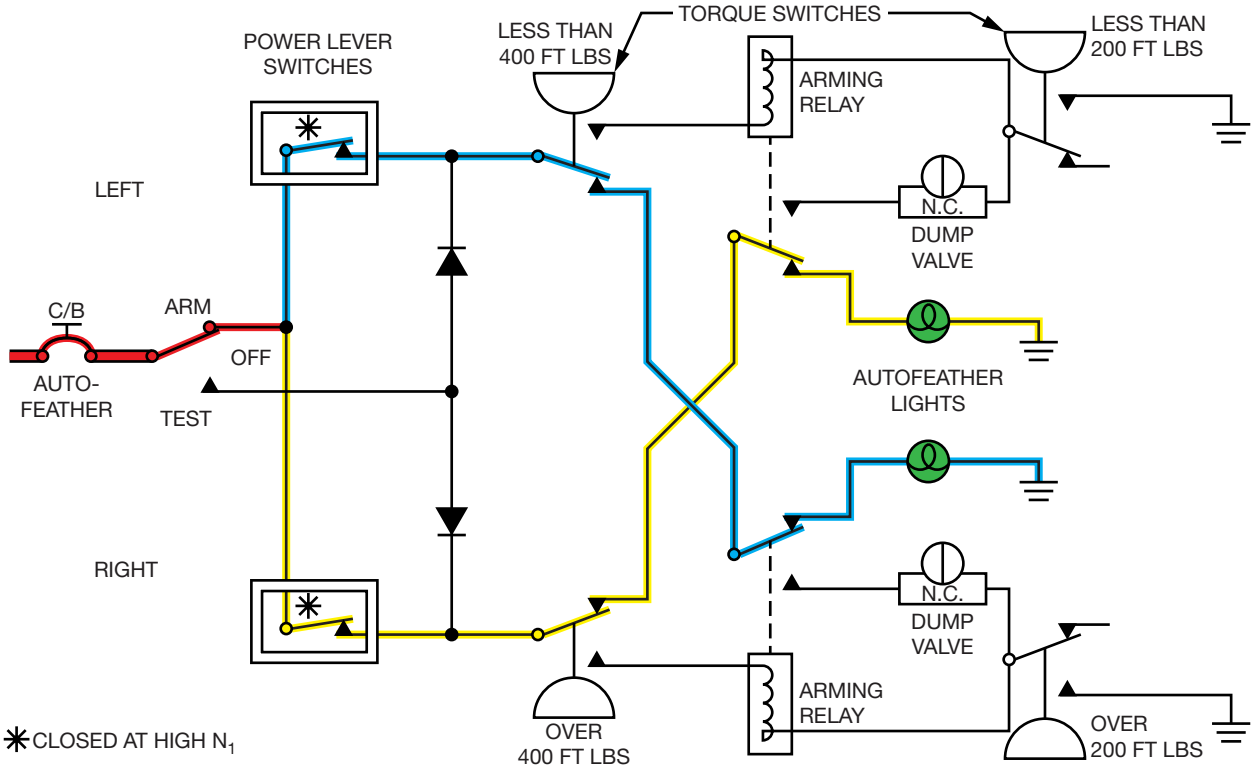


Figure 7 GT-38. Autofeather System Diagram—Armed

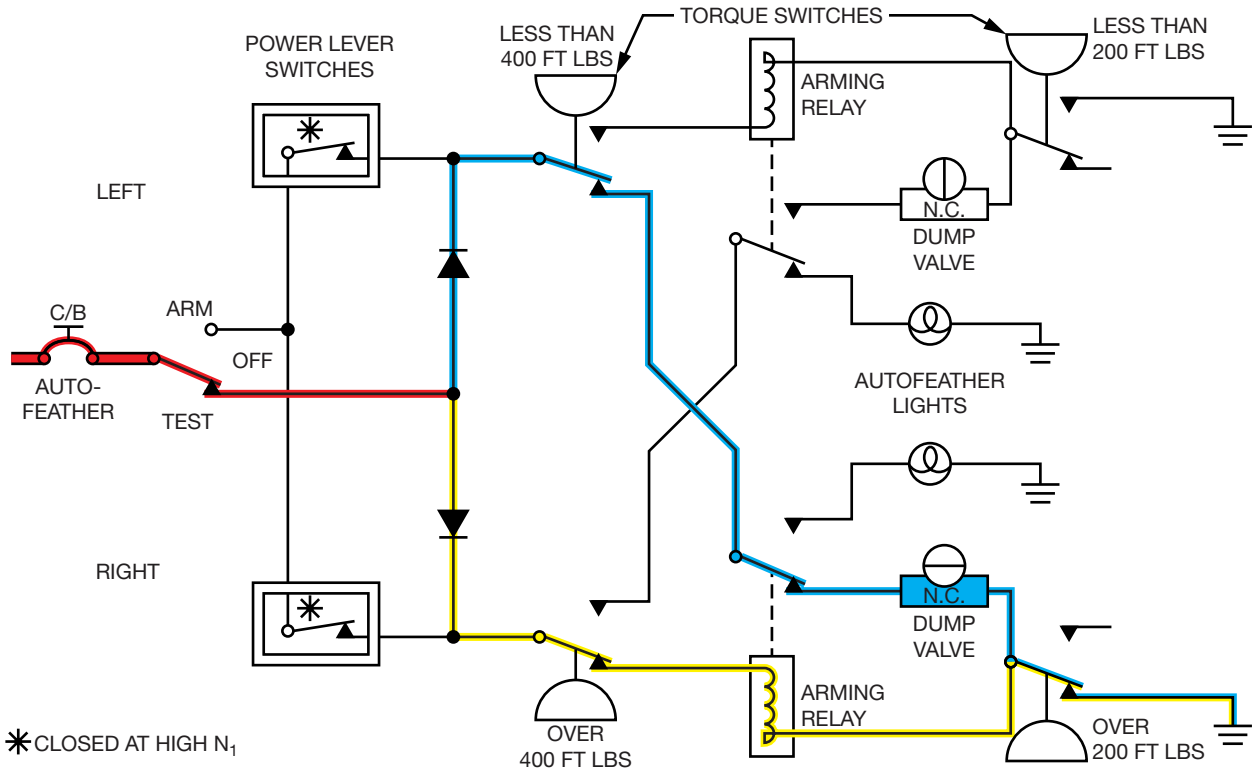


Figure 7 GT-39. Autofeather Test Diagram

PROPELLER SYNCHROPHASER SYSTEM

A Type II synchrophaser system is installed in the King Air C90GT. The propeller synchrophaser automatically matches the rpm of the two propellers and maintains the blades of one propeller at a predetermined relative position with the blades of the other propeller. The purpose of the system is to reduce propeller beat and cabin noise from unsynchronized propellers.

Synchrophaser Operation

The Type II synchrophaser system (Figure 7 GT-40) is an electronic system, certificated for takeoff and landing. It is not a master-slave system, and it functions to match the rpm of both propellers and establish a blade phase relationship between the left and right propellers to reduce cabin noise to a minimum.

The system cannot reduce rpm of either propeller below the datum selected by the propeller control lever. Therefore, there is no indicating annunciator light associated with the Type II system.

To prevent either propeller from losing excessive rpm if the other propeller is feathered while the synchrophaser is on, the synchrophaser has a limited range of authority from the manual governor setting. In no case will the rpm fall below that selected by the propeller control lever. Normal governor operation is unchanged, but the synchrophaser will continuously monitor propeller rpm and reset either governor as required. Propeller rpm and position is sensed by a magnetic pickup mounted adjacent to each propeller spinner bulkhead. This magnetic pick-up will transmit electrical pulses once per revolution to a control box installed forward of the pedestal.

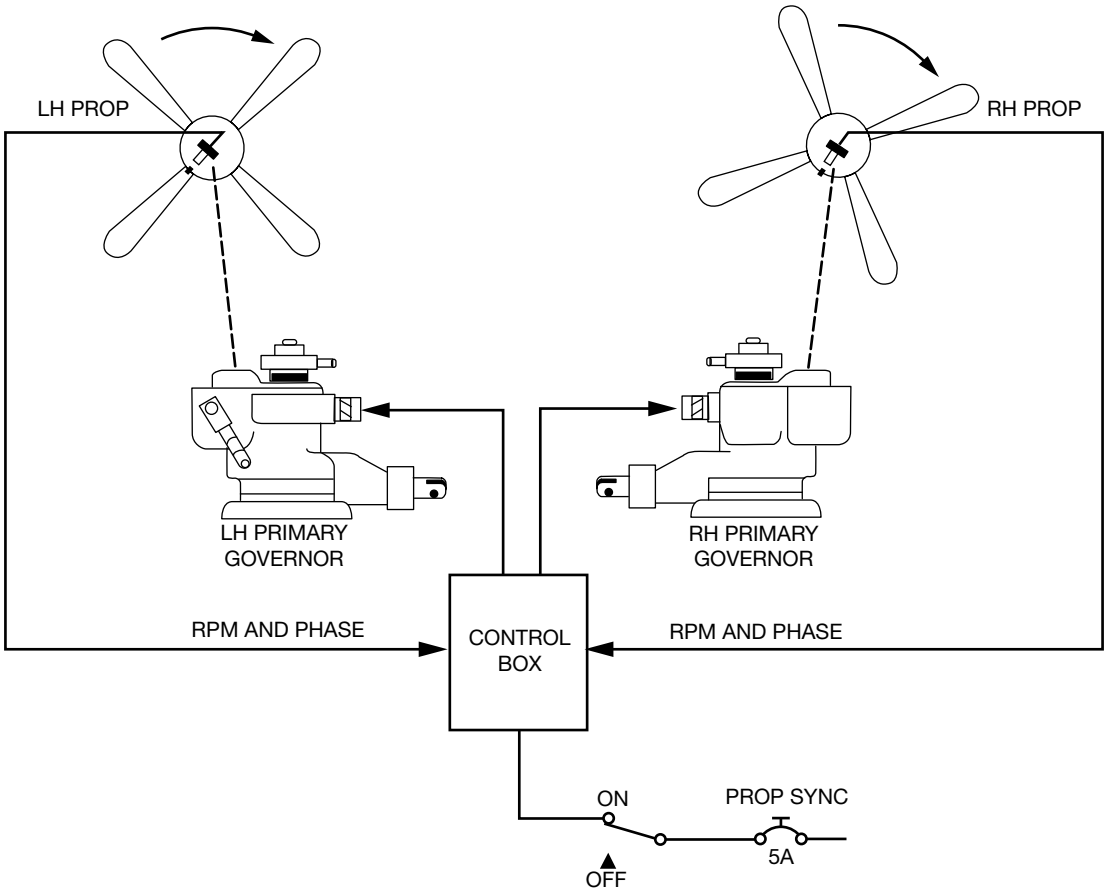


Figure 7 GT-40. Propeller Synchrophaser

The control box converts any pulse rate differences into correction commands, which are transmitted to coils mounted close to the flyweights of each primary governor. By varying the coil voltage, the governor speed settings are biased until the prop rpm's exactly match. A toggle switch installed adjacent to the synchroscope turns the system on. In the synchrophaser OFF position, the governors operate at the manual speed settings selected by the pilot. To operate the synchrophaser system, synchronize the propellers manually or establish a maximum of 10 rpm difference between the engines, then turn the synchrophaser on. The system may be on for takeoff and landing.

To change rpm with the system on, adjust both propeller controls at the same time. If the synchrophaser is on but does not adjust the prop rpm to match, the system has reached the end of its range. Increasing the setting of the slow prop, or reducing the setting of the fast prop, will bring the speeds within the limited synchrophaser range. If preferred, turn the synchrophaser switch off, resynchronize manually, and turn the synchrophaser on.

PROPELLER SYNCHROSCOPE

A propeller synchroscope (Figure 7 GT-41) is located to the left of the oil pressure/temperature indicators and gives the status of propeller synchronization. The face of the synchroscope has a black and white cross pattern which can spin either left or right. If the right propeller rpm is greater than the left, the face turns clockwise or right. With the left propeller rpm greater than the right, the face turns counterclockwise or left. No rotation of the face indicates that both propellers are synchronized.

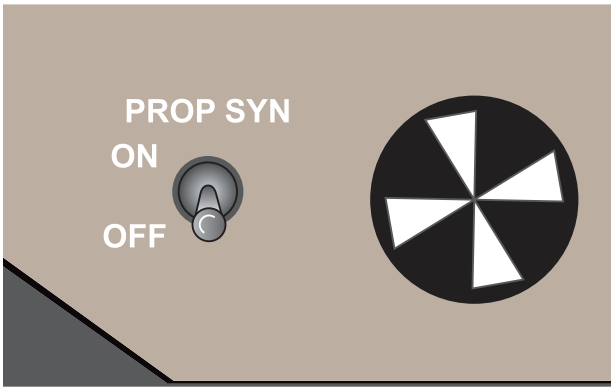


Figure 7 GT-41. Propeller Synchroscope

QUESTIONS

1. The PT6A engine power section consists of:

A. One compression stage and four turbine stages.
B. A single-stage reaction turbine.
C. A single-stage turbine and a centrifugal compressor.
D. Twin-spool, single-stage turbines.

2. The function of the reduction gear system is to provide gear reduction:

A. Between the compressor and the power turbine.
B. For the airplane's accessory drive section.
C. For the propeller.
D. Between the compressor and the compressor turbine.

3. If a CHIP DETECT annunciator illuminates, you must do one of the following:

A. Continue normal flight operations and have the filter checked after landing.
B. Reduce torque to 500 foot-pounds for the remainder of the flight.
C. Shut the engine down and land as soon as practical.
D. Check engine instruments and, if normal, no action is required.

4. During ground operation at LO IDLE, you note that ITT is exceeding 685°C. Which of the following actions would you consider best to reduce ITT?

A. Move the propeller control lever to the low rpm position
B. Move the power lever into the ground fine (Beta)/reverse range
C. Shut down and have the propeller LO IDLE stops checked
D. Reduce accessory load or increase N_1 rpm

5. When using maximum reverse power with the prop lever full-forward, you would expect a maximum propeller rpm of:

A. 1,900 rpm.
B. 1,825 rpm.
C. 1,750 rpm.
D. 2,000 rpm.

6. During a ground start of the right engine, the IGNITION ON annunciator should illuminate:

A. At 10% N_1 rpm.
B. When the condition lever is moved to LO IDLE.
C. At a stabilized 12% N_1.
D. When the start switch is moved to the IGNITION and ENGINE START position.

7. When the AUTO-IGNITION switch is in the ARM position, ignition is:

A. Continuous.
B. Inactive but armed, if torque is greater than 400 foot-pounds.
C. Controlled by the stall warning system.
D. Continuous when torque is greater than 400 foot-pounds.

8. After lift-off, if an autofeather is initiated, the immediate requirement is to:

A. Continue to fly the airplane and allow the propeller to feather.
B. Move the power lever to idle.
C. Move the condition lever to cutoff.
D. Reduce electrical loads.

9. Which of the following is the most accurate definition of Engine Torque Readout?

A. Power developed by the gas generator
B. Thrust supplied by the propeller
C. Ratio of compressor inlet to exhaust outlet
D. Power delivered to the propeller

CHAPTER 8
FIRE PROTECTION

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CHAPTER 8
FIRE PROTECTION

INTRODUCTION

The aircraft fire protection system consists of engine fire detection and fire extinguishing systems. Cockpit controls and indicators monitor and operate the system.

GENERAL

The fire protection chapter of the training manual presents a description and discussion of the airplane fire protection system and components. The location and purpose of switches and indicators are described.

FIRE DETECTION SYSTEM

The fire detection system (Figure 8-1) is designed to provide immediate warning in the event of fire in either engine compartment. The detection system is operable whenever the generator buses are active.

The system consists of the following: three photoconductive cells for each engine; a control amplifier for each engine; two red warning lights on the warning annunciator panel, one placarded L ENG FIRE, the other R ENG FIRE; a test switch on the copilot's left subpanel; and a circuit breaker placarded FIRE DET on the right side panel.

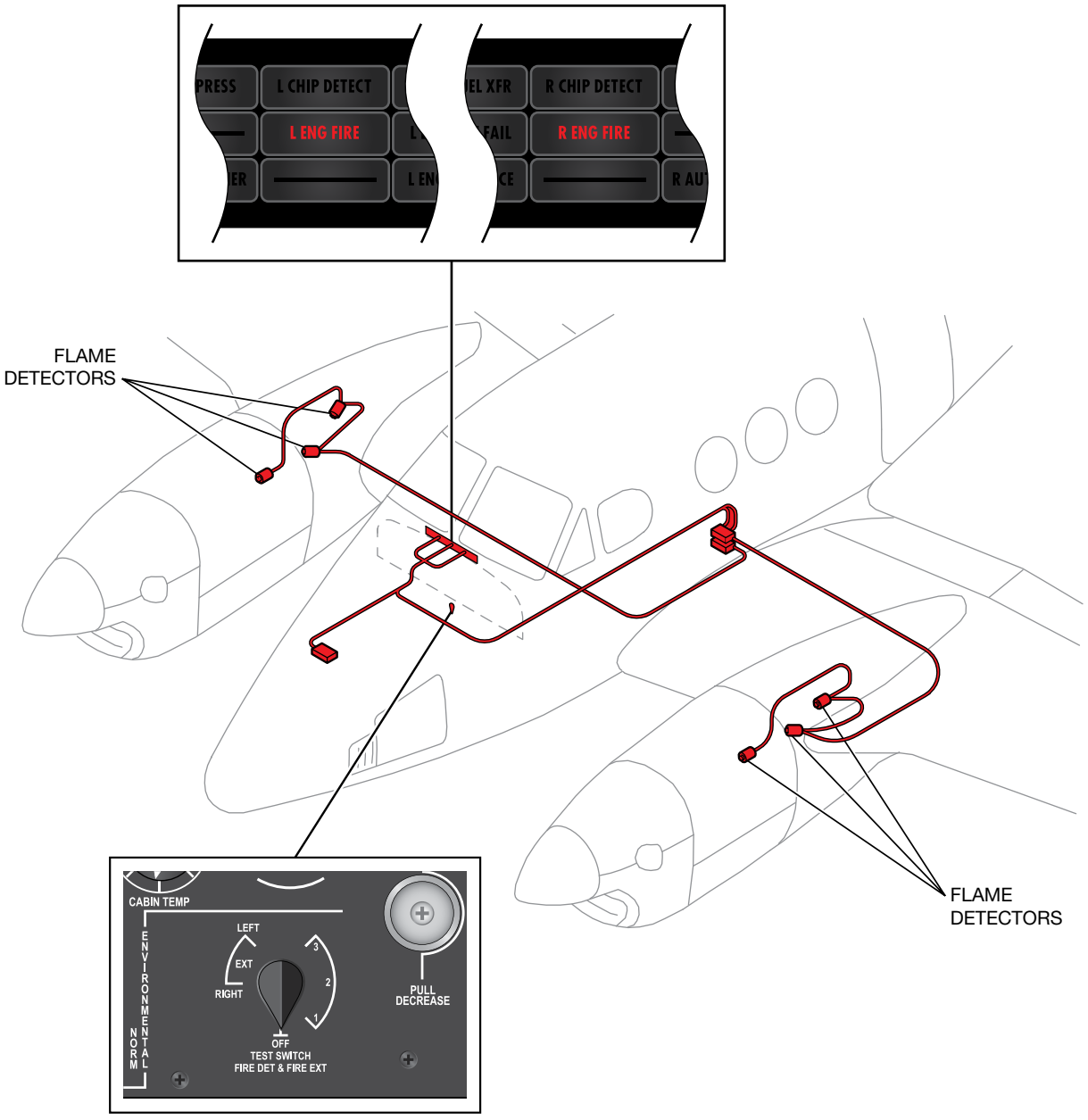

Figure 8-1. Fire Detection System

The six photoconductive-cell flame detectors are sensitive to infrared radiation. They are positioned in each engine compartment so as to receive both direct and reflected infrared rays, thus monitoring the entire compartment with only three photocells. Temperature level and rate of temperature rise are not controlling factors in the sensing method.

Conductivity through the photocell varies in direct proportion to the intensity of the infrared radiation striking the cell. As conductivity increases, the amount of current from the electrical system flowing through the flame detector increases proportionally. To prevent stray light rays from signaling a false alarm, a relay in the control amplifier closes only when the signal strength reaches a preset alarm level. When the relay closes, the appropriate left or right warning annunciators illuminate. When the fire has been extinguished, the cell output voltage drops below the alarm level and the relay in the control amplifier opens. No manual resetting is required to reactivate the fire detection system.

Fire Detection Test System

The rotary switch on the copilot's left subpanel, placarded TEST SWITCH-FIRE DET, has four positions: OFF–3–2–1. (If the optional engine fire extinguishing system is installed, the switch is placarded TEST SWITCH–FIRE DET & FIRE EXT and the left side of the test switch will include LEFT–EXT–RIGHT positions.)

The three test positions for the fire detector system are located on the right side of the switch. When the switch is rotated from OFF (down) to any one of these three positions, the output voltage of a corresponding flame detector in each engine compartment is increased to a level sufficient to signal the amplifier that a fire is present.

The following should illuminate as the selector is rotated through each of the three positions: the FAULT WARNING flasher, the L ENG FIRE and R ENG FIRE warning annunciators and, if the optional engine fire extinguishing System is installed, the red lenses placarded L ENG FIRE EXT–PUSH and R ENG FIRE EXT -PUSH on the fire-extinguisher activation switches. The system may be tested anytime, either on the ground or in flight. The TEST SWITCH should be placed in all three positions, in order to verify that the circuitry for all six fire detectors is functional. Illumination failure of all the fire detection system annunciators when the TEST SWITCH is in any one of the three flame-detector-test positions indicates a malfunction in one or both of the two detector circuits (one in each engine) being tested by that particular position of the TEST SWITCH.

FIRE EXTINGUISHING SYSTEM

The optional engine fire extinguishing system (Figure 8-2) incorporates an explosive cartridge inside the extinguisher of each engine. Each engine has its own self-contained extinguishing system, which can be used only once between rechargings. This system cannot be crossfed. When the activation valve is opened, the pressurized extinguishing agent is discharged through a plumbing network which terminates in strategically located spray nozzles.

The fire extinguisher control switches used to activate the system are located below the glareshield annunciator panel on the C90A. The control switches are on either side of the annunciator panel on the C90B. Their power is derived from the hot battery bus. The detection system is operable whenever the generator buses are active. But the extinguishing system can be discharged at any time, since it is operated from the hot battery bus. Therefore, even though the airplane may be parked with the engines off, the fire extinguishing system may be discharged.

Each push-to-actuate switch incorporates three indicator lenses. The red lens, placarded L (or) R ENG FIRE EXT–PUSH, warns of the presence of fire in the engine. The amber lens, placarded D, indicates that the system has been discharged and the supply cylinder is empty. The green lens, placarded OK, is provided only for the preflight test function.

To discharge the cartridge, raise the break-away wired clear plastic cover and press the face of the lens. This is a one-shot system and will be completely expended upon activation. The amber D light will illuminate and remain illuminated, regardless of battery switch position, until the pyrotechnic cartridge has been replaced.

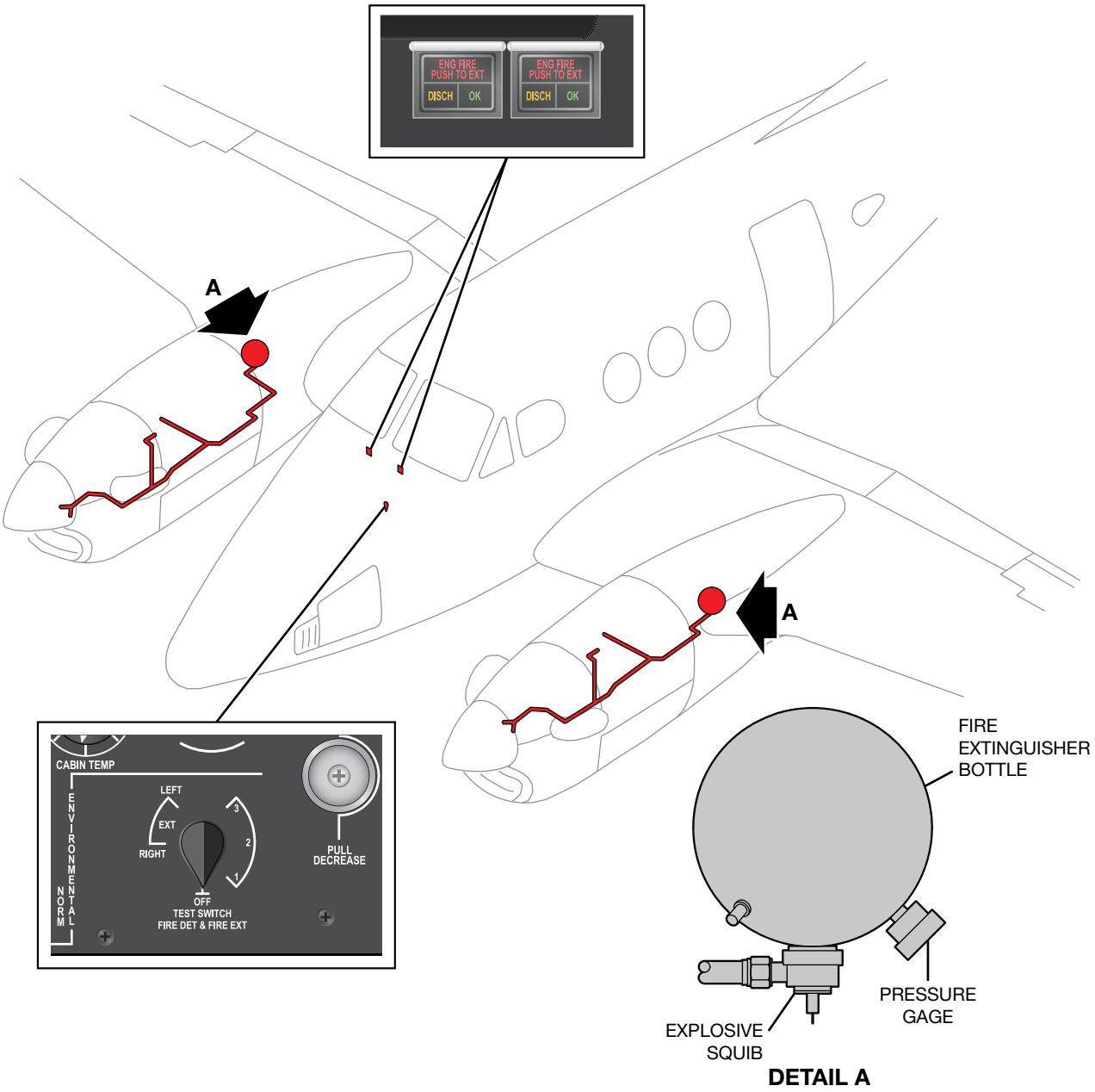

Figure 8-2. Fire Extinguishing System

Fire Extinguisher Test System

The fire extinguisher system test functions, incorporated in the rotary TEST SWITCH–FIRE DET & FIRE EXT, test the circuitry of the fire extinguisher system. During preflight, the pilot should rotate the TEST SWITCH to each of the two positions (RIGHT EXT and LEFT EXT) and verify the illumination of the amber D light and the green OK light on each fire-extinguisher activation switch below the glareshield. Illumination during this check indicates that the bottle charge detector circuitry and squib firing circuitry are operational and that the squib is in place.

A gage, (Figure 8-3) calibrated in psi, is provided on each supply cylinder for determining the level of charge. The gages should be checked during preflight. The cylinder and gages are located in the main wheel wells.

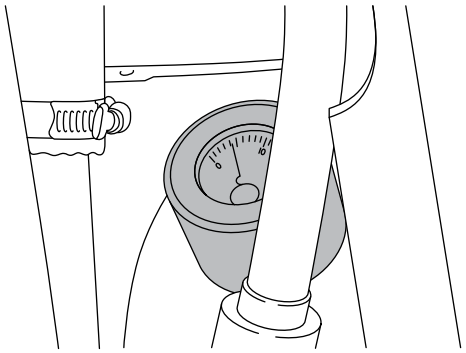

Figure 8-3. Fire Extinguisher Cylinder Pressure Gage

QUESTIONS

1. The fire-extinguishing system is on which bus?

A. The left generator bus
B. The center bus
C. The triple fed bus
D. The hot battery bus

2. Between supply cylinder recharges, each fire-extinguishing system can be fired:

A. One time.
B. Two times.
C. Three times.
D. Four times.

3. An engine fire is detected by:

A. A continuous loop filled with gas.
B. A continuous loop filled with a salt.
C. Photoconductive-cell flame detectors.
D. Heat sensing detectors.

CHAPTER 9
PNEUMATICS

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CHAPTER 9
PNEUMATICS

INTRODUCTION

The pneumatic and vacuum systems are necessary for the operation of surface deicers, instrument air, production of vacuum, rudder boost, flight hourmeter, cabin door seal, pressurization controller, and pressurization outflow and safety valves. Pilots need to know how the bleed air is distributed and controlled for these various uses. This section identifies these systems and covers the pneumatic manifold and controls in detail.

DESCRIPTION

The Pneumatic and Vacuum Systems section of the training manual presents a description and discussion of pneumatic and vacuum systems.

The sources for pneumatic air, vacuum, and acceptable gage readings are discussed.

ENGINE BLEED AIR PNEUMATIC SYSTEM

The pneumatic system in Beechcraft King Airs provides support for several operations on the airplane. These operations include surface deice, rudder boost, escape hatch seal, and the door seal. Pneumatic pressure is used to create a vacuum source for the air-driven gyros, pressurization control, and deflation of the deice boots.

High-pressure bleed air from each engine compressor section, regulated at 18 psi, supplies pressure for the surface deice system, rudder boost, escape hatch and door seals, and vacuum source (Figure 9-1). Vacuum for the flight instruments is derived from a bleed-air ejector. One engine can supply sufficient bleed air for all these systems.

During single-engine operation, a check valve in the bleed air line from each engine prevents flow back through the line on the side of the inoperative engine. A suction gage calibrated in inches of mercury, on the copilot's subpanel, indicates instrument vacuum (GYRO SUCTION). To the right of the suction gage is a PNEUMATIC PRESSURE gage, calibrated in pounds per square inch, which indicates the air pressure available.

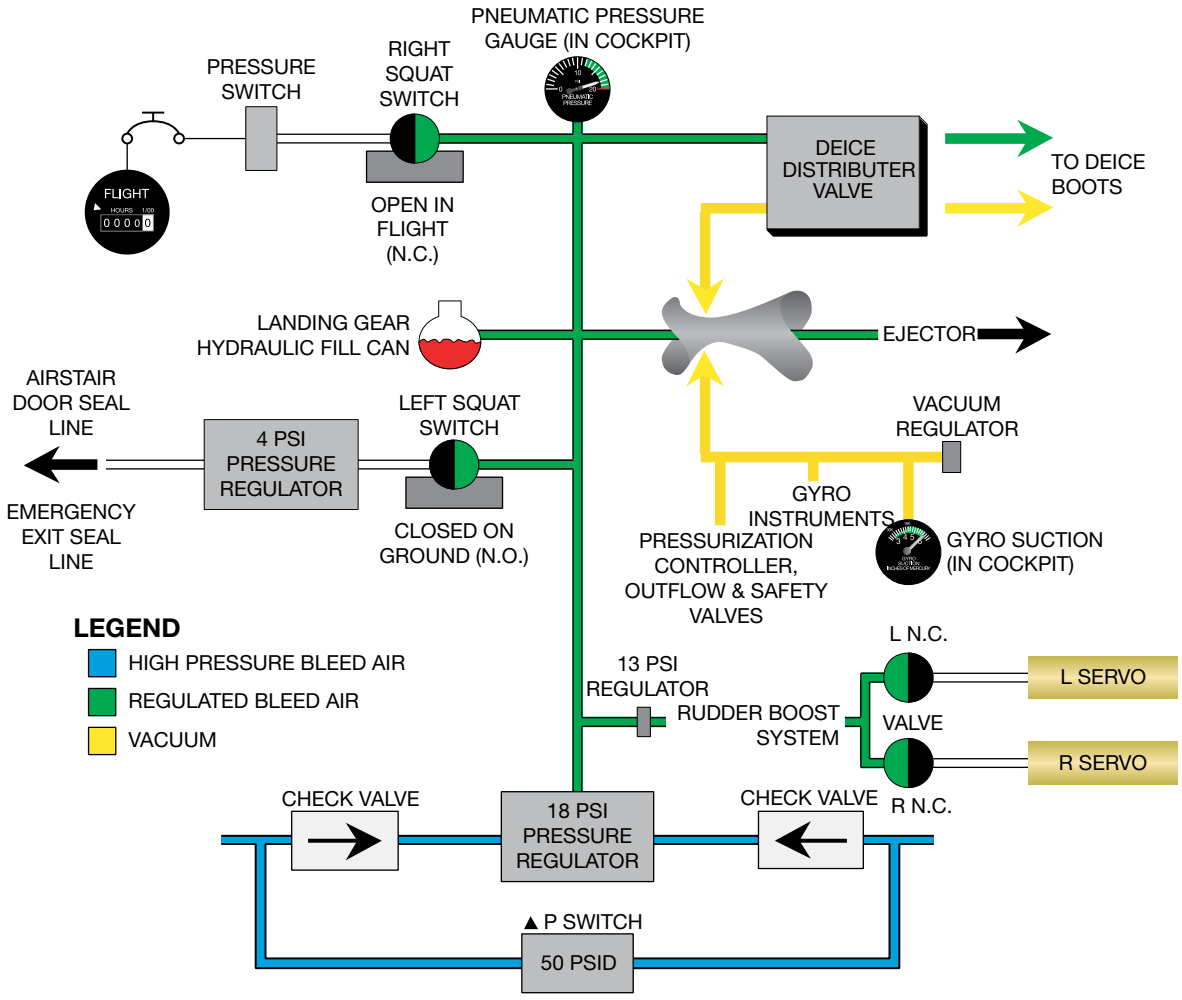

Figure 9-1. Pneumatic System Diagram

PNEUMATIC AIR SOURCE

Bleed air at a maximum rate of 90 to 120 psi pressure is obtained from both engines, and flows through pneumatic lines to a common manifold in the fuselage. Check valves prevent reverse flow during single engine operation.

Downstream from the manifold, the bleed air passes through an 18 psi regulator which incorporates a relief valve set to operate at 21 psi in case of regulator failure. This regulated bleed air is used to supply pneumatic pressure to inflate the surface deicers, escape hatch and door seals, and to provide flow and pressure for the vacuum ejector.

Bleed air is extracted from the P_3 tap of the engine at a temperature of approximately 450°F. It is cooled to approximately 70° above ambient temperature at the manifold in the fuselage due to heat transfer in the pneumatic plumbing.

Ordinarily, the pressure regulator valve, which is under the right seat deck immediately forward of the main spar, will provide 18 +1 psi with the engine running at 70 to 80% N_1. The PNEUMATIC PRESSURE gage on the copilot's right subpanel is provided to allow monitoring of the system pressure (Figure 9-2).

VACUUM AIR SOURCE

Vacuum is obtained from the vacuum ejector. The ejector is capable of supplying from 15 inches Hg vacuum at sea level, to 6 inches Hg vacuum at 31,000 feet. The ejector supplies vacuum for the pressurization control system at a regulated 4.3 to 5.9 inches Hg through a regulator valve.

The vacuum regulator is in the nose compartment on the left side of the pressure bulkhead. The valve is protected by a foam filter.

With one engine running at 70 to 80% N_1, the vacuum gage on the copilot's right subpanel normally should read approximately 5.9 +0/-0.2 inches Hg.

The vacuum line for the instruments is routed through a suction relief valve that is designed to admit into the system the amount of air required to maintain sufficient vacuum for proper operation of the instruments. A GYRO SUCTION gage (Figure 9-3), which is calibrated in inches Hg and is on the copilot's right subpanel, indicates instrument vacuum.

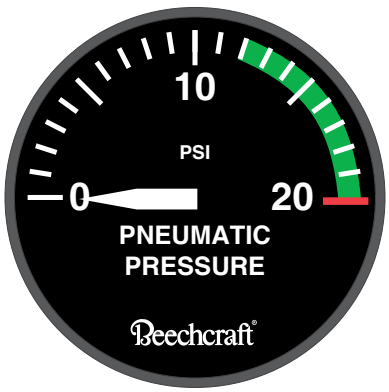

Figure 9-2. Pneumatic Pressure Gage

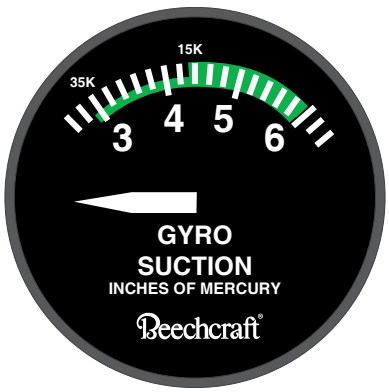

Figure 9-3. Gyro Suction Gage

CABIN DOOR SEAL

The entrance door to the cabin and the escape hatch uses air from the pneumatic system to inflate the seals after the airplane lifts off the ground. Pneumatic air is tapped off the manifold downstream of the 18 psi pressure regulator. The regulated air then passes through a 4 psi regulator and to the normally-open valve that is controlled by the left landing gear safety switch. When the airplane lifts off, the landing gear switch opens the valve to the door and hatch seals, and the seals inflate.

SURFACE DEICE SYSTEM

The leading edges of the wings and horizontal stabilizer are protected against an accumulation of ice buildup (Figure 9-4). Inflatable boots attached to these surfaces are inflated when necessary by pneumatic pressure to break away the ice accumulation, and are deflated by vacuum. The vacuum is always supplied while the boots are not in use and are held tightly against the wing. Vacuum pressure is overcome by pneumatic pressure when the boots are inflated.

Each wing has a leading-edge boot. The horizontal section of the tail has boots on the left and right segments of the horizontal stabilizer and on the vertical stabilizer.

The surface deice system removes ice accumulations from the leading edges of the wings and stabilizers. Ice is removed by alternately inflating and deflating the deice boots (Figure 9-5). Pressure-regulated bleed air from the engines supplies pressure to inflate the boots. A venturi ejector, operated by bleed air, creates a vacuum to deflate the boots and hold them down while not in use. To assure operation of the system in the event of failure of one engine, a check valve is incorporated in the bleed-air line from each engine to prevent loss of pressure through the compressor of the inoperative engine. Inflation and deflation phases are controlled by a distributor valve.

Figure 9-4. Surface Deice Boot Installation

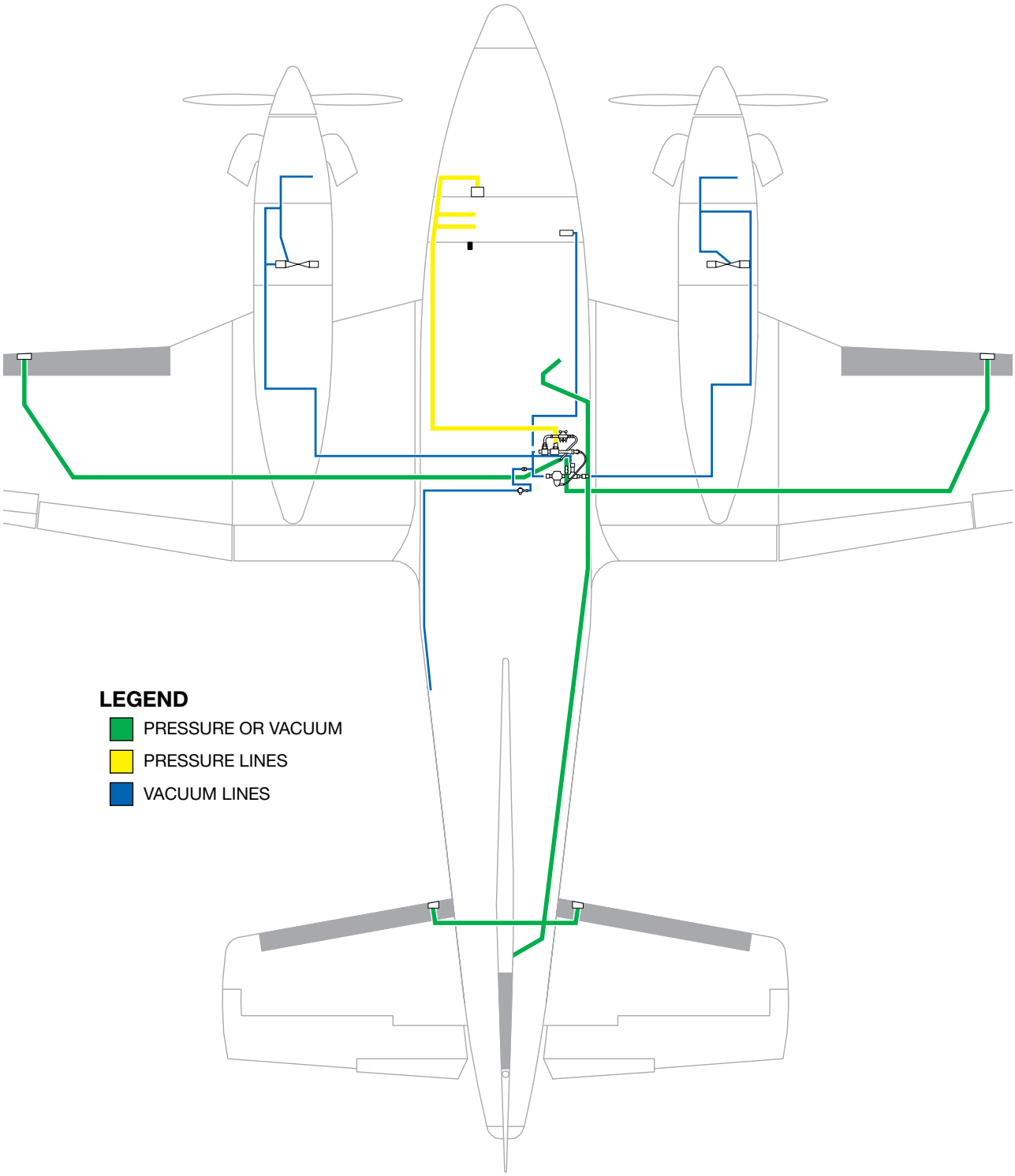

Figure 9-5. Surface Deice System Diagram

A three-position switch in the ICE PROTECTION group on the pilot's subpanel, placarded SURFACE DEICE–SINGLE–OFF MANUAL, controls the deicing operation (Figure 9-6). The switch is spring-loaded to return to the OFF position from SINGLE or MANUAL. When the SINGLE position is selected, the distributor valve opens to inflate the boots. After an inflation period of approximately seven seconds, an electronic timer switches the distributor to deflate the boots. When these boots have inflated and deflated, the cycle is complete. On LJ1138 and after, the wings and tail inflate separately. The wings inflate for six seconds then the tail inflates for four seconds.

When the switch is held in the MANUAL position, all the boots will inflate simultaneously and remain inflated until the switch is released. The switch will return to the OFF position when released. After the cycle, the boots will remain in the vacuum hold-down condition until again actuated by the switch.

Electrical power to the boot system is required for the control valve to inflate the boots in either single-cycle or manual operation. With a loss of this power, the vacuum will hold them tightly against the leading edge.

A single circuit breaker on the copilot's side panel, receiving power from the center bus, supplies the electrical operation of both boot systems. Should the timer fail in the inflated position, the surface deice circuit breaker may be used as a manual control. Pull the circuit breaker out to deflate the boots, and push in to inflate them. Treat the circuit breaker as a manual control.

For most effective deicing operation, allow at least 1/2 inch of ice to form before attempting ice removal. Very thin ice may crack and cling to the boots instead of shedding. Subsequent cyclings of the boots will then have a tendency to build up a shell of ice outside the contour of the leading edge, thus making ice removal efforts ineffective.

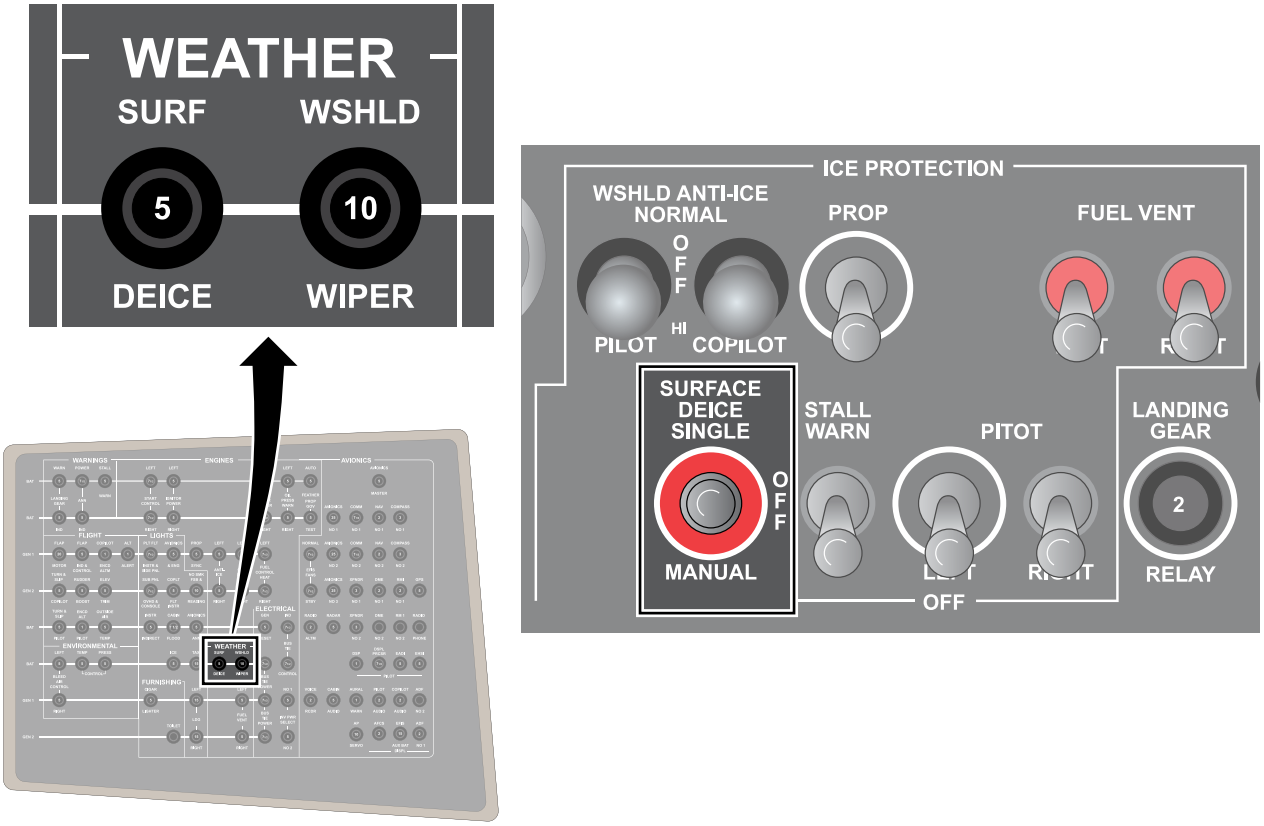

Figure 9-6. Surface Deice Controls

QUESTIONS

1. What systems and components use pneumatic bleed air from the engines?

A. Electrical and hydraulics
B. Air data computer
C. Surface deice, rudder boost, and emergency exit and door seals
D. Windshield heat, radiant heat, flight controls, and stall warning

2. How is negative pressure created to power the vacuum system?

A. An 18 psi regulator
B. A bleed-air ejector
C. An electric vacuum pump
D. A 4 psi regulator

3. What prevents loss of pneumatic pressure during single engine operations?

A. The 4 psi regulator in the pneumatic bleed-air line
B. A reverse current relay
C. A check valve
D. A manually operated valve in each bleed-air line

4. What surfaces are protected by inflatable pneumatic boots?

A. The propeller blade, spinner, and upper wing surfaces
B. The radom surface
C. The surface of the engine inlet lips
D. The leading edge of the wings, and vertical & horizontal stabilizers

5. What component allows 4 psi air to inflate the door and escape hatch seal?

A. A normally open electric valve
B. A normally closed electric valve
C. A manually controled valve
D. An automatic one way check valve

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ICE AND RAIN PROTECTION

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CHAPTER 10
ICE AND RAIN PROTECTION

INTRODUCTION

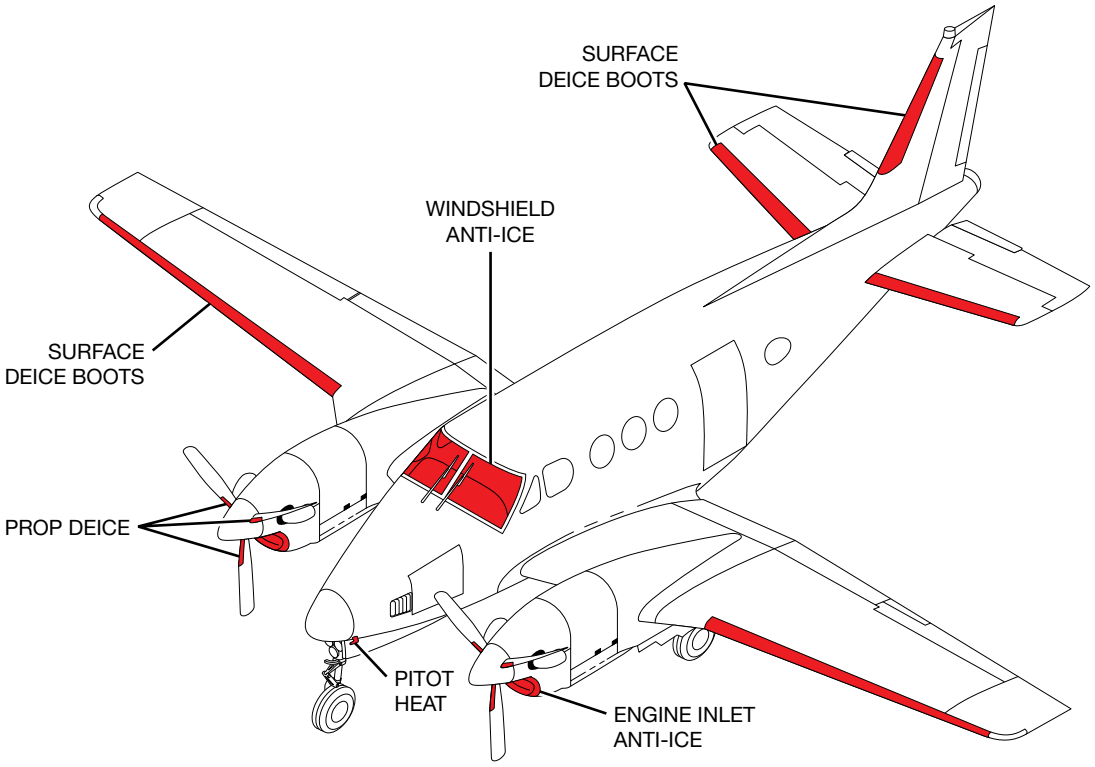
Flight in known icing conditions requires knowledge of conditions conducive to icing, and of all anti-ice and deice systems available to prevent excessive ice from forming on the airplane. This section identifies these systems with their controls and best usage.

GENERAL

This chapter presents a description and discussion of the airplane ice and rain protection systems. All of the anti-ice and deice systems in this airplane are described, showing location, controls, and how they are used.

The purpose of this chapter is to acquaint the pilot with all the systems available for flight in icing or heavy rain conditions, along with their controls. Procedures in case of malfunction in any system are included. This also includes information concerning preflight deicing and defrosting.

The Beechcraft King Air C90A and C90B are FAA-approved for flight in known icing conditions when the required equipment is installed and operational (Figure 10-1). The Required Equipment for Various Conditions of Flight List, contained in the "Limitations" section of the *Pilot's Operating Handbook*, lists the necessary equipment.



SYSTEM AND/OR COMPONENT	VFR DAY	VFR NIGHT	IFR DAY	IFR NIGHT	ICING CONDITIONS	
ICE AND RAIN PROTECTION						
1. ALTERNATE STATIC AIR SYSTEM	0	0	1	1	1	
2. ENGINE AUTO-IGNITION SYSTEM AND ANNUNCIATOR	2	2	2	2	2	
3. ENGINE ANTI ICE SYSTEM AND ANNUNCIATORS	2	2	2	2	2	
4. HEATED FUEL VENT	0	0	2	2	2	
5. HEATED WINDSHIELD (LEFT)	0	0	0	0	1	
6. PITOT HEAT	0	0	2	2	2	
7. PNEUMATIC PRESSURE INDICATOR	0	0	1	1	1	
8. STALL WARNING HEATER	0	0	0	0	1	
9. SURFACE DEICER SYSTEM	0	0	0	0	1	
10. PROPELLER DEICER SYSTEM	0	0	0	0	1	
11. WING ICE LIGHT (LEFT)	0	0	0	0	1	

Figure 10-1. Ice and Rain Protection Required Equipment

The ice and rain protection controls are grouped on the pilot's and copilot's subpanels, except the windshield wiper control, which is overhead (Figure 10-2).

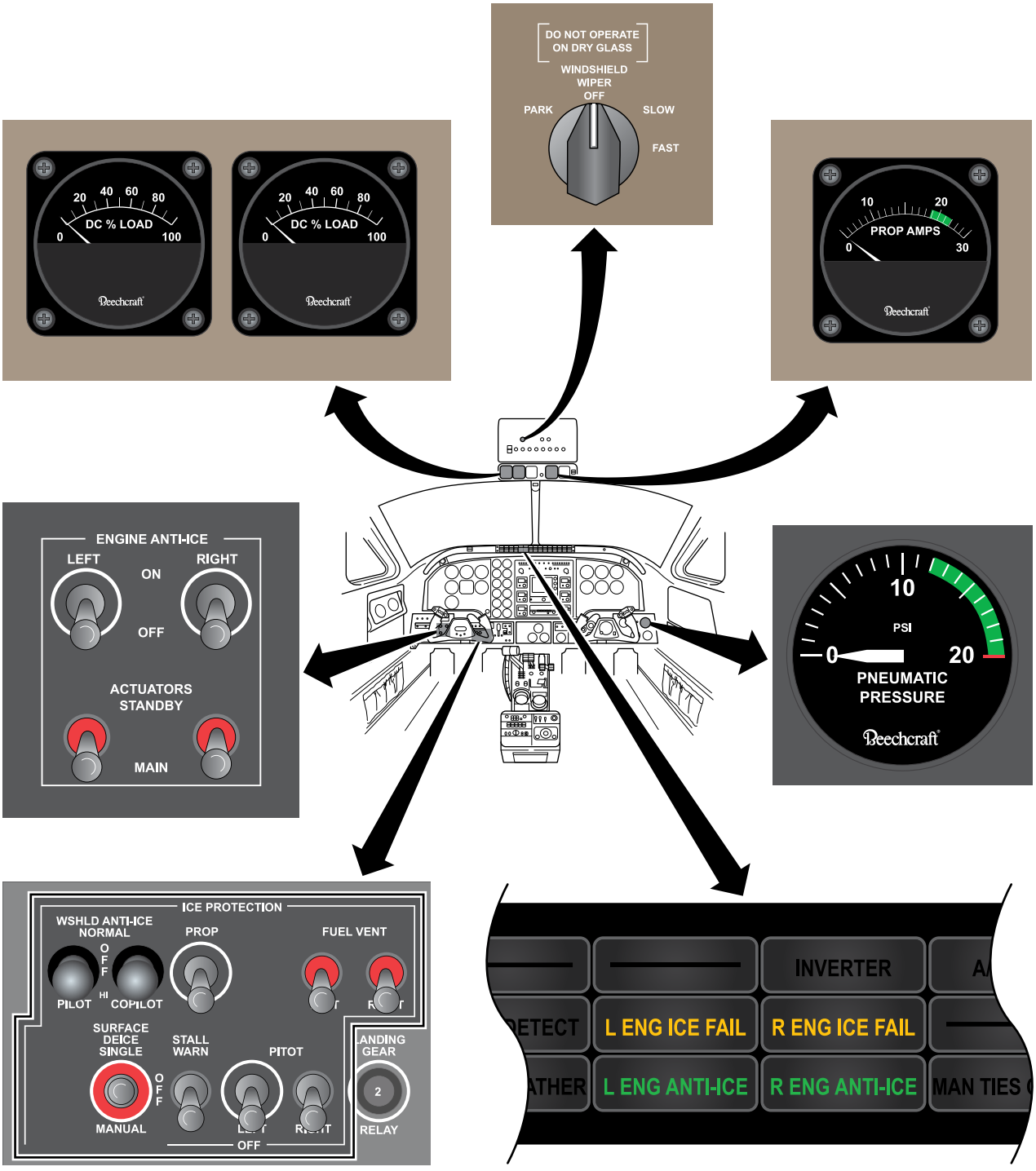


Figure 10-2. Ice and Rain Proctection Controls

ICE PROTECTION SYSTEMS

DESCRIPTION AND OPERATION

There are seven pilot-controlled anti-ice/deice systems:

- Windshield anti-ice
- Surface deice (leading-edge boots)
- Inertial separators (ice vanes)
- Pitot heat
- Propeller deice
- Stall warning heat
- Fuel vent heat

The airplane is equipped with a variety of ice and rain protection systems that can be utilized during operation under inclement weather conditions. Electrical heating elements embedded in the windshield provide adequate protection against the formation of ice, while air from the cabin heating systems prevents fogging, to ensure visibility during operation under icing conditions. Heavy-duty windshield wipers for both the pilot and copilot provide further visibility during rainy flight and ground conditions.

Pneumatic deicer boots on the wings and on the vertical and horizontal stabilizers remove the formation of ice during flight. Regulated bleed-air pressure and vacuum are cycled to the pneumatic boots for the inflation-deflation cycle. The selector switch that controls the system permits automatic single-cycle operation or manual operation.

Ice protection for the engine is provided by an inertial separation system utilizing an electrical actuator. Should the main electrical actuator motor fail, a standby actuator motor is provided. The leading-edge lip of the engine air inlet is continuously anti-iced by engine exhaust air. The propellers are protected against icing by electrothermal boots on each blade that automatically cycle to prevent the formation of ice.

A heating element in both pitot masts prevents the pitot openings from becoming clogged with ice. The heating elements are connected to the airplane electrical system through two 5-ampere circuit-breaker switches.

PITOT HEAT

Two pitot masts located on the nose of the aircraft contain heating elements to protect against ice accumulation (Figure 10-3). The pitot masts are electrically heated to ensure proper airspeed is indicated during icing conditions. Pitot heat is controlled by two circuit-breaker switches located on the pilot's right subpanel. The two switches

Figure 10-3. Pitot Mast and Heat Controls

placarded "PITOT," one for the left mast and one for the right, are located next to the stall warning anti-ice switch. They are two-position switches, with down being OFF and up being ON.

The pitot heat system should not be operated on the ground, except for testing or for short intervals to remove snow or ice from the mast. Pitot heat should be turned on for takeoff and can be left on in flight during icing conditions, or whenever icing conditions are expected. If during flight at altitude there is a gradual reduction in airspeed indication, there may be pitot icing. If turning on the pitot heat restores airspeed, leave the pitot heat on because icing conditions exist. With many pilots, it is standard practice to keep the pitot heat on during all flights at higher altitudes to prevent pitot icing.

STALL WARNING VANE

The stall warning vane and plate (Figure 10-4) is provided with heat to ensure against freeze-up during icing conditions. The stall warning plate is activated by a two-position switch located just to the right of the surface deicer cycle switch on the pilot's right subpanel. The down position is OFF, and the up position is ON. The vane is heated through the battery switch, so it is heated when the battery switch is ON.

A safety switch on the left landing gear limits the current flow to approximately 12 volts to prevent the vane from overheating while the airplane is on the ground. In flight, after the left strut extends, the full 28-volt current is applied to the stall warning vane. The heating elements protect the lift transducer vane and face plate from ice. A buildup of ice on the wing may change or disrupt the airflow and prevent the system from accurately indicating an imminent stall. Remember that the stall speed increases whenever ice accumulates on any airplane.

FUEL SYSTEM ANTI-ICE

There are several anti-ice systems to protect fuel flow through the fuel lines to the engine (Figure 10-5). Without heat, moisture in the fuel could freeze and diminish or cut off the fuel flow to the engines in freezing temperatures.

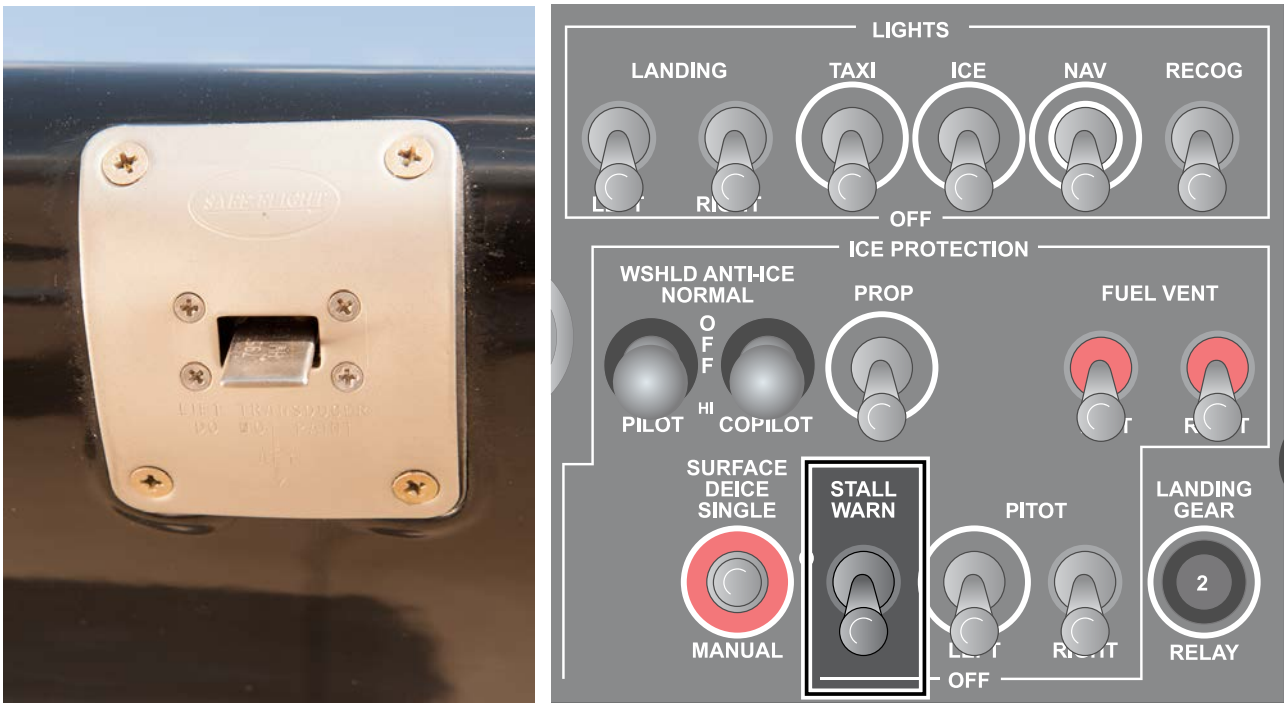


Figure 10-4. Stall Warning Vane and Heat Control

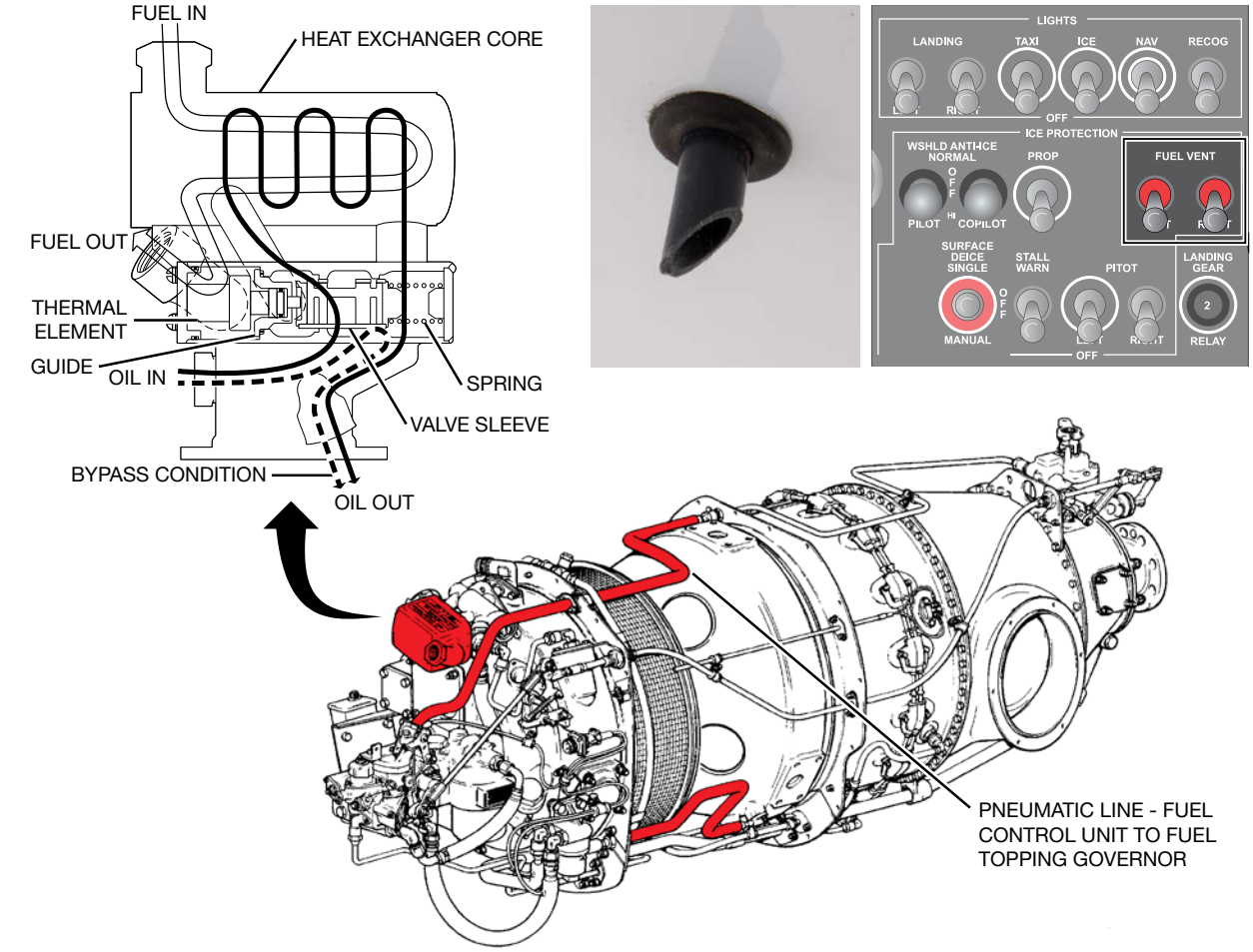


Figure 10-5. Fuel System Anti-ice

Ice formation in the fuel vent system is prevented by electrically heated vents in each wing. The fuel vent heat is operated by left and right switches located in the ICE PROTECTION group on the pilot's right subpanel. These switches should be turned on whenever ice is anticipated or encountered.

A portion of the fuel control unit ice protection is provided by an oil-to-fuel heat exchanger, mounted on the engine's accessory section. An engine oil line within the heat exchanger is located around the fuel line. Heat transfer occurs through conduction. This heat melts ice particles which may have formed in the fuel. This operation is automatic whenever the engines are running. Refer to the *POH* "Limitations" section for temperature limitations concerning the oil-to-fuel heat exchanger.

The pneumatic line, from the engine to the FCU and the pneumatic line from the FCU to the fuel topping governor, is protected by an electrically heated jacket. This heat is automatically applied when the condition levers move out of the fuel cutoff range. No other action is required.

WINDSHIELD WIPERS

Separate windshield wipers are mounted on the pilot's and copilot's windshield. The dual wipers are driven by a mechanism operated by a single electric motor, all located forward of the instrument panel.

The windshield wiper control is located on the overhead light control panel (Figure 10-6). It provides the wiper mechanism with SLOW, FAST, and PARK positions. The wipers may be used either on the ground or in flight, as required. The wipers must not be operated on a dry windshield. The windshield wiper circuit breaker is on the copilot's right-side circuit-breaker panel in the WEATHER group.

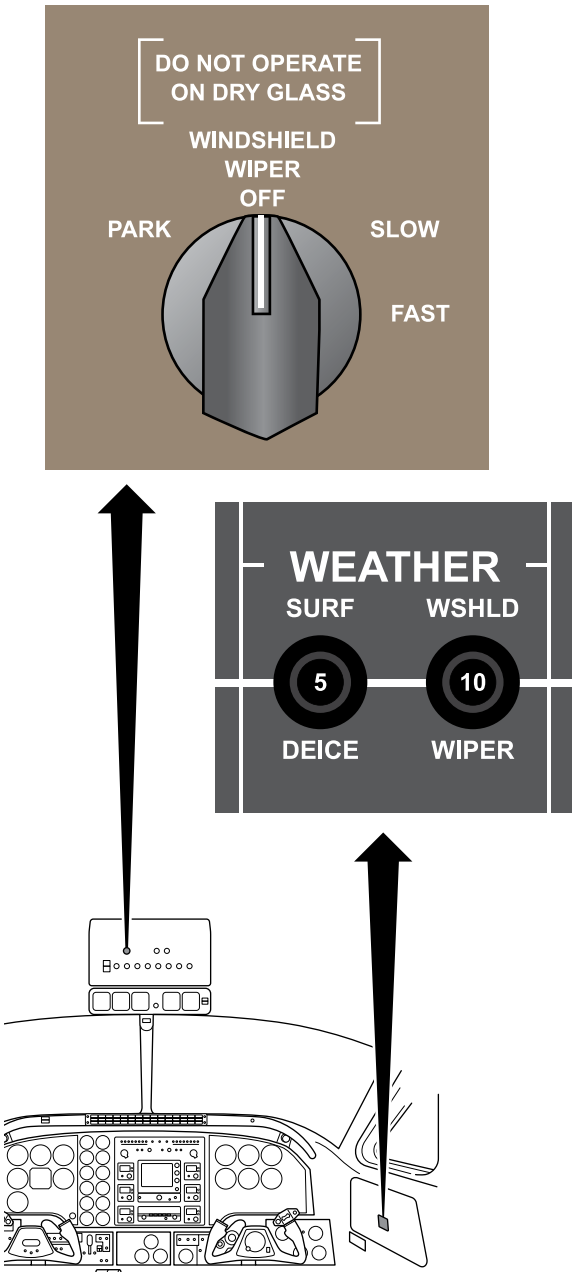


Figure 10-6. Windshield Wipers

WINDSHIELD ANTI-ICE

The pilot's and copilot's windshields each have independent controls and heating circuits. The control switch allows the pilot to select a high or a low intensity heat level.

The windshields are composed of three physical layers (Figure 10-7). The inner layer is a thick panel of glass that acts as the structural member. The middle layer is a polyvinyl sheet which carries the fine wire heating grids. The outer layer is a protective layer of glass bonded to the first two layers. The outside of the windshield is treated with a static discharge film called a "NESA coating."

Figure 10-7. Windshield Installation

The windshields are protected against icing by electrical heating elements (Figure 10-8). The heating elements are connected at terminal blocks in the corner of the glass to the wiring leading to the control switches mounted in the pilot's right subpanel.

A transparent material (usually stannic oxide) which has high electrical resistance is incorporated in the laminations of each windshield, pilot's and copilot's. Each windshield is also fitted with electrical connections for the resistive material and for temperature-sensing elements. The resistive material is arranged so as to provide primary heated surfaces and secondary heated surfaces.

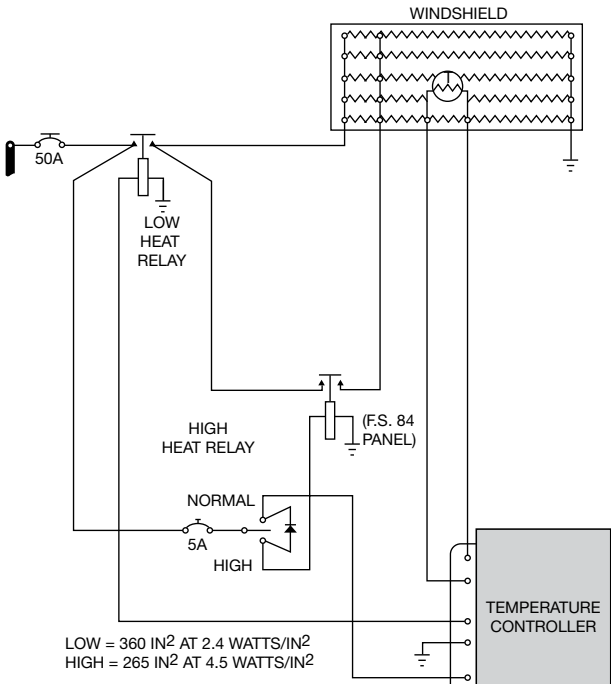


Figure 10-8. Windshield Anti-ice Diagram

PILOT and COPILOT WSHLD ANTI-ICE switches in the ICE PROTECTION group on the pilot's inboard subpanel are used to control windshield heat (Figure 10-9). They have positions labeled "NORMAL," "OFF," and "HI." When the PILOT and COPILOT switches are in the NORMAL (up) position, the secondary areas of the windshields are heated. When the switches are in the HI (down) position, the primary areas are heated. The primary areas are smaller areas and are heated faster to the same temperatures as the NORMAL position.

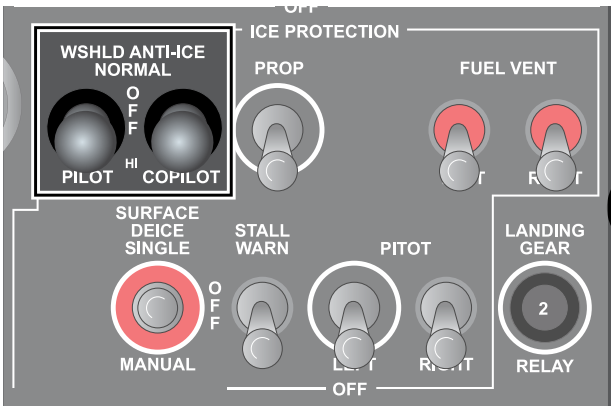


Figure 10-9. Windshield Anti-ice Switches

Each switch must be lifted over a detent before it can be moved into the HI position. This lever-lock feature prevents inadvertent selection of the HI position when moving the switches from NORMAL to the OFF (center) position.

Windshield temperature is controlled automatically by the use of a temperature-sensing element embedded in each windshield, and a temperature controller in each windshield circuit. The temperature controllers operate between 90° and 110° F to maintain the desired mean temperature of the windshield heating surfaces.

When the low level of heating is selected, an automatic temperature controller senses the windshield and attempts to maintain it at approximately 90° to 110° F. It does so by energizing the "low" heat relay as necessary. In this mode, the entire windshield is heated (Figure 10-10).

When the high level of heating is selected, the same temperature controller senses the windshield temperature and attempts to maintain it at 90° to

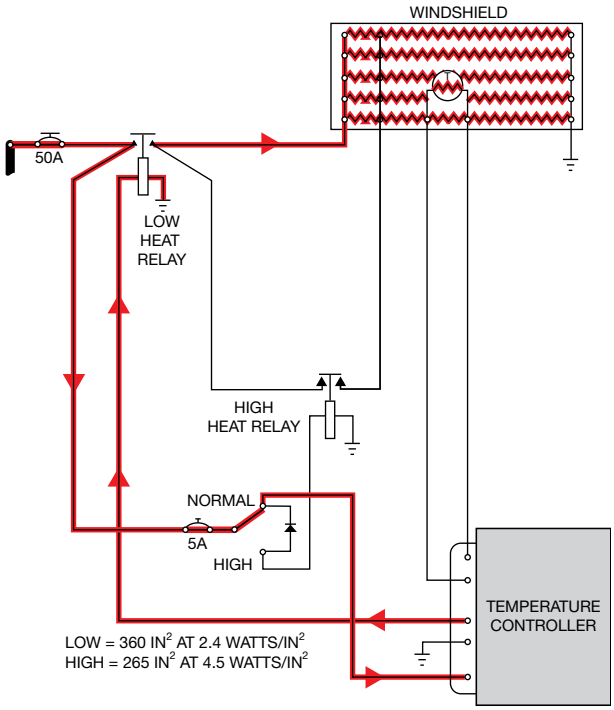


Figure 10-10. Windshield Anti-ice Diagram—Normal Heat

110º F. In this mode, however, the controller will energize the high heat relay switch, which applies the electrical heat to a more concentrated but more essential viewing area of the windshield. In high, approximately two-thirds of the windshield is heated at the outboard portion (Figure 10-11).

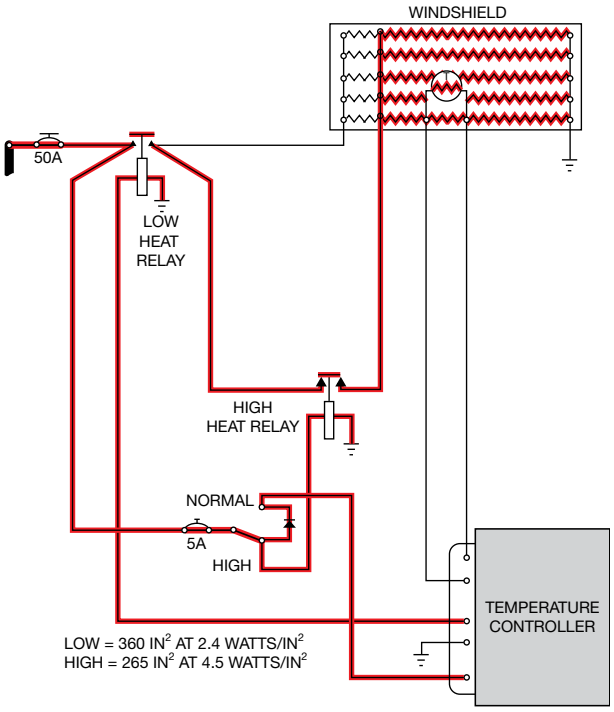


Figure 10-11. Windshield Anti-ice Diagram—High Heat

The power circuit of each system is protected by 50-ampere current limiters located in the power distribution panel. Windshield heater control circuits are protected with 5-ampere circuit breakers located on a panel mounted on the forward pressure bulkhead (forward of the pilot's left subpanel).

Windshield heat may be used at any time and in any combination. Use of windshield heat, however, may cause erratic operation of the magnetic compass because of the electrical field created by the heating elements.

CAUTION

In the event of windshield icing during sustained icing conditions, it may be necessary to reduce the airspeed in order to keep the windshield ice-free.

ENGINE AIR INLET LIP HEAT

The lip around each air inlet is heated by hot exhaust gases to prevent the formation of ice during inclement weather (Figure 10-12).

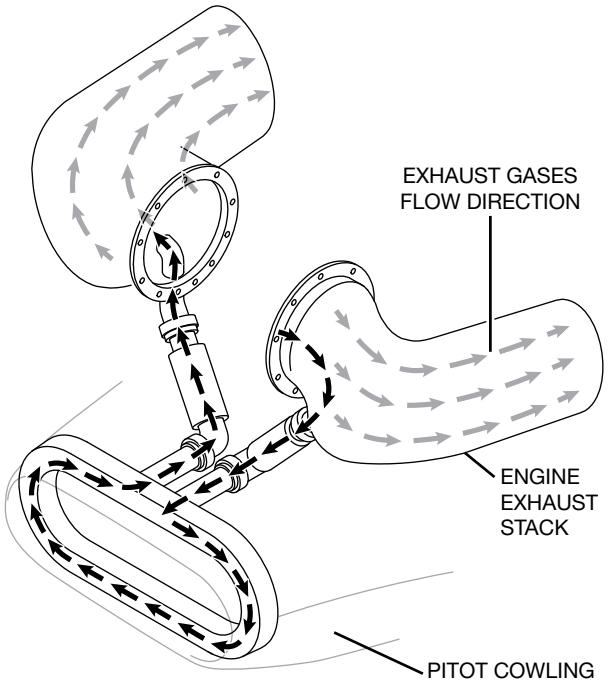


Figure 10-12. Engine Air Inlet Lip Heat

A scoop in the left engine exhaust stack deflects the hot exhaust gases downward into the hollow lip tube that encircles the engine air inlet. The gases are expelled through a line into the right exhaust stack, where they move out with the engine exhaust gases.

Heat will flow through the inlet whenever the engine is running.

ENGINE INERTIAL SEPARATORS

An inertial vane system of separators is installed on each engine to prevent ice, or other foreign objects such as dust or gravel, from entering the engine inlet plenum or ice accumulating on the engine inlet screen. A movable vane and a bypass door are closed (retracted) for normal flying conditions (Figure 10-13).

At temperatures above +5° C, the ice vane and door should be in the retract position, as ice formation is unlikely.

When in icing conditions with the ice vane in the extend position (Figure 10-14), the ice vane is positioned to create a venturi effect and introduces a sudden turn into the engine. At the same time the bypass door in the lower cowling at the aft end of the air duct is open.

As the ice particles or water droplets enter the air inlet, the airstream with these particles is accelerated by the venturi effect. Due to their greater mass, and therefore greater momentum, the frozen moisture particles accelerate past the screen area and are discharged overboard through the bypass door. The airstream, however, makes the sudden turn free of ice particles and enters the engine through the inlet screen.

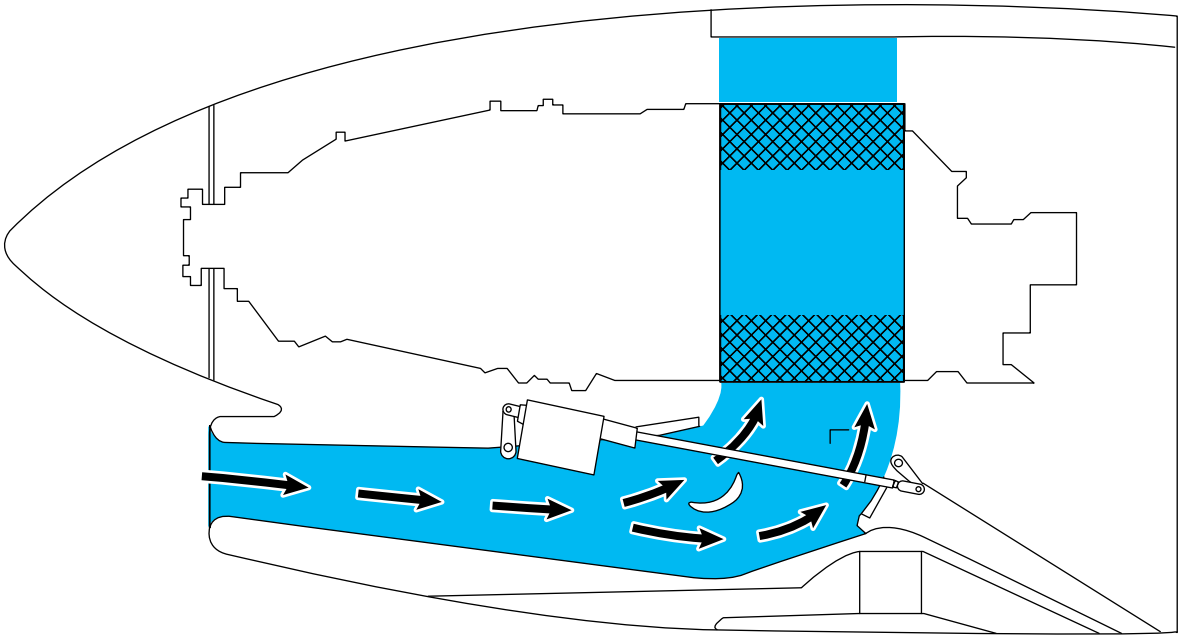


Figure 10-13. Inertial Separator in Retract Position

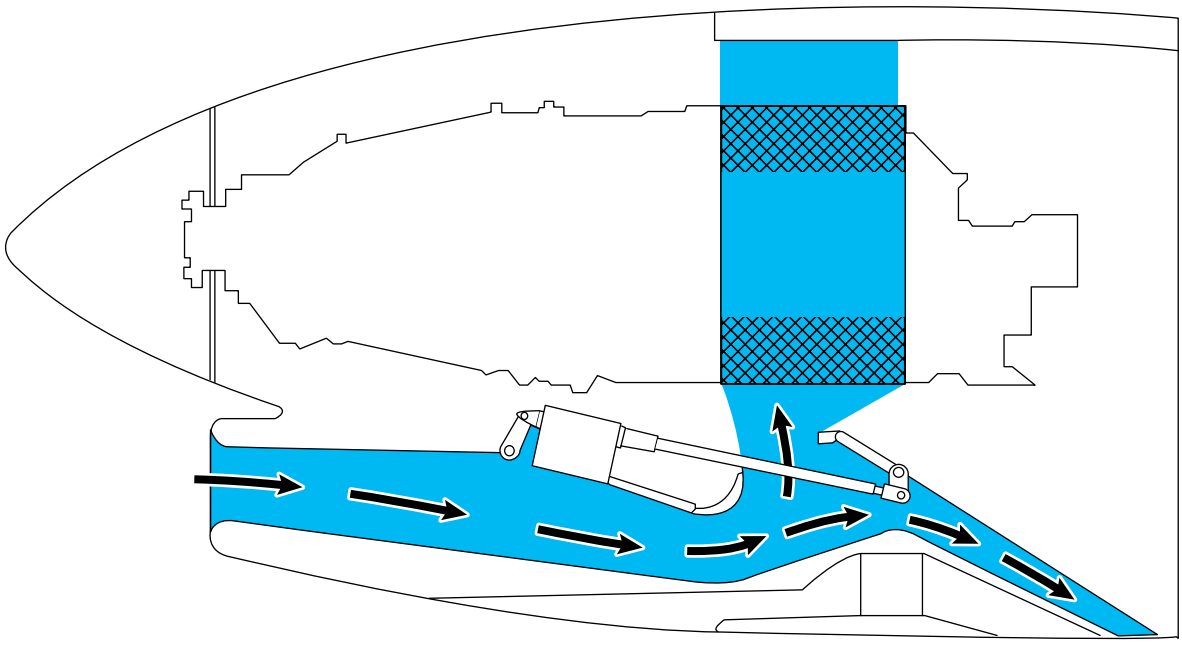


Figure 10-14. Inertial Separator in Extend Position

ICE VANE CONTROLS

The ice vane and bypass doors are extended or retracted simultaneously through a linkage system connected to electric actuators. The actuators are energized through switches in the ICE PROTECTION group located on the pilot's left subpanel (Figure 10-15). The ICE VANE switches extend the separators in the on position and retract them in the OFF position, which is used for all normal flight operations.

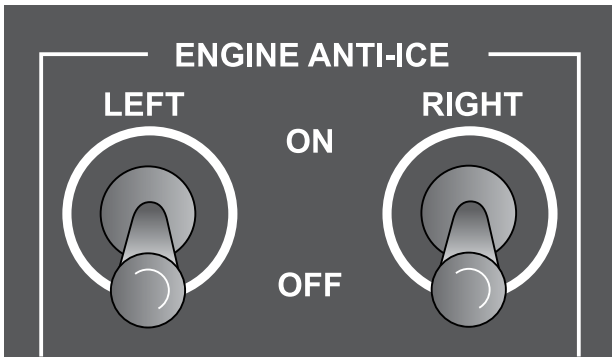


Figure 10-15. Ice Vane Controls

The ice vanes should be extended whenever there is visible moisture at +5º C. When the ice vanes are extended, the two green advisory annunciators will illuminate, and because the airflow into the engine will be restricted, there will be a drop in torque and a slight increase in ITT. When the ice vanes and bypass doors are retracted, the annunciators will extinguish, torque will be restored, and ITT will decrease.

The anti-ice vanes are controlled by switches located on the left subpanel. The LEFT and RIGHT ENGINE ANTI-ICE switches have positions labeled "ON" and "OFF," while the ACTUATORS switch has positions labeled "STANDBY" and "MAIN."

The actuators have dual motors to provide a redundant system. The ACTUATORS switch allows the selection of either the MAIN or STANDBY actuator motor. The main and standby actuators have different circuitry but share the same torque tube drive system.

The vanes have only two positions; there are no intermediate positions. The system is monitored by L and R ENG ANTI-ICE (green) and L and R ENGICE FAIL (yellow) annunciators (Figure 10-16). Illumination of the L and R ENG ANTIICE annunciators indicate that the system is actuated.

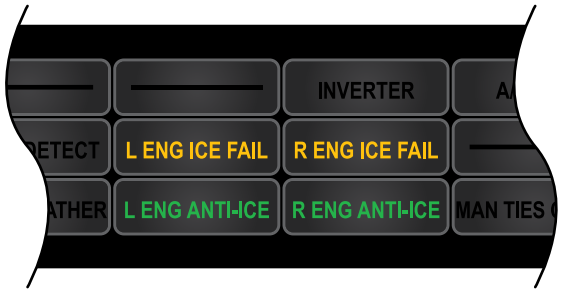


Figure 10-16. Caution and Advisory Annunciators

Illumination of the L or R ENG ICE FAIL annunciator indicates that the system did not operate to the desired position. Immediate illumination of the L or R ENG ICE FAIL annunciator indicates loss of electrical power, whereas delayed illumination indicates an inoperative actuator.

The yellow ENG ICE FAIL annunciator circuit compares the ANTI-ICE switch position to the microswitches checking ice vane open or closed. After a 35-second delay, the annunciator will illuminate if the switch position and microswitches do not agree. In addition, if the power source for the actuator system selected (MAIN or STANDBY) is removed, the ICE VANE FAIL light will illuminate immediately. In either event, the STANDBY actuator should be selected.

ENGINE AUTOIGNITION SYSTEM

The engine autoignition system provides automatic ignition to attempt a restart should a flame-out occur. Once armed, the system ensures ignition during takeoff, landing, turbulence, and penetration of icing or precipitation conditions. Should ice or rain cause an engine flameout, autoignition will automatically reignite the engine.

The switches used to arm the autoignition system are located on the pilot's left subpanel, above the ice vane switches and just to the left of the control column (Figure 10-17). The system is activated by moving the switches into the up or ARM position. Each switch must be lifted over a lock-out barrier before it can be moved into, or out of, the

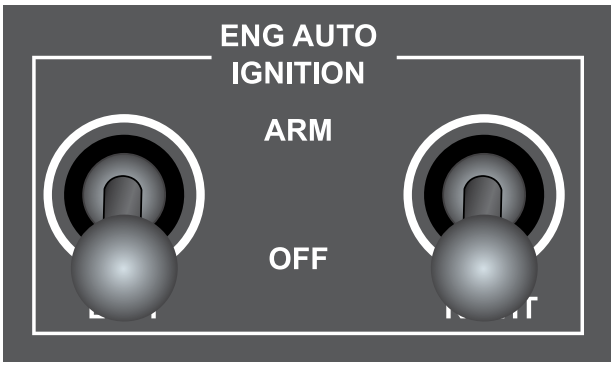


Figure 10-17. Engine Autoignition Switches

ARM position. This lever-lock feature prevents inadvertent movement to the OFF position.

If, for any reason, engine torque falls below four hundred foot-pounds, electrical power is provided to energize the engine igniters. As this happens, the green IGNITION ON annunciator on the panel will illuminate, indicating that the ignition system is energized. During ground operation, the system should be turned off to prolong the life of the igniter units.

PROPELLER ELECTRIC DEICE SYSTEM

The propeller electric deicer system includes: an electrically heated boot for each propeller blade, slip rings, brush assemblies, timer, on-off switch, and an ammeter (Figure 10-18).

When the switch is turned on, the ammeter registers the amount of current (14 to 18 amperes on the C90A or 18 to 24 amperes on the C90B) passing through the system. If the current rises beyond the limitations, a circuit-breaker switch or current limiter will shut off power to the deicer timer. The current flows from the timer through the brush assemblies to the slip rings, where it is distributed to the individual propeller deicer boots.

Heat produced by the heating elements in the deicer boots reduces the adhesion of the ice. Adhesion thus reduced, the ice is removed by the centrifugal effect of the propeller and the blast of the airstream.

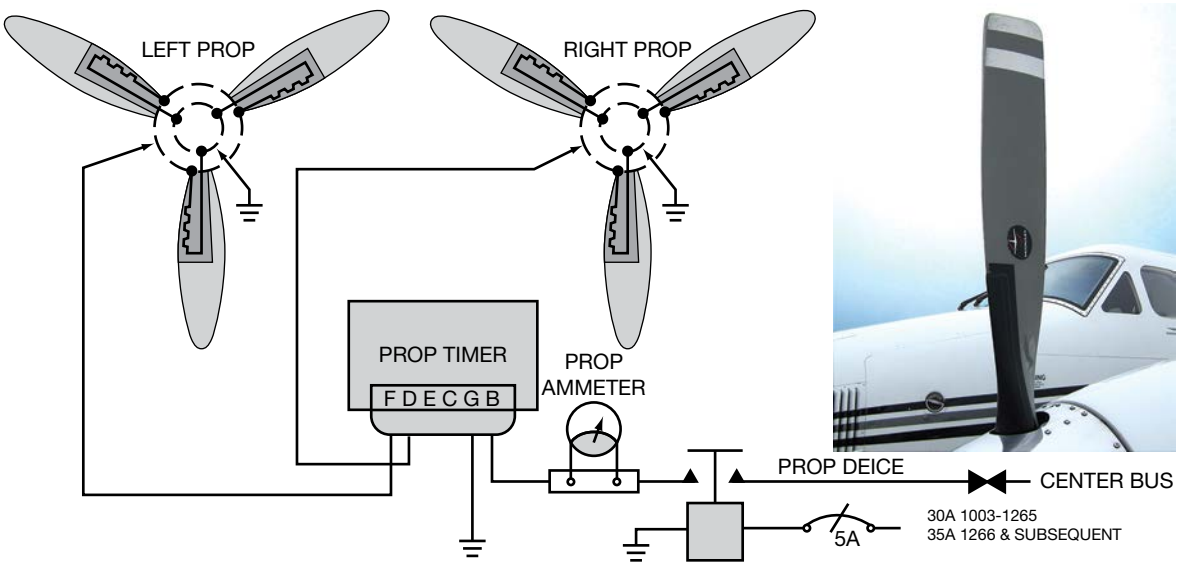


Figure 10-18. Propeller Electric Deice System

NOTE

The heating sequences for the deicer boots noted in the following section are the sequences which are in evidence during normal operation.

Power to the deicer boots is cycled in 90-second phases. The first 90-second phase heats all the deicer boots on the RH propeller. The second phase heats all the deicer boots on the LH propeller. The deicer timer completes one full cycle every three minutes. As the deicer timer moves from one phase to the next, a slight momentary deflection of the propeller ammeter needle may be noted. Propeller deice must not be operated when the propellers are static.

WING ICE LIGHTS

Wing ice lights are provided to light the wing leading edges to determine ice buildup in icing conditions. The wing lights are located on the outboard side of each nacelle. The circuit-breaker switch is located on the pilot's right subpanel in the LIGHTS group above the ICE protection group (Figure 10-19).

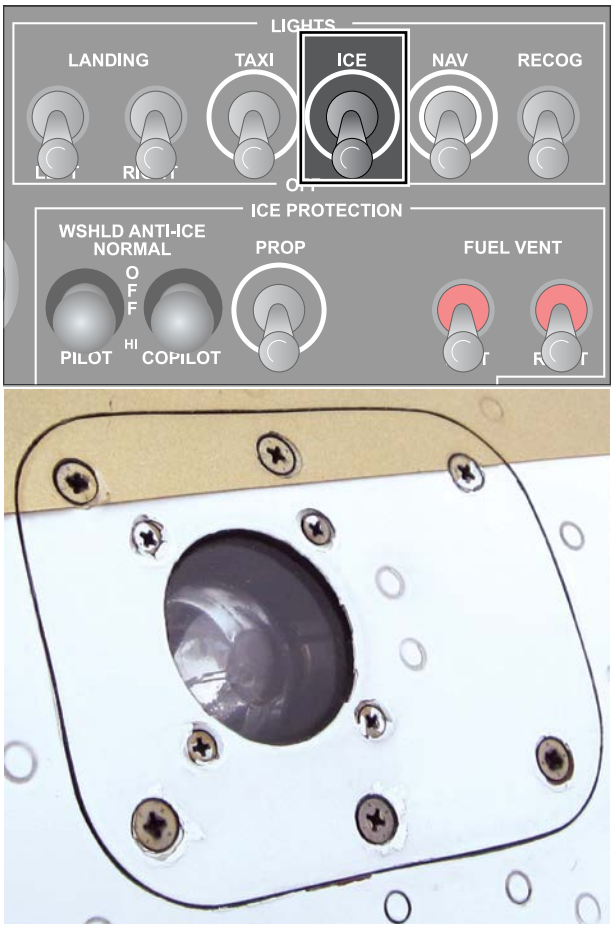


Figure 10-19. Wing Anti-ice Lights

The wing ice lights should be used as required in night flight to check for wing ice accumulation. The wing ice lights operate at a high temperature and therefore should not be used for prolonged periods while the airplane is on the ground. All ice lights installed must be operational for flights into known or forecast icing conditions at night.

PRECAUTIONS DURING ICING CONDITIONS

There are some precautions which prevail during winter or icing conditions. An airplane needs special care and inspection before operation in cold or potential icing weather. In addition to the normal exterior inspection, special attention should be paid to areas where frost and ice may accumulate.

Pilots should be familiar with the potential harm a harmless-looking, thin layer of frost can cause. It is not the thickness of the frost that matters; it is the texture. A slightly irregular surface can substantially decrease proper airflow over the wings and stabilizers. Never underestimate the damaging effects of frost. All frost should be removed from the leading edges of the wings, stabilons, stabilizers, and propellers before the airplane is moved.

Control surfaces, hinges, the windshield, pitot masts, fuel tank caps, and vents should also be free of frost. Deicing fluid should be used when needed.

Fuel drains should be tested for free flow. Water in the fuel system has a tendency to condense more readily during winter months, and if left unchecked, large amounts of moisture may accumulate in the fuel tanks. Moisture does not always settle at the bottom of the tank. Occasionally a thin layer of fuel gets trapped under a large mass of water, which may deceive the tester. Make sure a good-sized sample of fuel is taken.

It is also important to add only the correct amount of anti-icing additive to the fuel. A higher concentration of anti-icer does not ensure lower fuel freezing temperatures and may hinder the airplane's performance. Consult the "Normal Procedures" section of the *Pilot's Operating Handbook* to determine the correct blend.

The brakes and tire-to-ground contact should be checked for lockup. No anti-ice solution containing oil-based lubricant should be used on the brakes. If tires are frozen to the ground, use undiluted defrosting fluid or a ground heater to melt ice around the tires, then move the airplane as soon as the tires are free. Heat applied to tires should not exceed 160°F or 71°C.

Tiedowns for propellers should be installed to ensure against damage to internal engine components not lubricated when the engine is not operating. Spinning propellers can also be a source of danger to crew, passengers, and ground support personnel. Propeller blades held in their tiedown position channel moisture down the blades, past the propeller hub, and off the lower blade more effectively than in other positions or when left spinning. During particularly icy ground conditions, the propeller hubs should also be inspected for ice and snow accumulation.

Pitot masts should always be covered while the airplane is resting. Once the covers are removed, make sure both masts and drains are free of ice or water. Faulty readings could be obtained if they are clogged.

During extended periods of taxiing or ground holding, the autoignition system should be turned off until right before takeoff. This will help to prolong the service life of the igniter units.

Snow, slush, or standing water on the runway degrade airplane performance whether landing or taking off. During takeoff, more runway is needed to achieve necessary takeoff speed, while landing roll is longer because of reduced braking effectiveness.

Only the surface deicers are true deicers. The rest are really anti-icers and should be used to prevent the formation of ice, not melt ice already present. Accumulated ice on even the best-equipped airplane will degrade its performance and ruin at least the time and fuel calculations used for flight planning. A minimum speed of 140 KIAS is necessary to prevent ice formation on the underside of the wing, which cannot be adequately deiced.

Due to distortion of the wing airfoil, stalling airspeeds should be expected to increase as ice accumulates on the airplane. For the same reason, stall warning devices are not accurate and should not be relied upon. Maintain a comfortable margin of airspeed above the normal stall airspeed when ice is on the airplane. In order to prevent ice accumulation on unprotected surfaces of the wing, maintain a minimum of 140 knots during operations in sustained icing conditions. In the event of windshield icing, it may be necessary to reduce airspeed.

While in flight, the engine ice vanes must be extended and the appropriate annunciator lights monitored:

- Before visible moisture is encountered at OAT +5ºC and below
- At night when freedom from visible moisture is not assured and the OAT is +5ºC or below

During flight in icing conditions, fuel vent heat, pitot heat, prop deice, windshield heat, and stall warning heat should all be ON.

QUESTIONS

1. How many pilot-controlled anti-ice/deice systems are installed?

A. 5
B. 6
C. 7
D. 8

2. When should the pitot heat be turned on?

A. Immediately after starting engines
B. At the beginning of taxi operations
C. Just prior to takeoff
D. Only during IMC conditions

3. What should the pilot consider if there is an accumulation of ice on the leading edge of the wing?

A. The stall warning vane is always reliable

B. The stall warning vane is not to be considered reliable

C. The stall warning vane circuit breaker switch will turn off
D. The stall warning vane circuit breaker on the right breaker panel will open

4. What does the oil-to-fuel heat exchanger provide?

A. Ice free fuel entering the fuel control unit
B. Ice free fuel in the fuel tanks
C. Ice free bleed air lines feeding the fuel control unit
D. Ice free bleed air from entering the main bearings in the engine

5. When should the windshield wipers be operated?

A. Only when on the ground during taxi
B. Only when in flight
C. Only when above 5,000 AGL
D. Only when there is water on the windshield

6. What condition may be observed if windshield heat is selected to NORMAL or HI?

A. Erratic operation of the magnetic compass
B. Erratic operation of the windshield wipers
C. Erratic operation of the magnetic flux detectors
D. Erratic operation of the course deviation indicator

7. What provides engine inlet heat?

A. Hot bleed air
B. Electric heating elements embedded in a rubber coating
C. Pneumatic air supplied by a heat pump
D. Hot turbine engine exhaust gases

8. When should the engine ice vanes be extended?

A. Anytime the OAT is above 5°C
B. Anytime the OAT is below 5°C
C. Anytime the OAT is below 5°C and flight free of visible moisture cannot be assured
D. Anytime the OAT is above 5°C and flight free of visible moisture cannot be assured

9. When should the pilot activate the Engine Autoignition System?

A. Immediately after starting the engine
B. Beginning taxi operations
C. Only when flight free of rain cannot be assured
D. Just prior to takeoff

10. How long does the automatic propeller deice system heat a propeller blade?

A. 30 seconds
B. 45 seconds
C. 60 seconds
D. 90 seconds

11. What lights must be operational in known or forecast icing at night?

A. One landing light and one ice light
B. Both landing lights and one ice light
C. One ice light and a flashlight with two "D" cell batteries
D. All ice lights installed

12. What is the minimum speed for flight in sustained icing conditions?

A. 110 KIAS
B. 120 KIAS
C. 130 KIAS
D. 140 KIAS

CHAPTER 11
AIR CONDITIONING

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CHAPTER 11
AIR CONDITIONING

INTRODUCTION

Passenger comfort and safety is of prime importance. The task is to teach participants to operate the environmental systems effectively and within the system's limitations.

DESCRIPTION

The Environmental System section of the training manual presents a description and discussion of the air conditioning, bleed-air heating, and fresh air systems. Each system includes general description, principle of operation, controls, and emergency procedures.

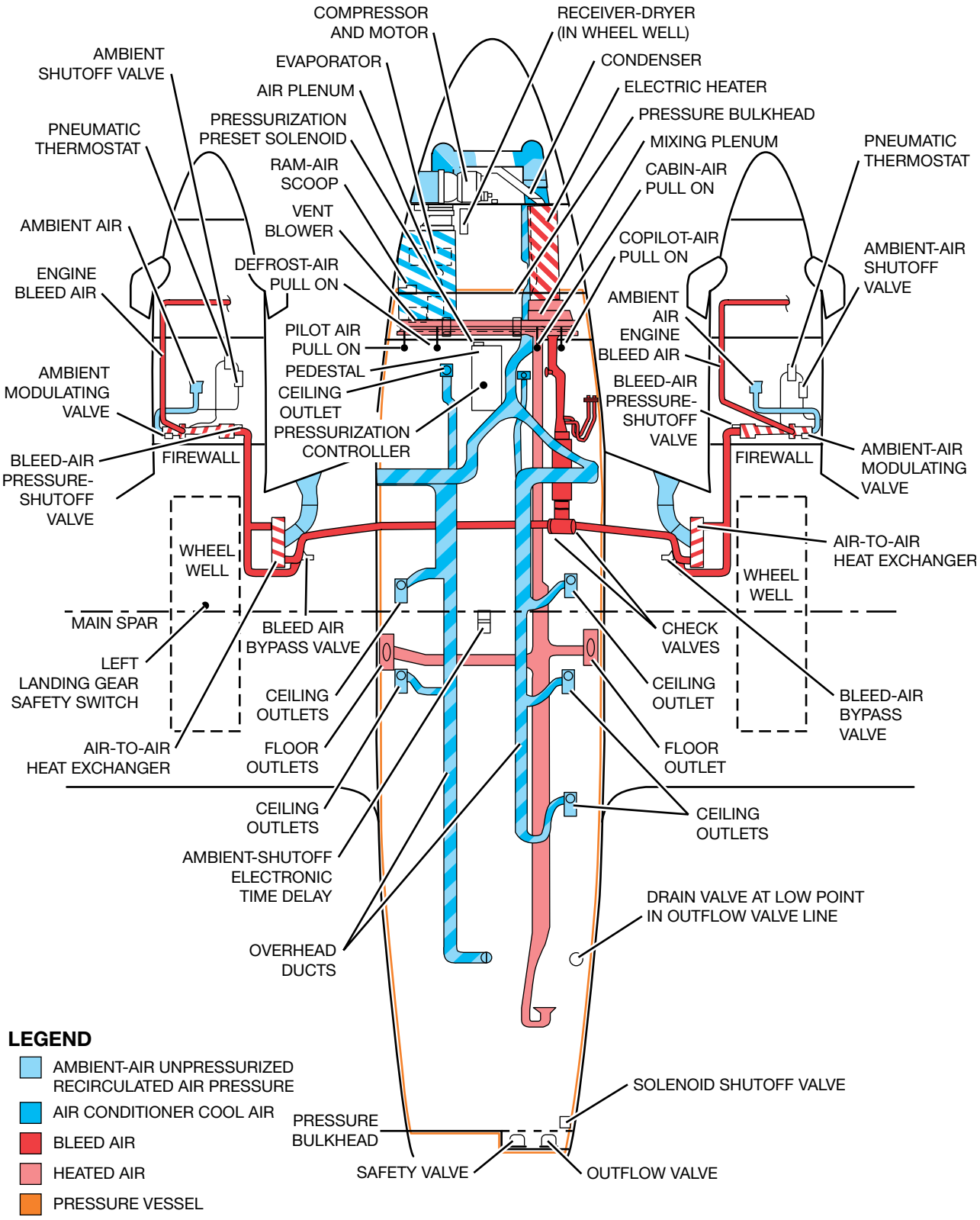

Figure 11-1. Environmental System Schematic

ENVIRONMENTAL SYSTEM

"Environmental System" refers to the devices which control the pressure vessel's environment. Along with insuring the circulation of air, this system controls temperature by utilizing heating and cooling devices as needed.

The environmental system consists of bleed-air pressurization, heating and cooling systems and their associated controls. The Beechcraft King Air series environmental system (Figure 11-1) uses turbine engine bleed air for cabin pressurization and cabin heating. The air conditioning system, driven by the electrical system, provides cool air to the airplane cabin.

The ENVIRONMENTAL control section on the copilot's left subpanel (Figure 11-2) provides for automatic or manual control of the system. This section contains all the major controls of the environmental function:

- Bleed-air valve switches
- Vent blower control switch
- Manual temperature switch for control of the bypass valves in the air-to-air heat exchangers
- Cabin-temperature-level control
- Cabin temperature mode selector switch for selecting automatic heating or cooling, manual heating or cooling
- Electric heat control switch

Four additional manual controls (Figure 11-3 through Figure 11-6) on the main instrument subpanels may be utilized for partial regulation of cockpit comfort when the cockpit partition curtain is closed and the cabin comfort level is satisfactory. They are: pilot's air, defroster air,

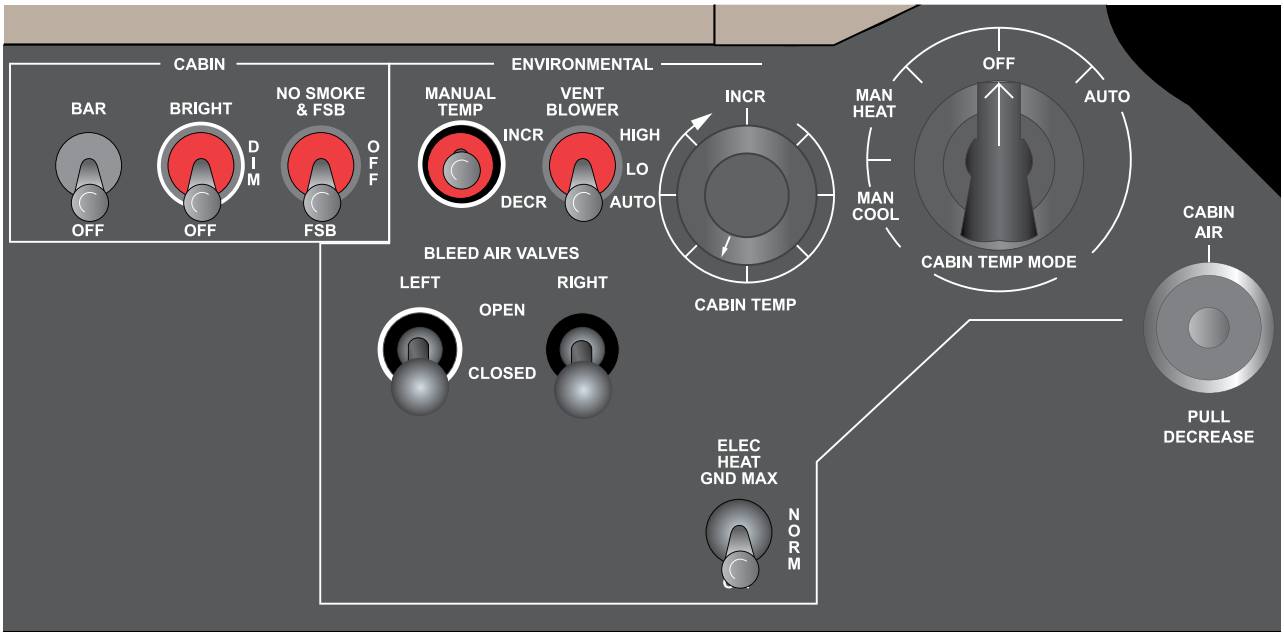

Figure 11-2. Environmental Group Switches and Knobs

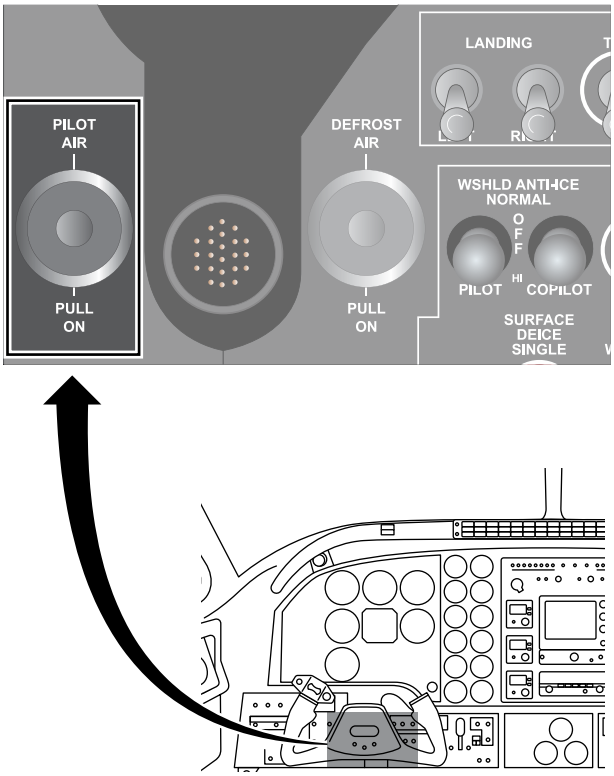

Figure 11-3. Air Control Knobs—Pilot Air

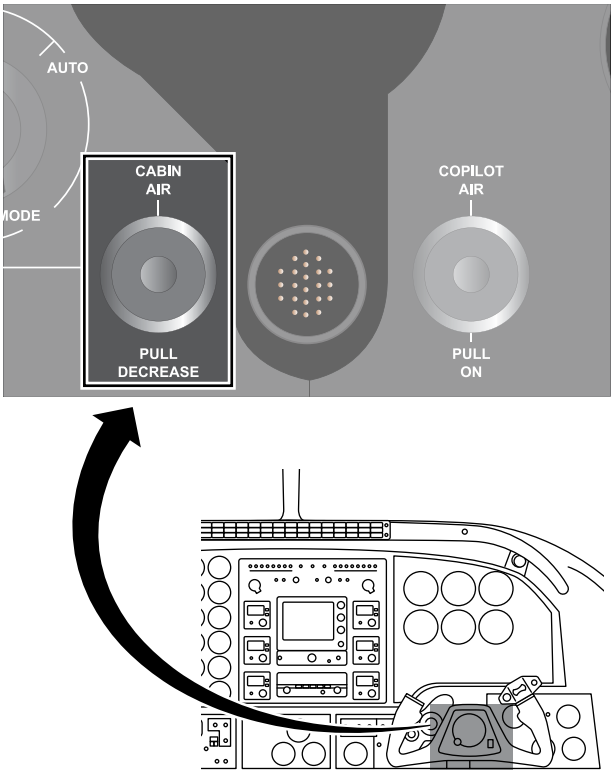

Figure 11-5. Air Control Knobs—Cabin Air

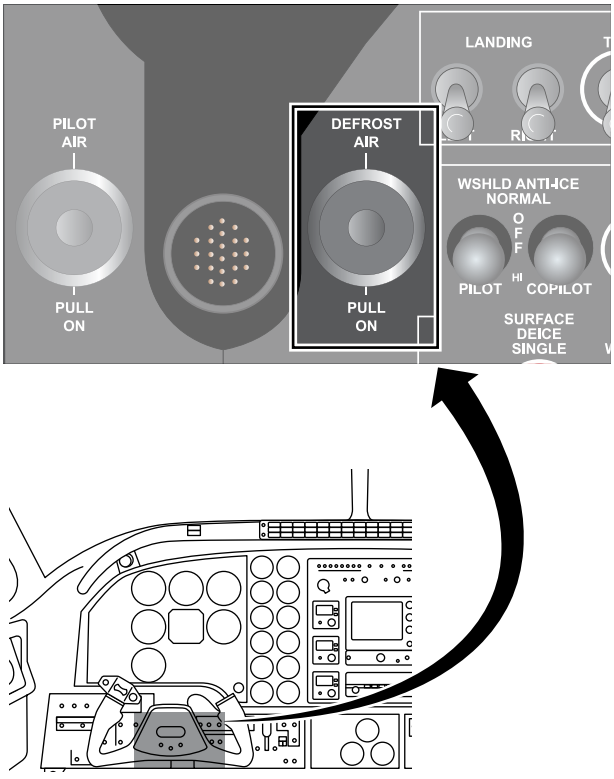

Figure 11-4. Air Control Knobs—Defrost Air

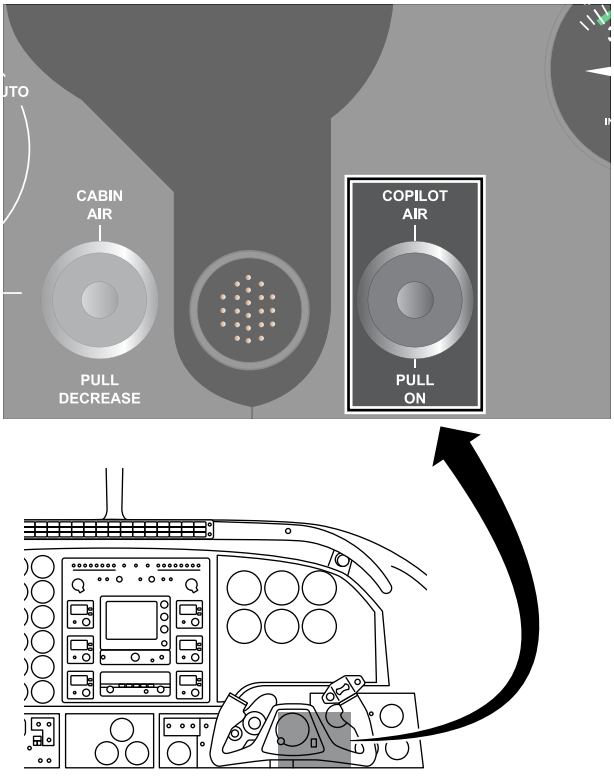

Figure 11-6. Air Control Knobs—Copilot Air

cabin air, and copilot's air control knobs. The fully out position of all these controls will provide the maximum heating to the cockpit, and the fully in position will provide minimum heating to the cockpit.

The pressurization, heating, and air conditioning systems operate in conjunction with each other or as separate systems to maintain the desired cabin pressure altitude and cabin air temperature. Occupied compartments are pressurized, heated, or cooled through a common ducting arrangement. Ventilation can be obtained on demand during nonpressurized flight through a ram-air scoop on the left side of the nose (Figure 11-7).

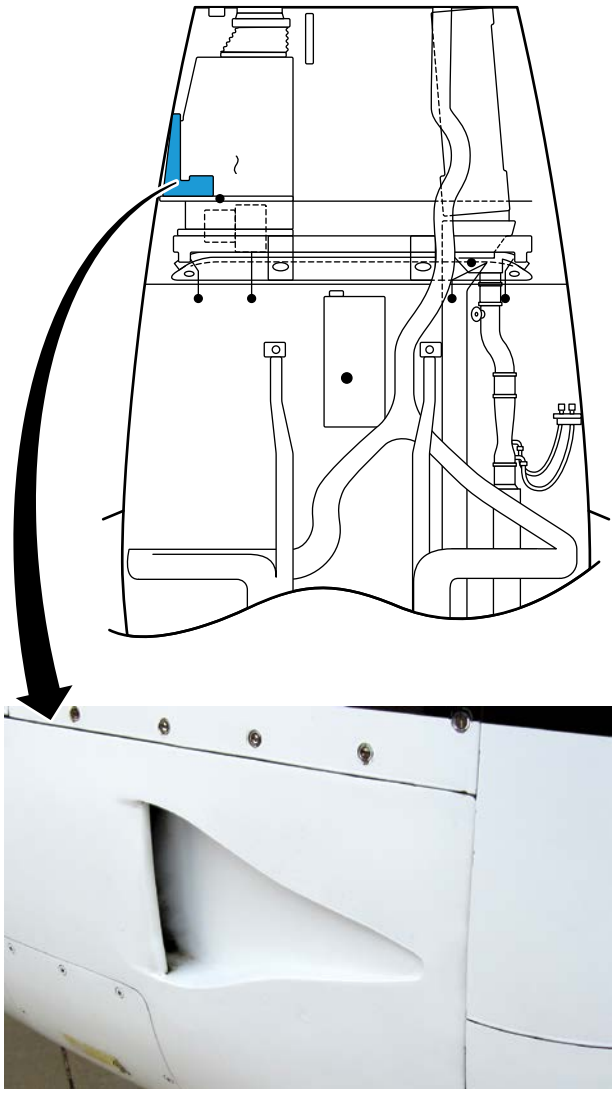

Figure 11-7. Ram-Air Scoop

UNPRESSURIZED VENTILATION

Fresh-air ventilation is provided from two sources. One source, which is available during both the pressurized and the unpressurized mode, is the bleed-air heating system. This air mixes with recirculated cabin air and enters the cockpit through glareshield "eyeball" outlets (Figure 11-8) and the cabin through the floor registers (Figure 11-9). The volume of air from the floor registers is regulated by using the cabin air control knob located on the copilot's subpanel.

Figure 11-8. Glareshield "Eyeball" Outlets

Figure 11-9. Cabin Floor Outlets

The second source of fresh air, which is available during the unpressurized mode only, is ambient air obtained from a ram-air scoop (Figure 11-10) on the nose (left side) of the airplane. During pressurized operation, an electromagnet, in addition to cabin pressure, forces the ram-air flapper door closed. During the unpressurized mode, ram air enters the evaporator plenum through the ram-air door when the electromagnet releases. Recirculated cabin air forced into the evaporator plenum by a blower, mixes with ram air from outside, is ducted around the electric heater and mixing plenum and into the ceiling-outlet duct. Air ducted to each individual cabin (Figure 11-11) or cockpit (Figure 11-12) ceiling eyeball outlet can be directionally controlled by moving the eyeball in the socket. Volume is regulated by twisting the outlet to open or close the outlet.

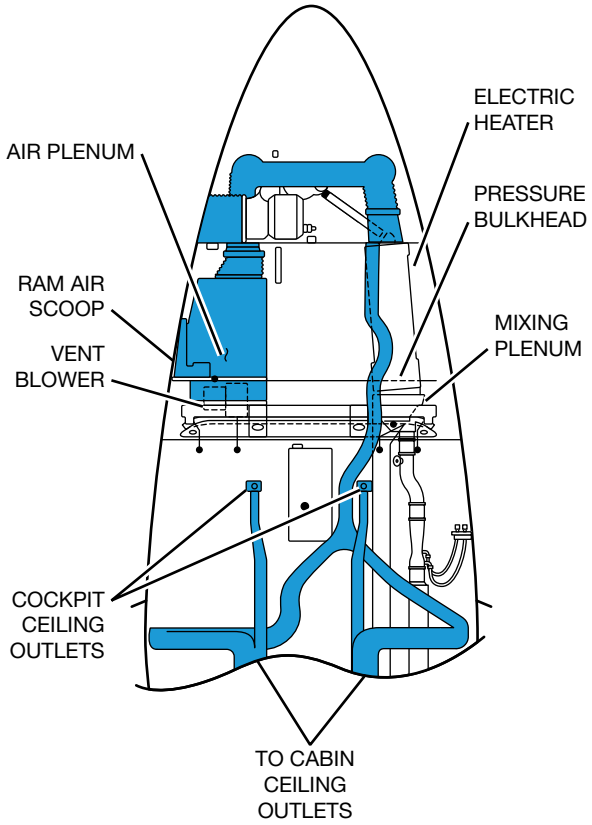

Figure 11-10. Fresh Air Source (Unpressurized Mode)

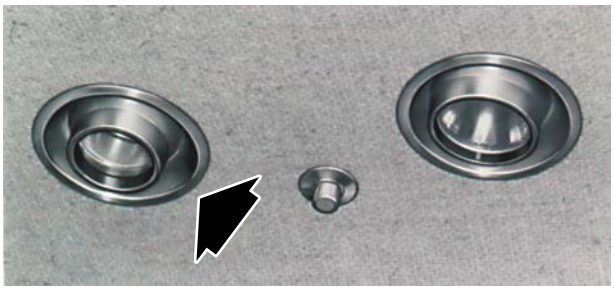

Figure 11-11. Cabin "Eyeball" Outlets

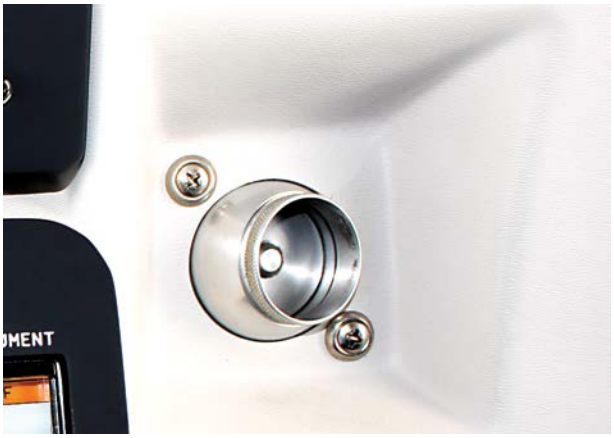

Figure 11-12. Cockpit "Eyeball" Outlets

BLEED-AIR HEATING SYSTEM

Air pressure for cabin pressurization, heating the cabin and cockpit, and for operating the instruments, rudder boost, and surface deice is obtained by bleeding air from the compressor stage (P_3) of each engine. When air is compressed, its temperature increases. Therefore, the bleed air extracted from the compressor section of each engine for pressurization purposes is hot. This heat is utilized to warm the cabin.

Engine bleed air is ducted from the engine to the flow control unit mounted on the firewall. The bleed air from either engine will continue to provide adequate air for pressurization and heating, and for the deicer system and instruments, should one engine fail. The bleed air and ambient air from the cowling intake are mixed together by the flow control units, and are routed aft through the firewall along the inboard side of each nacelle, and inboard to the center section forward of the main spar.

When the left landing gear safety switch is in the on-the-ground position, the ambient air valve (Figure 11-13) in each flow control unit is closed. Consequently, only bleed air is delivered to the environmental bleed-air duct when the airplane is on the ground. The exclusion of ambient air allows faster cabin warmup during cold weather operation. In flight, the ambient air valve is open when temperature is above –30°F, and ambient air is mixed with the engine bleed air in the flow control unit. During warm weather ground operation, the engine bleed air into the cabin can be shut off by placing the bleed-air valve switches on the copilot's subpanel to the CLOSED position. Closing the bleed-air valves prevents warm bleed air from entering the cabin area, maximizing the air conditioner operation.

The heat in the air may either be retained for cabin heating or dissipated for cooling purposes as the air passes through the center section to the fuselage. If the environmental bleed-air mixture is too warm for cabin comfort, the cabin temperature control bypass valve (Figure 11-14) routes some or all of it through the air-to-air heat exchanger in the wing center section. The position of the damper in the cabin temperature control bypass valve is determined by positioning of the controls in the ENVIRONMENTAL group on the copilot's subpanel. An air intake on the leading edge of the inboard wing brings ram air into the heat exchanger to cool the bleed air.

Depending upon the position of the cabin temperature control bypass valves, a greater or lesser volume of the bleed-air mixture will be routed through or around the heat exchanger. The temperature of the air flowing through the heat exchanger is lowered as heat is transferred to cooling fins, which are in turn cooled by ram air-flow through the fins of the heat exchanger. After leaving the heat exchanger, the ram air is ducted overboard through louvers on the underside of the wing.

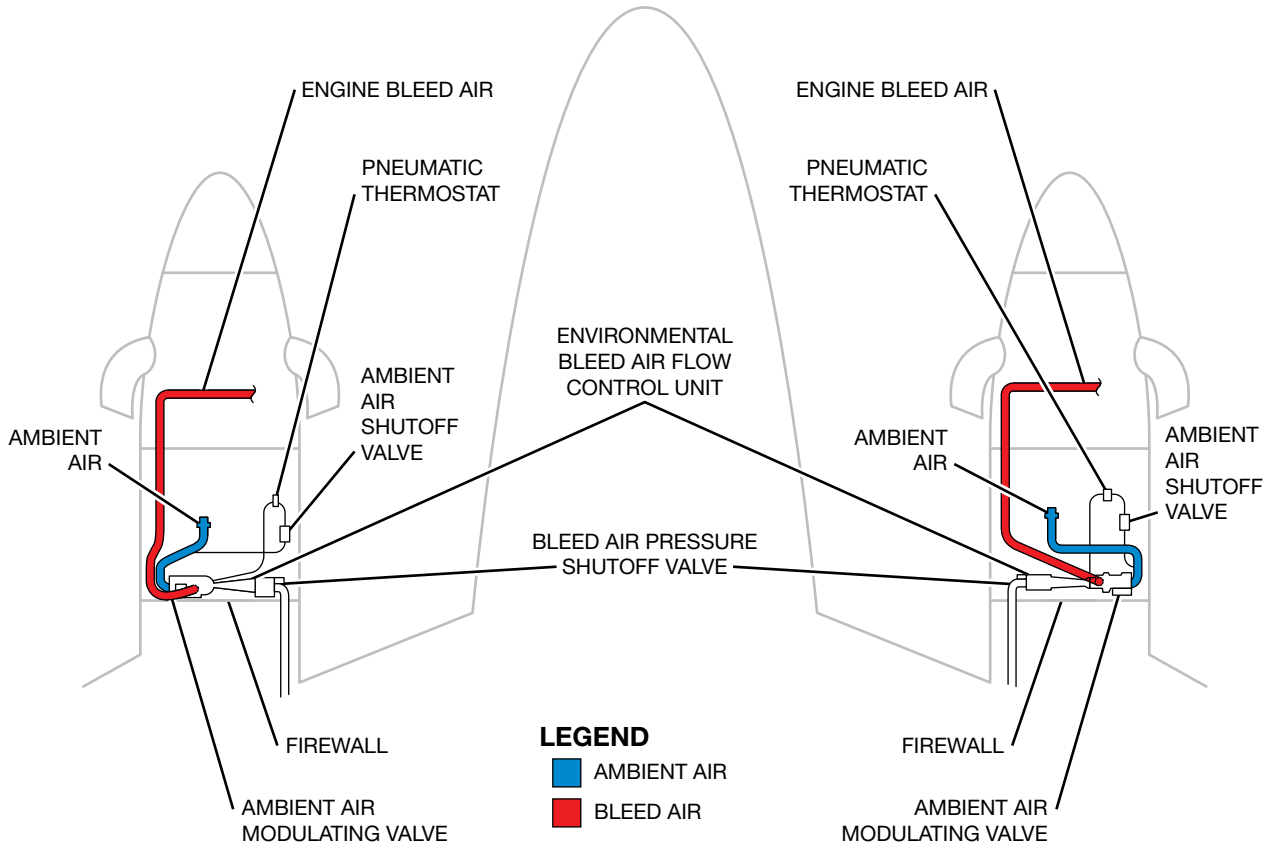

Figure 11-13. Ambient and Bleed Air Flow Forward of Firewalls

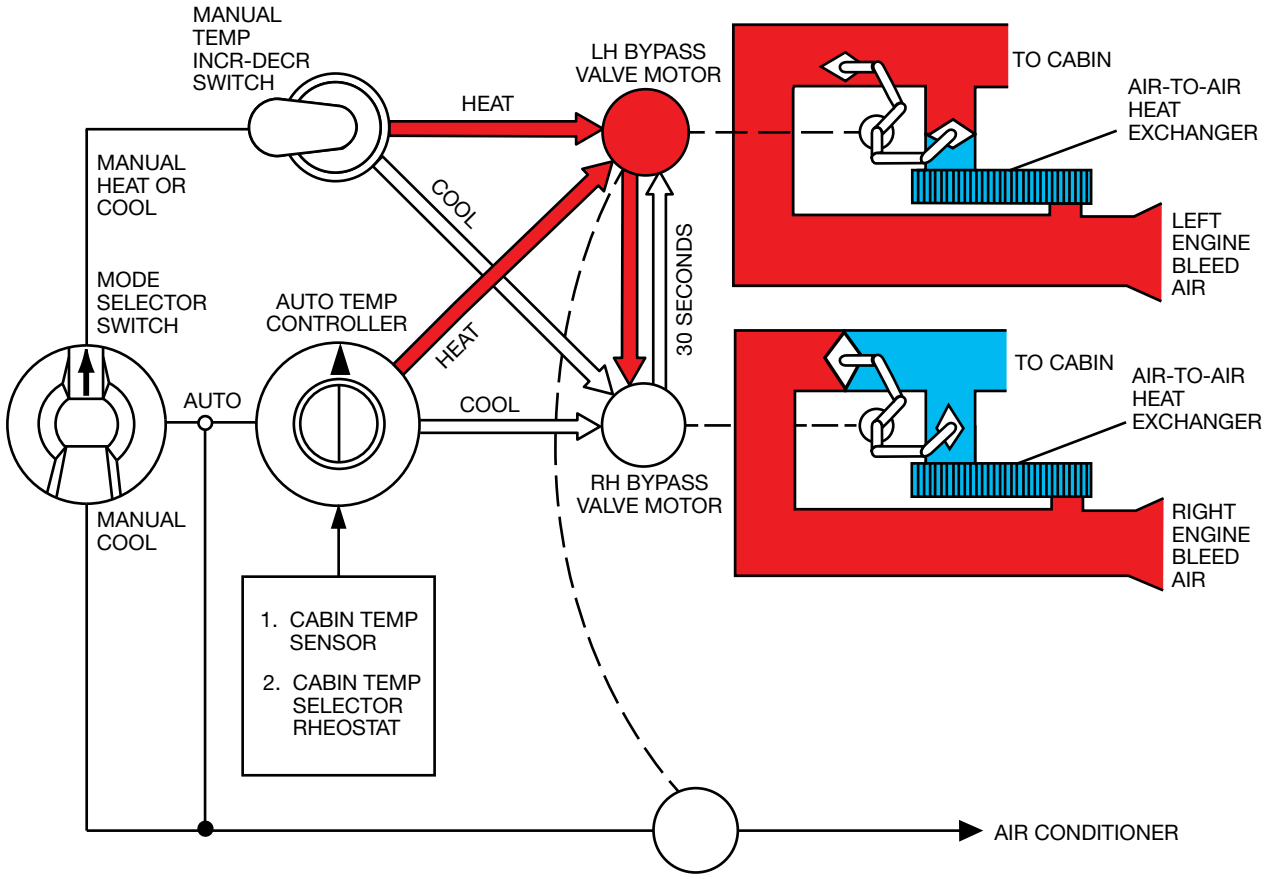

Figure 11-14. Air Conditioning System Control Diagram

The bleed air leaving both (left and right) cabin temperature control bypass valves is then ducted into a single muffler under the right floorboard forward of the main spar, which insures quiet operation of the environmental bleed-air system. The air mixture is then ducted from the muffler into the mixing plenum under the copilot's floorboard.

A partition divides the mixing plenum into two sections. One section supplies the floor-outlet duct, and the other supplies the ceiling outlet duct. Both sections receive recirculated cabin air from the vent blower. The air passes through the forward evaporator, so it will be cooled if the air conditioner is operating. Even in the event the vent blower becomes inoperative, some air will still be circulated, due to the duct design in the discharge side of the mixing plenum.

The environmental bleed-air duct is routed into the floor-duct section of the mixing plenum, then curves back to discharge the environmental bleed air toward the aft end of the floor duct section of the mixing plenum. Forward of the discharge end of the environmental bleed-air duct (Figure 11-15), warm air is tapped off and ducted up through the top of the mixing plenum and is delivered to the pilot/copilot heat duct, which is below the instrument panel. An outlet at each end of this duct is provided to deliver warm air to the pilot and copilot. A mechanically controlled damper in each outlet permits the volume of airflow to be regulated. The pilot's damper is controlled by the PILOT AIR (see Figure 11-3) knob, on the pilot's left subpanel, just outboard of the control column. The copilot's damper is controlled by the COPILOT AIR (see Figure 11-6) knob, on the copilot's right subpanel, just outboard of the control column. The DEFROST AIR control knob (see Figure 11-4) is on the

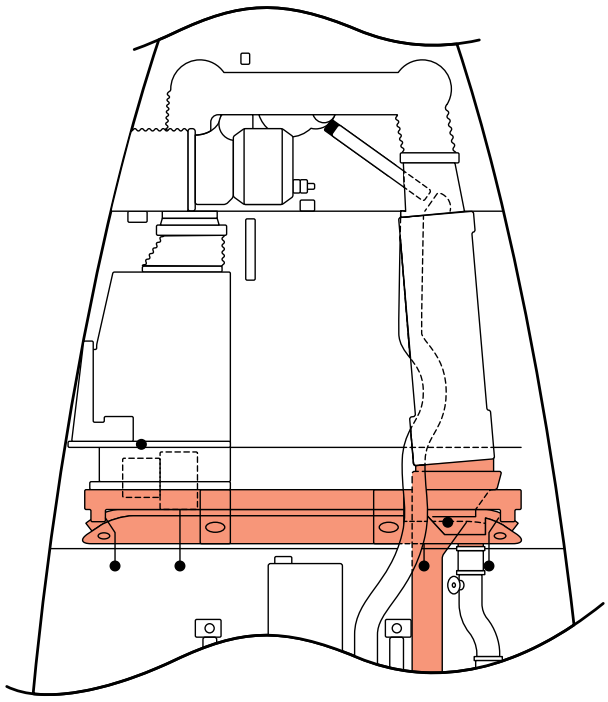

Figure 11-15. Mixing Plenum

pilot's right subpanel, just inboard of the control column. This knob controls a valve at the forward side of the pilot/copilot heat duct which admits air to two ducts that deliver the warm air to the defroster, just below the windshields in the top of the glareshield. An air plenum built into the glareshield feeds air to "eyeball" outlets on the left and right sides. Defrost air is the air source for the pilot and copilot glareshield "eyeball" outlets; thus, the use of the DEFROST AIR control knob also controls air to these eyeball outlets.

The remainder of the air in the environmental bleed-air duct is discharged into the floor-outlet duct section of the mixing plenum and mixed with recirculated cabin air. This air mixture passes through the cabin air control valve. This valve is controlled by the CABIN AIR control knob (see Figure 11-5) on the copilot's subpanel, just below and inboard of the control column. When this knob is pulled out to the stop, only a minimum amount of air will be permitted to pass through the valve, thereby increasing the amount of air available to the pilot and copilot outlets, and to the defroster. When this knob is pushed fully in, the valve is open and the air in the duct will be directed to the floor-outlet registers in the cabin.

ELECTRIC HEAT

Additional heating is available from an electrical heater (Figure 11-16) containing eight heating elements rated at 1,000 watts each. The eight electrical heating elements (Figure 11-17) are divided into two sets with four elements in each set. One set provides heat for NORMAL HEAT operation and both sets combine for GROUND MAX HEAT operation. The maximum output is available during ground operation and only four elements are available during flight. The airplane electrical system is protected against an overload by a lockout circuit that prevents use of the electrical heater during operation of the propeller deicers or windshield heat

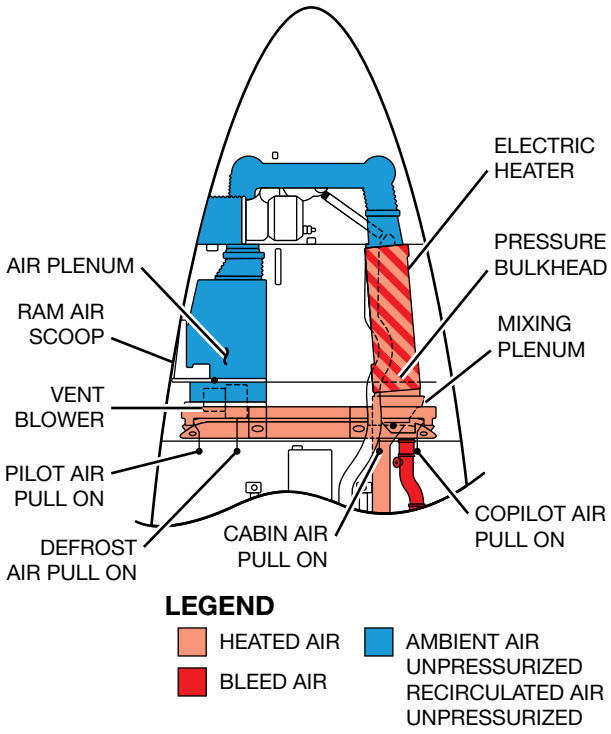

Figure 11-16. Electric Heater

The ELEC HEAT switch (Figure 11-18), in the ENVIRONMENTAL group in the copilot's sub-panel, has three positions: GND MAX–NORM–OFF. This switch is solenoid-held in GND MAX position on the ground and drops to NORM position when the landing gear safety switch is opened at lift-off. It provides maximum

Figure 11-17. Grid Heating Elements

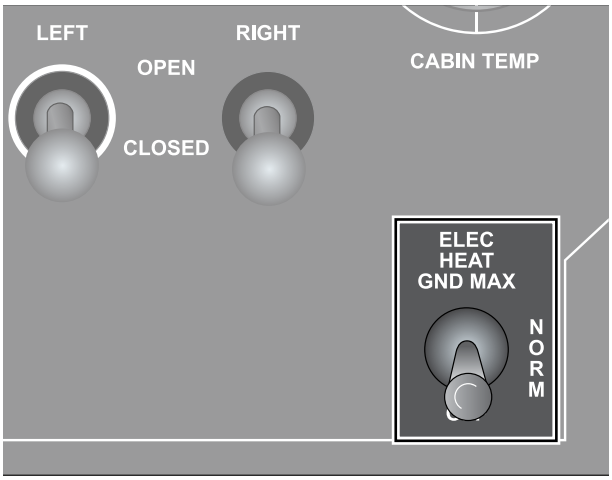

Figure 11-18. Elec Heat Switch

electric heat for initial warmup of the cabin. If use of all electrical heating elements is not desired for initial warmup, as in the GND MAX position, the switch may be placed in the NORM position, using only four elements. In the NORM position the four heating elements automatically supplement bleed-air heating, in conjunction with the cabin thermostat. The OFF position turns off all electric heat, leaving only bleed air to supply cabin heat.

COOLING SYSTEM

Cabin cooling is provided by a refrigerant-gas vapor-cycle refrigeration system consisting of:

- Belt-driven compressor, installed in the nose
- Condenser coil
- Condenser blower
- Evaporator
- Receiver-dryer
- Expansion valve
- Cabin heat control valve

It is routed (Figure 11-19) to the condenser coil, receiver-dryer, expansion valve, cabin heat control valve, and evaporator, which are all in the nose of the airplane. The rated output of the standard installation in the fuselage nose is 16,000 BTU.

The evaporator utilizes a solenoid-operated, hot-gas-cabin heat control valve to prevent icing. A 33°F thermal switch on the evaporator controls the valve solenoid.

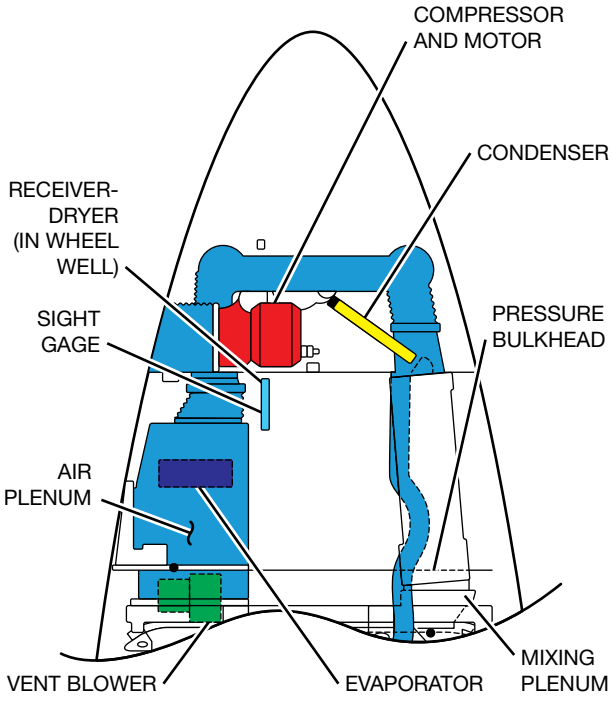

Figure 11-19. Cooling System Components in Nose

The vent blower blows recirculated cabin air through the evaporator, into the mixing plenum, and into both the floor-outlet and ceiling outlet ducts. If the cooling mode is operating, refrigerant will be circulating through the evaporator and the air leaving it will be cool. All the air entering the ceiling-outlet duct will be cool. This air is discharged through "eyeball" outlet nozzles in the cockpit and cabin. Each nozzle is movable, so that the airstream can be directed as desired. When the nozzle is twisted, a damper opens or closes to regulate airflow volume.

Cool air will enter the floor-outlet duct, but in order to provide cabin pressurization, warm environmental bleed air will also enter the floor-outlet duct anytime either BLEED AIR valve is OPEN. Therefore, pressurized air discharged from the floor registers will always be warmer than that discharged at the ceiling outlets, no matter what temperature mode is in use.

A condenser blower in the nose section draws ambient air through the condenser when the air conditioner is operating. The receiver-dryer and sight gage (Figure 11-20) are in the upper portion of the nose wheel well.

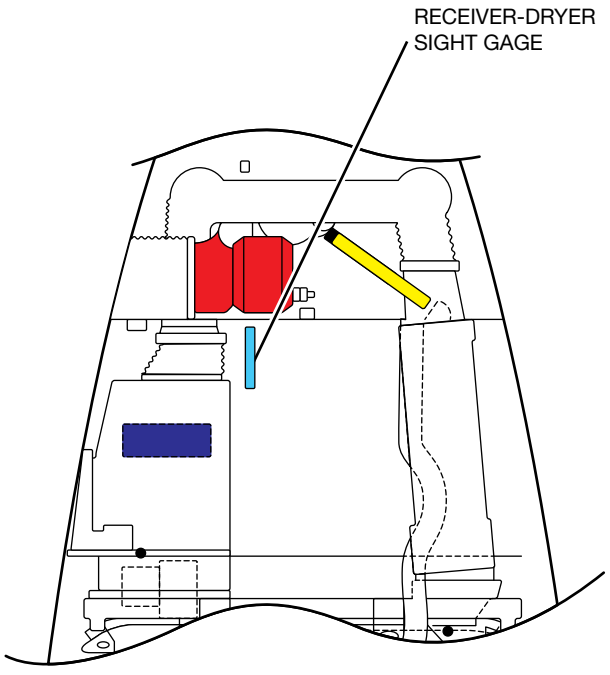

Figure 11-20. Receiver-Dryer Sight Gage

ENVIRONMENTAL CONTROLS

The ENVIRONMENTAL control section on the copilot's subpanel (see Figure 11-2) provides for automatic or manual control of the system. This section contains all the major controls of the environmental function:

- Bleed-air valve switches
- Vent blower control switch
- Manual temperature switch for control of the bypass valves in the air-to-air heat exchangers
- Cabin-temperature-level control
- Cabin temperature mode selector switch, for selecting automatic heating or cooling, manual heating or cooling, or off
- Electric heat control switch

Four additional manual controls on the main instrument subpanels may be utilized for partial regulation of cockpit comfort when the cockpit partition curtain is closed and the cabin comfort level is satisfactory. They are: pilot's air, defroster air, cabin air, and copilot's air control knobs. The fully out position of all these controls will provide the maximum heating to the cockpit, and the fully in position will provide maximum heating to the cabin.

For warm flights, such as short, low-altitude flights in summer, all the cabin floor registers and ceiling outlets should be fully open for maximum cooling. For cold flights, such as high-altitude flights, night flights, and flights in cold weather, the ceiling outlets should all be closed and the floor outlets fully open for maximum heating in the cabin.

AUTOMATIC MODE CONTROL

When the CABIN TEMP MODE selector switch (Figure 11-21) on the copilot's subpanel is in the AUTO position, the heating and air conditioning systems operate automatically. The systems are connected to a control box by means of a balanced bridge circuit. If a warmer cabin temperature has

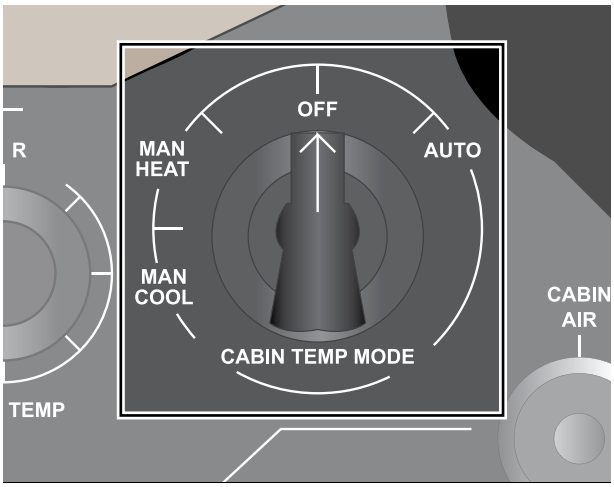

Figure 11-21. Cabin Temp Mode Selector Switch

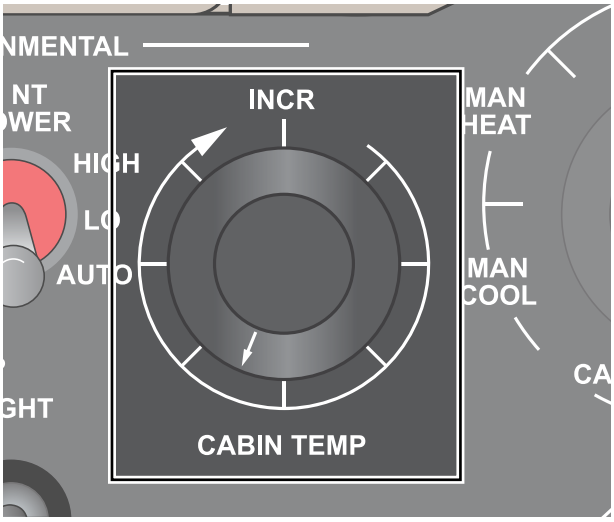

Figure 11-22. Cabin Temp Level Control

been selected, the automatic temperature control modulates the cabin heat control valves one at a time to allow heated air to bypass the air-to-air heat exchangers in the wing center sections. This warm bleed air is then brought into the cabin where it is mixed with recirculated cabin air in the floor ducting under the copilot floor area. The automatic temperature control system will then modulate the bypass valves to maintain the proper temperature of the incoming bleed air.

When the automatic control drives the environmental system from a heating mode to a cooling mode, the bypass valves move toward the cool position (bleed air passes through the air-to-air heat exchanger). When the left valve reaches the full cold position, the air-conditioning system will begin cooling. When the left bypass valve is moved approximately 30° toward the heat position the air-conditioning system will turn off preventing unnecessary recycling of the air-conditioning system.

The CABIN TEMP–INCR (Figure 11-22) control provides regulation of the temperature level in the automatic mode. A temperature-sensing unit in the cabin, in conjunction with the control setting, initiates a heat or cool command to the temperature controller, requesting the desired pressure-vessel environment.

MANUAL MODE CONTROL

When the CABIN TEMP MODE selector is in the MAN HEAT or MAN COOL position, regulation of the cabin temperature is accomplished manually by momentarily holding the MANUAL TEMP switch (Figure 11-23) to either the INCR or DECR position as desired. When released, this switch will return to the center (no change) position. Moving this switch to the INCR or DECR position results in modulation of the bypass valves in the bleed-air lines. Allow approximately 30 seconds per valve (one minute total time) for the valves to move to the full heat or full cold position. Only one valve moves at a time. Movement of these valves varies the amount of bleed air routed through

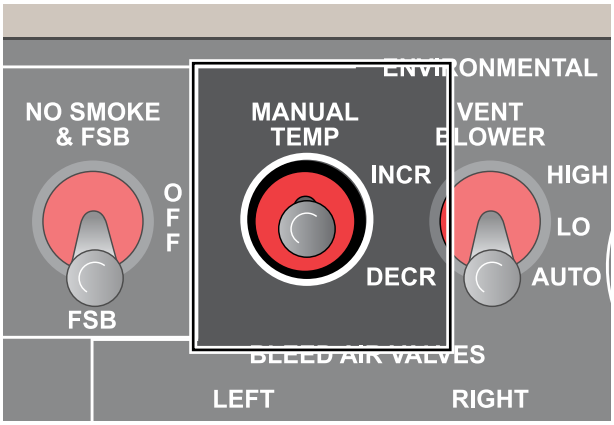

Figure 11-23. Manual Temp Switch

the air-to-air heat exchanger. Consequently, the temperature of the incoming bleed air will vary. This bleed air mixes with recirculated cabin air (which will be air conditioned if the refrigeration system is operating) in the mixing plenum, and is then ducted to the floor registers. As a result, the cabin temperature will vary according to the position of the bypass valves, whether or not the air conditioner is operating.

When the CABIN TEMP MODE selector is in the MAN COOL position, the air-conditioning system will operate, provided the bypass valves are positioned full cool, until turned off, or the evaporator reaches 33°F when the thermal sensor turns air conditioning off.

BLEED-AIR CONTROL

Bleed air entering the cabin is controlled by the two switches (Figure 11-24) placarded BLEED AIR VALVES–OPEN–CLOSED. When the switch is in the OPEN position, the environmental flow control units are open. When the switch is in the CLOSED position, the environmental flow control unit is closed. For maximum cooling on the ground, turn the bleed-air valve switches to the CLOSED position.

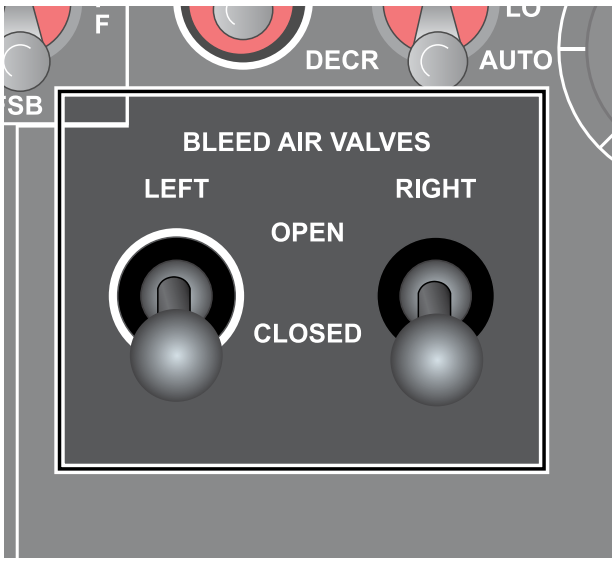

Figure 11-24. Bleed Air Valve Switches

VENT BLOWER CONTROL

The forward vent blower is controlled by a switch in the ENVIRONMENTAL group (Figure 11-25) placarded VENT BLOWER–HIGH–LO–AUTO. When this switch is in the AUTO position, the vent blower will operate at low speed if the CABIN TEMP MODE selector switch is in any position other than OFF (i.e., MANual COOL, MANual HEAT, or AUTOmatic), with one exception. The vent blower will operate in high if GND MAX HEAT is selected.

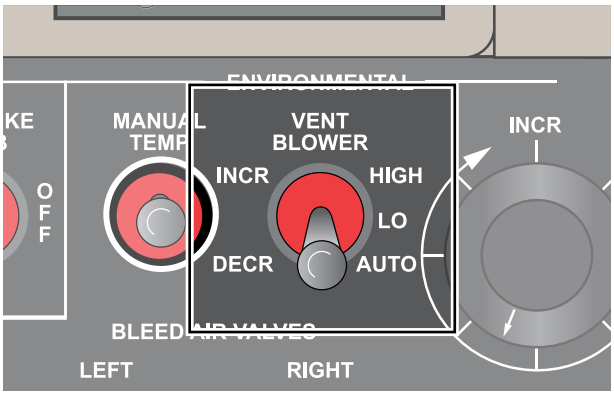

Figure 11-25. Vent Blower Switch

When the VENT BLOWER switch is in the AUTO position and the CABIN TEMP MODE selector switch is in the OFF position, the blower will not operate. Anytime the VENT BLOWER switch is in the LO position, the vent blower will operate at low speed, even if the CABIN TEMP MODE selector switch is OFF. Anytime the VENT BLOWER switch is in the HIGH position, the vent blower will operate at high speed, regardless of the position of the CABIN TEMP MODE selector switch (i.e., MAN COOL, MAN HEAT, OFF, or AUTO).

QUESTIONS

1. How is the airstream adjusted on the "eyeball" outlets?

A. By twisting the nozzle
B. By pushing in the nozzle
C. By moving a sliding lever
D. By positioning the VENT BLOWER switch to LO

2. What control regulates the air discharge rate from the glareshield "eyeball" outlets?

A. PILOT AIR control knob
B. CABIN AIR control knob
C. DEFROST AIR control knob
D. COPILOT AIR control knob

3. What control knob regulates the amount of air discharging from the floor-outlets in the environment bleed-air duct system?

A. PILOT AIR control knob
B. CABIN AIR control knob
C. DEFROST AIR control knob
D. COPILOT AIR control knob

4. What is the source of fresh air to achieve ventilation on demand during unpressurized flight?

A. Ram air through a fresh air scoop
B. The Bleed-air heating system
C. Refrigerant air, ram air
D. Refrigerant air, and the bleed-air heating system

5. How many heating elements are in the Electric Heat System?

A. 2
B. 4
C. 6
D. 8

6. How many heating elements will be used if the Electric Heat System is on during flight?

A. 2
B. 4
C. 6
D. 8

7. How long must a pilot hold the MANUAL TEMP INCR/DECR switch in the DECR position to have both by-pass valves to the full closed position?

A. 15 seconds
B. 30 seconds
C. 45 seconds
D. 60 seconds

CHAPTER 12
PRESSURIZATION

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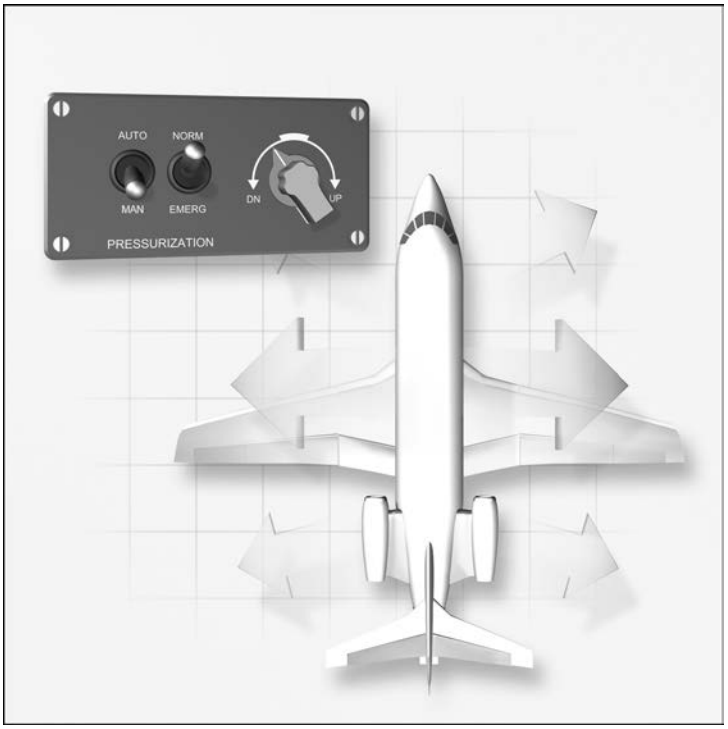
CHAPTER 12
PRESSURIZATION



INTRODUCTION

Pressurization is desirable in an airplane because it allows the altitude of the cabin to be lower than the altitude of the airplane, thus decreasing or eliminating the need for supplementary oxygen. In this section, the pilot learns how the system operates, is controlled, and how to handle malfunctions of the system.

DESCRIPTION

The Pressurization System section of the training manual presents a description of the pressurization system. The function of various major components, their physical location, and operation of the pressurization system controls are discussed. Where necessary, references are made to the environmental system as it affects pressurization.

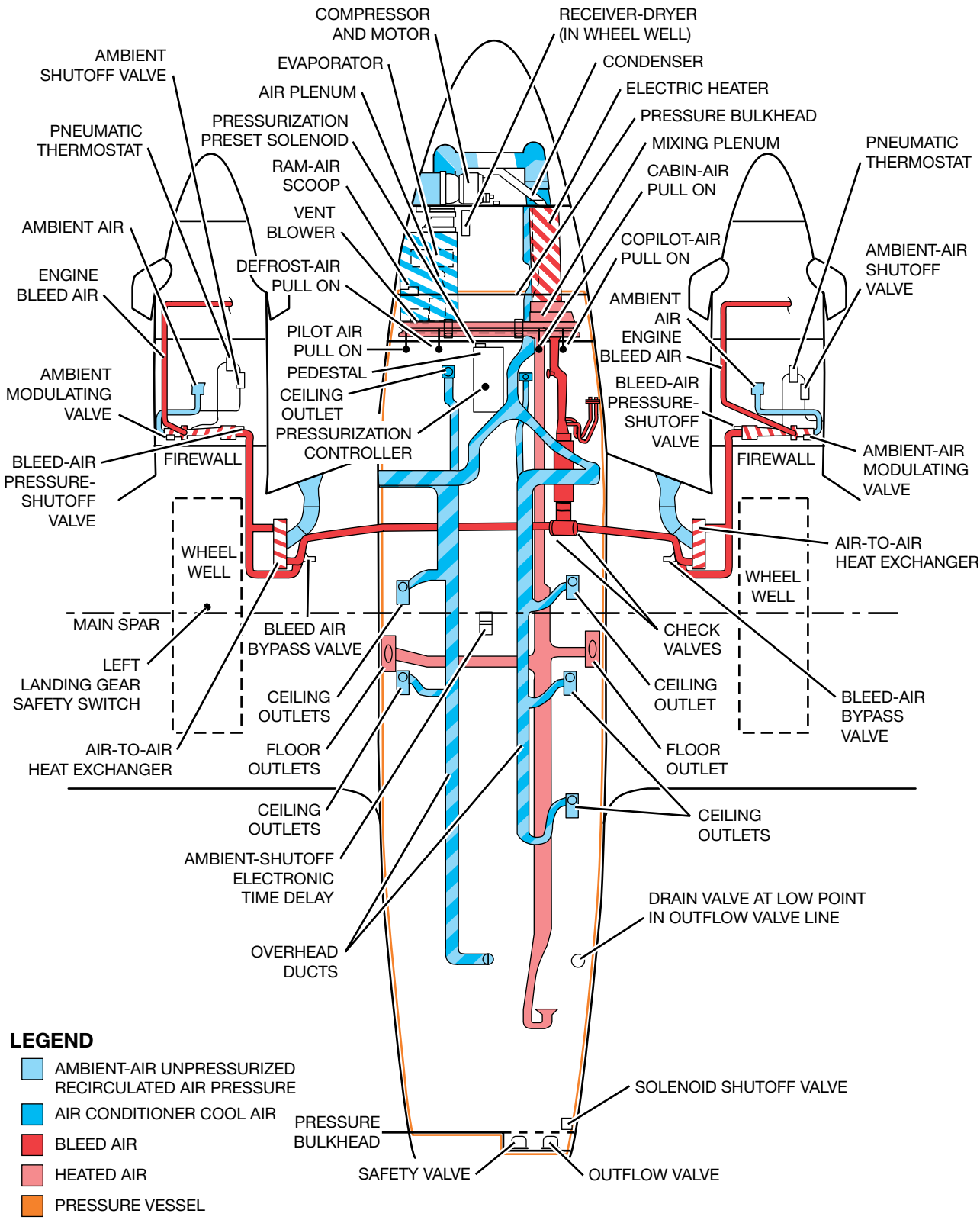


Figure 12-1. Pressurization and Air Conditioning Distribution System

PRESSURIZATION SYSTEM

The pressurization system (Figure 12-1) is designed to provide a cabin environment with sufficient oxygen for normal breathing, regardless of the airplane altitude, up to its design ceiling. As the airplane altitude increases, the outside ambient air pressure decreases until, at approximately 12,500 feet, it cannot support normal respiration. The pressurization system maintains a proportionally lower inside cabin altitude. The pressure differential between the inside cabin pressure and the outside ambient air pressure is measured in pounds per square inch.

As the cabin altitude chart shows (Figure 12-2), whenever cabin altitude and airplane altitude are the same, no pressure differential exists. Whenever cabin pressure is the greater of the two, pressure differential is a positive number. If cabin pressure is less than that of the outside ambient air, pressure differential is a negative number. Maximum differential is defined as a measure of the highest positive differential pressure the airplane structure can safely withstand for an extended period of time.

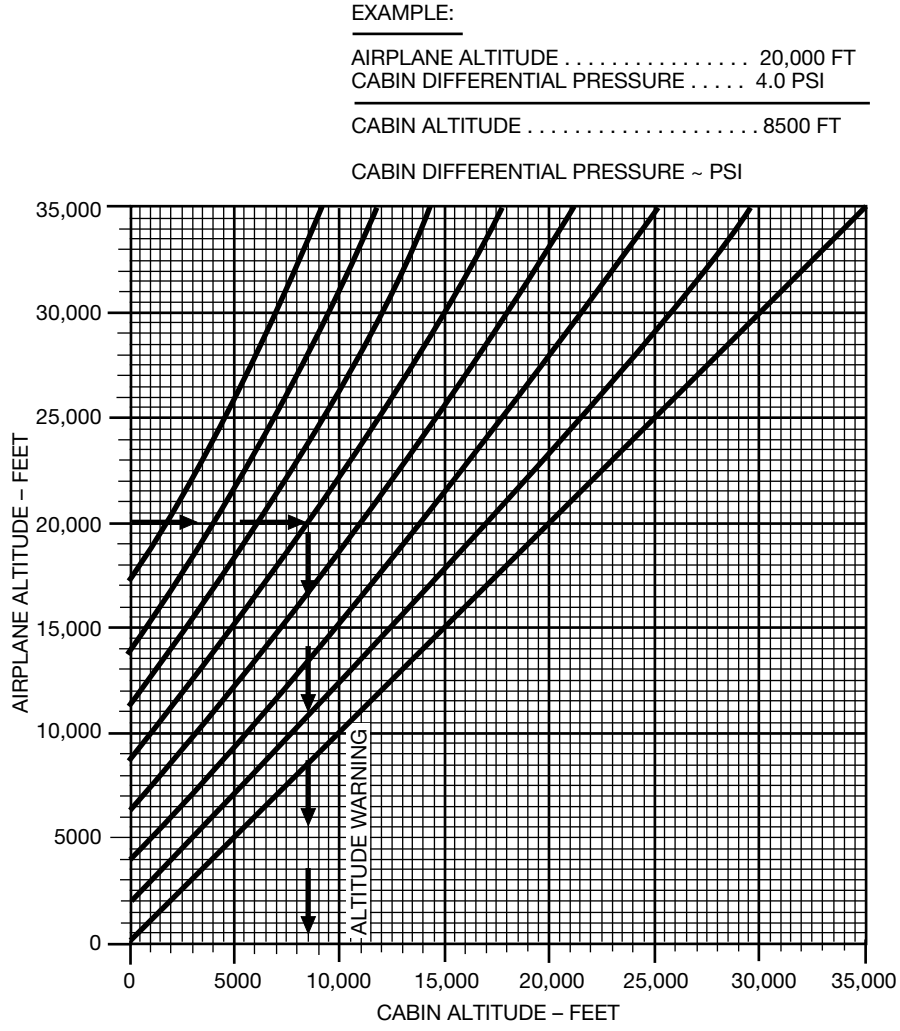


Figure 12-2. Cabin Altitude for Various Airplane Altitudes Graph

The King Air C90A and C90B, equipped with PT6A-21 engines maintain a 5.0 ±0.1 psi differential and provides a cabin pressure altitude of approximately 6,000 feet at an airplane altitude of 20,000 feet; and 12,000 feet at 30,000 feet. Although the King Air's pressure vessel is designed to withstand a maximum differential greater than 5.0 psi, the airplane structure is not designed to withstand a negative differential.

The pressurization and environmental systems (Figure 12-1) operate in conjunction with each other or as separate systems to maintain the desired cabin pressure altitude and cabin air temperature. Occupied compartments are pressurized, heated, or cooled through a common ducting arrangement.

"Pressure vessel" means that portion of the aircraft designed to withstand the pressure differential. In the King Air, the pressure vessel extends from a forward pressure bulkhead, between the cockpit and nose section to a rear pressure bulkhead, just aft of the cabin baggage compartment, with exterior skins making up the outer seal. Windows are round for maximum strength. All cables, wire bundles, and plumbing passing through the pressure vessel boundaries are sealed to reduce leaks.

AIR DELIVERY SYSTEM

Bleed air from the compressor section of each engine is utilized to pressurize the pressure vessel. A flow control unit in the nacelle of each engine controls the flow of the bleed air and mixes ambient air with it to provide an air mixture suitable for the pressurization function. The mixture flows to the environmental bleed air shutoff valve, which is a normally closed solenoid. This solenoid is controlled by a switch placarded BLEED AIR VALVES–LEFT (or) RIGHT OPEN–CLOSED in the ENVIRONMENTAL controls group (Figure 12-3) on the copilot's left subpanel. When this switch is in the CLOSED position, the solenoid is closed and no bleed air can enter the flow control unit or the cabin. When the BLEED AIR VALVE switch is in the OPEN position, the solenoid is electrically held open and the air mixture flows through the valve to the flow control package. Electricity is required

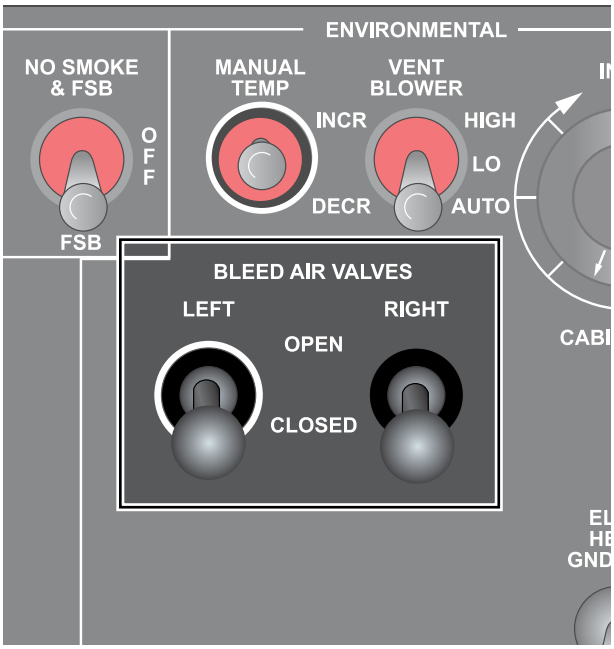


Figure 12-3. Bleed Air Valves Switches

to keep the flow control solenoid open. If there were a complete electrical failure, the solenoid would fail to the closed position. No more bleed air would enter the pressure vessel and the cabin pressure would leak out.

The air entering the airplane flows through the environmental bleed air duct (Figure 12-1). The air from the environmental bleed air duct is mixed with recirculated cabin air (which may or may not be air conditioned) in the mixing plenum, ducted upward into the crew heat duct, then routed into the floor outlet duct. This pressurized air is then introduced into the cabin through the floor registers. This air may be recirculated through the air conditioning system. Finally the air flows out of the pressure vessel through the outflow valve (Figure 12-4), located on the aft pressure bulkhead. A silencer on the outflow and safety/dump valves (Figure 12-5) ensures quiet operation. The mixture from both flow control units is delivered to the pressure vessel at a rate of approximately 14 pounds per minute, depending upon ambient temperature and pressure altitude. Pressure within the cabin and the rate of cabin pressure changes are regulated by pneumatic modulation of the outflow valve (Figure 12-6), which controls the rate at which air can escape from the pressure vessel.

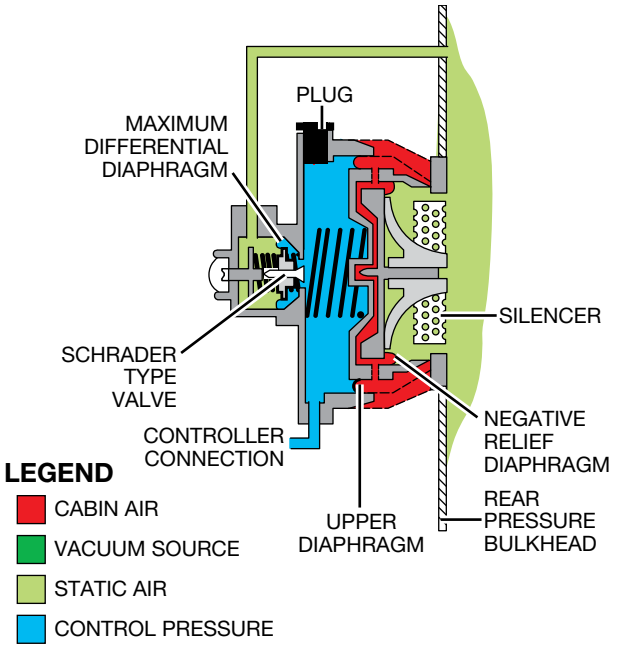


Figure 12-4. Cabin Air Outflow Valve

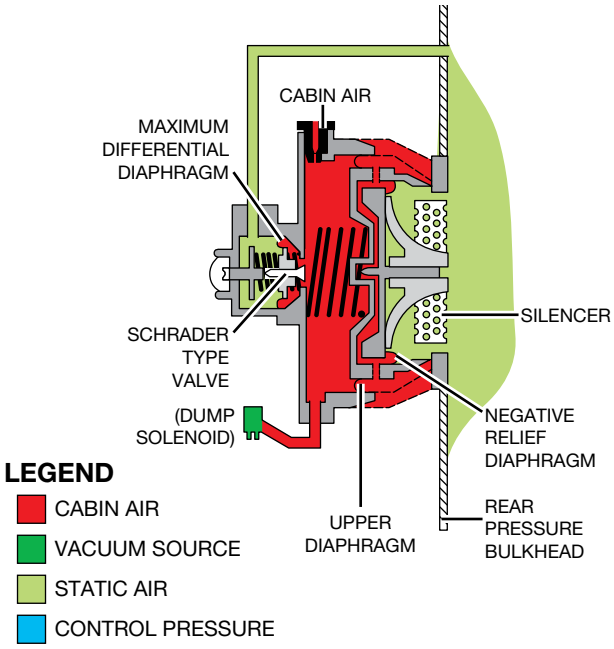


Figure 12-5. Cabin Air Safety Valve

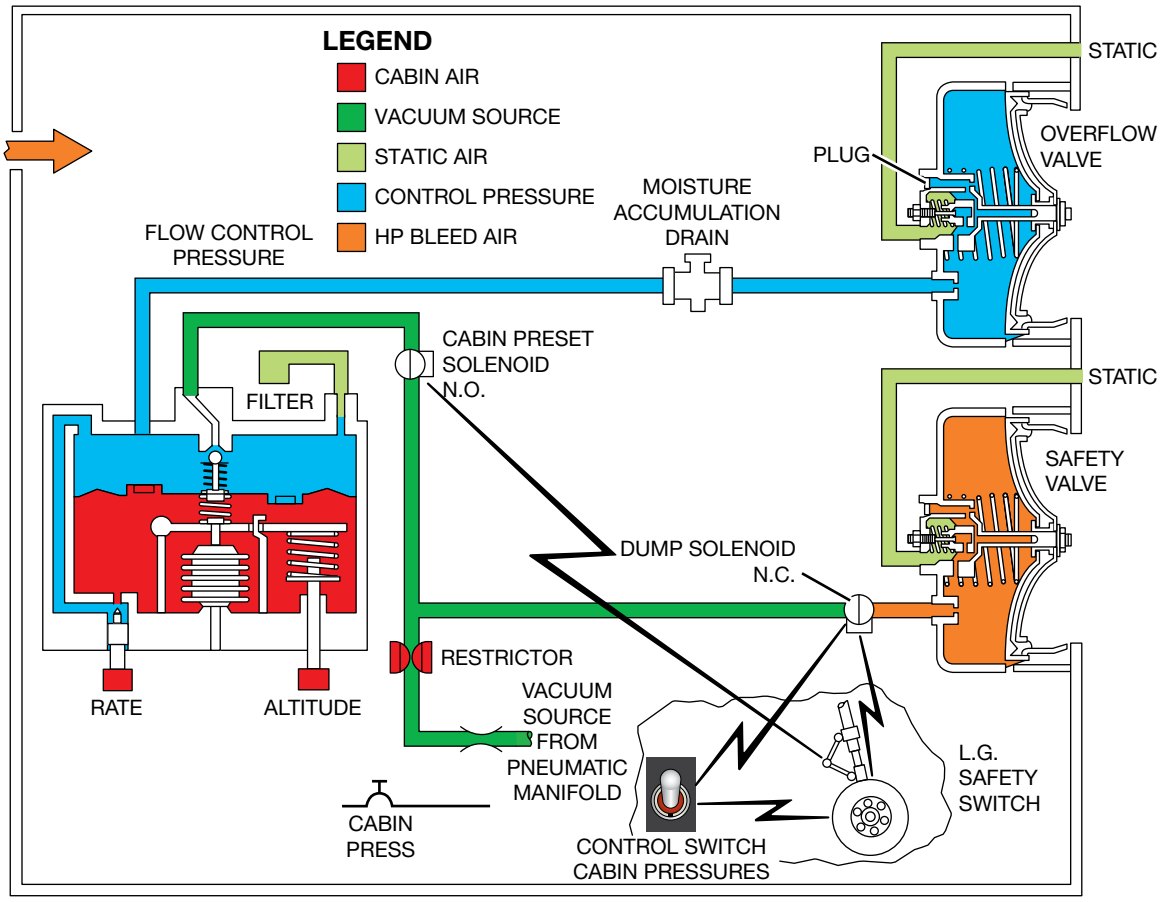


Figure 12-6. Pressurization Controls Schematic

A vacuum-operated safety valve is mounted adjacent to the outflow valve on the aft pressure bulkhead. It is intended to serve three functions:

- Provide pressure relief in the event of malfunction of the normal outflow valve
- Allow depressurization of the pressure vessel whenever the cabin pressure switch is moved into the DUMP position
- Keep the pressure vessel unpressurized while the airplane is on the ground, with the left landing gear safety switch compressed

A negative-pressure relief function is also incorporated into both the outflow and the safety valves. This prevents outside atmospheric pressure from exceeding cabin pressure by more than 0.1 psi during rapid descents, even if bleed-air inflow ceases.

When the BLEED AIR VALVE switches on the copilot's left subpanel are OPEN (up), the air mixture from the flow control units enters the pressure vessel. While the airplane is on the ground, a left landing gear safety switch-actuated solenoid valve (Figure 12-7) in each flow control unit keeps the ambient air modulating valve closed, allowing only bleed air to be delivered into the pressure vessel. At lift-off, the safety valve closes and the ambient air shutoff solenoid valve in the left flow control unit opens; approximately 6 seconds later, the solenoid in the right flow control unit opens. Consequently, by increasing the volume of airflow into the pressure vessel in stages, excessive pressure bumps during takeoff are avoided.

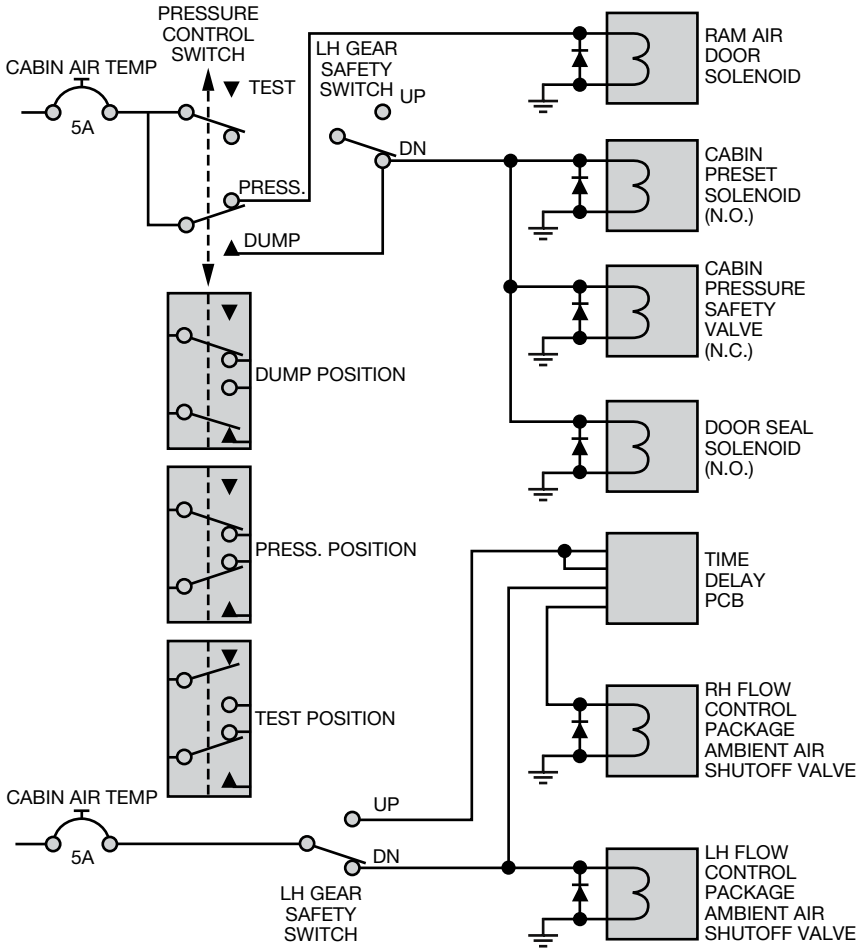


Figure 12-7. Bleed Air Control (Pressurization and Pneumatics)

CABIN PRESSURE CONTROL

An adjustable cabin pressurization controller (Figure 12-8) is mounted in the pedestal. It commands modulation of the outflow valve. A dual-scale indicator dial is mounted in the center of the pressurization controller. The outer scale (CABIN ALT) indicates the cabin pressure altitude which the pressurization controller is set to maintain. The inner scale (ACFT ALT) indicates the maximum ambient pressure altitude at which the airplane can fly without causing the cabin pressure altitude to climb above the value selected on the outer scale (CABIN ALT) of the dial. The indicated value on each scale is read opposite the index mark at the forward (top) position of the dial. Both scales rotate together when the cabin altitude selector knob, placarded CABIN ALT is turned.

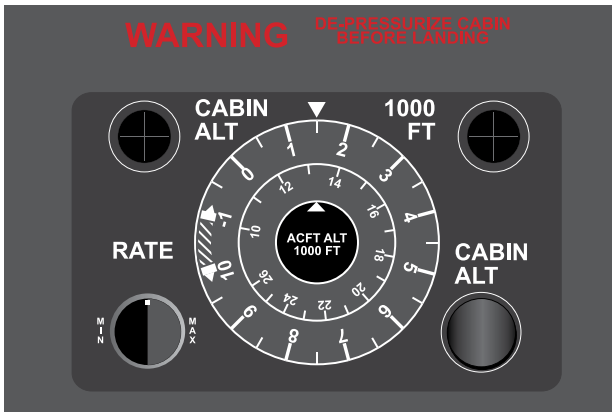


Figure 12-8. Pressurization Controller

Cabin altitude is obtained by setting the controller to the desired cruising altitude, and observing the cabin altitude on the scale. The maximum cabin altitude selected may be anywhere from -1,000 to +10,000 feet MSL. The rate control selector knob is placarded RATE–MIN–MAX. The rate at which the cabin pressure altitude changes from the current value to the selected value is controlled by rotating the rate control selector knob. The rate of change selected may be from approximately 200 to approximately 2,000 feet per minute. Normal setting on the rate knob will be from 9 o'clock to 12 o'clock.

The actual cabin pressure altitude (outer scale) and cabin differential (inner scale) is continuously indicated by the cabin altimeter (Figure 12-9), which is mounted in the right side of the panel that is located above the pedestal. Immediately to the left of the cabin altimeter is the cabin vertical speed (CABIN CLIMB) indicator (Figure 12-10), which continuously indicates the rate at which the cabin pressure altitude is changing.

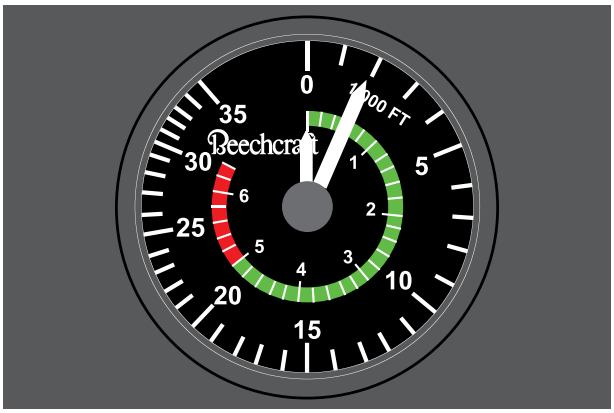


Figure 12-9. Cabin Altimeter

Figure 12-10. Cabin Climb Indicator

The cabin pressure switch (Figure 12-11), to the left of the pressurization controller on the pedestal, is placarded CABIN PRESS–

DUMP–PRESS–TEST. When this switch is in the DUMP (forward lever locked) position, the safety valve is held open, so that the cabin will depressurize and/or remain unpressurized. When it is in the PRESS (center) position, the safety valve is normally closed in flight, and the outflow valve is controlled by the pressurization controller, so

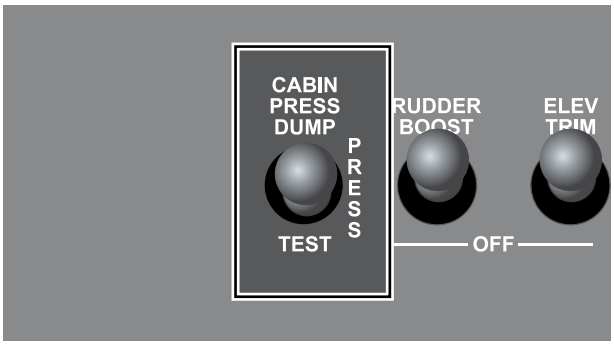


Figure 12-11. Cabin Pressure Switch

that the cabin will pressurize. When the switch is held in the spring-loaded TEST (aft) position, the safety valve is held closed, bypassing the landing gear safety switch, to facilitate testing of the pressurization system on the ground. Circuit breakers for the system (Figure 12-12) are on the copilot's side panel under the heading ENVIRONMENTAL.

PREFLIGHT CHECK

During runup, the pressurization system may be functionally checked using the cabin pressurization switch. With both bleed-air valves OPEN, adjust the cabin altitude selector knob so that the CABIN ALT dial indicates an altitude 500 feet BELOW field pressure altitude. Rotate the rate control selector knob to place the index between the 9 and 12 o'clock positions. Move both condition levers to high idle. Hold the cabin pressurization switch to the TEST position and check the CABIN CLIMB indicator for a descent indication. Release the pressurization switch to the PRESS position when pressurizing is confirmed and move both condition levers to their original position.

Prior to takeoff, the CABIN ALT selector knob should be adjusted so that the ACFT ALT scale on the indicator dial indicates an altitude approximately 500 feet above the planned cruise pressure altitude prior to takeoff. The rate control selector knob should be adjusted as desired; setting the index mark between the 9 and 12 o'clock positions will provide the most comfortable cabin rate of climb. The cabin pressure switch should be checked to ensure that it is the PRESS position.

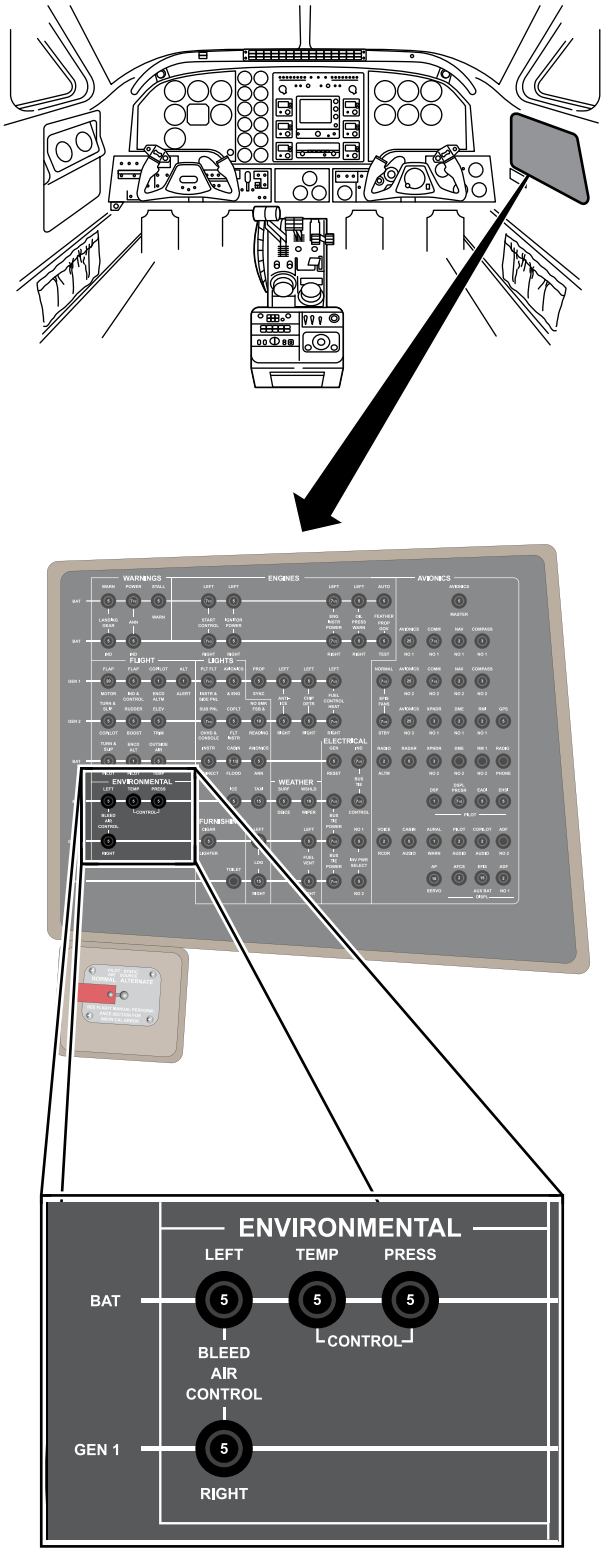


Figure 12-12. Environmental System Circuit Breakers

IN FLIGHT

As the airplane climbs, the cabin pressure altitude climbs at the selected rate of change until the cabin reaches the selected pressure altitude. The system then maintains cabin pressure altitude at the selected value. If the airplane climbs to an altitude higher than the value indexed on the ACFT ALT scale of the dial on the face of the controller, the pressure differential will reach the pressure relief setting of the outflow valve and safety valve. Either or both valves will then override the cabin pressurization controller in order to limit the pressure differential to the maximum pressure differential. If the cabin pressure altitude should reach a value of 10,000 feet (12,500 for LJ 1353 and later), a pressure-sensing switch will close. This causes the red ALTITUDE WARN annunciator light to illuminate, warning the pilot of operation requiring the use of oxygen. During cruise operation, if the flight plan calls for an altitude change of 1,000 feet or more, reselect the new altitude plus 500 feet on the CABIN ALT dial.

DESCENT

During descent and in preparation for landing, set the cabin altitude selector to indicate a cabin altitude of approximately 500 feet above the landing field pressure altitude (Table 12-1), and adjust the rate control selector as required to provide a comfortable cabin-altitude rate of descent. Control the airplane rate of descent so that the airplane altitude does not catch up with the cabin pressure altitude until the cabin pressure altitude reaches the selected value, which may happen before the airplane reaches the selected altitude. Then as the airplane descends to and reaches the cabin pressure altitude the negative pressure relief function opens the out-flow and safety valve poppets toward the fully open position, thereby equalizing the pressure inside and outside the pressure vessel. As the airplane continues to descend below the preselected cabin pressure altitude, the cabin will be unpressurized and will follow the airplane rate of descent to touchdown.

Table 12-1. PRESSURIZATION CONTROLLER SETTING FOR LANDING

CLOSEST ALTIMETER SETTING	ADD TO AIRPORT ELEVATION
28.00	+ 2,400
28.10	+ 2,300
28.20	+ 2,200
28.30	+ 2,100
28.40	+ 2,000
28.50	+ 1,900
28.60	+ 1,800
28.70	+ 1,700
28.80	+ 1,600
28.90	+ 1,500
29.00	+ 1,400
29.10	+ 1,300
29.20	+ 1,200
29.30	+ 1,100
29.40	+ 1,000
29.50	+ 900
29.60	+ 800
29.70	+ 700
29.80	+ 600
29.90	+ 500
30.00	+ 400
30.10	+ 300
30.20	+ 200
30.30	+ 100
30.40	0
30.50	- 100
30.60	- 200
30.70	- 300
30.80	- 400
30.90	- 500

FLOW CONTROL UNIT

A flow control unit, mounted in each nacelle on the forward side of the firewall, controls the bleed air from the engine for use in pressurization, heating, and ventilation. The function of the flow control unit (Figure 12-13) is to vary the flow and balance of bleed air and ambient air to the cabin pressure vessel. This is done by means of temperature and pressure sensors and their related modulating valves.

When the BLEED AIR switches on copilot's left subpanel are OPEN a bleed-air shutoff electric solenoid valve on each flow control unit opens to allow the bleed air into the unit. The flow control unit will then adjust the flow of bleed air mixed with ambient air into the pressure vessel. Ambient air is allowed to enter the flow control

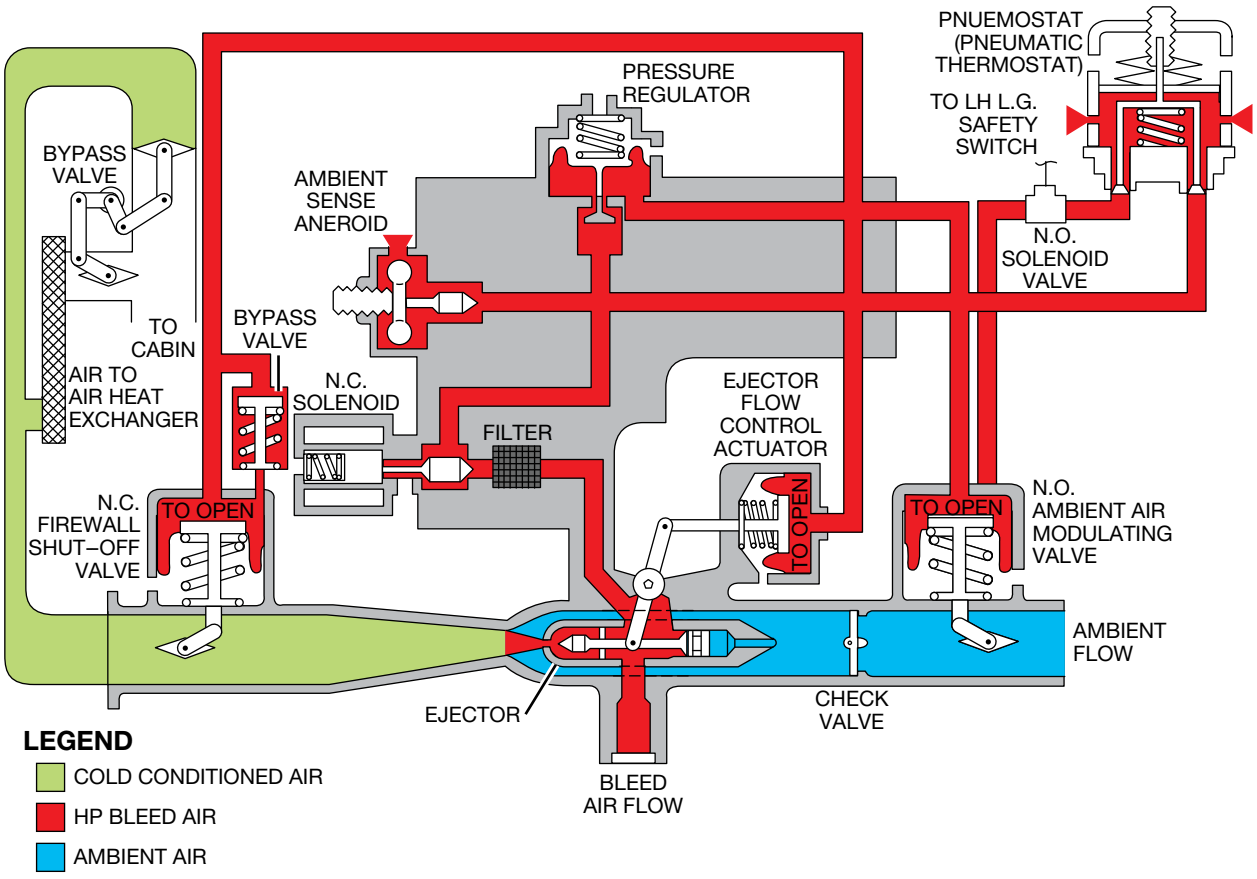


Figure 12-13. Flow Control Unit

unit through a normally-open modulating valve, and serves to add air mass and some cooling to the bleed air flow.

The ambient air valve, associated with the temperature sensing device, is also controlled by the left landing gear safety switch. When the aircraft is on the ground, the valve is directed to shut off the ambient air source from the flow control valve. The exclusion of ambient air allows faster cabin warm-up during cold weather operation.

After takeoff, the landing gear safety switch signals the ambient air modulating valves to open. They do so sequentially to prevent the simultaneous opening of the modulating valves and a sudden pressure surge into the cabin.

The pneumostat (pneumatic thermostat) provides temperature input to the flow control unit, which modulates the amount of ambient air entering the flow unit for blending. Warmer outside air opens the modulating valve and allows more ambient air in for blending. Cold air closes the valve until it closes completely at a preset temperature. At this point, bleed air will be providing all air for pressurization. A check valve prevents air from leaking out the ambient air input.

An aneroid near the bleed air ejector flow control actuator influences the amount of bleed air entering the flow control unit. The aneroid provides altitude sensing information to the flow control unit, and combined with the pneumostat, provides accurate bleed-air input into the pressure vessel.

The quantity of bleed-air flow into the pressure vessel is influenced directly by ambient temperature and ambient pressure.

QUESTIONS

1. What is the maximum cabin pressure differential?

A. 5.3 ±0.1 PSID
B. 5.0 ±0.1 PSID
C. 4.9 ±0.1 PSID
d. 4.6 ±0.1 PSID

2. What indicator reflects the rate of cabin pressure altitude change?

A. Aircraft Altimeter
B. Cabin Climb indicator
C. Cabin Altimeter
D. Pressurization Controller

3. Which position on the RATE control knob provides the most comfortable rate of climb?

A. Index mark set at MIN
B. Index mark set between the 2 o'clock and 6 o'clock positions
C. Index mark set between the 6 o'clock and 9 o'clock positions
D. Index mark set between the 9 o'clock and 12 o'clock positions

4. The rate of change selected on the RATE control knob may be from approximately:

A. 100 to 1,000 fps.
B. 200 to 2,000 fpm.
C. 200 to 2,500 fps.
D. 50 to 5,000 fpm.

5. What should the Pressurization Controller be set to if the planned cruise altitude is 22,000 feet?

A. 22,000 feet
B. 22,500 feet
C. 23,000 feet
D. 23,500 feet

CHAPTER 13
HYDRAULIC POWER SYSTEM

See Chapter 14—"Landing Gear and Brakes," for information on the hydraulic power system.

CHAPTER 14
LANDING GEAR AND BRAKES

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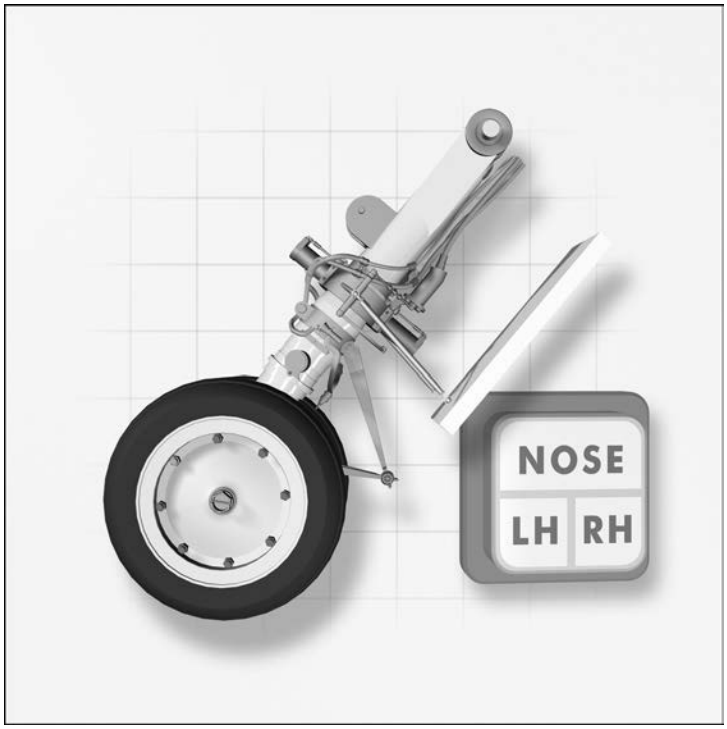

INTRODUCTION

An understanding of the landing gear system will aid the pilot in proper handling of landing gear operation and emergency procedures. This chapter, in addition to describing the system, identifies inspection points and abnormal conditions to be considered. This chapter also includes brakes, since an understanding of the brake system will help the pilot operate the brakes safely and with minimum wear. In addition to system description, operating and servicing procedures are covered.

GENERAL

This chapter presents a description and discussion of the landing gear system, landing gear controls, and limits. The indicator system and emergency landing gear extension are also described.

This chapter also presents a description and discussion of the wheel brake system. Correct use of the brakes and parking brakes, brake system description, and what to look for when inspecting brakes are also detailed.

LANDING GEAR SYSTEM

LANDING GEAR ASSEMBLIES

Components

Each landing gear assembly (main and nose) consists of a shock strut, torque knee (scissors), drag leg, actuator, wheel, and tire. Brake assemblies are located on the main gear assemblies; the shimmy damper is mounted on the nose gear assembly (Figure 14-1 and Figure 14-2).

Operation

The upper end of the drag legs and two points on the shock struts are attached to the airplane structure. When the gear is extended, the drag braces are rigid components of the gear assemblies.

The landing gear incorporates Beech air/oil shock struts that are filled with both compressed air and hydraulic fluid. Airplane weight is borne by the air charge in the shock struts. At touchdown, the lower portion of each strut is forced into the upper cylinder; this moves fluid through an orifice, further compressing the air charge and thus absorbing landing shock. Orifice action also reduces bounce during landing. At takeoff, the lower portion of the strut extends until an internal stop engages.

A torque knee connects the upper and lower portions of the shock strut. It allows strut compression and extension but resists rotational forces, thereby keeping the wheels aligned with the longitudinal axis of the airplane. On the nose gear assembly, the torque knee also transmits steering motion to the nosewheel, and nosewheel shimmy motion to the shimmy damper.

The shimmy damper, mounted on the right side of the nose gear strut, is a balanced hydraulic cylinder that bleeds fluid through an orifice to dampen nosewheel shimmy.

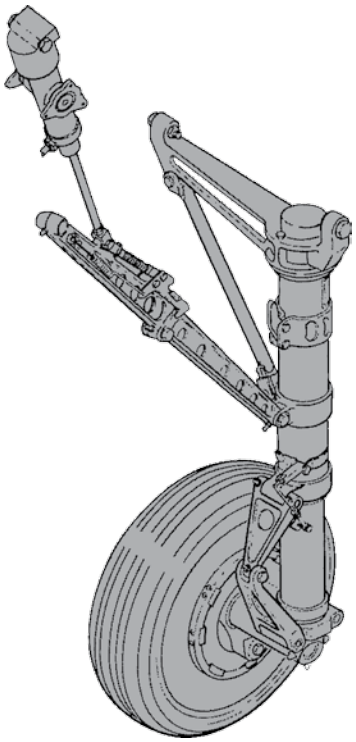

Figure 14-1. Main Gear Assembly

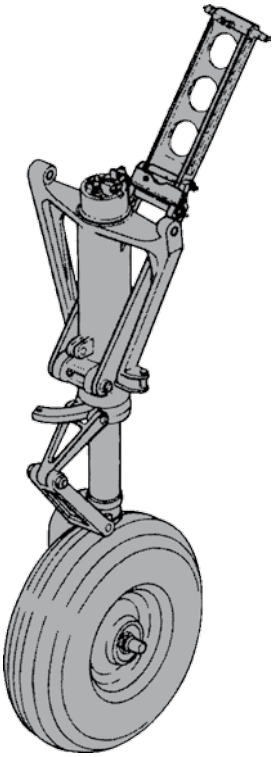

Figure 14-2. Nose Gear Assembly

WHEEL WELL DOOR MECHANISMS

The landing gear doors consist of one set of nose gear doors and two sets of main gear doors. Landing gear doors are mechanically actuated by gear movement during extension and retraction.

The nose gear doors are hinged at the sides and are spring-loaded to the open position. As the landing gear is retracted, a roller on each side of the nose gear assembly engages a cam assembly on each door, and draws the doors closed behind the gear. The reverse action takes place, and spring-loading takes effect as the nose gear is extended.

The main gear doors are hinged at the sides and are connected to a landing-gear, door-actuator torque tube assembly (Figure 14-3) with two push-pull links. The torque tube assembly also contains an uplock roller support assembly which, when contacted by the uplock cam on the main gear shock cylinder, rotates the torque tube to pull the doors closed upon gear retraction, or push the doors open upon gear extension.

Roller movement is transmitted through linkage to close the doors. During extension, roller action reverses cam movement to open the doors. When the cam has left the roller, springs pull the linkage over-center to hold the doors open.

STEERING

Direct linkage to the rudder pedals permits nosewheel steering when the nose gear is down. One spring-loaded link in the system absorbs some of the force applied to any of the interconnected rudder pedals until the nosewheel is rolling. At this time the resisting force is less, and more pedal motion results in more nosewheel deflection. Since motion of the pedals is transmitted via cables and linkage to the rudder, rudder deflection occurs when force is applied to the rudder pedals. With the nose landing gear retracted, some of the force applied to any of the rudder pedals is absorbed by the spring-loaded link in the steering system, so that there is no motion at the nosewheel but rudder deflection still occurs. The nosewheel is self-centering upon retraction.

When force on the rudder pedal is augmented by a main wheel braking action, the nosewheel deflection can be considerably increased.

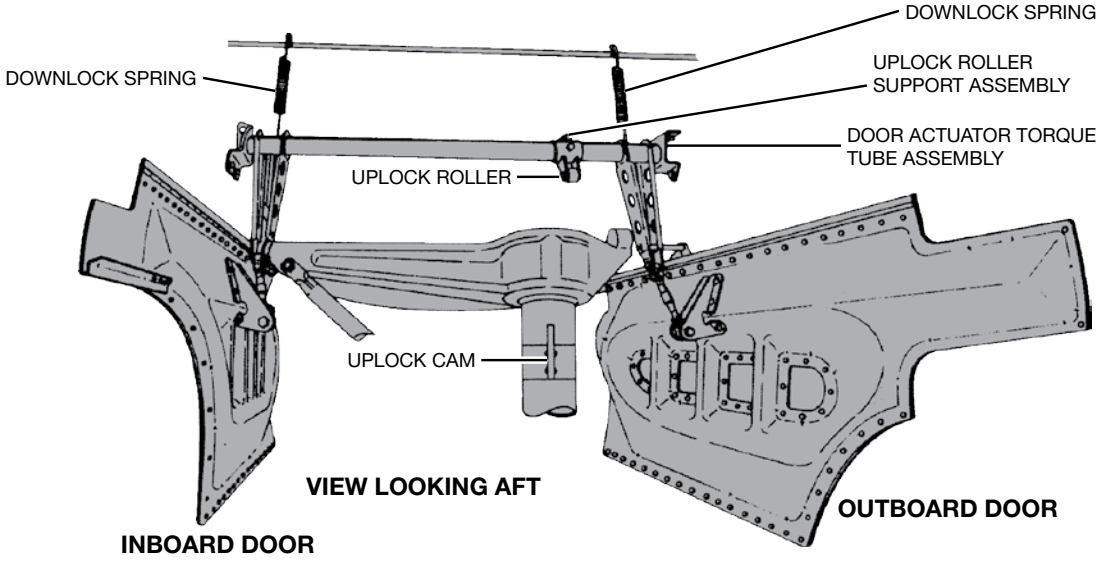

Figure 14-3. Main Gear Door Mechanism

HYDRAULIC LANDING GEAR

The retractable tricycle landing gear is electrically controlled (Figure 14-4) and hydraulically actuated. The system utilizes folding braces, called "drag legs," that lock in place when the gear is fully extended.

The individual landing gear actuators incorporate internal/mechanical downlocks to hold the gear in the fully extended position. The landing gear is held in the up position by hydraulic pressure.

Hydraulic pressure to the system is supplied by a hydraulic power pack (Figure 14-5). A hydraulic reservoir located in the left center wing section provides hydraulic fluid to the power pack. The reservoir incorporates a dipstick to provide a visual check of fluid level.

An electrically actuated selector valve controls the flow of hydraulic fluid to the individual gear actuators. The selector valve receives electrical power through the landing gear control switch.

Accidental retraction of the landing gear is prevented through safety switches located on the main landing gears.

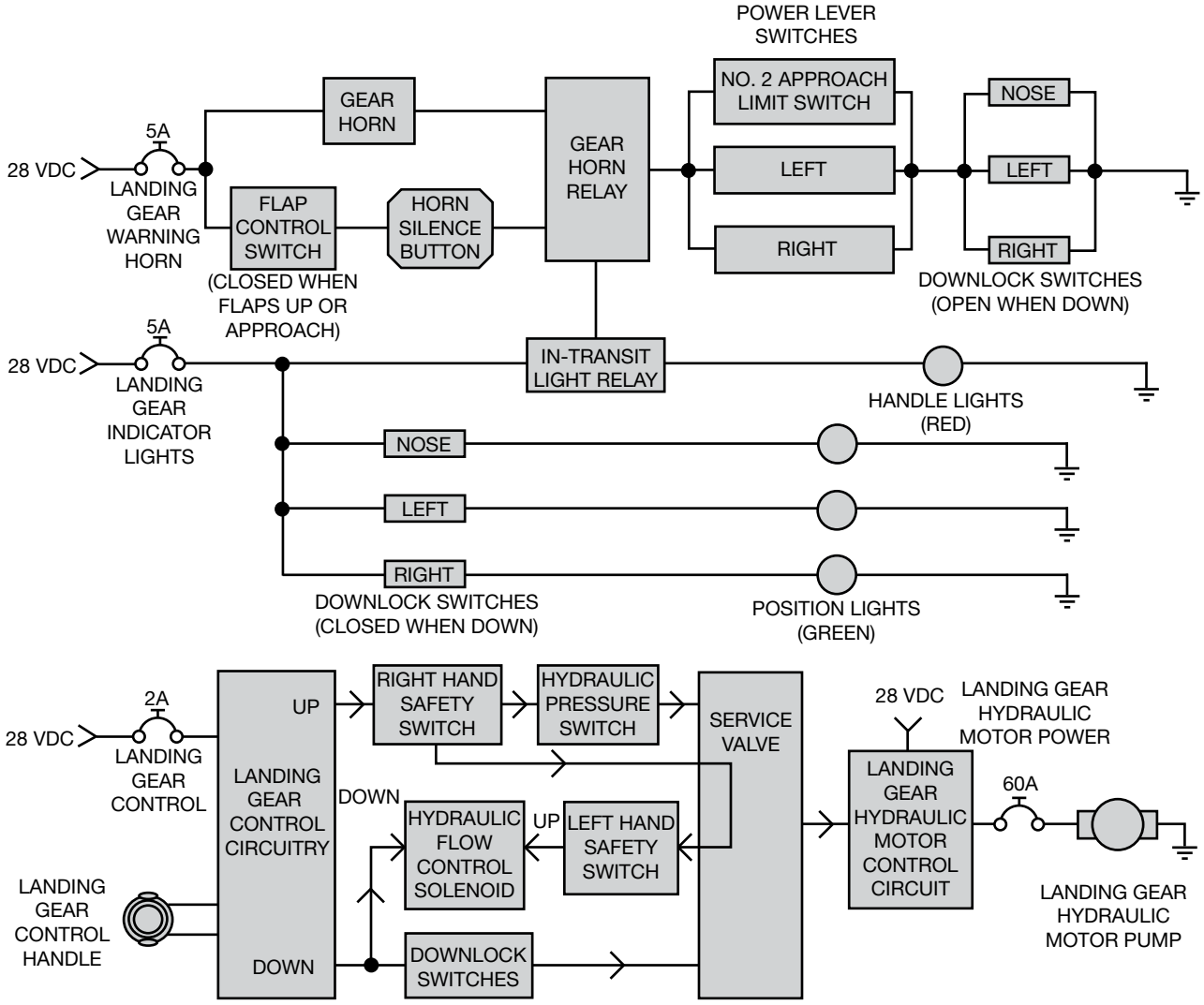

Figure 14-4. Landing Gear Electrical Schematic

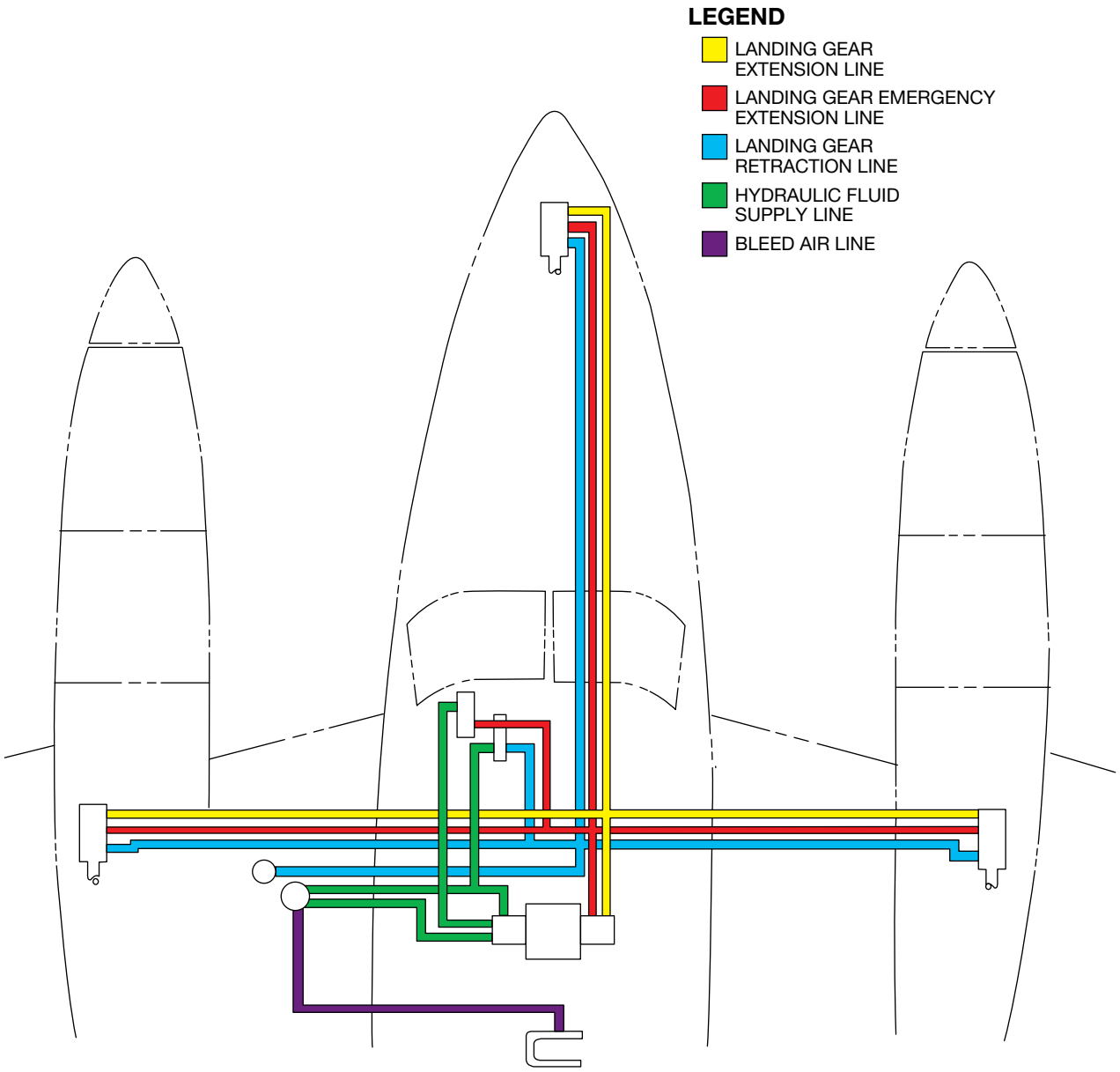

Figure 14-5. Hydraulic Landing Gear Plumbing Schematic

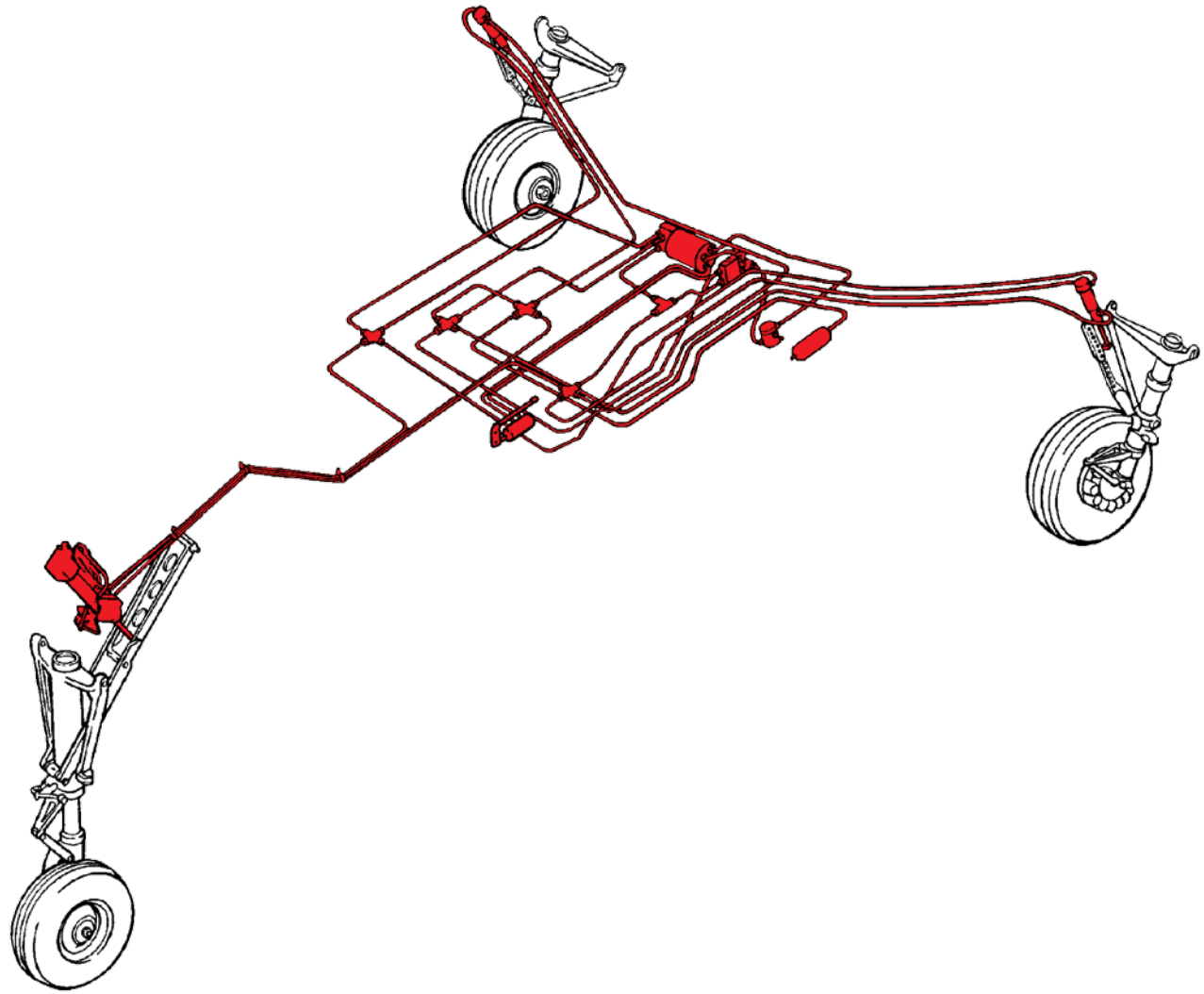

Figure 14-6. Hydraulic Landing Gear Diagram

LANDING GEAR EXTENSION AND RETRACTION

The nose and main landing gear assemblies are extended and retracted by a hydraulic power pack in conjunction with hydraulic actuators (Figure 14-6). The hydraulic power pack is located in the center of the center section, just forward of the main spar. One hydraulic actuator is located at each landing gear.

The power pack (Figure 14-7) consists of: a hydraulic pump, a 28-VDC motor, a two-section fluid reservoir, filter screens, a four-way gear selector valve, an up selector solenoid, a fluid level sensor, and an uplock pressure switch. For manual extension the system has a hand-lever-operated pump. The pump handle is located on the floor between the pilot's seat and the pedestal.

Three hydraulic lines (one for normal extension and one for retraction, routed from the power pack, and one for emergency extension routed from the hand pump) are routed to the nose and main gear actuators. The normal extension lines and the manual extension lines are connected to the upper end of each hydraulic actuator. The hydraulic lines for retraction are fitted to the lower ends of the actuators. Hydraulic fluid under

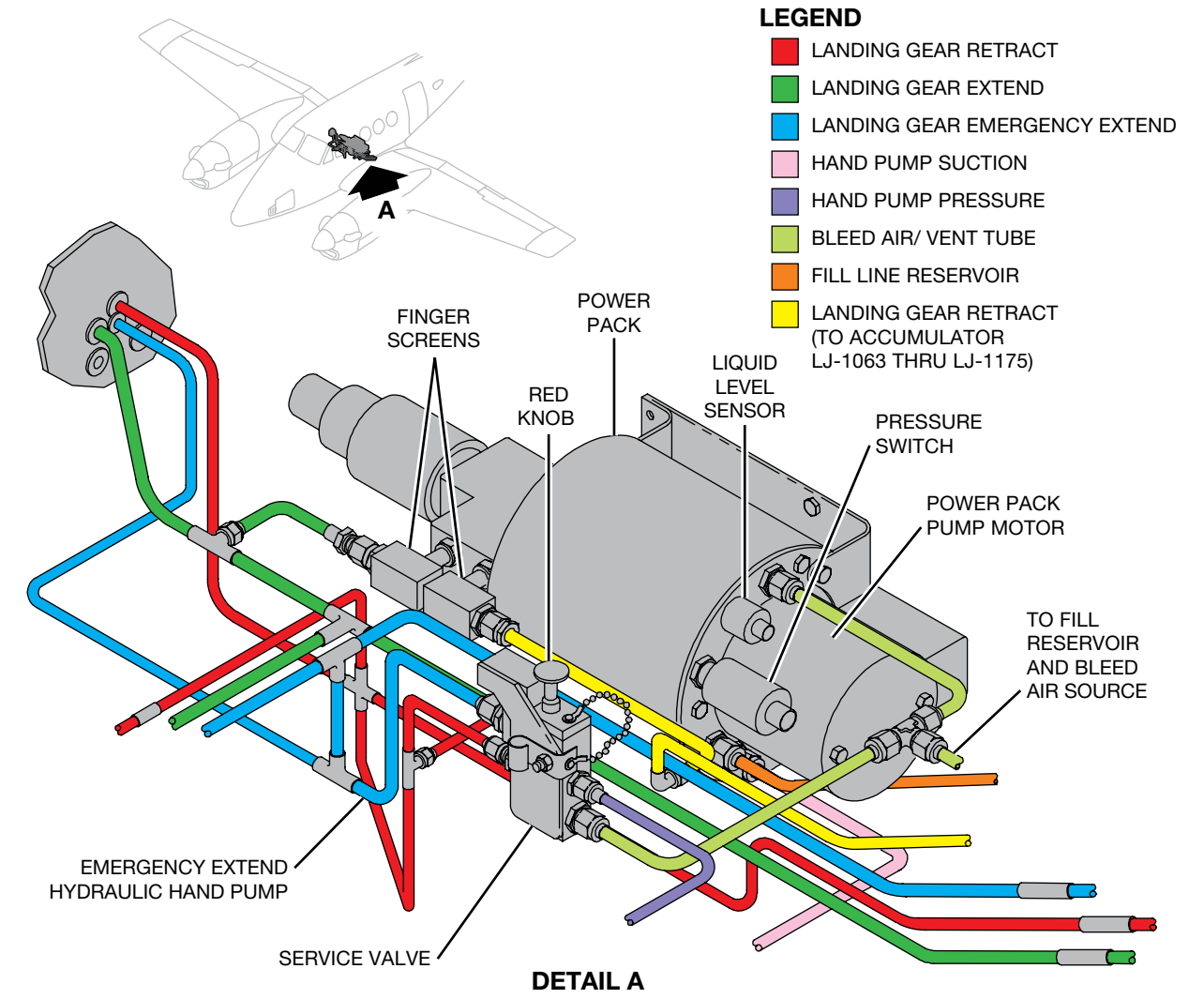

Figure 14-7. Hydraulic Landing Gear Power Pack

pressure (generated by the power pack pump and contained in the accumulator) acts on the piston faces of the actuators (which are attached to folding drag braces), resulting in the extension or retraction of the landing gear.

When the actuator pistons are repositioned to fully extend the landing gear, an internal mechanical lock in the nose gear actuator and the over-center action of the nose gear drag leg assembly lock the nose gear in the down position. In this position, the internal locking mechanism in the nose gear actuator will actuate the actuator downlock switch to interrupt current to the pump motor. The motor will continue to run until all three landing gears are down and locked. A spring-loaded downlock assembly is fitted to each main gear upper drag leg, providing positive downlock action for the main gear.

In flight, with the LDG GEAR CONTROL (Figure 14-8) in the DN position, as the landing gear moves to the fully down position, the downlock switches are actuated, and they cause the landing gear relay to interrupt current to the pump motor. When the red GEAR-IN-TRANSIT lights in the LDG GEAR CONTROL switch handle extinguish, and the green NOSE-L-R indicators illuminate, the landing gear is in the fully down-and-locked position.

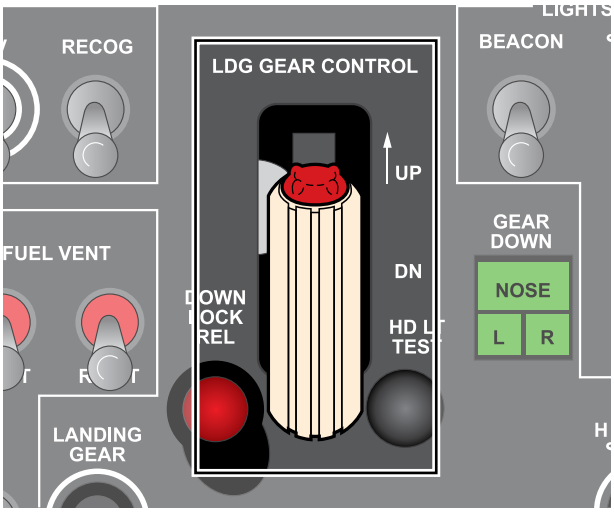

Figure 14-8. Landing Gear Control Switch Handle

A solenoid mounted on the valve body end of the pump is energized when the LDG GEAR CONTROL is in the UP position and actuates the gear select valve, allowing system fluid to flow to the retract side of the system. The gear select valve is spring-loaded in the down position and will move to the up position only when energized. The nose gear actuator will unlock when 200 to 400 psi of hydraulic pressure is applied to the retract port of the nose gear actuator. The landing gear will begin to retract after the nose gear actuator is unlocked.

Hydraulic system pressure performs the uplock function, holding the landing gear in the retracted position. When the hydraulic pressure reaches approximately 1,850 psi, the uplock pressure switch will cause the landing gear relay to open and interrupt the current to the pump motor. The same pressure switch will cause the pump to actuate should the hydraulic pressure drop to approximately 1,600 psi.

The landing gear control circuit is protected by a 2-ampere circuit breaker located on the pilot's inboard subpanel. Power for the pump motor is supplied through the landing gear motor relay and a 60-ampere circuit breaker, both of which are located under the cabin floor in the wing center section. The motor relay is energized by current from the 2-ampere circuit breaker and the down-lock switches.

HYDRAULIC FLUID LEVEL INDICATION SYSTEM

A caution annunciator placarded "HYD FLUID LOW" (Figure 14-9), in the annunciator panel, will illuminate (yellow) whenever the hydraulic fluid level in the landing gear power pack reservoir is low. The annunciator is tested by pressing the HYD FLUID SENSOR TEST button located on the pilot's subpanel.

If the HYD FLD LOW annunciator comes on, normal extension may be attempted, but the pilot should be prepared for an emergency manual extension.

Control

The landing gear hydraulic power pack motor is controlled by the landing gear switch handle placarded "LDG GEAR CONTROL" with UP and DN positions, located on the pilot's right subpanel (Figure 14-9). The switch handle must be pulled out of a detent before it can be moved from either the UP or DN position.

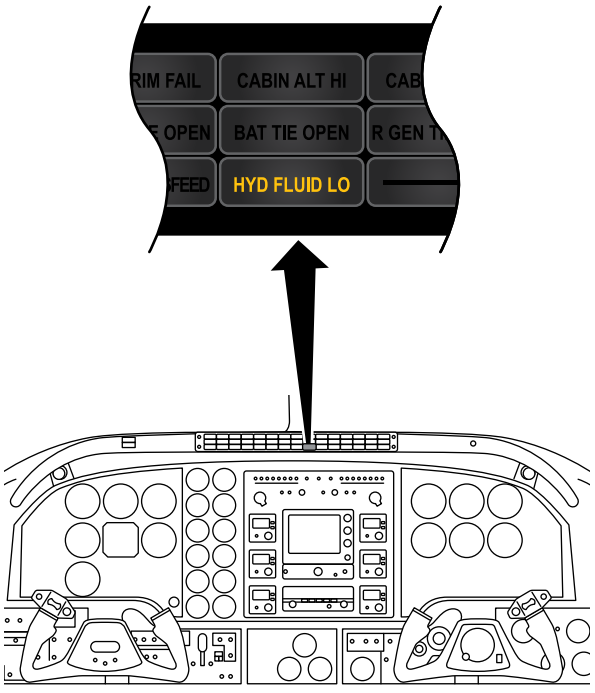

Figure 14-9. Hydraulic Fluid Indicator

Safety switches (Figure 14-10) called "squat" switches, on the main gear shock strut, open the control circuit when the oleo strut is compressed. The squat switches must close to actuate a solenoid, which moves a downlock hook on the LDG GEAR CONTROL switch to the released position. This mechanism prevents the LDG GEAR CONTROL switch handle from being placed in the UP position when the airplane is on the ground. The downlock hook automatically unlocks when the airplane leaves the ground.

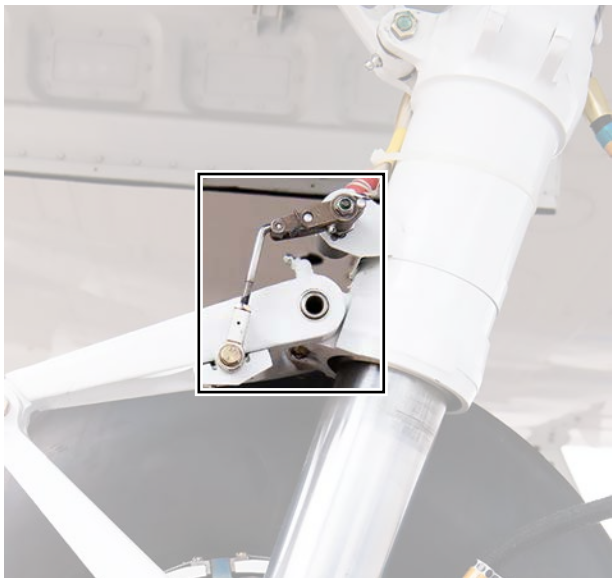

Figure 14-10. Safety Switch

The downlock hook disengages when the airplane leaves the ground because the squat switches close and a circuit is completed through the solenoid that moves the hook. In the event of a malfunction of the downlock solenoid or the squat switch circuit, the downlock hook can be overridden by pressing downward on the red DOWN LOCK REL button. The release button is located just left of the LDG GEAR CONTROL switch handle.

The LDG GEAR CONTROL handle should never be moved out of the DN detent while the airplane is on the ground. If it is, the landing gear warning horn will sound intermittently, and the red gear-in-transit lights in the LDG GEAR CONTROL switch handle will illuminate (provided the MASTER SWITCH is ON), warning the pilot to return the handle to the DN position.

Position Indicators

Landing gear position is indicated by an assembly of three lights in a single unit located on the pilot's right subpanel (Figure 14-11). The unit has a light transmitting cap that is marked as follows: "NOSE-L-R." Light bulbs in each segment, when illuminated, make the segment appear green and indicate that particular gear is down and locked. Absence of illumination may indicate an unsafe gear indication (Figure 14-12). The green position indicator lights may be checked by pushing on the light housing. Two red parallel-wired indicator lights, located in the LDG GEAR CONTROL switch handle, illuminate to show that the gear is in-transit (Figure 14-13) or unlocked. Gear UP is indicated when the red lights go out. The red lights in the handle also illuminate when the landing gear warning system is activated.

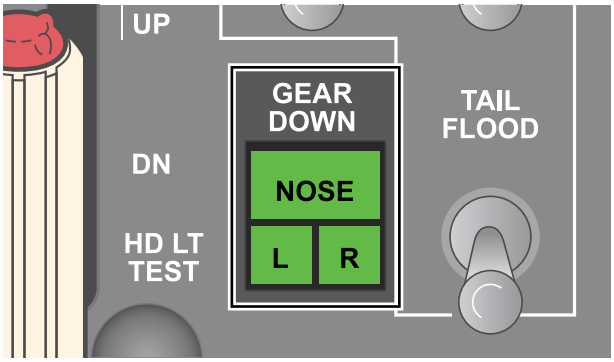

Figure 14-11. Gear Position Indicator

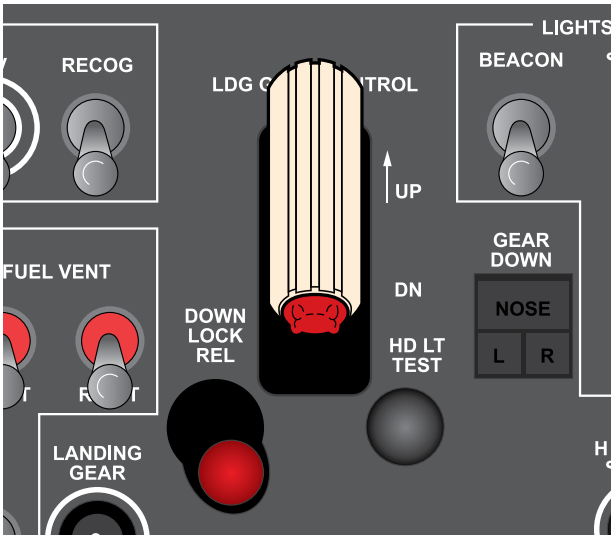

Figure 14-12. Gear Position Indicator—No Illumination

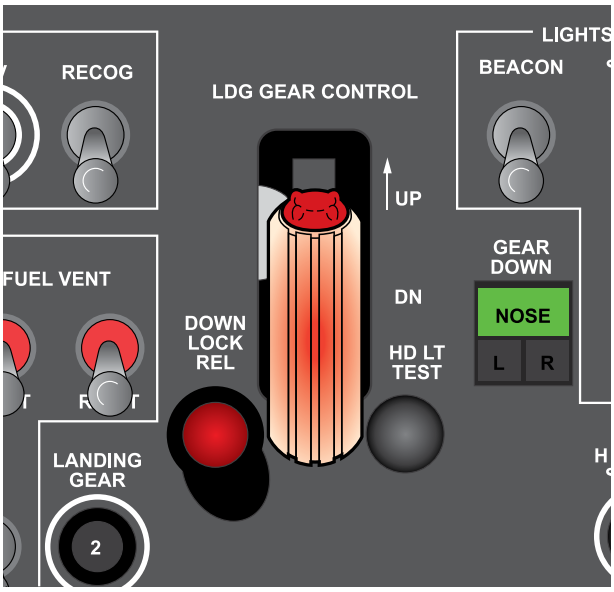

Figure 14-13. Landing Gear Control Switch Handle—Red In-Transit Indicators

The red control handle lights may be checked by pressing HD LT TEST button (Figure 14-14) located adjacent to the LDG GEAR CONTROL switch handle.

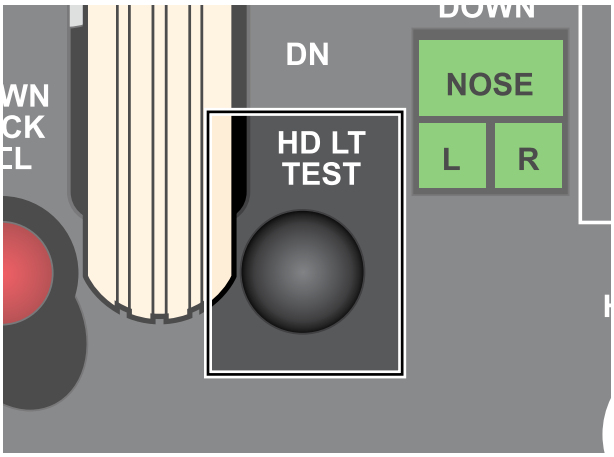

Figure 14-14. Handle Light Test

Each normally closed, up-position switch is located in the upper portion of its respective wheel well. When the gear is in the fully retracted position, each strut actuates its respective up-position switch to open the circuit from the in-transit light to ground. As soon as the gear moves from the fully retracted position, each strut actuates its respective up-position switch to illuminate the in-transit light by providing a path to ground through the down-position switch. The in-transit light goes out when the drag brace in each landing gear passes over-center to actuate its respective down-position switch to the momentary contacts. In this position, the switch opens the circuit to the in-transit light and completes a path to ground for the down-position lights. The down-position switch on each landing gear also functions as a warning switch for the system.

The landing gear in-transit light will indicate one or all of the following conditions:

- Landing gear handle is in the UP position, and the airplane is on the ground with weight on the landing gear.
- With flaps up or approach and one or both power levers retarded below approximately 79 ±2% N_1, one or more landing gears are not down and locked.
- Any landing gear is not in the fully retracted position.
- Flaps are beyond the APPROACH position (36% or more) with any gear not down, regardless of power lever position.

Thus, the function of the landing gear in-transit light is to indicate that the landing gear is in transit.

The up indicator, down indicator, and warning horn systems are essentially independent systems. A malfunction in any one system will probably leave the other two systems unaffected.

LANDING GEAR WARNING SYSTEM

The landing gear warning system is provided to warn the pilot that the landing gear is not down and locked during specific flight regimes. Various warning modes result, depending upon the position of the flaps.

With the flaps in the UP or APPROACH position and either or both power levers retarded below about 79% N_1, the warning horn will sound intermittently. The horn can be silenced by pressing the GEAR WARN SILENCE button adjacent to the LDG GEAR CONTROL switch handle. On the C90B (LJ-1353 and After), the warning horn is silenced by pressing the silence button located on the left power lever. The landing gear warning system will be rearmed if the power levers are advanced sufficiently.

With the FLAPS beyond the APPROACH position, the warning horn activates regardless of the power lever settings and cannot be canceled.

Landing gear warning horn operation is shown in Table 14-1

MANUAL LANDING GEAR EXTENSION

A hand pump handle, placarded "LANDING GEAR ALTERNATE EXTENSION" (Figure 14-15), is located on the floor between the pilot's seat and the pedestal. The pump is located under the floor, below the handle, and is used when emergency extension of the gear is required.

To engage the system, pull the LANDING GEAR RELAY circuit breaker (Figure 14-16), located below and to the left of the LDG GEAR CONTROL switch handle on the pilot's sub-panel, and ensure that the LDG GEAR CONTROL handle is in the DN position. Remove the pump handle from the securing clip, and pump the handle up and down until the green NOSE-L-R gear-down indicator lights illuminate and further resistance is felt. Place the handle in the fully down position and secure in the retaining clip.

WARNING

If for any reason the green GEAR DOWN lights do not illuminate (e.g., in case of an electrical system failure or in the event an actuator is not locked "down"), continue pumping until sufficient resistance is felt to ensure that the gear is down and locked. Do not stow pump handle. The landing gear cannot be manually retracted in flight.

WARNING

After a manual landing gear extension has been made, do not move any landing gear controls or reset any switches or circuit breakers until the airplane is on jacks.

Table 14-1. LANDING GEAR WARNING HORN OPERATION

GEAR POSITION	FLAPS	POWER	HORN	SILENCE MODE
Up	Up	+77 to 81%	No	N/A
Up	Up	–77 to 81%	Yes	Silence button
Up	Approach	–77 to 81%	Yes	Silence button
Up	Past approach	Any	Yes	Lower gear

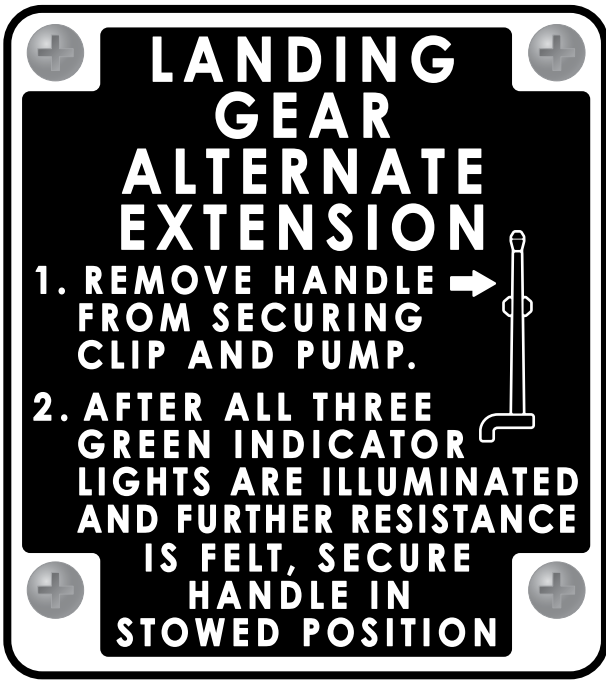

Figure 14-15. Landing Gear Alternate Extension Placard

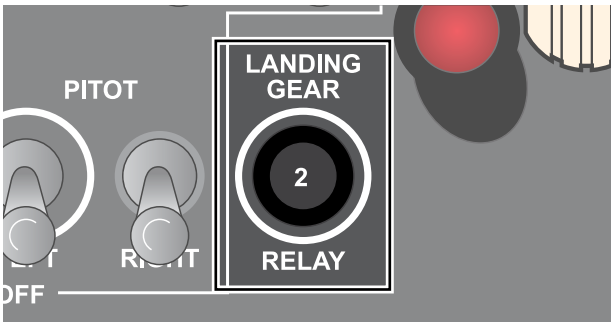

Figure 14-16. Landing Gear Relay Circuit Breaker

After a practice manual extension of the landing gear, the gear may be retracted hydraulically by pushing the LANDING GEAR RELAY circuit breaker in and moving the LDG GEAR CONTROL handle to the UP position.

HYDRAULIC SCHEMATICS

The hydraulic gear schematics shown are for the gear extended, gear retracted, hand pump emergency extension, and gear maintenance retraction modes. Power is available to the contacts of the landing gear remote power relay.

When the relay is open, power comes down from the 2-amp gear control circuit breaker to the landing gear control assembly switch and on to the three downlock switches. Each gear is down and locked, so these three switches are open and no circuit passes through them. This is the static condition of the system after a normal gear extension.

Landing Gear Retraction

When the aircraft is airborne, the pilot selects GEAR UP (Figure 14-17). Circuits are made from the gear selector switch to the uplock pressure switch. The pressure switch is closed at this time, so the circuit is complete to the gear up main switch and landing gear remote power relay. This relay now closes and provides the power circuit to the hydraulic pump motor. Backing up to the pressure switch, a circuit is made to the hydraulic selector valve up-solenoid. Power to this solenoid will position the selector valve body to route hydraulic fluid in the appropriate direction to retract the gear.

After approximately six seconds the retraction cycle is complete. Once the landing gear reaches full-up travel, each actuator physically bottoms out. The pressure on the retract line builds rapidly until pressure reaches approximately 1,850 psi. The uplock pressure switch opens at this time, breaking the power circuit to the pump motor and stopping the hydraulic pump. This pressure switch will close periodically when pressure drops to approximately 1,600 psi, due to the normal system pressure leak-down, and reenergize the pump to restore needed uplock pressure. Consequently, when the gear is retracted, pressure will be maintained between approximately 1,600 and 1,850 psi to keep the gears in their retracted position. An accumulator pre-charged to 800 psi, located in the left wing inboard of the nacelle, is designed to aid in maintaining the system pressure in the gear-up mode.

Landing Gear Extension

For normal gear extension, a pilot selects GEAR DOWN (Figure 14-18), and circuits are made from the landing gear control assembly through any one of the three actuator downlock switches, back through the landing gear control assembly,

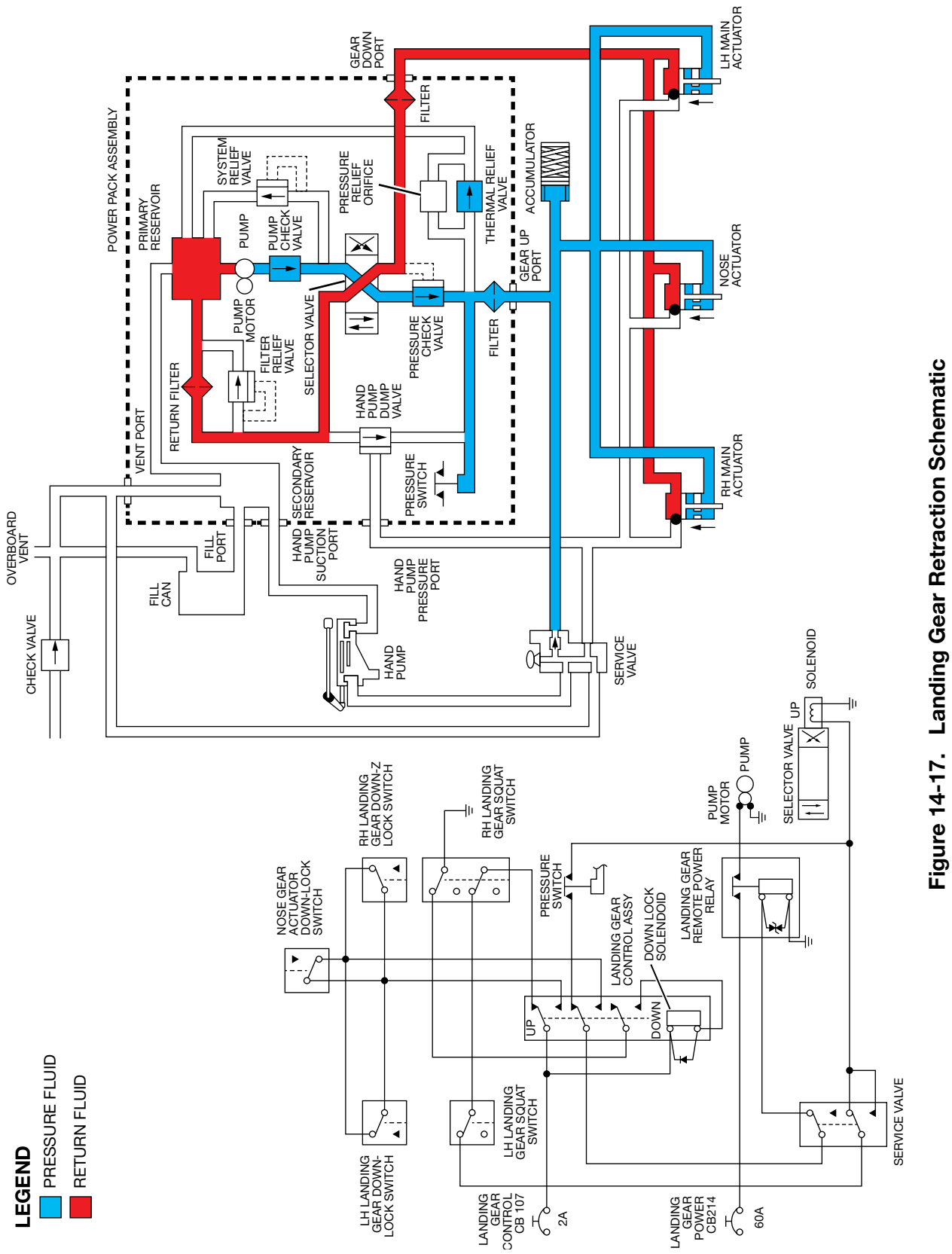

Figure 14-17. Landing Gear Retraction Schematic

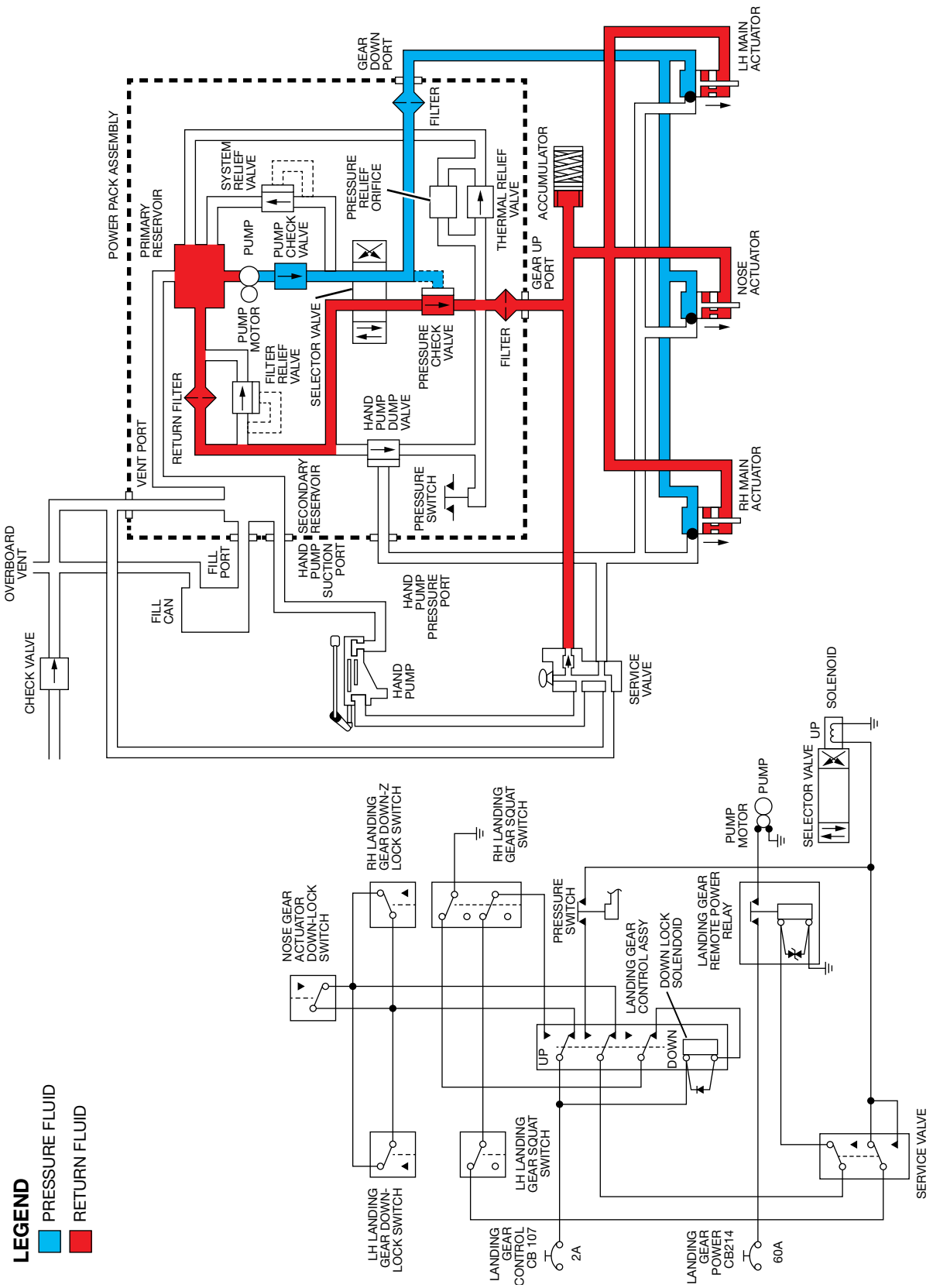

Figure 14-18. Landing Gear Extension Schematic

the service valve, and finally to the landing gear remote power relay. The power relay closes and provides a power circuit to the pump motor. The selector valve is not being powered at this time. Thus, fluid under pump pressure is routed through the selector valve body in the appropriate direction to extend the landing gear.

The gear comes down under fluid pressure until each main gear downlock and the nose gear actuator downlock switches are depressed. When all three gears are down and locked, the control circuit to the pump motor is broken, and the pump stops. Notice that no pressure switches are involved. Consequently, there is no downlock pressure maintained. The mechanical downlocks on each main gear drag brace, and an internal mechanical lock in the nose gear actuator, prevent gear retraction.

Hand Pump Emergency Extension

A hand-pump handle, placarded "LANDING GEAR ALTERNATE EXTENSION," is located on the floor between the pilot's seat and the pedestal. The pump is located under the floor below the handle and is used when emergency extension of the gear is required.

To engage the system, pull the LANDING GEAR RELAY circuit breaker, located on the pilot's inboard subpanel, and place the LDG GEAR CONTROL switch handle in the DN position (Figure 14-19). Remove the pump handle from the securing clip, and pump the handle up and down until the green NOSE-L-R gear down indicator lights illuminate. Place the pump handle in the fully down position and secure in the retaining clip.

After a practice manual extension of the landing gear, the gear may be retracted hydraulically by pushing the LANDING GEAR RELAY circuit breaker in and moving the LDG GEAR CONTROL switch handle to the UP position.

If an alternate landing gear extension becomes necessary, there is no limit to the amount of cycles the hydraulic gear may be pumped. During a complete or partial electrical failure, the gear down lights, in-transit lights, and gear warning horn may not be operating. A positive method of checking that the gear is down is through resistance when pumping the extension handle. When all three gears are extended, hydraulic pressure is built up until the pressure relief valve opens, relieving the pressure built up by the handle. This can be felt by the pilot as increased resistance while pumping, followed by a give as the relief valve opens.

Landing Gear Maintenance Retraction

A service valve (Figure 14-20), located forward of the power pack assembly, may be used in conjunction with the hand pump to raise the gear for maintenance purposes. With the aircraft on jacks and an external electrical power source attached, unlatch the hinged retainer and pull up on the red knob located on top of the service valve. The hand pump can then be used to raise the gear to the desired position. After the required maintenance has been performed, push the red knob down, and use the hand pump to lower the gear. The valve is not accessible to the pilot.

CAUTION

If the red knob on the service valve is pushed down while the landing gear is retracted, the electrical power on, and the landing gear control handle is in the down position, the landing gear will extend immediately.

A fill reservoir, located just inboard of the left nacelle and forward of the front spar, contains a cap and dipstick assembly to facilitate maintenance of the system fluid level. A line plumbed to the upper portion of the fill reservoir is routed overboard to act as a vent.

TIRES

The nose landing gear wheel is equipped with a 6.50 x 10, 6-ply-rated, tubeless, rim-inflation tire. Each main landing gear wheel is equipped with an 8.50 x 10, 8-ply-rated, tubeless, rim-inflation tire.

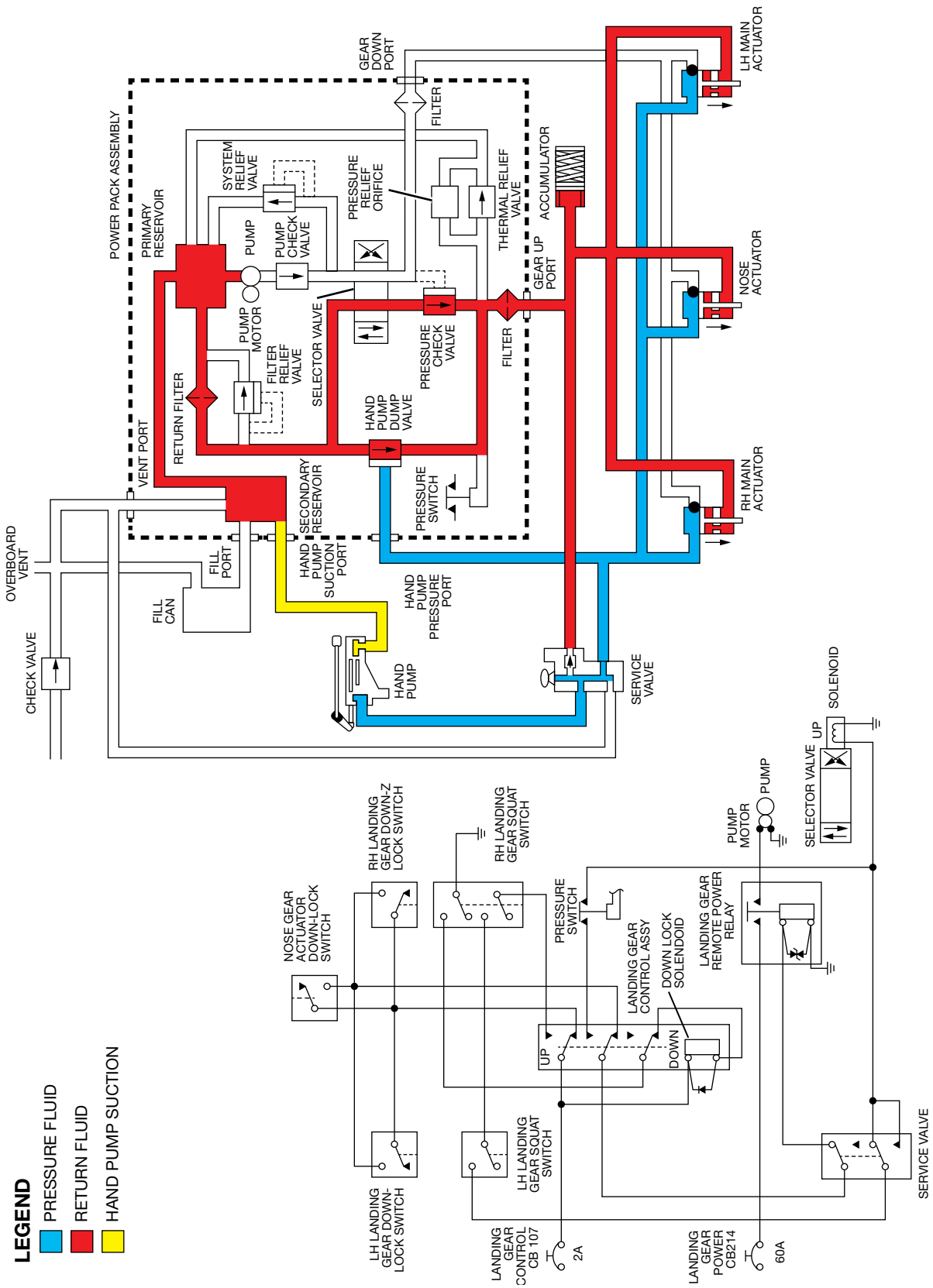

Figure 14-19. Hand Pump Emergency Extension Schematic

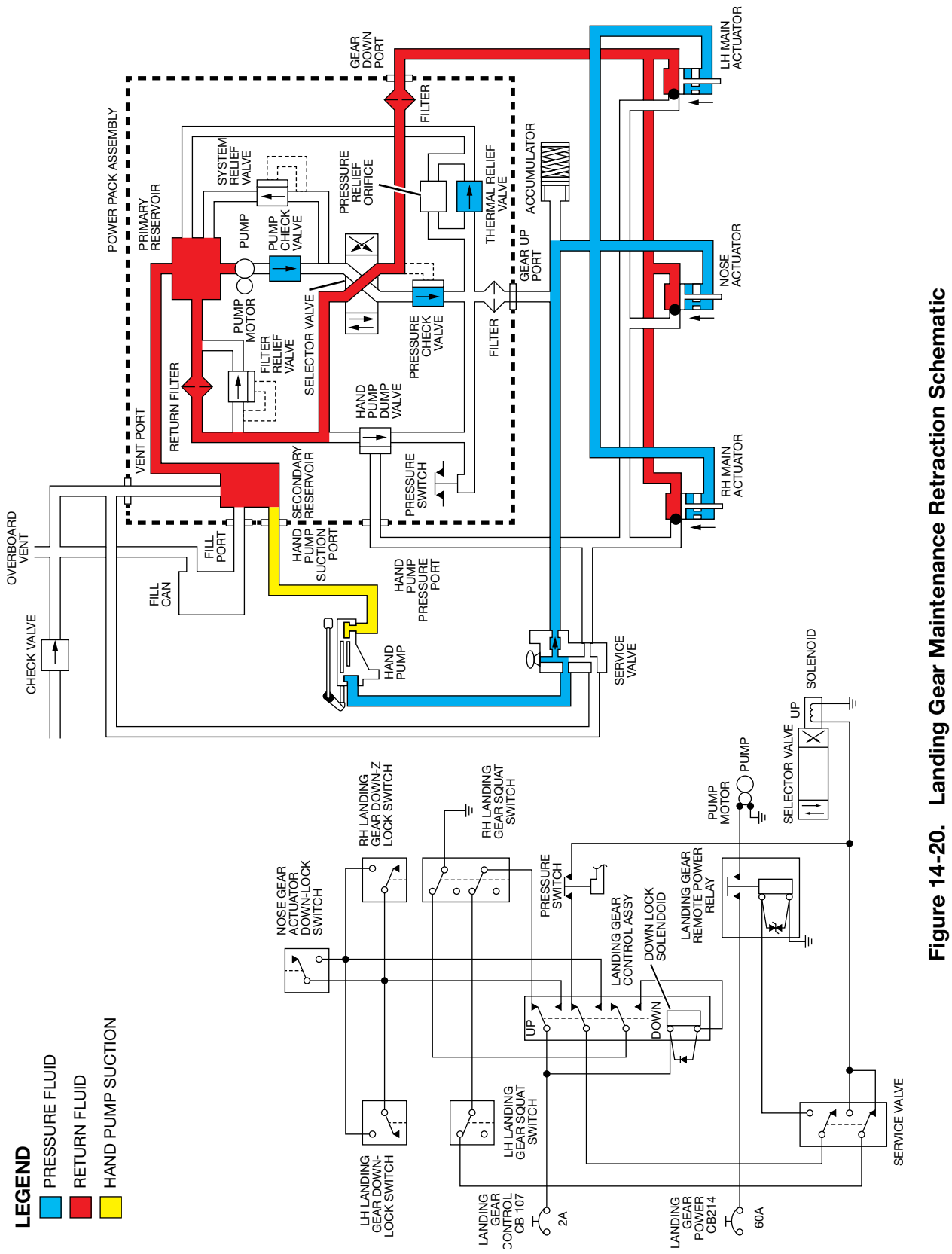

Figure 14-20. Landing Gear Maintenance Retraction Schematic

For increased service life, 10-ply-rated tires of the same size may be installed. Check the *Pilot's Operating Handbook* for correct tire pressure.

SHOCK STRUTS

Shock struts should always be properly inflated. Do not over- or under-inflate, and never tow or taxi an aircraft when any strut is flat. Correct inflation is approximately 3 inches for the main strut and 3.0 to 3.5 inches for the nose strut.

LANDING GEAR OPERATING LIMITS

The landing gear operating limits are shown in Table 14-2 below.

KING AIR WHEEL BRAKES

The King Air series brakes are a non-assisted hydraulic brake system. The main landing gear wheels are equipped with multi-disc dual hydraulic brakes. These brakes are actuated by toe pressure on the rudder pedals by either the pilot or copilot. The depression of either set of pedals compresses the piston rod in the master cylinder attached to each pedal. The hydraulic pressure resulting from the movement of the pistons in the master cylinders is transmitted through flexible hoses and fixed aluminum tubing to the disc brake assemblies on the main landing gear wheels. This pressure forces the brake pistons on the wheel to press against the multiple linings and discs of the brake assembly.

As with any airplane, proper traction and braking control cannot be expected until the landing gear is carrying the full weight of the airplane. Use extreme care when braking to prevent skidding and the resulting flat sections on tires caused by skidding. Braking should be smooth and even all the way to the end of ground roll.

SERIES BRAKE SYSTEM

The dual brakes are plumbed in series (Figure 14-21). Each rudder pedal is attached to its own master cylinder. The pilot's master cylinders are plumbed through the copilot's master cylinders, thus allowing either set of pedals to perform the braking action. The pilot's and copilot's right rudder pedals control the brake in the right main landing gear. Similarly, the pilot's and copilot's left rudder pedals control braking in the left main gear. This arrangement allows differential braking for taxiing and maneuvering on the ground.

PARKING BRAKE

The parking brake utilizes the regular brakes and a set of valves (Figure 14-22). Dual parking brake valves are installed adjacent to the rudder pedals between the master cylinders of the copilot's rudder pedals and the wheel brakes. The two lever-type valves are located just aft of the flight compartment under the center aisle floorboard. A push-pull cable from the valve control levers runs to the pedestal, terminating with a knob. The control knob for the parking brake valves, placarded "PARKING BRAKE-PULL ON," is below the lower left corner of the pilot's subpanel.

Table 14-2. LANDING GEAR OPERATING LIMITS

AIRSPEED	KIAS	REMARKS
Maximum landing gear operation (V_{LO})		Do not extend or retract the landing gear above this speed.
• Extension	182	
• Retraction	163	
Maximum Landing gear extended (V_{LE})	182	Do not exceed this speed with the landing gear extended.

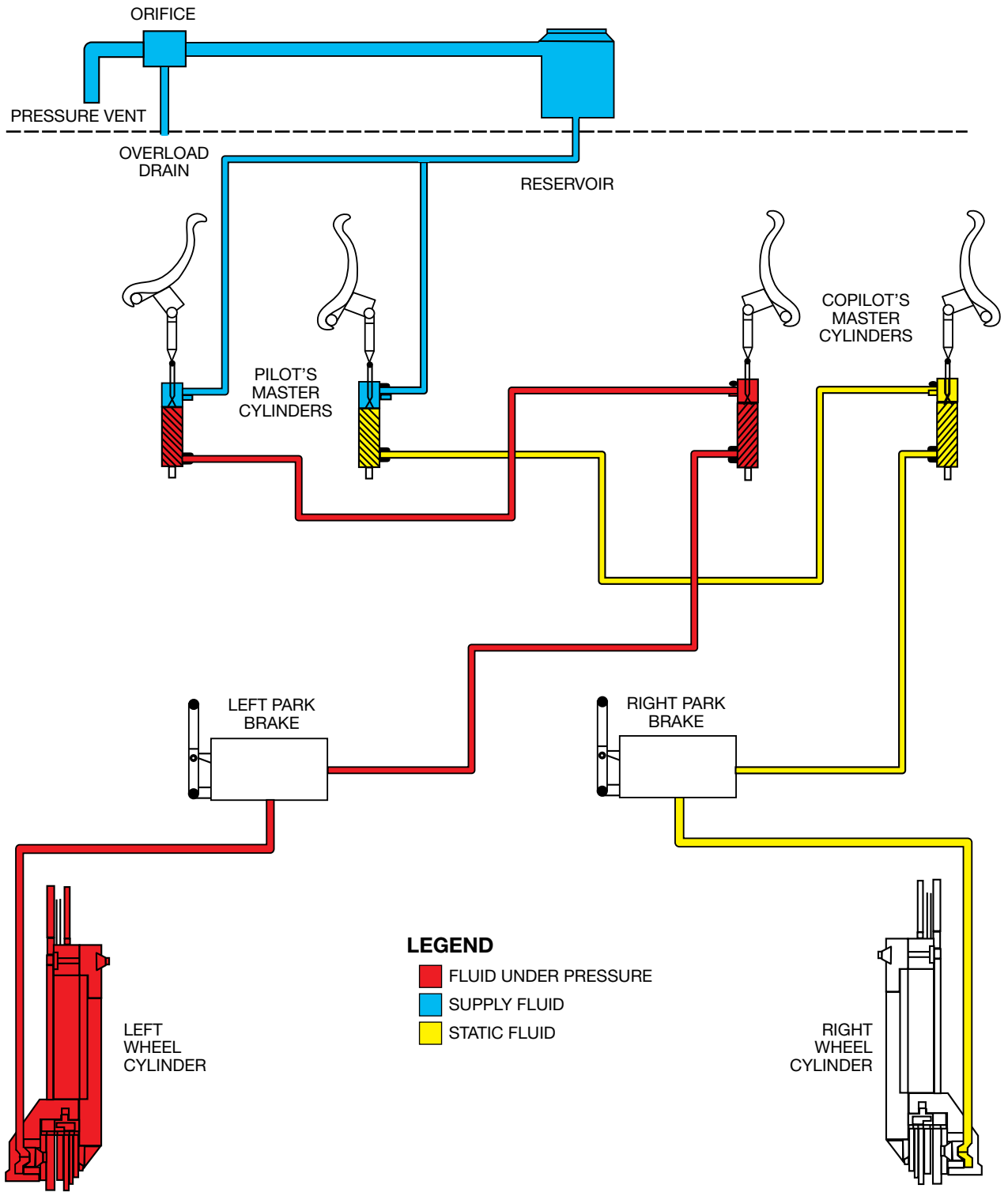

Figure 14-21. Brake System Schematic

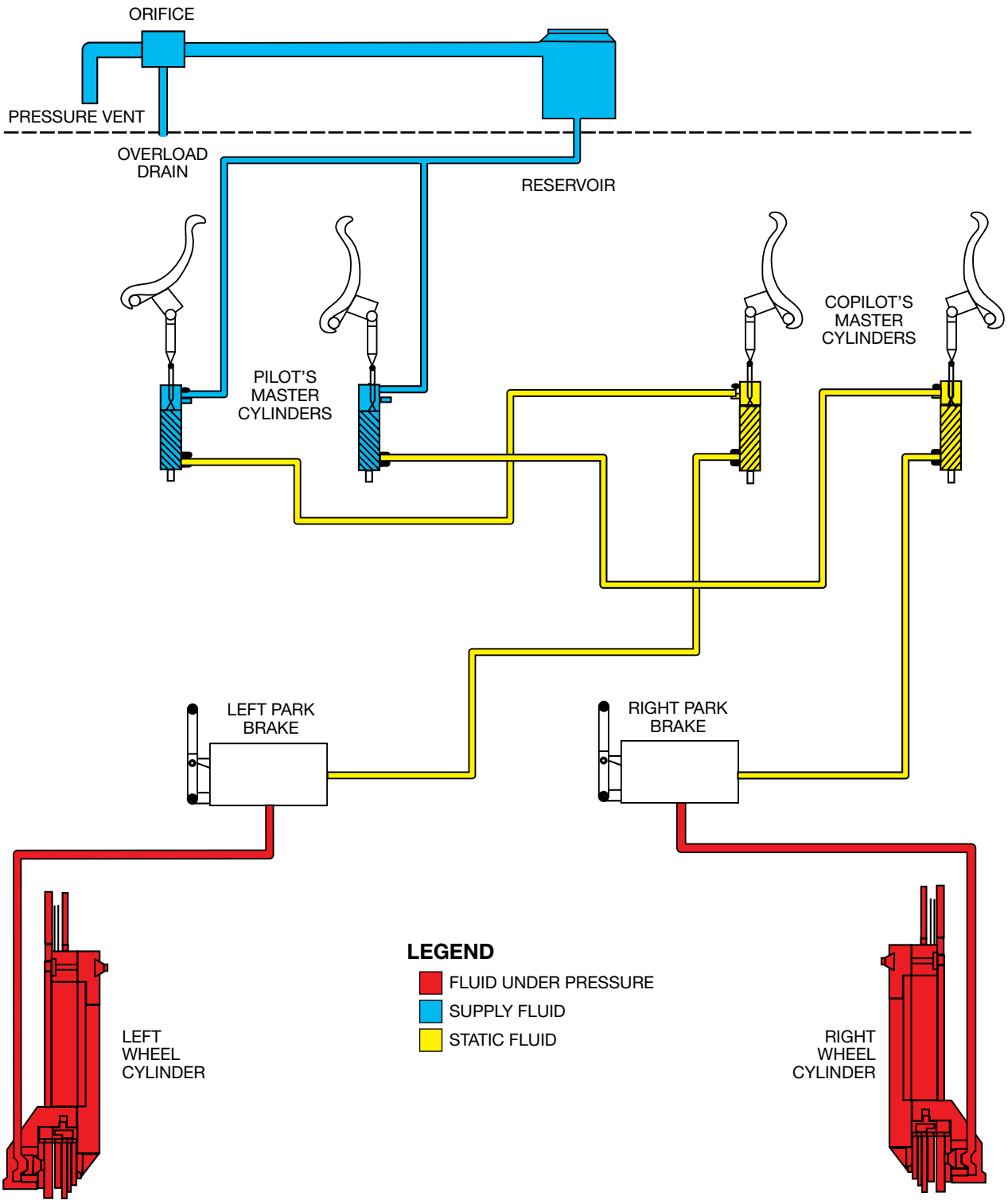

Figure 14-22. Parking Brake Schematic

To set the parking brake: depress the brake pedals to build up pressure in the brake system, then depress the button in the center of the parking brake control, and pull the control handle aft or ON. This procedure closes both parking brake valves simultaneously. The parking brake valves should retain the pressure previously pumped into the system.

The parking brake can be released from either the pilot's or copilot's side when the brake pedals are depressed briefly to equalize the pressure on both sides of the valves, and the PARKING BRAKE handle is pushed in to allow the parking brake valves to open.

To avoid damage to the parking brake system, tires, and landing gear, the parking brake should be left off and wheel chocks or tiedowns installed if the airplane is to be left unattended, because the airplane may be moved by ground personnel in the pilot's absence. Also, ambient temperature changes can expand or contract the brake fluid, causing excessive brake pressure or brake release.

BRAKE SERVICE

Brake fluid is supplied to the master cylinders from a reservoir accessible through the nose avionics compartment door, prior to LJ-1531. On LJ-1531 and subsequent the door was replaced with an access panel (Figure 14-23). The brake fluid reservoir is located on the upper corner of the left side of the nose avionics compartment.

Brake system servicing is limited primarily to maintaining the hydraulic fluid level in the reservoir. A dipstick is provided for measuring the fluid level. When the reservoir is low on fluid, add a sufficient quantity of MIL-H-5606 hydraulic fluid to fill the reservoir to the full mark on the dipstick. Check all hydraulic landing gear connections for signs of seepage and correct if necessary. Do not check while the parking brake is deployed.

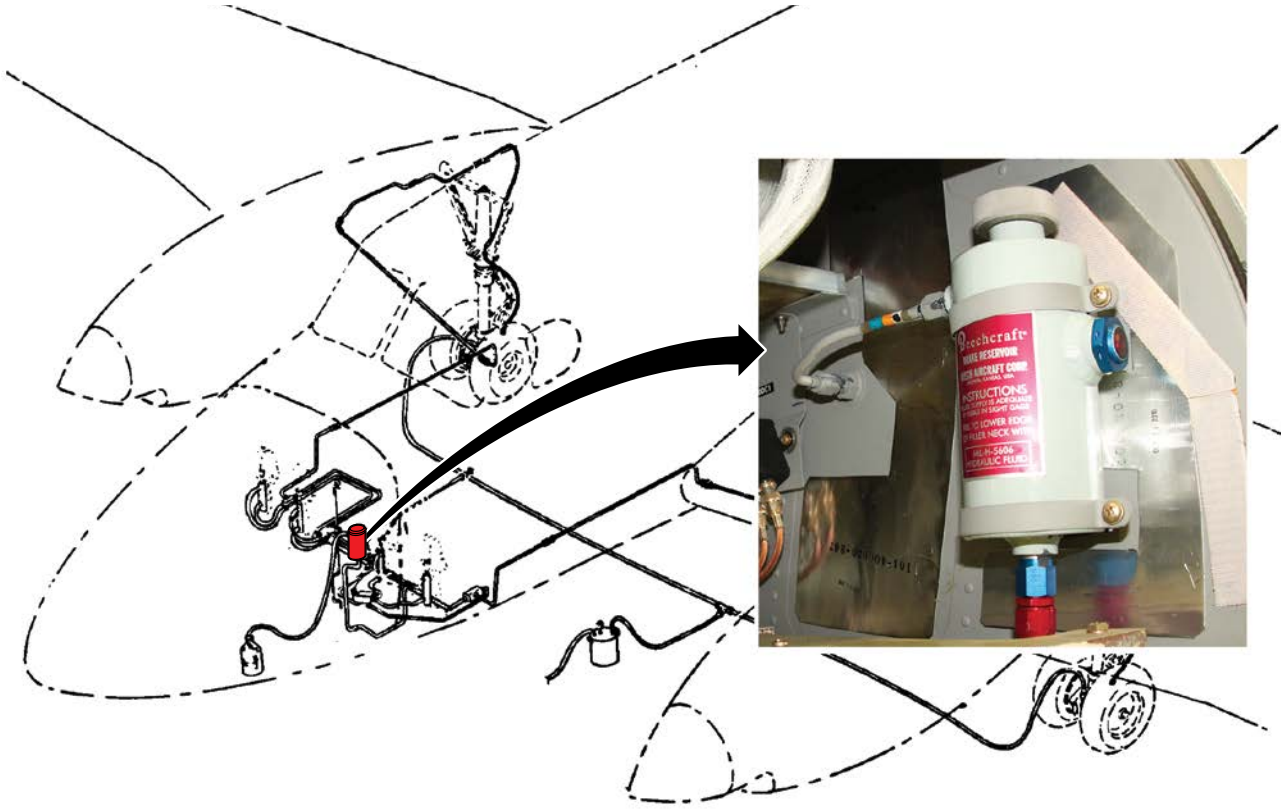

Figure 14-23. Brake Fluid Reservoir

Standard brakes used on this airplane are equipped with automatic brake adjusters. The automatic brake adjusters reduce brake drag, thereby allowing unhampered roll. Airplanes with the automatic adjusters tend to exhibit a softer pedal and a somewhat longer pedal stroke.

BRAKE WEAR LIMITS

Brake lining adjustment is automatic, eliminating the need for periodic adjustment of the brake clearance. Check brake wear periodically to assure that dimension "A," in the Brake Wear Diagram (Figure 14-24), does not reach zero. When it reaches zero, refer to the Beechcraft servicing and maintenance instructions for King Air brakes and wheels. The parking brake must be set (pressure on the brakes) before this can be done.

COLD WEATHER OPERATION

When operating in cold weather, check the brakes and the tire-to-ground contact for freeze lock-up. Anti-ice solutions may be used on the brakes or tires if freeze-up occurs. No anti-ice solution which contains a lubricant, such as oil, should be used on the brakes. It will decrease the effectiveness of the brake friction areas.

When possible, taxiing in deep snow or slush should be avoided. Under these conditions the snow and slush can be forced into the brake assemblies. Keep flaps retracted during taxiing to avoid throwing snow or slush into the flap mechanisms and to minimize damage to flap surfaces.

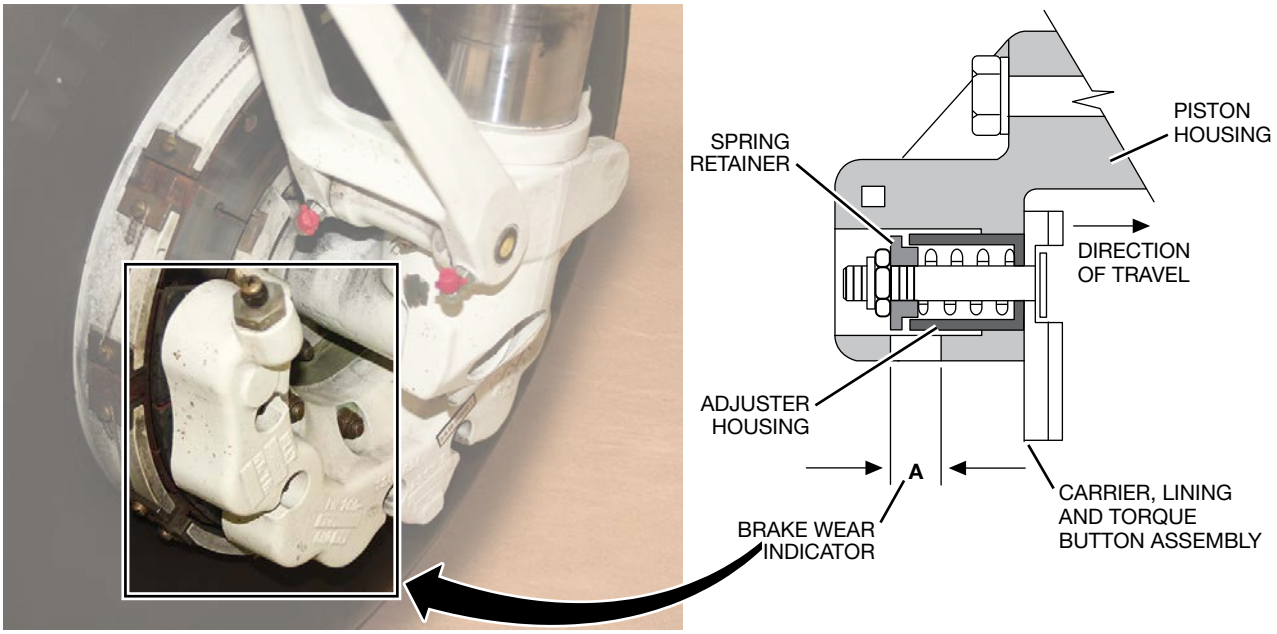

Figure 14-24. Brake Wear Diagram

QUESTIONS

1. Which of the following statements is correct about the retractable tricycle landing gear?

A. The gear is electrically controlled and mechanically actuated
B. The gear is mechanically controlled and electrically actuated
C. The gear is electrically controlled and hydraulically actuated
D. The gear is hydraulically controlled and electrically actuated

2. What holds the landing gear in the up position?

A. Hydraulic pressure
B. Mechanical up locks
C. Friction in the system
D. Electrical up locks

3. What prevents the landing gear handle from being positioned to the UP position when the aircraft is on the ground?

A. A downlock hook
B. A magnetic lock
C. Down pressure on the mechanical locks
D. The lack of hydraulic pressure

4. What is the approximate N_1 setting that will activate the landing gear warning horn when one or both of the power levers are retarded?

A. 58%
B. 62%
C. 69%
D. 79%

5. What is the maximum landing gear extension speed?

A. 148 KIAS
B. 163 KIAS
C. 182 KIAS
D. 184 KIAS

6. What is the maximum landing gear retraction speed?

A. 148 KIAS
B. 163 KIAS
C. 182 KIAS
D. 184 KIAS

7. Which of the following statements is correct concerning the brake system?

A. Depress the brake pedal to build up pressure and pull the parking brake control aft or on
B. Pull the parking brake control aft or on and depress the brake pedal to build up pressure
C. The parking brake may be used as a chock when the aircraft is left unattended
D. Differential braking should never be used to maneuver the aircraft on the ground

8. Where is the brake reservoir located?

A. In the avionics compartment on the right side
B. In the left center wing section beside the landing gear power pack
C. In the utility area aft of the aft pressure bulkhead
D. In the avionics compartment on the left side

9. When the landing gear is fully retracted, the electrically driven hydraulic pump:

A. Stops, and does not start again.
B. Stops, but cycles as required.
C. Operates continuously.
D. Continues to operate for five minutes, then stops.

CHAPTER 15
FLIGHT CONTROLS

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CHAPTER 15
FLIGHT CONTROLS

INTRODUCTION

Familiarization with the flap system operation and limits is necessary to provide optimum performance in takeoff, approach, and landing modes. This chapter identifies and describes flap action so the pilot will understand their operation, controls, and limits.

A basic understanding of how the rudder boost system works, and its value in engine-out situations, will assist the pilot in making full use of its advantages. This chapter also presents familiarization with and operation of the rudder boost system.

DESCRIPTION

This chapter presents a description and discussion of flap system. The four-segment Fowler-type system, its controls and limits are considered with reference to operation as outlined in the *Pilot's Operating Handbook*.

The rudder boost system section of this chapter presents a description and discussion of the rudder boost system. This system is designed to reduce pilot effort in single-engine flight configurations.

FLAPS SYSTEM

The flaps, two panels on each wing, are driven by an electric motor through a gearbox mounted on the forward side of the rear spar (Figure 15-1). The motor incorporates a dynamic braking system through the use of two sets of motor windings. This system helps to prevent overtravel of the flaps. The gearbox drives four flexible driveshafts, each of which is connected to a jackscrew actuator at each flap.

The flaps are operated by a sliding lever located just below the condition levers on the pedestal (Figure 15-2). Flap travel, from 0% (UP) to 100% (DOWN), is registered at 20, APPROACH, 40, 60, and 80 and DOWN in percentage of travel on an electric indicator on top of the pedestal (Figure 15-3).

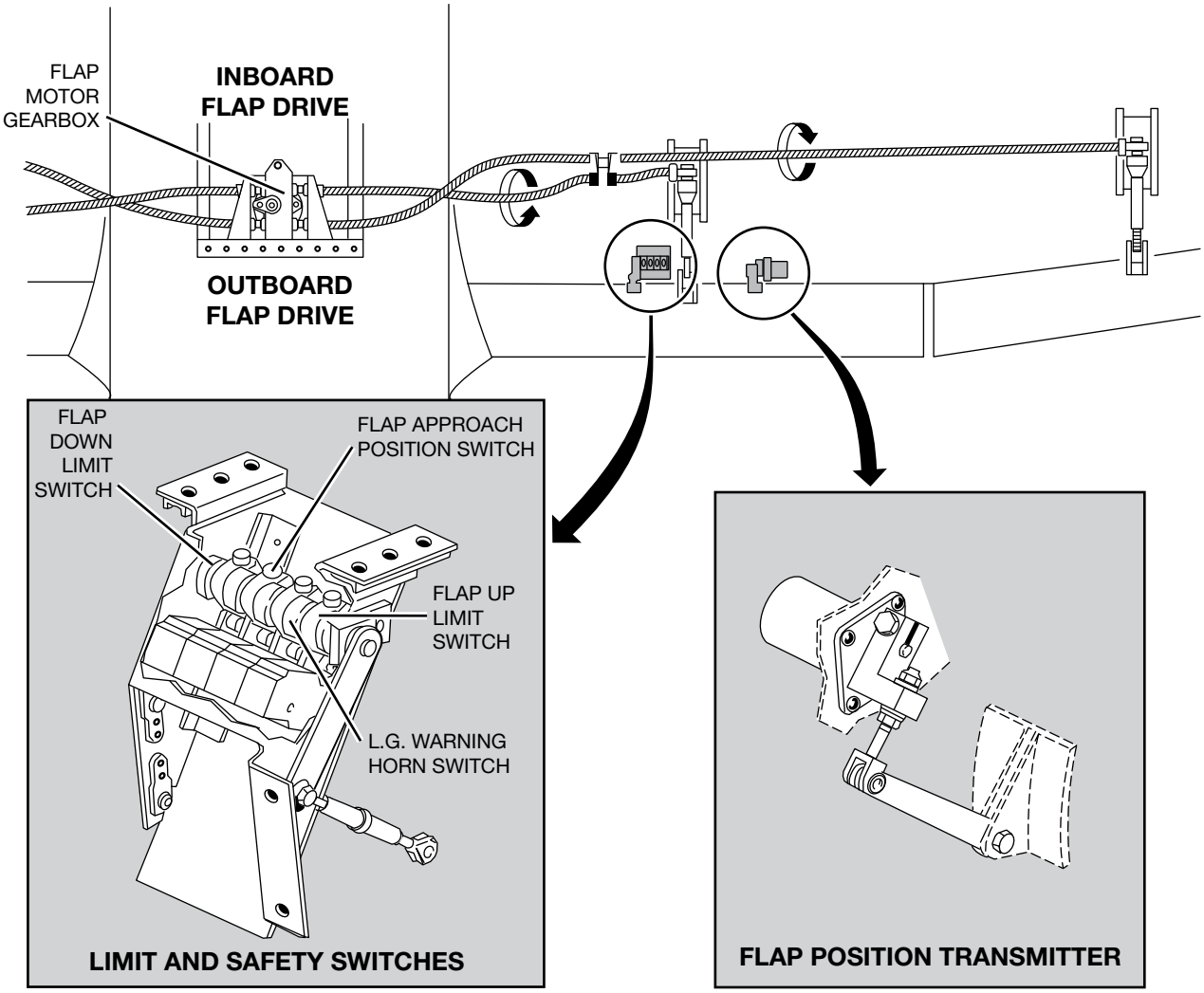


Figure 15-1. Flap Control System

Figure 15-2. Flap Control Lever

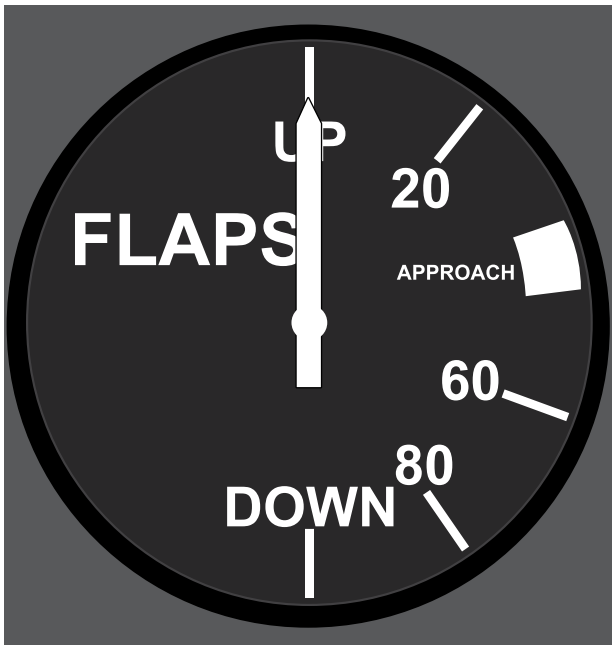


Figure 15-3. Flap Position Indicator

The flap control has a position detent provided for quick selection of 30% (15°) flaps for APPROACH. Full flap deflection is approximately 43°. The indicator is operated by a potentiometer driven by the right hand inboard flap. Flap position limit switches are also driven by the RH inboard flap.

The flap motor power circuit is protected by a 20-ampere circuit breaker placarded FLAP MOTOR, located on the right hand circuit breaker panel. A 5-ampere circuit breaker, placarded FLAP IND & CONTROL, for the flap control circuit is also located on this panel (Figure 15-4).

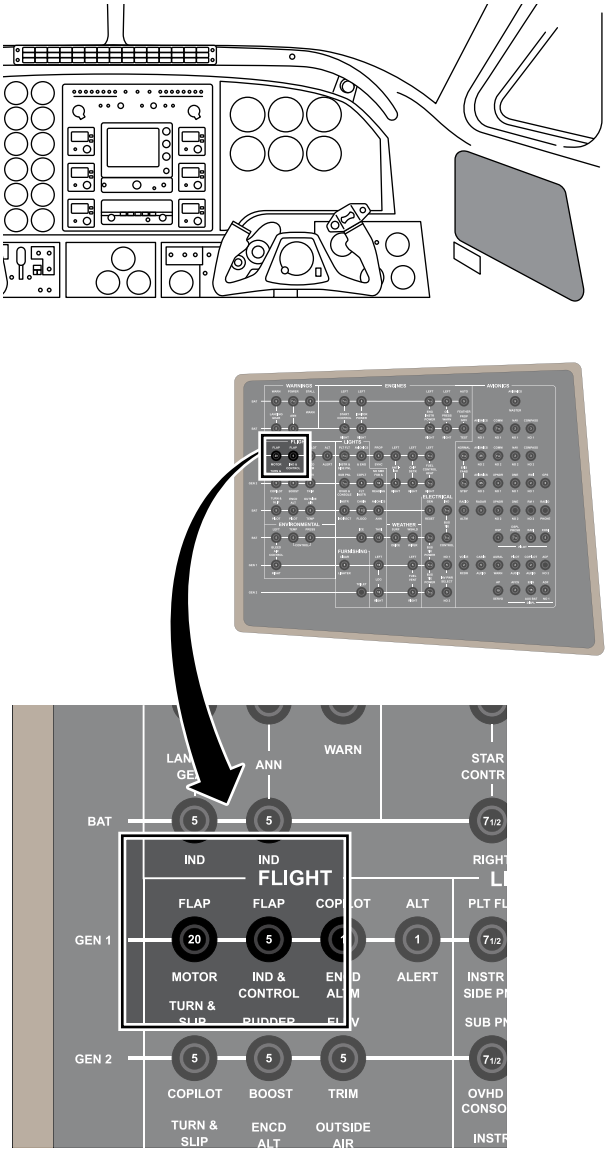


Figure 15-4. Flap System Circuit Breaker

C90A FLAP OPERATION

From the UP position to the APPROACH position, the flaps cannot be stopped at an intermediate point. Between APPROACH and DOWN, the flaps may be stopped as desired by moving the handle to DOWN position until the flaps have moved to the desired position, then moving the handle back to APPROACH. In like manner, the flaps may be raised to any position between DOWN and APPROACH by raising the handle to UP until the desired setting is reached, then returning the handle to APPROACH. The APPROACH detent acts as a stop for any position greater than 35%. Moving the handle from DOWN to APPROACH will not retract the flaps. When the flaps are at APPROACH and the handle is moved from APPROACH position to the UP position, the flaps retract completely and cannot be stopped in between.

C90B FLAP OPERATION

Flaps are selectable to 3 positions: up, approach (15°), and down (43°). If a go-around is initiated with flaps fully extended, retraction to either approach or full-up positions can be accomplished with a single switch position selection.

LANDING GEAR WARNING SYSTEM

The landing gear warning system is provided to warn the pilot that the landing gear is not down and locked during specific flight regimes. The warning horn will sound continuously when the flaps are lowered beyond the APPROACH (30%) position, regardless of the power lever setting, until the landing gear is extended or the flaps are retracted. Although the landing gear warning system is affected by the flap position, this subject is discussed more completely in the LANDING GEAR section of this training manual.

FLAP AIRSPEED LIMITS

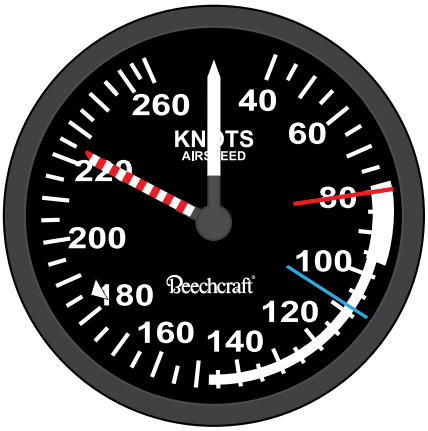
Airspeed indicator (Figure 15-5) markings show the maximum speeds and operating range of the flaps VFE). The white triangle indicates maximum flaps-to or at-approach (30%) speed. The upper limit of the narrow white arc is the maximum speed permissible with flaps extended beyond APPROACH (more than 30%). Approach speed (flaps 30%) is 184 KIAS. Beyond APPROACH position, the maximum speed is 148 KIAS.

Lowering the flaps will produce these results:

- Attitude-Nose up
- Airspeed-Reduced
- Stall speed-Lowered

NOTE

All illustration needles may not reflect normal indications.



C90A AIRSPEED INDICATOR MARKINGS

- 226 KIAS MAXIMUM (V_{MO})
- 90 KIAS MINIMUM SINGLE-ENGINE CONTROL SPEED (V_{MCA})
- 107 KIAS SINGLE-ENGINE BEST RATE OF CLIMB SPEED (V_{YSE})
- 76-148 KIAS FULL FLAP OPERATING RANGE
- 184 KIAS MAXIMUM FLAPS TO/AT APPROACH SPEED

Figure 15-5. Airspeed Indicator

RUDDER BOOST SYSTEM

A rudder boost system (Figure 15-6) is provided to aid the pilot in maintaining directional control in the event of an engine failure or a large variation of power between the engines. Incorporated into the rudder cable system are two pneumatic rudder-boosting servos that actuate the cables to provide rudder pressure to help compensate for asymmetrical thrust.

The rudder boost system consists of pneumatic actuators in the empennage which provide the required rudder deflection upon loss of an engine. A differential pressure switch, mounted on the pneumatic manifold, senses engine P_3 pressures. Upon sensing a loss of P_3 on one engine, this pressure switch will energize a solenoid to direct pneumatic manifold air to the appropriate actuator.

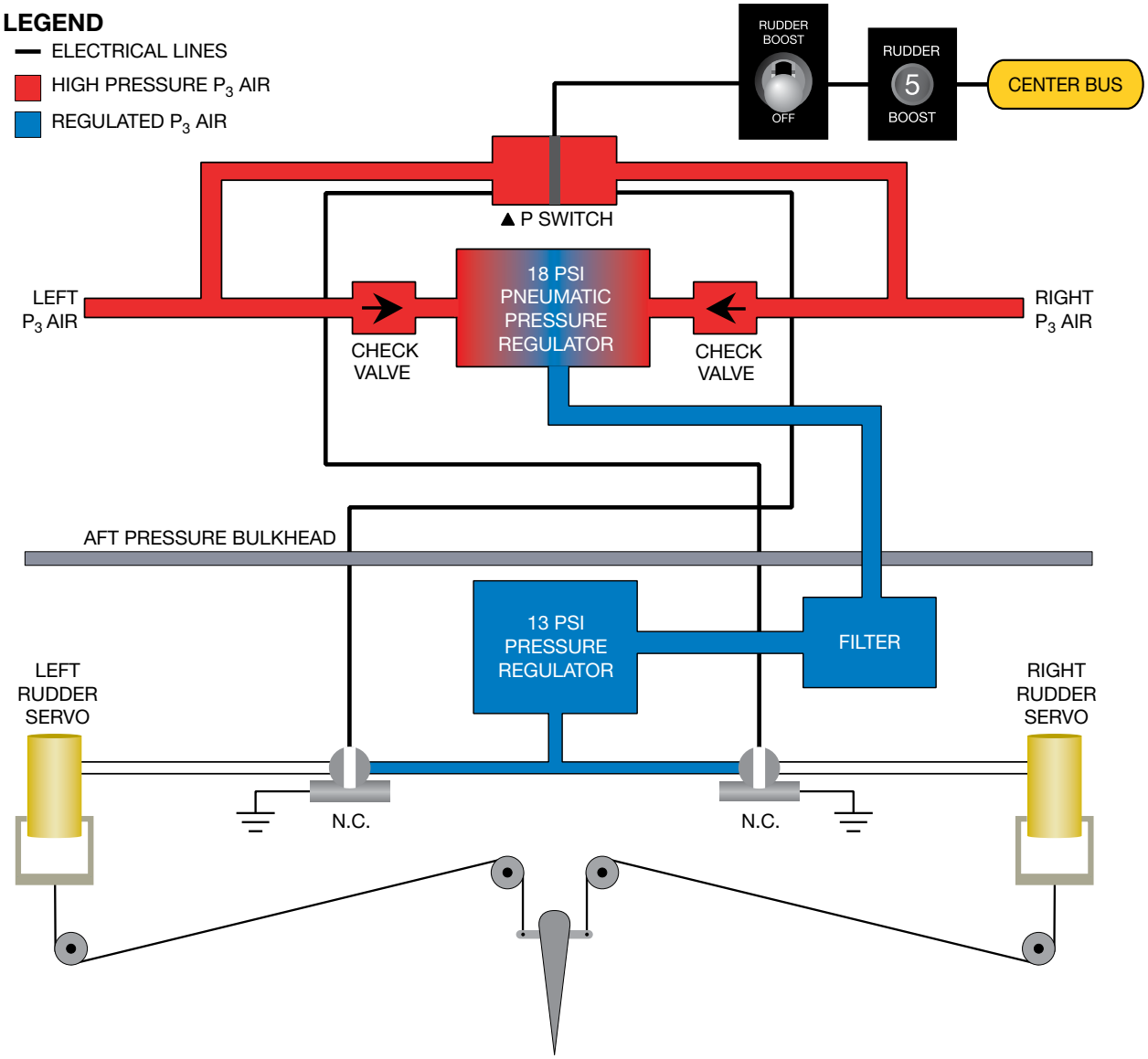


Figure 15-6. Rudder Boost System Diagram

During operation, a differential pressure switch senses bleed air pressure differences between the engines. If the bleed air pressure differential exceeds about 50 psi differential pressure, a signal from the differential pressure switch to one of the lines to the rudder boost servos causes the solenoid valve to open, and one of the servos is actuated. The pressurized servo will then pull on one of the rudder cables. Tension springs in the connection between the servos and the rudder cables take up the slack in the rudder cable when one or the other of the servos is actuated.

A drop in bleed air pressure from the left engine will actuate the appropriate servo and the right rudder pedal will move forward. A drop in bleed air pressure from the right engine will cause the left rudder pedal to move forward. Pedal rigging causes the opposite pedal to move in the opposite direction. This system is intended to help compensate for asymmetrical thrust only. Appropriate trimming is to be done with the trim controls.

The system is controlled by a toggle switch (Figure 15-7), placarded RUDDER BOOST– OFF, located on the pedestal below the aileron trim control knob. The switch is to be in RUDDER BOOST position before flight.

The circuit is protected by the 5-ampere RUDDER BOOST circuit breaker on the right side panel.

A preflight check of the system can be performed during the run-up by retarding the power on one engine to idle, and advancing power on the opposite engine until the power difference between the engines is great enough to close the switch that activates the rudder boost system. Movement of the appropriate rudder pedal (left engine idling, right rudder pedal moves forward) will be noted when the switch closes, indicating the system is functioning properly for low engine power on that side. Repeat the check with opposite power settings to check for movement of the opposite rudder pedal.

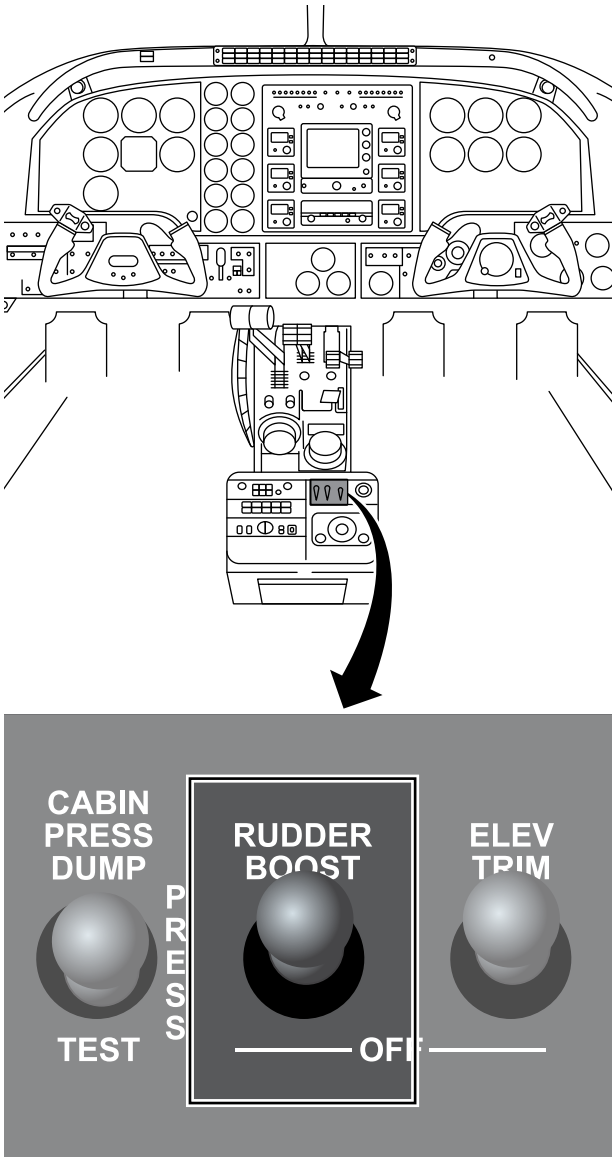


Figure 15-7. Rudder Boost Switch

QUESTIONS

1. What is the maximum allowable altitude with yaw damper inoperative?

A. No limit applies
B. 10,000 ft
C. 20,000 ft
D. 25,000 ft

2. What happens when the FLAP handle is moved from the DOWN to the APPROACH position?

A. The flaps will bypass the APPROACH position and retract fully
B. The flaps will not retract
C. The flaps will retract to the APPROACH position
D. The flaps will retract completely, then return to the APPROACH position

3. The elevator electric trim initiated by?

A. The pilot or the copilot moving either element of his PITCH TRIM switch
B. The pilot and the copilot moving both elements of their PITCH TRIM switches in the same direction simultaneously
C. Either the pilot or the copilot moving both elements of his PITCH TRIM switches simultaneously
D. Both the pilot and copilot moving either element of their PITCH TRIM switches in the same direction simultaneously

4. Why should the rudder control lock be removed prior to towing the airplane?

A. So the airplane can be steered with the rudder pedals
B. So the brakes can be applied
C. To prevent damage to the steering linkage
D. It is not necessary to remove the rudder control lock prior to towing

5. How can the rudder boost system be checked for proper operation during the engine runup?

A. Increasing the power on an engine until the rudder on the same side moves forward
B. Increasing the power on an engine until the rudder on the opposite side moves forward
C. Rudder boost operation cannot be checked during engine runup
D. Reducing power on an engine and noting that neither rudder pedal moves forward

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CHAPTER 16
AVIONICS

INTRODUCTION

This section describes the standard avionics installation and how it operates. Earlier avionics installations are also included. The manufacturers' *Pilot's Guides* and *Pilot's Operating Manual Supplements* are to be relied on for the most accurate information regarding the use of the equipment.

DESCRIPTION

This chapter explains the standard C90B avionics including the Collins Pro Line II remote mounted package, Collins EFIS-84, and Collins APS-65 Autopilot system. Also disscussed are the optional Garmin 400 and Bendix/King KLN-90B GPS systems.

The King Silver Crown II line of avionics equipment is installed on many earlier King Airs. The main units typically installed are addressed.

King Air avionics controls, and optional multifunction or navigation displays, are mounted on an isolation panel in the center of the instrument panel and are easily accessible to the pilot or copilot (Figure 16-1). Installed across the top of the panel are individual audio switches which control audio to the speakers or headphones.

Figure 16-1. NAV/COM Control Panel

AVIONICS POWER DISTRIBUTION

All avionics equipment may be turned on and off by the avionics master switch (Figure 16-2 and Figure 16-3). In the event that this switch fails, power may be restored by pulling the avionics master circuit breaker, located in the upper right-hand corner of the main circuit breaker panel (Figure 16-4).

The King Air C90A/B/GT has three avionics buses (Figure 16-5) to feed DC power to the various types of avionics equipment. To determine specifically what equipment is being fed from a specific bus or power source, refer to the wiring diagram entitled "DC Power Distribution" which is supplied with each airplane. There are, however, some general rules of thumb which usually apply.

For example:

- Items numbered one (e.g., comm 1, nav 1, etc.) are fed by the number one avionics bus, which in turn is fed from the electrical system triple-fed bus. It is important to note that in the event of a dual generator failure, the items fed by the number one avionics bus would continue to operate for a limited period of time, being fed directly by the battery.
- Items numbered two (e.g., comm 2, nav 2, etc.) are fed by the number 2 avionics bus. The number two avionics bus is fed by the left generator bus.
- Additional avionics items which are not fed by the previous buses are fed by the number 3 avionics bus. The number 3 avionics bus is fed by the right generator bus.

During a normal engine starting sequence, when a generator is brought on line, both generator bus ties close. Therefore, assuming the avionics master switch is turned ON, all avionics systems will receive power from their respective buses under normal circumstances. Also, when running equipment checks on the ground with the external power switch ON and an APU connected, all three avionics buses will be powered. In these instances, the bus ties are automatically closed.

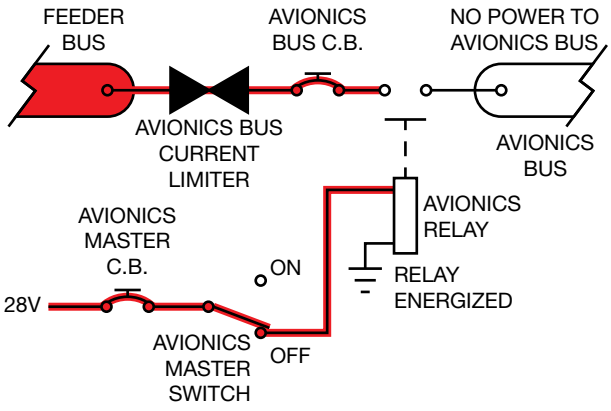

Figure 16-2. Avionics Master Switch OFF

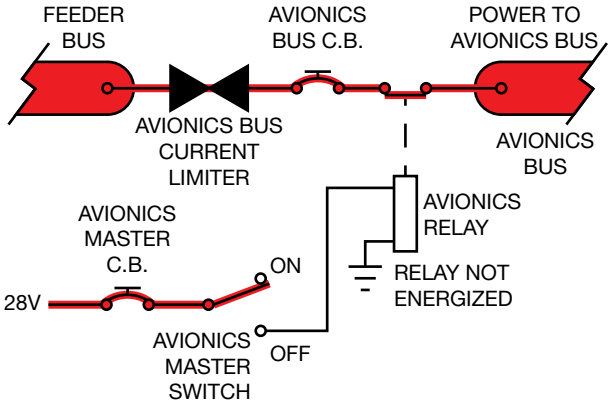

Figure 16-3. Avionics Master Switch ON

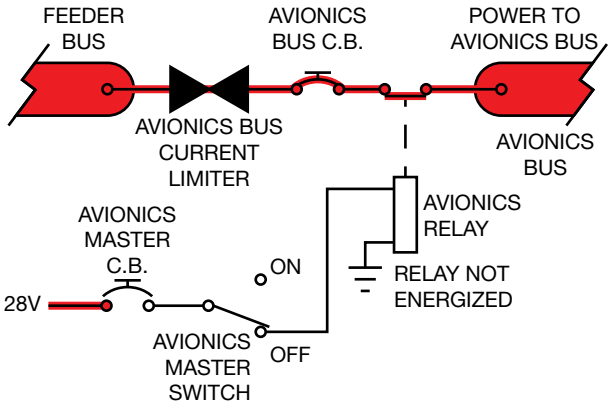

Figure 16-4. Alternate Avionics Bus Power

However, consider the need to make a quick ground check of comm 2, prior to starting engines, and without an APU connected. In this situation, manually close the bus ties with the appropriate switch located on the pilot outboard subpanel.

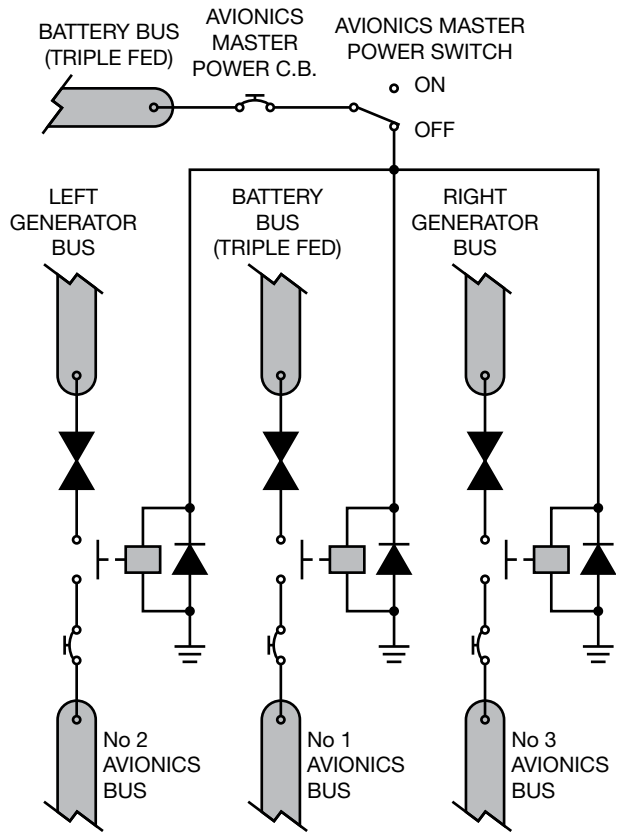

Figure 16-5. Avionics Buses

An APU should be considered essential for running avionics equipment on the ground. For electronic flight instrument system (EFIS) equipped airplanes, the avionics equipment and one of the inverters require approximately fifty amperes of current from the battery. This amount of current drain would deplete the battery in a short period of time.

Also, especially with EFIS equipment installed, it is desirable to have the avionics nose compartment doors removed to allow sufficient cross ventilation and cooling of the equipment. Particularly during practice sessions with the avionics equipment which exceed fifteen minutes in duration.

AC power is available from either of two 400 Hz inverters. Under normal circumstances, the number one inverter is fed from the left generator bus and the number two inverter is fed from the right generator bus (Figure 16-6). However, in the event that the operating inverter loses power from its appropriate bus, the inverter automatically switches over to the center bus as its power source.

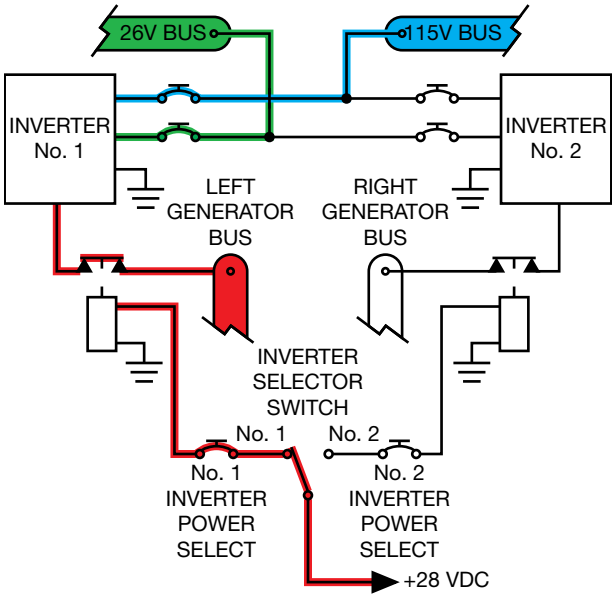

Figure 16-6. Inverter Power Supply

COLLINS EFIS-84

ELECTRONIC FLIGHT INSTRUMENT SYSTEM

The Collins EFIS-84 system is made up of four components: The Electronic Attitude Direction Indicator (EADI), Electronic Horizontal Situation Indicator (EHSI), Display Processor Unit (DPU) and the Display Select Panel (DSP). The EADI and EHSI are panel mounted, the DPU is remote-mounted and the DSP is mounted on the center pedestal. Refer to Figure 16-7 for a block diagram of how the units are interconnected. An optional configuration allows weather radar to be diplayed on the EHSI.

The EADI and EHSI, being CRT displays, are susceptible to image burn-in and should be dimmed when not in use on the ground, and just bright enough that the information can be clearly seen in flight. The EADI/EHSI DIM control knob (Figure 16-8) is located to the left of the EHSI display.

COMPONENTS

Display Processor Unit (DPU)

The Display Processor Unit processes all input from the various aircraft systems and transforms it into a video signal to be displayed on the appropriate display (EADI or EHSI).

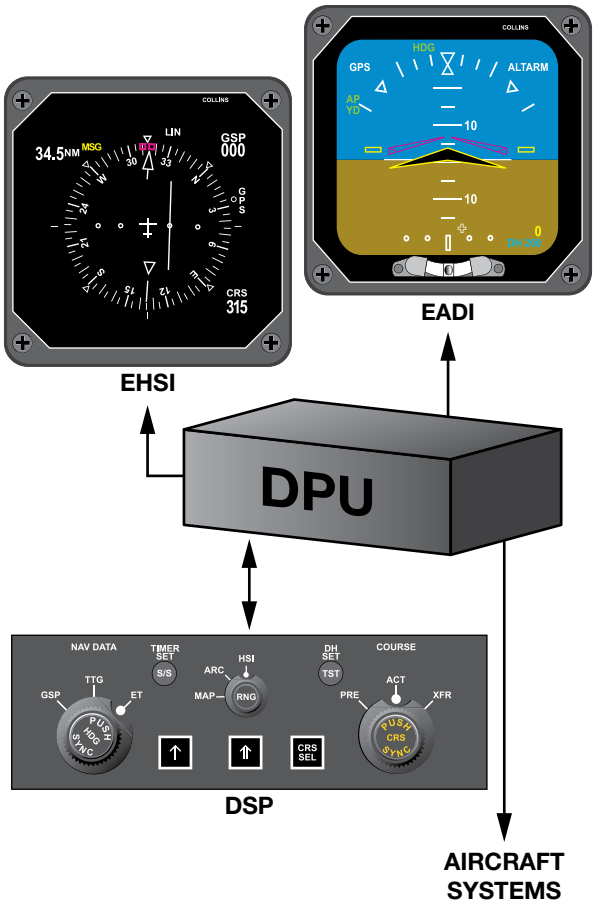

Figure 16-7. EFIS Block Diagram

Figure 16-8. EADI/EHSI DIM Control Knob

Electronic Attitude Director Indicator (EADI)

The EADI (Figure 16-9) displays aircraft attitude, Flight Director commands and annunciators, CDI and glideslope information, as well as, other information. The information is displayed through the use of a multi-colored CRT.

Figure 16-9. Electronic Attitude Director Indicator

Electronic Horizontal Situation Indicator (EHSI)

The EHSI displays the aircraft's navigation information in one of five display formats through the use of a multi-colored CRT. The five display formats are HSI, ARC, MAP and ARC or MAP with weather overlay (Figure 16-10).

Figure 16-10. Electronic Horizontal Situation Indicator

Display Select Panel (DSP)

The display select panel allows the pilot to select the display format of the EHSI, as well as, selecting various other information to be displayed on both the EHSI and the EADI. The DSP also allows the pilot to select the desired heading and course information (Figure 16-11).

EFIS AUX POWER SYSTEM

A standby EFIS power supply system (Figure 16-12) is provided to prevent the pilot EFIS displays from blanking during momentary drops in voltage. EFIS AUX POWER is intended for only short periods of use and should not be relied upon to maintain operation of the EFIS if the normal power supply should fail. Refer to the *Pilot's Operating Handbook* for testing and operation.

EADI Display

Attitude Display

A blue sky and brown ground area displays pitch attitude. This area contains the pitch scale and the aircraft symbol. The pitch scale is graduated into 5° increments up to ±15°. From ±20° up to ± 90° expanded markings are shown (10° increments) (Figure 16-13).

Roll attitude is shown by a moving sky pointer with reference to a fixed roll attitude scale along the top portion of the display. Roll attitude reference markings are shown at 0°, 10°, 20°, 30°, 45°, and 60° bank.

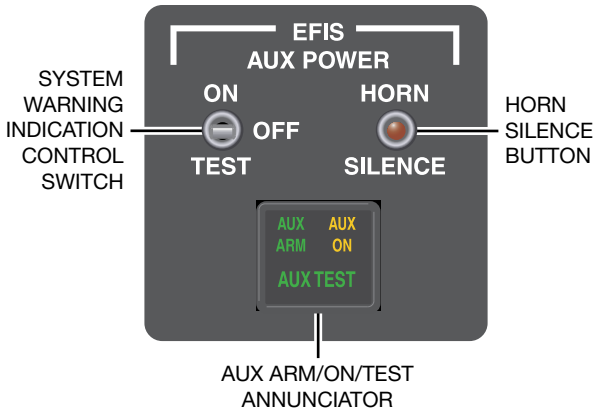

Figure 16-12. EFIS AUX Power Panel Controls

Excessive Attitude Display

An excessive attitude display (Figure 16-14) is shown when the pitch attitude exceeds 30° nose up or 20° nose down, or when the angle of bank exceeds 65° left or right. When in excessive attitudes, all information except the aircraft symbol and the attitude display are removed from view.

The display is restored once the attitude returns to a pitch of less than 25° pitch up or 15° pitch down, or 60° of bank or less. Also during excessive pitch attitudes, red chevrons will appear pointing in the direction that the airplane needs to be pitched to return to a normal attitude. There are two downward pointing chevrons and three upward pointing chevrons.

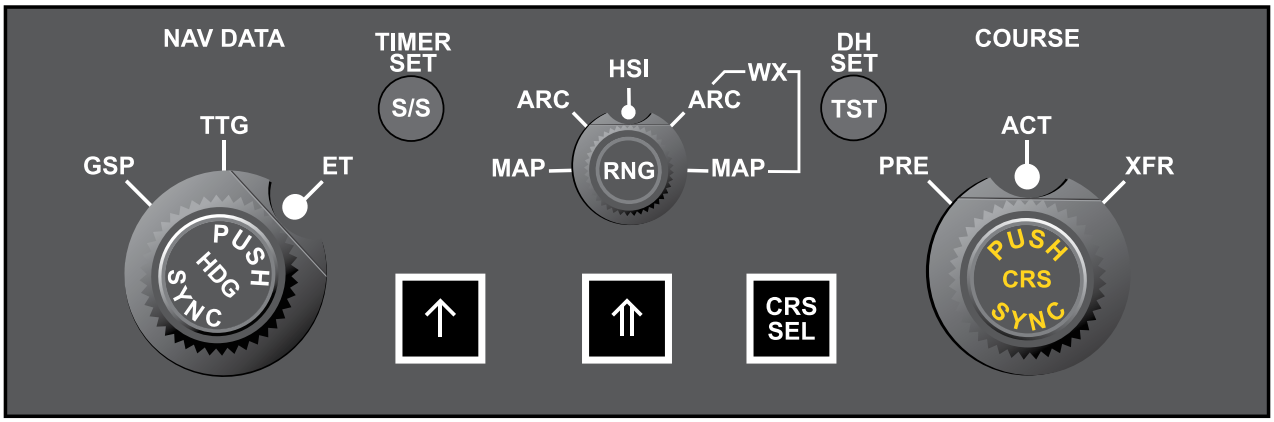

Figure 16-11. Display Select Panel

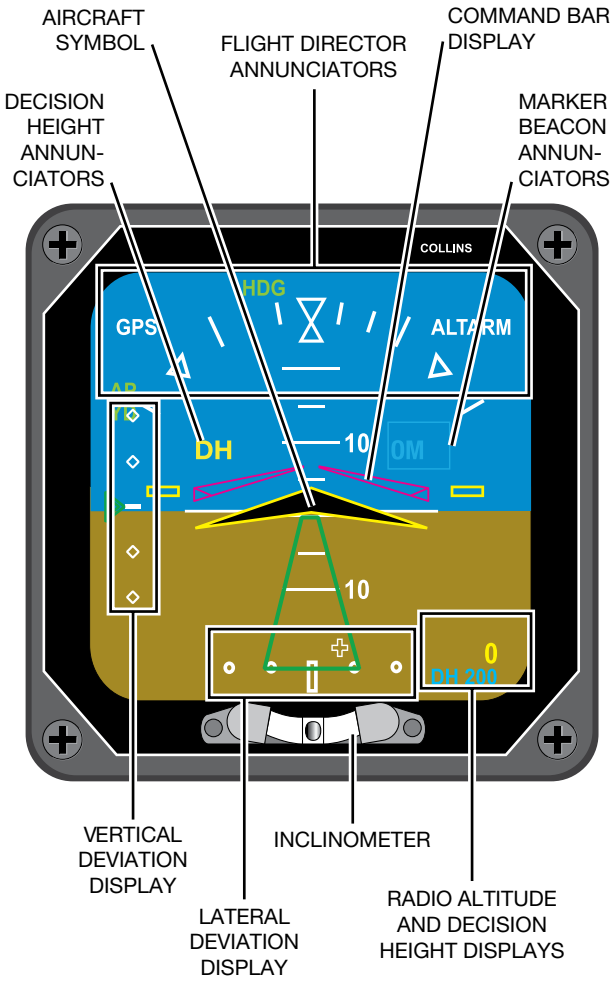

Figure 16-13. EADI Displays

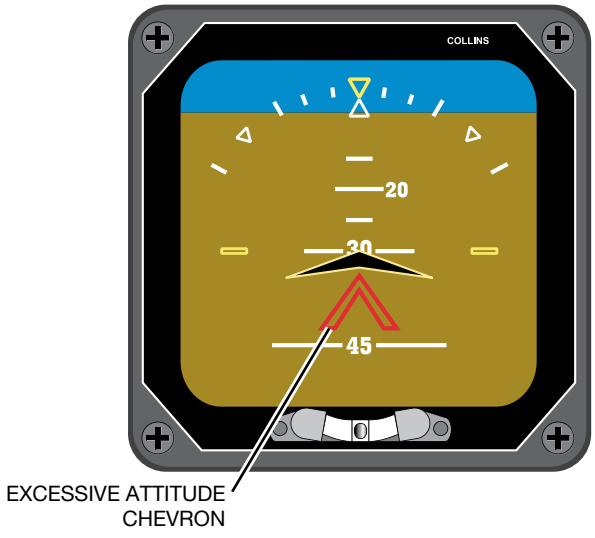

Figure 16-14. Excessive Attitude Display

Flight Director Annunciators

The Flight Director annunciators are displayed on the top portion of the display (see Figure 16-13). The lateral modes and the Autopilot and yaw damper engage/disengage annunciators are displayed in the upper left-hand corner. The vertical modes are displayed in the upper right-hand corner. The active modes, green in color, are shown towards the center. Modes that are armed, white in color, are shown along the outer edges of the display. When a new active mode is selected or automatically captured, the annunciation will flash for 5 seconds before becoming steady.

Decision Height Annunciator

The Decision Height (DH) annunciator illuminates in the upper left center portion of the attitude display (see Figure 16-13). It will flash for 5 seconds and then become steady when the decision height is reached.

Marker Beacon Display

See Figure 16-13.

Radio Altitude and Decision Height Displays

The green colored radio altitude and the cyan colored decision height are displayed in the lower right hand corner (see Figure 16-13).

Command Bar Display

The magenta colored Flight Director command bars are displayed in the center portion of the EADI when the Flight Director is being used (see Figure 16-13). The display disappears when the Flight Director is turned off or flagged, or when the aircraft is in an excessive attitude.

Aircraft Symbol

The aircraft symbol remains stationary representing the position of the aircraft relative to the pitch and roll attitude display (see Figure 16-13).

Vertical Deviation Display

A scale and pointer are located on the left-hand side of the EADI (see Figure 16-13). The scale consists of two white dots above and below a center index. The double-triangle pointer moves

up and down the scale to indicate the aircraft position relative to on-glidepath. When a sensor other than an ILS is selected, the scale and pointer are removed from view.

Lateral Deviation Display

A scale and pointer are displayed in the bottom center portion of the EADI (see Figure 16-13). The scale consists of two white diamonds on either side of a center index. The pointer moves from side to side indicating the aircraft position relative to an on-course condition. The pointer is displayed in three different fashions dependent upon which nav source is selected. GPS will display a white plus sign, VOR will display a green circle, and a localizer will display a green trapezoid that turns into a rising runway. At 200 feet radio altitude the runway symbol will begin growing in size such that at 0 feet radio altitude the top edge of the symbol is touching the bottom of the aircraft symbol. If the radio altitude is off or inoperative, the runway symbol will continue to show lateral deviation but will not expand vertically. During a GPS approach the letters APPR will appear above the left-hand portion of the scale. As the sensitivity of the CDI changes with the approach the word "SCALING" will appear above the right-hand portion of the scale.

Inclinometer

The inclinometer or slip indicator is mounted on the bottom center portion of the EADI (see Figure 16-13).

EADI Warning Flags

The warning flags will appear as the appropriate red letters bound by a red box (Figure 16-15). This does not apply to the DPU FAIL flag. The DPU FAIL flag will appear as red letters only. When an ATT, RA, FD, or DSP flag appears, it will flash for 10 seconds before becoming steady.

Display Processor Unit Flag (DPU)

In the event of a DPU failure a steady "DPU FAIL" annunciation will appear in the center of the display (see Figure 16-15). If the flag remains in view for more than 5 seconds, the entire display will disappear except for the "DPU FAIL" annunciator.

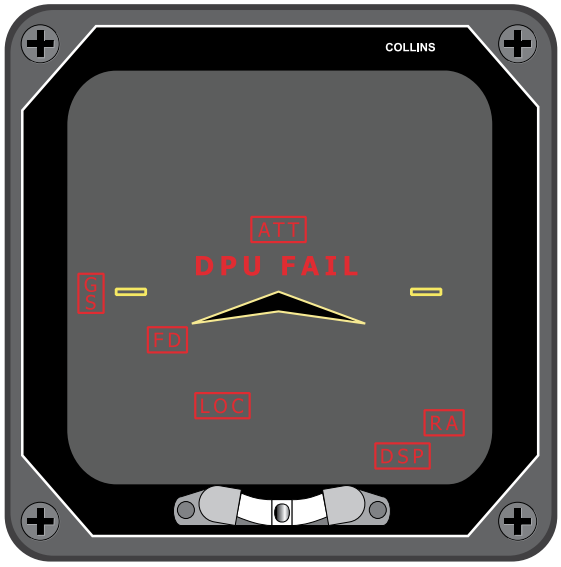

Figure 16-15. EADI Warning Flags

Display Select Panel Flag (DSP)

When a failure of the Display Select Panel is detected, The "DSP" flag will appear (see Figure 16-15). The flag will replace the DH display and the Flight Director mode annunciators will also be removed from view.

Attitude Flag (ATT)

If a failure of the attitude sensor is detected, the pitch and roll scales, roll pointer, command bars, and the pitch display will all disappear and the "ATT" flag will appear above the aircraft symbol (see Figure 16-15).

Vertical Deviation Flag (GS)

When a sensor failure of the glide slope is detected the vertical deviation scale and pointer are removed from view and are replaced by the "GS" flag (see Figure 16-15).

Lateral Deviation Flags (LOC, VOR, GPS)

When a sensor failure is detected the scale and pointer are removed from view and the appropriate letters will appear advising the pilot as to which sensor failed.

Radio Altimeter Flag (RA)

When a failure of the radio altimeter is detected the "RA" flag will appear replacing the normal RA readout (see Figure 16-15). The DH display and annunciator will be removed from view.

DISPLAY SELECT PANEL OPERATION

EHSI Format Switch

The EHSI format switch selects which of the 5 available formats to display (Figure 16-16).

HSI

The HSI format will display a full compass rose similar to a conventional HSI display (Figure 16-17).

ARC

The ARC format will display an expanded compass rose across the top of the display comprising of the upper 60° of the compass. An expanded course needle will also be displayed. An aircraft symbol will be displayed at the bottom center (Figure 16-18).

Figure 16-17. HSI

Figure 16-18. ARC

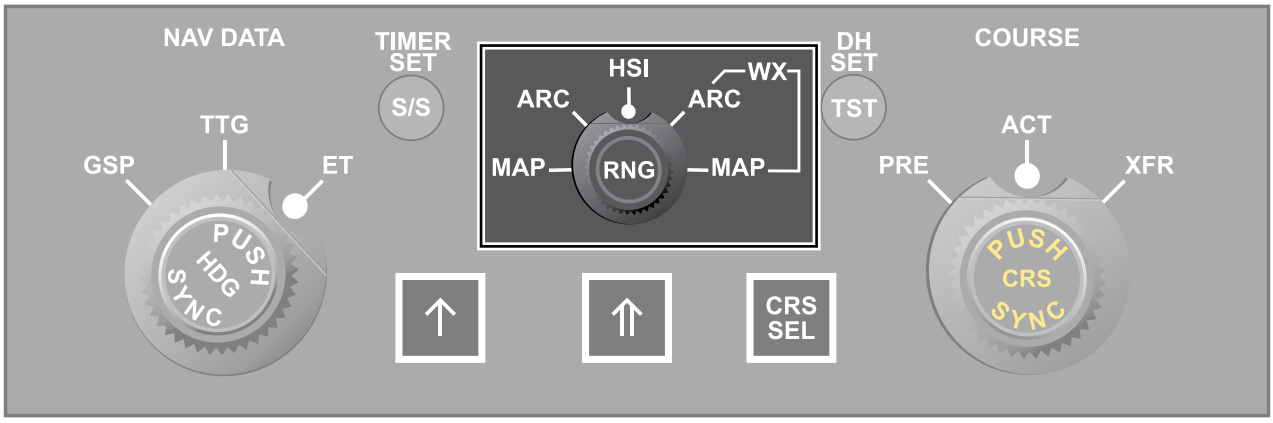

Figure 16-16. Range (RNG) Knob and Format Switch

MAP

The MAP format displays an expanded compass rose and airplane symbol, as with the ARC format, but without a course needle. The format will also display VOR and/or waypoint symbols and course lines in the proper position with respect to the airplane symbol and selected range (Figure 16-19).

Figure 16-19. MAP

ARC/WX

The ARC/WX format is the same as the ARC format with the exception that weather radar data is added to the display (Figure 16-20).

Figure 16-20. ARC Weather

MAP/WX

The MAP/WX format is the same as the MAP format with the exception that the weather radar data is added to the display (Figure 16-21).

Figure 16-21. MAP Weather

Range (RNG) Knob

The range knob (see Figure 16-16) selects the range in the ARC and MAP formats on the EHSI. A cyan arc is displayed in the center of the display representing one-half of the full-scale range. The one-half range distance is shown at the right side of the arc. The available full-scale ranges are 5, 10, 25, 50, 100, 200, 300, and 600 nm. The range knob does not have any effect in the ARC/WX and MAP/WX formats. The range buttons on the radar itself must be used in these modes.

NAV DATA Switch

The Nav Data Switch (Figure 16-11) allows the pilot to select whether Ground Speed (GSP), Time To Go (TTG) or Elapsed Timer (ET) information is displayed in the upper right hand corner of the EHSI. GSP and TTG information is derived from the active nav source (GPS or VOR). If the information is invalid, the display will be replaced with dashed lines. The elapsed timer is controlled by the TIMER SET knob, just above and to the right of the NAV DATA Switch and the red timer button to the left of the EHSI. After beginning the timer, GSP or TTG may be selected and displayed without disturbing the timer. The GSP range is 0 to 3999 knots with a 1-knot resolution and the TTG range is 0 to 409 minutes with a 1-minute resolution.

TIMER SET Knob and S/S (Start/Stop/Reset) Button

The Timer Set knob and button (Figure 16-11) are used to control the elapsed timer when it is selected with the NAV DATA Switch. If the timer is at zero, pressing the S/S button will start the count-up timer. Pressing the button a second time will stop the timer and pressing it a third time will reset it to zero. Rotating the knob will allow the pilot to preset a time to count down. Pressing the S/S button will begin the count down. Upon reaching zero, the digits will flash for 5 seconds and then change to dashes. Any time from 0 seconds to 9 hours 59 minutes may be selected. Selecting a time greater than 59 minutes 59 seconds will change the format to hours and minutes only.

DH SET (Decision Height) Knob

The DH SET Knob (Figure 16-11) allows the pilot to set a decision height to be displayed in the lower right-hand corner of the EADI. This sets the DH on the radar altimeter (RA). When the RA reaches the preset value +50 feet, the digits will flash until the DH is reached. The digits will remain steady until a RA of DH+100 feet is reached. The digits will also stop flashing and remain steady once a RA of less than 6 feet is reached. A DH annunciator is also associated with the Radar Altimeter, located in the left center portion of the display. The annunciator will flash for 5 seconds when the DH is reached and then become steady. It will turn off at a RA or less than 6 feet, DH + 100 feet or if the RA fails.

The DH SET Knob also has a RA Test function. Press the knob to begin the RA self-test. During the self-test, the RA display will count to 50 feet and the DH annunciator will illuminate if the DH SET value is greater than 50 feet.

Bearing Pointer Buttons

The Bearing Pointer buttons allow the pilot to display either a single or a double needle on the EHSI representing an RMI (Figure 16-22). Each needle has a nav source identifier near the arrow end. The possible identifiers are VOR=V, ADF=A and GPS=N. Repeatedly pressing the button will step through the nav sources available to be displayed by the appropriate needle. Each time the button is pressed and released an annunciation of the selected nav source is displayed in the lower left-hand corner of the EHSI. The bearing pointer may be removed from view by pressing and holding the appropriate button for approximately 1/2 second. The needle may then be recalled by again pressing and holding the button for approximately 1/2 second.

The possible nav sources may vary with each installation. The standard sources for the single needle (cyan in color) are VOR, ADF, GPS, and “blank”. The standard sources of the double needle (magenta in color) are ADF, GPS or “blank”. “blank” will remove the needle from the display. If the selected nav source’s reception is lost, the needle and title will turn red.

Course Select (CRS SEL) Button

The CRS SEL Button (Figure 16-11) allows the pilot to select the navigation (nav) source for the active or preset course needle. Pushing the CRS SEL button repeatedly will step through the menu of available nav sources. Once the desired navigation source is displayed in the menu, pressing the button for at least 1/2 second will activate the selected nav source. GPS is displayed as white in color and VOR or localizer (LOC) is displayed as green. The green VOR or LOC display is dependent on the NAV 1 active frequency. If desired, the pilot may choose to display a basic

Figure 16-22. Bearing Pointers

Directional Gyro (DG) by selecting the "blank" selection, denoted by a hollow rectangle, in the menu with the CRS SEL button. If no source is selected within 5 seconds of the last push of the button, while stepping through the menu, it will default to the previously active navigation source.

While in the preset mode, it is not necessary to press and hold the button for at least 1/2 second to select a navigation source. Pressing and holding the CRS SEL button while in the Active (ACT) mode without stepping through the menu will default to the VOR/LOC.

Nav source data is stored if power is lost or the system is turned off.

COURSE Switch

The Course switch position (Figure 16-11) determines whether the active (ACT) or preset (PRE) course needle is being controlled. The switch will also transfer (XFR) the preset course to the active course.

The ACT position allows the pilot to set up the active course needle (nav source and course). With the switch in the ACT position a small circle is displayed next to the nav source annunciator on the right side of the EHSI.

The PRE position allows the pilot to preset a course to be used later. With the switch in the PRE position, the small circle moves to just below its location when in the ACT position. This indicates that the PRE position is selected. The nav source "menu" will appear next to the circle. At the same time, a cyan colored dashed double course needle will appear on the EHSI. The nav source can be selected with the CRS SEL button and a course selected with the CRS knob. The needle, though preset, will act normally indicating course deviation and a To/From flag. However, the course needle is not to be used for navigation.

The spring-loaded XFR position allows the pilot to transfer the course needle in the PRE position to the ACT position. The ACT needle will then disappear and the PRE needle will change to a single needle in the appropriate color. The ACT needle is stored as the new PRE needle; however, it is removed from view. It may be recalled as the preset course at any time by moving the Course switch to the PRE position and pressing and holding the CRS SEL button for at least ½ second. If it is not held for the required ½ second, then each press of the button will resume stepping through the available nav sources.

Heading Select (HDG) Knob

The HDG knob (Figure 16-11) controls the position of the heading bug on the EHSI. Rotating the knob will change the position. The heading value is stored if power is lost or the system is turned off.

Heading Synchronization (PUSH HDG SYNC) Button

The PUSH HDG SYNC button, located on the top of the Heading Select knob (Figure 16-11), rotates the heading bug to the current aircraft heading when pushed.

Course Select (CRS) Knob

The CRS knob (Figure 16-11) changes the position of the course needle on the EHSI. The knob will control either the active or preset course needle, whichever is selected with the Course switch. The course value is stored if power is lost or the system is turned off.

Course Direct-To (PUSH CRS DIRECT) Button

The PUSH CRS DIRECT button (Figure 16-22) will rotate the course needle to a "direct to" course to the VOR station. This function will only work if VOR is the navigation source selected.

EHSI DISPLAY

Compass Display and Heading Sensor Type

The compass display (Figure 16-23) is a typical 360° compass rose with letters at the cardinal headings, numbers every 30°, and reference marks around the outer edge every 45°. The 45° reference markings can be very useful in procedure turns. The current aircraft heading is displayed under the lubber line. During normal operations, no heading source annunciation is present next to the lubber line. If the pilot sets the compass system in the DG or "free" mode the DG annunciation will be displayed. The compass

is shown as an expanded compass sector with a ±30° segment shown at the top of the display in the ARC and MAP formats. The airplane symbol is also placed at the bottom of the display with the needles and heading bug rotating around the symbol.

Selected Heading Display

The selected heading is shown by two adjacent magenta rectangles (the heading bug, Figure 16-23) with respect to the compass card and is controlled by the HDG knob on the Display Select Panel. Pressing the PUSH HDG SYNC portion of the HDG knob will rotate the heading bug to the current heading. In either the ARC or MAP format, when the heading bug is selected to a heading not displayed on the segment, a digital readout of the heading will appear above the appropriate end of the segment. As the heading passes through 180° of the current heading the readout will switch to the other side. A magenta line will also connect the bug and the airplane symbol in this situation. The magenta line will appear continuously if no active course is selected while in the ARC or MAP format.

LNAV Annunciation

When LNAV (GPS) is the active selected course, MSG, LIN and APPR annunciations may appear as received from the GPS (Figure 16-23). LIN (white in color) appears, continuously, to the right of the lubber line indicating that the CDI is in linear deviation and not angular deviation as with a VOR. MSG (yellow in color) appears to the left of the lubber line when the GPS has a message. APPR (white in color) appears in the upper left center of the compass rose and the lower left center of the EADI indicating that the GPS approach has become active and is scaling to ±0.3 nm. The change of scaling is also indicated by a SCALING annunciation in the right center portion of the EHSI and EADI displays. The SCALING annunciation will also appear when the approach arms and is scaling to ±1 nm.

Active Selected Course Display

As with a typical HSI, the aircraft position relative to the active selected course is shown by the relationship of the aircraft symbol to the solid single course arrow with respect to the compass card. The active

Figure 16-23. EHSI Displays (1 of 4)

selected course is numerically shown in the lower right hand corner of the display with the letters CRS above the readout (Figure 16-24). The active nav source is displayed at the right center of the display. The nav source is changed, as described in the operation of the DSP, by the CRS SEL button and the COURSE switch in the ACT position. The color of the active selected course needle is dependent on the nav source. A green needle indicates VOR or LOC and a white needle indicates GPS. A line drawn through the appropriate waypoint symbol in the MAP format indicates the active selected course. If the "to" waypoint is off-screen, a course line with an arrow pointing in the proper direction will indicate where the waypoint is located. The waypoint name will also be displayed next to the course line. If the waypoint comes on-screen it will flash alerting the pilot to the fact that it is on-screen. The waypoint name will flash if the waypoint goes off-screen. The course line will be solid on the "to" side of the waypoint and dashed on the "from" side.

Preset Course Display

The preset course is shown in the same manner as the active course except for the fact that the needle is a double-dashed line in appearance and is always cyan in color (Figure 16-24). The needle will only appear

when a course has been preset. The preset course is numerically shown in the same fashion as the active course except that the letters PRE replace the letters CRS above the readout. A cyan course line drawn through a waypoint symbol indicates the preset course while in the MAP format. If the preset nav source is a VOR, the symbol will be an octagon, GPS will be a plus sign, and localizer will not display a course line or a way-point symbol. The line, on the "to" side of the waypoint, will be solid and dashed on the "from" side. The preset course is not to be used for navigation.

To/From Arrow

The To-From arrow (Figure 16-24) indicates that the selected course is either "to" or "from" the selected navaid. The arrow will always show "to" when using the GPS unless auto-sequencing has been disabled. Also, the arrow will flash when the waypoint alert has been activated when the GPS is in use. The arrow will be removed from view if a localizer has been selected. The words "TO" or "FROM" at the lower right-center of the display indicate "to" or "from" while in the ARC format (see Figure 16-18 or Figure 16-20). An arrow pointing in the proper direction on the course line indicates "to" or "from" while in the Map mode (see Figure 16-19 or Figure 16-21).

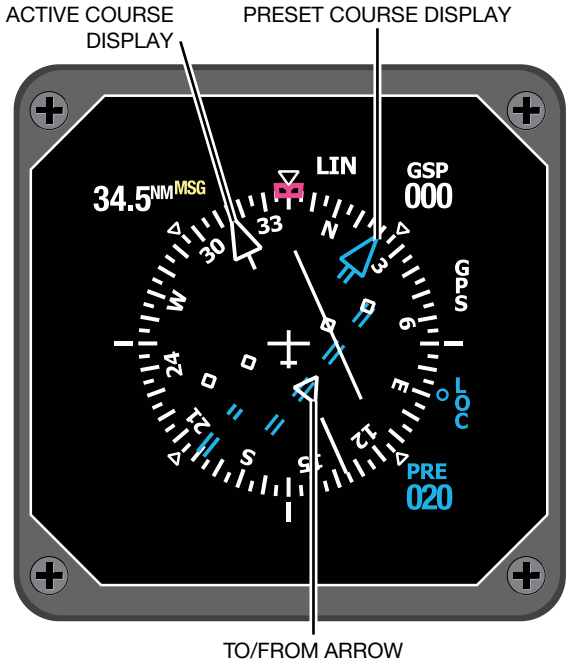

Figure 16-24. EHSI Displays (2 of 4)

Weather Annunciation

Weather annunciation in the ARC-WX (see Figure 16-20) or MAP-WX (see Figure 16-21) format is shown by the same type radar echo display shown on the radar unit. The display consists of four colors: green, yellow, red and magenta. The range must be changed on the radar unit and cannot be changed with the RNG knob on the DSP.

The target alert feature provides a target alert "window" from 50 to 150 nm range and ±15° of the aircraft heading regardless of the range selected on the radar unit. The target alert notifies the pilot when a return of contour strength is detected within the target alert "window". The TGT button on the radar unit selects the target alert. A boxed letter T, colored yellow, to the right of the lubber line indicates selection of the mode. The annunciator will flash when a target is identified and the radar is not in the OFF, STBY, or MAP mode.

Data Display

The NAV DATA switch on the DSP selects whether ground speed (GS), time-to-go (TTG), or elapsed time (ET) is displayed in the upper right-hand corner of the display (Figure 16-25). The information is displayed in white letters and numbers.

Waypoint or DME Identifier and Waypoint Alert

The VOR, LOC or GPS identifier is displayed in white letters in the upper left-hand corner of the display (Figure 16-25). Flashing of the "to" arrow indicates waypoint alert.

Distance Display

The distance to the VOR, Waypoint or LOC/DME is displayed numerically just below the waypoint identifier (Figure 16-25). The letters "NM" indicating that the distance is in nautical miles follows the digits. Resolution of the readout is 0.1 nm below 100 nm and 1nm above 100 nm. If the associated DME has been placed in hold a yellow-colored "H" will appear adjacent to the

readout. A red-colored "D" will appear, along with dashes replacing the readout, if the DME information becomes unreliable.

Active Course Lateral Deviation Bar (CDI)

The center portion of the active course needle moves laterally with respect to the course arrow to indicate lateral deviation from the course. A deviation scale is

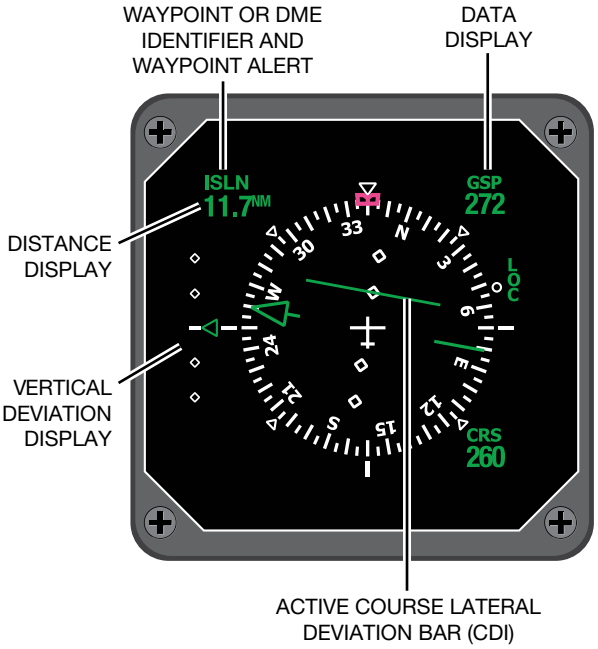

Figure 16-25. EHSI Displays (3 of 4)

indicated by two white dots displayed on either side of the course needle (Figure 16-25). The CDI operates in the same fashion as the CDI on a typical HSI. If a localizer frequency is active, LOC is selected as the active nav source, and the aircraft heading is more than 105° from the selected course, a "B/C" (back-course) annunciation will appear to the right of the lubber line (Figure 16-26). This annunciation indicates that the course deviation has automatically reversed to compensate for reverse sensing. This will provide correct CDI deviation when executing a localizer B/C approach. The course deviation bar may be partially or completely out of view with some selected courses while in the ARC format. The MAP format does not show a course needle. The course will still be digitally read out in the lower right-hand corner, however. Approaches in the MAP format are not authorized. The deviation, in all formats, is displayed at the bottom of the EADI.

Vertical Deviation Display

A scale and pointer, located on the left-hand side of the display (see Figure 16-25), indicate deviation from the glideslope. The scale consists of two white dots above and below a white colored center index. The scale operates in the same fashion as a glideslope indicator on a typical HSI. When a back-course localizer is detected the glides-lope scale is removed from view.

EHSI WARNING FLAGS

When a HDG, DSP, or bearing pointer flag appears (Figure 16-27), it will flash for 10 seconds before becoming steady. The other flags will not flash. All flags, except for the distance and data flags, are red in color.

Distance Flag (----)

If invalid distance data is received, the distance digits and the "NM" annunciation are replaced by dashes that are the same color as the active selected course (Figure 16-27). The dashes or digits will disappear if the DME is inactive (non-DME VOR).

Figure 16-26. EHSI Displays (4 of 4)

Data Flags

If the time-to-go or groundspeed is invalid, the digits will be replaced with dashed that are the same color as the active selected course. The digits or dashes will be removed from view if the data is inactive.

Heading Flag (HDG)

If a failure of the heading system occurs, the letters HDG, bounded by a red box, will appear to the left of the lubber line (Figure 16-27).

Display Processor Unit Flag (DPU)

In the event of a DPU failure a steady "DPU FAIL" annunciation will appear in the center of the display (Figure 16-27). If the flag remains in view for more than 5 seconds, the entire display will disappear except for the "DPU FAIL" annunciator.

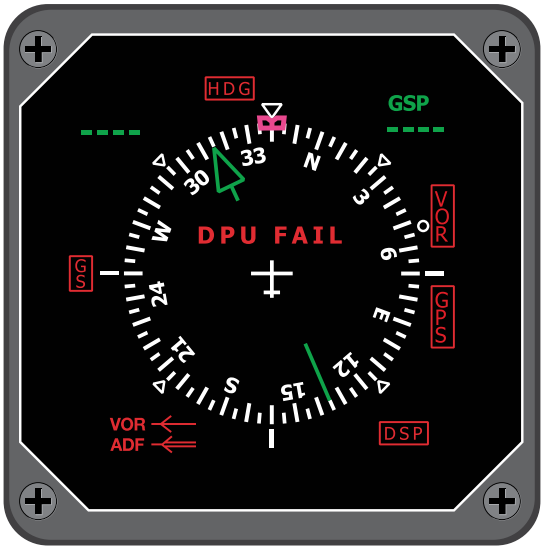

Figure 16-27. EHSI Warning Flags

Display Select Panel Flag (DSP)

When a failure of the Display Select Panel is detected, The "DSP" flag will appear in the lower right-hand corner of the display, replacing the digital course readout (Figure 16-27). The EHSI will continue to operate in the last mode selected prior to the failure.

Vertical Deviation Flag (GS)

When a failure of the glide slope is detected the "GS" flag, bounded by a red box, will appear replacing the Vertical Deviation Scale (Figure 16-27).

Active Selected Course Flag

When a sensor failure is detected the course sensor annunciator will become red in color and be bounded by a red box (Figure 16-27). If a VOR is active, the CDI will center, and the "to/from" indicator will be removed from view. Nothing else will be removed from view. If the selected nav source is GPS, the course arrow will freeze and the "to/from" indicator will be removed from view. If waypoint symbols are being displayed they, and the course lines, will be removed from view, as well.

Preset Course Flag (HSI Format)

A nav source failure, while selected as a preset course, is indicated by the annunciator turning red and being bounded by a red box and the preset course being removed from view.

Preset Course Flag (ARC and MAP Formats)

A nav source failure, while selected as a preset course, is indicated by the annunciator turning red and being bounded by a red box and the preset course being removed from view (ARC format). The waypoint symbol (ARC format) or waypoint symbols and course lines (MAP format) will also be removed from view.

LIMITATIONS

1. The Collins *Pilot's Guide for the EFIS84 Electronic Flight Instrument System (2-Tube)* must be immediately available to the crew.
2. The copilot attitude indicator and navigation instrument must be operational for takeoff.
3. The pilot EADI and EHSI must be operational in the NORMAL mode for takeoff.
4. Instrument approaches in the MAP mode not authorized.

COLLINS APS-65H

FLIGHT DIRECTOR AND AUTOPILOT

The Collins APS-65H system is made up of two distinct entities: The Flight Director and the Autopilot. The Flight Director can be operated with or without the Autopilot engaged. The Autopilot, however, must have the Flight Director to operate in more than a two-dimensional mode (Attitude Hold Mode). There are two types of modes for the Flight Director: Lateral and Vertical. As a rule, the lateral modes are on the top row of buttons on the Flight Control Panel (FCP) with the exception of the last button and the vertical modes are on the bottom row. All Flight Director modes, except for Go-Around, are selected on the FCP (Figure 16-28). The Flight Director Modes are annunciated on the FCP and the Electronic Attitude Director Indicator (See Figure 16-28 and Figure 16-29).

Pressing a mode button, after it has already been selected, will cancel that mode. Doing so will cause the Flight Director to revert back to normal Pitch Hold mode for the vertical modes and to Heading mode for the lateral modes, except for pressing the heading button where it will revert back to Roll Hold mode. The object to flying with the Flight Director, without the Autopilot engaged, it to position the aircraft such that the airplane symbol on the EADI is "tucked up" tightly inside the Command Bars (see Figure 16-29). When the Autopilot is engaged, the Autopilot servos actuate the flight controls to maneuver the airplane as commanded by the Flight Director.

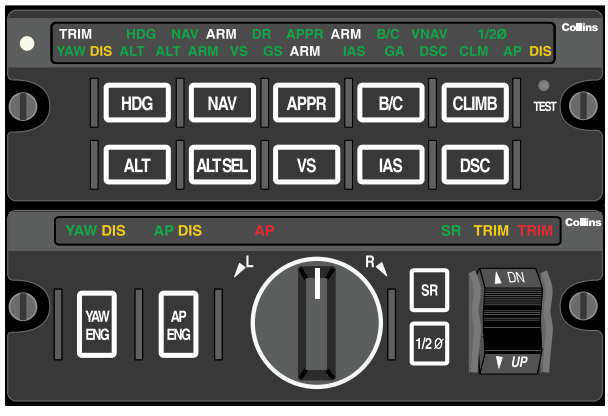

Figure 16-28. APS-65H Flight Director and Autopilot System

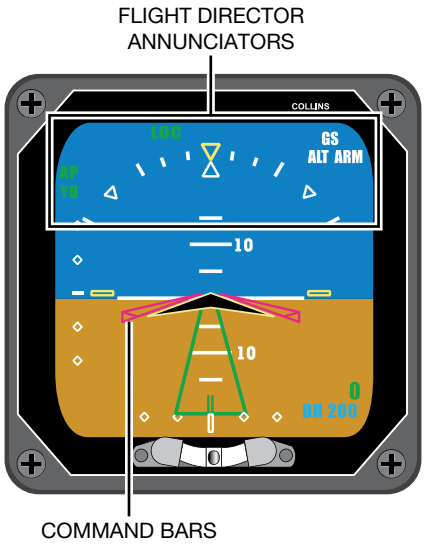

Figure 16-29. Flight Director Annunciators and Command Bars

MODES OF OPERATION

Flight Director

HDG (Heading)

The HDG mode of operation will instruct the command bars to maintain the selected heading, as selected by the heading bug on the Electronic Horizontal Situation Indicator (EHSI).

NAV (Navigation)

The NAV mode will capture and then track the selected course on the EHSI (VOR, LOC or GPS). The mode will ARM when the button is first pushed and will then capture the course once the aircraft is maneuvered to within a standard rate turn of being on course.

APPR (Approach)

The APPR mode will capture a course in the same fashion as the NAV mode, as well as, capture a vertical course (i.e. glide slope). The APPR mode is more sensitive, so as to prevent "chasing" of the course during an approach.

BC (Back Course)

The BC mode should be used if a localizer back course approach is to be flown. Pressing the BC button will command the Flight Director to fly the back course portion of the localizer rather than the front course. Pressing the BC button will automatically arm the approach mode; however, the glide slope receiver will be disabled.

CLIMB

The CLIMB mode will climb the aircraft at 150 KIAS up to 10,000 feet and then decrease at a rate of 2 KIAS per 1,000 feet there after. If the aircraft is below the appropriate airspeed, the aircraft will maintain the present pitch attitude until that airspeed is attained. If the mode is selected above 10,000 feet, the Flight Director will command a climb at the appropriate airspeed as per the cruise climb schedule in the *Pilot's Operating Manual.* CAUTION: Operation of the Climb mode above 10,000 feet in icing conditions may result in airspeeds below the minimum flight in icing conditions airspeed of 140 KIAS.

ALT (Altitude)

The ALT mode will maintain the altitude at which the aircraft is located at the moment the mode is selected. If a new barometric pressure setting is input, the Flight Director will automatically adjust to retain the desired altitude. CAUTION: The Flight Director will not reference airspeed possibly resulting in a stall or an overspeed.

ALT SEL (Altitude Select)

The ALT SEL mode will arm or disarm the pre-selected altitude as selected in the altitude pre-select head. The Autopilot will then capture and hold the altitude with the ALT mode.

IAS (Indicated Airspeed)

The IAS mode will maintain the indicated airspeed present at the moment the mode is selected.

VS (Vertical Speed)

The VS mode will maintain the present vertical speed as displayed on the VSI at the moment the mode is selected. CAUTION: The Flight Director will not reference airspeed possibly resulting in a stall or an overspeed condition.

DSC (Descend)

The DSC mode will begin and maintain a descent at a rate of 1500 feet per minute. CAUTION: The Flight Director will not reference airspeed possibly resulting in an overspeed condition.

PITCH SYNC / CWS

The Pitch Synchronization/Control Wheel Steering button, located on the pilot and copilot yokes, will synchronize the command bars to the current pitch, altitude, bank, vertical speed or indicated airspeed dependent on which mode has been selected. The Pitch Sync is operational with or without the Autopilot engaged. The Control Wheel Steering (CWS) function of the button acts as a momentary interrupt for the Autopilot. Depressing the button will momentarily take the Autopilot offline allowing the pilot to maneuver the airplane as desired. Once the button is released, the Autopilot will then hold the new attitude dependent upon what modes are selected on the Flight Director.

G/A (Go-Around)

The Go-Around mode is used to transition the aircraft from the approach to the missed approach. The G/A button, located on the left power lever, activates this mode. The mode will select a +7° pitch and wings level attitude removing any active Flight Director mode. The Autopilot, but not the yaw damper, will disengage when the G/A mode is activated. The G/A mode is disengaged by engagement of the Autopilot, the Pitch SYNC/CWS button, or any other mode selection. Engaging the Autopilot will cause the Flight Director to revert to the Attitude Hold mode.

Autopilot

YD (Yaw Damper)

The yaw damper is an electrical input into mechanical actuators attached to the rudder cables designed to dampen any oscillations about the vertical axis. The yaw damper may be used independently from the Autopilot but must be operational for the Autopilot to function. The yaw damper will engage automatically if the Autopilot is engaged. CAUTION: The yaw damper should

be OFF for takeoff and landing as it may interfere with the directional control of the aircraft, especially in a crosswind.

AP (Autopilot)

The Autopilot button will engage the Autopilot and yaw damper. The Autopilot, when engaged, will follow the commands of the Flight Director. Engaging the Autopilot, with no modes selected on the Flight Director, will result in the default Attitude Hold mode.

SR (Soft Ride)

The Soft Ride mode will dampen the corrections of the Autopilot, in turbulence, to maintain the selected modes. The Soft Ride mode is disengaged by the Approach mode.

1/2 Ø (Half Bank)

The Half Bank mode will limit the Flight Director to one-half the normal bank angle for the situation. This mode gives passengers the appearance of a smoother ride, as turns are less noticeable. Capture of the NAV or APPR modes will cancel the 1/2 Ø mode. The mode can then be reselected, if desired.

Turn Control

The Turn Control Knob allows the pilot to turn the aircraft without the use of the HDG mode. The knob is turned left or right to begin a turn to the left or the right. The angle of bank is proportionate to the amount of deflection left or right of the knob, up to the maximum bank angle of 30°. Use of the Turn Control will cancel any lateral mode selected, except APPR. The Autopilot may be engaged with the Turn Control out of the center detent, but the control must be returned to center before it will be active again.

Vertical Control Switch

The Vertical Control switch is used to incrementally adjust certain vertical modes of the Flight Director. The switch can continuously adjust the pitch reference while in the Pitch Hold mode. Pushing and holding the Vertical Control switch disengages the current vertical mode, the Flight Director will enter the Pitch Hold mode, and the pitch reference will change at a rate of 1°/second.

Each momentary actuation (up or down) will result in the following adjustments (Table 16-1).

Table 16-1. MOMENTARY ACTUATION ADJUSTMENTS

MODE	INCREMENT
Pitch Hold	±1/2 Degree
ALT	±25 Feet
IAS	±1 KIAS
VS	±200 FPM
DSC	±200 FPM
CLIMB	±1 KIAS

Attitude Hold, Roll Hold, Pitch Hold Modes

The Attitude Hold mode is selected when the Autopilot is engaged with no Flight Director modes selected. If a vertical mode is selected with no lateral mode selected then Roll Hold will be active. Likewise, if a lateral mode is selected with no vertical mode selected, then Pitch Hold will be active. If the Autopilot is engaged with no modes selected, then the Attitude Hold mode will be active.

FLIGHT DIRECTOR OPERATION

Heading Mode (HDG)

Set the heading bug to the desired heading on the EHSI. Press the HDG button (Figure 16-30).

Figure 16-30. Heading Mode

The Flight Director command bars will command a turn to capture and hold the selected heading. Bank angles, while in the HDG mode, are limited to ±25°.

Altitude Hold Mode (ALT)

To Hold an Altitude:

1. Maneuver the airplane to the desired altitude.
2. Press the ALT button. The Flight Director command bars will command pitch changes to hold the new altitude.
3. If a new barometric pressure is input, the Flight Director will automatically adjust to maintain the desired altitude (Figure 16-31).

Figure 16-31. Altitude Hold Mode

To Change Altitude:

1. Press and hold the CWS/Pitch SYNC button.
2. Maneuver to the desired altitude.
3. Release the CWS/Pitch SYNC button.
4. The Flight Director will now hold the new altitude.
5. Incremental changes of ±25 feet are made by momentarily pressing the Vertical Control switch.

NOTE

An alternate method of making adjustments to the altitude is to use the method as described per ALT SEL.

Altitude Select (ALT SEL)

ALT SEL can be armed by either pressing the ALT SEL button or by setting a new altitude in the Altitude Preselect Controller (Aircraft LJ1353 and after) (Figure 16-32).

To Capture an Altitude:

1. Set the new altitude in the Altitude Preselect Controller. The ALT ARM annunciator illuminates, indicating that the selected altitude has been armed.
2. Select a vertical mode to intercept the selected altitude.
3. At 1000 feet prior to the selected altitude, the ALT ALERT annunciators on the altitude preselector, the pilot altimeter, and the copilot panel illuminate and an aural alert sounds. At 200 feet or 300 feet* prior to the selected altitude, the annunciators extinguish
4. Upon reaching the calculated capture point, the ALT ARM annunciator will extinguish and the ALT annunciator will begin flashing as the altitude is being captured. Upon capture of the altitude, the ALT annunciator will be illuminated steady.

*200 feet on LJ-1567 and after 300 feet on airplanes prior to LJ-1567.

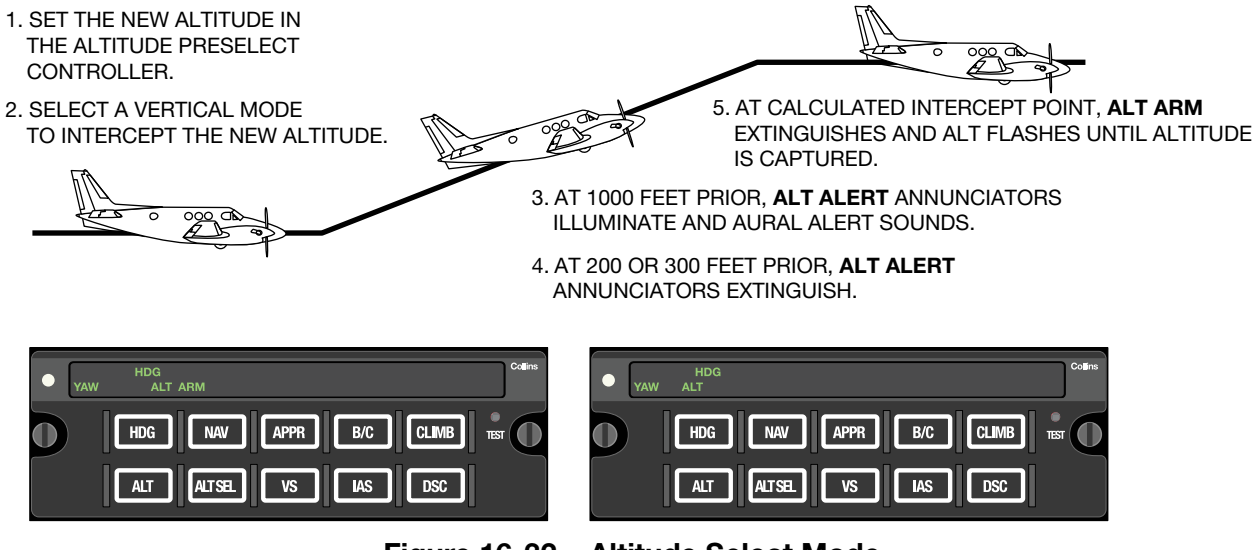

Figure 16-32. Altitude Select Mode

NOTE

Pressing the Pitch SYNC/CWS button or changing the selected altitude while the ALT annunciator is flashing (capturing) will cancel the mode and cause the Flight Director to revert back to PITCH HOLD mode.

Climb Mode (CLM)

To Select the Climb Mode:

1. Press the CLM button
2. Airplane will begin to climb at 150 KIAS up to 10,000 feet then decrease at a rate of 2 KIAS / 1000 feet.
3. Incremental changes of ±1 KIAS are made by momentarily pressing the vertical control switch (Figure 16-33).

NOTE

The PITCH SYNC is not operative in CLM mode. The rate of climb is dependent on excess power. Some conditions may result in no rate of climb.

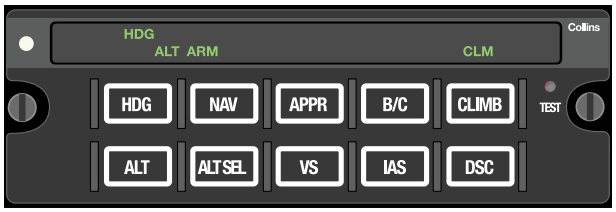

Figure 16-33. Climb Mode

Descend Mode (DSC)

To Select the Descend Mode:

1. Press the DSC button.
2. Flight Director will smoothly command a nose down attitude to begin a descent at a rate of 1500 feet per minute.
3. Incremental changes of ±200 FPM may be made by momentarily pressing the vertical control switch (Figure 16-34).

NOTE

Another method to adjust vertical speed is to press and hold the Pitch SYNC/CWS button while maneuvering the airplane to the desired vertical speed. Once at the new vertical speed, release the button. The Flight Director will maintain the new vertical speed.

Figure 16-34. Descend Mode

Vertical Speed Hold (VS)

To hold a Vertical Speed:

1. Maneuver to the desired vertical speed.
2. Press the VS button. The Flight Director will command pitch changes to maintain the desired vertical speed.
3. Incremental changes of ±200 FPM may be made by momentarily pressing the vertical control switch (Figure 16-35).

NOTE

An alternate method of making adjustments to the vertical speed is to press and hold the Pitch SYNC/CWS button while maneuvering the airplane to the desired vertical speed. Once at the new vertical speed, release the button and the Flight Director will then maintain the new vertical speed.

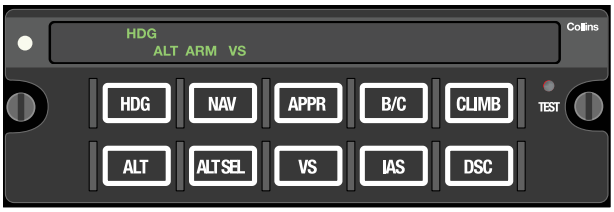

Figure 16-35. Vertical Speed Mode

Indicated Airspeed Hold (IAS)

To Hold an Airspeed:

1. Maneuver to desired airspeed.
2. Press the IAS button. The Flight Director command bars will command pitch changes to hold the new airspeed.
3. Incremental changes of ±1 KIAS may be made by momentarily pressing the vertical control switch (Figure 16-36).

NOTE

An alternate method of making adjustments to the indicated airspeed is to press and hold the Pitch SYNC/ CWS button while maneuvering the airplane to the desired indicated airspeed. Once at the new indicated airspeed, release the button and the Flight Director will then maintain the new indicated airspeed.

Figure 16-36. Indicated Airspeed Mode

Navigation Mode (NAV)

1. Tune the desired frequency into NAV 1 or select the desired waypoint in the GPS, whichever will be used for navigation.
2. Set the desired course on the EHSI.
3. Set the heading bug to a heading that will give the desired intercept angle.
4. Press the NAV button. The NAV ARM annunciator on the FCP will illuminate indicating that the NAV mode is now armed. If HDG was not already selected, selecting the NAV Mode will automatically activate HDG if the airplane is outside of the computed capture point.
5. At the computed capture point: HDG cancels, NAV ARM changes to NAV, and the Flight Director commands a turn to intercept and track the selected course (Figure 16-37).

NOTE

For optimum operation, select an intercept angle that will allow the system to maintain straight and level flight in the NAV ARM mode for a minimum of 30 seconds.

The maximum recommended capture angle for NAV mode is 90°.

As the airplane approaches the VOR, the Autopilot will enter the Dead Reckoning (DR) mode. While in the DR mode, course changes of 30° or less may be made with the course knob. Course changes of more than 30° must be made using the HDG mode to set up a new NAV capture.

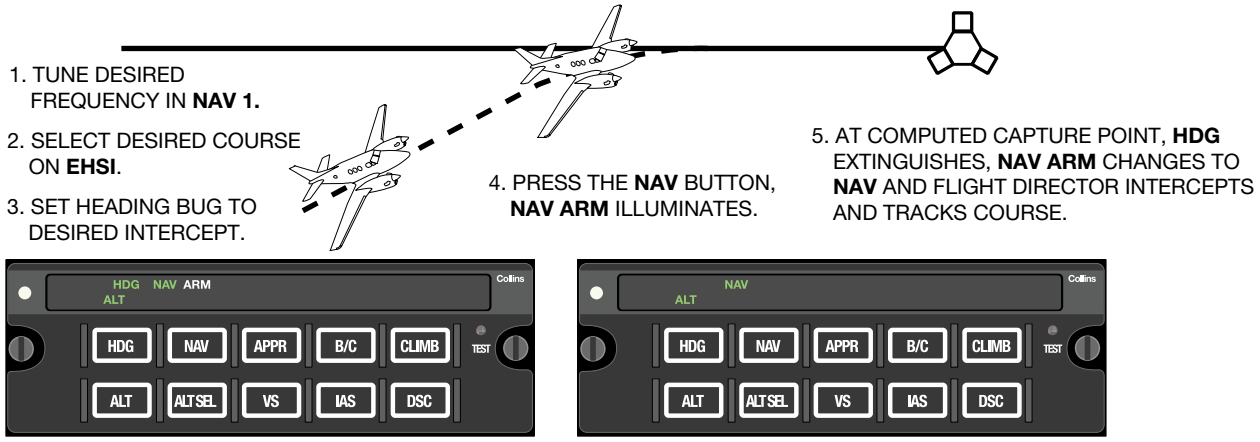

Figure 16-37. Navigation Mode

Approach Mode (APPR)

VOR Approaches

1. Tune the desired frequency into NAV 1.
2. Set the desired course on the EHSI.
3. Set the heading bug to a heading that will give the desired intercept angle.
4. Press the APPR button. The APPR ARM annunciator on the FCP will illuminate indicating that the APPR mode is now armed. If HDG was not already selected. Selecting the APPR Mode will automatically activate HDG if the airplane is outside of the computed capture point.
5. At the computed capture point: HDG cancels, APPR ARM changes to APPR, and the Flight Director commands a turn to intercept and track the selected course.
6. Upon reaching the Final Approach Fix select a vertical mode, as required, to begin the descent (Figure 16-38).

NOTE

The maximum recommended capture angle for APPR mode is 60°.

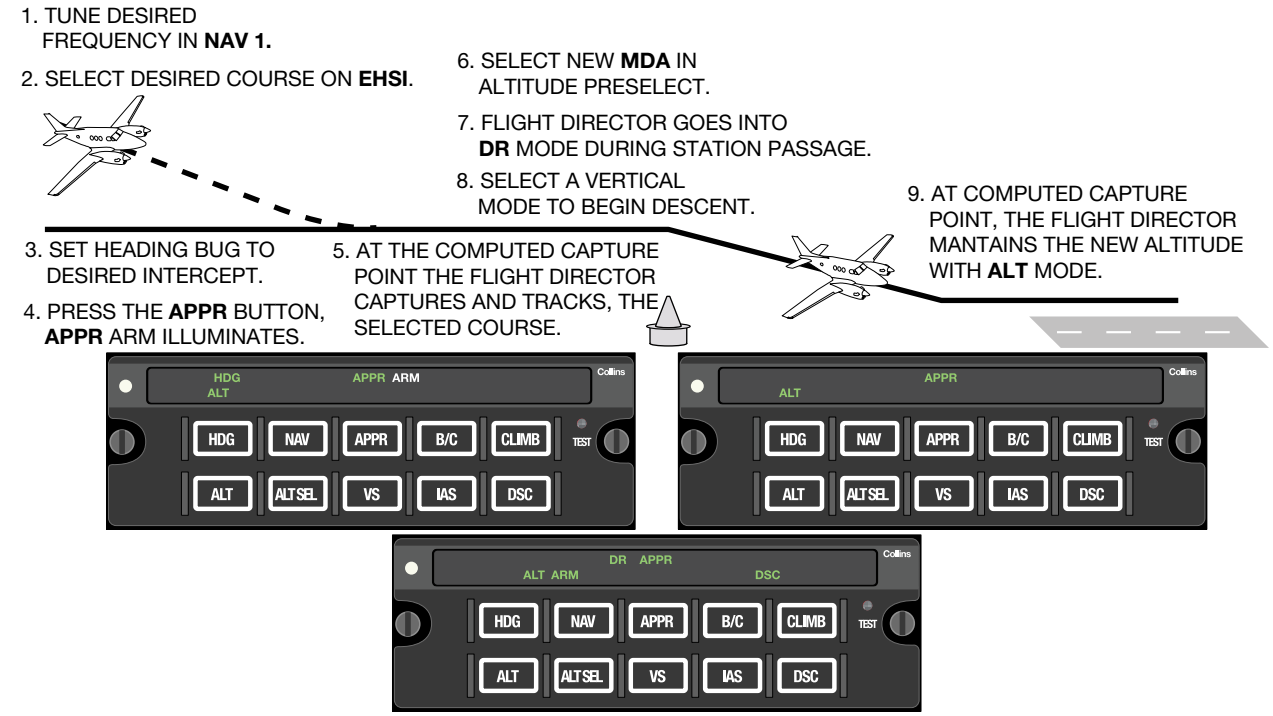

Figure 16-38. VOR Approach Mode

During a VOR approach, the Autopilot will automatically go to DR (Dead-Reckoning) mode if a VOR station is crossed. If the VOR approach crosses the VOR, the HDG mode should be used for station passage and recapturing the outbound course.

Use of the VOR Approach mode is limited to within 10 nm of the VOR Station.

Localizer Approaches

1. Tune the desired frequency into NAV 1.
2. Set the localizer front course on the EHSI.
3. Set the heading bug to a heading that will give the desired intercept angle.
4. Press the APPR button. The APPR ARM annunciator on the FCP will illuminate indicating that the APPR mode is now armed. If HDG was not already selected. Selecting the APPR Mode will automatically activate HDG if the airplane is outside of the computed capture point.
5. At the computed capture point: HDG cancels, APPR ARM changes to APPR, and the Flight Director commands a turn to intercept and track the localizer.
6. Upon reaching the Final Approach Fix select a vertical mode, as required, to begin the descent (Figure 16-39).

NOTE

The maximum recommended capture angle for APPR mode is 60°.

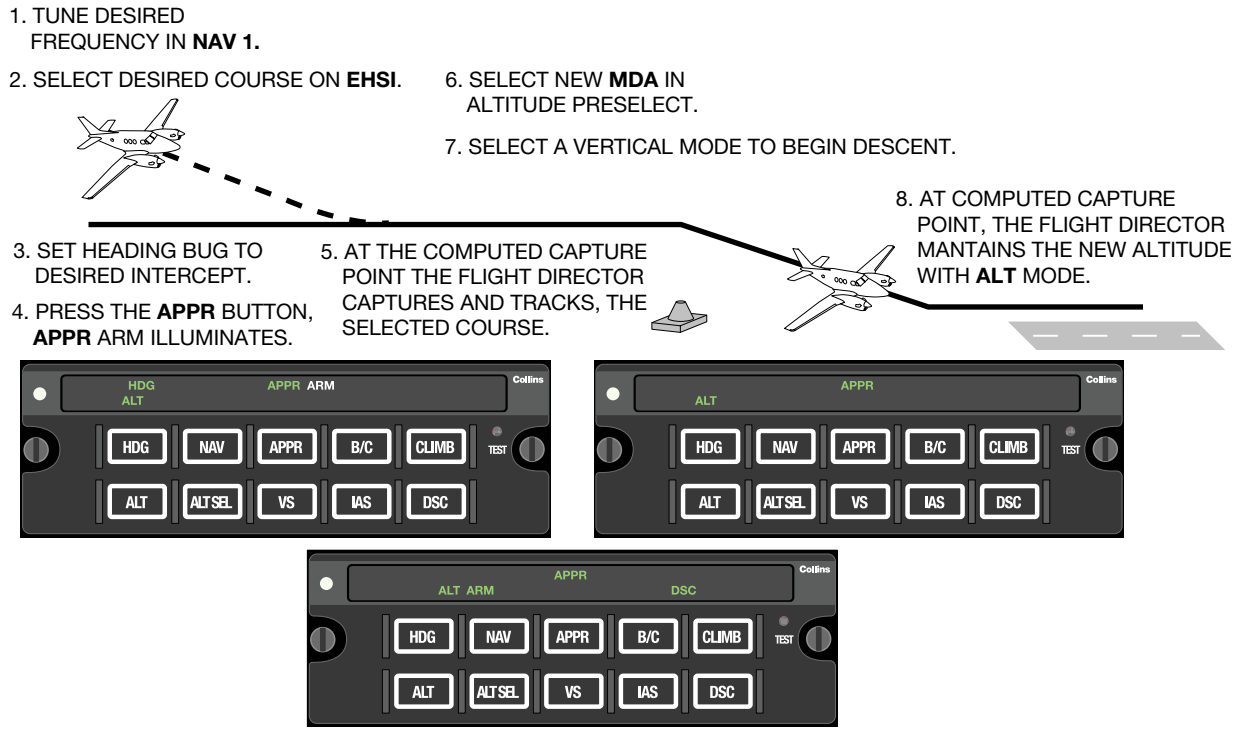

Figure 16-39. Localizer Approach Mode

Back Course Approach (BC)

1. Tune the desired frequency into NAV 1.
2. Set the course on the EHSI to the localizer front course (prevents reverse sensing).
3. Set the heading bug to a heading that will give the desired intercept angle.
4. Press the B/C button. The APPR ARM and B/C annunciators on the FCP will illuminate indicating that the B/C APPR mode is now armed. If HDG was not already selected. Selecting the B/C Mode will automatically activate HDG if the airplane is outside of the computed capture point. B/C will command the Flight Director to follow the back course instead of the front course
5. At the computed capture point: HDG cancels, APPR ARM changes to APPR B/C, and the Flight Director commands a turn to intercept and track the localizer back course.
6. Upon reaching the Final Approach Fix select a vertical mode, as required, to begin the descent (Figure 16-40).

NOTE

The maximum recommended capture angle for APPR mode is 60°.

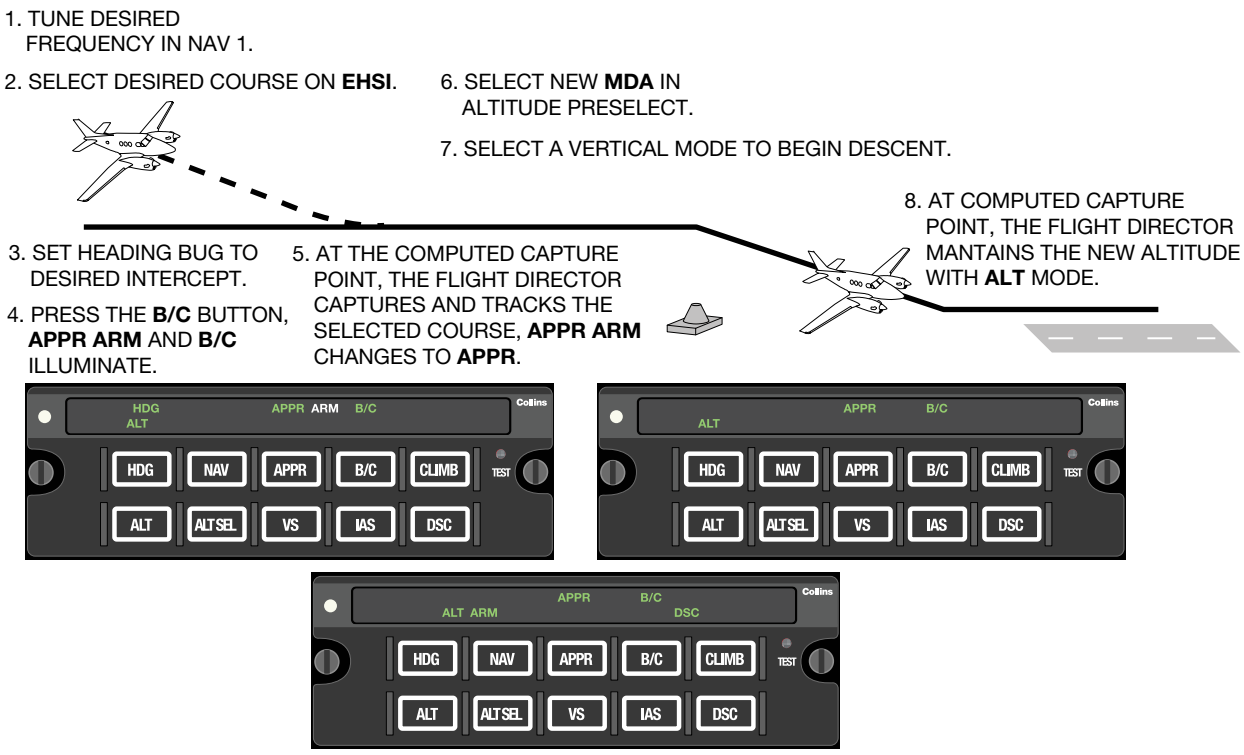

Figure 16-40. Localizer Back Course Approach Mode

ILS Approach (APPR)

1. Tune the desired frequency into NAV 1.
2. Set the localizer front course on the EHSI.
3. Set the heading bug to a heading that will give the desired intercept angle.
4. Press the APPR button. The APPR ARM and GS annunciators on the FCP will illuminate indicating that the APPR and GS modes are now armed. If HDG was not already selected. Selecting the APPR Mode will automatically activate HDG if the airplane is outside of the computed capture point. Note: the GS ARM annunciator will illuminate once the localizer is captured and the glide slope signal is received.
5. At the computed localizer capture point: HDG cancels, APPR ARM changes to APPR, and the Flight Director commands a turn to intercept and track the selected course.
6. At the computed glideslope capture point, the active vertical mode is disengaged and the Flight Director commands a pitch change to capture and track the glideslope.

NOTES

The maximum recommended capture angle for APPR mode is 60°.

Glideslope capture will not occur if the localizer has NOT been captured and/or the aircraft is above the glideslope.

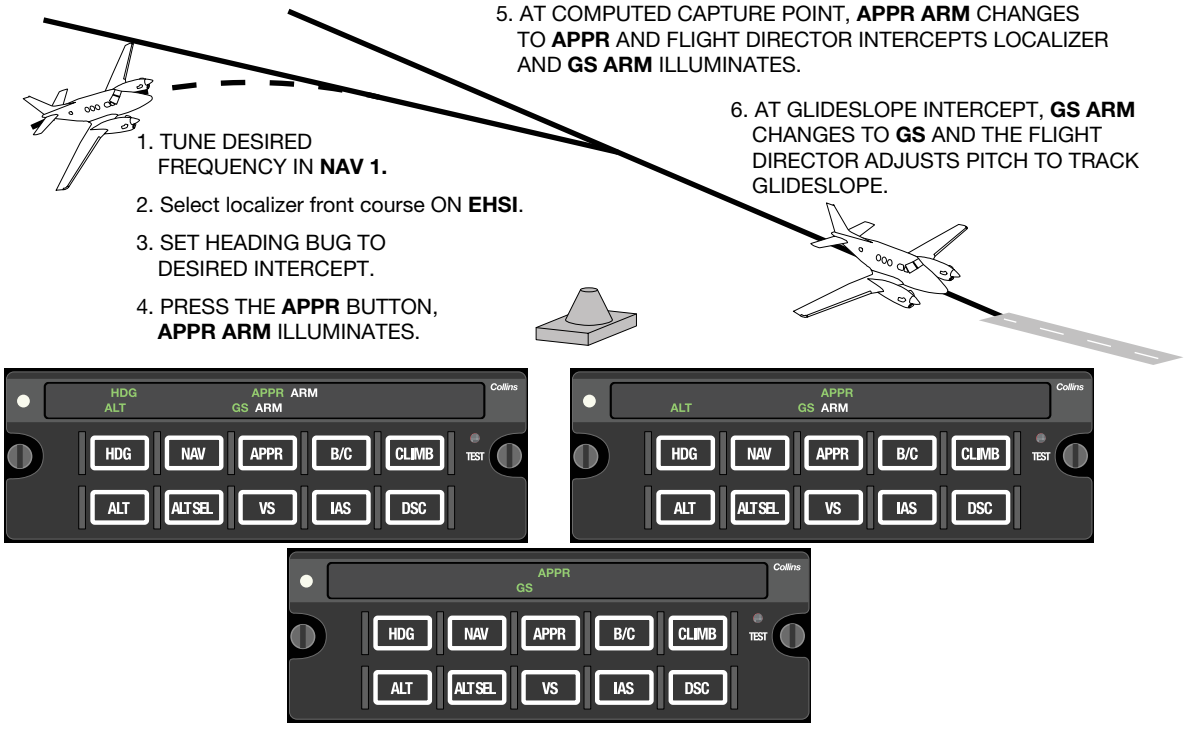

Figure 16-41. ILS Approach Mode

Go-Around Mode (GA)

1. Press the GA button on the left power lever.
2. The Autopilot disengages, the disconnect aural warning sounds, and the Flight Director commands a wings-level, 7° nose-up attitude.
3. The GA mode is disengaged by engagement of the Autopilot, the Pitch SYNC/CWS button, or other mode selections.

Figure 16-42. Go-Around Mode

LIMITATIONS

1. Maximum speed limit for Autopilot operation is V_{MO}/M_{MO}.
2. Do not use the Autopilot below the following altitudes:
   a. Approach – 200 feet above ground level.
   b. Climb – 500 feet above ground level.
   c. All other operations – 1000 feet above ground level.
3. Do not use the Autopilot or yaw damper during takeoff or landing.
4. Pilot must be seated at the controls with the seat belt fastened during Autopilot operations.
5. The Autopilot preflight check must be conducted and found satisfactory prior to each flight on which the Autopilot is to be used.
6. Use of the VOR Approach mode is limited to an area within 10 nm of the VOR station.

GARMIN GPS 400 GLOBAL POSITIONING SYSTEM

GENERAL INFORMATION

Display

The GPS 400 uses a color liquid crystal type display. The display consists of an annunciator area located along the bottom-left portion, a group/page indicator along the bottom-right portion and either a map with data blocks or a full page located in the remainder of the screen (Figure 16-43).

Figure 16-43. Garmin GPS 400

Knobs, Keys and Annunciators

The unit has one set of concentric knobs. The knobs control which group or page is displayed and cursor position or value. Each group contains one or more pages. The outer knob is used to scroll through the groups and the inner knob is used to scroll through the pages within the selected group. If the cursor is activated the outer knob changes the position of the cursor and the inner knob changes the value within it. This movement of the knobs is used to enter waypoint names or data when needed.

Twelve keys are located on the unit. Pressing the desired key will activate its associated function. For example, pressing the NRST key will display the nearest airport information.

Key Names and Functions

CRSR

The CRSR key, activated by pressing the inner knob, will activate the cursor in a field on the display to enter data and/or make a selection from a list of options.

NRST

The NRST key displays the Nearest group page 1 depicting the 9 nearest airports.

OBS

The OBS key is used to select manual or automatic waypoint sequencing of the waypoints in the flight plan.

MSG

The MSG key will display the Message page. If a message is active, the MSG annunciator in the annunciator portion of the screen will flash. Once the MSG key has been pressed and the message viewed pressing the button a second time will return the display back to the original display. If a message is still active (i.e. airspace alert while the airplane is still within the airspace) the MSG annunciator will remain illuminated.

FPL

The FPL key will display the flight plan page allowing the creation, editing, or activation of flight plans.

PROC

The PROC key will display the Procedures page of the Airport group allowing the selection or removal of departures, arrivals, and approaches from the flight plan.

Left-hand Knob

The left-hand knob controls the power to the unit. Rotating it clockwise will turn the unit on.

RNG

The RNG key allows the map to be scaled to the desired range using the up and down arrow sides of the key. The available ranges are from 500 ft to 2000 nm.

D→

The direct-to key establishes a course direct-to the active waypoint or any other as input by the user.

MENU

The menu key displays a menu list of options, if available, for the page being displayed.

CLR

The clear key will erase information or cancel an entry. Pressing and holding this key will display the Default Navigation Page.

ENT

The enter button is used to confirm data or an action.

One external annunciator (MSG) is associated with the GPS 400. It is displayed on the EFIS.

OPERATION

Initialization

On power-up, the unit will begin a self-test displaying a series of four pages. The first page displayed is the welcome page (Figure 16-44) followed by the land data page (Figure 16-45). Next, the database confirmation page will appear showing the date of

Figure 16-44. Garmin GPS 400 Initialization—Welcome Page

Figure 16-45. Garmin GPS 400 Initialization—Data Land Page

the database currently in the unit (Figure 16-46). The database must be updated every 28 days to maintain the IFR certification. The user must accept the database with the ENT key prior to the initialization continuing. The last page to be displayed is the satellite status page at which point the unit will begin to collect satellite information (Figure 16-47). A "bar graph" shows signal strength of the satellites being received. If the receiver is not able to receive enough satellites to pass the built-in integrity test then an INTEG annunciator will appear at the bottom of the left-hand data fields. Once the unit acquires enough satellites to calculate a position, the Map page will appear.

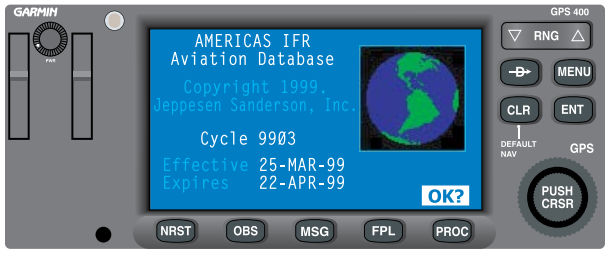

Figure 16-46. Garmin GPS 400 Initialization—Database Confirmation Page

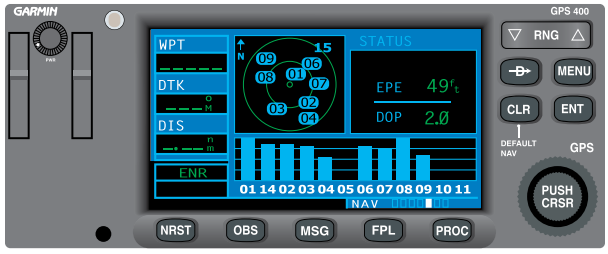

Figure 16-47. Garmin GPS 400 Initialization—Satellite Status Page

During the self-test the following indications should be indicated on the EFIS, provided the GPS is the active nav source:

- DME = 10 nm
- Course Deviation =1/2 scale deflection to the left
- TO/FROM flag = TO
- Bearing = 135°
- Groundspeed = 150 kts.
- Time to destination = 4 minutes
- Desired track = 149.5°

GROUPS AND PAGES

The GPS 400 has four groups. Each group has numerous pages within it. The groups are: NAV, WPT, AUX and NRST.

NAV (NAVIGATION)

The NAV group contains six or seven pages, depending on the installation. The Traffic/Weather page is optional. It will exist if a STORMSCOPE and/or SKYWATCH/ RYAN TCAD are installed in the aircraft.

- Default Nav (Figure 16-48)
- Map (Figure 16-49)
- Traffic/Weather (Optional)
- NAVCOM
- Position Satellite Status
- VNAV

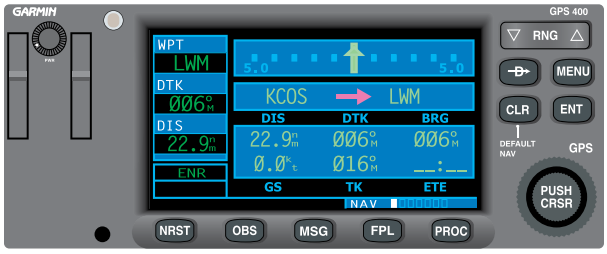

Figure 16-48. GPS 400 Default NAV Page

Figure 16-49. GPS 400 MAP Page

WPT (WAYPOINT)

The waypoint group contains ten pages dedicated to detailed information regarding airports, navigational facilities, intersections and user waypoints stored in the database.

- Airport Location
- Airport Runway
- Airport Frequency
- Airport Approach
- Airport Arrival
- Airport Departure
- Intersection
- NDB
- VOR
- User Waypoint

AUX (AUXILIARY)

Two Setup pages are contained within the AUX Group allowing the user to customize the operation and units of the GPS. The group also contains a Flight Planning and a Utility page.

- Flight Planning
- Utility
- Setup 1
- Setup 2

NRST (NEAREST)

The Nearest group contains eight pages dedicated to detailed information regarding the nearest airports, intersections, NDBs, VORs, user waypoints, airspace and ATC facilities within 200 nm of the present position.

- Nearest Airport (Figure 16-50)
- Nearest Intersection
- Nearest NDB
- Nearest VOR
- Nearest User
- Nearest Center
- Nearest Flight Service
- Nearest Airspace

DIRECT-TO NAVIGATION

Direct-to navigation allows the user to navigate on a course directly from the present position to the desired waypoint. There are numerous ways to go direct-to a particular waypoint. The simplest is to press the D→ key and enter the name of the waypoint desired and then press the ENT key twice to acknowledge it. If the desired waypoint is the current active waypoint, simply press the D→ key followed by the ENT key twice. Another option is to highlight the desired waypoint in the flight plan or the appropriate NRST page and press the D→ key followed by the ENT key twice. The last option is to display the desired waypoint on the appropriate WPT page and press the D→ key followed by the ENT key twice.

Cancelling Direct-To Navigation

Direct-to navigations are cancelled by pressing the D→ key followed by the MENU key to display the direct-to options list. Select "Cancel Direct-To NAV?" and press the ENT key.

Manually Setting a Direct-To Course

The user can set a course, other than a direct-to course. This may be useful if intercepting a particular radial off a VOR or any other instance where a course is to be something other than direct-to from the present position. To do so, press the D→ key and enter the desired waypoint followed by the ENT key. Next, highlight the CRS field in the lower right-hand corner of the display and enter the desired course and press the ENT key once more.

FLIGHT PLANS

Up to 19 flight plans can be stored in the memory of the GPS 400 with up to 30 waypoints each. Pressing the FPL key along the bottom of the unit accesses the Flight Plan pages. The two flight plan pages are the Active Flight Plan (Figure 16-51) and the Flight Plan Catalog (Figure 16-52). If a stored flight plan is to be actively used it must be moved to the FP 00 position first. FP 00 is always the active flight plan.

Figure 16-50. GPS 400 Nearest Airport

Figure 16-51. GPS 400 Active Flight Plan

Figure 16-52. GPS 400 Flight Plan Catalog

Entering a Flight Plan

To enter a flight plan, press the FPL key to access the Flight Plan pages and select the Flight Plan Catalog (second page). Next, press the MENU key to display the flight plan options, select "create new flight plan?" with the outer concentric knob and press the ENT key. A blank flight plan with the cursor active in the first waypoint position will be displayed. Enter the first waypoint using the concentric knobs and press the ENT key. Continue this operation until all desired waypoints in the flight plan have been entered. Note that the flight plan number (in the upper left-hand corner) is the next empty flight plan in the Flight Plan Catalog. A copy of the flight plan will remain saved in that position unless moved by the user.

Activating a Flight Plan

To activate the flight plan press the MENU key and select "ACTIVATE FLIGHT PLAN?". This moves the flight plan to the FP 00 position (active flight plan) while still keeping a copy in the Flight Plan Catalog. Any changes made, once the flight plan is active, will not effect the stored flight plan. If a flight plan was entered into FP 00 originally or changes were made and the stored flight plan needs to be updated the user can copy the flight plan into the desired stored position.

Copying a Flight Plan

To copy a flight plan to a stored location, display the flight plan and press the MENU key. Select "Copy Flight Plan?" and the unit will ask if it should copy it to the next empty catalog position. If this is not where it should go select highlight the flight plan number, enter the desired number and press the ENT key twice, provided the desired location is an empty position.

Inverting a Flight Plan

To invert a flight plan, select the desired flight plan an activate it. Press the MENU key and select "Invert Flight Plan?". The flight plan is now inverted; however, the stored flight plan will remain unchanged.

Editing a Flight Plan

To edit a flight plan, select the desired flight plan and turn on the cursor by pressing the inner concentric knob. Position the cursor at the desired location, enter the waypoint identifier and press the ENT key. If the waypoint is to be inserted between two existing way-points, highlight the later of the two and enter the waypoint name. The new waypoint will be inserted in front of the highlighted waypoint.

Activating a Flight Plan Leg

Any leg of a flight plan can be activated as long as the flight plan is active. To activate a specific leg of a flight plan, highlight the waypoint at the end of the desired leg, press the D▸ key twice and press the ENT key. This function can also be selected through the menu on the Active Flight Plan page by pressing the MENU key, highlighting the "Activate Leg?" followed by pressing the ENT key.

OBS AND AUTO MODES

The GPS 400 has two modes of operation, OBS and AUTO. AUTO, the default mode, is characterized by the fact that the unit automatically selects the course and has automatic waypoint sequencing through the flight plan. No automatic waypoint sequencing and the ability to manually select the desired course, on the other hand, characterize the OBS mode.

AUTO should be used for normal waypoint-to-waypoint operations. As the aircraft passes the active waypoint in the flight plan the unit will activate the next waypoint in the flight plan with a course direct from the waypoint that was just passed and the next waypoint. If the EHSI has GPS as the active nav source the course needle will automatically slew to the new course. When the airplane passes abeam the active waypoint the unit will recognize the fact that the waypoint is going to be bypassed and will sequence to the next waypoint with a course from the bypassed waypoint to the new waypoint. The user can also manually sequence to the next course, from the bypassed waypoint to the new waypoint, by one of two methods. One method is to go to the flight plan, press the MENU key, and then select "Activate Leg?". The other method is to press the Direct key twice followed by enter.

OBS is used any time the active waypoint is to be passed more than once (holding, radar vectors, procedure turns, etc…) or when the user wants to input a desired course manually. As mentioned above, the unit will sequence automatically through the waypoints when each waypoint is passed. This will not work when flying a holding pattern. To fly a published hold not associated with a missed approach, the unit will allow one turn in the hold without selecting OBS. However, the OBS key must be pressed each time around the hold prior to reaching the waypoint after the first turn. A hold associated with a missed approach does not require the OBS key to be pressed other than to sequence the GPS from the Missed Approach waypoint to the Holding waypoint. A procedure turn in an approach will also allow for one procedure turn. Upon passing the waypoint inbound, the unit will automatically sequence to the next waypoint. With a hold not associated with an approach at all, the OBS key must be pressed prior to reaching the waypoint the first time, as well as, every other time. OBS does not need to be used during all radar vectors. One instance where it must be used is vectors for an approach where the user has not specified VECTORS as the desired transition. Another instance is vectors that will cause the waypoint to be passed during the vector other than inbound on the desired course.

One way to look at OBS is that it will change any waypoint into a sort of pseudo-VOR without the drawbacks of a true VOR. While in OBS the user should treat flying with the GPS basically the same as flying a VOR.

GPS APPROACHES

When properly installed, the GPS 400 is approved for GPS based non-precision approaches. Flying a GPS approach is very similar, yet different from flying a standard non-precision approach. The GPS loads the approach from the database, therefore, the database must be current. If an approach is loaded with an outdated database, the approach will load normally but the GPS will not automatically sequence beyond the Initial Approach Fix.

Two approach modes, TERM (terminal) and APR (approach), are incorporated in to the unit. TERM is automatically selected once the aircraft is within 30 nm of an airport with an approach loaded and activated for that airport. APR is also automatically selected once the aircraft is within 2 nm prior to the FAF but not beyond it.

The two differences between TERM and APR are the increased integrity monitoring and the increased CDI needle sensitivity. Prior to the TERM mode, the unit is in the ENR (Enroute) mode. This gives the CDI a sensitivity of ±5 nm full-scale deflection. Once in the TERM mode, the integrity monitoring will increase to a higher level and the CDI sensitivity will increase to ±1 nm. Then, once the APR mode is active the CDI sensitivity will further increase to ±0.3 nm (Figure 16-53).

LOADING THE APPROACH

The GPS approaches can be loaded into the flight plan using one of two methods. The easiest is to press the PROC key and select "Select Approach?"

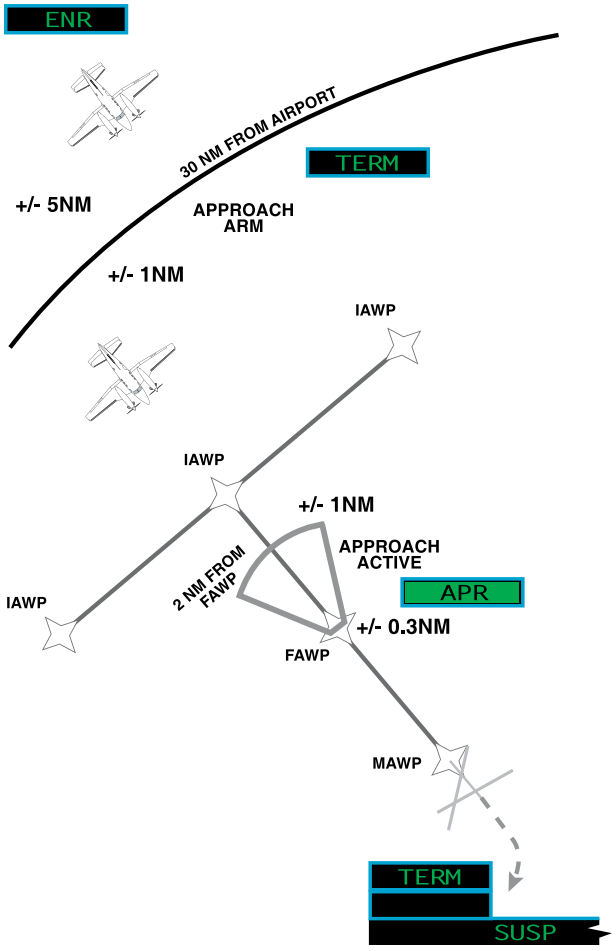

Figure 16-53. GPS 400 Approach Scaling

(Figure 16-54). The approach page for the next airport in the flight plan will appear with the APR menu displayed (Figure 16-55). If the approach page displayed is not the desired airport place the cursor over the airport identifier and enter the correct identifier. Use the outer concentric knob to highlight the desired approach and press the ENT key. The unit will prompt for the selection

Figure 16-54. GPS 400 Approach Loading (1 of 4)

Figure 16-55. GPS 400 Approach Loading (2 of 4)

of the desired transition to be used. The options will include Vectors and any IAFs available for the approach (Figure 16-56). Press the ENT key, select either "Load?" or "Activate?" with the outer concentric knob and press the ENT key.

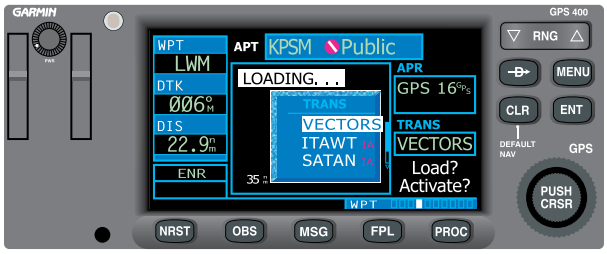

Figure 16-56. GPS 400 Approach Loading (3 of 4)

Selecting "Load?" will load the approach at the end of the active flight plan but it will not be active. The user will have to manually activate the approach when desired. If "Activate?" is selected the approach will be loaded in the flight plan in the correct position relative to the other waypoints. The approach will not have to be manually activated in this situation (Figure 16-57). The benefit of loading the approach and not immediately activating it is ability to navigate further on the flight plan until ready for the approach. Also, if "Activate?" is

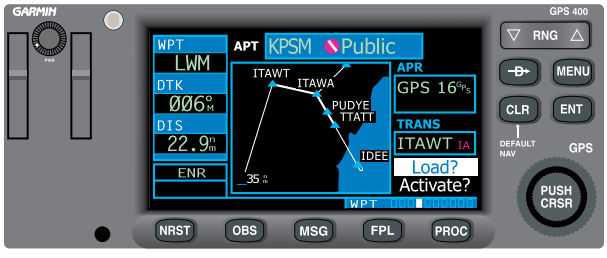

Figure 16-57. GPS 400 Approach Loading (4 of 4)

selected, depending on the position of the aircraft, the approach may be loaded into the flight plan as the next way-points and command navigation to the IAF when the user may not be ready to do so.

The approaches are listed by the primary navigation method and the runway number. For instance, a stand-alone GPS approach to runway 26 will be listed as "GPS 26" and a GPS overlay approach (VOR or GPS 14) will be listed as "VOR 14 GPS" The GPS denotes that it is an overlay approach. All approaches for the airport are listed, including all non-GPS approaches. If a non-GPS approach is loaded, a message will appear alerting the user to the fact that it is designed solely for monitoring and the appropriate navigational aid MUST be used as the primary navigational source.

Once the approach has been loaded the Active Flight Plan page will be displayed showing where in the flight plan it was loaded.

UNDERSTANDING THE DISPLAY

With the approach loaded into the active flight plan the waypoints in the approach will be listed below the approach title (Figure 16-58). Some way-points will also have a suffix at the end of the waypoint name. These are the most common to be seen in the flight plan.

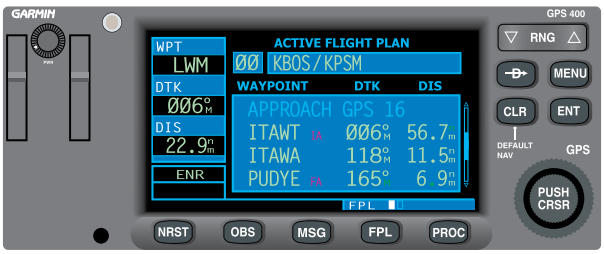

Figure 16-58. GPS 400 Approach Display

IA = The Initial Approach Fix.

FA = The Final Approach Fix.

MA = The Missed Approach Point.

MH = The Missed Approach Holding Point

Also seen in the flight plan are "proc. turn" "hold" or "dme arc".

The GPS 400 uses special "terminal way-points" for approach operations. The way-points are not usually shown on the approach charts but are required for the GPS approach.

FFyyy .....F = Final Approach Fix

IAyyy ..... I = Initial Approach Fix

CFyyy.....C = Course Fix

MAyyy .....M = Missed Approach Point

x will be either an "A" or "F" if the runway has only one approach. If it has more than one approach, x will be "V" for VOR, "N" for NDB, etc…

yyy will be either the runway identifier for straight-in approaches, or the inbound course for circling approaches.

RWzzz RW = Runway fix which is usually the Missed Approach Point

zzz = the runway number which may include L for Left, C for Center, R for Right, or B for Both.

Daaab D = DME arc waypoints

aaa = the radial that the fix is on from the reference VOR.

b = a letter corresponding to the distance from the VOR. For example, G = 7 nm since G is the seventh letter of the alphabet.

FLYING THE MISSED APPROACH

The unit will not automatically sequence from the Missed Approach Point to the Missed Approach Holding Point in the flight plan. This is due to the fact that the missed approach procedure may require specific actions (e.g. runway heading to 3000' then right direct…) prior to going to the Missed Approach Holding Point. The unit will display a "SUSP" (suspend) annunciator directly above the OBS key alerting the user to this fact once the Missed Approach Point has been passed. The way-points must be manually sequenced in

this situation. The simplest way is to press the OBS key once ready to proceed direct to the holding fix. The GPS will sequence with a course direct to the Missed Approach Holding Point. To activate the missed approach portion of the approach prior to the Missed Approach Fix display the active flight plan and press the D↠ key twice followed by the ENT key. Once the Holding Point is reached a timer will appear on the Default NAV page counting down from one minute. The counter will reset upon intercepted inbound and passing the holding fix outbound. The unit will switch to the SUSP mode once crossing the holding fix allowing the user to hold as many times as necessary. This alleviates the user from having to select OBS. Just prior to reaching the holding fix, the unit will display a message in the lower right-hand corner of the display recommending a method of entry based on the current position. However, the GPS will not fly the holding pattern.

Changing or Deleting an Approach

Waypoints cannot be added to or deleted from an approach. To load another approach, press the PROC key and select "Change Approach?". Selecting another approach will automatically delete the previously loaded approach. To delete an approach, press the PROC key and select "Delete Approach?" (Figure 16-59).

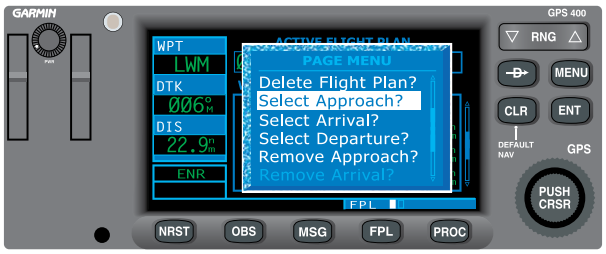

Figure 16-59. GPS 400 Change or Delete Approach

MAP PAGE

The second page of the NAV group is the Map page (Figure 16-60). The Map page displays a full screen color moving map. The left side of the display contains three user-selectable data blocks. Four additional ones can be displayed on the right side of the display, if so desired (Figure 16-61).

Figure 16-60. GPS 400 MAP Page

Figure 16-61. GPS 400 MAP Page—Dual Data Fields

Understanding the Display

The map displays a varying amount of data contained in layers. The default display shows the aircraft present position (indicated by an airplane symbol), desired track (indicated by a white or magenta line), airports, navigational aids, airspace, user-waypoints, and geographical data.

The layers of data can be removed from the display by pressing the CLR key. Each time the key is pressed one layer of information is removed beginning with layer one. A -1, -2, or -3 will appear next to the range display indicating the level of "decluttering" as each layer is removed. The layers are:

- Layer 1—Geographical Data
- Layer 2—Airspace
- Layer 3—Non-flight plan aviation data

The range of the map is selectable from 500 feet to 2000 nm using the RNG key. An Autozoom feature automatically decreases the range, dependent on the distance from the destination, down to a final range of 1 nm. The Autozoom feature can be disabled on the Map Setup menu. If the map is zoomed down to a range less than that used in the designing of the map, "OVERZOOM"

will appear above the range display indicating that the geographical data may not be completely accurate. If the map is further zoomed in "OVERZOOM" will be replaced with "NO MAP" as the geographical detail is removed from view. The aviation data will remain in view, however. In both scenarios the aviation data remains accurate.

The waypoints in the flight plan and the desired track between each one will be displayed on the map. The desired track to the active waypoint is depicted in a magenta color. The other "inactive" desired tracks are white in color.

Pressing the inner knob will activate a pointer on the map. The pointer is panned around the map by rotating the inner and the outer knobs. If the pointer is moved off-screen, the map will pan with it. Once the cursor is turned off the map will return to the aircraft position. At the top of the map, a box displays the current position of the cursor in latitude and longitude, as well as, bearing and distance from the aircraft. As the pointer moves over an object on the map, that object will be highlighted and the name of the object will appear. Pressing the ENT key, at this point, will display the appropriate waypoint page. If the object is something other than a database or user waypoint pressing the ENT key will display the user waypoint page to allow the user to create a user waypoint at that position. Press the CLR key to return to the Map page. Pressing the D→ key will allow the user to activate that waypoint as a direct-to waypoint.

Pressing the MENU key will display the Page Menu. Selecting "Setup Map?" displays the setup menu allowing the user to select the desired orientation of the map, Autozoom on or off and land or aviation data on or off. The other selections on the Page Menu are "Measure Distance?", "Data Fields Off?", "Change Fields?" and "Restore Defaults?". Selecting "Measure Distance?" allows the user to measure the distance and bearing between any two points on the map using the cursor pointer.

BENDIX/KING KLN-90B GLOBAL POSITIONING SYSTEM

GENERAL INFORMATION

Display

Three display options are available depending on which chapters and pages have been selected. The standard display format is a split-screen with three segments along the bottom designed to display the selected chapter and page, on each side, and then an annunciator display area in the center (Figure 16-62). The chapter and page are displayed as a chapter name and a page number. If the selected page has information that continues on to a sub-page, the page number will have a "+" next to it. The split-screen (Figure 16-63) enables the user to display different chapters and pages on the left and the right sides of the display. The other option is for a "Super" page. Two "Super" pages are available, Super NAV 1 and Super NAV 5. NAV chapter page 1 or page 5 have the ability to be displayed on one

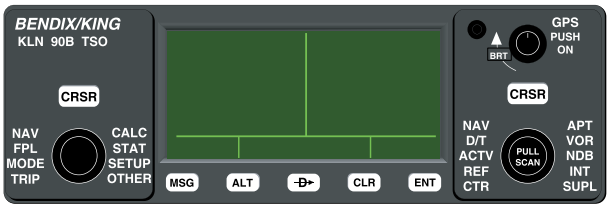

Figure 16-62. KLN-90B Displays—Standard Format

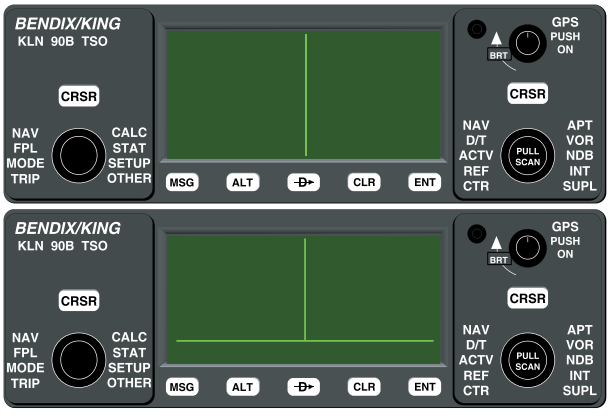

Figure 16-63. KLN-90B Displays—Split Screen Formats

side or the other, however, if NAV 1 or NAV 5 is displayed on both sides at the same time the display becomes a "Super" page meaning the display is no longer a split-screen (Figure 16-64).

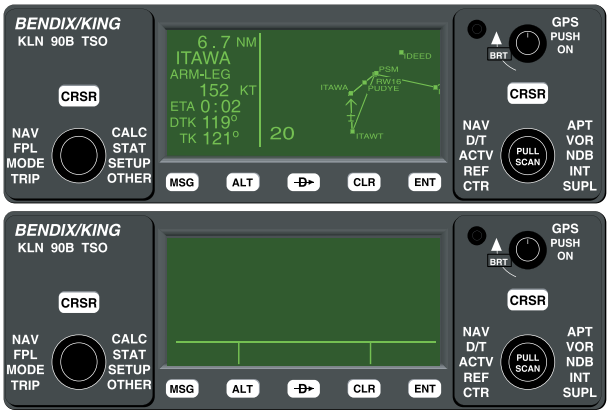

Figure 16-64. KLN-90B Displays—Super Pages

Knobs, Buttons, and Annunciators

There are two concentric sets of knobs. Each set of knobs has the names of the individual chapters that it controls surrounding it. The chapters contain one or many pages within them. The outer knobs scroll through the chapters. The inner knobs scroll through the pages within the selected chapter. If the cursor is activated on either side, that side's outer knob will change the position of the cursor and the inner knob will change the value displayed within the cursor. This movement of the knobs is used to "type" waypoint names or data when needed.

Seven buttons are located along the bottom center of the unit and just above each set of knobs. Pressing the desired button will activate its associated function. For example, pressing the left cursor button will activate the cursor on the left side of the display.

Button Names and Functions

CRSR

The cursor buttons will activate the cursor in a field on the button's respective side of the display.

MSG

The message button will display the Message page. If a message is active, the MSG annunciator in the annunciator portion of the screen will flash. Once the MSG button has been pressed and the message viewed pressing the button a second time will return the display back to the original display. If a message is still active (i.e. airspace alert while the airplane is still within the airspace) the MSG annunciator will remain illuminated steady.

ALT

The altitude button will display the Altitude Reference page on the left-hand screen and the NAV 4 page (VNAV) on the right-hand screen. The Altitude Reference page is used to input the current barometric pressure. The page will also allow the user to select whether the altitude alert is turned on or off and the buffer for the alert if it is turned on. Press ALT again to return to the previous page.

D➔

The Direct-To button is used to go "direct-to" a waypoint. Pressing the button, entering the waypoint name, and then pressing the ENT button does this.

CLR

The clear button will clear the data in a field highlighted by the cursor, remove inactive waypoints from the moving map, and delete data from the flight plan. The button can also be used as a "cancel" button if a mistake has been made.

ENT

The enter button is used to confirm data or an action.

Six external annunciators are associated with the KLN-90B. Two (MSG and WPT) are displayed on the EFIS and the other four are displayed on two annunciator display/switches located above the pilot altimeter. The four include one display labeled GPS CRS and the other labeled GPS APR. The GPS CRS annunciator will illuminate either OBS or LEG, whichever has been selected by the user. The GPS APR annunciator display

will illuminate either ARM or ACTV, whichever is active, during a GPS approach procedure. All of the annunciators will be discussed in more depth later in this manual.

OPERATION

Initialization

On power-up, the first page to be displayed is informational in nature indicating that the unit has begun its self-test (Figure 16-65). This page will also display a personal message that can be set by the user. Upon completion of the self-test the unit will automatically display the second page.

Figure 16-65. KLN-90B Initialization—Powerup Page

The second page displays the data outputs to the screen as well as to the HSI (Figure 16-66). The cursor is automatically turned on highlighting the barometric pressure prompting to the user to input the current pressure. Enter the pressure and press the ENT button to accept it, when the cursor highlights the word "APPROVE?" press the ENT button a second time to accept the data. The data includes:

- A distance readout of 34.5 NM
- The CDI showing 1/2 scale deflection to the right
- OBS IN _ _ _° and OUT 315°
- RMI 130°
- ANNUN ON.

If GPS has already been selected as the active nav source then the EHSI will show the 1/2 scale deflection of the CDI and the appropriate DME readout. The bearing pointer, if selected, will also show a bearing of 130°.

Figure 16-66. KLN-90B Initialization—Data Outputs Page

The unit will next display the Database Verification page (Figure 16-67). Verify that the database is current, by the date of expiration displayed, and press the ENT button to accept it. If the page appears as a message stating "WARNING SYSTEM IS IN OBS MODE PRESS GPS CRS BUTTON TO CHANGE TO LEG MODE" then press the button as stated to change to LEG mode. The database information will then appear as normal. The initialization process is now complete. The next page to be displayed will be a split screen with the present position displayed on the left side and the waypoint page for the last waypoint to be active, prior to shutdown, displayed on the right side (Figure 16-68). If the last active waypoint was an airport the APT 4 page (airport communications) will be displayed. The external annunciators will be illuminated during the entire initialization until the database has been accepted.

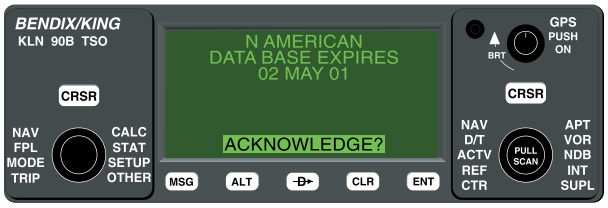

Figure 16-67. KLN-90B Initialization—Database Verification Page

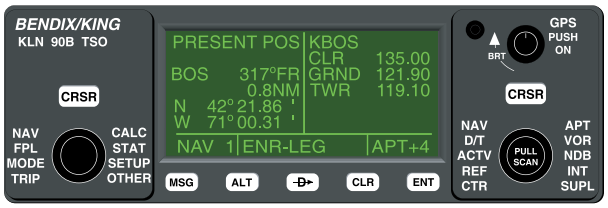

Figure 16-68. KLN-90B Initialization—Process Completed Page

WAYPOINT CHAPTERS AND PAGES

Five waypoint chapters are accessible with the right-hand outer knob; Airport (APT), VOR, NDB, Intersection (INT), and Supplemental (SUPL). Supplemental way-points may also be known as user-defined waypoints. These chapters are used to display information about the selected waypoint, as well as, to select SIDs, STARS and Approaches.

The user may use any of five ways to select a waypoint from one of the chapters.

1. Select a chapter, APT for instance, and turn on the right-hand cursor. This will highlight the identifier field. Type the desired identifier and then turn the cursor off.
2. Turn the right-hand cursor on and pull out the right-hand inner knob. This will activate the scan function of the knob enabling the user to scan through the waypoints without having to type. The user can also type as many letters/numbers of the identifier as are known andthen scan to find the rest.
3. Move the cursor down to the full waypoint name field and type in the nameof the waypoint.
4. Do the same as number 3, but scanthrough the waypoint list with theright-hand inner knob in the same wayas in number 2.
5. Select the Active (ACT) chapter, provided the desired waypoint is already in the active flight plan. The user can thenscan through the active flight plan way-points using the right-hand inner knob(pulled out).

DIRECT-TO NAVIGATION

The GPS enables the user to proceed on acourse direct-to any waypoint from the presentposition. Pressing the D→ button will display the Direct-To page with the current active waypoint as the direct-to waypoint. If this waypoint is not the desired waypoint enter the correct identifier and press the ENT button twice. If it is the desired waypoint simply press the ENT button once and the direct-to course will be activated. Another way to select a direct-to waypoint is to pull out the right-hand inner knob, while on the Super NAV 5 page, and rotate it until the desired waypoint in the flight plan is displayed in the lower right-hand corner of the display. Then, press the D→ button followed by the ENT key.

Cancelling a Direct-To Navigation

Any direct-to operation can be cancelled by pressing the D→, CLR and then ENT buttons. Doing so will cancel the direct-to navigation and revert back to the proper location in the flight plan. This will not work if a flight plan is not active.

Manually Setting a Direct-To Course

A direct-to course can be manually set through the use of the OBS function. To do so, select the OBS mode, select direct-to the desired waypoint then set the desired course with the course knob or on the Super NAV 5 page. Leg mode can be reselected at this point if waypoint auto sequencing is desired. The OBS mode is described in further detail on the next page.

FLIGHT PLANS

Up to 25 flight plans can be stored in the memory of the KLN-90B with up to 25 waypoints each (Figure 16-69). Each of the 25 stored flight plans is stored in the flight plan (FP) 1 through 25 positions. If a stored flight plan is to be actively used it must be moved to the FP 0 position first. FP 0 is always the active flight plan.

Figure 16-69. KLN-90B Flight Plan Page

Entering a Flight Plan

To enter a flight plan, choose either FP 0 (if it is not to be immediately stored) or FP 1 through 25 (where it is to be stored) and then turn on the left-hand cursor and begin "typing" with the left-hand concentric knobs. After a waypoint is "typed" press ENT and the way-point information will be displayed on the right-hand side of the screen

for verification. If the information is correct then press ENT again to accept it. The cursor will then move to the next space below. If the waypoint is not correct press CLR and re-enter the data. Continue the process until all waypoints are entered into the flight plan. If during the process two or more waypoints share the same identifier, the unit will first give the user the ability to select which of the waypoints is desired by highlighting it with the cursor and pressing ENT. The unit will then display the waypoint information for the usual waypoint verification.

Once a flight plan has been entered into any of FP 1 through 25 it is automatically stored for future use. To actively use any of the flight plans, turn on the left-hand cursor highlighting the first waypoint, and scroll the cursor up to highlight either “ACTIVATE” or “ACTIVATE INVERT”, as desired, and press ENT. The flight plan is then automatically loaded and displayed in FP 0. Once the flight plan is in FP 0 any changes made will not effect the stored flight plan. If a flight plan was entered into FP 0 originally or changes were made and the stored flight plan needs to be updated the user can transfer the data into FP 1 through 25, as desired. To do so scroll through FP 1 through 25 until the desired location is displayed and turn on the cursor. When activated, the cursor will high-light the first waypoint location. Scroll the cursor up to highlight “LOAD FPL 0?” and press ENT. FP 0 is then loaded into the selected location and stored for future use.

NEAREST WAYPOINTS

The nearest airport can be identified one of two ways. The quickest is to press the MSG button followed by the ENT button. The nearest airport will appear on the right side of the screen (Figure 16-70). “NR 1” will be flashing in the upper right hand corner identifying it as the nearest airport.

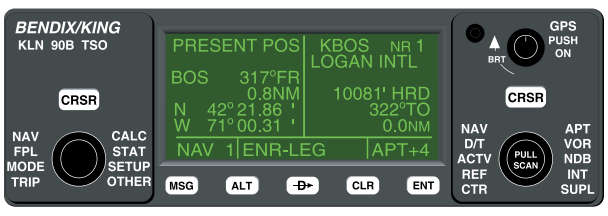

Figure 16-70. KLN-90B Nearest Airport Page

The display will give the airport identifier, airport name, runway length, and bearing and distance to the airport from the present position. The information is on page APT1, therefore, the user can select any of theother eight pages to obtain the normal airport information. To identify the next eight near-est airports, pull out the right-hand inner knob to scan and rotate it clockwise to view the other airports. The alternate method is to goto the APT chapter and pull out the right-handinner knob, to scan, and rotate it counter-clockwise until at the beginning of the air-port list. The criteria for the nearest airport selection is set on the SET 3 page. This page allows the user to define the minimum runway length and hard, or hard and soft surface run-ways.

The nearest 9 VORs and NDBs can be displayed by selecting the VOR or NDB chapter and pulling out the right-hand inner knob and scrolling to the beginning of the list.

OBS AND LEG MODES

The KLN 90B has two modes of operation, OBS and LEG. LEG, the default mode, is characterized by the fact that the unit auto-matically selects the course and has automatic waypoint sequencing through the flight plan. The OBS mode, on the other hand, is characterized by the ability of the user to manually select the desired course and no automatic waypoint sequencing.

LEG should be used for normal waypoint-to-waypoint operations. As the aircraft passes the active waypoint in the flight plan the unit will activate the next waypoint in the flight plan with a course direct from the waypoint that was just passed to the new waypoint. If the EHSI has GPS as the active nav source the course needle will automatically slew to the new course. When the airplane passes abeam the active waypoint the unit will recognize the fact that the waypoint is going to be bypassed and will sequence to the next waypoint with a course from the bypassed waypoint to the new waypoint.

OBS is used any time the active waypoint is to be passed more than once (holding, radar vectors, procedure turns, etc…) or when the user wants to input a desired course manually. As mentioned

above, the unit will sequence automatically through the waypoints as each waypoint is passed. This will not work when flying a holding pattern. Therefore, the OBS mode should be used allowing the user to set in the inbound course of the hold on the EHSI and have the unit not sequence each time the waypoint is passed. Once inbound towards the waypoint for the last time, select LEG so that the unit will resume the auto sequencing. The same procedure should be used for procedure turns or radar vectors. OBS does not need to be used during all radar vectors, just vectors that will cause the waypoint to be passed during the vector other than inbound on the desired course. One way to look at OBS is that it will change any waypoint into a sort of pseudo-VOR without the drawbacks of a true VOR. While in OBS the user should treat flying with the GPS basically the same as flying a VOR.

GPS APPROACHES

When properly installed, the KLN 90B is approved for GPS based non-precision approaches. Flying a GPS approach is very similar, yet different from flying a standard non-precision approach. The GPS loads the approach from the database. Therefore, the database must be current. If an approach is loaded with an outdated database, a status-line message: "OUTDATED DB" will appear in the bottom center portion of the screen as a reminder.

Two approach modes, ARM and ACTV, are incorporated in to the KLN 90B. ARM can be selected one of two ways. Once within 30 nm of an airport with an approach loaded for the same airport the unit will automatically ARM the approach. ARM can also be manually selected, while greater than 30 nm from the airport, with the external annunciator/switch labeled "GPS APR". If the switch is pressed when within 30 nm from the airport the approach will be disarmed. Simply pressing the switch again will rearm the approach. ACTV (active) can only be automatically selected by the KLN 90B. This is done within 2 nm of the FAF but not beyond it. Pressing the "GPS APR" switch, with the approach ACTV, will return the unit to the ARM mode. ACTV can be regained by pressing the switch again. This is only true when not beyond the FAF. If the FAF has already been passed a missed approach and flying back to the FAF is the only way to regain the ACTV mode.

The two differences between ARM and ACTV are increased integrity monitoring and increased CDI needle sensitivity. Prior to the approach being armed, the unit is in the Enroute (ENR) mode. This gives the CDI a sensitivity of ±5 nm full-scale deflection. Once the approach is armed, the integrity monitoring will increase to a higher level and the CDI sensitivity will increase to ±1 nm over a 30second period of time. Then, once the approach is ACTV the CDI sensitivity will increase to ±0.3 nm (Figure 16-71).

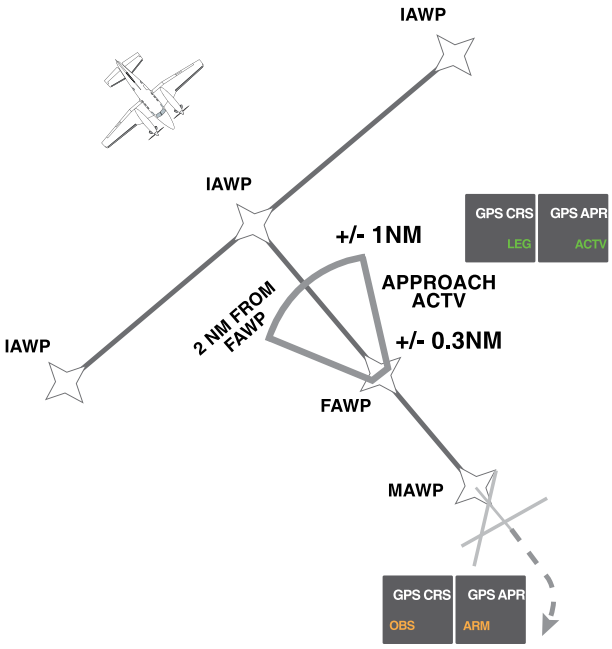

Figure 16-71. KLN-90B Approach Scaling

LOADING THE APPROACH

GPS approaches are loaded into the flight plan from the APT 8 page (Figure 16-72). This page lists all of the GPS based approaches in the database for the particular airport. The approaches are listed by the primary navigation method and the runway number. For

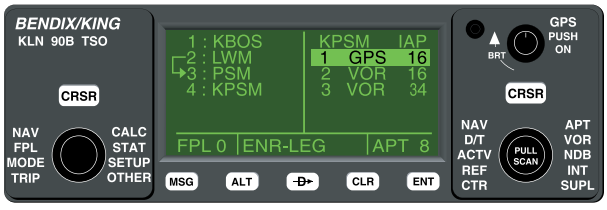

Figure 16-72. KLN-90B Approach Loading (1 of 3)

instance, a stand-alone GPS approach to runway 26 will be listed as "GPS 26" and a GPS overlay approach (VOR or GPS 14) will be listed as "VOR 14". From the APT 8 page, turn on the right-hand cursor and select the desired approach and press ENT. If more than one IAF is available for the approach the unit will prompt you to select one. Press ENT to select it once it is highlighted (Figure 16-73). A list of the waypoints in the approach will be displayed and "LOAD IN FPL" will be highlighted. Verify that the waypoints are correct and that the correct approach was loaded and press ENT (Figure 16-74). If the airport at which the approach is being selected is not already in the flight plan then the unit will ask if it should be added. It must be in the flight plan for an approach to be loaded. The FP 0 page will be displayed on the left-hand side of the screen, at this point, displaying the way-points in the approach (Figure 16-75).

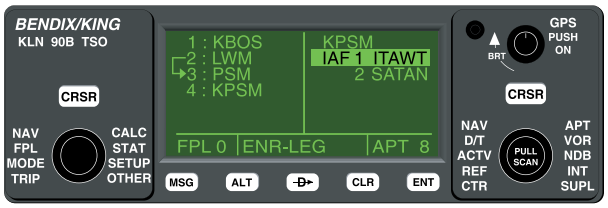

Figure 16-73. KLN-90B Approach Loading (2 of 3)

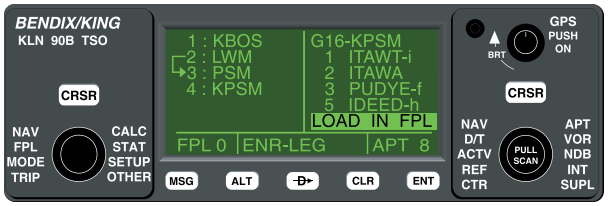

Figure 16-74. KLN-90B Approach Loading (3 of 3)

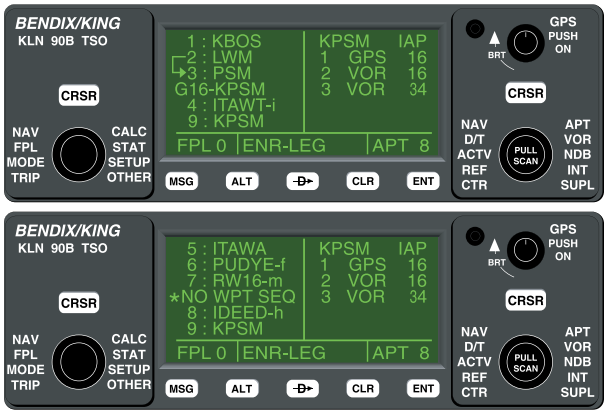

Figure 16-75. Approach Flight Plan

UNDERSTANDING THE DISPLAY

With the approach loaded into FP 0 the way-points in the approach will be listed below the approach title. The first letter of the title is the type of approach being flown (e.g. V for VOR) next is the runway number followed by the airport identifier. An example of the title would be "N01R –KICT".

Some waypoints will have a suffix at the end of the waypoint identifier. These are the most common to be seen in the flight plan.

i = The Initial Approach Fix.

f = The Final Approach Fix.

m = The Missed Approach Point.

h = The Missed Approach Holding Point

The KLN 90B uses special "terminal way-points" for approach operations. The way-points are not usually shown on the approach charts but are required for the GPS approach.

Fxyyy .....F = Final Approach Fix

Ixyyy ..... I = Initial Approach Fix

Cxyyy.....C = Course Fix

Mxyyy ..... M = Missed Approach Point

X will be either an "A" or "F" if the runway has only one approach. If it has more than one approach, x will be "V" for VOR, "N" for NDB, etc.

yyy will be either the runway identifier for straight-in approaches, or the inbound course for circling approaches.

RWzzz RW = Runway fix which is usually the Missed Approach Point

zzz = the runway number which may include L for Left, C for Center, R for Right, or B for Both.

Daaab D = DME arc waypoints

aaa = the radial that the fix is on from the reference VOR.

b = a letter corresponding to the distance from the VOR. For example, G = 7 nm since G is the seventh letter of the alphabet.

"*NO WPT SEQ" will be displayed between the Missed Approach Point and the Missed Approach Holding Point in the flight plan. This alerts the user that the automatic waypoint sequencing will be inhibited at this point.

FLYING THE MISSED APPROACH

The unit will not automatically sequence from the Missed Approach Point to the Missed Approach Holding Point in the flight plan. This is due to the fact that the missed approach procedure may require specific actions (e.g. runway heading to 3000' then right direct…) prior to going to the Missed Approach Holding Point. The unit will maintain the Missed Approach Point as the active waypoint after it has been passed. The waypoints must be manually sequenced in this situation. The simplest way is to press the D➔ button followed by the ENT button once ready to proceed direct to the holding fix. The GPS will sequence with a course direct to the Holding Fix, the GPS APR mode will switch from ACTV to ARM and the scaling will change back to ±1 nm. To activate the missed approach portion of the approach prior to the Missed Approach Fix display the active flight plan, highlight the Holding Fix with the cursor press the D button followed by the ENT button. The OBS mode must be selected prior to reaching the Holding Fix if the holding pattern is to be flown.

CHANGING OR DELETING AN APPROACH

Waypoints cannot be added to or deleted from an approach. To load another approach, turn on the left-hand cursor and place it over the approach title (approach title changes to "CHANGE APPROACH") (Figure 16-76). Press the ENT button and the APT 8 page will appear on the right side of the screen. Using the right-hand outer and inner knobs in the normal manner, it is now possible to select a different airport, approach or IAF. To delete the existing approach, turn on the left-hand cursor and place it over the approach title. Press the CLR button (approach title changes to "DELETE APR?") and press the ENT button (Figure 16-77).

Figure 16-76. KLN-90B Change Approach

Figure 16-77. KLN-90B Delete Approach

SUPER NAV 5 PAGE

The Super NAV 5 page is displayed by placing the left and right sides in the Nav 5 page (Figure 16-78). The "Super" refers to the fact that the display goes from the standard split screen to a full-screen display. The Super NAV 5 page displays a moving map and a limited amount of data. The data is displayed on the far left-hand portion of the display. The data consists of:

- Distance To Waypoint
- Waypoint Identifier
- Current GPS mode
- Estimated Time Enroute (ETE), Cross-Track Error (XTK) or Vertical Navigation (VNAV)
- Desired Track (DTK), Bearing to Active Waypoint (BRG) and Radial from Active Waypoint (RAD)
- Actual Track (TRK), Bearing to Active Waypoint (BRG) and Radial from Active Waypoint (RAD)

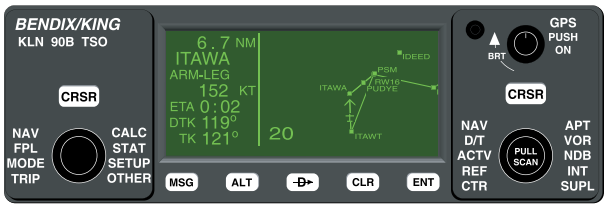

Figure 16-78. KLN-90B Super Nav 5 Display

The scale of the moving map display is shown in the bottom left-hand corner of the map. Turning on the left-hand cursor to highlight the range value and rotating the inner left-hand knob to the desired range changes the scale. The available ranges are 1 to 1000 nm and AUTO. The AUTO scale adjusts the scale to keep the active waypoint "in-view" at all times. The map display is customized through the use of a menu. The menu is displayed in the upper right-hand portion of the display when the right-hand cursor is turned on. The menu allows the user to determine what if any VORs are displayed. Whether or not NDBs or Airports are displayed and the map orientation can also be selected in the menu.

- Terminal, Low and High
- Low and High
- High
- Off
- The map orientation options are:
- North Up (N↑)
- Desired Track Up (DTK↑)
- Track Up (TRK↑)

Selecting Track Up will not display a map until the aircraft begins to move. The other orientations will display the map constantly.

The map display itself depicts the waypoints in the flight plan or the active waypoint if a flight plan is not in use and course lines. It also displays any other waypoints as selected in the menu by the user. Especially at larger scales, the map can become very cluttered due to the large number of waypoints displayed. If the map is too cluttered for the user's preference go to the menu and alter the displayed way-points, as desired. A quick way to do so is to press the CLR button. Doing so will delete all waypoints from the display except those that are in the flight plan. Pressing the CLR button a second time will redisplay them.

COLLINS PRO LINE II EQUIPMENT

The Pro Line II family presently consists of a VHF comm (VHF-22), a VOR/LOC/GS/MB receiver (VIR-32), and a DME (DME-42). These units employ many state-of-the-art features, including extensive self-diagnostic capabilities and multiple frequency storage. Some of the features of this equipment will be described here. For additional information, see the current *Collins Pro Line II Pilot's Guide*.

A new series of Collins CTL control heads is used for the nav, comm, ADF, and Transponder. The ADF-60 and TDR-90 (ADF and transponder units respectively), however, are retained from the earlier Collins Pro Line System.

The comm and nav units have many features in common; therefore, we will use the features of the nav (VIR-32/CTL-32) as a building block for the comm, which will be described next. Features common to both will be described under the nav explanation; differences will be pointed out under the comm explanation.

NAV SYSTEM

The VIR-32/CTL-32 nav system is comprised of a VOR/localizer receiver, a glide slope receiver, and a marker beacon receiver, all contained in one "black box" located in the nose avionics compartment.

The nav receiver (Figure 16-79) may be tuned to the correct frequency in any one of three ways:

Standby Tuning

Normally, the pilot uses the frequency select knob to change the frequency in the lower frequency window. The frequency may be transferred up to the ACTive window by holding the XFR/MEM switch to the XFR position momentarily.

Direct Tuning

The ACTive frequency may be tuned directly by first holding down the ACT push button for approximately three seconds. The lower (PREset) frequency display will be dashed out. The two concentric frequency select knobs will now directly channel the ACTive frequency. Features such as DME hold, preset channels, etc., are still operable in this situation.

Preset Tuning

The PREset (standby) frequency may be initially selected and displayed in the lower frequency window. It may be necessary to cancel the direct tuning mode (described above) by again holding the ACT pushbutton for approximately three seconds. Once the PREset frequency is displayed in the lower window, it may be transferred up to the ACTive window by holding the XFR/MEM switch to the XFR position momentarily.

Up to four frequencies may be placed into the four channel slots of the memory. This is done by repeatedly pressing the XFR/MEM switch to the MEM position until the desired channel number appears in the upper (ACTive) window (e.g., CH-1). Now the frequency may be selected using the two concentric frequency select knobs and will be displayed in the lower (PREset) window. Once selected, the frequency may be stored by simply pressing the STOre button twice. Subsequent frequencies/channels maybe stored in a similar fashion.

DME Hold

DME hold may be selected by placing the mode selector switch in the HLD position. This topic will be further discussed under the topic of DME, to be covered later in this section.

DME SYSTEM

By using frequency scanning techniques, the DME-42 is capable of working with up to three DME stations simultaneously (Figure 16-80). It can display DME distance (NM), ground speed (GS), time to station (MIN), and station identification to any one of these stations; however, the DME stays locked onto all three stations.

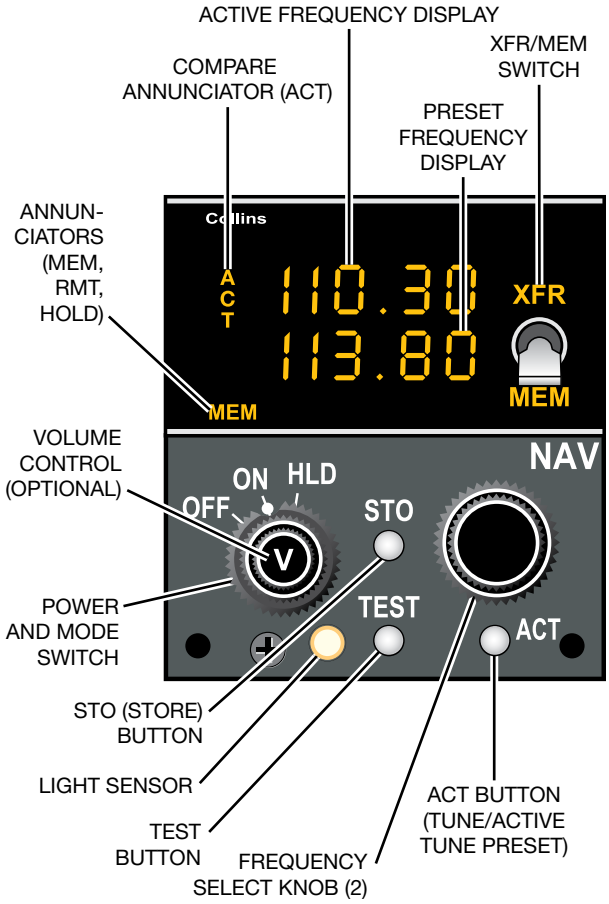

Figure 16-79. Pro Line II NAV Control

Figure 16-80. Pro Line II DME Control

In a single DME-42 installation, the three frequencies or channels are connected to the nav 1 and nav 2 control heads (CTL-32s) (Figure 16-81).

In a dual DME-42 installation, the number one DME-42 is only connected to the number one nav control head. Likewise, the number two DME-42 is only connected to the number two nav control head. In this configuration, each DME-42 is purposefully limited to displaying only two channels (Figure 16-82).

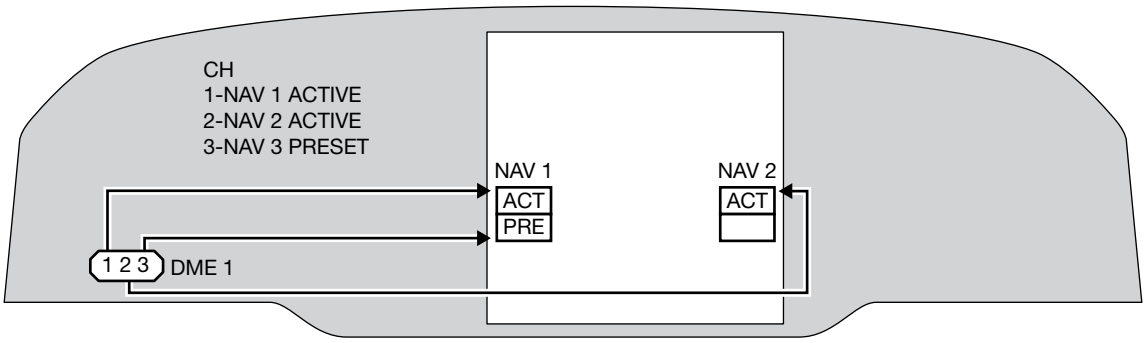

Figure 16-81. Pro Line II Single DME Installation

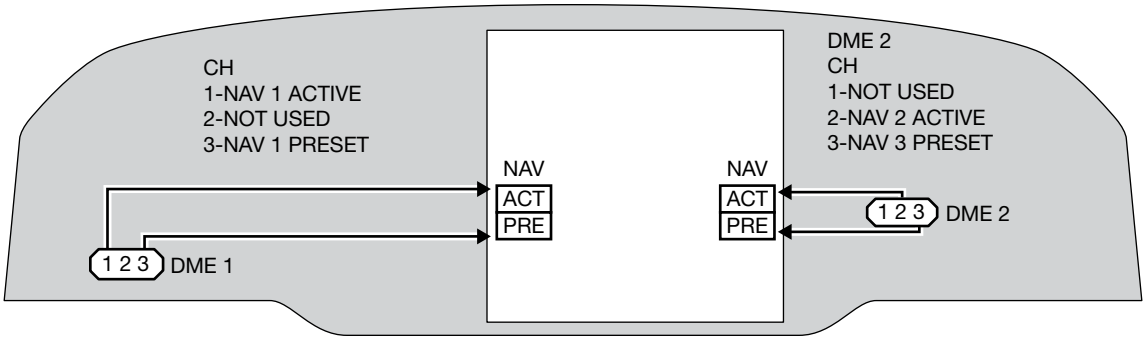

Figure 16-82. Pro Line II Dual DME Installation

COMM SYSTEM

In most respects, the VHF-22 comm works just like the features previously explained on the VIR-32 navigation receiver (Figure 16-83). The primary differences are as follows:

- There are six frequency memory positions instead of four.
- In place of the HLD annunciator, there is a TX (unit transmitting) annunciator.
- SQ OFF (squelch off) replaces HLD on the mode selector.
- Two short tones indicate a fault. Push the test button to display fault code.
- Continued turning of the small knob results in 50 kHz steps. When reversed one click, however, a 25 kHz step results.

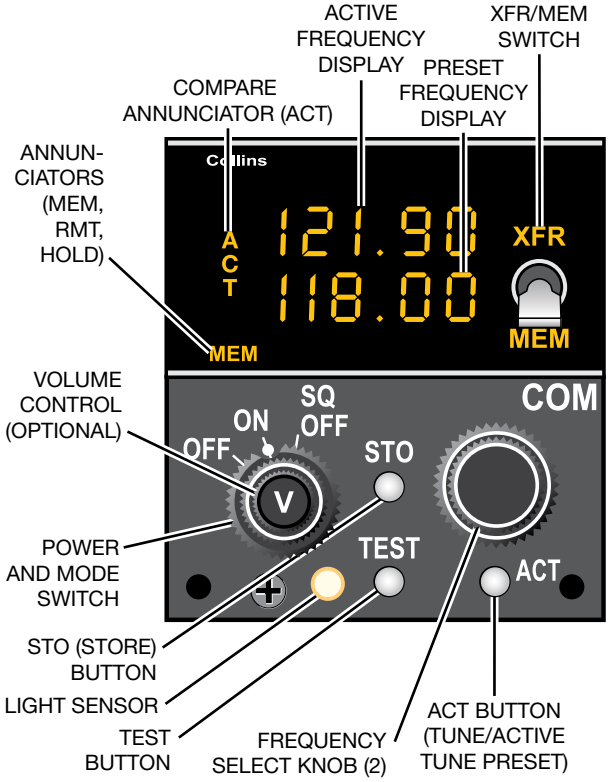

Figure 16-83. Pro Line II COM Control

ADF SYSTEM

The ADF control head also works like the nav control head in many respects, however, the modes on the mode selector switch are appropriate to an ADF (Figure 16-84).

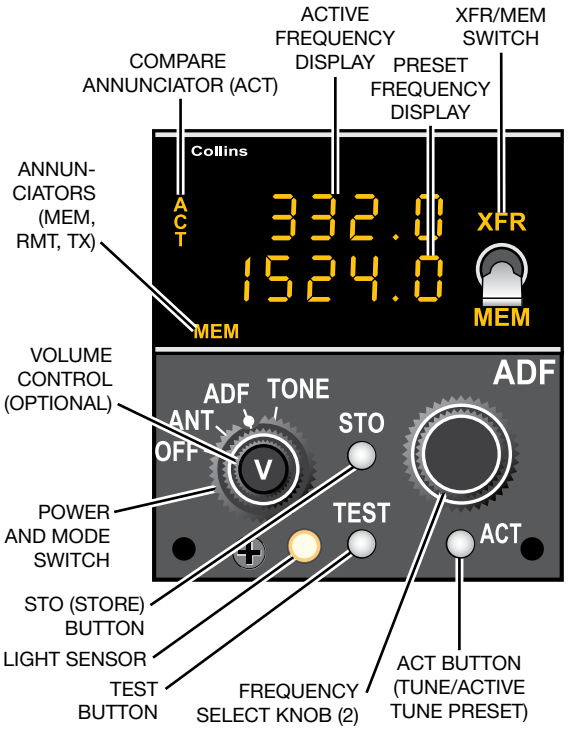

Figure 16-84. Pro Line II ADF Control

Although the ADF utilizes one of the new Pro Line II control heads (CTL-62), the actual ADF unit is of an older generation and it does not display test codes.

TRANSPONDER SYSTEM

Like the ADF, the transponder unit is of an earlier generation has a new Pro Line II control head (CTL-92), and does not display test codes (Figure 16-85).

The transponder control head can store one preselected code, such as 1200, ready for use at the push of the PRE button.

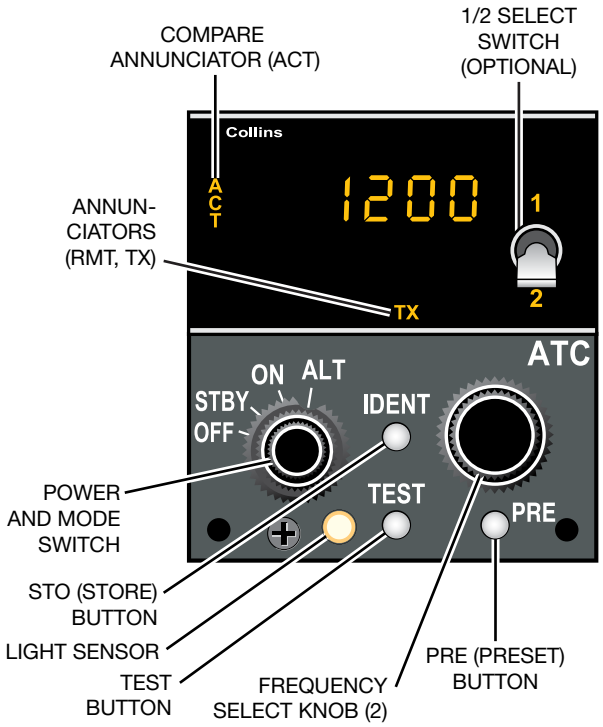

Figure 16-85. Pro Line II Transponder Control

AUDIO SYSTEM

The majority of the King Air C90s built to date have a version of the DB-Series audio system installed. Although other optional audio systems may be installed, the standard DB-Series system is the one described in this section.

COMPONENTS

The audio system consists of the following:

- An audio control panel (Figure 16-86)
- Two flight compartment speakers with jacks for pilot and copilot headphones and microphones
- Dual audio amplifiers
- Passenger speaker amplifier
- Aural warning tone generator
- Ground Communication Power system

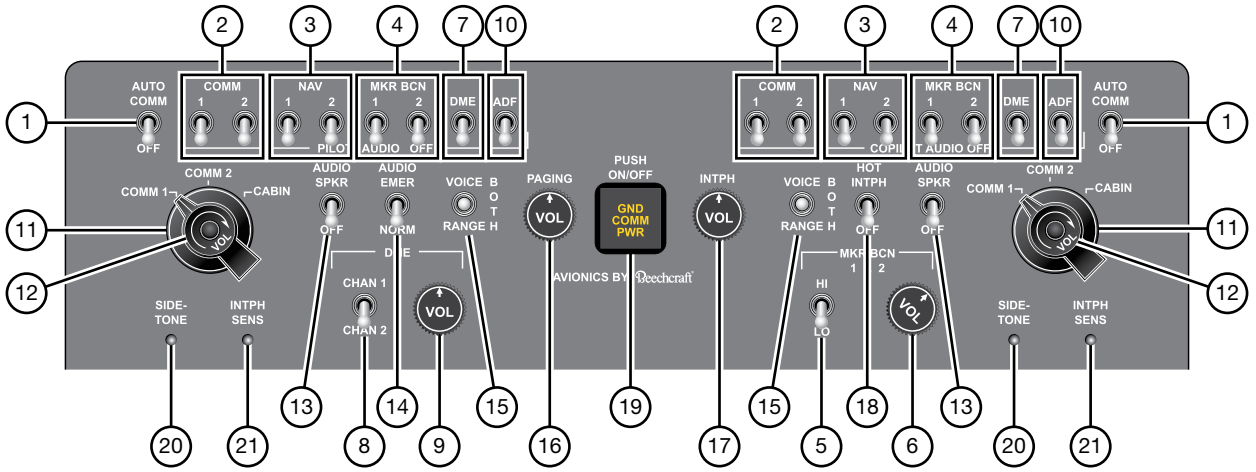

Figure 16-86. Pro Line II Audio Panel Controls

Table 16-2. PRO LINE II AUDIO PANEL CONTROLS

REF. No.	CONTROL	FUNCTION/DESCRIPTION
1	AUTO COMM	Automatically selects the received audio of the transceiver selected by the Transmit Select switch when AUTO COMM selected on.
2	COMM 1 / 2 audio	Allows COMM 1 and/or 2 to be monitored by the pilot and/or copilot side audio system.
3	NAV 1 / 2 audio	Allows NAV 1 and/or 2 to be monitored by the pilot and/or copilot side audio system.
4	MKR BCN 1 / 2 audio	Allows MKR BCN 1 and/or 2 to be monitored by the pilot and/or copilot side audio system.
5	MKR BCN 1 and 2 HI/LO	Toggle switch to HI or LO position to select on side maker beacon receiver sensitivity.
6	MKR BCN 1 and 2 VOL	Adjusts Marker Beacon 1 and 2 volume level.
7	DME 1 or 2 audio	Allows DME 1 or 2 to be monitored by the pilot and/or copilot side audio system.
8	DME CHAN 1 / 2 audio	Toggle switch to select DME channel 1 or 2.
9	DME VOL	Adjust DME volume level.
10	ADF 1 or 2 audio	Allows ADF 1or 2 to be monitored by the pilot and/or copilot side audio system.
11	Transmit Select	Selects the transceiver or PA system for the pilot or copilot side to transmit: COMM 1 Selects VHF No.1 transceiver. COMM 2 Selects VHF No.2 transceiver. CABIN Selects cabin public address system.
12	VOL Master	Controls all receiver audio volume for the pilot or copilot side headphone and cockpit speaker.
13	AUDIO SPKR	Allows the selected sources to be broadcast over the pilot or copilot side speaker.
14	AUDIO EMER/NORM	Selects normal or emergency audio path for both pilot and copilot audio. NORM Receives audio controlled by audio amplifiers. EMER Bypasses audio amplifier and receives direct audio from all audio sources.
15	VOICE/BOTH/ RANGE	Controls voice/morse filter for NAV and ADF received audio. VOICE Audio is filtered to enhance the audio voice content. BOTH Both Voice and Morse code may be heard (No filtering). RANGE Audio is filtered to enhance the Morse code identification.
16	PAGING VOL	Adjusts volume level of PA audio provided to the cabin speakers.
17	INTPH VOL	Adjusts interphone audio volume level.
18	HOT INTPH	Selects cockpit interphone on or off.

Table 16-2. PRO LINE II AUDIO PANEL CONTROLS

REF. No.	CONTROL	FUNCTION/DESCRIPTION
19	GND COMM PWR	Activates ground communication system when battery switch is off.
20	SIDETONE	Adjusts onside Sidetone volume level (maintenance function).
21	INTPH SENS	Adjusts onside interphone mic threshold level (maintenance function).

The audio control panel provides control over transmission and reception of all communication and navigation equipment (Table 16-2). ON–OFF switches, selector switches, and volume controls are provided for control of each individual audio system. As part of the audio control system, microphone key PTT switches are provided on respective control wheels. In addition, microphone and headphone jacks are located at each pilot and copilot station.

Two audio control amplifiers operate independently for the pilot and copilot systems. The audio amps provide audio from the radio receivers to the headphones and speakers and, also, provide interphone communication between the crew. Microphone audio is recorded at all times on the cockpit voice recorder.

The passenger speaker amplifier provides chime tones and paging audio to the cabin speakers.

The aural warning tone generator generates aural warning tones to the cockpit headphones and speaker for stall warning, landing gear warning, etc. The DB 600 aural warning tones are as follows:

- Landing gear warning tone
- Stall warning tone
- Autopilot disconnect warning tone
- Altitude alert warning tone
- Decision height warning tone
- Overspeed warning tone

Parallel aural warning outputs are provided for the pilot and copilot audio control amplifiers.

Ground Communications and Audio System Power

A ground communication system, powered by the Hot Battery Bus allows the crew to operate COMM 1 radio without powering up the main electrical system. With all electrical power off in the aircraft, pushing the GRD COMM PWR switch activates the COMM 1 radio and the pilot and copilot audio amps. The annunciator in the switch illuminates to indicate system operation. Pressing the GND COMM PWR switch off or turning the battery master or external power switch on deactivates the system. The annunciator in the GND COMM PWR switch extinguishes.

NOTE

Ensure the ground communication power switch is off before leaving the aircraft to prevent drain on the battery.

AUDIO SYSTEM OPERATION

The audio systems are totally independent of each other except that there is only one emergency/normal switch on the radio panel which affects both the pilot and copilot audio systems. Therefore, if the emergency/ normal switch is in either position, both systems will be in that mode of operation. See Table 16-2 for descriptions of normal audio panel operations. When in the *emergency mode* of operation, the following operating rules apply:

- All audio sources (COMM 1, NAV 2, ADF, etc.) are connected directly to the headphones.
- To eliminate any specific audio source, turn down the volume control on that audio source (e.g., NAV 1). This rule does not apply to COMM 1 and COMM 2.
- The volume control located on the microphone selector switch has no function in the emergency mode.

If the emergency/normal switch should fall for any reason, the audio systems may still be placed into the emergency mode of operation by pulling the two circuit breakers labeled PLT AUDIO and COPLT AUDIO located directly beneath the avionics master circuit breaker, on the main circuit breaker panel.

SLAVED COMPASS SYSTEMS

The standard compass system for the King Air C90B/GT uses two Collins MCS-65 systems. The King KCS-55A, Sperry C-14A-43 or the Collins MCS-103 systems could also be installed, especially on prior C90s. As far as the pilot is concerned, all of these systems operate in a similar manner. They will be treated as one common system in the following discussion.

Occasionally, a Collins PN-101 system will be installed on the copilot side. This system operates in a slightly different manner and will be discussed separately in this section.

MCS-65, KCS-55A, MCS-103, AND C-14A-43 SYSTEMS

From an operational standpoint, all three of these systems may be treated identically. They all require 400 Hz electrical power from an inverter. In the unlikely event that both inverters fail, these systems would be inoperative.

Each system has the following components (Figure 16-87):

Flux Sensor

Also called a flux gate or flux valve, the function of this device is to sense the earth's magnetic field relative to the airplane and convert that information into an electrical signal which represents the airplane's magnetic heading.

Slaving Amplifier

The magnetic heading signal from the flux sensor is too weak to be used directly; therefore, it is amplified (made larger or stronger) by the slaving amplifier. The output signal is then strong enough to directly drive a torque motor in the directional gyro and thus maintain the gyro rotor in alignment with magnetic north.

Directional Gyro

Once the gyro rotor is aligned with magnetic north, it will have a natural tendency to stay there for a short period of time, due to gyroscopic rigidity in space.

COLLINS MC-65, MC-103, KING KCS 55A AND SPERRY C-14A-43 (AC POWERED)

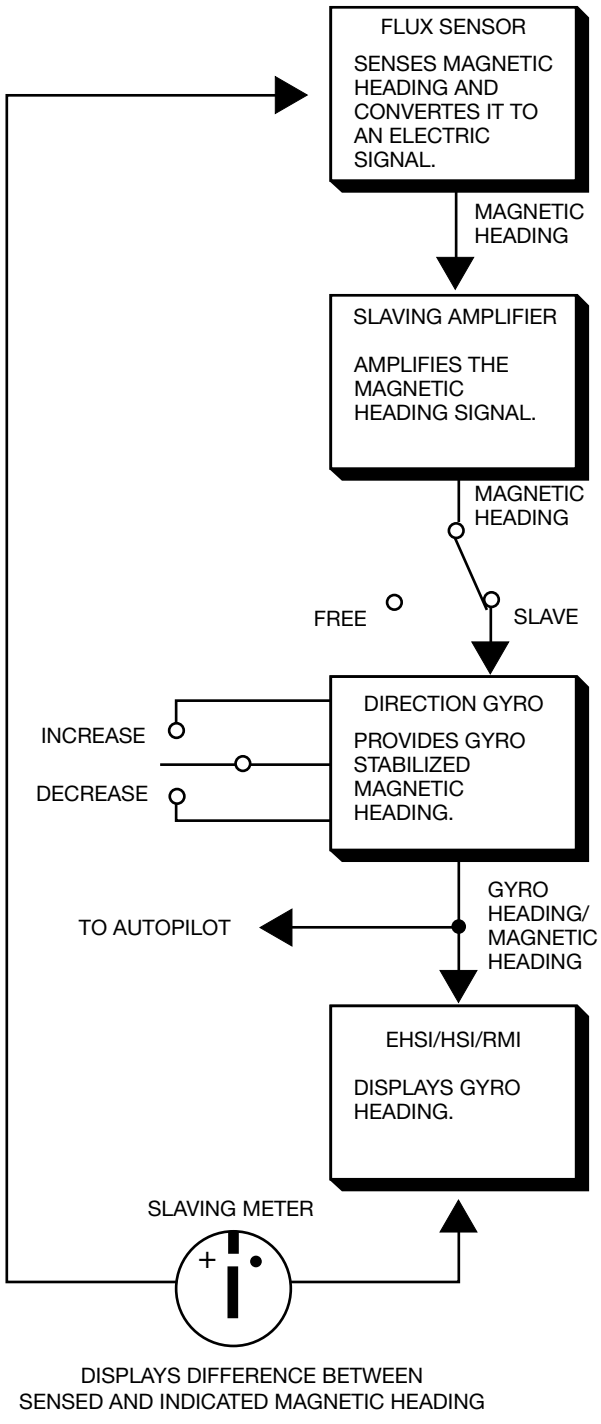

Figure 16-87. Slaved Compass System Block Diagram

This force will continue to keep the gyro mechanism in relatively good alignment as long as the gyro rotor continues to turn at its design speed. When the gyro drifts out of alignment (precesses) the condition will be sensed, and the magnetic heading reference information from the slaving amplifier will again drive the gyro rotor back into alignment with magnetic north, using the torque motor previously described.

Horizontal Situation Indicator

The gyro heading information (which should be the same as magnetic heading) is sent to a compass card on the EFIS system EHSI or standard HSI to display the magnetic heading to the pilot. This heading information is also sent to the compass card on the opposite radio magnetic indicator (RMI) (Figure 16-88). In this way, gyro-stabilized, magnetic heading information is displayed in front of each pilot from two independent sources, the pilot and the copilot compass systems (Figure 16-87).

Slaving Meter

The slaving meter (Figure 16-89) compares the sensed magnetic heading at the flux sensor (system input) to the displaced magnetic heading at the HSI (system output). The difference, if any, is displayed on the slaving meter by displacement of the slaving needle from the center position (which indicates synchronization or zero error). It is normal for this needle to deviate occasionally due to precession, however, it should always come back to center. If it is displaced to one side for more than approximately one minute the gyro may be precessing excessively and/or the slaving system may not be doing its job. In any case, the accuracy of the compass system should be checked by

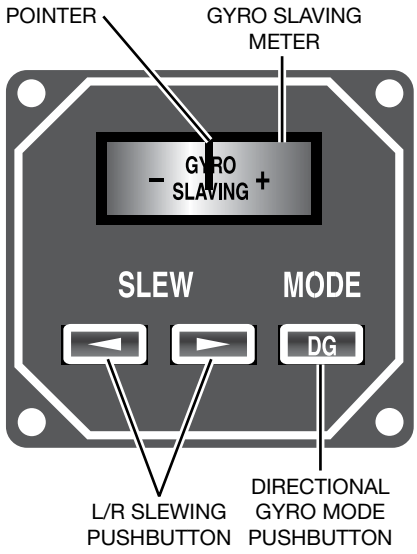

Figure 16-89. Compass Compensation Unit

cross referencing the heading information from the opposite system and/or the magnetic compass.

Slave or Free Mode Selection

The DG push-button (Figure 16-89) or lever-locking SLAVE/FREE switch is used to select either the slaved or the free mode of operation for the compass system.

This switch should normally remain in the slaved mode of operation. In this mode, when power is initially applied to the system, it will automatically "slave" itself to the correct magnetic heading and remain there throughout the flight, correcting for precession as necessary.

The free (DG) mode of operation is generally reserved for occasions when the slaved (automatic)

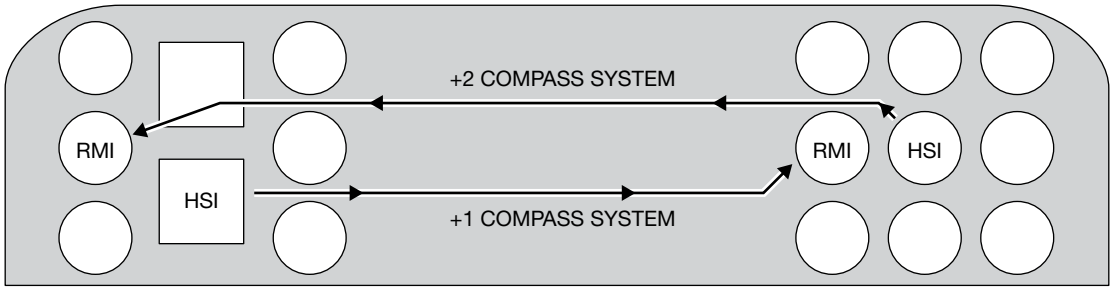

Figure 16-88. Compass System Connections

mode of operation has failed and the pilot wishes to revert to a directional gyro mode of operation. This mode may also be used for flight in polar regions where extreme levels of magnetic variation exist. In this mode of operation, the flux sensor and the slaving amplifier are disconnected from the rest of the system. The result is that the pilot now has a directional gyro (which will precess and must be corrected manually using the increase/decrease switch) which uses the HSI to display the heading information from the directional gyro. The MCS-65 "DG" mode button is lit when the free/DG mode is selected and a DG 1 or 2 annunciator appears on the EHSI display driven by that compass system. Going from DG to slaved mode causes the gyro to fast slave to the current magnetic heading.

Directional Gyro Correction

Slewing Pushbuttons

The right and left SLEW pushbuttons (Figure 16-89) are momentary action buttons that are operational in both the DG and slaved modes. Pushing either SLEW button causes the directional gyro to slew toward the selected direction.

Increase/Decrease switch

This is a toggle switch which is spring loaded to the center (OFF) position. The switch may be used when in the free mode of operation to manually change the directional gyro to the left or right, thus increasing or decreasing the displayed heading information. When in the slaved mode of operation, momentarily holding this switch in either position causes the system to "reset" itself to the fast-slave mode of operation, thereby correcting any displayed error at a rapid rate. This could be helpful if for any reason the gyro had tumbled or precessed excessively.

COLLINS PN-101 SYSTEM

This compass system is sometimes installed on the copilot side (especially prior to the C90B). It has the advantage of being directly powered by the 28-volt DC electrical system. If both inverters fail, the system would continue to operate. However, the PN-101 system does not have a manual back-up mode of operation (FREE) if the slaving system (flux sensor and/or slaving amplifier) fails. The PN-101 system does have a fast-slave switch which may be momentarily held in the UP position to initiate the fast-slaving sequence (see fast-slaving explanation under increase/decrease switch above). Except for the differences mentioned here, the basic operation of this system is virtually identical to that of the Collins MCS-65, MCS-103 and the Sperry C-14A-43 systems described previously.

PITOT-STATIC SYSTEM

DESCRIPTION

This section presents a description and discussion of the pitot-static system. The dual pitot-static system is vital to airspeed, altitude and vertical speed indications in the airplane. The principle of operation, sources of static and pitot pressure, instruments that depend on the system, and the alternate static air source are covered.

PITOT AND STATIC SYSTEM

The pitot and static system (Figure 16-90) provides a source of impact air and static air for operation of the flight instruments.

A heated pitot mast is located on each side of the lower portion of the nose. Tubing from the left pitot mast is connected to the pilot airspeed indicator and tubing from the right pitot mast is connected to the copilot airspeed indicator. The pilot system is completely independent of the copilot system.

The normal static system provides two separate sources of static air: one for the pilot flight instruments and one for the copilot. Each of the normal static air lines opens to the atmosphere through two static air ports. One port is on each side of the aft fuselage with four total ports.

An alternate static airline is also provided for the pilot flight instruments. In the event of a failure of the pilot normal static air source (if for example, ice accumulations should obstruct the static air ports), the alternate source can be selected by lifting the red spring clip retainer of the PILOT STATIC AIR SOURCE valve handle, located on the right side panel

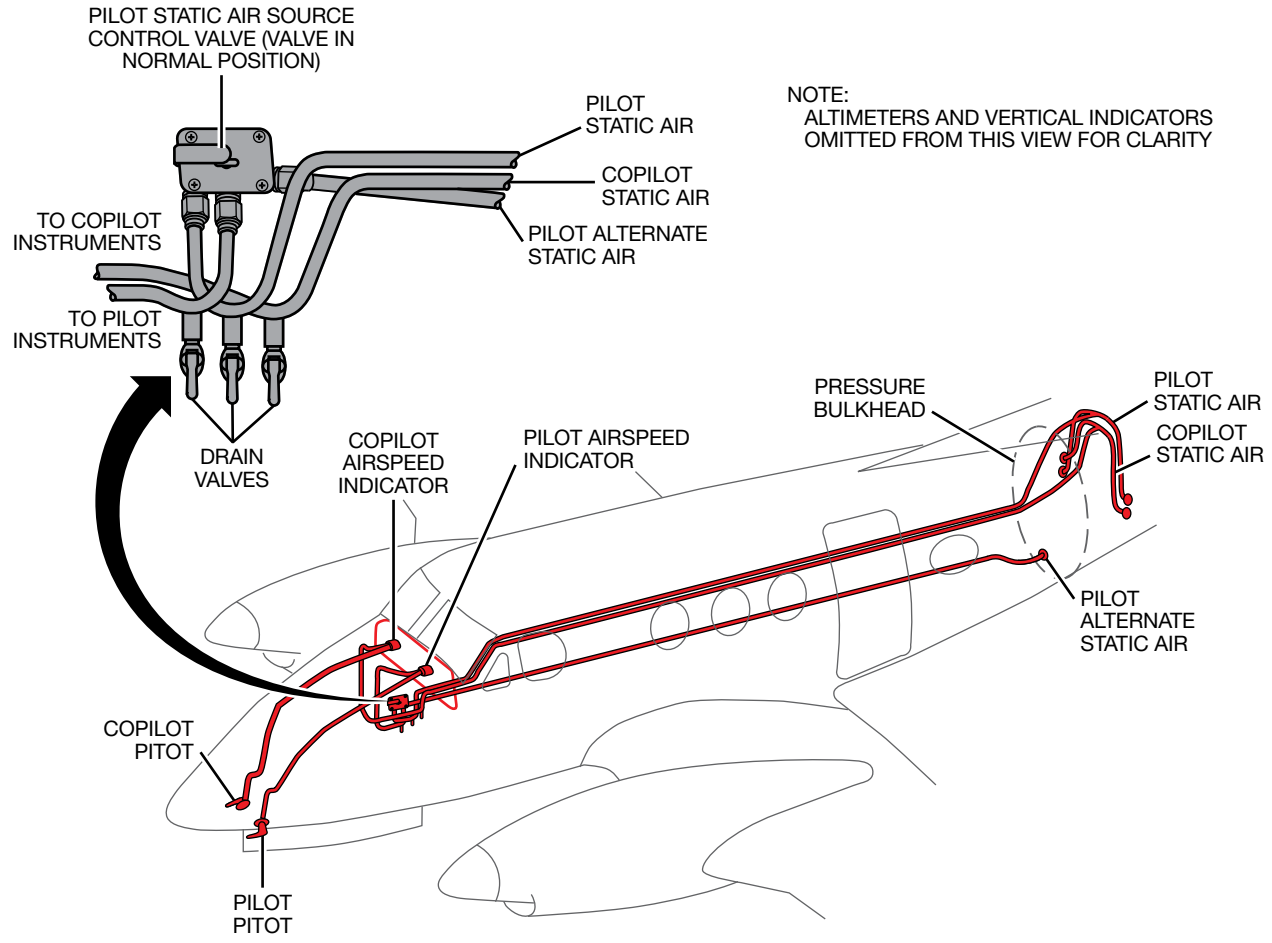

Figure 16-90. Pitot and Static System Schematic

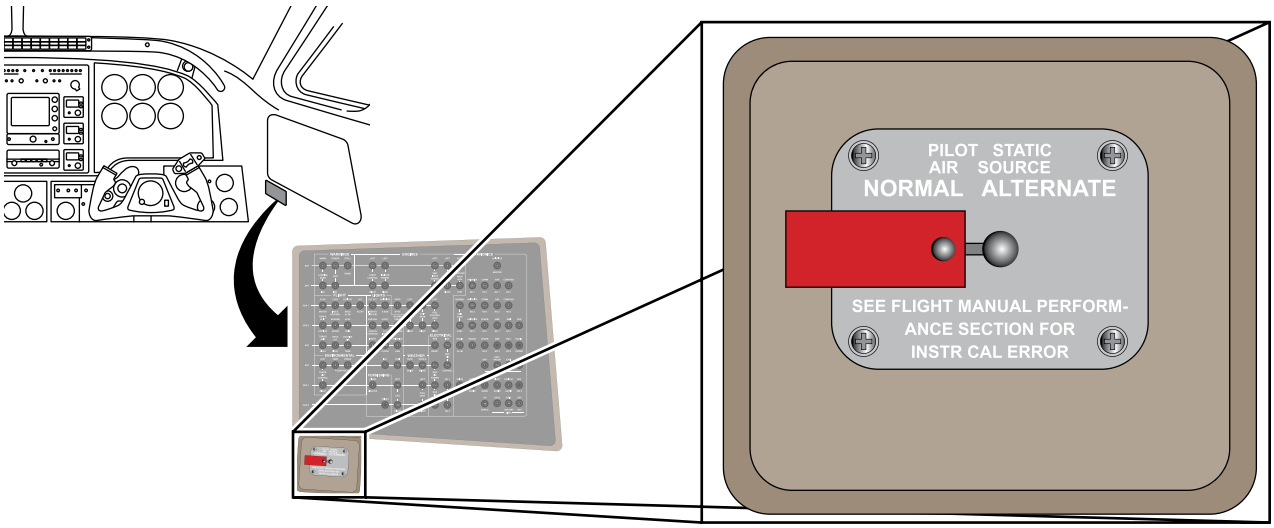

Figure 16-91. Pitot-Static Normal Alternate Air Source Valve

(Figure 16-91), and moving the handle aft to the ALTERNATE position. This will connect the alternate static air line to the pilots flight instruments. The alternate line obtains static air just aft of the rear pressure bulkhead, from inside the unpressurized area of the fuselage.

The pilot altimeter, vertical speed indicator, and airspeed indicator are connected to the pilot static air source. When the system is switched to the pilot alternate air source, the pilot altimeter and vertical airspeed indicator are affected, as well as the pilot airspeed indicator. With alternate air, the pilot airspeed indicator and altimeter will read higher that actual, and the vertical speed indicator will show a momentary climb. The copilot airspeed indicator, altimeter, and vertical speed indicator are all on the copilot static air source and cannot be switched to the alternate source (Figure 16-92).

Refer to the Airspeed Calibration-Emergency System, and the Altimeter Correction-Emergency System graphs in the Performance section of the *Pilot Operation Handbook* for operation when the alternate static air source is in use.

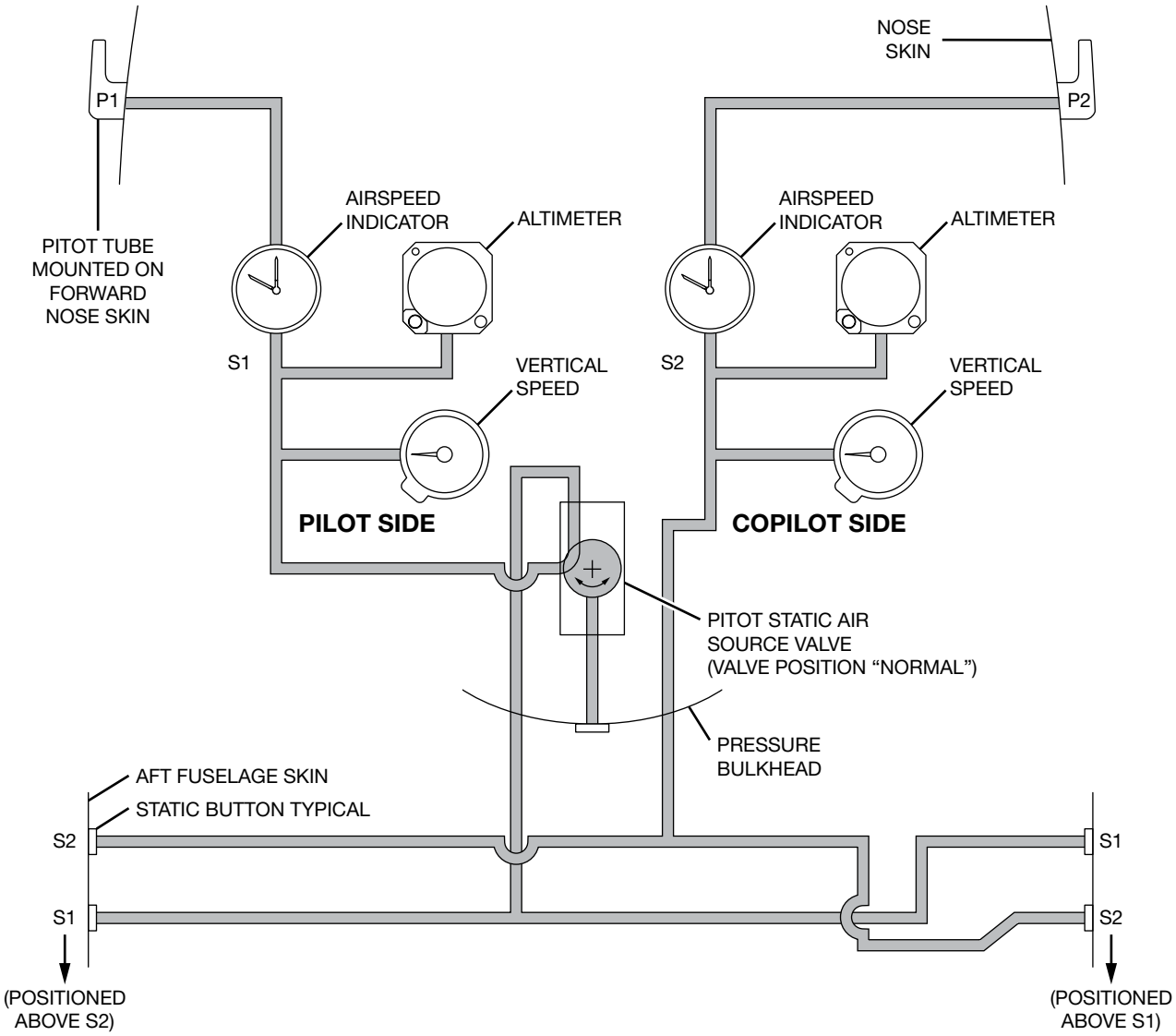

Figure 16-92. Schematic Diagram of Pitot and Static System

When the pilot system is switched to ALTERNATE, use these graphs to determine the required Indicated Airspeed to maintain a desired Calibrated Airspeed and the required Indicated Altitude to maintain a desired Pressure Altitude.

When the alternate static air source is not needed, ensure that the PILOT STATIC AIR SOURCE valve handle is held in the forward (NORMAL) position by the spring clip retainer.

Three petcocks are provided to facilitate draining moisture from the static air lines. They are located behind an access cover below the circuit breakers on the right side panel. These are primarily intended for maintenance personnel and the drain valves should be opened to release any trapped moisture at each 100-hour inspection, and after exposure to visible moisture on the ground. They must be closed after draining. If the drains were to be opened in-flight, pressurized air form the cabin would rush into the static ports of the instruments, resulting in possible instrument damage.

The pitot masts can be heated electrically for flight in icing conditions. The pitot heat switches are located in the lower right hand corner of the ICE PROTECTION control panel in the right hand pilot subpanel (Figure 16-93). It is customary, as a precautionary measure, to have the pitot heat ON during flight when there is visible moisture and temperatures are near or below +5°C. There is no restriction on use of pitot heat except not to use it excessively on the ground where there is no air flow around the masts.

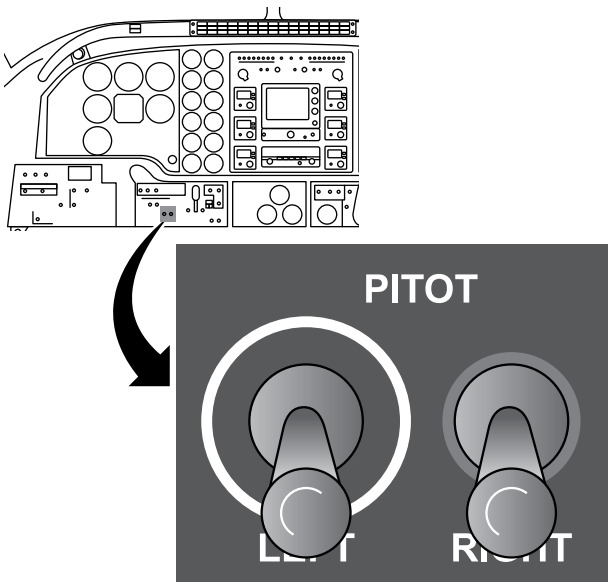

Figure 16-93. Ice Protection Switches

KING SILVER CROWN II EQUIPMENT

The King Silver Crown II line of panel-mounted avionics equipment is installed on many King Airs. Although not all equipment types in the Silver Crown II line will be discussed here, all of the main units typically installed in a King Air will be addressed. For additional information on each system, please consult the appropriate *Pilot's Guide*.

AUDIO CONTROL SYSTEM

The KMA 24 Audio Control System (Figure 16-94) consists of a rotary microphone selector switch, speaker and phone switches for each receiver installed in the aircraft, and an integral marker beacon receiver with marker beacon lights.

The microphone selector switch connects the microphone to each transmitter installed on the aircraft. On versions of the KMA 24 which do not have the capability of handling audio from a Number 2 ADF, the proper comm receiver audio switch may be automatically selected by simply pushing either the speaker or phone AUTO switch to the ON position and placing the microphone selector switch to the desired transmitter.

Versions of the KMA 24 which have the capability of handling audio from a second ADF do not have the AUTO buttons and therefore the appropriate comm receiver must be manually selected each time the microphone selector switch is changed to a different transmitter.

Each receiver may be connected to either the speaker and/or the phones by pushing the appropriate alternate action pushbutton switch to the in or ON position.

The marker beacon receiver has a HIGH (button pushed in) and a LOW (button out) sensitivity position. The marker beacon lights may also be tested by pushing and holding the test button in. A built in photo cell automatically adjusts the lighting intensity depending on ambient lighting conditions.

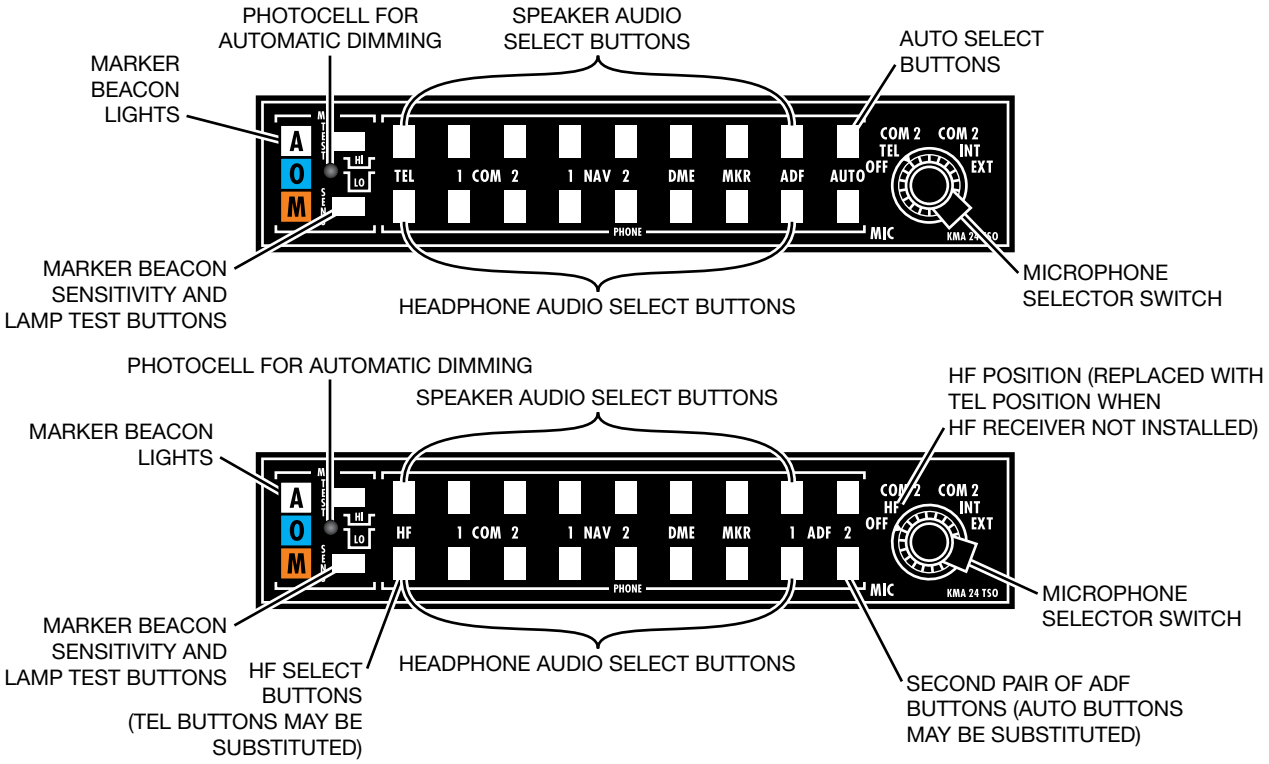

Figure 16-94. King Audio Control System

COMMUNICATIONS TRANSCEIVER SYSTEM

The KY 196 (Figure 16-95) is capable of transmitting and receiving a frequency range of 118.0 through 135.975 MHz in either 25 or 50 kHz steps.

The large frequency knob changes the frequency to the left of the decimal point while the smaller knob changes the frequency to the right of the decimal point. The smaller knob makes 50 kHz changes when pushed in and 25 kHz changes when pulled out.

The ON/OFF/VOLUME control switch turns the unit on when rotated clockwise past the initial detent. Further clockwise rotation increases the volume level. Pulling out this control "opens up" the receiver squelch circuit, enabling the pilot to hear weaker stations. This might be an appropriate action when attempting to receive a weak transmitter from a distance, such as listening to an ATIS at a distant point.

The left frequency display indicates the frequency to which the transceiver is actively tuned. The right display indicates the "standby" frequency. In order to transfer or swap the two frequencies, the pilot pushes the transfer button momentarily. (The frequency selector knob only changes the "standby" frequency.)

Transmitter operation is annunciated by the illumination of the letter "T" located between the active and standby frequencies.

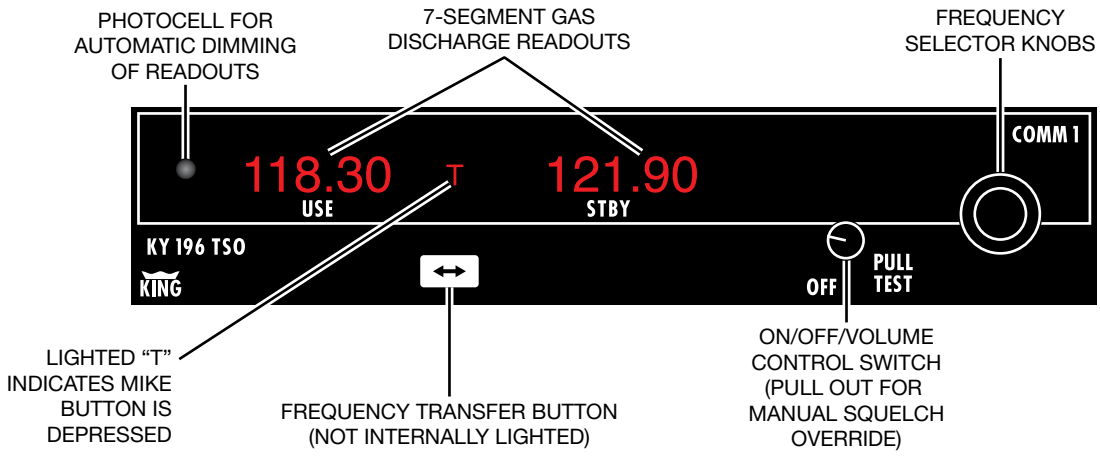

Figure 16-95. King Communications Transceiver

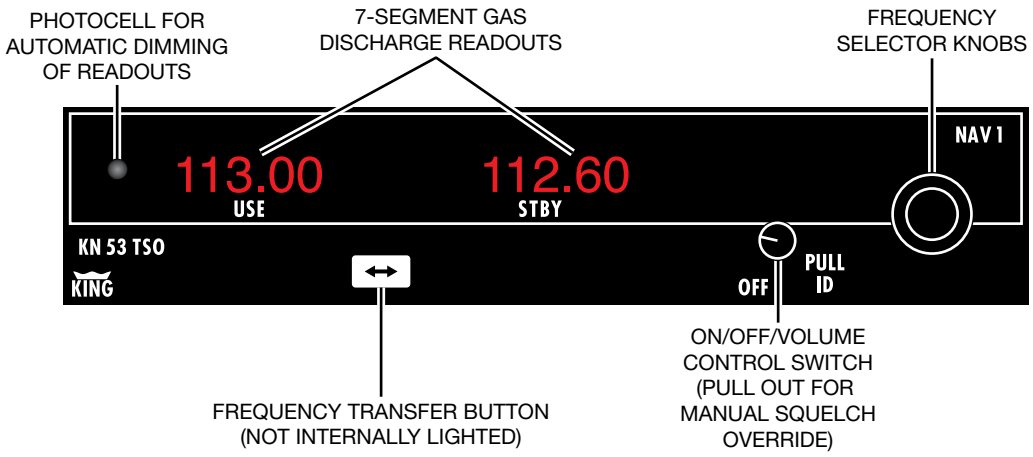

Figure 16-96. King VOR/LOC/GS Receiver

VOR/LOC/GS RECEIVER SYSTEM

Operation of the KN 53 (Figure 16-96) is virtually identical to that of the KY 196 comm transceiver with the following exceptions:

- Pulling out on the volume control knob activates the Morse code identification circuit, thus allowing the "ident" to be heard through the audio system.
- There is no transmit annunciator on this system.

DME SYSTEM

The remote-mounted KN 63 DME with the KDI 572 panel-mounted indicator operates in a straightforward manner. The indicator is capable of displaying DME distance, ground speed, and time to station simultaneously (Figure 16-97).

The mode selector allows the unit to be channeled by either nav 1 or nav 2. Selecting the HLD (hold) position allows the DME to remain channeled to the previously selected frequency, and is annunciated by either H1 or H2 depending on whether nav 1 or nav 2 was previously used. The mode selector also allows the DME to be turned off.

RNAV SYSTEM

For operational information on the KNS 81 system, refer to the appropriate flight manual supplement.

ADF SYSTEM

The KR 87 ADF (Figure 16-98) has two basic modes of operation, ANT (antenna) and ADF. In the ANT mode, the bearing pointer in the RMI/ADF indicator will not point to the station but provides improved audio reception. The ADF mode is used for navigation purposes, allowing the bearing pointer to point to the station. The ADF mode is selected by pushing the alternate action pushbutton in, and the ANT mode is selected by allowing the same pushbutton to remain in the "out" position. The selected mode is annunciated on the left side of the ADF display.

This unit incorporates a BFO (beat frequency oscillator) circuit which allows non-directional beacons to be identified which are not modulating the carrier with audio. These types of stations are sometimes used outside of the United States. The BFO circuit, when activated by pushing the BFO pushbutton to the "in" position, generates a 1020 Hz tone which will be heard each time the NDB transmitter is turned on. This allows the Morse code to be identified in a normal fashion.

As with the KY 196 comm and KN 53 nav, two frequencies may be displayed on the KR 87 ADF. The frequency on the left is always the frequency in use, however, the right display window is shared by several different functions. Like the comm and the nav, the right window may display the standby frequency. However, pushing the

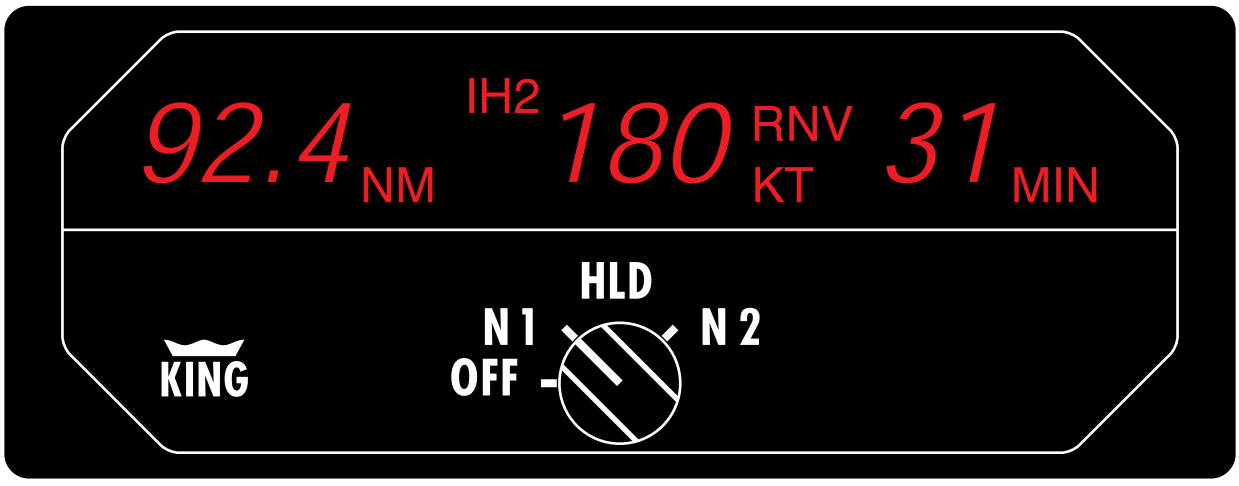

Figure 16-97. King DME System

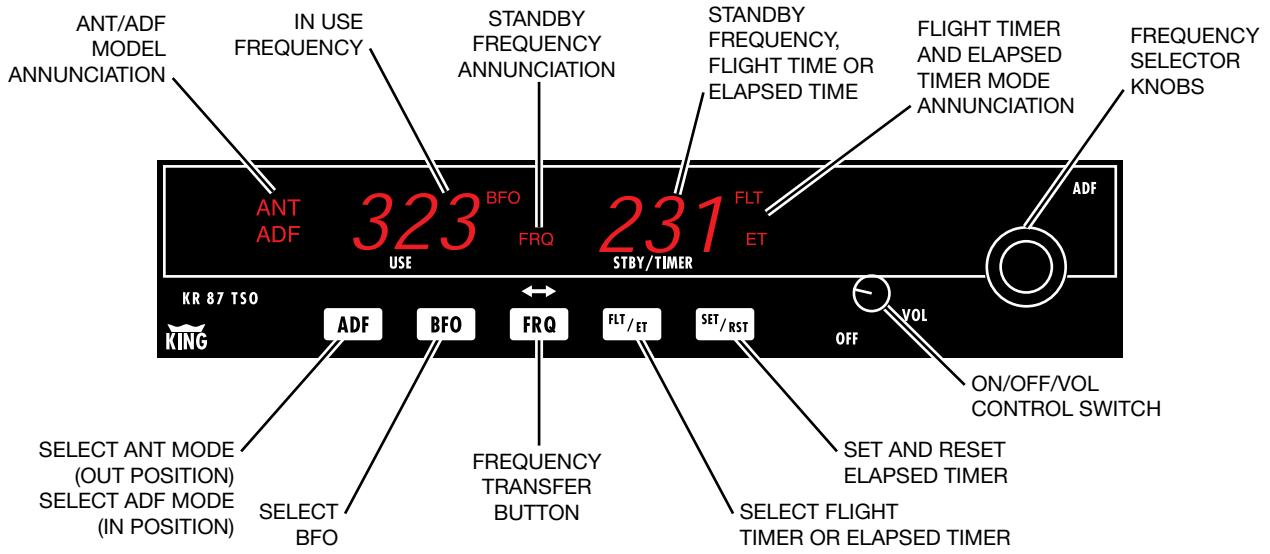

Figure 16-98. King ADF System

FLT/ET alternate action pushbutton changes the function of the right display window.

When FLT is annunciated to the right of the right display window, the display is being used to display flight time. Initially, the flight timer begins operation when the unit is turned on. Then, during takeoff, the flight timer is reset to zero and begins counting again when the weight of the aircraft is off the landing gear "squat switch." The flight timer continues to operate until the weight of the aircraft is once again on the landing gear squat switch, at which time the display is "frozen," and will remain so until power is removed or the aircraft takes off again.

Pushing the FLT/ET button again will switch the right window to display an elapsed time function. The elapsed timer may be reset to zero at any time by momentarily pushing the SET/RST (set/reset) button. Elapsed time will continue to accumulate until the SET/RST button is pushed again or power is removed.

The elapsed timer also has a "countdown" mode of operation, which may be initiated by holding the SET/RST button in for approximately three seconds, or until the ET annunciator begins to flash. Now, the countdown time (in minutes and seconds) may be set into the right display by rotating the two concentric knobs which are normally used to change the frequency. Set the minutes with the large knob and the seconds with the small knob.

In order to start the countdown cycle (as when passing the final approach fix) push the SET/RST button. Time remaining will now be continuously displayed until the timer reaches zero, at which time it will revert to a count up mode of operation and will now automatically display the elapsed time above and beyond that which was originally set in. Additionally, when the countdown mode switches to the count up mode, the right display window will flash for fifteen seconds in order to alert the pilot to the fact that he has gone beyond the originally preset time.

With both the flight and elapsed timer, the display will initially be read in minutes and seconds (up to 59 minutes and 59 seconds (59:59). After the first hour, these timers will display hours and minutes.

Pushing the FREQ (frequency transfer) button initially changes the right window back to the standby frequency display. Subsequent pushes of the FREQ button transfers the standby and in-use frequencies back and forth (flip-flops).

QUESTIONS

1. If the Avionics Master switch fails, how can the pilot restore avionics power?

A. Reset the Avionics Master switch
B. Select the other inverter
C. Pull the Avionics Master circuit breaker
D. Avionics power cannot be restored

2. The active lateral and vertical flight director mode annunciations are _____.

A. White
B. Green
C. Amber
D. Red

3. The armed lateral and vertical flight director mode annunciations are _____.

A. White
B. Green
C. Amber
D. Red

4. When GPS is the ACTIVE navigation source on the HSI, the needle color will be _____.

A. Green
B. Magenta
C. White
D. Cyan

5. When VOR or LOC is the PRESET navigation source on the HSI, the needle color will be _____.

A. Green
B. Magenta
C. White
D. Cyan

6. When VOR or LOC is the ACTIVE navigation source, the needle color will be _____.

A. Green
B. Magenta
C. White
D. Cyan

7. How many memory frequencies can be stored in each of the NAV radios?

A. 2
B. 4
C. 6
D. 8

8. How many memory frequencies can be stored in each of the COMM radios?

A. 2
B. 4
C. 6
D. 8

9. With the VOICE/BOTH/RANGE switch selected to RANGE, what will the pilot hear when listening to a NAV radio source?

A. The NAV sources Morse code identification only
B. The NAV sources DME identification
C. All information transmitted over the NAV frequency
D. Nothing, unless the NAV is out of service

10. If loss of radio communications is suspected, communication may be regained by:

A. Selecting the EMER position on the AUDIO switch.
B. Selecting the NORM position on the AUDIO switch.
C. Pulling the Avionics Master circuit breaker.
D. Resetting the Avionics Master switch.

11. When using alternate static air, the pitot-static air is being supplied from:

A. The upper static port on the pilot side.
B. The lower static port on the pilot side.
C. The upper static port on the copilot side.
D. Inside the unpressurized area of the empennage.

CHAPTER 17
MISCELLANEOUS SYSTEMS

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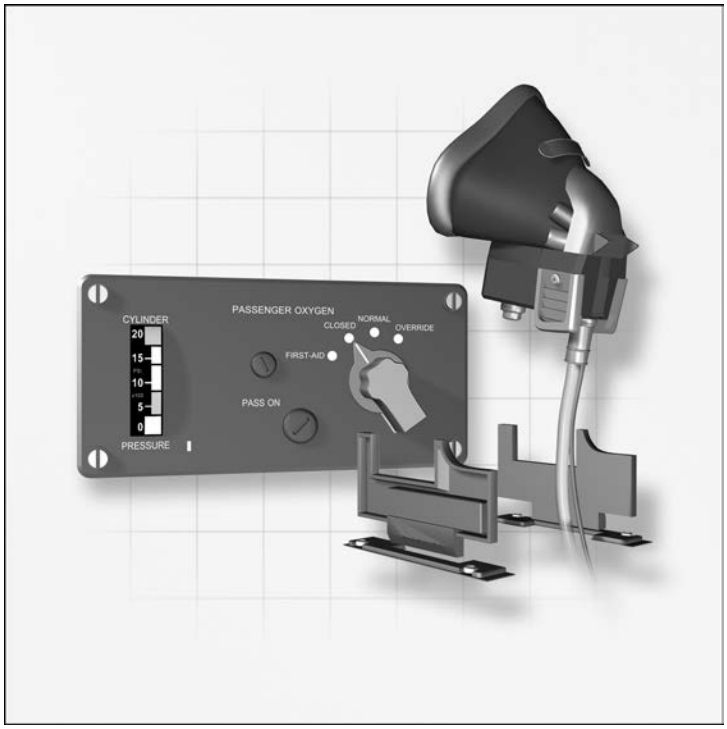
CHAPTER 17
MISCELLANEOUS SYSTEMS



INTRODUCTION

Pilot and passenger comfort and safety are of prime importance in operating this airplane. The task is to teach flight crewmembers to use the oxygen system safely and effectively, when required, within the requirements of applicable FARs.

DESCRIPTION

This chapter presents a description and discussion of the oxygen system. It includes general description, principle of operation, controls, and emergency procedures. Use of the oxygen duration chart involves working simulated problems under various flight conditions. FAR requirements for crew and passenger needs are part of the discussion, as well as the types and availability of oxygen masks. Local servicing procedures referenced in the *Pilot's Operating Handbook* are also included.

OXYGEN SYSTEM

Current FARs require that anytime an aircraft flies above 25,000 feet, oxygen must be immediately available to the crew and passengers. The King Air C90A and C90B systems comply with this requirement.

The oxygen system (Figure 17-1) provides an adequate flow for an altitude of 30,000 feet. The masks and Oxygen Duration chart (Normal Procedures section of the *POH*) are based on 3.7 LPM-NTPD. The only exception is the diluter-demand crew mask when used in the 100% mode. For oxygen duration computation, each diluter-demand mask being used in the 100% mode is counted as two masks at 3.7 LPMNTPD each.

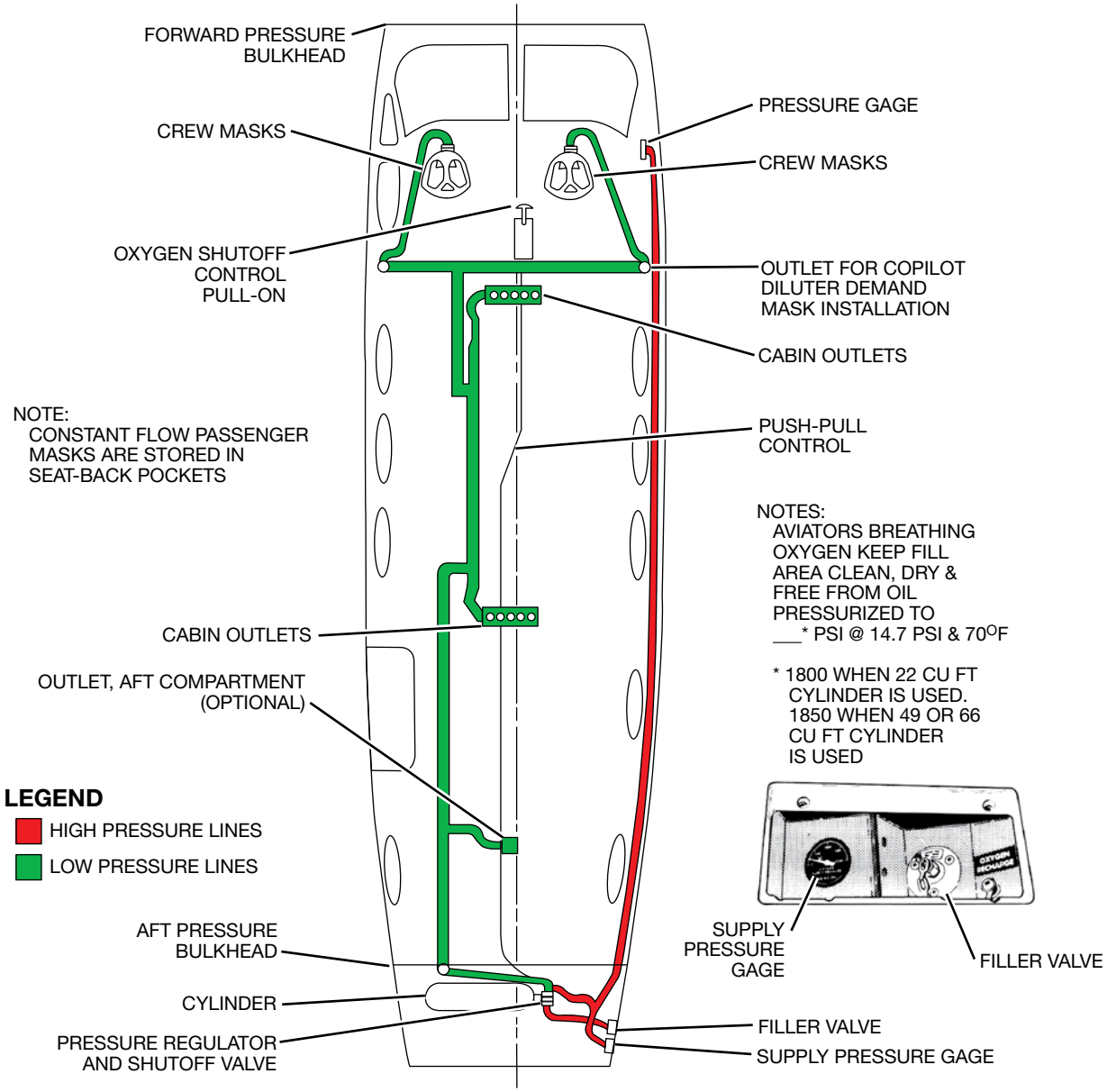


Figure 17-1. Oxygen System Schematic—Typical C90A

MANUAL PLUG-IN SYSTEM

The manual plug-in system is of the constant-flow type (Figures 17-2 and 17-3). Each mask plug is equipped with its own regulating orifice. The pilot and copilot oxygen masks are quick-donning oxygen masks and are connected to the oxygen supply lines at all times (Figure 17-4). When the diluter demand masks are not in use, one hangs from a bracket (on the stub partition) behind the pilot's head and one hangs from a bracket behind the copilot's head.

Figure 17-2. Plug-in Type Oxygen Mask

Passenger masks are kept in seatback pockets except in the couch installation, in which case they are stored under the couch. The cabin outlets are located at both the forward and aft ends of the cabin. All masks are easily plugged in by pushing the orifice in firmly and turning clockwise approximately one-quarter turn. Unplugging is easily accomplished by reversing the motion.

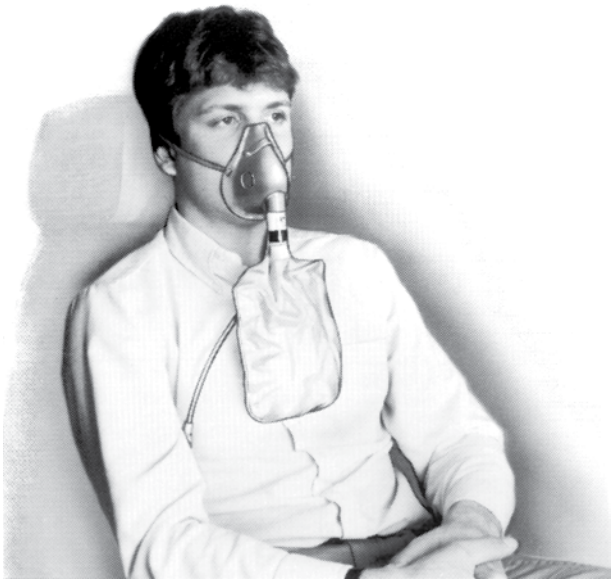


Figure 17-3. Oxygen Mask Donned

Figure 17-4. Crew Oxygen Mask

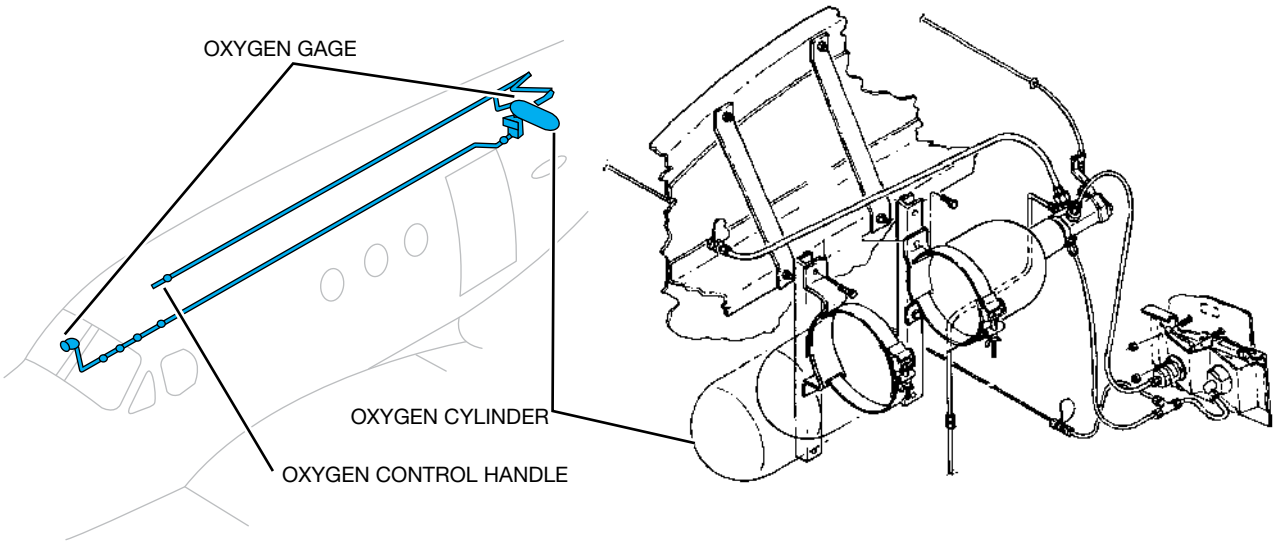


Figure 17-5. Oxygen Cylinder Installation

The oxygen supply cylinder is in the aft unpressurized area of the fuselage (Figure 17-5). The oxygen system pressure regulator and control valve are attached to the cylinder, and are activated by a remote push/pull knob located to the rear of the cockpit overhead light control panel (Figure 17-6). When this control is pushed in, no oxygen supply is available anywhere in the airplane. When this control is pulled out, the oxygen system is charged with oxygen ready for use provided the oxygen supply cylinder is not empty. The oxygen supply pressure gage is located in the copilot's right subpanel (Figure 17-7).

Figure 17-6. Oxygen System Control Handle

Figure 17-7. Oxygen Pressure Gage

DILUTER-DEMAND CREW OXYGEN MASKS

The crew are provided with diluter-demand, quick-donning oxygen masks (see Figure 17-4). These masks hang on the aft cockpit partition behind and outboard on the pilot and copilot seats. They are held in the armed position by spring tension clips, and can be donned immediately with one hand. The diluter-demand crew masks deliver oxygen to the user only upon inhalation.

Consequently, there is no loss of oxygen when the masks are plugged in and the PULL ON handle is pulled out, even though oxygen is immediately available upon demand.

A small lever on each diluter-demand oxygen mask permits the selection of two modes of operation: NORMAL and 100%. In the NORMAL position, air from the cockpit is mixed with the oxygen supplied through the mask. This reduces the rate of depletion of the oxygen supply, and it is more comfortable to use than 100% aviator's breathing oxygen. However, in the event of smoke or fumes in the cockpit, the 100% position should be used to prevent the breathing of contaminated air. For this reason, the selector levers should be left in the 100% position when the masks are not in use so the masks are always ready for maximum emergency use.

PLUG-IN MASKS

The plug-in oxygen masks in the cabin (see 17-2) are designed to be adjustable to fit the average person with minimum leakage of oxygen. To don the mask, fit the nose and mouth piece over the face and adjust the elastic headband over the head to hold the mask firmly in place. Insert the fitting in one of the oxygen outlets in the overhead cavity, push in firmly, and turn clockwise approximately one-quarter turn to lock it in place. If oxygen is available (the system is turned on and the oxygen cylinder charged), the red flow indicator will move and the green portion will come into view. The mixing bag will inflate with breathing. Breath normally. System efficiency is determined by the fit of the oxygen mask. Make certain the masks fit properly and are in good condition. The hose plug must be disconnected to stop the flow of oxygen.

There are certain important considerations any time oxygen is in use. Do not use combustible products near oxygen. Common items such as chapstick, lipstick, women's makeup, or mustache wax could spontaneously ignite in the presence of oxygen. These items should be removed before using oxygen. No smoking should be allowed in the airplane when oxygen is in use.

OXYGEN SUPPLY CYLINDER

Oxygen for flight at high altitudes is supplied by a cylinder mounted behind the aft pressure bulkhead. The cylinder is filled by a valve accessible through an access door on the right side of the aft fuselage. The high-pressure system has two pressure gages, one on the copilot's RH sub-panel in the cockpit for in-flight use (see Figure 17-7), and one adjacent to the filler valve for checking the pressure of the system during filling (Figure 17-8). The cylinder is available in three different capacities: 22 cubic feet, 49 cubic feet, or 66 cubic feet.

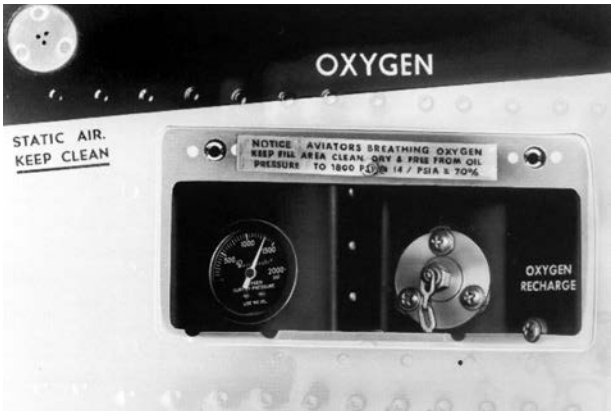


Figure 17-8. Oxygen Pressure Gage

OXYGEN SYSTEM CONTROLS

A shutoff valve regulator in the cylinder is actuated by its a push-pull shutoff control located overhead between the pilot and copilot seats (see Figure 17-6). Pushing in the handle deactivates the oxygen supply, while pulling out the handle actuates the oxygen supply. The regulator is a constant-flow type which supplies low-pressure oxygen through aluminum plumbing to the outlets.

OXYGEN DURATION

A preflight requirement is to check the oxygen available, considering the number of crew and passengers, to assure that it is sufficient for descent to 12,500 feet, or until loss of pressure in the airplane can be corrected and cabin altitude pressure restored. Full oxygen system

pressure is 1800 ±50 psi at 70° F for the 22 cubic feet cylinder, and 1850 ±50 psi for the larger cylinders. First, read the oxygen pressure gage and note the pressure. Determine from the OXYGEN AVAILABLE WITH PARTIALLY FULL BOTTLE graph the percent of usable capacity. To obtain the duration in minutes of the supply, obtain the duration for a full bottle from the Oxygen Duration table, considering the number of persons aboard. Multiply the full bottle duration by the percent of full bottle available to obtain the available oxygen duration in minutes.

On the C90A or C90B airplane, oxygen duration is for a Puritan-Zep oxygen system which must use the red, color-coded, plug-in mask, rated at 3.7 standard liters per minute–normal temperature pressure (SLPM–NTPD) flow. Both aircraft are approved for altitudes up to 30,000 feet.

OXYGEN DURATION COMPUTATION

In this sample computation, oxygen duration is computed for a Puritan-Zep oxygen system which utilizes the red, color-coded, plug-in mask rated at 3.7 standard liters per minute (SLPM) flow and is approved for altitudes up to 30,000 feet. This table is also used for the quick-donning, diluter-demand crew oxygen masks. When selected to the 100% mode, the number of crew masks in use should be doubled for computation. To compute oxygen duration for four passengers and two crew members using their masks in 100% mode, consider eight people using oxygen.

To compute the duration in minutes of available oxygen for eight people, assume the pressure gage shows 1,500 pounds. Enter the Percent of Usable Oxygen Capacity chart (Figure 17-9) at 1,500 pounds and read across to intersect the 32° F diagonal, then down to read 85% of usable capacity. To compute the duration available, enter the Oxygen Duration chart (Table 17-1) at the 8-people-using column and read down to 55 minutes available for a 66 cubic-foot supply bottle. Now take 85% of 55 and find the current oxygen duration available of approximately 46 minutes.

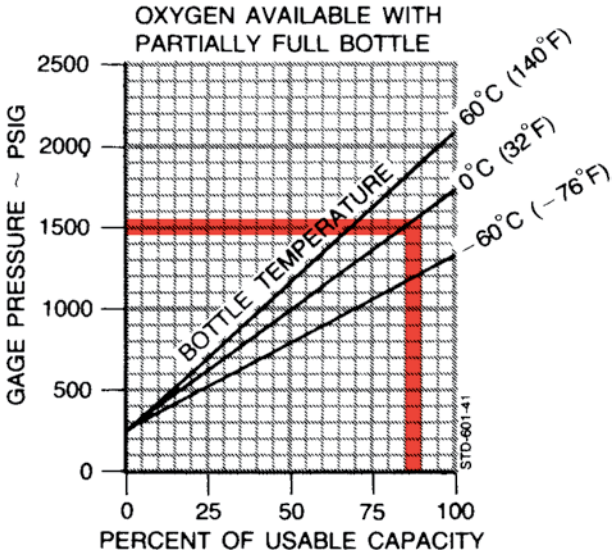


Figure 17-9. Percent of Usable Oxygen Capacity

Table 17-1. OXYGEN DURATION (MINUTES)

CYL VOL CU FT	NUMBER OF PEOPLE USING*											
	1	2	3	4	5	6	7	8	9	10	11	12
22	151	75	50	37	30	25	21	18	16	15	13	12
49	334	167	111	83	66	55	47	41	37	33	30	27
66	445	222	148	111	89	74	63	55	49	44	40	37

TIME OF USEFUL CONSCIOUSNESS

In the event of decompression at altitude, the primary need is for oxygen to prevent hypoxia. Hypoxia is a lack of the oxygen needed to keep the brain and other body tissues functioning properly. The early symptoms of hypoxia, such as an increased sense of well-being, quickly give way to slow reactions, impaired thinking ability, unusual fatigue and a dull headache. Therefore, the crew must act quickly to don oxygen masks and supply oxygen to the passengers before the onset of hypoxia.

The ALTITUDE WARN annunciator illuminates when cabin altitude exceeds 10,000 feet (12,500 for LJ-1353 and later), should the red ALTITUDE WARN annunciator illuminate due to inadequate cabin pressure, or loss of pressurization at high altitudes, crew and passengers should don oxygen masks immediately and descend to a safe altitude.

The Time of Useful Consciousness table (Table 17-2) shows the average time of useful consciousness available at various altitudes. This is the time from the onset of hypoxia until loss of effective performance. Individuals may differ from that shown in the table. Using the Emergency Descent procedure in the Emergency Procedures section of the *POH*, a very rapid descent can minimize the exposure to hypoxia.

Table 17-2. TIME OF USEFUL CONSCIOUSNESS

ALTITUDE	TIME
30,000 feet	1 to 2 minutes
28,000 feet	2-1.2 to 3 minutes
25,000 feet	3 to 5 minutes
22,000 feet	5 to 10 minutes
12 to 18,000 feet	30 minutes or more

PHYSIOLOGICAL TRAINING

WHAT IS IT?

Physiological training is a program directed toward understanding and surviving in the flight environment. It covers the problems of both high and low altitudes and recommends procedures to prevent or minimize the human factor errors which occur in flight.

WHO NEEDS IT?

The course is primarily of benefit to pilots. It is also recommended for other air crew personnel, air traffic controllers, aviation medical examiners and other personnel from the national aviation system.

WHERE CAN YOU GET IT?

A resident physiological training course at the FAA's Aeronautical Center in Oklahoma City is devoted entirely to problems in civil aviation (Figure 17-10). Many military installations, and the National Aeronautics and Space Administration (NASA) in Houston, Texas, conduct a resident program for non-government personnel.

HOW LONG IS THE COURSE?

The course takes one full day.

WHAT IS CONTAINED IN THE COURSE?

Many topics are covered. They include the environment to which the flyer is exposed, physiological functions of the body at ground level, and alteration of some of these functions by changes in the environment. The higher one flies, the more critical becomes the need for supplemental oxygen. This need is discussed so that the trainee will understand why a pilot cannot fly safely at altitudes in excess of 12,500 feet for a prolonged period without some aid, either supplemental oxygen or a pressurized aircraft. Both oxygen equipment and pressurization are discussed.

Figure 17-10. FAA Altitude Chamber

When humans are confronted with certain stressful situations, there is a tendency to breathe too rapidly. This topic (hyperventilation) and methods of control are discussed. Ear pain on descent and other problems with body gases and procedures to prevent or minimize gas problems are explained. Alcohol, tobacco, and drugs are also discussed as they apply to flying. Pilot vertigo is discussed and demonstrated so that the trainee will understand why a non-current instrument pilot should never attempt to fly in clouds and other weather situations where visibility is reduced. Resident courses include an altitude chamber flight where the trainees experience individual symptoms of oxygen deficiency as well as decompression. This flight will demonstrate that:

1. Proper oxygen equipment and its use will protect an individual from oxygen deficiency.
2. An individual can experience and recognize symptoms that will be the same as those found in actual flight and therefore take the necessary action to prevent loss of judgment and consciousness.
3. Decompression is not dangerous provided proper supervision is present, and proper actions are planned and taken when necessary.

WHAT ARE THE PREREQUISITES FOR TRAINING?

Personnel must have a valid FAA medical certificate. A fee of twenty dollars is required. The applicant must be eighteen years of age or older.

HOW DO YOU APPLY FOR TRAINING?

All requests for the training course must be coordinated with:

FAA Airman Education Section (AAC–142)

Civil Aeromedical Institute

P.O. Box 25082

Oklahoma City, Oklahoma 73125

HOW CAN YOU GET FURTHER INFORMATION?

Write to the Airman Education Section at the above address, or phone (405) 686-4837.

SERVICING THE OXYGEN SYSTEM

The oxygen system is serviced by a filler valve accessible by removing an access plate on the right side of the aft fuselage (see Figure 17-8). The system has two pressure gages, one on the right subpanel in the crew compartment for in-flight use, and one adjacent to the filler valve for checking system pressure during filling. A shutoff valve and regulator on the cylinder control the flow of oxygen to the crew and passenger outlets. The shutoff valve is actuated by a push-pull control located aft of the overhead light control panel in the cockpit. The regulator is a constant-flow type which supplies low-pressure oxygen through system plumbing to the outlets.

The following precautions should be observed when purging or servicing the oxygen system:

1. Avoid any operation that could create sparks. Keep all burning cigarettes or fire away from the vicinity of the airplane when the outlets are in use.
2. Inspect the filler connection for cleanliness before attaching it to the filler valve.
3. Make sure that your hands, tools, and clothing are clean, particularly of grease or oil stains. These contaminants are extremely dangerous in the vicinity of oxygen.
4. As a further precaution against fire, open and close all oxygen valves slowly during filling.

FILLING THE OXYGEN SYSTEM

When filling the oxygen system, only use aviator's breathing oxygen (MIL-0-27210).

WARNING

DO NOT USE MEDICAL OXYGEN. It contains moisture which can cause the oxygen valve to freeze.

Fill the oxygen system slowly by adjusting the recharging rate with the pressure regulating valve on the servicing cart, because the oxygen, under high pressure, will cause excessive heating of the filler valve. Fill the cylinder (22-cubic-foot cylinder installation) to a pressure of 1,800 ±50 psi at a temperature of 70°F. This pressure may be increased an additional 3.5 psi for each degree of increase in temperature; similarly, for each degree of drop in temperature, reduce the pressure for the cylinder by 3.5 psi. The oxygen system, after filling, will need to cool and stabilize for a short period before an accurate reading on the gage can be obtained. The 49- or 66-cubic-foot cylinders may be charged to a pressure of 1,850 ±50 psi at a temperature of 70° F.

When the system is properly charged, disconnect the filler hose from the filler valve and replace the protective cap on the filler valve.

KING AIR C90A AND C90B CAPACITY

Oxygen for unpressurized, high-altitude flight is supplied by a cylinder in the compartment immediately aft of the pressure bulkhead (see Figure 17-5). A 22-, 49-, or 66-cubic-foot cylinder may be installed.

OXYGEN CYLINDER RETESTING

Oxygen cylinders used in the airplane are of two types. Lightweight cylinders, stamped "3HT" on the plate on the side, must be hydrostatically tested every three years and the test date stamped on the cylinder. This bottle has a service life of 4,380 pressurizations or 15 years, whichever occurs first, and then must be discarded. Regular weight cylinders, stamped "3A," or "3AA," must be hydrostatically tested every five years and stamped with the retest date. Service life on these cylinders is not limited.

QUESTIONS

1. When selected to 100%, the number of crew masks in use, to be used for computing oxygen duration is:

A. Halved
B. Counted once
C. Doubled
D. Tripled

2. The crew diluter-demand, quick-donning mask should be set to NORMAL:

A. At all times.
B. At altitudes below 20,000 ft.
C. At altitudes above 20,000 ft.
D. Anytime there is smoke in the cockpit.

3. The passenger masks are deployed:

A. Automatically when the cabin altitude exceeds 12,500 ft.
B. By pulling the PASSENGER MANUAL DROPOUT handle.
C. Manually by the passengers
D. Automatically when the Oxygen system is armed.

APPENDIX A
SYMBOLS, ABBREVIATIONS, AND TERMINOLOGY

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APPENDIX A
SYMBOLS, ABBREVIATIONS, AND TERMINOLOGY

AIRSPEED

CAS—Calibrated airspeed is the indicated airspeed of an airplane corrected for position and instrument error. Calibrated airspeed is equal to true airspeed in standard atmosphere at sea level.

GS—Groundspeed is the speed of an airplane relative to the ground.

IAS—Indicated airspeed is the speed of an airplane as shown on the airspeed indicator when corrected for instrument error. IAS values published in this training manual assume zero instrument error.

KCAS—Calibrated airspeed expressed in knots.

KIAS—Indicated airspeed expressed in knots.

M—Mach number is the ratio of true airspeed to the speed of sound.

TAS—True airspeed is the airspeed of an airplane relative to undisturbed air, which is the CAS corrected for altitude, temperature, and compressibility.

VYSE—Best single-engine rate-of-climb speed.

V_A—Maneuvering speed is the maximum speed at which application of full available aerodynamic control will not overstress the airplane.

VF—Design flap speed is the highest speed permissible at which wing flaps may be actuated.

V_{FE}—Maximum flap extended speed is the highest speed permissible with wing flaps in a prescribed extended position.

V_{LE}—Maximum landing gear extended speed is the maximum speed at which an airplane can be safely flown with the landing gear extended.

V_{LO}—Maximum landing gear operating speed is the maximum speed at which the landing gear can be safely extended or retracted.

V_{MCA}—Air minimum control speed is the minimum flight speed at which the airplane is directionally controllable, as determined in accordance with Federal Aviation Regulations. The airplane certification conditions include: one engine becoming inoperative and windmilling, a 5° bank toward the operative engine, takeoff power on operative engine, landing gear up, flaps in takeoff position, and most rearward CG. For some conditions of weight and altitude, stall can be encountered at speeds above VMCA, as established by the certification procedure described above, in which event stall speed must be regarded as the limit of effective directional control.

V_{MCG}—Ground minimum control speed.

V_{MO}/M_{MO}—Maximum operating limit speed is the speed limit that may not be deliberately exceeded in normal flight operation. V is expressed in knots and M in Mach number.

V_R—Decision speed/rotation speed.

V_S—Stalling speed or the minimum steady flight speed at which the airplane is controllable.

V_{SO}—Stalling speed or the minimum steady flight speed at which the airplane is controllable in the landing configuration.

V_{SSE}—Intentional one-engine-inoperative speed is a speed above both VMCA and stall speed, selected to provide a margin of lateral and directional control when one engine is suddenly rendered inoperative. Intentional failing of one engine below this speed is not recommended.

V_X—Best angle-of-climb speed is the airspeed which delivers the greatest gain of altitude in the shortest possible horizontal distance.

V_Y—Best rate-of-climb speed is the airspeed which delivers the greatest gain in altitude in the shortest possible time.

METEOROLOGICAL

Altimeter setting—Barometric pressure corrected to sea level.

Indicated pressure altitude—The number actually read from an altimeter when the barometric subscale has been set to 29.92 inches of mercury (1013.2 millibars).

IOAT—Indicated outside air temperature is the temperature value read from an indicator.

ISA—International standard atmosphere in which:

- Air is a dry, perfect gas.
- Temperature at sea level is 59° Fahrenheit (15° Celsius).
- Pressure at sea level is 29.92 inches of mercury (1013.2 millibars).
- Temperature gradient from sea level to the altitude at which the temperature is –69.7°F (–56.5°C), is –0.003566°F (–0.00198°C) per foot, and is zero above the altitude.

OAT—Outside air temperature is the free air static temperature, obtained either from the temperature indicator (IOAT) adjusted for compressibility effects or from ground meteorological sources.

Pressure altitude—Altitude measured from standard sea level pressure (29.92 inches Hg) by a pressure (barometric) altimeter. It is the indicated pressure altitude corrected for position and instrument error. In this training manual, altimeter instrument errors are assumed to be zero. Position errors may be obtained from the altimeter correction graphs.

Station pressure—Actual atmospheric pressure at field elevation.

Temperature compressibility effects—An error in the indication of temperature caused by airflow over the temperature probe. The error varies, depending on altitude and airspeed.

Wind—The wind velocities recorded as variables on the charts of this training manual are to be understood as the headwind or tailwind components of the reported winds.

POWER

Beta range—The range of propeller blade angle control from the primary low-pitch-stop blade angle setting to the full-reverse blade angle setting.

Cruise climb—Cruise climb is the maximum power approved for normal climb. These powers are torque or temperature (ITT) limited.

High idle—High idle is obtained by placing the condition lever in the HIGH IDLE position. This limits the power operation to a minimum of 70% of N_1 rpm.

Low idle—Low idle is obtained by placing the condition lever in the LOW IDLE position. This limits the power operation to a minimum of 51% of N_1 rpm.

Maximum continuous power—Maximum continuous power is the highest power rating not limited by time. Use of this rating is at the discretion of the pilot.

Maximum cruise power—Maximum cruise power is the highest power rating for cruise and is not time-limited.

Reverse—Reverse thrust is obtained by lifting the power levers and moving them into the Beta plus power range.

SHP—Shaft horsepower.

Minimum takeoff power—Minimum takeoff power is the minimum power which must be available for takeoff without exceeding the engine limitations.

Takeoff power—Takeoff power is the maximum power rating. Use of this rating should be limited to normal takeoff operations and emergency situations.

CONTROL AND INSTRUMENT

Condition lever (fuel shutoff lever)—The fuel shutoff lever actuates a valve in the fuel control unit which shuts off the fuel at the fuel control outlet and regulates the idle range from low to high idle.

ITT (interstage turbine temperature)—Eight probes, wired in parallel, sense the temperature between the compressor and power turbines, and send the reading to the ITT indicator in degrees centigrade x 100.

N1 tachometer (gas generator rpm)—The N1 tachometer registers the rpm of the gas generator in percent, with 100% representing a gas generator speed of approximately 37,500 rpm.

Power lever (gas generator N1 rpm)—The power lever serves to modulate engine power from full reverse thrust to takeoff. The position for idle represents the lowest recommended level of power for flight operation.

Propeller control lever (NP rpm)—The propeller control is used to control the rpm setting of the propeller governor. Movement of the lever results in an increase or decrease in propeller rpm. Propeller feathering is the result of lever movement beyond the detents at the low rpm end of the lever travel.

Propeller governor—The propeller governor senses changes in rpm and hydraulically changes propeller blade angle to compensate for the changes in rpm. Constant propeller rpm is thereby maintained at the selected rpm setting.

Torquemeter—The torquemeter system indicates the shaft output torque. Differential pressure from the mechanism within the reduction gearcase causes a bellows and servo system to indicate torque on a meter. Instrument readout is in foot-pounds.

GRAPH AND TABULAR

Accelerate-go-Accelerate—go is the distance to accelerate to takeoff decision speed (V_R), experience an engine failure, continue accelerating to liftoff, then climb and accelerate in order to achieve takeoff safety speed (V_{YSE}) at 50 feet above the runway, for C90A aircraft and 35 feet for C90B aircraft.

Accelerate-stop—Accelerate-stop is the distance to accelerate to takeoff decision speed (V_R) and then bring the airplane to a stop.

AGL—Above ground level.

Best angle-of-climb—The best angle-of-climb delivers the greatest gain of altitude in the shortest possible horizontal distance with gear and flaps up.

Best rate-of-climb—The best rate-of-climb delivers the greatest gain of altitude in the shortest possible time with gear and flaps up.

Clearway—A clearway is an area beyond the airport runway not less than 500 feet wide, centrally located about the extended centerline of the runway, and under the control of the airport authorities. The clearway is expressed in terms of a clear plane, extending from the end of the runway with an upward slope not exceeding 1.25%, above which no object nor any terrain protrudes. However, threshold lights may protrude above the plane if their height above the end of the runway is 26 inches or less and if they are located to each side of the runway.

Climb gradient—Climb gradient is the ratio of the change in height during a portion of a climb to the horizontal distance traversed in the same time interval.

Demonstrated crosswind—Demonstrated crosswind is the demonstrated crosswind component for which adequate control of the airplane during takeoff and landing was actually demonstrated during certification; however, this is not considered a limitation.

MEA—Minimum enroute altitude.

Net gradient of climb—Net gradient of climb is the gradient of climb with the flaps in the takeoff position and the landing gear retracted. "Net" indicates that the actual gradients of climb have been reduced by 8% to allow for turbulence and pilot technique. The net gradient of climb graphs are constructed so that the value(s) obtained using the airport pressure altitude and outside air temperature will be the average gradient from 35 feet above the runway up to 1,500 feet above the runway.

Route segment—Route segment is a part of a route. Each end of that part is identified by a:

- Geographic location, or
- Point at which a definite radio fix can be established

Takeoff flight path—Takeoff flight path is the minimum gradient of climb required to clear obstacles in excess of 50 feet, measured horizontally from reference zero and vertically at the altitude above the runway. Reference zero is the point where the airplane has reached 50 feet above the runway, as determined from the accelerate-go graphs.

WEIGHT AND BALANCE

Approved loading envelope—Those combinations of airplane weight and center of gravity which define the limits beyond which loading is not approved.

Arm—Arm is the distance from the center of gravity of an object to a line about which moments are to be computed.

Basic empty weight—Basic empty weight is the weight of an empty airplane, including full engine oil and unusable fuel. This equals empty weight plus the weight of unusable fuel, and the weight of all the engine oil required to fill the lines and tanks. Basic empty weight is the basic configuration from which loading data is determined.

Center of gravity—Center of gravity is the point at which the weight of an object may be considered concentrated for weight and balance purposes.

CG limits—CG limits are the extreme center-of-gravity locations within which the airplane must be operated at a given weight.

Datum—Datum is a vertical plane perpendicular to the airplane's longitudinal axis from which fore and aft (usually aft) measurements are made for weight and balance purposes.

Empty weight—Empty weight is the weight of an empty airplane before any oil or fuel has been added. This includes all permanently installed equipment, fixed ballast, full hydraulic fluid, full chemical toilet fluid, and all other operating fluids full, except that the engines, tanks, and lines do not contain any engine oil or fuel.

Engine oil—Total system oil, including that portion of the engine oil which cannot be drained from the engine.

Jack point—Jack points are points on the airplane identified by the manufacturer as suitable for supporting the airplane for weighing or other purposes.

Landing weight—Landing weight is the weight of the airplane at landing touchdown.

Leveling points—Leveling points are those points which are used during the weighing process to level the airplane.

Maximum weight—Maximum weight is the greatest weight allowed by design, structural, performance, or other limitations.

Maximum zero fuel weight—Any weight above the value given must be loaded as fuel.

Moment—Moment is a measure of the rotational tendency of a weight, about a specified line, mathematically equal to the product of the weight and the arm.

Payload—Payload is the weight of occupants, cargo, and baggage.

PPH—Pounds per hour.

Ramp weight—Ramp weight is the airplane weight at engine start, assuming all loading is completed.

Station—Station is the longitudinal distance from some point to the zero datum or zero fuselage station.

Takeoff weight—Takeoff weight is the weight of the airplane at liftoff from the runway.

Tare—Tare is the apparent weight of any items (wheel chocks, jack stands, etc.) used on the scales but which are not a part of the airplane weight.

Unusable fuel—Unusable fuel is the fuel remaining after consumption of usable fuel.

Usable fuel—Usable fuel is that portion of the total fuel which is available for consumption as determined in accordance with applicable regulatory standards.

Useful load—Useful load is the difference between the airplane ramp weight and the basic empty weight.

Zero fuel weight—Zero fuel weight is the airplane ramp weight minus the weight of fuel on board.

AVIONICS

ADF mode—A mode of automatic direction finder operation allowing the ADF needle to point to the station.

NOTE

In this mode of operation, on many receivers the audio fidelity is severely limited.

Air data computer—An electronic system primarily designed to gather information for an autopilot flight director system with outputs relating to pitot and static data. Possible information from this system includes: pressure altitude, indicated airspeed, total air temperature, static air temperature, and other information related to autopilot operation.

Altitude alert light—An amber light associated with an altitude alerter system. This light will be illuminated prior to intercepting a preselected altitude, or if for any reason the aircraft strays beyond a preset limit from the selected altitude once the aircraft has intercepted the altitude.

Altitude preselector—An autopilot flight director subsystem that allows a pilot to preselect the altitude to which he desires to climb or descend. The controlling mechanism for an altitude preselect system is normally combined with the same device which controls the altitude alerter system.

Amplifier—A basic type of electronic device that seeks to make an electrical signal greater in strength. A public address system, for instance, is a type of amplifier. Amplifying devices are typically tubes or transistors.

Analog—A type of electronic circuitry that is characterized by smooth, continuous operation rather than discrete steps, as would be observed with digitally operated equipment.

Angle-of-attack (AOA) indicator—A supplemental flight instrumentation system that attempts to read out to the pilot the angle-of-attack or deck angle information. Several variations of this system are available.

Angular deviation—A means of showing displacement from a selected course either to or from a VOR station, TACAN, or NDB, showing displacement from the desired course in terms of angle. This is commonly used with the VOR system having a normal course width of 10° on each side of the course.

Annunciator—An indicator light with a message. An annunciator makes an announcement as to the specific status of a system or subsystem.

ANT (antenna) mode—This mode of ADF operation allows improved audio fidelity in order to listen to the music or voice programs of an AM broadcast station. However, the ADF needle operation is defeated in this mode of operation.

Area navigation system (RNAV)—A system of direct point-to-point navigation having four further subclassifications:

- **Course line computer**—A computer, utilizing information from VOR and colocated DME stations, that allows the operator to change the location of the VOR station from its physical position to wherever the operator wants.
- **OMEGA/VLF** system—See related definition.
- **Inertial navigation system**—See related definition.
- **Loran system**—Operationally similar to an OMEGA/VLF system.

Asymptotic—A design characteristic of an autopilot or flight director system. The function of this characteristic is to allow the autopilot to attempt to intercept a given course or altitude without overshooting. This is done by continuously reducing the intercept angle as the aircraft approaches the selected ground track or altitude.

Attenuation—The process of electrically reducing the size of a radio or audio signal (i.e., to turn down or make smaller.)

Attitude director indicator (ADI) (flight director indicator)—This instrument combines the basic functions of an attitude indicator with the steering commands received from the flight director system.

Attitude indicator (artificial horizon)—A gyroscopically controlled instrument used to display the aircraft's pitch and roll attitude relative to the earth's surface. The gyro used to display this information may be contained within the case of the displayed instrument, or it may receive its information from a remotely located attitude gyro.

Audio filters—An electronic means of removing a portion of the audio which the pilot does not desire to listen to. The pilot may choose to remove either the voice portion or the Morse code identifier of a VOR or an ADF system.

Audio selector switches—The system of switches which allows one or several audio systems to be "piped-in" to the speaker or headphones of an aircraft.

Audio system—The electronic system that serves as a switchboard and amplification system for the varied receivers that require the audio to be funneled to the speaker or headphones.

Autopilot/flight director modes:

- **Vertical mode**s—Vertical modes control changes in the pitch attitude of the aircraft using the elevator servo. Examples of vertical modes are: altitude hold, altitude preselect, indicated airspeed hold, and vertical speed hold.
- **Lateral modes**—Lateral modes control autopilot operation by controlling the aileron and rudder servos. Examples of lateral modes are: heading hold, navigation modes and submodes (e.g., en route nav tracking, approach, backcourse, etc.).

- **Autopilot/flight director submode**—Generally speaking, this concept represents two subclassifications of operation within a given mode. Namely:
- **Arm**—The process of activating a system or preparing it to operate at a future time. For instance, if you push the Nav button to track a specific radial from a VOR station, but the CDI needle is displaced full-scale to the left or right at the moment you push the Nav button, then the autopilot flight director system will initially be activated in the nav-arm mode while the aircraft continues to intercept the selected radial.
- **Capture**—A submode allowing the autopilot flight director system to track a specified lateral or vertical reference (e.g., altitude or glide slope as a vertical mode; VOR or localizer course as a lateral mode).

Avionics master circuit breaker—The circuit breaker that supplies power to the avionics master switch in Beechcraft factory-installed avionics packages on Baron, Bonanza, Duke, and King Air installations. This circuit breaker serves as a backup means of activating the avionics system should the avionics master switch fail for any reason.

Avionics master switch—A central on/off power switch for the entire avionics package in an aircraft. This switch conveniently allows the pilot to turn on the entire avionics package by turning on only one switch.

Beat frequency oscillator (BFO)—A device used on an ADF receiver that generates a tone allowing the pilot to identify the Morse code being transmitted by some nondirectional beacons.

NOTE

This type of transmitter is not employed in the United States.

Carrier—That portion of the transmitted radio energy which "carries" the useful information (i.e., modulation).

Compass system slaving—The process of automatically aligning the directional gyro in a compass system with the earth's magnetic field to display the aircraft's magnetic heading. When the compass system is initially powered, slaving occurs at a fast rate to quickly align the compass system with magnetic north. Once the fast-slaving rate is accomplished, the system automatically goes into a slow-slaving rate for continuous operation. It will correct for precession errors of the compass system up to a maximum error of about 3° per minute.

Concentric—Two or more knobs mounted on one common system of shafts having the same axis. For example, most frequency selector knobs used in all avionics systems employ concentric knobs in the interest of conserving panel space.

Course deviation indicator (CDI)—An indicator used with a VOR/localizer receiver that shows only left/right deviation and to/from information. This instrument has a knob called an OBS knob, meaning "omni bearing selector," which allows the pilot to choose the course to or from a VOR station.

Course knob—The name applied to the omni bearing selector on an HSI type of instrument. The course knob is attached to the resolver and a course pointer on the HSI indicator.

Course width—Displacement left or right of the desired course:

- **Angular**—Degrees left or right of the desired course. Course width using the VOR system is 10° on each side of the desired course.
- **Linear**—In the "en route mode" most course line computer RNAVs have a course width of ±5 nautical miles. In the approach mode most course line computer RNAVs have a course width of ±1 1/4 nautical miles on each side of the centerline.

Cross sidetone—Sending sidetone audio across the cockpit from one side to the other; for example, this allows the pilot to hear what the copilot is saying on the transmitter. Cross sidetone may be heard through either the phones or the speaker.

Digital—A type of electronic circuitry technology that operates in specific steps, as opposed to the smooth, sweeping type of operation employed in analog.

Double-cue flight director system (cross pointer)—A command presentation system using one vertical bar to indicate commanded roll-attitude instructions and one horizontal bar to indicate commanded pitch instructions.

Electronic flight instrument system (EFIS)—A type of flight instrumentation system employing cathode ray tubes (television screens) to display information.

Emergency/normal switch—In the event of the failure of the audio system, this switch (when placed in the emergency position) allows audio from the aircraft receivers to be "piped" directly to the headphones.

Fast erect—A mode of operation whereby an attitude indicator may be quickly realigned with the earth's horizon if for any reason the gyro has precessed or tumbled.

Flux valve (flux gate)—A component of a slaved compass system that senses the earth's magnetic field and converts this information into an electrical signal representing magnetic north.

Free operation—A mode of operation for a slaved compass system whereby the directional gyro is disconnected from the slaving system. Normally this would be used when the slaving system fails or for operation in the polar regions where the earth's magnetic field will not permit normal slaved operation. The concept here is that the directional gyro is free of its master, magnetic north.

Gain—The relative amount of amplification of a radio receiver. A gain control is commonly used on a radar indicator to control the relative amount of amplification of the received radar echo. This allows the pilot to optimize the information displayed, especially when the radar is used for terrain mapping purposes.

Go-around mode—An autopilot flight director mode intended to be used during a missed approach. This mode will command a pitch-up attitude appropriate for a climbout with an associated wings-level command. The autopilot may or may not remain engaged during the go-around mode, depending upon the type of autopilot installed in a specific aircraft.

Gyro erection—The process of an attitude gyro becoming aligned with the earth's horizon or, viewed in another way, aligned with true vertical. This happens automatically when the system first receives power.

Half bank—An autopilot mode of operation whereby the bank angle is limited during turns in such a way that the aircraft will only bank approximately half as much as normal. This is designed to give the passengers the perception of a smoother ride with no steep banks.

Heading bug—An adjustable marker used on a heading indicator to direct an autopilot and/or flight director system according to the magnetic heading the pilot desires to fly. Also, this device may be used simply as a reminder to the pilot of what heading he is to fly when not using the autopilot flight director system.

Heading indicator (directional gyro)—A gyroscopically controlled instrument used to display an aircraft's heading relative to magnetic north. The compass card of this indicator may be receiving the information which it displays from a remotely located gyro and an associated slaved compass system.

Hertz—The unit of measure used to describe the number of cycles of alternating current per second.

Horizontal situation indicator (HSI)—This instrument, alternately called a CDI by some manufacturers, displays heading information from a compass system, left/right and to/from information from a VOR/localizer receiver, and deviation above and below a glide slope from a glide-slope receiver. The pilot's workload is reduced by integrating these displays onto one indicator.

Inertial navigation system—This system allows direct point-to-point navigation via a great circle route. This system is completely self-sufficient, utilizing a group of gyros and accelerometers to sense movement along the earth's surface.

Integrated autopilot/flight director system—A system utilizing both autopilot and flight direction information to respond to selected modes.

Interrogation—In the secondary surveillance radar system the ground-based radar unit is said to "interrogate" the transponders of all aircraft flying within reception range of that radar. Once a transponder has been interrogated, it should reply to the ground radar unit by sending a brief transmission of radio energy. For general aviation aircraft a transponder may be interrogated in both modes. A mode supplies azimuth and distance information, and altitude information is provided through mode C.

Keying—The process of turning on the transmitter by means of the push-to-talk button located on the microphone or the control wheel.

Latitude—The angular displacement of a geographic location north or south of the equator. This is normally expressed in terms of degrees, minutes, and tenths of minutes.

Linear deviation—A means of showing lateral displacement from the desired navigational course calibrated in miles. Linear deviation allows for parallel course boundaries whether far away from or near a station.

Longitude—The angular displacement of a geographic location east or west of the prime meridian located in Greenwich, England. This is normally expressed in terms of degrees, minutes, and tenths of minutes.

Magnetic bearing—The direction of a nondirectional beacon (NDB) or VOR station relative to magnetic north.

Meter movement—An application of an ammeter used in any instrumentation system to show deviation such as left/right, to/from, slaving indicator, etc.

Mode—One of several operating conditions of a system. For instance, most airborne weather radars have both weather mapping and terrain mapping modes of operation.

Mode A—That portion of the transponder reply which transmits azimuth and distance information for the radar controller.

Mode C—The portion of a transponder reply containing the pressure altitude of an aircraft as provided by an encoding altimeter.

Modulation—The addition of useful information to the carrier wave that is emitted from a transmitter; for example, talking into the microphone or the transmission of the Morse code identification from a VOR station.

Muting—The silencing of incoming receiver audio while one is transmitting.

Nonintegrated AP/FD system—Two separate flight control systems, each using its own computer. Information coming from these two systems may or may not agree at any given time.

OMEGA/VLF system—A world-wide navigation system that allows direct great circle flight from one point to another. This system utilizes

U.S. Navy VLF communication transmitters and the OMEGA system of navigation.

Parallax error—A problem that can cause inaccurate interpretations of an instrument reading. It is caused by the user's viewing angle not being directly in line with the instrument.

Parallax error adjustment—An adjustment of some single-cue flight director systems which allows the command bars to be adjusted up or down in order to "nestle" just above the aircraft symbol.

Parked or stowed ADF needle—The process of placing the ADF needle at the 3 o'clock relative bearing location to indicate that the ADF unit is in the antenna mode and that the needle is not operating.

Radio magnetic indicator (RMI)—The combined display of magnetic heading from a compass system and relative bearing to a nondirectional beacon or VOR, which results in displaying the product, called "magnetic bearing," to or from the station.

Range filter—An audio filter designed to remove the Morse code identification from a radio transmission. Actually, range is something of a misnomer as pilots know it today-you should think of this as being an "ident" filter.

Relative bearing—The direction of a nondirectional beacon relative to the longitudinal axis of the aircraft.

Relay—An application of an electromagnet to perform switching duties. A relay may be used to switch large quantities of current. A multiple pole relay will allow a single pole switch to switch many circuits from a remote location.

Remote mounted avionics—Avionic equipment which is not fully self-contained and mounted on the instrument panel. Typically, the "black boxes" for these systems are located in the forward avionics compartment, forward of the front pressure bulkhead, or aft of the rear pressure bulkhead.

Resolver—The electronic device to which the course knob or OBS knob is attached. This device communicates the desired course, which the pilot selects, to the VOR receiver.

Servo system—Using an electric motor in any one of several applications to reduce pilot workload or allow automatic operation of some systems; for example: autopilot servos, electric elevator trim servos, servoed altimeters, compass systems, etc.

Sidetone—The ability to hear oneself talk while transmitting. The sidetone may be heard through either the headphones or the speaker. Additionally, sidetone may be considered as a means of verifying normal transmitter and receiver operation. If the receiver and transmitter are working properly, the sidetone will sound "normal." If either the transmitter or receiver is malfunctioning, the sidetone will sound weak or garbled.

Single-cue flight director system (V-bar system)—A command display system using a pair of bars which work in unison to display the commanded attitude to the pilot.

Slant/range correction—A means of correcting for the inherent error in raw slant/range data which will result in a true lateral distance from the aircraft to the DME station. Many of the more sophisticated RNAV computers provide slant/range correction.

Slant/range distance—Conventional, uncorrected DME distance to the station.

Slaved compass system—A directional gyro system that is automatically synchronized to the magnetic heading of the aircraft. The concept of this system is that magnetic north is the master; therefore, the compass system is its slave.

Slaved operation—The normal mode of a slaved compass system whereby the directional gyro automatically remains synchronized to magnetic north. This type of operation continually compensates for gyro precession and other compass system errors. The concept is that the compass system is a slave to magnetic north.

Slaving amplifier—An amplifier which takes the weak signal representing magnetic north, coming from the flux valve, and boosts that signal to a usable level in order to drive the directional gyro to the proper magnetic heading.

Slaving indicator—A meter used in some slaved compass systems that displays the difference between sensed magnetic heading and displayed magnetic heading. If the needle on this indicator is centered, there is no error between sensed and indicated magnetic heading. If the needle is

off to the left or right, a small amount of error is indicated. Normal operation of the compass system causes the needle to sway to the left and right because of gyro precession and other factors.

Soft ride—A mode for an autopilot whereby the responsiveness of the autopilot to rough air is altered in such a way that the ride is perceived to be much smoother than it is.

Squelch—A silencing circuit employed in communication receivers that allows undesirable background noise to be omitted. Only a strong incoming signal from a transmitter will be heard.

Transponder code—A specific four-digit code that may be selected by the pilot on his transponder to identify his specific aircraft.

NOTE

A common misconception is that this control changes the transponder reply frequency. The transponder always operates on the same frequency.

Voice filter—An audio filter designed to remove the voice portion of a received radio transmission.

Voice terrain advisories—Voice callouts of pertinent altitude-above-ground information. The information announced will be determined by the type of system installed.

Waypoint—The geographic location of navigational fix used in area navigation. This may be used in either a VLF/OMEGA system or a VOR/DME system utilizing a course line computer.

Waypoint address—The radial and distance of a waypoint from a VORTAC.

Waypoint coordinates—The latitude and longitude of the waypoint used with a VLF/OMEGA system.

Yaw damperA system connected to the rudder servo that seeks to dampen or reduce oscillations of the aircraft about the yaw axis. The yaw damper system significantly reduces the level of motion sickness experienced by passengers. This system should be engaged soon after takeoff and under normal operations should remain engaged until just prior to landing.

APPENDIX B
ANSWERS TO QUESTIONS

Section 2—Electrical

1. C
2. C
3. D
4. B
5. C
6. B
7. A
8. B
9. D
10. B
11. C
12. C
13. C
14. B
15. C

Section 3—Lighting

1. D
2. C
3. A
4. C
5. C
6. C

Section 4—Master Warning

1. C
2. D
3. B
4. A

Section 5—Fuel

1. C
2. C
3. A
4. C
5. D
6. B
7. B
8. C
9. B

Section 7—Powerplant

1. B
2. C
3. D
4. D
5. B
6. D
7. B
8. A
9. D

Section 8—Fire Protection

1. D
2. A
3. C

Section 9—Pneumatics

1. C
2. B
3. C
4. D
5. A

Section 10—Ice and Rain

1. C
2. C
3. B
4. A
5. D
6. A
7. D
8. C
9. D
10. D
11. D
12. D

Section 11—Air Conditioning

1. A
2. C
3. B
4. A
5. D
6. B
7. D

Section 12—Pressurization

1. B
2. B
3. D
4. B
5. C

Section 14—Landing Gear

1. C
2. A
3. A
4. D
5. C
6. B
7. A
8. D
9. B

Section 15—Flight Controls

1. A
2. C
3. C
4. C
5. A

Section 16—Avionics

1. C
2. B
3. A
4. C
5. D
6. A
7. B
8. C
9. D
10. A
11. D

Section 17—Oxygen

1. C
2. B
3. C

ANNUNCIATORS

The Annunciators section presents a color representation of all the annunciator lights in the airplane.

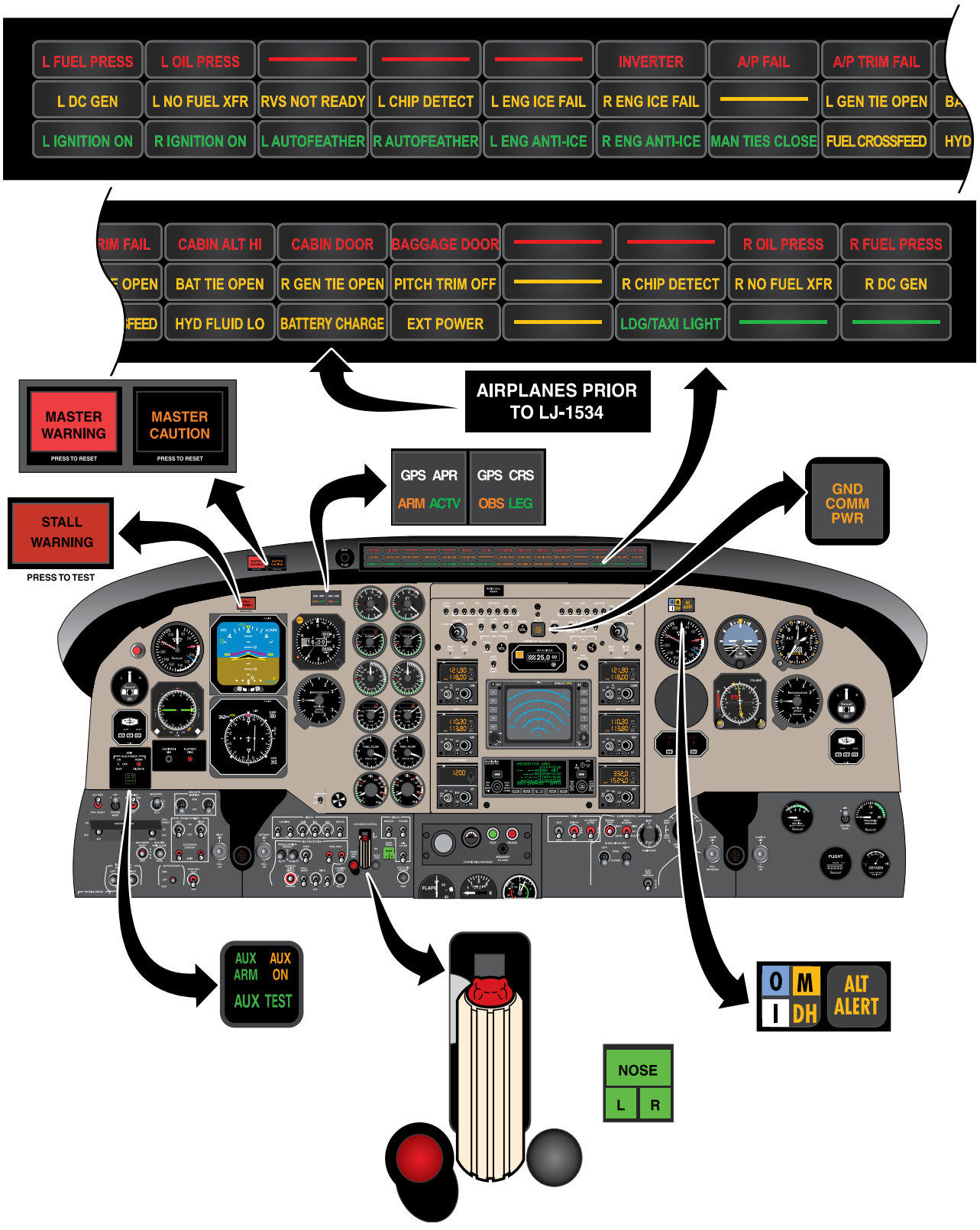

Figure ANN-1. Annunciators—King Air C90B (SNs LJ-1353 to LJ-1533)